Introduction to Statistics: A Simulation-First Approach

An Open Educational Resource

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2026-03-24

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Welcome

This is an open educational resource (OER) for introductory statistics, built on a **simulation-first** pedagogy. Students develop statistical reasoning through hands-on simulation before encountering formal mathematical theory.

Why simulation first?

Traditional introductory statistics courses begin with probability rules and distribution theory, then use those tools to build confidence intervals and hypothesis tests. Students memorize formulas without understanding *why* they work.

This textbook reverses that order:

1. **Explore data** — See patterns, describe distributions, visualize relationships.
2. **Simulate inference** — Use resampling (bootstrap, permutation) to understand *what inference is* and *why it works*.
3. **Approximate with theory** — Learn that the normal and t-distributions are mathematical shortcuts for what simulation does directly.
4. **Formalize probability** — Study probability rules and random variables *after* students already have deep intuition about randomness.
5. **Model relationships** — Apply all inference tools to regression and categorical data.

By the time students encounter the Central Limit Theorem, they have already *built* sampling distributions hundreds of times. The theorem becomes an elegant confirmation of what they've seen, not an abstract declaration to accept on faith.

How to use this book

Throughout this text, **StatLens** activities invite you to explore statistical concepts interactively. Look for boxes like this one. Each activity links to a browser-based simulation tool — no software installation required.

Launch StatLens to explore all available tools.

Guided Practice exercises appear throughout each chapter with solutions in footnotes — try them before peeking.

Worked Examples walk through complete analyses step by step.

Definitions highlight key statistical vocabulary.

About this book

This textbook remixes material from two outstanding open-source statistics textbooks:

- Introduction to Modern Statistics (2nd edition) by Mine Çetinkaya-Rundel and Johanna Hardin
- OpenIntro Statistics by David Diez, Mine Çetinkaya-Rundel, and Christopher Barr

Both are licensed under Creative Commons BY-SA 3.0, and this remix is shared under the same license.

Interactive simulations are powered by StatLens, an open-source statistics visualization toolkit developed at the University of Wisconsin-La Crosse.

License

Preface

For instructors

This textbook is designed around the **simulation-based inference** (SBI) approach popularized by Lock, Lock, Lock, Lock, and Lock in *Statistics: Unlocking the Power of Data*. The pedagogical sequence is:

EDA → Simulation inference → Theoretical inference → Probability → Modeling

The book contains more material than most one-semester courses will cover. The 28 chapters are organized into six parts, and instructors should select the chapters that fit their curriculum. A typical 15-week course might cover:

- **Part I: Explore** (Chapters 1–5): Weeks 1–3
- **Part II: Simulate** (Chapters 6–9): Weeks 4–6
- **Part III: Calculate** (Chapters 10–13): Weeks 7–9
- **Part IV: Relationships** (Chapters 14–20): Weeks 10–13
- **Part V: Foundations** (Chapters 21–25): Select chapters as needed
- **Part VI: Design** (Chapters 26–28): As time permits

Parts V (probability foundations) and VI (power/sample size) are included for courses that need them but can be omitted without loss of continuity.

StatLens integration

StatLens is a free, browser-based statistics toolkit that supports the simulation-first approach. The text features StatLens throughout but is written to be self-contained — instructors who prefer StatKey, R's `infer` package, or other tools can substitute freely.

StatLens activities in this text use URL parameters for reproducibility. For example:

```
https://datascienceuwl.github.io/statlens/simulate/bootstrap-mean/?seed=42&n=30
```

The `seed` parameter ensures every student who uses the same link gets identical results, enabling graded simulation-based assignments.

Accessibility

This textbook is designed to meet WCAG 2.1 AA accessibility standards:

- **Text**: Atkinson Hyperlegible font, designed for readers with low vision
- **Math**: MathJax rendering with screen reader support
- **Color**: All color choices meet 4.5:1 contrast ratio; information is never conveyed by color alone

- **Structure**: Semantic HTML with proper heading hierarchy
- **Images**: All figures include descriptive alt text
- **Navigation**: Full keyboard accessibility with visible focus indicators

If you encounter accessibility barriers, please file an issue on the book's GitHub repository.

Acknowledgments

This book remixes and augments material from two open-source textbooks published by the OpenIntro project: *Introduction to Modern Statistics* (IMS) by Mine Çetinkaya-Rundel and Johanna Hardin, and *OpenIntro Statistics* by David Diez, Mine Çetinkaya-Rundel, and Christopher Barr. Their commitment to open education made this work possible.

The simulation-first pedagogy is inspired by the Lock family's pioneering work in *Statistics: Unlocking the Power of Data*.

StatLens was developed at the University of Wisconsin-La Crosse with support from the Department of Mathematics and Statistics.

PART I

Explore

Before we can draw conclusions from data, we need to understand what data *is*, where it comes from, and what it looks like. We develop the vocabulary and visual tools for exploring data — identifying variable types, visualizing distributions, and describing relationships. These skills form the foundation for everything that follows.

Chapter 1

Introduction to Data

Scientists seek to answer questions using rigorous methods and careful observations. These observations — collected from the likes of field notes, surveys, and experiments — form the backbone of a statistical investigation and are called **data**. Statistics is the study of how best to collect, analyze, and draw conclusions from data. In this first chapter, we focus on both the properties of data and on the collection of data. We will introduce the building blocks of statistical thinking: observations, variables, data frames, and the types of questions that statistics can help us answer.

1.1 Case study: Using stents to prevent strokes

In this section we introduce a classic challenge in statistics: evaluating the efficacy of a medical treatment. The terms and ideas introduced here will be revisited throughout the course. For now, the goal is simply to get a sense of the role statistics can play in practice.

A stent is a small mesh tube that is placed inside a narrow or weak artery to assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. This leads to the principal question the researchers hoped to answer:

> Does the use of stents reduce the risk of stroke?

The researchers who asked this question conducted an experiment with 451 at-risk patients.[1] Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group.** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group.** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group. In this study, the control group provides a reference point against which we can measure the medical impact of stents in the treatment group.

[1] Chimowitz MI, Lynn MJ, Derdeyn CP, et al. 2011. Stenting versus Aggressive Medical Therapy for Intracranial Arterial Stenosis. *New England Journal of Medicine* 365:993–1003.

1.1.1 Examining the data

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment. Patient outcomes were recorded as "stroke" or "no event," representing whether the patient had a stroke during that time period.

It would be difficult to answer the research question by looking at individual patient records one at a time. Instead, we organize the results into a summary table that lets us quickly compare the two groups.

Table 1.1: Descriptive statistics for the stent study.

	30 days: Stroke	30 days: No event	365 days: Stroke	365 days: No event
Treatment	33	191	45	179
Control	13	214	28	199
Total	46	405	73	378

Of the 224 patients in the treatment group, 45 had a stroke by the end of the first year. Using these two numbers, compute the proportion of patients in the treatment group who had a stroke by the end of their first year. (Note: answers to all Guided Practice exercises are provided in footnotes!)

Show answer

The proportion of the 224 patients who had a stroke within 365 days: $45/224 = 0.20 = 20\%$.

1.1.2 Drawing conclusions from the data

We can compute **summary statistics** from Table 1.1 to give us a better idea of how the impact of the stent treatment differed between the two groups.

A **summary statistic** is a single number that summarizes data from a sample. Summary statistics condense a large amount of information into a form that is easy to digest and compare.

For instance, the primary results of the study after one year could be described by two summary statistics: the proportion of people who had a stroke in the treatment and control groups.

- Proportion who had a stroke in the treatment (stent) group: $45/224 = 0.20 = 20\%$.
- Proportion who had a stroke in the control group: $28/227 = 0.12 = 12\%$.

These two summary statistics are useful in looking for differences between the groups, and we are in for a surprise: an additional 8% of patients in the treatment group had a stroke! This is important for two reasons. First, it is contrary to what doctors expected, which was that stents would *reduce* the rate of strokes. Second, it leads to a statistical question: do the data show a "real" difference between the groups?

This second question is subtle. Suppose you flip a coin 100 times. While the chance a coin lands heads in any given coin flip is 50%, we probably will not observe exactly 50 heads. This type of variation is part of almost any type of data generating process. It is possible that the 8% difference in the stent study is due to this natural variation. However, the larger the difference we observe (for a particular sample size), the less believable it is that the difference is due to chance. So what we are really asking is the following: if in fact stents have no effect, how unlikely is it that we would observe such a large difference?

While we do not yet have the statistical tools to fully address this question on our own, we can state the conclusion of the published analysis: there was compelling evidence of harm by stents in this study of stroke patients.

Be careful about overgeneralizing. Do not generalize the results of this study to all patients and all stents. This study looked at patients with very specific characteristics who volunteered to be part of the study and who may not be representative of all stroke patients. In addition, there are many types of stents, and this study considered only the self-expanding Wingspan stent (Boston Scientific). However, the study does leave us with an important lesson: we should keep our eyes open for surprises.

Does this study prove that stents *cause* strokes? What features of the study design might strengthen or weaken this conclusion?

Show answer

Because patients were *randomly assigned* to treatment and control groups, this is a randomized experiment. Random assignment helps ensure the groups are similar in all respects except the treatment itself. This strengthens the case for a causal conclusion. However, we should still ask: could the difference (20% vs. 12%) have arisen by chance alone? We will develop tools to answer this question rigorously later in the course.

This case study highlights a key principle that will guide much of our work in this course: **a single study's results can be surprising, and statistics gives us the tools to evaluate whether an observed result is real or could be due to random chance**. We will return to this idea again and again as we develop the tools of statistical inference.

1.2 Data basics

Effective presentation and description of data is a first step in most analyses. This section introduces one structure for organizing data as well as some terminology that will be used throughout the course.

1.2.1 Observations, variables, and data matrices

An **observation** (also called a **case** or **unit of observation**) is a single entity in a dataset — one person, one county, one transaction. A **variable** is a characteristic that is measured or recorded for each observation. A **data frame** (or **data matrix**) organizes observations as rows and variables as columns. Each cell holds a single value.

Consider data for 50 randomly sampled loans offered through a peer-to-peer lending company. Each row in the table represents a single loan — that is the observation, or unit of observation. The columns represent characteristics of each loan, and each column is a variable. For example, the first row might represent a loan of $22,000 with an interest rate of 10.90%, where the borrower is based in New Jersey (NJ) and has an income of $59,000.

Table 1.2: Six observations from a loan dataset.

	loan_amount	interest_rate	term	grade	state	total_income	homeownership
1	22,000	10.90	60	B	NJ	59,000	rent
2	6,000	9.92	36	B	CA	60,000	rent
3	25,000	26.30	36	E	SC	75,000	mortgage
4	6,000	9.92	36	B	CA	75,000	rent
5	25,000	9.43	60	B	OH	254,000	mortgage
6	6,400	9.92	36	B	IN	67,000	mortgage

In practice, it is especially important to ask clarifying questions to ensure important aspects of the data are understood. For instance, it is always important to be sure we know what each variable means and its units of measurement. Table 1.3 describes each variable in the loan dataset.

Table 1.3: Variables and their descriptions for the loan dataset.

Variable	Description
loan_amount	Amount of the loan received, in US dollars.
interest_rate	Interest rate on the loan, in an annual percentage.
term	The length of the loan, which is always set as a whole number of months.
grade	Loan grade, which takes values A through G and represents the quality of the loan and its likelihood of being repaid.
state	US state where the borrower resides.
total_income	Borrower's total income, including any second income, in US dollars.
homeownership	Indicates whether the person owns, owns but has a mortgage, or rents.

What is the grade of the first loan in Table 1.2? And what is the homeownership status of the borrower for that first loan?

Show answer

The loan's grade is B, and the borrower rents their residence.

1.2.2 Tidy data

A data frame where each row is a unique case (observational unit), each column is a variable, and each cell is a single value is commonly referred to as **tidy data**.

Tidy data is a standard way of mapping the meaning of a dataset to its structure. A dataset is tidy when:

1. Each row represents a single observation.
2. Each column represents a single variable.
3. Each cell contains a single value.

When recording data, use a tidy data frame unless you have a very good reason to use a different structure. This structure allows new cases to be added as rows or new variables as new columns and facilitates visualization, summarization, and other statistical analyses.

The grades for assignments, quizzes, and exams in a course are often recorded in a gradebook that takes the form of a data frame. How might you organize a course's grade data using a data frame? Describe the observational units and variables.

Show answer

There are multiple strategies that can be followed. One common strategy is to have each student represented by a row, and then add a column for each assignment, quiz, or exam. Under this setup, it is easy to review a single line to understand the grade history of a student. There should also be columns to include student information, such as one column to list student names.

1.2.3 A larger dataset: US counties

We consider data for 3,142 counties in the United States, which includes the name of each county, the state where it resides, its population in 2017, the population change from 2010 to 2017, poverty rate, and several additional characteristics.

How might these data be organized in a data frame? What are the observations and variables?

Show answer

Each county may be viewed as a case, and there are multiple pieces of information recorded for each case. A table with 3,142 rows and 14 columns could hold these data, where each row represents a county and each column represents a particular piece of information.

Table 1.4: Six observations and six variables from a US county dataset.

name	state	pop2017	pop_change	unemployment_rate	median_edu
Autauga County	Alabama	55,504	1.48	3.86	some_college
Baldwin County	Alabama	212,628	9.19	3.83	some_college
Barbour County	Alabama	25,270	-6.22	6.44	hs_diploma
Bibb County	Alabama	22,668	0.73	4.39	hs_diploma
Blount County	Alabama	58,013	0.68	4.02	hs_diploma
Bullock County	Alabama	10,309	-2.28	7.72	hs_diploma

Table 1.5: Variables and their descriptions for the county dataset.

Variable	Description
name	Name of county.
state	Name of state.
pop2000	Population in 2000.
pop2010	Population in 2010.
pop2017	Population in 2017.
pop_change	Population change from 2010 to 2017 (in percent).
poverty	Percent of population in poverty in 2017.
homeownership	Homeownership rate, 2006–2010.
multi_unit	Percent of housing units that are in multi-unit structures, 2006–2010.
unemployment_rate	Unemployment rate in 2017.
metro	Whether the county contains a metropolitan area (yes or no).
median_edu	Median education level (2013–2017): below_hs, hs_diploma, some_college, or bachelors.
per_capita_income	Per capita (per person) income (2013–2017).
median_hh_income	Median household income.
smoking_ban	Type of county-level smoking ban in place in 2010: none, partial, or comprehensive.

1.3 Types of variables

Examine the variables `unemployment_rate`, `pop2017`, `state`, and `median_edu` in the county dataset. Each of these variables is inherently different from the other three, yet some share certain characteristics. Understanding these distinctions is fundamental to choosing the right analysis tools.

1.3.1 Numerical variables

A **quantitative variable** (also called a **numerical variable**) takes on values where arithmetic operations like adding, subtracting, or averaging make sense.

Consider `unemployment_rate`, which can take a wide range of numerical values, and it is sensible to add, subtract, or take averages with those values. On the other hand, we would not classify a variable reporting telephone area codes as quantitative since the average, sum, and difference of area codes does not have any clear meaning.

Quantitative variables can be further subdivided:

- **Continuous variables** can take any value within a range, including decimals. The `unemployment_rate` is continuous — it could be 3.86%, 4.127%, or any value in between.
- **Discrete variables** can only take counting values (typically whole numbers: 0, 1, 2, ...). The variable `pop2017` (population count) is discrete — you cannot have 55,504.7 people.

The key test for quantitative vs. categorical. Ask yourself: does it make sense to compute the average of this variable? If you can meaningfully average the values, it is quantitative. If averaging does not make sense (as with zip codes or phone numbers), the variable is categorical, even if it happens to be recorded using digits.

1.3.2 Categorical variables

A **categorical variable** (also called a **qualitative variable**) takes on values that are category names or labels. The possible values of a categorical variable are called its **levels**.

The variable `state` can take up to 51 values (after accounting for Washington, DC): Alabama, Alaska, ..., and Wyoming. Because the responses themselves are categories, `state` is a categorical variable, and the possible values (states) are the variable's levels.

Categorical variables can be further subdivided:

- **Nominal variables** have categories with no natural ordering. The variable `state` is nominal — there is no inherent reason to rank Alabama above Alaska or Wyoming.
- **Ordinal variables** have categories with a meaningful order. The variable `median_edu` takes values `below_hs`, `hs_diploma`, `some_college`, or `bachelors`. These categories have a natural ordering (less education to more education), making `median_edu` ordinal.

In most of this course, we will treat categorical variables as nominal (unordered). The distinction between nominal and ordinal becomes more important in advanced courses when special methods are used to take advantage of the ordering.

1.3.3 Variable type hierarchy

The breakdown of variable types is summarized in the diagram below.

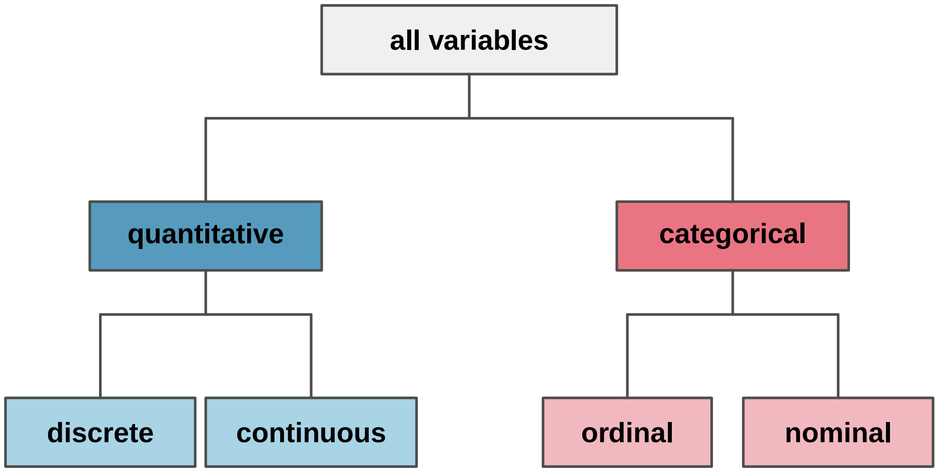

Figure 1.1: Breakdown of variables into their respective types. All variables are either quantitative or categorical. Quantitative variables are further classified as discrete or continuous. Categorical variables are further classified as ordinal or nominal.

Data were collected about students in a statistics course. Three variables were recorded for each student: number of siblings, student height, and whether the student had previously taken a statistics course. Classify each of the variables as continuous quantitative, discrete quantitative, or categorical.

The number of siblings and student height represent quantitative variables. Because the number of siblings is a count, it is discrete. Height varies continuously, so it is a continuous quantitative variable. The last variable classifies students into two categories — those who have and those who have not taken a statistics course — which makes this variable categorical.

An experiment is evaluating the effectiveness of a new drug in treating migraines. A `group` variable is used to indicate the experiment group for each patient: treatment or control. The `num_migraines` variable represents the number of migraines the patient experienced during a 3-month period. Classify each variable as either quantitative or categorical.

Show answer

The `group` variable can take just one of two group names, making it categorical. The `num_migraines` variable describes a count of the number of migraines, which is an outcome where basic arithmetic is sensible, which means this is a quantitative outcome; more specifically, since it represents a count, `num_migraines` is a discrete quantitative variable.

Classify each variable as continuous quantitative, discrete quantitative, nominal categorical, or ordinal categorical:

a. Number of text messages sent per day
b. Favorite type of music
c. Temperature in Fahrenheit
d. Satisfaction rating (1 = very unsatisfied, ..., 5 = very satisfied)

Show answer

a. Discrete quantitative — it is a count. b. Nominal categorical — the categories (rock, jazz, classical, etc.) have no natural ordering. c. Continuous quantitative — temperature can take any value in a range. d. Ordinal categorical — the numbers have a meaningful order, but the intervals between them may not be equal, and averaging satisfaction ratings is debatable.

1.4 Relationships between variables

Many analyses are motivated by a researcher looking for a relationship between two or more variables. A social scientist might like to answer questions such as:

> Does a higher-than-average increase in county population tend to correspond to counties with higher or lower median household incomes?
>
> If homeownership in one county is lower than the national average, will the percent of housing units that are in multi-unit structures in that county tend to be above or below the national average?
>
> How much can the median education level explain the median household income for counties in the US?

To answer these questions, data must be collected, such as the county dataset shown in Table 1.4. Examining summary statistics can provide numerical insights about the specifics of each of these questions. Alternatively, graphs can be used to visually explore the data, potentially providing more insight than a summary statistic.

1.4.1 Scatterplots

A **scatterplot** is a graph that displays the relationship between two quantitative variables. Each observation is represented by a point, with one variable on the horizontal axis and the other on the vertical axis.

The figure below displays the relationship between `homeownership` and `multi_unit` (the percent of housing units that are in multi-unit structures such as apartments and condos) for US counties. Each point on the plot represents a single county. For instance, one highlighted point corresponds to Chattahoochee County, Georgia, which has 39.4% of housing units in multi-unit structures and a homeownership rate of 31.3%.

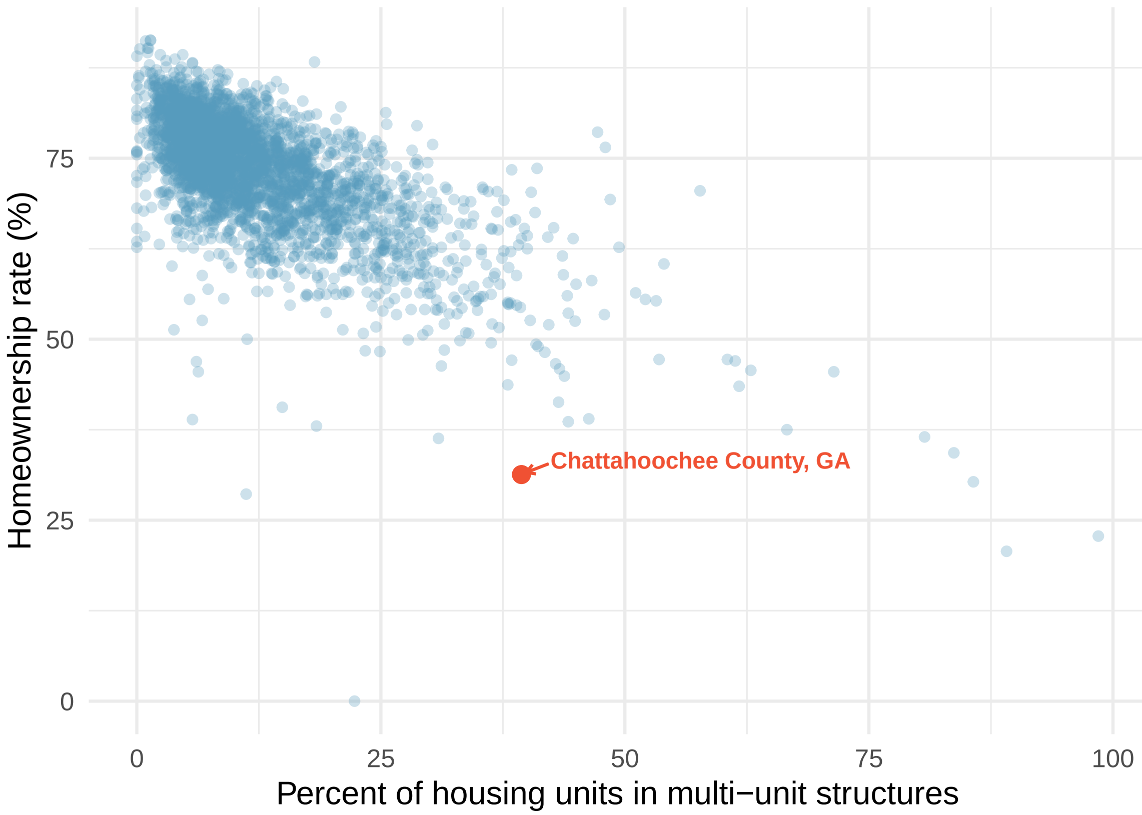

Figure 1.2: A scatterplot of homeownership rate versus the percent of housing units in multi-unit structures for US counties. There is a clear negative trend: as the multi-unit rate increases, homeownership tends to decrease. Chattahoochee County, Georgia is highlighted.

The scatterplot suggests a relationship between the two variables: counties with a higher rate of housing units in multi-unit structures tend to have lower homeownership rates. We might brainstorm as to why this relationship exists and investigate each idea to determine which are the most reasonable explanations.

1.4.2 Associated and independent variables

When two variables show some connection or pattern with one another, they are called **associated** variables (or **related** variables). If two variables are not associated — there is no evident relationship between them — they are said to be **independent**.

Because there is a downward trend in the homeownership vs. multi-unit scatterplot — counties with more multi-unit housing are associated with lower homeownership — these variables are said to be **negatively associated**.

A **positive association** means that as one variable increases, the other tends to increase as well. A **negative association** means that as one variable increases, the other tends to decrease.

Consider the relationship between the percent change in population from 2010 to 2017 and median household income for counties, as shown in the scatterplot below. Are these variables associated?

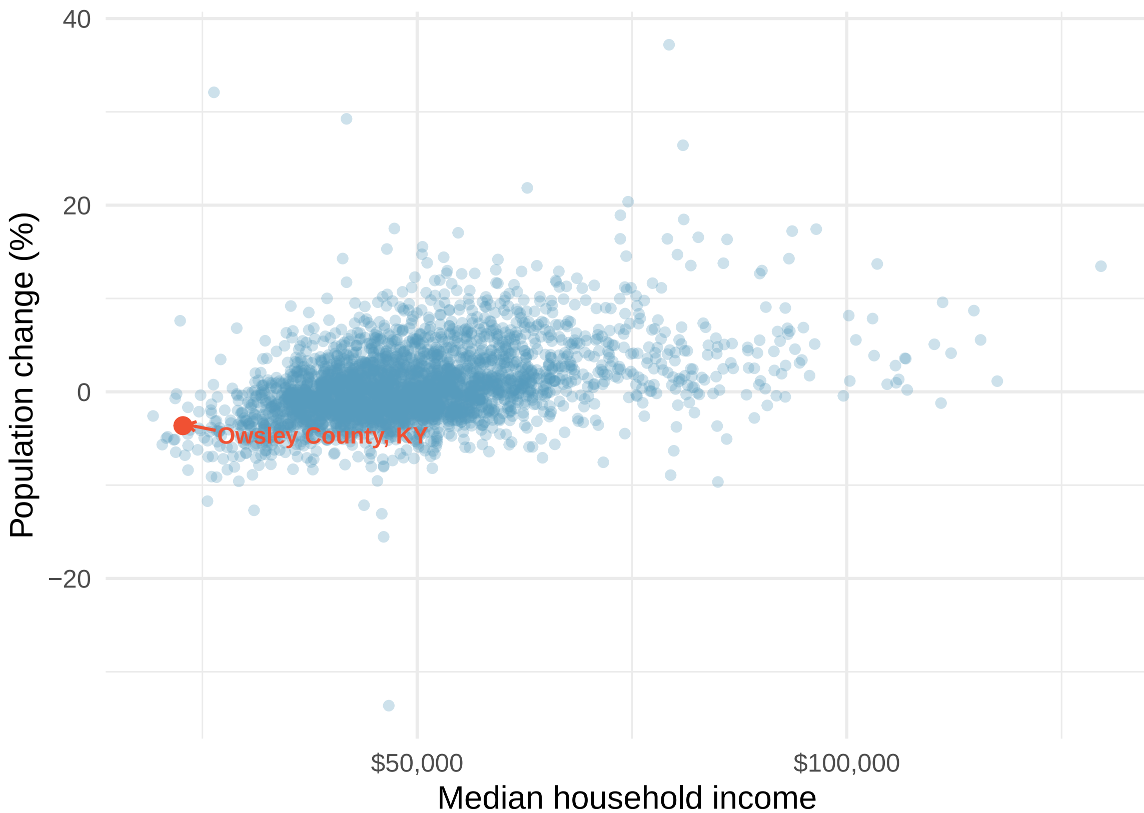

Figure 1.3: A scatterplot showing population change against median household income for US counties. There is a positive trend: counties with higher median household income tend to have higher population growth. Owsley County, Kentucky is highlighted.

The larger the median household income for a county, the higher the population growth observed for the county. While it is not true that every county with a higher median household income has higher population growth, the trend in the plot is evident. Since there is some relationship between the variables, they are associated. Because both variables tend to increase together, this is a positive association.

Associated or independent, not both. A pair of variables are either related in some way (associated) or not (independent). No pair of variables is both associated and independent.

Examine the variables in the loan dataset, which are described in Table 1.3. Create two questions about possible relationships between variables in the loan data that are of interest to you.

Show answer

Two example questions: (1) What is the relationship between loan amount and total income? (2) If someone's income is above the average, will their interest rate tend to be above or below the average?

1.5 Explanatory and response variables

When we ask questions about the relationship between two variables, we sometimes also want to determine if the change in one variable causes a change in the other. Consider the following rephrasing of an earlier question about the county dataset:

> If there is an increase in the median household income in a county, does this *drive* an increase in its population?

In this question, we are asking whether one variable affects another.

When we suspect one variable might causally affect another, we label the first variable the **explanatory variable** and the second the **response variable**.

$$\text{explanatory variable} \rightarrow \textit{might affect} \rightarrow \text{response variable}$$

We also use the terms explanatory and response to describe variables where the response might be *predicted* using the explanatory variable, even if there is no causal relationship.

For many pairs of variables, there is no hypothesized relationship, and these labels would not be applied to either variable.

In the stent study from Section 1.1, identify the explanatory and response variables.

The explanatory variable is the treatment group assignment (stent or control), because this is the factor that the researchers manipulated. The response variable is whether the patient had a stroke, because this is the outcome the researchers measured to evaluate the treatment's effect.

Labeling one variable as "explanatory" does not guarantee that it *causes* changes in the response. A formal evaluation of whether one variable causes changes in another requires an experiment. We discuss this distinction further in Section 1.6.

Some disciplines use the terms **independent variable** and **dependent variable** instead of explanatory and response. We avoid this language because the word "independent" has a different technical meaning in statistics — it refers to variables that are not associated. Using "independent" in two different ways can cause confusion.

A researcher is studying whether hours spent studying for an exam are related to exam scores. Identify the explanatory and response variables.

Show answer

The explanatory variable is hours spent studying, because this is the factor that might influence the outcome. The response variable is exam score, because this is the outcome being measured.

1.6 Observational studies and experiments

There are two primary types of data collection: experiments and observational studies. Understanding the difference is critical, because the type of study determines what conclusions we can draw.

1.6.1 Experiments

An **experiment** is a study in which the researcher actively imposes treatments on the subjects in order to observe their responses. When subjects are randomly assigned to treatment groups, the study is called a **randomized experiment**.

For instance, we may suspect that drinking a high-calorie energy drink will improve performance in a race. To check if there really is a causal relationship between the explanatory variable (whether the runner drank an energy drink or not) and the response variable (the race time), researchers identify a sample of individuals and split them into groups. The individuals in each group are *assigned* a treatment. Random assignment organizes the participants into groups that are roughly equal on all aspects, thus allowing us to control for any other variables that might affect the outcome (such as fitness level, racing experience, or age).

For example, each runner in the experiment could be randomly assigned — perhaps by flipping a coin — into one of two groups: the first group receives a **placebo** (a fake treatment, in this case a no-calorie drink that looks identical to the energy drink) and the second group receives the high-calorie energy drink.

A **placebo** is a fake treatment that is made to look like the real treatment. Placebos are used so that subjects in the control group have the same experience as those in the treatment group (except for the actual treatment), which helps researchers isolate the effect of the treatment itself.

See the stent case study in Section 1.1 for another example of an experiment, though that study did not employ a placebo.

1.6.2 Observational studies

An **observational study** is a study in which the researcher collects data without interfering with how the data arise. In an observational study, the researcher merely observes what happens naturally.

Researchers perform observational studies when they collect data via surveys, review medical or company records, or follow a **cohort** (a group of similar individuals) over time to form hypotheses about why certain outcomes might occur. In each of these situations, researchers merely observe the data that arise — they do not assign treatments to subjects.

A researcher wants to know if smoking causes lung cancer. She follows 10,000 adults over 20 years, recording whether each person smokes and whether they develop lung cancer. Is this an experiment or an observational study?

This is an observational study. The researcher did not assign anyone to smoke or not smoke — she simply observed the participants' behavior and health outcomes. While the study might reveal an association between smoking and lung cancer, it cannot by itself prove causation, because there may be other differences between smokers and non-smokers (such as diet, exercise habits, or genetics) that contribute to the outcome.

1.6.3 The key distinction: association vs. causation

In general, observational studies can provide evidence of a naturally occurring association between variables, but they cannot by themselves show a causal connection. This is because in observational studies, researchers cannot control for all the other variables that might be influencing the outcome.

Association does not imply causation.

In general, association does not imply causation. An advantage of a randomized experiment is that it is easier to establish causal relationships, because random assignment tends to balance out the effects of other variables across treatment groups. Establishing causation from observational studies requires advanced statistical methods that are beyond the scope of this course. We revisit these ideas when we discuss study design in Chapter 2.

Determine whether each of the following is an experiment or an observational study:

a. A pharmaceutical company randomly assigns 200 patients to receive either a new drug or a placebo and measures their cholesterol levels after 6 months.
b. A university surveys its graduates to determine whether those who participated in study-abroad programs earn higher salaries five years after graduation.

c. A teacher assigns students in her morning class to use a new learning app and students in her afternoon class to study using traditional methods, then compares exam scores.

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a. This is an experiment — the company randomly assigns patients to treatments.
b. This is an observational study — the university simply collects data on students who chose (or chose not) to study abroad. The students were not randomly assigned.
c. This is tricky. The teacher assigns groups, so it might seem like an experiment. However, because students were not randomly assigned (they ended up in morning vs. afternoon classes for their own reasons), this is better described as a quasi-experiment or an observational study. Differences in outcomes could be due to the app, or they could be due to systematic differences between the types of students who take morning vs. afternoon classes.

1.6.4 Putting it all together

The table below summarizes the key differences between experiments and observational studies.

	Experiment	Observational Study
Researcher's role	Assigns treatments	Observes what happens naturally
Random assignment?	Yes (in a randomized experiment)	No
Can establish causation?	Yes	Generally no
Example	Randomly assign patients to drug vs. placebo	Survey people about their exercise habits and health

1.7 Chapter review

1.7.1 Summary

This chapter introduced you to the world of data. Here are the key ideas:

- **Data** are observations collected about the world around us. **Statistics** is the science of collecting, analyzing, and drawing conclusions from data.
- A **data frame** organizes data so that each row represents an observation and each column represents a variable. Data organized this way is called **tidy data**.
- Variables come in two fundamental types: **quantitative** (continuous or discrete) and **categorical** (nominal or ordinal).
- When two variables show a pattern together, they are **associated**; if not, they are **independent**. Associations can be **positive** (both increase together) or **negative** (one increases while the other decreases).
- The **explanatory variable** is the variable we think might influence the **response variable**.
- **Experiments** randomly assign subjects to treatments and can establish causation. **Observational studies** merely observe what happens and can identify associations, but generally cannot establish causation.
- **Association does not imply causation** — this is one of the most important principles in statistics.

Many of the ideas from this chapter will be revisited as we move on to doing end-to-end data analyses. In the next chapter, you will learn about how we can design studies to collect the data we need to make conclusions with the desired scope of inference.

1.7.2 Key terms

The following terms were introduced in this chapter. You should be able to define and give an example of each.

Term	Term	Term
associated	data	data frame
case	categorical variable	cohort
continuous	dependent variable	discrete
experiment	explanatory variable	independent
level	negative association	nominal
quantitative variable	observation	observational study
ordinal	placebo	positive association
randomized experiment	response variable	summary statistic
tidy data	unit of observation	variable

1.8 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Marvel Cinematic Universe films.** The data frame below contains information on Marvel Cinematic Universe films through the Infinity saga (a movie storyline spanning from Ironman in 2008 to Endgame in 2019). Box office totals are given in millions of US Dollars. How many observations and how many variables does this data frame have?

		Length				Gross	
	Title	Hrs	Mins	Release Date	Opening Wknd US	US	World
1	Iron Man	2	6	5/2/2008	98.62	319.03	585.8
2	The Incredible Hulk	1	52	6/12/2008	55.41	134.81	264.77
3	Iron Man 2	2	4	5/7/2010	128.12	312.43	623.93
4	Thor	1	55	5/6/2011	65.72	181.03	449.33
5	Captain America: The First Avenger	2	4	7/22/2011	65.06	176.65	370.57
...	...	...	...	...	...	...	...
23	Spiderman: Far from Home	2	9	7/2/2019	92.58	390.53	1131.93

2. **Cherry Blossom Run.** The data frame below contains information on runners in the 2017 Cherry Blossom Run, which is an annual road race that takes place in Washington, DC. Most runners participate in a 10-mile run while a smaller fraction take part in a 5k run or walk. How many observations and how many variables does this data frame have?

						Time			
	Bib	Name	Sex	Age	City / Country	Net	Clock	Pace	Event
1	6	Hiwot G.	F	21	Ethiopia	3217	3217	321	10 Mile
2	22	Buze D.	F	22	Ethiopia	3232	3232	323	10 Mile
3	16	Gladys K.	F	31	Kenya	3276	3276	327	10 Mile
4	4	Mamitu D.	F	33	Ethiopia	3285	3285	328	10 Mile
5	20	Karolina N.	F	35	Poland	3288	3288	328	10 Mile
...	...	...	...	...	...	...	...	...	...
19961	25153	Andres E.	M	33	Woodbridge, VA	5287	5334	1700	5K

3. **Air pollution and birth outcomes, study components.** Researchers collected data to examine the relationship between air pollutants and preterm births in Southern California. During the study, air pollution levels were measured by air quality monitoring stations. Specifically, levels of carbon monoxide were recorded in parts per million, nitrogen dioxide and ozone in parts per hundred million, and inhalable particulate matter (PM_{10}) in $\mu g/m^3$. Length of gestation data were collected on 143,196 births between the years 1989 and 1993, and air pollution exposure during gestation was calculated for each birth. The analysis suggested that increased ambient PM_{10} and, to a lesser degree, CO concentrations may be associated with the occurrence of preterm births. (Ritz et al. 2000)

a. Identify the main research question of the study.

b. Who are the subjects in this study, and how many are included?

c. What are the variables in the study? Identify each variable as numerical or categorical. If numerical, state whether the variable is discrete or continuous. If categorical, state whether the variable is ordinal.

4. **Cheaters, study components.** Researchers studying the relationship between honesty, age and self-control conducted an experiment on 160 children between the ages of 5 and 15. Participants reported their age, sex, and whether they were an only child or not. The researchers asked each child to toss a fair coin in private and to record the outcome (white or black) on a paper sheet, and said they would only reward children who report white. (Bucciol and Piovesan 2011)

a. Identify the main research question of the study.

b. Who are the subjects in this study, and how many are included?

c. The study's findings can be summarized as follows: *"Half the students were explicitly told not to cheat and the others were not given any explicit instructions. In the no instruction group probability of cheating was found to be uniform across groups based on child's characteristics. In the group that was explicitly told to not cheat, girls were less likely to cheat, and while rate of cheating didn't vary by age for boys, it decreased with age for girls."* How many variables were recorded for each subject in the study in order to conclude these findings? State the variables and their types.

5. **Gamification and statistics, study components.** Gamification is the application of game-design elements and game principles in non-game contexts. In educational settings, gamification is often implemented as educational activities to solve problems by using characteristics of game elements. Researchers investigating the effects of gamification on learning statistics conducted a study where they split college students in a statistics class into four groups: (1) no reading exercises and no gamification, (2) reading exercises but no gamification, (3) gamification but no reading exercises, and (4) gamification and reading exercises. Students in all groups also attended lectures. Students in the class were from two majors: Electrical and Computer Engineering (n = 279) and Business Administration (n = 86). After their assigned learning experience, each student took a final evaluation comprised of 30 multiple choice question and their score was measured as the number of questions they answered correctly. The researchers considered students' gender, level of studies (first through fourth year) and academic major. Other variables considered were expertise in the English language and use of personal computers and games, both of which were measured on a scale of 1 (beginner) to 5 (proficient). The study found that gamification had a positive effect on student learning compared to traditional teaching methods involving lectures and reading exercises. They also found that the effect was larger for females and Engineering students. (Legaki et al. 2020)

a. Identify the main research question of the study.

b. Who were the subjects in this study, and how many were included?

c. What are the variables in the study? Identify each variable as numerical or categorical. If numerical, state whether the variable is discrete or continuous. If categorical, state whether the variable is ordinal.

6. **Stealers, study components.** In a study of the relationship between socio-economic class and unethical behavior, 129 University of California undergraduates at Berkeley were asked to identify themselves as having low or high social-class by comparing themselves to others with the most (least) money, most (least) education, and most (least) respected jobs. They were also presented with a jar of individually wrapped candies and informed that the candies were for children in a nearby laboratory, but that they could take some if they wanted. After completing some unrelated tasks, participants reported the number of candies they had taken. (Piff et al. 2012)

a. Identify the main research question of the study.

b. Who were the subjects in this study, and how many were included?

c. The study found that students who were identified as upper-class took more candy than others. How many variables were recorded for each subject in the study in order to conclude these findings? State the variables and their types.

7. **Migraine and acupuncture.** A migraine is a particularly painful type of headache, which patients sometimes wish to treat with acupuncture. To determine whether acupuncture relieves migraine pain, researchers conducted a randomized controlled study where 89 individuals who identified as female diagnosed with migraine headaches were randomly assigned to one of two groups: treatment or control. Forty-three (43) patients in the treatment group received acupuncture that is specifically designed to treat migraines. Forty-six (46) patients in the control group received placebo acupuncture (needle insertion at non-acupoint locations). Twenty-four (24) hours after patients received acupuncture, they were asked if they were pain free. Results are summarized in the contingency table below. Also provided is a figure from the original paper displaying the appropriate area (M) versus the inappropriate area (S) used in the treatment of migraine attacks. (Allais et al. 2011)

	Pain free?	
Group	No	Yes
Control	44	2
Treatment	33	10

a. What percent of patients in the treatment group were pain free 24 hours after receiving acupuncture?

b. What percent were pain free in the control group?

c. In which group did a higher percent of patients become pain free 24 hours after receiving acupuncture?

d. Your findings so far might suggest that acupuncture is an effective treatment for migraines for all people who suffer from migraines. However this is not the only possible conclusion. What is one other possible explanation for the observed difference between the percentages of patients that are pain free 24 hours after receiving acupuncture in the two groups?

e. What are the explanatory and response variables in this study?

8. **Sinusitis and antibiotics.** Researchers studying the effect of antibiotic treatment for acute sinusitis compared to symptomatic treatments randomly assigned 166 adults diagnosed with acute sinusitis to one of two groups: treatment or control. Study participants received either a 10-day course of amoxicillin (an antibiotic) or a placebo similar in appearance and taste. The placebo consisted of symptomatic treatments such as acetaminophen, nasal decongestants, etc. At the end of the 10-day period, patients were asked if they experienced improvement in symptoms. The distribution of responses is summarized below. (Garbutt et al. 2012)

	Improvement	
Group	No	Yes
Control	16	65
Treatment	19	66

a. What percent of patients in the treatment group experienced improvement in symptoms?

b. What percent experienced improvement in symptoms in the control group?

c. In which group did a higher percentage of patients experience improvement in symptoms?

d. Your findings so far might suggest a real difference in the effectiveness of antibiotic and placebo treatments for improving symptoms of sinusitis. However this is not the only possible conclusion. What is one other possible explanation for the observed difference between the percentages patients who experienced improvement in symptoms?

e. What are the explanatory and response variables in this study?

9. **Daycare fines, study components.** Researchers tested the deterrence hypothesis which predicts that the introduction of a penalty will reduce the occurrence of the behavior subject to the fine, with the condition that the fine leaves everything else unchanged, by instituting a fine for late pickup at daycare centers. For this study, they worked with 10 volunteer daycare centers that did not originally impose a fine to parents for picking up their kids late. They randomly selected 6 of these daycare centers and instituted a monetary fine (of a considerable amount) for picking up children late and then removed it. In the remaining 4 daycare centers no fine was introduced. The study period was divided into four: before the fine (weeks 1–4), the first 4 weeks with the fine (weeks 5-8), the last 8 weeks with fine (weeks 9–16), and the after fine period (weeks 17-20). Throughout the study, the number of kids who were picked up late was recorded each week for each daycare. The study found that the number of late-coming parents increased discernibly when the fine was introduced, and no reduction occurred after the fine was removed. (Gneezy and Rustichini 2000)

center	week	group	late_pickups	study_period
1	1	test	8	before fine
1	2	test	8	before fine
1	3	test	7	before fine
1	4	test	6	before fine
1	5	test	8	first 4 weeks with fine
...	...	...	...	...
10	20	control	13	after fine

a. Is this an observational study or an experiment? Explain your reasoning.

b. What are the cases in this study and how many are included?

c. What is the response variable in the study and what type of variable is it?

d. What are the explanatory variables in the study and what types of variables are they?

10. **Efficacy of COVID-19 vaccine on adolescents, study components.** Results of a Phase 3 trial announced in March 2021 show that the Pfizer-BioNTech COVID-19 vaccine demonstrated 100% efficacy and robust antibody responses on 12 to 15 years old adolescents with or without prior evidence of SARS-CoV-2 infection. In this trial 2,260 adolescents were randomly assigned to two groups: one group got the vaccine (n = 1,131) and the other got a placebo (n = 1,129). While 18 cases of COVID-19 were observed in the placebo group, none were observed in the vaccine group. (Pfizer 2021)

a. Is this an observational study or an experiment? Explain your reasoning.

b. What are the cases in this study and how many are included?

c. What is the response variable in the study and what type of variable is it?

d. What are the explanatory variables in the study and what types of variables are they?

11. **Palmer penguins.** Data were collected on 344 penguins living on three islands (Torgersen, Biscoe, and Dream) in the Palmer Archipelago, Antarctica. In addition to which island each penguin lives on, the data contains information on the species of the penguin (*Adelie*, *Chinstrap*, or *Gentoo*), its bill length, bill depth, and flipper length (measured in millimeters), its body mass (measured in grams), and the sex of the penguin (female or male). (Gorman, Williams, and Fraser 2014)

a. How many cases were included in the data?

b. How many numerical variables are included in the data? Indicate what they are, and if they are continuous or discrete.
c. How many categorical variables are included in the data, and what are they? List the corresponding levels (categories) for each.

12. **Smoking habits of UK residents.** A survey was conducted to study the smoking habits of 1,691 UK residents. Below is a data frame displaying a portion of the data collected in this survey. A blank cell indicates that data for that variable was not available for a given respondent.

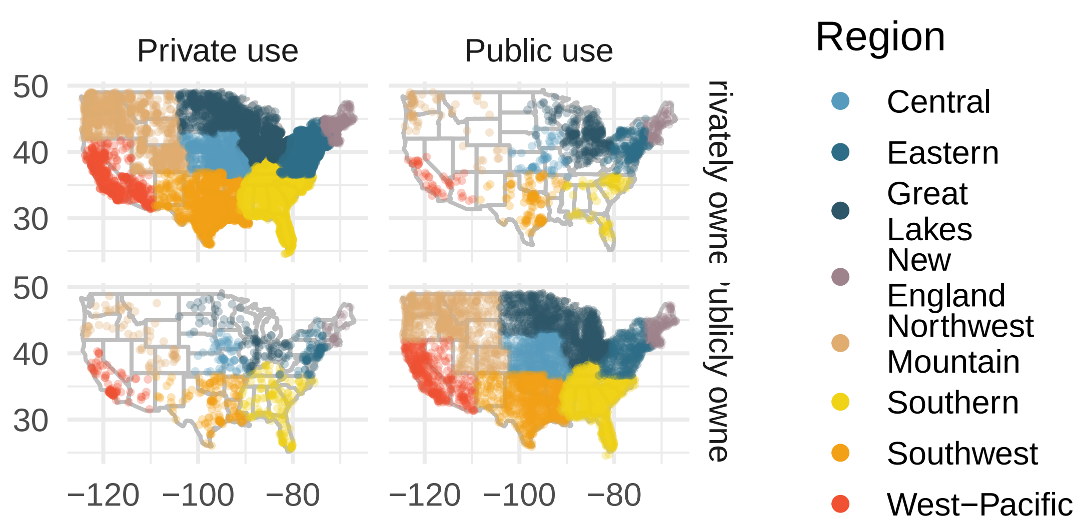
						amount	
	sex	age	marital_status	gross_income	smoke	weekend	weekday
1	Female	61	Married	2,600 to 5,200	No	NA	NA
2	Female	61	Divorced	10,400 to 15,600	Yes	5	4
3	Female	69	Widowed	5,200 to 10,400	No	NA	NA
4	Female	50	Married	5,200 to 10,400	No	NA	NA
5	Male	31	Single	10,400 to 15,600	Yes	10	20
...	...	...	...	...	...	NA	NA
1691	Male	49	Divorced	Above 36,400	Yes	15	10

a. What does each row of the data frame represent?

b. How many participants were included in the survey?

c. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

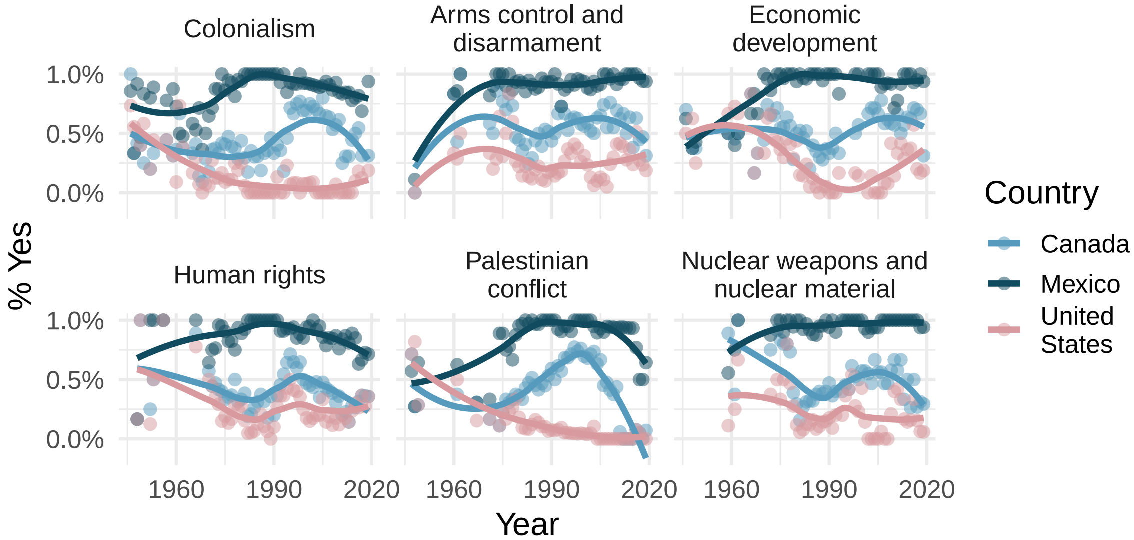
13. **US Airports.** The visualization below shows the geographical distribution of airports in the contiguous United States and Washington, DC. This visualization was constructed based on a dataset where each observation is an airport.



a. List the variables you believe were necessary to create this visualization.

b. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

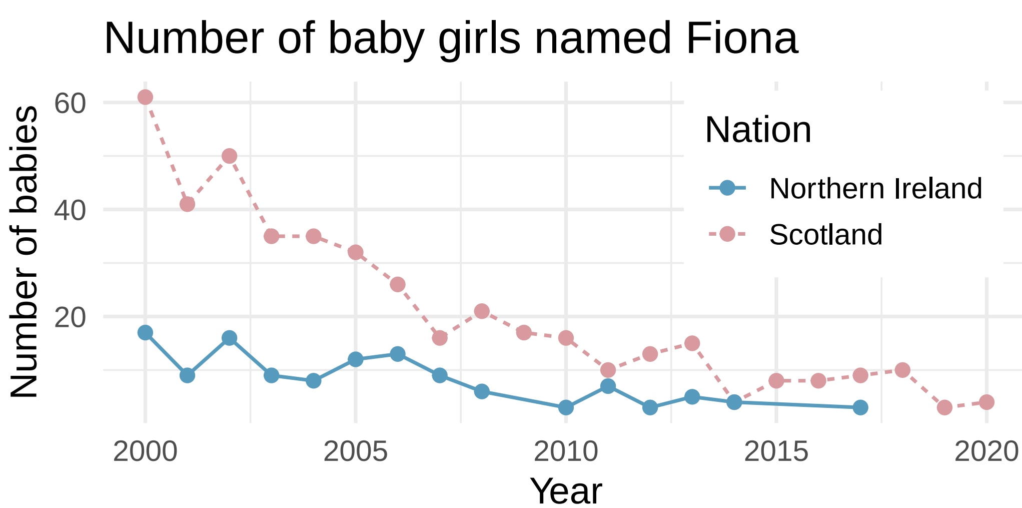
14. **UN Votes.** The visualization below shows voting patterns in the United States, Canada, and Mexico in the United Nations General Assembly on a variety of issues. Specifically, for a given year between 1946 and 2019, it displays the percentage of roll calls in which the country voted yes for each issue. This visualization was constructed based on a dataset where each observation is a country/year pair.



a. List the variables used in creating this visualization.

b. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

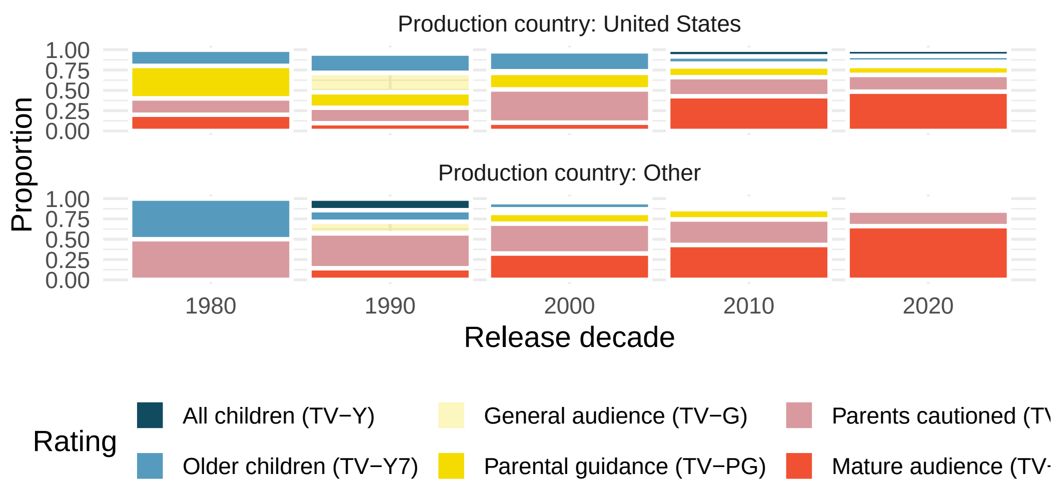
15. **UK baby names.** The visualization below shows the number of baby girls born in the United Kingdom (comprised of England & Wales, Northern Ireland, and Scotland) who were given the name "Fiona" over the years.



a. List the variables you believe were necessary to create this visualization.

b. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

16. **Shows on Netflix.** The visualization below shows the distribution of ratings of TV shows on Netflix (a streaming entertainment service) based on the decade they were released in and the country they were produced in. In the dataset, each observation is a TV show.



a. List the variables you believe were necessary to create this visualization.

b. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

17. **Stanford Open Policing.** The Stanford Open Policing project gathers, analyzes, and releases records from traffic stops by law enforcement agencies across the United States. Their goal is to help researchers, journalists, and policy makers investigate and improve interactions between police and the public. The following is an excerpt from a summary table created based off of the data collected as part of this project. (Pierson et al. 2020)

		Driver		Car	
County	State	Race / Ethnicity	Arrest rate	Stops / year	Search rate
Apache County	AZ	Black	0.016	266	0.077
Apache County	AZ	Hispanic	0.018	1008	0.053
Apache County	AZ	White	0.006	6322	0.017
Cochise County	AZ	Black	0.015	1169	0.047
...	...	...	...	...	...
Wood County	WI	Hispanic	0.029	27	0.036
Wood County	WI	White	0.029	1157	0.033

a. What variables were collected on each individual traffic stop in order to create the summary table above?

b. State whether each variable is numerical or categorical. If numerical, state whether it is continuous or discrete. If categorical, state whether it is ordinal or not.

c. Suppose we wanted to evaluate whether vehicle search rates are different for drivers of different races. In this analysis, which variable would be the response variable and which variable would be the explanatory variable?

18. **Space launches.** The following summary table shows the number of space launches in the US by the type of launching agency and the outcome of the launch (success or failure).

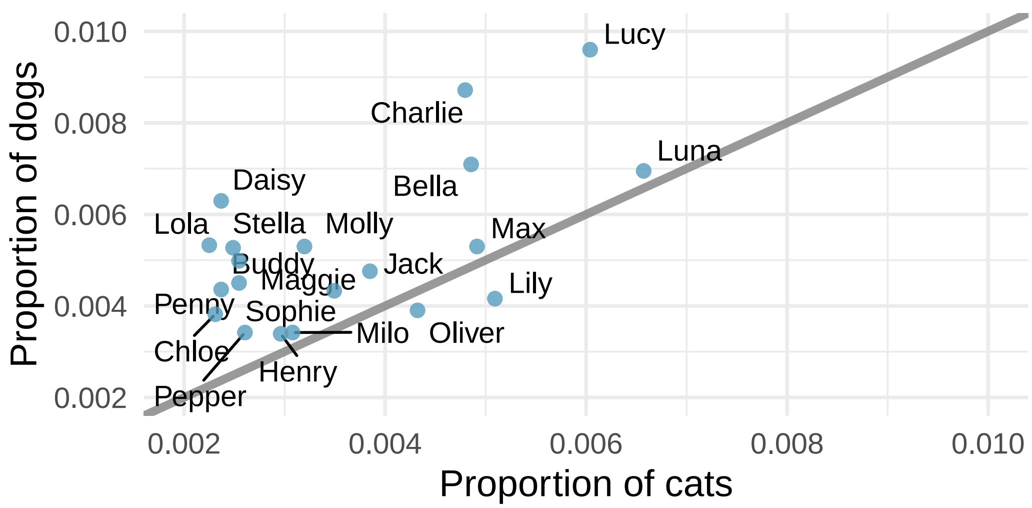
	1957 - 1999		2000-2018	
	Failure	Success	Failure	Success
Private	13	295	10	562
State	281	3751	33	711
Startup	0	0	5	65

a. What variables were collected on each launch in order to create to the summary table above?

b. State whether each variable is numerical or categorical. If numerical, state whether it is continuous or discrete. If categorical, state whether it is ordinal or not.

c. Suppose we wanted to study how the success rate of launches vary between launching agencies and over time. In this analysis, which variable would be the response variable and which variable would be the explanatory variable?

19. **Pet names.** The city of Seattle, WA has an open data portal that includes pets registered in the city. For each registered pet, we have information on the pet's name and species. The following visualization plots the proportion of dogs with a given name versus the proportion of cats with the same name. The 20 most common cat and dog names are displayed. The diagonal line on the plot is the $x = y$ line; if a name appeared on this line, the name's popularity would be exactly the same for dogs and cats.



a. Are these data collected as part of an experiment or an observational study?

b. What is the most common dog name? What is the most common cat name?

c. What names are more common for cats than dogs?

d. Is the relationship between the two variables positive or negative? What does this mean in context of the data?

20. **Stressed out in an elevator.** In a study evaluating the relationship between stress and muscle cramps, half the subjects are randomly assigned to be exposed to increased stress by being placed into an elevator that falls rapidly and stops abruptly and the other half are left at no or baseline stress.

a. What type of study is this?

b. Can this study be used to conclude a causal relationship between increased stress and muscle cramps?

Datasets: `mcu_films` (openintro) | `run17` (cherryblossom) | `migraine` (openintro) | `sinusitis` (openintro) | `daycare_fines` (openintro) | `biontech_adolescents` (openintro) | `smoking` (openintro) | `usairports` (airports) | `ukbabynames` (ukbabynames) | `netflix_titles` (tidytuesdayR) | `seattlepets` (openintro)

Chapter 2

Data Collection and Study Design

> **Draft Chapter.** This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Before digging into the details of working with data, we pause to think about how data come to be. If data are to be used to draw broad conclusions, then it is important to understand who or what the data represent. One important aspect is **sampling** — knowing how the observational units were selected from a larger group allows us to generalize back to the population from which the data were drawn. Additionally, by understanding the structure of the study, we can separate causal relationships from mere associations. A good question to ask before analyzing any dataset is: *How were these observations collected?* You will learn a lot about the data by understanding its source.

2.1 Populations and samples

The first step in conducting research is to identify the question to be investigated. A clearly stated research question helps identify what subjects or cases should be studied and what variables are important. It is also important to consider *how* data are collected so that they are reliable and help achieve the research goals.

Consider the following three research questions:

1. What is the average mercury content in swordfish in the Atlantic Ocean?
2. Over the last five years, what is the average time to complete a degree for UW–La Crosse undergrads?
3. Does a new drug reduce the number of deaths in patients with severe heart disease?

The **population** is the entire group of individuals or cases about which we want information. A **sample** is a subset of the population from which we actually collect data. A **census** is a study that attempts to collect data from every member of the population.

Each research question refers to a target population. In the first question, the target population is all swordfish in the Atlantic Ocean, and each fish represents a case. Oftentimes it is not feasible to collect data for every case in a population. A census is difficult because it may be too expensive, or it might be difficult or impossible to even identify every member of the population. Instead, we take a sample. Ideally, a sample is a manageable fraction of the population. For instance, 60 swordfish might be selected, and this sample data may be used to estimate the population average mercury content.

For the second and third research questions above, identify the target population and what represents an individual case.

Show answer

For the question about time to complete a degree, the population includes all UW–La Crosse undergraduates who graduated in the last five years (we can only compute an average for students who actually finished). Each graduate is an individual case. For the question about the heart disease drug, the population includes all people with severe heart disease, and each person represents a case.

2.1.1 Parameters and statistics

In most statistical analyses, the research question boils down to understanding a numerical summary — perhaps a quantity you already know (like the average) or one you will learn about in this course.

A numerical summary can be calculated on either the sample or the entire population. However, measuring every unit in the population is usually impossible. So we calculate the summary from a sample and use it to estimate the corresponding population value.

A **population parameter** is a numerical summary calculated from (or defined for) the entire population. A **sample statistic** is a numerical summary calculated from the sample data.

We use specific terms to distinguish between numbers calculated from a sample (sample statistics) and numbers that describe the entire population (population parameters). These terms will be used extensively in later chapters when we discuss making inferences about populations.

A poll of 1,000 likely voters finds that 54% plan to vote for Candidate A. Identify the population parameter and the sample statistic.

Show answer

The sample statistic is 54% — this is the proportion observed in the sample of 1,000 voters. The population parameter is the true proportion of *all* likely voters who plan to vote for Candidate A, which is unknown. The sample statistic (54%) is our best estimate of that unknown parameter.

2.1.2 Anecdotal evidence

Consider the following possible responses to the three research questions from above:

1. A man on the news got mercury poisoning from eating swordfish, so the average mercury concentration in swordfish must be dangerously high.
2. I met two students who took more than 7 years to graduate from UW–La Crosse, so it must take longer to graduate here than at many other schools.
3. My friend's dad had a heart attack and died after they gave him a new heart disease drug, so the drug must not work.

Each conclusion is based on data. However, there are two problems. First, the data only represent one or two cases. Second, and more importantly, it is unclear whether these cases are actually representative of the population. Data collected in this haphazard fashion are called **anecdotal evidence**.

Be careful of anecdotal evidence. Such evidence may be true and verifiable, but it may only represent extraordinary cases and therefore not be a good representation of the population.

Anecdotal evidence typically consists of unusual cases that we recall based on their striking characteristics. For instance, we are more likely to remember the two people who took 7 years to graduate than the six others who graduated in four years. Instead of focusing on the most unusual cases, we should examine a sample of many cases that better represent the population.

2.2 Sampling from a population

We might try to estimate the time to graduation for UW–La Crosse undergraduates by collecting a sample of graduates. All graduates in the last five years represent the *population*, and graduates who are selected for review are collectively called the *sample*. In general, we always seek to *randomly* select a sample from a population. The most basic type of random selection is equivalent to how raffles are conducted. For example, we could write each graduate's name on a raffle ticket and draw 10 tickets. The selected names would represent a random sample of 10 graduates.

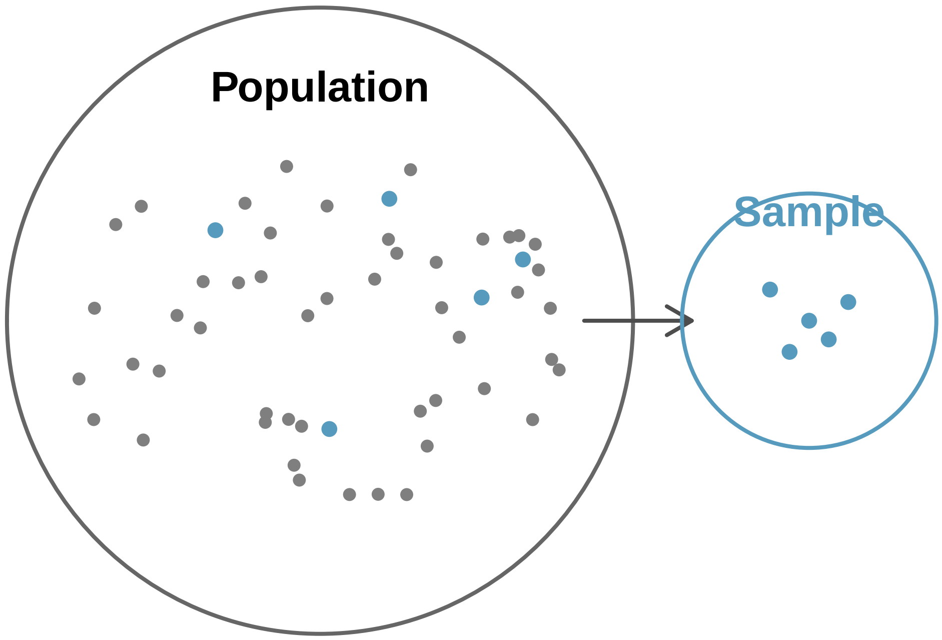


Figure 2.1: Sampling from a population. Individuals are randomly selected from the population (large circle) to be included in the sample (small circle).

Suppose we ask a student who happens to be majoring in nutrition to select several graduates for the study. Which students do you think they might pick? Do you think their sample would be representative of all graduates?

They might pick a disproportionate number of graduates from health-related fields. When selecting samples by hand, we run the risk of picking a **biased** sample, even if our bias is unintentional.

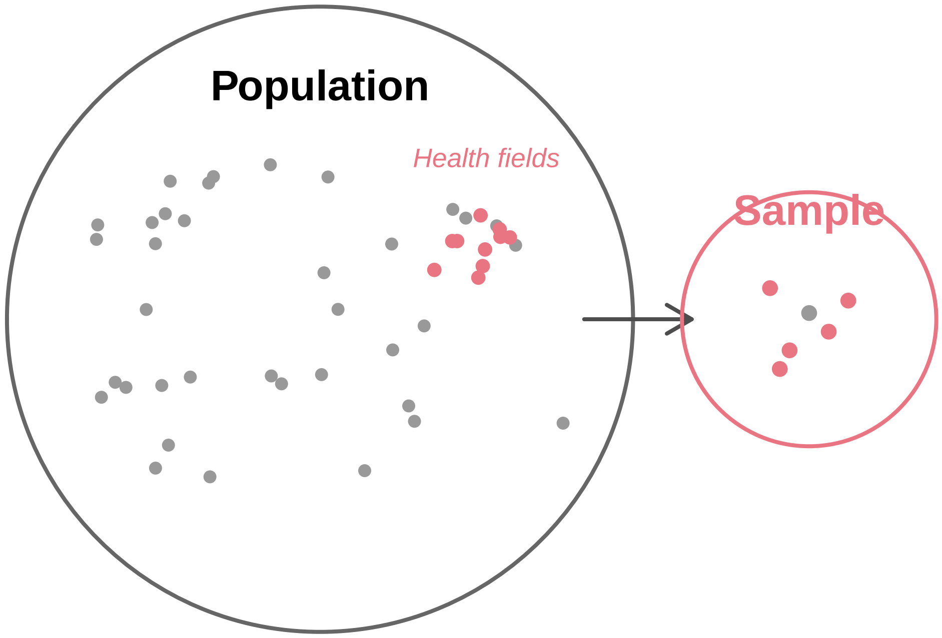


Figure 2.2: A biased sample. A nutrition major asked to select graduates might inadvertently pick a disproportionate number from health-related fields.

If someone is permitted to pick and choose exactly which individuals are included in the sample, it is entirely possible that the sample will overrepresent that person's interests — even if the bias is unintentional. Sampling randomly helps address this problem.

A **simple random sample** (**SRS**) is a sample drawn so that each case in the population has an equal chance of being included, and the selection of any one case is independent of the selection of any other case. This is equivalent to drawing names out of a hat.

The act of taking a simple random sample helps minimize bias. However, bias can crop up in other ways. Even when people are selected at random — for example, for surveys — caution must be exercised if the **non-response rate** is high. If only 30% of the people randomly sampled for a survey actually respond, then it is unclear whether the results are **representative** of the entire population. This **non-response bias** can skew results.

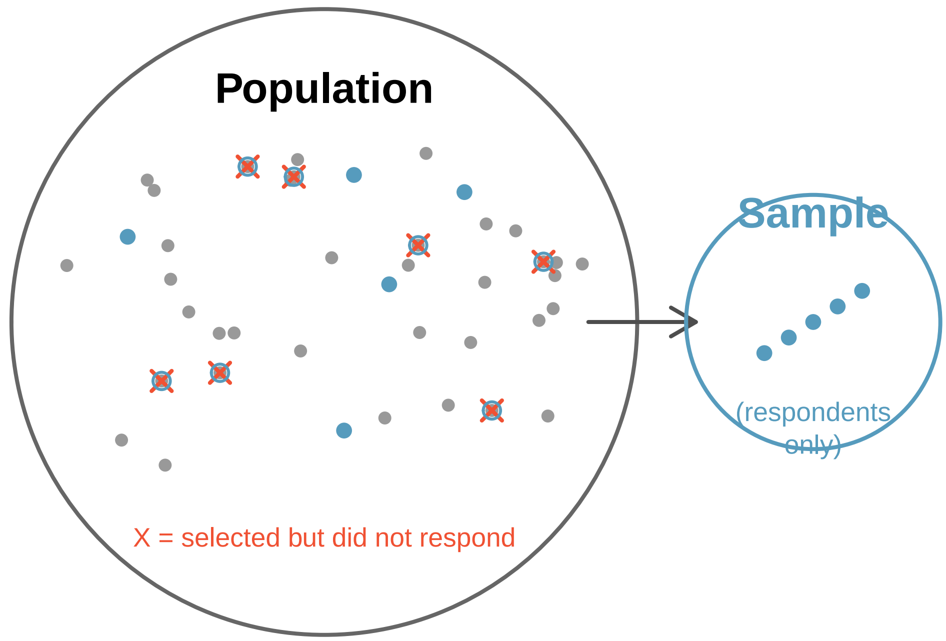


Figure 2.3: Non-response bias. Survey studies may only reach a certain group within the population. Non-response can make even a random sample unrepresentative.

Another common problem is a **convenience sample**, where individuals who are easily accessible are more likely to be included. For instance, if a political survey is conducted by stopping people on a single street corner, the results will not represent the broader city. It is often difficult to discern what sub-population a convenience sample actually represents.

We can easily access ratings for products, sellers, and companies on websites like Amazon. These ratings come only from people who go out of their way to provide a rating. If 50% of online reviews for a product are negative, do you think this means that 50% of buyers are dissatisfied with the product? Why or why not?

Show answer

Probably not. People tend to be more motivated to leave a review when they have a negative experience than when a product meets expectations. This creates a negative bias in product ratings — dissatisfied customers are overrepresented among reviewers. This is a form of non-response bias (satisfied customers are less likely to respond).

2.3 Sampling methods

Almost all statistical methods are based on the notion of randomness. If data are not collected using a random process from a population, then the estimates and the errors associated with those estimates are not reliable. Here we consider four random sampling techniques: simple random sampling, stratified sampling, cluster sampling, and multistage sampling.

2.3.1 Simple random sampling

Simple random sampling is the most intuitive form of random sampling. Consider the salaries of Major League Baseball (MLB) players, where each player belongs to one of the league's 30 teams. To take a simple random sample of 120 baseball players and their salaries, we could write the names of all players onto slips of paper, drop the slips into a bucket, mix them thoroughly, then draw out 120 slips. In general, a sample is called "simple random" if each case in the population has an equal chance of being included in the final sample *and* knowing that a particular case is included tells us nothing about which other cases are included.

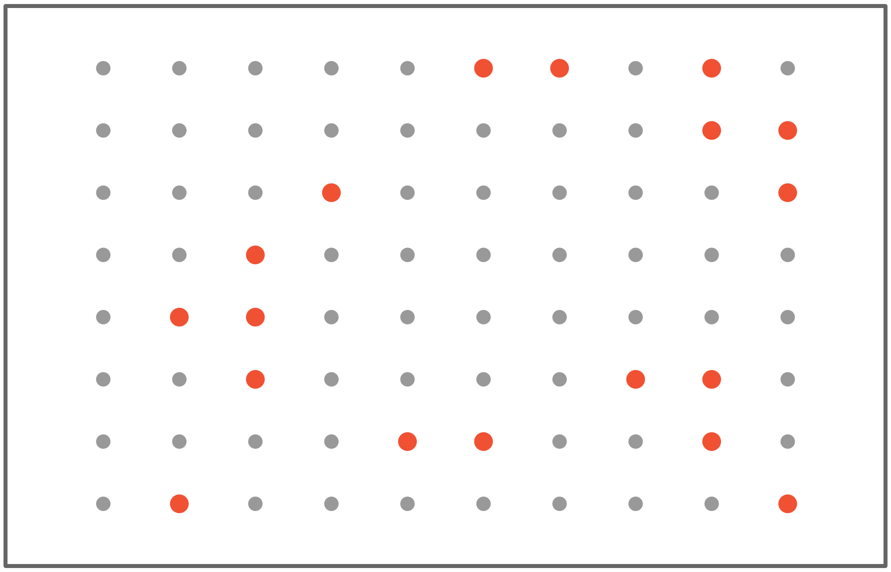


Figure 2.4: Simple random sampling. 18 cases (shown in red) are randomly selected from the population.

2.3.2 Stratified sampling

Stratified sampling is a divide-and-conquer strategy. The population is divided into groups called **strata**. The strata are chosen so that similar cases are grouped together, then a second sampling method (usually simple random sampling) is used within each stratum. In the baseball salary example, each of the 30 teams could represent a stratum, since some teams have much larger payrolls than others. We might randomly sample 4 players from each team to get our sample of 120.

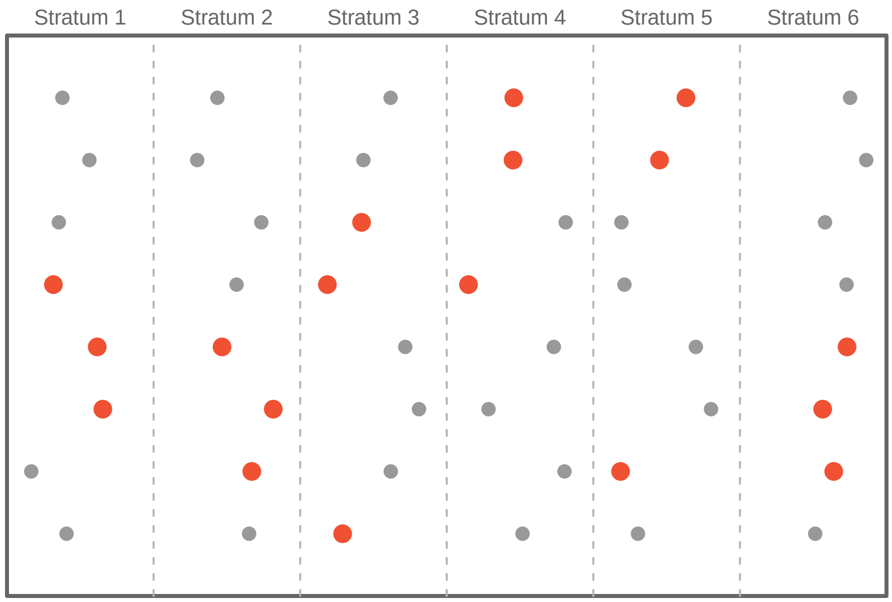


Figure 2.5: Stratified sampling. Cases are first grouped into strata, then simple random sampling is used within each stratum to select 3 cases per stratum.

Stratified sampling is especially useful when cases within each stratum are very similar with respect to the outcome of interest.

Why is it beneficial for cases within each stratum to be very similar?

If cases within a stratum are similar, we can get a very stable (precise) estimate for each subgroup. When we combine these stratum-level estimates into a single estimate for the full population, the overall estimate will tend to be more precise because each individual component is itself more precise.

2.3.3 Cluster sampling

In a **cluster sample**, we break up the population into many groups called **clusters**, then randomly select a fixed number of clusters and include *all* observations from each selected cluster in the sample.

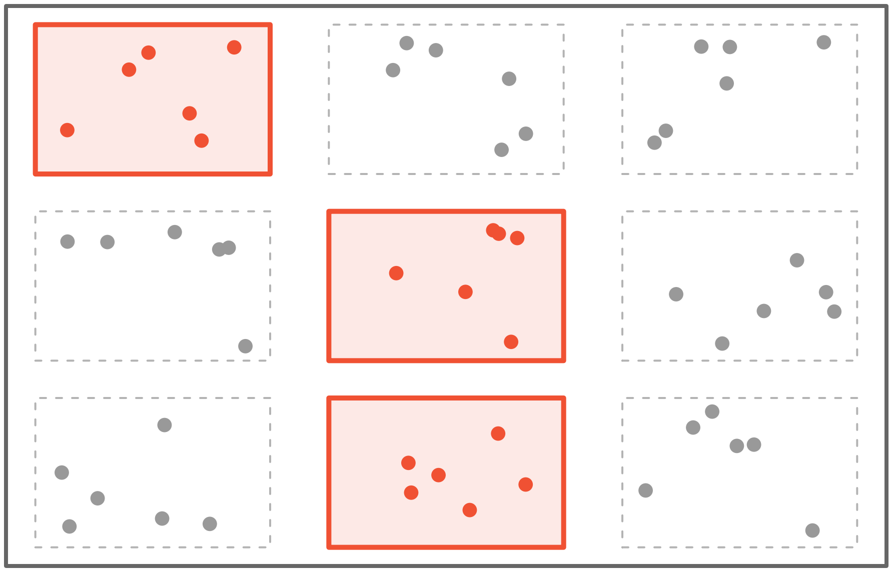


Figure 2.6: Cluster sampling. The population is divided into nine clusters. Three clusters are randomly selected, and all observations within those clusters are included in the sample.

2.3.4 Multistage sampling

A **multistage sample** is like a cluster sample, but instead of keeping all observations in each selected cluster, we take a random sample *within* each selected cluster.

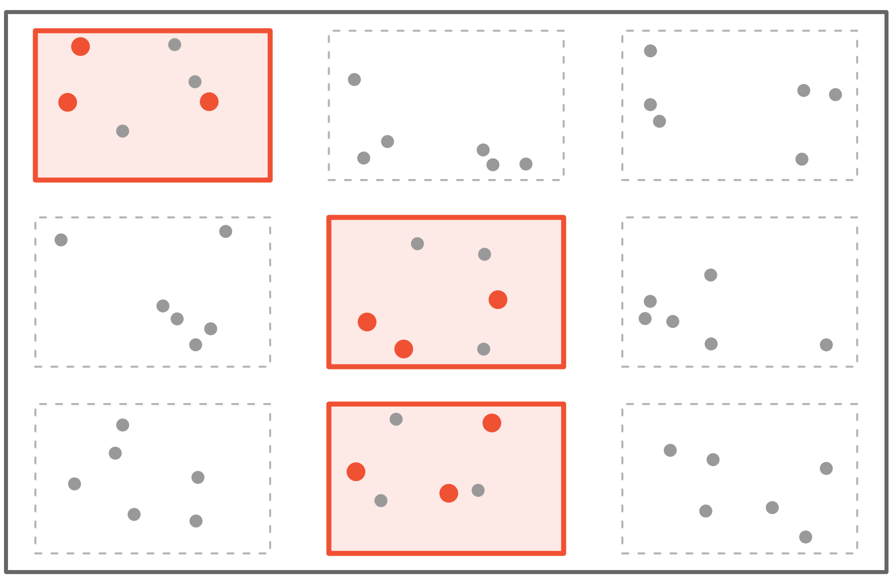


Figure 2.7: Multistage sampling. Three clusters are randomly selected, then a random sample of observations is drawn from within each selected cluster.

Cluster and multistage sampling can be more economical than other techniques. Unlike stratified sampling, these approaches work best when there is a lot of case-to-case variability *within* a cluster but the clusters themselves are fairly similar to one another. For example, if neighborhoods represent clusters, then cluster sampling works best when the residents within each neighborhood are diverse.

Suppose we want to estimate the malaria rate in a densely tropical portion of rural Indonesia. There are 30 villages in the region, each more or less like the next, but the distances between villages are substantial. We want to test 150 individuals for malaria. What sampling method should we use?

A simple random sample would likely draw individuals from all 30 villages, making data collection very expensive given the travel distances. Stratified sampling would be difficult because it is unclear how to group individuals into similar strata. However, cluster or multistage sampling are excellent choices. With multistage sampling, we could randomly select, say, 15 of the 30 villages, then randomly select 10 people from each village. This would substantially reduce data collection costs while still yielding reliable information.

2.3.5 Comparing the four methods

The table below summarizes the key features of each sampling method:

Method	How it works	Best used when...
Simple random	Every case has an equal chance of selection	Population is relatively homogeneous and accessible
Stratified	Divide into similar groups (strata), then SRS within each	Known subgroups differ on the outcome of interest
Cluster	Randomly select entire groups (clusters)	Clusters are internally diverse but similar to each other; travel/cost is a concern
Multistage	Randomly select clusters, then SRS within each	Same as cluster, but clusters are large

2.4 Experiments

An **experiment** is a study where the researchers actively assign treatments to cases. When this assignment uses randomization (e.g., a coin flip to decide which treatment a patient receives), it is called a **randomized experiment**. Randomized experiments are fundamentally important when trying to establish a causal connection between two variables.

2.4.1 Principles of experimental design

Well-designed experiments incorporate four key principles:

1. **Controlling.** Researchers assign treatments to cases and do their best to **control** any other differences between the groups. For example, when patients take a drug in pill form, some patients might take the pill with a sip of water while others drink a full glass. To control for the effect of water consumption, a doctor might instruct every patient to drink a 12-ounce glass of water with the pill.

2. **Randomization.** Researchers randomly assign subjects to treatment groups to account for variables that cannot be controlled. For example, some patients may be more susceptible to a disease than others due to their dietary habits. Randomly assigning patients to treatment or control groups helps even out such differences across groups.

3. **Replication.** The more cases researchers observe, the more accurately they can estimate the effect of the explanatory variable on the response. In a single study, we **replicate** by collecting a sufficiently large sample. At a minimum, we want multiple subjects per treatment group. Replication also refers to repeating an entire study to verify earlier findings. (The **replication crisis** refers to the ongoing problem in several scientific disciplines where past findings have failed to be replicated in new studies.)

4. **Blocking.** Researchers sometimes know or suspect that variables other than the treatment influence the response. They may first group individuals based on this variable into **blocks** and then randomize cases within each block to the treatment groups. For instance, in studying the effect of a drug on heart attacks, researchers might first split patients into low-risk and high-risk blocks, then randomly assign half the patients from each block to the treatment and half to the control group.

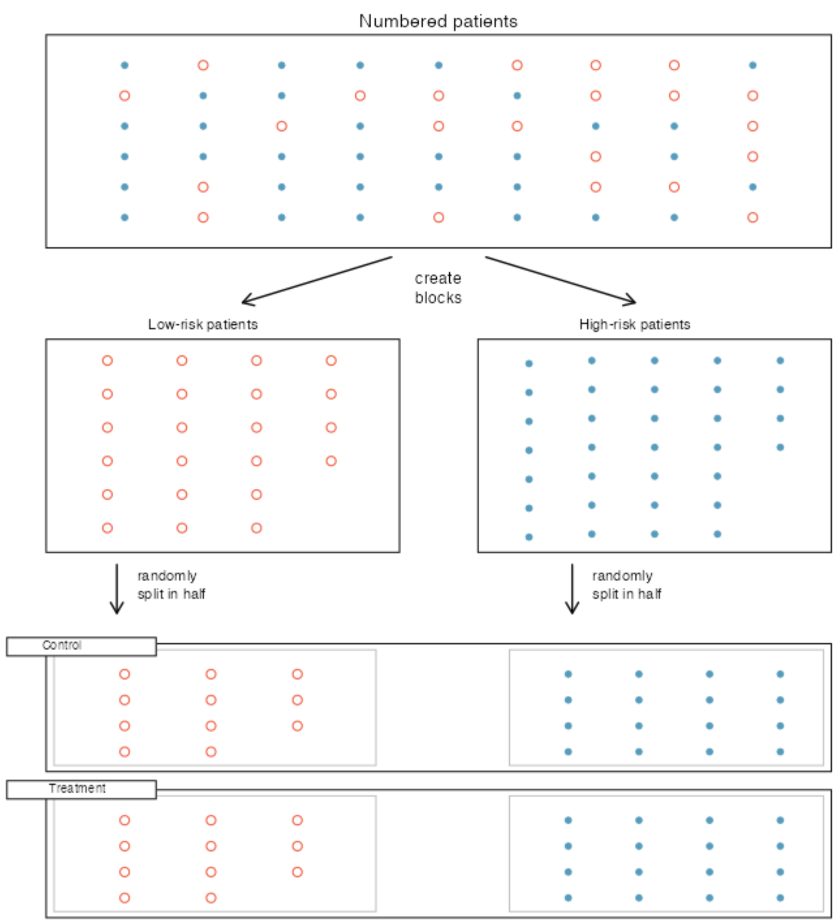


Figure 2.8: Blocking for patient risk. Patients are first divided into low-risk and high-risk blocks, then patients in each block are evenly randomized into treatment groups.

It is important to incorporate at least the first three principles into any study. Blocking is a slightly more advanced technique that adds precision when relevant grouping variables are known.

2.4.2 Reducing bias in human experiments

Randomized experiments are considered the gold standard for establishing cause-and-effect, but they do not automatically eliminate all bias. Human studies are a perfect example of where bias can unintentionally arise.

Consider a study where a new drug is tested to see if it reduces deaths in heart attack patients. Researchers designed a randomized experiment: study volunteers were randomly placed into two groups. The **treatment group** received the drug. The **control group** did not receive any drug treatment.

Put yourself in the place of a person in the study. If you are in the treatment group, you receive a promising new drug and may feel hopeful. A person in the control group receives nothing and might feel anxious. These emotional effects could influence health outcomes independently of the drug itself. This means there are actually two effects at play: the drug's real effectiveness, and the emotional impact of knowing (or not knowing) that you are receiving treatment.

A study is **blind** (or **single-blind**) if the patients do not know which treatment group they are in. A **placebo** is a fake treatment (such as a sugar pill) given to the control group so that patients cannot tell whether they are receiving the real treatment. The **placebo effect** is the phenomenon where patients show improvement simply because they believe they are receiving treatment. A study is **double-blind** if neither the patients nor the doctors and researchers interacting with them know who is receiving which treatment.

To prevent emotional effects from biasing the study, researchers want patients to be unaware of their group assignment — that is, the study should be *blind*. But if a patient in the control group receives nothing at all, they will know they are not getting treated. The solution is a placebo. An effective placebo is the key to making a study truly blind.

The patients are not the only ones who should be blinded. Doctors and researchers can also unintentionally bias a study. A doctor who knows a patient is receiving the real treatment might give that patient more attention or care. To guard against this, most modern studies use a **double-blind** setup.

Think back to the stent study described in Section 1.1. Is it an experiment? Was the study blinded? Was it double-blinded?

Show answer

The researchers assigned patients to treatment groups, so it is an experiment. However, patients could distinguish what treatment they received because a stent is a surgical procedure. There is no simple equivalent of a "placebo stent" (though sham surgeries do exist), so the study was likely not fully blinded. If the study was not blind, it could not be double-blind either.

For the stent study, could the researchers have employed a placebo? If so, what would it have looked like?

Show answer

In principle, researchers could use a **sham surgery** — the patient undergoes a surgical procedure but does not actually receive the stent. Sham surgeries do exist in medical research, but they raise serious ethical questions because they expose control-group patients to the risks of surgery without the potential benefit of the treatment.

There are always multiple viewpoints on experiments and placebos, and rarely is it obvious which approach is ethically "correct." Is it ethical to use a sham surgery that creates risk for the patient?

On the other hand, if we do not use one, we might promote a costly treatment that has no real effect, diverting resources from treatments that actually work.

2.5 Observational studies

An **observational study** is a study where no treatment has been explicitly applied or withheld by the researchers. The researchers simply observe and record what happens naturally.

Making causal conclusions based on experiments is often reasonable because researchers can randomly assign treatments. However, making causal conclusions from observational data is much more dangerous and is generally not recommended. Observational studies are typically only sufficient to show **associations** or to form hypotheses that can later be tested with experiments.

Suppose an observational study tracked sunscreen use and skin cancer, and found that people who used more sunscreen were *more* likely to have skin cancer. Does this mean sunscreen causes skin cancer?

No! Previous research tells us that sunscreen actually *reduces* skin cancer risk. The missing piece is **sun exposure**. People who spend a lot of time in the sun are more likely to use sunscreen *and* more likely to get skin cancer. Sun exposure is a **confounding variable** — it is associated with both sunscreen use and skin cancer, creating a misleading association between the two.

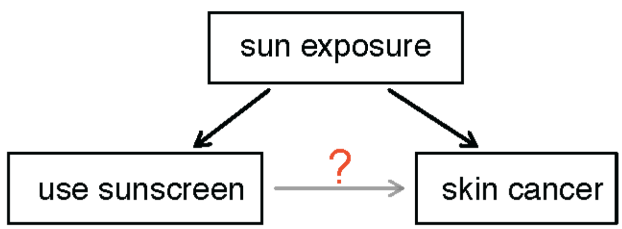


Figure 2.9: Sun exposure as a confounding variable. Sun exposure causes both increased sunscreen use and increased skin cancer risk, creating a spurious association between sunscreen use and skin cancer.

2.6 Confounding variables

A **confounding variable** is a variable that is associated with *both* the explanatory variable and the response variable. Because of this dual association, a confounding variable makes it impossible to determine whether the explanatory variable truly caused the observed response.

Confounding variables may or may not be measured as part of the study. Regardless, drawing cause-and-effect conclusions from observational studies is difficult because of the ever-present possibility of confounding.

Consider a classic example: ice cream sales and drowning rates both increase during summer months. If we looked only at the association between ice cream sales and drownings, we might (absurdly) conclude that ice cream causes drowning. The confounding variable is **temperature** (or season) — warm weather drives both increased ice cream consumption and increased swimming, which leads to more drowning incidents.

Suppose data show a negative association between homeownership rates and the percentage of housing units that are in multi-unit structures across U.S. counties. Is it reasonable to conclude that building more apartments *causes* homeownership rates to drop? Suggest a confounding variable.

Show answer

No, a causal conclusion is not warranted from observational data. A likely confounding variable is **population density**. Dense urban counties tend to have more multi-unit housing (apartments, condos) because of limited space, and they also tend to have lower homeownership rates because property values are higher. Population density is associated with both variables.

2.6.1 Prospective and retrospective studies

Observational studies come in two forms:

- A **prospective study** identifies individuals and collects information as events unfold going forward. For example, medical researchers might identify and follow a group of patients over many years to assess the influence of behavior on cancer risk. The famous Nurses' Health Study, which started in 1976 and has collected data on over 275,000 nurses, is a prospective study.
- A **retrospective study** collects data after events have already taken place — for example, by reviewing past medical records.

Some studies contain both prospective and retrospective elements. For instance, a medical study might gather information about participants' histories (retrospective) and then follow them forward in time (prospective).

2.7 Random assignment and causation

The distinction between random *sampling* and random *assignment* is crucial. They serve different purposes and affect what conclusions we can draw.

- **Random sampling** (selecting subjects randomly from a population) allows us to **generalize** our findings to the broader population.
- **Random assignment** (randomly assigning subjects to treatment groups in an experiment) allows us to draw **causal** conclusions.

Random sampling and random assignment are independent concepts. A study can have one, both, or neither. The type of randomization determines the scope of conclusions we can draw.

A company wants to know whether a new keyboard layout increases typing speed. They recruit 40 volunteers from their office, randomly assign 20 to train on the new layout and 20 to continue with the standard layout, then measure typing speed after one month.

Can the company conclude that the new layout *causes* faster typing? Can they generalize the results to all office workers?

Because the study used **random assignment** to treatment groups, the company can make a causal conclusion: any observed difference in typing speed can be attributed to the keyboard layout rather than to pre-existing differences between the groups. However, the 40 volunteers were not randomly selected from all office workers — they are a convenience sample of volunteers from one company. Therefore, the results may not generalize to all office workers.

2.8 Scope of inference

The scope of inference — what conclusions we can draw from a study — depends on how data were collected. The following table summarizes the four possible scenarios:

	Random assignment (to treatments)	**No random assignment**
Random sampling (from population)	Can generalize to the population **and** can draw causal conclusions	Can generalize to the population, but **cannot** draw causal conclusions
No random sampling	**Cannot** generalize to the population, but can draw causal conclusions (for subjects like those in the study)	**Cannot** generalize **and cannot** draw causal conclusions

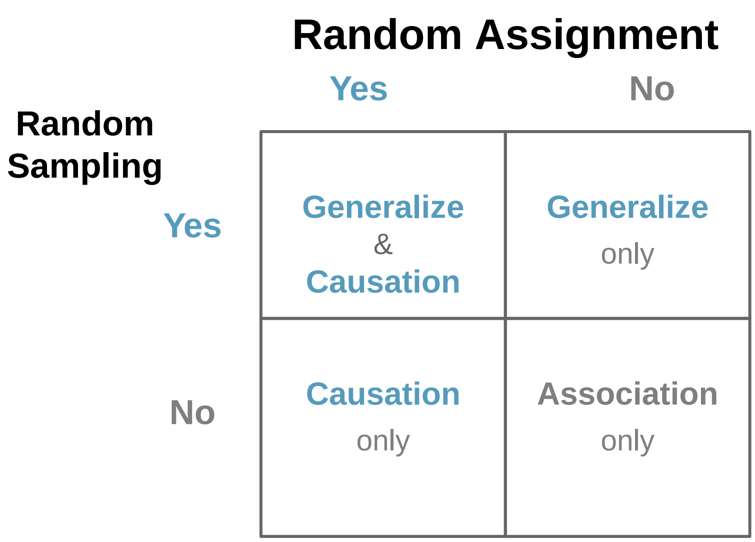


Figure 2.10: Scope of inference. The type of sampling determines whether results generalize to the population; the type of assignment determines whether causal conclusions are possible.

Very few studies achieve both random sampling *and* random assignment. In practice, experiments typically use volunteers (not random samples from the population), so they fall in the bottom-left cell: they can establish causation, but the results may not generalize beyond the type of people who volunteered. Observational studies based on random samples (top-right cell) can generalize to the population but cannot establish cause-and-effect.

A researcher randomly selects 500 adults from a city's voter rolls and surveys them about exercise habits and depression symptoms. She finds that people who exercise more report fewer depression symptoms. What conclusions can she draw?

Show answer

Because the researcher used **random sampling** (from the voter rolls), she can generalize her findings to the city's adult voter population. However, because she did not randomly **assign** people to exercise or not exercise (this is an observational study), she cannot conclude that exercise *causes* fewer depression symptoms. There may be confounding variables — for example, people with fewer depression symptoms may be more motivated to exercise in the first place.

A pharmaceutical company recruits 200 volunteers with chronic pain and randomly assigns half to receive a new pain medication and half to receive a placebo. The group receiving the medication reports significantly less pain. What is the scope of inference?

Show answer

Because the study used **random assignment**, the company can conclude that the medication *caused* the reduction in pain (a causal conclusion). However, because the participants were volunteers rather than a random sample of all chronic pain patients, the results may not generalize to the entire population of chronic pain sufferers — only to people similar to those who volunteered.

2.9 Chapter review

2.9.1 Summary

In this chapter, we explored how data are collected and why the method of collection matters for the conclusions we can draw.

Key concepts:

- The **population** is the entire group of interest; a **sample** is a subset from which we collect data. A **census** collects data on every member of the population but is usually impractical.
- **Anecdotal evidence** — conclusions drawn from one or a few non-representative cases — is unreliable.
- **Sampling methods** include simple random sampling, stratified sampling, cluster sampling, and multistage sampling. Each has advantages depending on the population's structure and practical constraints.
- **Bias** can enter through non-random sampling (selection bias), non-response, or convenience sampling.
- In an **experiment**, researchers actively assign treatments to subjects. In an **observational study**, researchers simply observe without intervening.
- **Random assignment** to treatment groups is the key to establishing **causation**.
- **Confounding variables** are associated with both the explanatory and response variables and can create misleading associations.
- **Blinding** (single and double) and **placebos** help reduce bias in human experiments.
- The **scope of inference** depends on whether the study used random sampling (which enables generalization) and random assignment (which enables causal conclusions).

2.9.2 Key terms

Term	Definition
Population	The entire group of individuals about which we want information
Sample	A subset of the population from which we collect data
Census	An attempt to collect data from every member of the population
Population parameter	A numerical summary of the entire population
Sample statistic	A numerical summary calculated from sample data
Anecdotal evidence	Data collected in a haphazard fashion, often representing unusual cases
Simple random sample (SRS)	A sample where every case has an equal chance of selection, independently

Term	Definition
Stratified sampling	Dividing the population into strata, then taking an SRS within each
Cluster sampling	Randomly selecting entire clusters and including all cases within them
Multistage sampling	Randomly selecting clusters, then taking an SRS within each selected cluster
Bias	Systematic favoritism in data collection that makes a sample unrepresentative
Non-response bias	Bias introduced when sampled individuals do not respond
Convenience sample	A sample where easily accessible individuals are overrepresented
Experiment	A study where the researchers assign treatments to subjects
Randomized experiment	An experiment where treatments are assigned using randomization
Observational study	A study where researchers observe without assigning treatments
Confounding variable	A variable associated with both the explanatory and response variables
Treatment group	The group receiving the experimental treatment
Control group	The group not receiving the treatment (or receiving a placebo)
Placebo	A fake treatment given to the control group
Placebo effect	Improvement in patients who believe they are receiving treatment
Blind (single-blind)	Subjects do not know their treatment assignment
Double-blind	Neither subjects nor researchers know the treatment assignments
Blocking	Grouping subjects by a known variable before random assignment
Prospective study	An observational study that follows subjects forward in time
Retrospective study	An observational study that looks back at past data
Scope of inference	The extent to which study results can be generalized and/or support causal claims

2.10 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Parameters and statistics.** Identify which value represents the sample mean and which value represents the claimed population mean.

a. American households spent an average of about $52 in 2007 on Halloween merchandise such as costumes, decorations and candy. To see if this number had changed, researchers conducted a new survey in 2008 before industry numbers were reported. The survey included 1,500 households and found that average Halloween spending was $58 per household.

b. The average GPA of students in 2001 at a private university was 3.37. A survey on a sample of 203 students from this university yielded an average GPA of 3.59 a decade later.

2. **Sleeping in college.** A recent article in a college newspaper stated that college students get an average of 5.5 hrs of sleep each night. A student who was skeptical about this value decided to conduct a survey by randomly sampling 25 students. On average, the sampled students slept 6.25 hours per night. Identify which value represents the sample mean and which value represents the claimed population mean.

3. **Air pollution and birth outcomes, scope of inference.** Researchers collected data to examine the relationship between air pollutants and preterm births in Southern California. During the study air pollution levels were measured by air quality monitoring stations. Length of gestation data were collected on 143,196 births between the years 1989 and 1993, and air pollution exposure during gestation was calculated for each birth. (Ritz et al. 2000)

a. Identify the population of interest and the sample in this study.

b. Comment on whether the results of the study can be generalized to the population, and if the findings of the study can be used to establish causal relationships.

4. **Cheaters, scope of inference.** Researchers studying the relationship between honesty, age and self-control conducted an experiment on 160 children between the ages of 5 and 15. The researchers asked each child to toss a fair coin in private and to record the outcome (white or black) on a paper sheet, and said they would only reward children who report white. Half the students were explicitly told not to cheat and the others were not given any explicit instructions. Differences were observed in the cheating rates in the instruction and no instruction groups, as well as some differences across children's characteristics within each group. (Bucciol and Piovesan 2011)

a. Identify the population of interest and the sample in this study.

b. Comment on whether the results of the study can be generalized to the population, and if the findings of the study can be used to establish causal relationships.

5. **Gamification and statistics, scope of inference.** Researchers investigating the effects of gamification (application of game-design elements and game principles in non-game contexts) on learning statistics randomly assigned 365 college students in a statistics course to one of four groups; one of these groups had no reading exercises and no gamification, one group had reading but no gamification, one group had gamification but no reading, and a final group had gamification and reading. Students in all groups also attended lectures. The study found that gamification had a positive impact on student learning compared to traditional teaching methods involving reading exercises. (Legaki et al. 2020)

a. Identify the population of interest and the sample in this study.

b. Comment on whether the results of the study can be generalized to the population, and if the findings of the study can be used to establish causal relationships.

6. **Stealers, scope of inference.** In a study of the relationship between socio-economic class and unethical behavior, 129 University of California undergraduates at Berkeley were asked to identify themselves as having low or high social-class by comparing themselves to others with the most (least) money, most (least) education, and most (least) respected jobs. They were also presented with a jar of individually wrapped candies and informed that the candies were for children in a nearby laboratory, but that they could take some if they wanted. After completing some unrelated tasks, participants reported the number of candies they had taken. It was found that those who were identified as upper-class took more candy than others. (Piff et al. 2012)

a. Identify the population of interest and the sample in this study.

b. Comment on whether the results of the study can be generalized to the population, and if the findings of the study can be used to establish causal relationships.

7. **Relaxing after work.** The General Social Survey asked the question, *"After an average work day, about how many hours do you have to relax or pursue activities that you enjoy?"* to a random sample of 1,155 Americans. The average relaxing time was found to be 1.65 hours. Determine which of the following is an observation, a variable, a sample statistic, or a population parameter.

a. An American in the sample.

b. Number of hours spent relaxing after an average work day.

c. 1.65.

d. Average number of hours all Americans spend relaxing after an average work day.

8. **Cats on YouTube.** Suppose you want to estimate the percentage of videos on YouTube that are cat videos. It is impossible for you to watch all videos on YouTube so you use a random video picker to select 1000 videos for you. You find that 2% of these videos are cat videos. Determine which of the following is an observation, a variable, a sample statistic, or a population parameter.

a. Percentage of all videos on YouTube that are cat videos.

b. 2%.

c. A video in your sample.

d. whether a video is a cat video.

9. **Course satisfaction across sections.** A large college class has 160 students. All 160 students attend the lectures together, but the students are divided into 4 groups, each of 40 students, for lab sections administered by different teaching assistants. The professor wants to conduct a survey about how satisfied the students are with the course, and he believes that the lab section a student is in might affect the student's overall satisfaction with the course.

a. What type of study is this?

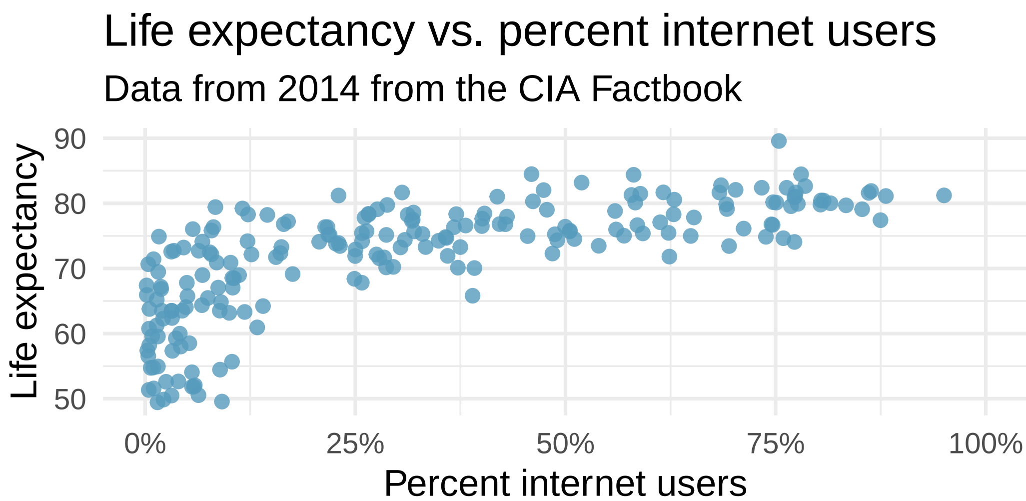
b. Suggest a sampling strategy for carrying out this study.

10. **Housing proposal across dorms.** On a large college campus first-year students and sophomores live in dorms located on the eastern part of the campus and juniors and seniors live in dorms located on the western part of the campus. Suppose you want to collect student opinions on a new housing structure the college administration is proposing and you want to make sure your survey equally represents opinions from students from all years.

a. What type of study is this?

b. Suggest a sampling strategy for carrying out this study.

11. **Internet use and life expectancy.** The following scatterplot was created as part of a study evaluating the relationship between estimated life expectancy at birth (as of 2014) and percentage of internet users (as of 2009) in 208 countries for which such data were available.



a. Describe the relationship between life expectancy and percentage of internet users.

b. What type of study is this?

c. State a possible confounding variable that might explain this relationship and describe its potential effect.

12. **Stressed out.** A study that surveyed a random sample of otherwise healthy high school students found that they are more likely to get muscle cramps when they are stressed. The study also noted that students drink more coffee and sleep less when they are stressed.

a. What type of study is this?

b. Can this study be used to conclude a causal relationship between increased stress and muscle cramps?

c. State possible confounding variables that might explain the observed relationship between increased stress and muscle cramps.

13. **Evaluate sampling methods.** A university wants to determine what fraction of its undergraduate student body support a new $25 annual fee to improve the student union. For each proposed method below, indicate whether the method is reasonable or not.

a. Survey a simple random sample of 500 students.

b. Stratify students by their field of study, then sample 10% of students from each stratum.

c. Cluster students by their ages (e.g., 18 years old in one cluster, 19 years old in one cluster, etc.), then randomly sample three clusters and survey all students in those clusters.

14. **Random digit dialing.** The Gallup Poll uses a procedure called random digit dialing, which creates phone numbers based on a list of all area codes in America in conjunction with the associated number of residential households in each area code. Give a possible reason the Gallup Poll chooses to use random digit dialing instead of picking phone numbers from the phone book.

15. **Haters are gonna hate, study confirms.** A study published in the *Journal of Personality and Social Psychology* asked a group of 200 randomly sampled participants recruited online using Amazon's Mechanical Turk to evaluate how they felt about various subjects, such as camping, health care, architecture, taxidermy, crossword puzzles, and Japan in order to measure their attitude towards mostly independent stimuli. Then, they presented the participants with information about a new product: a microwave oven. This microwave oven does not exist, but the participants didn't know this, and were given three positive and three negative fake reviews. People who reacted positively to the subjects on the dispositional attitude measurement also tended to react positively to the microwave oven, and those who reacted negatively tended to react negatively to it. Researchers concluded that *"some people tend to like things, whereas others tend to dislike things, and a more thorough understanding of this tendency will lead to a more thorough understanding of the psychology of attitudes."* (Hepler and Albarracín 2013)

a. What are the cases?

b. What is (are) the response variable(s) in this study?

c. What is (are) the explanatory variable(s) in this study?

d. Does the study employ random sampling? Explain your reasoning.

e. Is this an observational study or an experiment? Explain your reasoning.

f. Can we establish a causal link between the explanatory and response variables?

g. Can the results of the study be generalized to the population at large?

16. **Reading the paper.** Below are excerpts from two articles published in the *NY Times*:

a. An excerpt from an article titled *Risks: Smokers Found More Prone to Dementia* is below. Based on this study, can we conclude that smoking causes dementia later in life? Explain your reasoning. (Rabin 2010)

> "Researchers analyzed data from 23,123 health plan members who participated in a voluntary exam and health behavior survey from 1978 to 1985, when they were 50-60 years old. 23 years later, about 25% of the group had dementia, including 1,136 with Alzheimer's disease and 416 with vascular dementia. After adjusting for other factors, the researchers concluded that pack-a-day smokers were 37% more likely than nonsmokers to develop dementia, and the risks went up with increased smoking; 44% for one to two packs a day; and twice the risk for more than two packs."

b. An excerpt from an article titled *The School Bully Is Sleepy* is below. A friend of yours who read the article says, *"The study shows that sleep disorders lead to bullying in school children."* Is this statement justified? If not, how best can you describe the conclusion that can be drawn from this study? (Parker-Pope 2011)

> "The University of Michigan study, collected survey data from parents on each child's sleep habits and asked both parents and teachers to assess behavioral concerns. About a third of the students studied were identified by parents or teachers as having problems with disruptive behavior or bullying. The researchers found that children who had behavioral issues and those who were identified as bullies were twice as likely to have shown symptoms of sleep disorders."

17. **Sampling strategies.** A statistics student who is curious about the relationship between the amount of time students spend on social networking sites and their performance at school decides to conduct a survey. Various research strategies for collecting data are described below. In each, name the sampling method proposed and any bias you might expect.

a. They randomly sample 40 students from the study's population, give them the survey, ask them to fill it out, and bring it back the next day.

b. They give out the survey only to their friends, making sure each one of them fills it out.

c. They post a link to an online survey on Facebook and ask their friends to fill it out.

d. They randomly sample 5 classes and asks a random sample of students from those classes to fill out the survey.

18. **Family size.** Suppose we want to estimate household size, where a *"household"* is defined as people living together in the same dwelling, and sharing living accommodations. If we select students at random at an elementary school and ask them what their family size is, will this be a good measure of household size? Or will our average be biased? If so, will it overestimate or underestimate the true value?

19. **Light and exam performance.** A study is designed to test the effect of light level on exam performance of students. The researcher believes that light levels might have different effects on people who wear glasses and people who don't, so they want to make sure both groups of people are equally represented in each treatment. The treatments are fluorescent overhead lighting, yellow overhead lighting, no overhead lighting (only desk lamps).

   a. What is the response variable?

   b. What is the explanatory variable? What are its levels?

   c. What is the blocking variable? What are its levels?

20. **Vitamin supplements.** To assess the effectiveness of taking large doses of vitamin C in reducing the duration of the common cold, researchers recruited 400 healthy volunteers from staff and students at a university. A quarter of the patients were assigned a placebo, and the rest were evenly divided between 1g Vitamin C, 3g Vitamin C, or 3g Vitamin C plus additives to be taken at onset of a cold for the following two days. All tablets had identical appearance and packaging. The nurses who handed the prescribed pills to the patients knew which patient received which treatment, but the researchers assessing the patients when they were sick did not. No statistically discernible differences were observed in any measure of cold duration or severity between the four groups, and the placebo group had the shortest duration of symptoms. (Audera et al. 2001)

   a. Was this an experiment or an observational study? Why?

   b. What are the explanatory and response variables in this study?

   c. Were the patients blinded to their treatment?

   d. Was this study double-blind?

   e. Participants are ultimately able to choose whether to use the pills prescribed to them. We might expect that not all of them will adhere and take their pills. Does this introduce a confounding variable to the study? Explain your reasoning.

21. **Light, noise, and exam performance.** A study is designed to test the effect of light level and noise level on exam performance of students. The researcher believes that light and noise levels might have different effects on people who wear glasses and people who don't, so they want to make sure both groups of people are equally represented in each treatment. The light treatments considered are fluorescent overhead lighting, yellow overhead lighting, no overhead lighting (only desk lamps). The noise treatments considered are no noise, construction noise, and human chatter noise.

   a. What type of study is this?

   b. How many factors are considered in this study? Identify them, and describe their levels.

   c. What is the role of the wearing glasses variable in this study?

22. **Music and learning.** You would like to conduct an experiment in class to see if students learn better if they study without any music, with music that has no lyrics (instrumental), or with music that has lyrics. Briefly outline a design for this study.

23. **Soda preference.** You would like to conduct an experiment in class to see if your classmates prefer the taste of regular Coke or Diet Coke. Briefly outline a design for this study.

24. **Exercise and mental health.** A researcher is interested in the effects of exercise on mental health and they propose the following study: use stratified random sampling to ensure representative proportions of 18-30, 31-40 and 41- 55 year-olds from the population. Next, randomly assign half the subjects from each age group to exercise twice a week, and instruct the rest not to exercise. Conduct a mental health exam at the beginning and at the end of the study, and compare the results.
    a. What type of study is this?
    b. What are the treatment and control groups in this study?
    c. Does this study make use of blocking? If so, what is the blocking variable?
    d. Does this study make use of blinding?
    e. Comment on whether the results of the study can be used to establish a causal relationship between exercise and mental health, and indicate whether the conclusions can be generalized to the population at large.
    f. Suppose you are given the task of determining if this proposed study should get funding. Would you have any reservations about the study proposal?

25. **Chia seeds and weight loss.** Chia Pets – those terra-cotta figurines that sprout fuzzy green hair – made the chia plant a household name. But chia has since gained a reputation as a diet supplement. In one 2009 study, 38 men and 38 women were recruited and and divided each randomly into two groups: treatment or control. One group was given 25 grams of chia seeds twice a day, and the other was given a placebo. The subjects volunteered to be a part of the study. After 12 weeks, the scientists found no statistically discernible difference between the groups in appetite or weight loss. (Nieman et al. 2009)
    a. What type of study is this?
    b. What are the experimental and control treatments in this study?
    c. Has blocking been used in this study? If so, what is the blocking variable?
    d. Has blinding been used in this study?
    e. Comment on whether we can make a causal statement, and indicate whether we can generalize the conclusion to the population at large.

26. **City council survey.** A city council has requested a household survey be conducted in a suburban area of their city. The area is broken into many distinct and unique neighborhoods, some including large homes, some with only apartments, and others a diverse mixture of housing structures. For each part below, identify the sampling methods described, and describe the statistical pros and cons of the method in the city's context.
    a. Randomly sample 200 households from the city.
    b. Divide the city into 20 neighborhoods, and then sample 10 households from each neighborhood.
    c. Divide the city into 20 neighborhoods, randomly sample 3 neighborhoods, and then sample all households from those 3 neighborhoods.
    d. Divide the city into 20 neighborhoods, randomly sample 8 neighborhoods, and then randomly sample 50 households from those neighborhoods.
    e. Sample the 200 households closest to the city council offices.

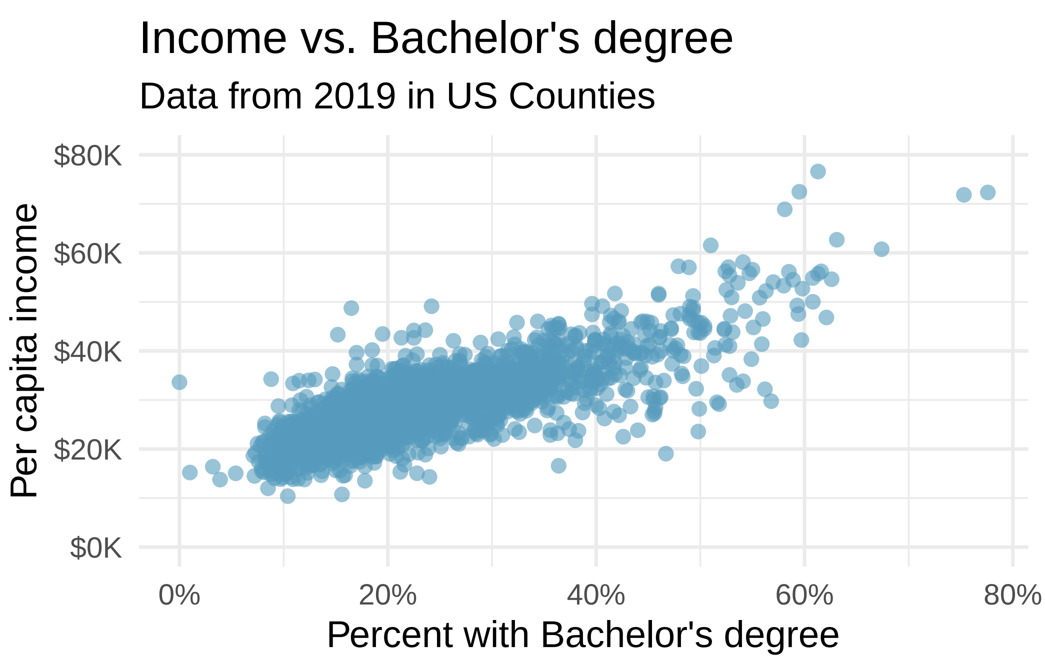
27. **Flawed reasoning.** Identify the flaw(s) in reasoning in the following scenarios. Explain what the individuals in the study should have done differently if they wanted to make such strong conclusions.

a. Students at an elementary school are given a questionnaire that they are asked to return after their parents have completed it. One of the questions asked is, *"Do you find that your work schedule makes it difficult for you to spend time with your kids after school?"* Of the parents who replied, 85% said *"no"*. Based on these results, the school officials conclude that a great majority of the parents have no difficulty spending time with their kids after school.

b. A survey is conducted on a simple random sample of 1,000 women who recently gave birth, asking them about whether they smoked during pregnancy. A follow-up survey asking if the children have respiratory problems is conducted 3 years later. However, only 567 of these women are reached at the same address. The researcher reports that these 567 women are representative of all mothers.

c. An orthopedist administers a questionnaire to 30 of his patients who do not have any joint problems and finds that 20 of them regularly go running. He concludes that running decreases the risk of joint problems.

28. **Income and education in US counties.** The scatterplot below shows the relationship between per capita income (in thousands of dollars) and percent of population with a bachelor's degree in 3,142 counties in the US in 2019.



a. What are the explanatory and response variables?

b. Describe the relationship between the two variables. Make sure to discuss unusual observations, if any.

c. Can we conclude that having a bachelor's degree increases one's income?

29. **Eat well, feel better.** In a public health study on the effects of consumption of fruits and vegetables on psychological well-being in young adults, participants were randomly assigned to three groups: (1) diet-as-usual, (2) an ecological momentary intervention involving text message reminders to increase their fruits and vegetable consumption plus a voucher to purchase them, or (3) a fruit and vegetable intervention in which participants were given two additional daily servings of fresh fruits and vegetables to consume on top of their normal diet. Participants were asked to take a nightly survey on their smartphones. Participants were student volunteers at the University of Otago, New Zealand. At the end of the 14-day study, only participants in the third group showed improvements to their psychological well-being across the 14-days relative to the other groups. (Conner et al. 2017)

   a. What type of study is this?

   b. Identify the explanatory and response variables.

   c. Comment on whether the results of the study can be generalized to the population.

   d. Comment on whether the results of the study can be used to establish causal relationships.

   e. A newspaper article reporting on the study states, "The results of this study provide proof that giving young adults fresh fruits and vegetables to eat can have psychological benefits, even over a brief period of time." How would you suggest revising this statement so that it can be supported by the study?

30. **Screens, teens, and psychological well-being.** In a study of three nationally representative large-scale datasets from Ireland, the United States, and the United Kingdom (n = 17,247), teenagers between the ages of 12 to 15 were asked to keep a diary of their screen time and answer questions about how they felt or acted. The answers to these questions were then used to compute a psychological well-being score. Additional data were collected and included in the analysis, such as each child's sex and age, and on the mother's education, ethnicity, psychological distress, and employment. The study concluded that there is little clear-cut evidence that screen time decreases adolescent well-being. (Orben and Baukney-Przybylski 2018)

   a. What type of study is this?

   b. Identify the explanatory variables.

   c. Identify the response variable.

   d. Comment on whether the results of the study can be generalized to the population, and why.

   e. Comment on whether the results of the study can be used to establish causal relationships.

Datasets: `cia_factbook` (openintro) | `county_complete` (openintro)

Chapter 3

Exploring Categorical Data

This chapter focuses on **categorical data** — variables whose values are categories or labels rather than numbers. We begin with visualizations for a single categorical variable (bar charts, pie charts, waffle charts), then move to tools for exploring the relationship between two categorical variables (contingency tables and stacked bar charts). We close with principles for creating effective, honest data visualizations.

StatLens tools for this chapter: Use the One Categorical Variable explorer for bar charts, pie charts, and waffle charts. Use the Categorical Data (Two Variables) explorer for contingency tables and stacked bar charts.

3.1 Visualizing one categorical variable

3.1.1 Bar charts

A **bar chart** is the most common way to display the distribution of a single categorical variable. Each category gets its own bar, and the height (or length) of the bar represents either the count or the proportion of observations in that category.

Consider a dataset of 10,000 loans from a lending platform. Each borrower's homeownership status is recorded as `rent`, `mortgage`, or `own`. The frequency table below summarizes the data:

Homeownership	Count
Rent	3,858
Mortgage	4,789
Own	1,353
Total	**10,000**

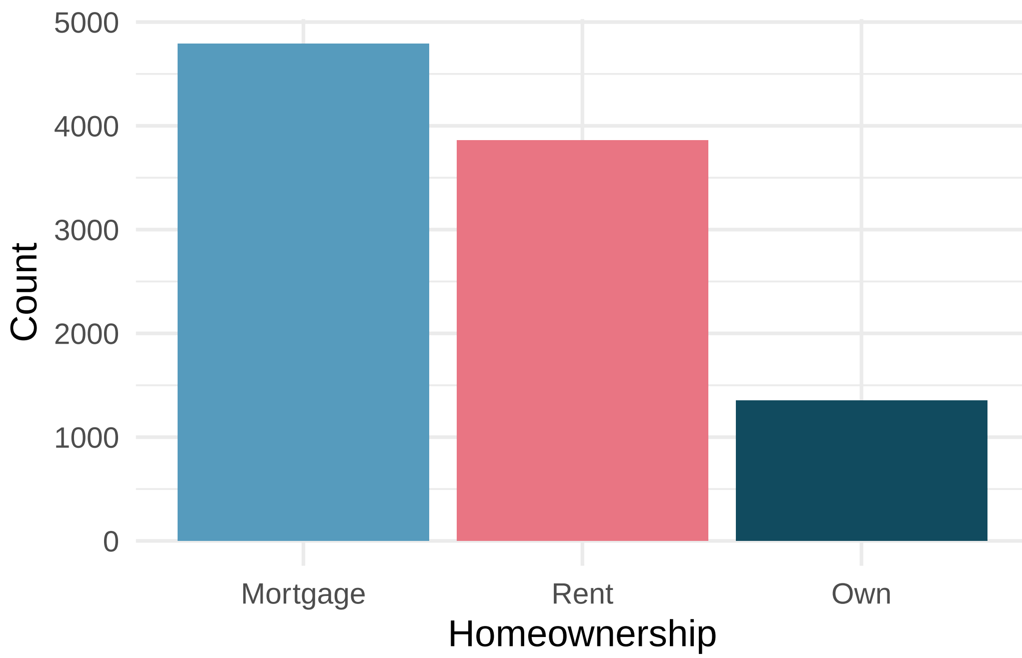


Figure 3.1: Bar chart showing counts of homeownership status. Mortgage is the tallest bar, followed by rent, then own.

A bar chart of proportions conveys the same information but rescales the vertical axis so the bars represent the fraction of observations in each category. In the homeownership example, about 47.9% of borrowers have a mortgage, 38.6% rent, and 13.5% own their home outright.

In the homeownership bar chart, which category is most common? Which is least common? About what fraction of borrowers rent?

Show answer

Mortgage is the most common category and own is the least common. About 38.6% of borrowers rent their home.

3.1.2 Pie charts

A **pie chart** divides a circle into slices, where each slice represents a category and the size of the slice corresponds to the proportion of observations in that category.

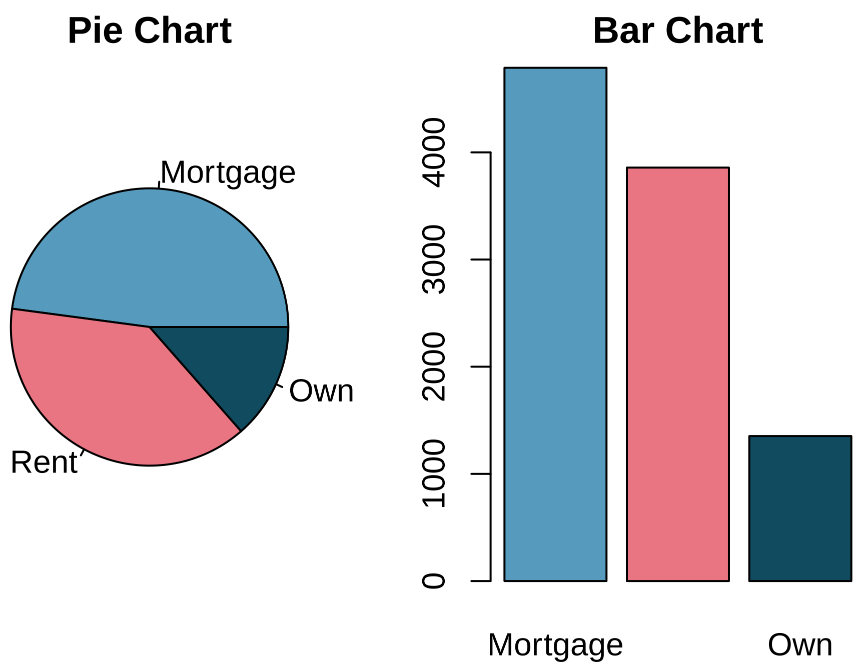


Figure 3.2: A pie chart and bar chart displaying the same homeownership data side by side.

Pie charts can give a quick high-level overview, especially when a variable has only a few levels or when each level represents a simple fraction (one-half, one-quarter, etc.). However, bar charts are almost always easier to read because our eyes are better at comparing lengths than angles or areas.

Bar charts vs. pie charts. When a categorical variable has many levels, bar charts are far superior to pie charts for making comparisons. Reserve pie charts for situations where the variable has very few categories and the goal is to show how a whole breaks into parts.

3.1.3 Waffle charts

A **waffle chart** is another way to visualize proportions. It uses a grid (often 10 by 10 = 100 squares) where each square represents 1% of the data. Squares are shaded according to the categories. Like pie charts, waffle charts work best when the number of categories is small, but they can make it easier to compare proportions that do not correspond to simple fractions.

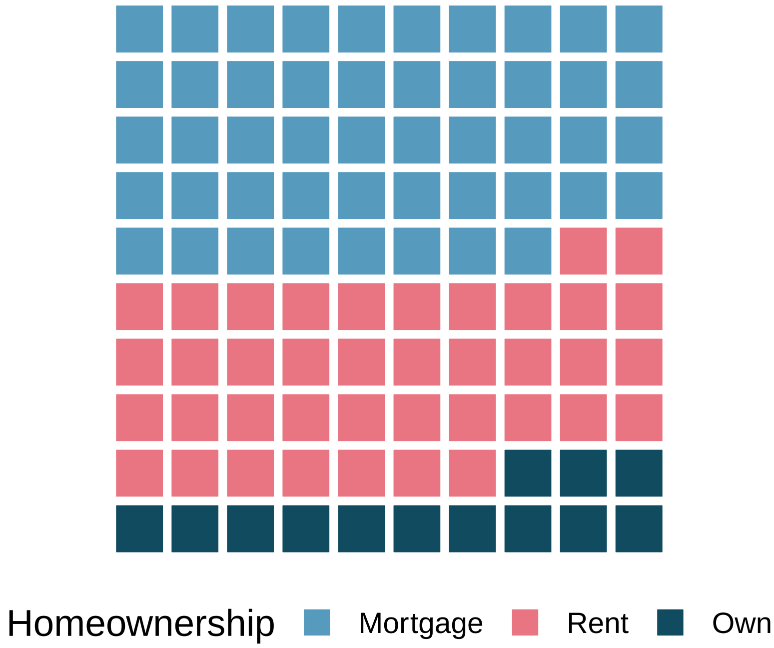


Figure 3.3: A waffle chart of homeownership status. Each square represents 1% of borrowers.

Why might a waffle chart be easier to read than a pie chart when comparing two categories with similar proportions (say, 38% and 42%)?

Show answer

In a waffle chart, you can count the number of shaded squares for each category, making it straightforward to see the difference. In a pie chart, the angle difference between 38% and 42% is very subtle and hard to judge visually.

Try it yourself. Open the Comparing Charts for Categorical Data activity to explore Brexit survey data interactively. Switch between bar charts, pie charts, and waffle charts to see which makes it easiest to compare five response categories.

3.2 Two categorical variables

When we want to explore the relationship between two categorical variables, we organize the data in a **contingency table** and display it with stacked or grouped bar charts.

3.2.1 Contingency tables

A **contingency table** summarizes data for two categorical variables. The rows represent the levels of one variable, the columns represent the levels of the other, and each cell contains the count of observations in that combination. **Row totals** and **column totals** give the marginal distributions.

Consider a dataset of 10,000 loans. Each loan has a homeownership status (`rent`, `mortgage`, or `own`) and an application type (`joint` or `individual`). The contingency table below summarizes these two variables:

	Rent	Mortgage	Own	Total
Joint	362	950	183	1,495
Individual	3,496	3,839	1,170	8,505
Total	**3,858**	**4,789**	**1,353**	**10,000**

Each cell represents a count. For example, 3,496 loans were made by individuals who rent their home. The row totals tell us that 1,495 loans were joint applications and 8,505 were individual applications. The column totals tell us the overall distribution of homeownership.

What does the value 950 in the table represent? What does the column total 4,789 represent?

Show answer

The value 950 represents the number of loans that were joint applications from borrowers with a mortgage. The column total 4,789 represents the total number of borrowers (both joint and individual) who have a mortgage.

3.2.2 Row and column proportions

Raw counts can be hard to compare when group sizes differ. Instead, we can compute **row proportions** or **column proportions** to see conditional relationships.

Row proportions are computed by dividing each count by its row total. They show how a variable is distributed *within each level* of the row variable.

Column proportions are computed by dividing each count by its column total. They show how a variable is distributed *within each level* of the column variable.

Both row and column proportions are examples of **conditional proportions —** the proportion of observations in one category, conditional on (given) the level of the other variable.

Row proportions (each count divided by its row total):

	Rent	Mortgage	Own	Total
Joint	0.242	0.635	0.122	1.000
Individual	0.411	0.451	0.138	1.000

For example, $3{,}496/8{,}505 = 0.411$, meaning 41.1% of individual applicants rent their home.

Column proportions (each count divided by its column total):

	Rent	Mortgage	Own
Joint	0.094	0.198	0.135
Individual	0.906	0.802	0.865
Total	1.000	1.000	1.000

The value 0.906 tells us that 90.6% of renters applied individually. This rate is higher than for borrowers with mortgages (80.2%) or who own their home (86.5%). Because these rates vary across the levels of homeownership, there is evidence that application type and homeownership may be **associated**.

What does 0.451 represent in the row proportions table? What does 0.802 represent in the column proportions table?

Show answer

0.451 represents the proportion of individual applicants who have a mortgage. 0.802 represents the fraction of applicants with mortgages who applied as individuals.

Data scientists use statistics to build email spam filters. One characteristic examined is email format (HTML or plain text). A contingency table of email format and spam status is shown below:

	HTML	Text	Total
Not spam	2,568	986	3,554
Spam	158	209	367
Total	**2,726**	**1,195**	**3,921**

Which would be more useful for classifying email: row or column proportions?

A data scientist would want to know how the proportion of spam changes depending on email format. This corresponds to **column proportions**: $209/1{,}195 = 17.5\%$ of plain text emails are spam, compared to $158/2{,}726 = 5.8\%$ of HTML emails. Plain text emails are more likely to be spam. Row or column proportions are not equivalent — before computing, consider which variable you want to "condition on" (usually the explanatory variable).

In the spam example, which variable would you consider the explanatory variable and which the response?

Show answer

Email format (HTML vs. text) is the explanatory variable — it is a characteristic of the email that might help explain or predict whether the email is spam. Spam status (spam vs. not spam) is the response variable.

3.2.3 Bar charts for two categorical variables

Bar charts can be extended to display the relationship between two categorical variables. There are three common variants:

A **stacked bar chart** stacks the bars for the second variable on top of each other within each level of the first variable. It shows both the total count and the breakdown by the second variable.

A **100% stacked bar chart** (also called a standardized or filled bar chart) is like a stacked bar chart, but the bars are scaled to the same height (100%). It shows the *proportion* of each category of the second variable within each level of the first variable.

A **side-by-side bar chart** (also called a grouped or dodged bar chart) places bars for the second variable next to each other within each level of the first variable. It makes it easy to compare counts across groups.

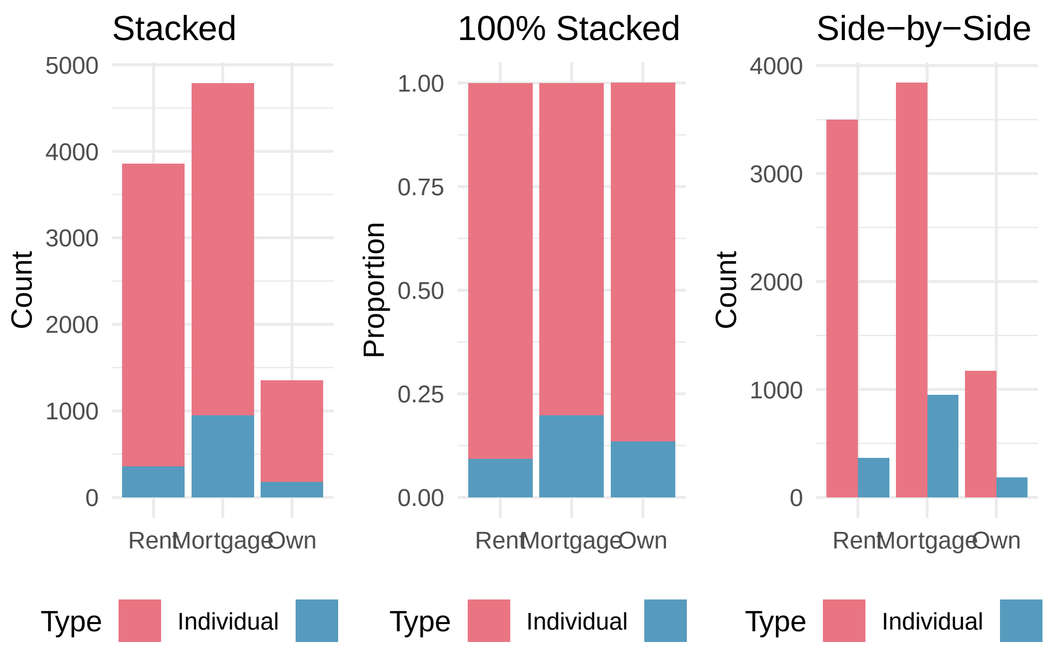


Figure 3.4: Three bar charts showing homeownership and application type: stacked, 100% stacked, and side-by-side.

When is each type of bar chart most useful?

- The **stacked bar chart** is most useful when you want to see both the total count for each group and the breakdown by the second variable.
- The **100% stacked bar chart** is helpful when comparing proportions across groups, especially when the groups have very different sizes. However, it sacrifices information about the total count.
- The **side-by-side bar chart** makes it easy to compare individual counts directly. It works well for showing both variables without implying one is the explanatory variable. However, it requires more horizontal space.

Detecting association in a 100% stacked bar chart. If the proportional breakdown of the response variable is the *same* across all levels of the explanatory variable, then the variables are **not** associated — the colored segments will line up at the same heights. If the breakdown *differs* across levels, the variables are associated.

In the 100% stacked bar chart above, how can you tell whether homeownership and application type are associated?

Show answer

If the variables were not associated, the division between joint and individual would occur at the same height in every bar. Since the bars are divided at different heights — mortgage holders have a higher proportion of joint applications than renters or owners — the variables appear to be associated.

Try it yourself. Open the Two-Variable Categorical Explorer in StatLens to explore contingency tables interactively. Switch between Stacked, Side-by-Side, and 100% Stacked bar charts to see how each reveals different aspects of the relationship.

3.3 Effective data visualization

Graphs can powerfully communicate ideas directly and quickly. However, there are times when a visualization conveys a message that is inaccurate or misleading. This section provides guiding principles for creating clear, honest, and effective visualizations.

3.3.1 Keep it simple

Colors should be used purposefully — to group items or differentiate levels in meaningful ways. Colors used only for decoration can be distracting.

Consider a company that has summarized its costs into five categories. A three-dimensional pie chart makes it hard to compare the sizes of the slices, and the unnecessary 3D effect distorts the proportions. A simple bar chart conveys the same information much more clearly.

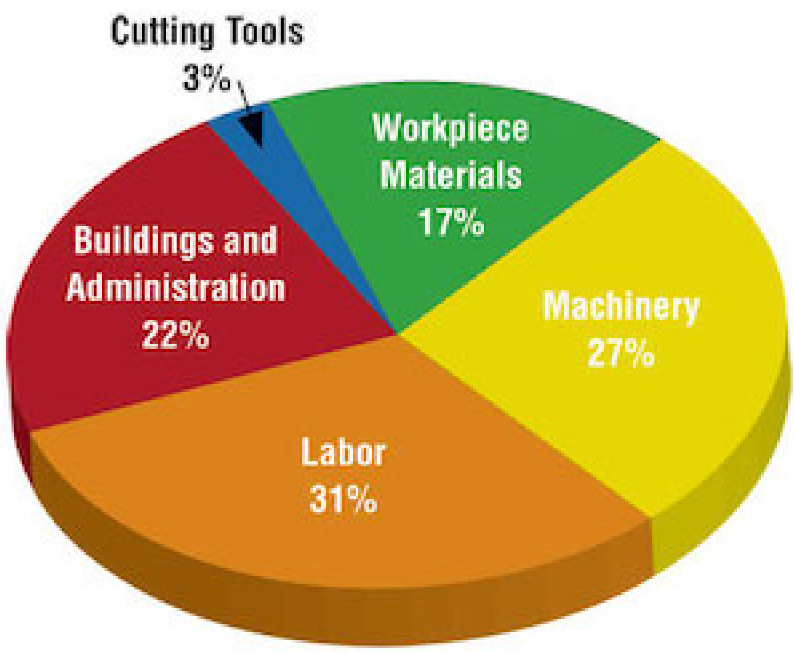


Figure 3.5: A 3D pie chart next to a bar chart, both showing the same cost breakdown. The bar chart is much easier to read.

3.3.2 Use color to draw attention

An important principle is to use color **to draw attention** to the most important part of your graphic. Default rainbow coloring often adds no meaning and can confuse readers.

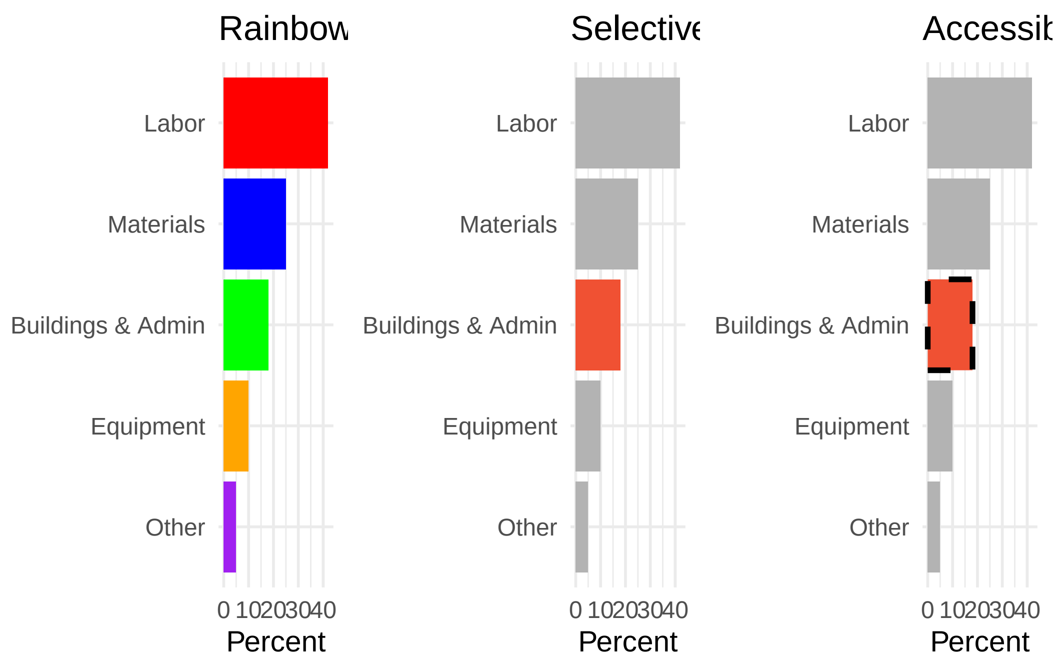


Figure 3.6: Three versions of the same bar chart. The first uses distracting rainbow colors. The second highlights one important bar in red against gray bars. The third adds both color and a dashed border for accessibility.

Accessibility matters. Not everyone perceives color the same way. When using color to draw attention, also consider adding a secondary visual cue (such as a pattern, line style, or border) so that the highlighted feature is distinguishable even for people with color vision deficiency.

3.3.3 Tell a story

Effective graphs often include **annotations** — text, lines, or labels that provide context beyond the raw data. For example, adding a note that "July is the start of the fiscal year" to a time series of monthly hiring can explain why there is a spike every July.

3.3.4 Order matters

The arrangement of categories in a bar chart can help or hinder understanding. Alphabetical order is rarely the most informative choice. Consider ordering bars by frequency (tallest to shortest) or by a meaningful sequence (such as the order the options appeared in a survey question).

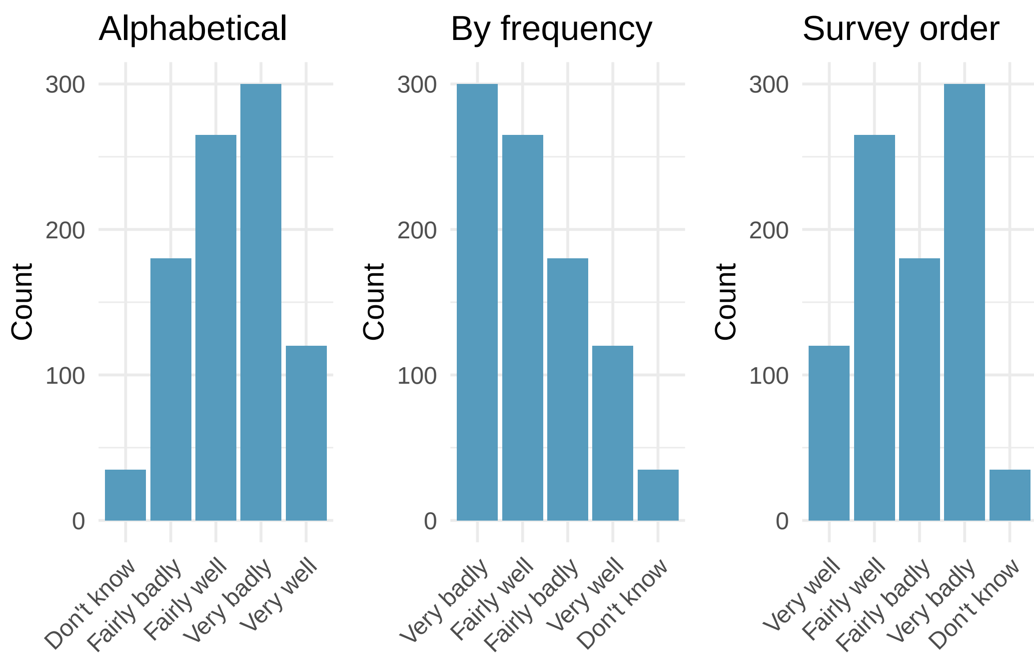


Figure 3.7: Three versions of a bar chart showing survey responses ordered alphabetically, by frequency, and by the original survey order.

3.3.5 Make labels readable

When category labels are long, consider using horizontal bars (where labels appear along the vertical axis with plenty of space) rather than vertical bars (where labels must be angled or truncated).

3.3.6 Pick a purpose

Every graphical decision should be made with a purpose. Before choosing a chart type, ask:

- What is the key comparison I want the reader to make?
- Which chart type makes that comparison easiest?
- Am I showing proportions, counts, distributions, or relationships?

A survey asks residents of five regions about their opinion on a policy, with responses ranging from "Very well" to "Very badly." You want to compare how opinions differ across regions. What visualization would you choose?

A 100% stacked bar chart would be useful if the main comparison is about proportions within each region. A side-by-side bar chart would work if comparing raw counts across regions matters. A faceted bar chart — one panel per region — would allow the most detailed comparison. The best choice depends on which comparison is most important to the story you are telling.

3.3.7 Select meaningful colors

Default or rainbow color schemes are not always the best choice. Perceptually uniform color scales (such as the viridis or cividis scales) are designed to be distinguishable by people with a variety of color vision abilities and to be readable when printed in grayscale. For ordinal data, a sequential color scale (light to dark) communicates the ordering effectively.

Summary of data visualization principles:

1. **Keep it simple —** avoid unnecessary 3D effects, decorative colors, and chart junk
2. **Use color with purpose —** to group, differentiate, or draw attention
3. **Include context —** titles, axis labels, annotations, and data sources
4. **Choose appropriate ordering —** by frequency, natural order, or meaningful sequence
5. **Make labels readable —** flip axes if labels are long; avoid angled text
6. **Consider your audience —** use accessible color schemes and secondary visual cues
7. **Match the chart to the question —** each chart type has strengths; choose deliberately

3.4 Chapter review

3.4.1 Summary

This chapter introduced tools for exploring categorical data. For a single categorical variable, **bar charts** are the primary visualization, with **pie charts** reserved for simple cases and **waffle charts** useful for estimating percentages. For two categorical variables, **contingency tables** organize counts, **row and column proportions** reveal conditional relationships, and **stacked**, **100% stacked**, and **side-by-side bar charts** provide visual comparisons. When the proportional breakdown of one variable differs across levels of the other, the variables are **associated**. Effective visualizations follow principles of simplicity, purposeful use of color, meaningful ordering, and accessibility.

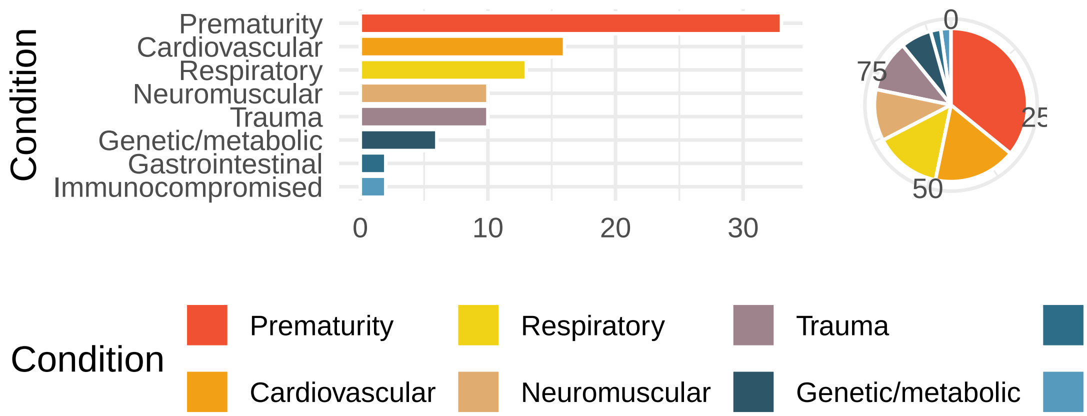
3.4.2 Key terms

- Bar chart, pie chart, waffle chart
- Frequency table, proportion
- Contingency table, row totals, column totals
- Row proportions, column proportions, conditional proportions
- Stacked bar chart, 100% stacked bar chart, side-by-side bar chart
- Association (between categorical variables)
- Explanatory variable, response variable

3.5 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Antibiotic use in children.** The bar plot and the pie chart below show the distribution of pre-existing medical conditions of children involved in a study on the optimal duration of antibiotic use in treatment of tracheitis, which is an upper respiratory infection.



a. What features are apparent in the bar plot but not in the pie chart?

b. What features are apparent in the pie chart but not in the bar plot?

c. Which graph would you prefer to use for displaying these categorical data?

2. **Views on immigration.** Nine-hundred and ten (910) randomly sampled registered voters from Tampa, FL were asked if they thought workers who have illegally entered the US should be (i) allowed to keep their jobs and apply for US citizenship, (ii) allowed to keep their jobs as temporary guest workers but not allowed to apply for US citizenship, or (iii) lose their jobs and have to leave the country. The results of the survey by political ideology are shown below.

Response	Conservative	Liberal	Moderate	Total
Apply for citizenship	57	101	120	278
Guest worker	121	28	113	262
Leave the country	179	45	126	350
Not sure	15	1	4	20
Total	372	175	363	910

a. What percent of these Tampa, FL voters identify themselves as conservatives?

b. What percent of these Tampa, FL voters are in favor of the citizenship option?

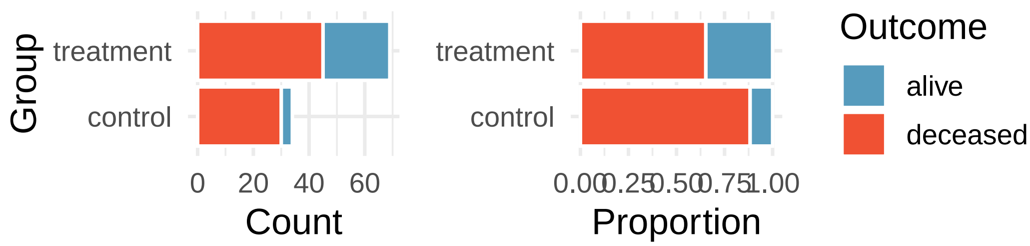
c. What percent of these Tampa, FL voters identify themselves as conservatives and are in favor of the citizenship option?

d. What percent of these Tampa, FL voters who identify themselves as conservatives are also in favor of the citizenship option? What percent of moderates share this view? What percent of liberals share this view?

e. Do political ideology and views on immigration appear to be associated? Explain your reasoning.

f. Conjecture other variables that might explain the potential relationship between these two variables.

3. **Heart transplant data display.** The Stanford University Heart Transplant Study was conducted to determine whether an experimental heart transplant program increased lifespan. Each patient entering the program was officially designated a heart transplant candidate, meaning that they were gravely ill and might benefit from a new heart. Patients were randomly assigned into treatment and control groups. Patients in the treatment group received a transplant, and those in the control group did not. The visualizations below display two different versions of the study results. (Turnbull, Brown, and Hu 1974)

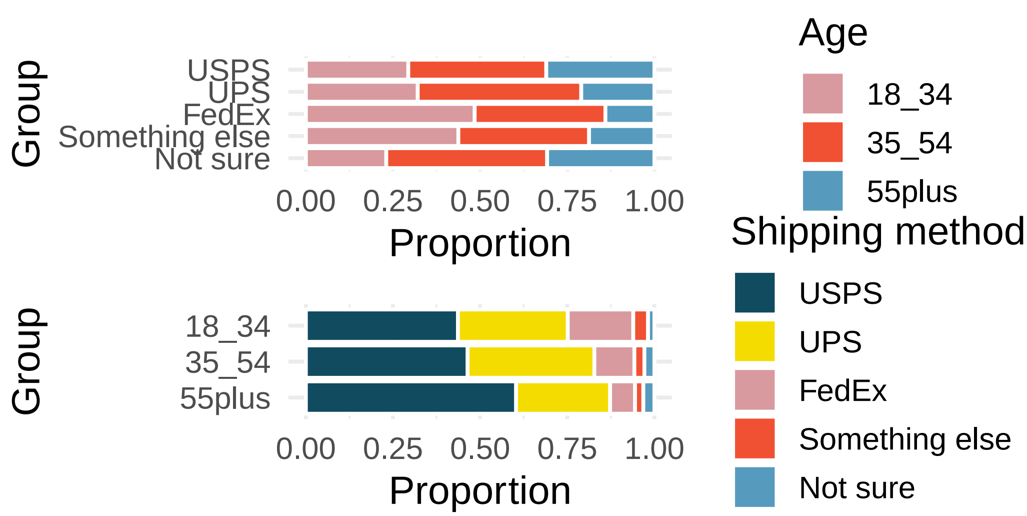


a. Provide one aspect of the two group comparison that is easier to see from the stacked bar plot (left)?

b. Provide one aspect of the two group comparison that is easeir to see from the standardized bar plot (right)?

c. For the Heart Transplant Study which of those aspects would be more important to display? That is, which bar plot would be better as a data visualization?

4. **Shipping holiday gifts data display.** A local news survey asked 500 randomly sampled Los Angeles residents which shipping carrier they prefer to use for shipping holiday gifts. The bar plots below show the distribution of responses by age group as well as distribution of responses by shipping method.



a. Which graph (top or bottom) would you use to understand the shipping choices of people of different ages? Explain.

b. Which graph (top or bottom) would you use to understand the age distribution across different types of shipping choices? Explain.

c. A new shipping company would like to market to people over the age of 55. Who will be their biggest competitor? Explain.

d. FedEx would like to reach out to grow their market share so as to balance the age demographics of FedEx users. To what age group should FedEx market?

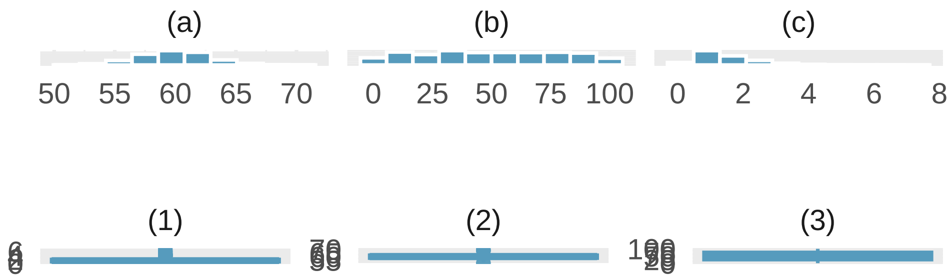
5. **Florence Nightingale.** Florence Nightingale was a nurse in the Crimean War and an early statistician. In her notes, she opined, "In comparing the deaths of one hospital with those of another, any statistics are justly considered absolutely valueless which do not give the ages, the sexes, and the diseases of all the cases." (Nightingale 1859)

a. Nightingale describes three confounding variables to consider when comparing death rates across hospitals. What are they? Describe what makes each variable potentially confounding.

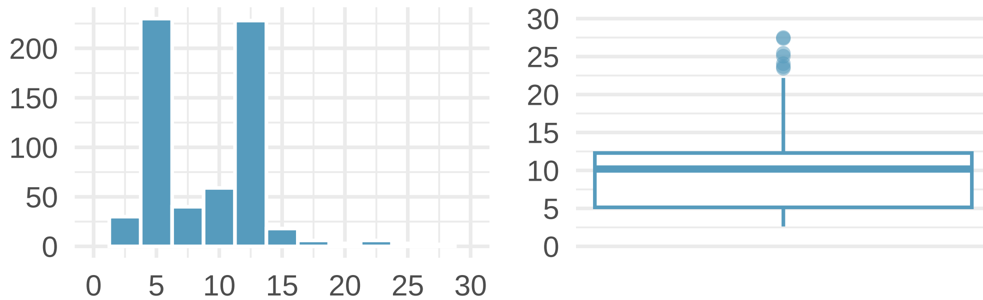
b. Provide two additional potential confounding variables for this situation. Check to make sure that the variables are associated with both the explanatory variable (hospital) and the response variable (death).

c. Why does Nightingale say that the statistics are "valueless" if given without being broken down by age, sex, and disease? Explain.

6. **Histograms and box plots.** Describe (in words) the distribution in the histograms below and match them to the box plots.



7. **Histograms vs. box plots.** Compare the two plots below. What characteristics of the distribution are apparent in the histogram and not in the box plot? What characteristics are apparent in the box plot but not in the histogram?



Datasets: `antibiotics` (openintro) | `immigration` (openintro) | `heart_transplant` (openintro)

StatLens Exercises

The following exercises use StatLens, an interactive statistics tool. Each exercise includes a link that opens a pre-loaded dataset — explore, interact, and answer the questions.

8. **Which chart for categorical data?** Open the jury racial composition data in StatLens. This dataset records the race of 275 jurors from a county. The chart opens with numeric labels hidden so you can focus on the visual patterns.

   a. Start with the **bar chart**. Using only the bar heights, rank the four racial groups from most to least represented. Which group is clearly the largest?

   b. Switch to a **pie chart**. Try to rank the four groups from largest to smallest using only the visual size of each slice. Is it easier or harder than with the bar chart? Check the **Show values** box to reveal the numbers — how close was your ranking?

   c. Switch to a **waffle chart** (uncheck **Show values** first). Count the squares for the White group to estimate their percentage. Then check **Show values** to see the actual number. Is estimating percentages easier from the waffle chart or the pie chart? Why?

   d. If a community advocacy group wanted to argue that the jury pool does not reflect the racial makeup of the county, which chart type would make the strongest visual case? Explain your reasoning.

9. **Does binning change the story?** Open the US county poverty rates in StatLens. This dataset contains poverty rates (%) for 3,137 US counties.

   a. Describe the shape of the distribution with the default number of bins. Is it symmetric, left-skewed, or right-skewed? Is it unimodal?

   b. Use the **Bins** control to reduce the number of bins to about 5–6. Does your conclusion about the shape change? What detail is lost?

   c. Now increase the bins to 30–40. Does your shape conclusion change? What new detail (if any) becomes visible?

   d. Based on your exploration, explain in your own words why two histograms of the same data can look somewhat different yet tell the same story.

10. **What do box plots hide?** Open the email character counts in StatLens. This dataset records the number of characters (in thousands) in 50 emails. Numeric labels are hidden so you can focus on visual patterns.

    a. View the data as a **histogram**. Describe the shape. Are there any gaps or clusters that stand out?

    b. Switch to a **dotplot**. With only 50 observations, can you see individual emails? Identify approximately how many emails have more than 25,000 characters.

    c. Now switch to a **boxplot**. Can you still tell whether the distribution has one peak or two? Can you identify the same gaps or clusters you saw in the histogram?

    d. A classmate claims "box plots and histograms show the same information, just in different ways." Do you agree? Use what you observed to explain why or why not.

11. **Choosing the right chart for the question.** Open the US county population data in StatLens. This dataset contains the 2017 population of 3,137 US counties.

    For each question below, state which chart type (**histogram**, **dotplot**, or **boxplot**) would be most useful, then switch to that chart type to answer it.

    a. What is the approximate median county population? Which chart type shows this most directly?

    b. Are there counties with populations over 5 million? How many? Which chart type makes this easiest to determine?

    c. Is the distribution symmetric, left-skewed, or right-skewed? Which chart type makes the skewness most visually obvious?

    d. Try increasing the number of histogram bins to 50 or more. What happens? Why does this dataset pose a challenge for histograms?

Chapter 4

Exploring Quantitative Data

In the previous chapter we explored categorical data. Now we turn to **quantitative data** — variables that take numerical values where arithmetic makes sense (heights, incomes, interest rates). This chapter teaches you to both *see* and *measure* distributions. We introduce dot plots and histograms alongside the mean and median, box plots alongside the IQR, and the standard deviation alongside histogram shapes. By the end, you will be able to describe any distribution in terms of its shape, center, spread, and unusual features.

StatLens tools for this chapter: Use the Descriptive Statistics Explorer to visualize distributions as histograms, dot plots, or box plots, and compute means, medians, standard deviations, and quartiles for any dataset.

4.1 Dot plots and measures of center

4.1.1 Dot plots

A **dot plot** is the simplest display for a single quantitative variable. Each observation is represented by a dot placed above its value on a number line. When observations share the same value (or are rounded to the same value), the dots stack vertically.

Consider the interest rates on 50 loans from a lending platform. In a dot plot, each of the 50 loans is a single dot. The red triangle marks the mean.

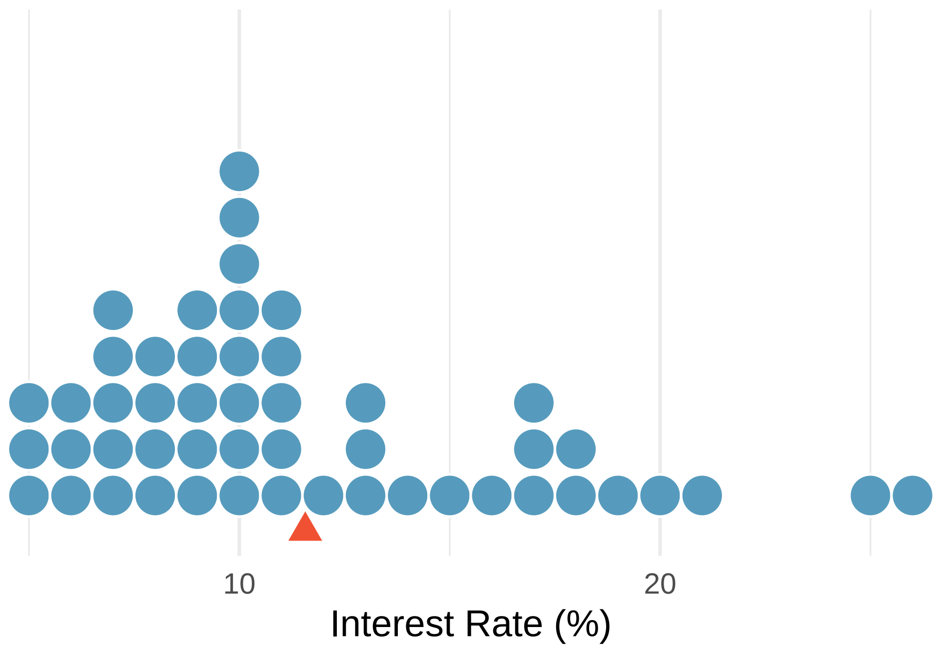


Figure 4.1: A dot plot of interest rates for 50 loans. Most values cluster between 5% and 15%, with a few high rates pulling the distribution to the right. The red triangle marks the mean (11.57%).

Dot plots are most useful for small datasets (up to about 50–100 observations) where you want to see every individual value. For larger datasets, histograms provide a better summary.

4.1.2 The mean

The **mean** (also called the **average**) is computed by adding up all the observed values and dividing by the number of observations.

Mean.

The sample mean can be calculated as the sum of the observed values divided by the number of observations:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

The notation $\bar{x}$ (read "x-bar") represents the sample mean. It's useful to think of the mean as the **balancing point** of a distribution. In the dot plot above, the mean of 11.57% is where the data would balance if placed on a seesaw — the few high interest rates on the right pull the balance point away from where most of the dots are concentrated.

The population mean is denoted by the Greek letter μ (pronounced "mew"). We often cannot measure μ directly because we rarely have access to every member of the population. Instead, we estimate μ using the sample mean $\bar{x}$.

In the formula for the mean, what does x_1 correspond to? And x_2? What does x_i represent in general?

Show answer

x_1 is the value of the variable for the first observation in the dataset, x_2 is the value for the second observation, and x_i is the value for the ith observation. For example, if the variable is interest rate, x_4 is the interest rate of the fourth loan.

Suppose we want to compute the average income per person in the US. We might think to take the mean of per capita incomes across all 3,142 counties. What is wrong with this approach?

Each county represents a different number of people. If we simply average the county-level incomes, we treat a county with 5,000 residents the same as one with 5,000,000 residents. Instead, we should compute the total income for each county, add up all counties' totals, and divide by the total population. This is called a **weighted mean**. Failing to weight by population would underestimate the true average because larger, higher-income urban counties would be underrepresented.

4.1.3 The median

The **median** is the value in the middle of the ordered data. It divides the dataset in half: 50% of the observations fall below the median and 50% fall above it.

Median: the number in the middle.

If the data are ordered from smallest to largest, the **median** is the observation right in the middle. If there are an even number of observations, the median is the average of the two middle values.

The dot plot below shows both the mean and the median marked on the interest rate data. Notice that the median (9.93%) is lower than the mean (11.57%) — the few loans with very high interest rates pull the mean to the right, but the median stays anchored in the middle of the data.

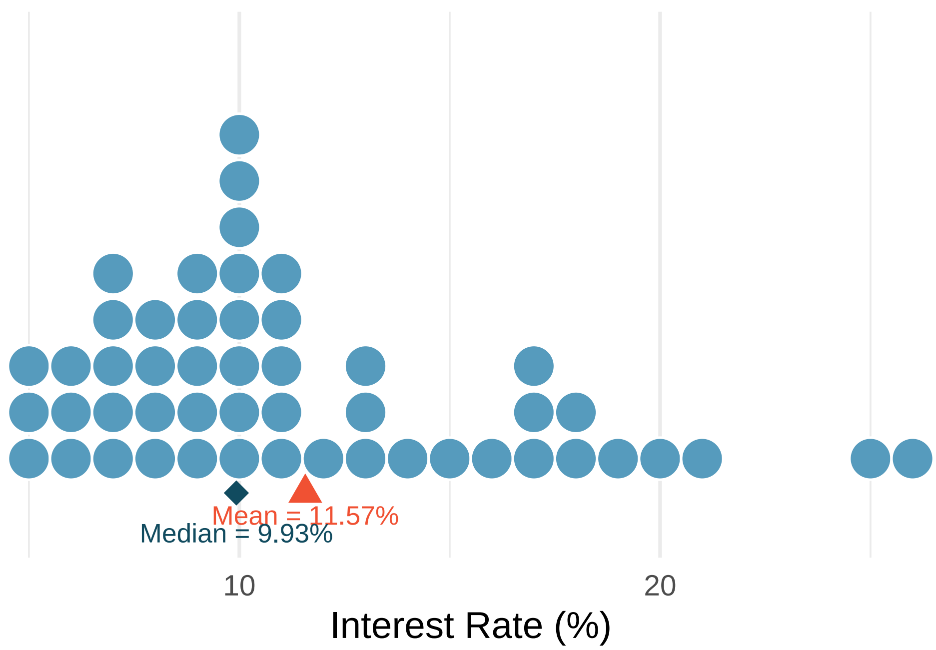


Figure 4.2: A dot plot of interest rates with the mean (red triangle, 11.57%) and median (blue diamond, 9.93%) marked. The mean is pulled to the right by the high-interest loans.

For a dataset with 7 observations arranged in order, which observation is the median?

Show answer

The 4th observation. With an odd number of observations, the median is the single middle value. There are 3 observations below it and 3 above it.

4.1.4 The mode

The **mode** is the most frequently occurring value in a dataset. While useful for categorical data (e.g., the most common eye color), the mode is less useful for continuous quantitative data because many values may occur only once.

In practice, when we refer to the "mode" of a quantitative distribution, we usually mean the location of a prominent *peak* in a histogram or density plot (as discussed in Section 4.2), rather than a single repeated value.

4.1.5 When to use the mean vs. the median

The mean and median measure different things. The choice between them depends on the context:

A company wants to understand its profit per customer. Should it use the mean or the median?

It depends on the question:

- If the concern is overall profitability (total profit divided by number of customers), the **mean** is the right statistic. A company could have a positive median profit per customer and still be unprofitable if a few customers generate large losses.
- If the concern is understanding what the *typical* customer experience looks like, the **median** tells you more about the profit from a representative customer.

Mean vs. median: which to choose?

- The **mean** is best when you care about the *total* or *aggregate* (e.g., total revenue, total time spent).
- The **median** is best when you care about the *typical* observation, especially when the distribution is skewed or contains outliers.
- For symmetric distributions without extreme outliers, the mean and median are approximately equal and either is appropriate.

The distribution of home prices in a city is strongly right skewed. If you wanted to describe the price of a typical home, should you report the mean or the median? Why?

Show answer

The median. In a right-skewed distribution, a few very expensive homes pull the mean upward, making it unrepresentative of a typical home price. The median is not affected by these extreme values and better represents what a typical buyer would encounter.

4.2 Histograms and distribution shape

4.2.1 Histograms

A **histogram** groups quantitative data into bins (intervals) and displays the count or proportion of observations in each bin as a bar. Unlike bar charts for categorical data, histogram bars touch each other because the bins represent a continuous range of values.

To build a histogram, we:

1. Choose a bin width (e.g., 2.5 percentage points for interest rates)
2. Count how many observations fall in each bin
3. Draw bars whose heights equal the counts

Interest rate bin	Count
(5.0% - 7.5%]	11
(7.5% - 10.0%]	15
(10.0% - 12.5%]	8
(12.5% - 15.0%]	4
(15.0% - 17.5%]	5
(17.5% - 20.0%]	4
(20.0% - 22.5%]	1
(22.5% - 25.0%]	1
(25.0% - 27.5%]	1

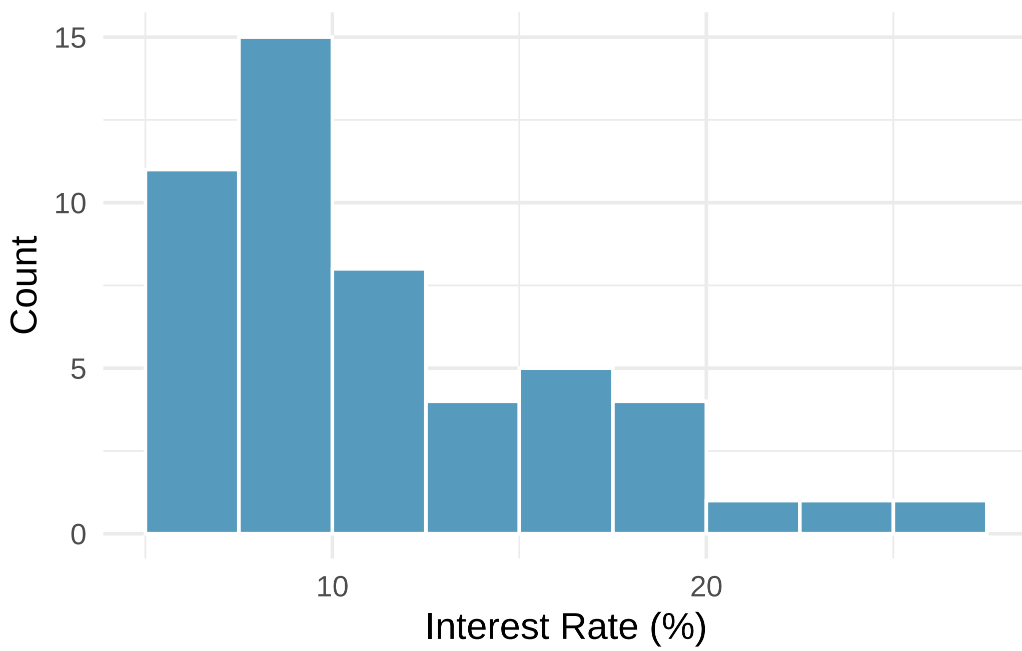


Figure 4.3: A histogram of interest rates with bin width 2.5 percentage points. The distribution is right skewed.

Histograms provide a view of the **data density**: higher bars indicate where data values are more common, and lower bars indicate where they are less common.

Choosing bin width. The bin width affects the appearance of a histogram. Too few bins (wide bins) can obscure important features; too many bins (narrow bins) can create a jagged display that is hard to interpret. There is no single "correct" bin width — experiment with different choices and look for the display that best reveals the shape of the data.

4.2.2 Density plots

A **density plot** is a smoothed version of a histogram. Instead of bars, it draws a smooth curve that estimates the distribution of the data. The area under the entire curve equals 1.

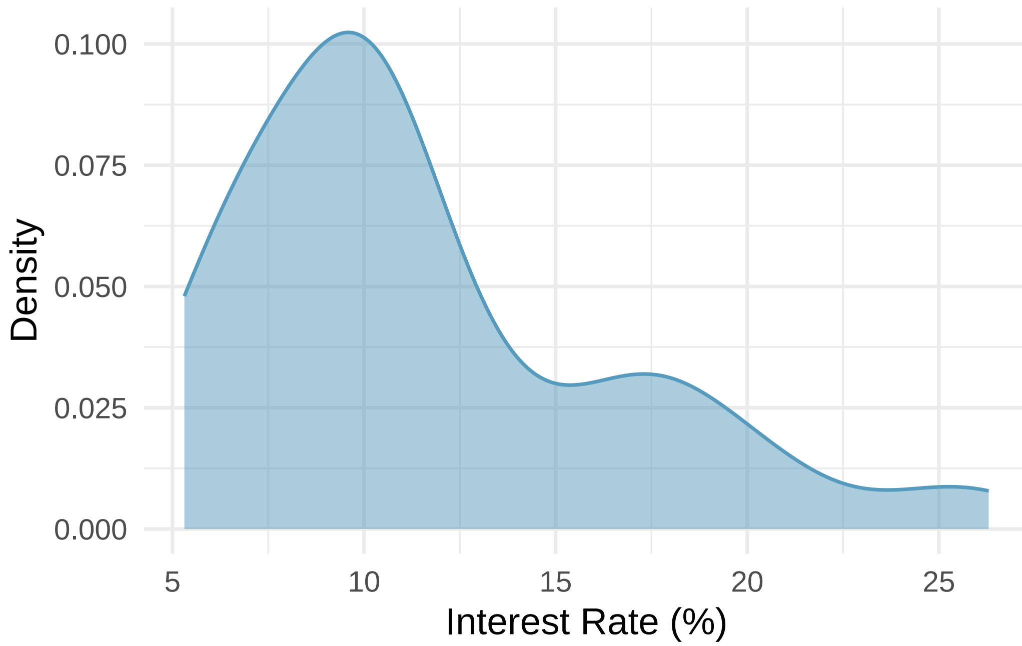


Figure 4.4: A density plot of interest rates, showing the same right-skewed shape as the histogram.

Density plots are particularly useful for comparing the shapes of distributions because overlapping curves are easier to read than overlapping histogram bars.

4.2.3 Skewness

Histograms are especially useful for identifying the **shape** of a distribution. One key characteristic is symmetry.

A distribution is **symmetric** if the left and right sides are approximately mirror images of each other.

A distribution is **right skewed** (or skewed to the right) if the right **tail** — the thinner end of the distribution — is longer. Most observations are on the left, with a few large values pulling the tail to the right.

A distribution is **left skewed** (or skewed to the left) if the left tail is longer. Most observations are on the right, with a few small values pulling the tail to the left.

Identifying skew. The direction of skew refers to the direction of the *long tail*, not the direction where most data are concentrated. A distribution with a long tail stretching to the right is right skewed, even though the bulk of the data is on the left side.

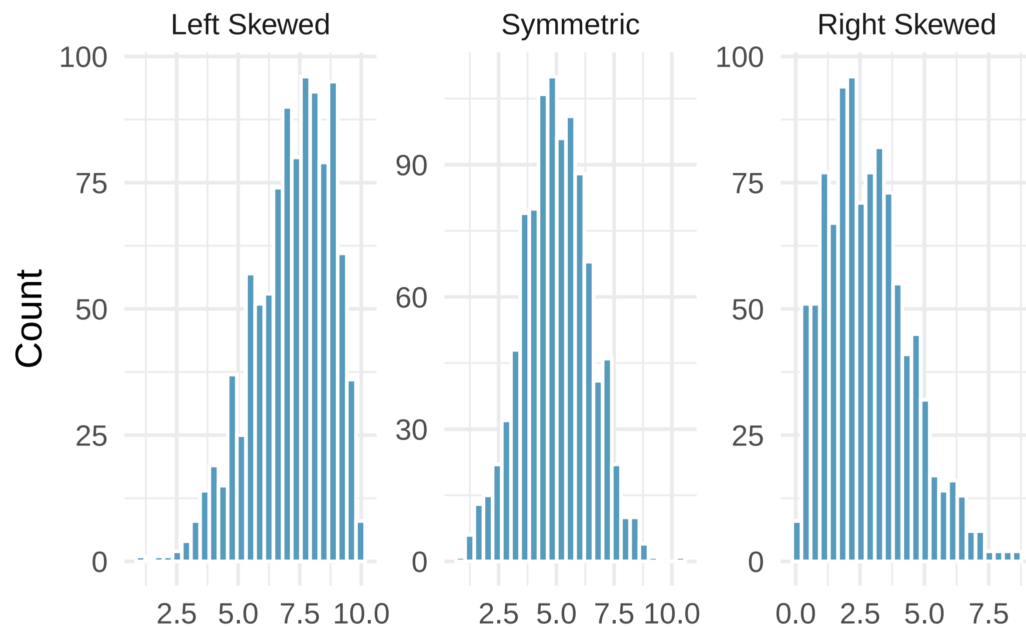


Figure 4.5: Three distributions illustrating left skewed, symmetric, and right skewed shapes.

Consider the distribution of home prices in a city. Would you expect this distribution to be symmetric, left skewed, or right skewed? Explain your reasoning.

Show answer

Home prices tend to be right skewed. Most homes fall in a moderate price range, but a small number of very expensive homes create a long right tail. This is common with income, wealth, and price data.

4.2.4 Modality

A **mode** is a prominent peak in a distribution. A distribution with a single prominent peak is called **unimodal**. A distribution with two prominent peaks is **bimodal**, and a distribution with more than two prominent peaks is **multimodal**.

A distribution with no prominent peaks — where all values are roughly equally likely — is called **uniform**.

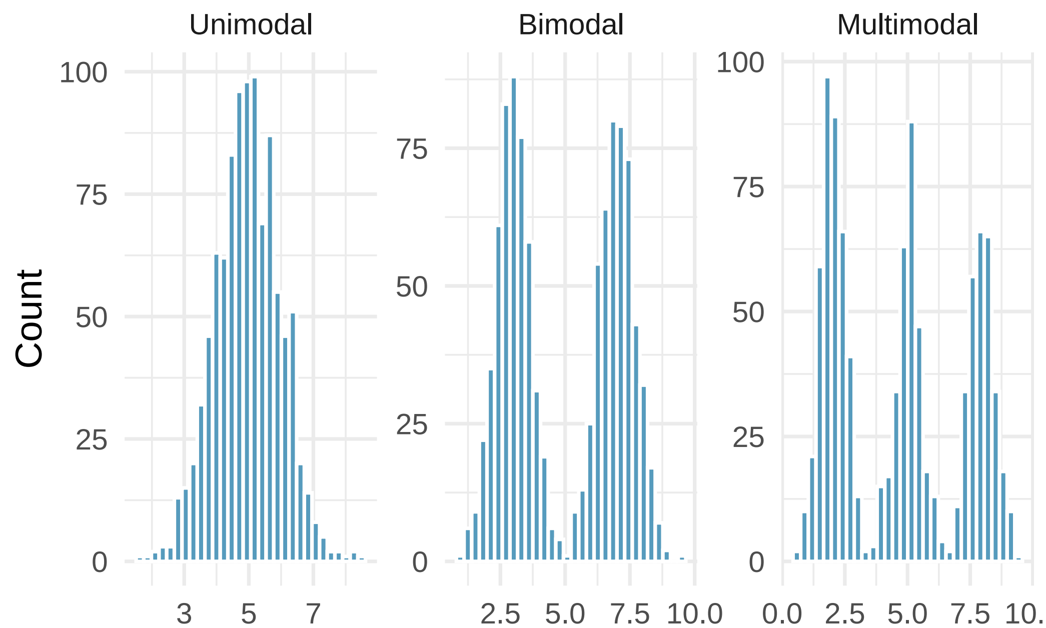


Figure 4.6: Three distributions: unimodal, bimodal, and multimodal.

Height measurements were collected from a group of young students and adult teachers at an elementary school. How many modes would you expect in a histogram of these heights?

You would likely expect two modes (bimodal): one centered on the typical height of children and another centered on the typical height of adults.

A histogram of interest rates shows one prominent peak near 7%. There is a very small bump near 20%, but it differs from its neighboring bins by only one or two observations. Would you call this distribution unimodal or bimodal?

Show answer

Unimodal. The small bump near 20% is not a prominent peak — it differs only slightly from its neighbors. We look for prominent peaks when identifying modes.

4.2.5 How skewness affects the mean and median

Now we can connect what we see in a histogram to the summary statistics we computed earlier.

In a **symmetric** distribution, the mean and median are approximately equal.

In a **right-skewed** distribution, the mean is pulled toward the right tail and is larger than the median.

In a **left-skewed** distribution, the mean is pulled toward the left tail and is smaller than the median.

Relationship between mean, median, and skewness.

- Right skewed: mean $>$ median
- Symmetric: mean $\approx$ median
- Left skewed: mean $<$ median

The direction of skew pulls the mean toward the tail. The median is more resistant to this pull.

The interest rate distribution is right skewed, with a mean of 11.57% and a median of 9.93%. How does the skewness explain the gap between these two values?

The few loans with very high interest rates (above 20%) pull the mean to the right. The median, which simply finds the middle value, is not affected by how extreme those high rates are. This is why the mean (11.57%) is larger than the median (9.93%) in a right-skewed distribution.

Try it yourself. Open the Descriptive Statistics Explorer to explore the interest rate data interactively. Switch between histograms, dot plots, and box plots, and compare the mean and median displayed in the summary statistics panel.

4.3 Box plots

A **box plot** (also called a box-and-whisker plot) provides a compact summary of a distribution using five numbers: the minimum, first quartile (Q_1), median, third quartile (Q_3), and maximum — with special handling for outliers.

4.3.1 Constructing a box plot

The components of a box plot are:

1. **The box**: Spans from Q_1 (the 25th percentile) to Q_3 (the 75th percentile). This box contains the middle 50% of the data. The length of the box is the **interquartile range** (IQR = $Q_3 - Q_1$).
2. **The median line**: A line inside the box marks the median (the 50th percentile), which divides the data in half.
3. **The whiskers**: Lines extending from the box toward the smallest and largest observations that are *not* outliers. Specifically, the whiskers extend to the most extreme data points within $1.5 \times$ IQR of the box.
4. **Outlier points**: Any observations beyond $1.5 \times$ IQR from the edges of the box are plotted individually as dots. These are potential outliers.

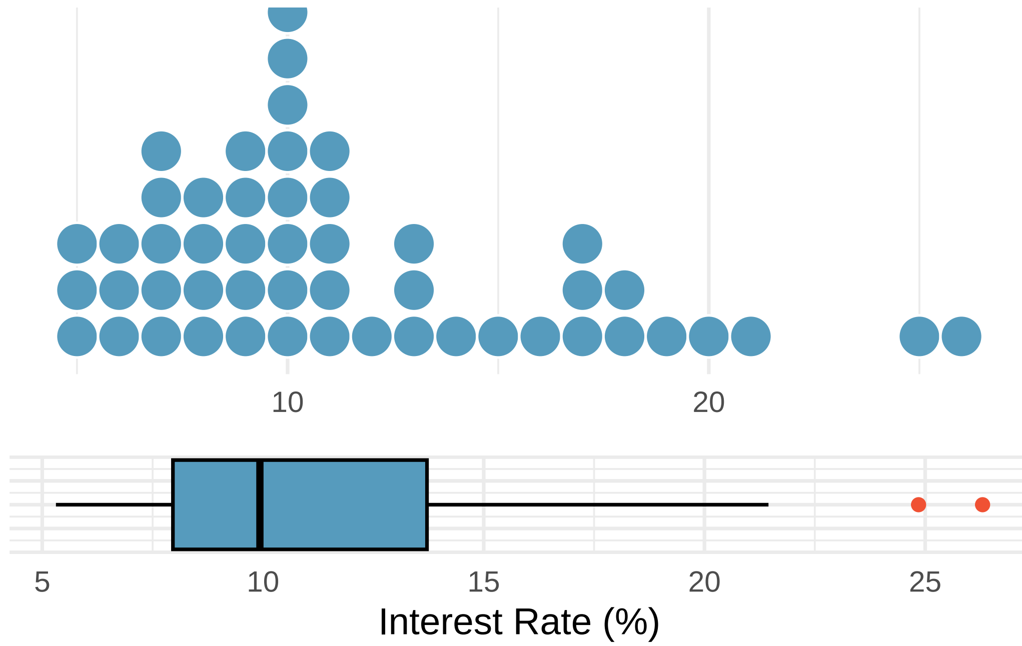


Figure 4.7: A dot plot and box plot of the same interest rate data, shown one above the other for comparison. The box plot summarizes the distribution with five numbers plus outliers.

The **first quartile** (Q_1) is the 25th percentile — 25% of the data fall below this value. The **third quartile** (Q_3) is the 75th percentile — 75% of the data fall below this value.

The **interquartile range** (IQR) is the range of the middle 50% of the data: $\text{IQR} = Q_3 - Q_1$.

4.3.2 Interpreting a box plot

A box plot of interest rates shows: the left edge of the box at about 8%, the median line at about 10%, and the right edge of the box at about 14%. Two dots appear beyond the right whisker at about 25% and 26%. What can we learn from this display?

The middle 50% of interest rates fall between about 8% and 14% (the IQR is about 6 percentage points). The median rate is about 10%. The right whisker extends to about 20%, and the two dots beyond the whisker at 25% and 26% indicate unusually high interest rates. The box plot also reveals right skew, since the median is closer to the left edge of the box and the right whisker is longer than the left whisker.

What percentage of observations fall between Q_1 and the median? What percentage fall between the median and Q_3?

Show answer

25% of the data fall between Q_1 and the median, and another 25% fall between the median and Q_3. Together, the middle 50% of the data lie within the box.

4.4 Measures of spread

Knowing the center of a distribution is important, but it tells only part of the story. Two distributions can have the same center but look very different if one is tightly clustered and the other is widely dispersed. We need measures of **variability** (also called spread or dispersion).

4.4.1 Range

The simplest measure of spread is the **range**: the difference between the maximum and minimum values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

While easy to compute, the range has a major weakness: it depends entirely on the two most extreme observations. A single outlier can dramatically increase the range, giving a misleading impression of the overall variability. For this reason, we usually prefer the IQR or standard deviation.

4.4.2 The interquartile range (IQR)

The **interquartile range** measures the spread of the middle 50% of the data. We already saw the IQR when constructing box plots — it is the length of the box.

Interquartile range (IQR).

The IQR is computed as:

$$\text{IQR} = Q_3 - Q_1$$

where Q_1 is the first quartile (25th percentile) and Q_3 is the third quartile (75th percentile).

For the 50-loan interest rate data, $Q_1 \approx 8\%$ and $Q_3 \approx 14\%$. What is the IQR, and what does it tell us?

IQR $= 14\% - 8\% = 6$ percentage points. This means the middle 50% of loans have interest rates spanning a range of 6 percentage points. Half of all loans have rates between about 8% and 14%.

4.4.3 Variance and standard deviation

The **standard deviation** is the most commonly used measure of spread. It roughly describes how far the typical observation is from the mean.

We start with the concept of a **deviation**: the distance of an observation from the mean.

$$\text{deviation of } x_i = x_i - \bar{x}$$

Some deviations are positive (observations above the mean) and some are negative (observations below the mean). If we simply averaged the deviations, the positives and negatives would cancel out, giving us zero. To avoid this, we *square* each deviation before averaging.

The **variance** (denoted s^2) is the average of the squared deviations, using $n - 1$ in the denominator:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n-1}$$

The **standard deviation** (denoted s) is the square root of the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

We divide by $n-1$ rather than n when computing a sample's variance. The mathematical reasons are beyond this course, but the practical effect is that this makes the statistic slightly more reliable as an estimate of the population variance.

Why do we square the deviations instead of just taking the absolute value? What are the two consequences of squaring?

Show answer

Squaring the deviations does two things: (1) it eliminates negative signs (since any number squared is non-negative), and (2) it makes large deviations relatively *much* larger than small ones. An observation that is 10 units from the mean contributes 100 to the sum of squared deviations, while one that is 2 units away contributes only 4. This means the variance and standard deviation are especially sensitive to outliers.

The following histogram shows the interest rate data with shaded bands marking one and two standard deviations from the mean. The standard deviation of 5.05 percentage points gives us a sense of how spread out the data are around the mean of 11.57%.

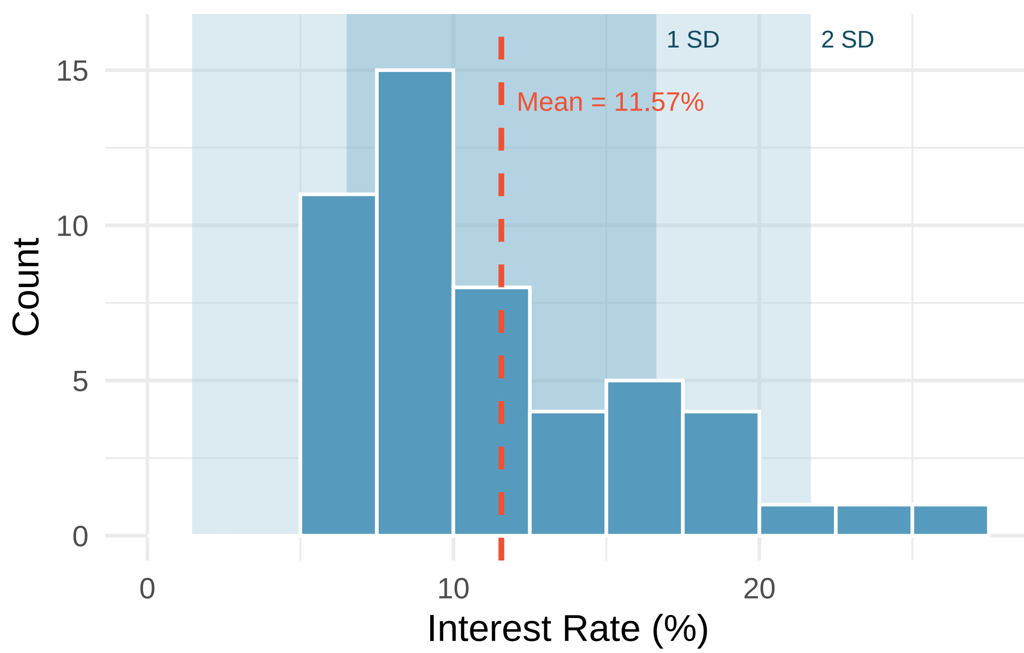


Figure 4.8: A histogram of interest rates with shaded regions showing one standard deviation (dark) and two standard deviations (light) from the mean.

For the 50-loan interest rate data, the variance is $s^2 = 25.52$ and the standard deviation is $s = \sqrt{25.52} = 5.05$ percentage points. What does a standard deviation of 5.05 percentage points mean in context?

The standard deviation tells us that interest rates typically deviate about 5 percentage points from the mean (11.57%). Most loans have rates that are within one standard deviation of the mean — roughly between 6.5% and 16.6%.

Like the mean, the population values for variance and standard deviation have special symbols: σ^2 for the variance and σ for the standard deviation. The Greek letter σ is pronounced "sigma."

Variance and standard deviation.

The variance is the average squared distance from the mean. The standard deviation is the square root of the variance. The standard deviation represents the *typical* deviation of observations from the mean. Often about 68% of the data will be within one standard deviation of the mean and about 95% will be within two standard deviations. However, these percentages are not strict rules.

4.5 Describing a distribution

A complete description of a distribution should address three things: **shape**, **center**, and **spread**.

A good description of a distribution should always:

1. Name the **shape**: symmetric or skewed (and direction), and the number of modes
2. Give the **center**: report the mean and/or median with units
3. Give the **spread**: report the standard deviation and/or IQR with units
4. Note any **unusual features**: outliers, gaps, clusters
5. Use **context**: refer to the actual variable and its units, not just abstract numbers

Example: "The distribution of commute times is unimodal and right skewed, with a median of 22 minutes and an IQR of 15 minutes. A few commuters have commute times exceeding 90 minutes."

Write a complete description of the interest rate distribution from the loan dataset.

The distribution of interest rates is **unimodal** and **right skewed**. The **center** is around 10–12% (median = 9.93%, mean = 11.57%). The **spread** is moderate: the standard deviation is 5.05 percentage points and the IQR is about 6 percentage points. There are a few **unusually high** interest rates above 20%, which contribute to the right skew and the gap between the mean and median.

A description says: "The mean is 15 and the standard deviation is 3." What is missing from this description?

Show answer

The description lacks context (what variable? what units?), shape information (symmetric? skewed? unimodal?), and any mention of unusual features. A better version might be: "The distribution of wait times (in minutes) is approximately symmetric and unimodal, with a mean of 15 minutes and a standard deviation of 3 minutes. There are no obvious outliers."

4.6 Outliers and robust statistics

4.6.1 Identifying outliers

An **outlier** is an observation that appears extreme relative to the rest of the data.

Examining data for outliers serves several purposes:

- Identifying strong skew in the distribution
- Identifying possible data collection or data entry errors
- Providing insight into interesting or unusual properties of the data

Not all outliers are errors, and not all errors are outliers. Some datasets naturally have long tails and outlying points are legitimate observations. **Never remove an outlier without a substantive reason** (e.g., confirmed data entry error). Blindly removing outliers can lead to misleading conclusions.

A common rule for identifying outliers in the context of box plots is the **1.5 IQR rule**: any observation more than $1.5 \times$ IQR below Q_1 or above Q_3 is flagged as a potential outlier. This is the rule used to determine which points appear as dots beyond the whiskers of a box plot.

4.6.2 Robust statistics

A statistic is called **robust** (or **resistant**) if extreme observations have little effect on its value.

Consider what happens when we change a single extreme observation in the interest rate data. The table below shows how the median, IQR, mean, and standard deviation respond when the most extreme value (26.3%) is moved:

Scenario	Median	IQR	Mean	SD
Original data	9.93	5.75	11.57	5.05
Move 26.3% to 15%	9.93	5.75	11.34	4.61
Move 26.3% to 35%	9.93	5.75	11.74	5.68

In the table above, the median and IQR are unchanged across all three scenarios, but the mean and standard deviation shift. Why?

The median and IQR are determined by the values near the middle of the ordered data (Q_1, the median, and Q_3). Moving a single extreme value does not change the positions of these middle values. The mean and standard deviation, however, use every data value in their calculations. The mean adds up all values, so changing one extreme value changes the sum. The standard deviation squares deviations from the mean, so extreme values have an outsized effect.

Robust vs. non-robust statistics.

Robust (resistant to outliers)	Not robust (sensitive to outliers)
Median	Mean
IQR	Standard deviation
	Range

When a distribution is skewed or contains outliers, robust statistics (median and IQR) often give a more representative summary of the "typical" observation. When a distribution is roughly symmetric with no extreme outliers, the mean and standard deviation work well and carry additional mathematical advantages.

The distribution of salaries at a company is right skewed because a few executives earn much more than most employees. Which pair of summary statistics would better represent the "typical" salary and its variability: (mean, standard deviation) or (median, IQR)?

Show answer

The median and IQR. In a right-skewed distribution, the mean is pulled upward by the high executive salaries, making it unrepresentative of the typical employee. The median is not affected by these extreme values, and the IQR captures the spread of the middle 50% of salaries.

The distribution of loan amounts is right skewed. If you wanted to understand the typical loan size, should you be more interested in the mean or the median?

Show answer

If the goal is to understand what a typical individual loan looks like, the median is probably more useful. However, if the goal is to understand something that scales — such as the total amount of money needed on hand to fund 1,000 loans — then the mean would be more useful.

4.7 The empirical rule and Chebyshev's theorem

4.7.1 The empirical rule (68-95-99.7 rule)

For distributions that are approximately **bell-shaped** (symmetric and unimodal), the standard deviation provides a useful ruler. Statisticians call this particular bell shape the **normal distribution —** it comes up so often in statistics that it has its own name. We'll study it formally later, but for now, just know that "bell-shaped" and "normal" mean the same thing.

The empirical rule.

For a roughly bell-shaped distribution:

- About **68%** of the data fall within 1 standard deviation of the mean: $(\bar{x} - s,\ \bar{x} + s)$
- About **95%** of the data fall within 2 standard deviations of the mean: $(\bar{x} - 2s,\ \bar{x} + 2s)$
- About **99.7%** of the data fall within 3 standard deviations of the mean: $(\bar{x} - 3s,\ \bar{x} + 3s)$

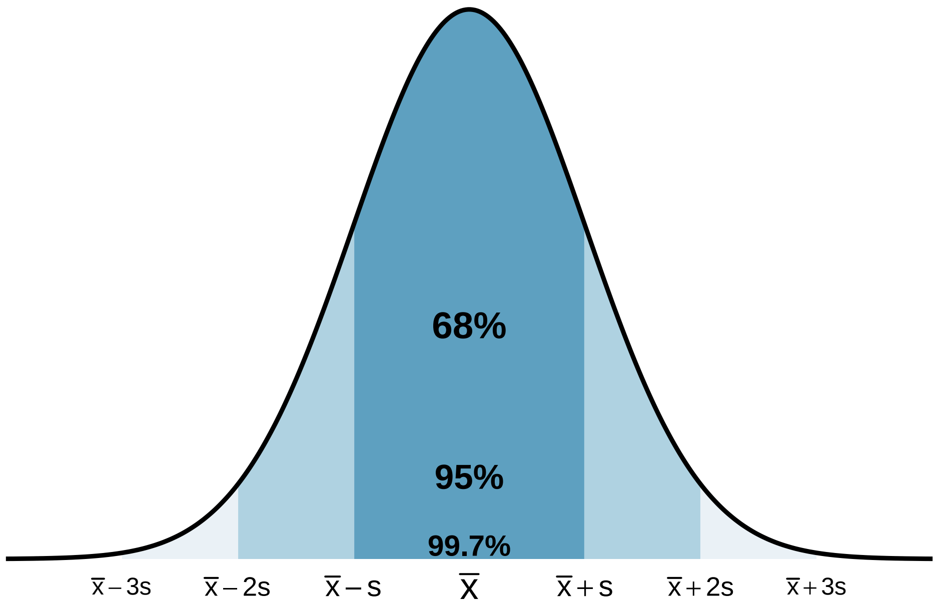


Figure 4.9: A bell-shaped distribution with shaded regions showing 68%, 95%, and 99.7% of the data within 1, 2, and 3 standard deviations of the mean, respectively.

The heights of adult women in the US are approximately bell-shaped with a mean of 64.5 inches and a standard deviation of 2.5 inches. Use the empirical rule to estimate the range of heights for the middle 95% of women.

By the empirical rule, about 95% of the data fall within 2 standard deviations of the mean:

$$64.5 - 2(2.5) = 59.5 \text{ inches} \quad \text{to} \quad 64.5 + 2(2.5) = 69.5 \text{ inches}$$

So about 95% of adult women in the US are between 59.5 inches (4'11.5") and 69.5 inches (5'9.5") tall.

Using the same height distribution (mean = 64.5 inches, SD = 2.5 inches), what percentage of women are shorter than 59.5 inches?

Show answer

By the empirical rule, about 95% of women are between 59.5 and 69.5 inches. That means about 5% are outside this range. Since the distribution is symmetric, about half of that 5% (i.e., about 2.5%) are below 59.5 inches.

The empirical rule is only an approximation, and it works best for distributions that are roughly bell-shaped. For skewed or multimodal distributions, the percentages can be quite different. For example, in a right-skewed distribution, more than 68% of the data may fall within one standard deviation of the mean because the bulk of the data is concentrated on one side.

Three distributions all have the same mean (0) and the same standard deviation (1), but one is bimodal, one is bell-shaped, and one is right skewed. Would the empirical rule work well for all three? Why or why not?

Show answer

No. The empirical rule works well only for the bell-shaped distribution. The bimodal distribution would have more data near the edges (near the two peaks) and less in the middle. The right-skewed distribution would have more data close to the mean on one side and a long tail on the other. In both cases, the 68-95-99.7 percentages would be poor approximations.

4.7.2 Chebyshev's theorem

While the empirical rule applies only to bell-shaped distributions, **Chebyshev's theorem** provides a guarantee that works for *any* distribution, regardless of shape:

Chebyshev's theorem.

For **any** distribution (regardless of shape), at least $\left(1 - \frac{1}{k^2}\right) \times 100\%$ of the data fall within k standard deviations of the mean, for any $k > 1$.

Standard deviations (k)	At least this % of data
2	75%
3	89%
4	93.75%

Chebyshev's theorem is weaker than the empirical rule (it guarantees "at least 75%" within 2 standard deviations, while the empirical rule says "about 95%" for bell-shaped data), but it has the advantage of applying to every distribution. It is most useful when you cannot assume the distribution is bell-shaped.

4.8 Chapter review

4.8.1 Summary

This chapter introduced the tools for exploring quantitative data — both visual and numerical. **Dot plots** show individual values for small datasets, **histograms** reveal the shape and density of a distribution, **density plots** provide smoothed alternatives, and **box plots** give a compact five-number summary. Distribution shape is described using **symmetry** (left skewed, symmetric, right skewed) and **modality** (unimodal, bimodal, multimodal). The **mean** and **median** measure center, with the mean being the balancing point and the median being the middle value. The **standard deviation** and **IQR** measure spread, with the standard deviation roughly representing the typical deviation from the mean and the IQR representing the range of the middle 50%. The **median** and **IQR** are robust to outliers, while the **mean** and **standard deviation** are sensitive to extreme values. The **empirical rule** connects the standard deviation to the distribution shape for bell-shaped data: about 68%, 95%, and 99.7% of observations fall within 1, 2, and 3 standard deviations of the mean.

4.8.2 Key terms

- Dot plot, histogram, density plot, box plot
- Data density, bin width
- Symmetric, left skewed, right skewed, tail
- Unimodal, bimodal, multimodal, uniform
- Mean ($\bar{x}$), median, mode
- Population mean (μ), weighted mean
- Deviation, variance (s^2), standard deviation (s)
- Population variance (σ^2), population standard deviation (σ)

- Range, interquartile range (IQR)
- First quartile (Q_1), third quartile (Q_3), percentile
- Outlier, robust statistic (resistant statistic)
- Empirical rule (68-95-99.7 rule), Chebyshev's theorem

4.9 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Make-up exam.** In a class of 25 students, 24 of them took an exam in class and 1 student took a make-up exam the following day. The professor graded the first batch of 24 exams and found an average score of 74 points with a standard deviation of 8.9 points. The student who took the make-up the following day scored 64 points on the exam.

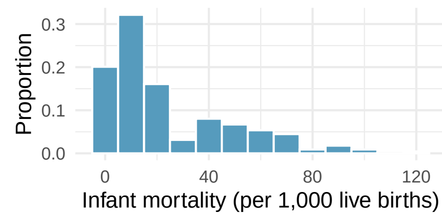
a. Does the new student's score increase or decrease the average score?

b. What is the new average?

c. Does the new student's score increase or decrease the standard deviation of the scores?

2. **Infant mortality.** The infant mortality rate is defined as the number of infant deaths per 1,000 live births. This rate is often used as an indicator of the level of health in a country. The relative frequency histogram below shows the distribution of estimated infant death rates for 224 countries for which such data were available in 2014.

   a. Estimate Q1, the median, and Q3 from the histogram.
   b. Would you expect the mean of this dataset to be smaller or larger than the median? Explain your reasoning.



3. **Days off at a mining plant.** Workers at a particular mining site receive an average of 35 days paid vacation, which is lower than the national average. The manager of this plant is under pressure from a local union to increase the amount of paid time off. However, he does not want to give more days off to the workers because that would be costly. Instead he decides he should fire 10 employees in such a way as to raise the average number of days off that are reported by his employees. In order to achieve this goal, should he fire employees who have the most number of days off, least number of days off, or those who have about the average number of days off?

4. **Medians and IQRs.** For each part, compare distributions A and B based on their medians and IQRs. You do not need to calculate these statistics; simply state how the medians and IQRs compare. Make sure to explain your reasoning. *Hint:* It may be useful to sketch dot plots of the distributions.

a. **A:** 3, 5, 6, 7, 9; **B:** 3, 5, 6, 7, 20

b. **A:** 3, 5, 6, 7, 9; **B:** 3, 5, 7, 8, 9

c. **A:** 1, 2, 3, 4, 5; **B:** 6, 7, 8, 9, 10

d. **A:** 0, 10, 50, 60, 100; **B:** 0, 100, 500, 600, 1000

5. **Means and SDs.** For each part, compare distributions A and B based on their means and standard deviations. You do not need to calculate these statistics; simply state how the means and the standard deviations compare. Make sure to explain your reasoning. *Hint:* It may be useful to sketch dot plots of the distributions.

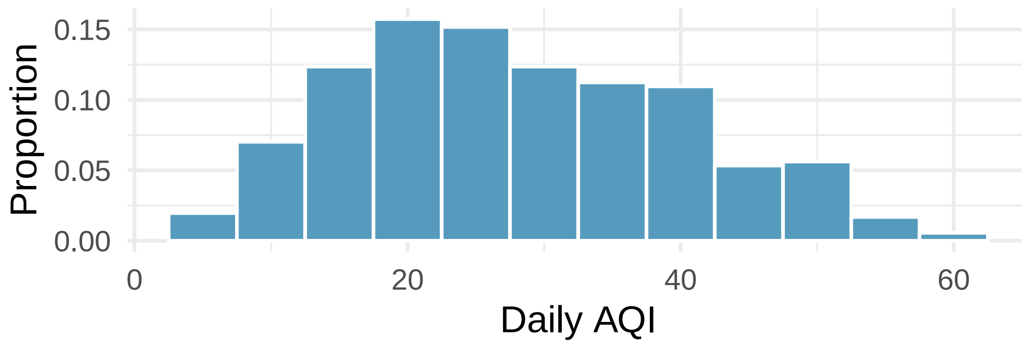
a. **A:** 3, 5, 5, 5, 8, 11, 11, 11, 13; **B:** 3, 5, 5, 5, 8, 11, 11, 11, 20

b. **A:** -20, 0, 0, 0, 15, 25, 30, 30; **B:** -40, 0, 0, 0, 15, 25, 30, 30

c. **A:** 0, 2, 4, 6, 8, 10; **B:** 20, 22, 24, 26, 28, 30

d. **A:** 100, 200, 300, 400, 500; **B:** 0, 50, 300, 550, 600

6. **Air quality.** Daily air quality is measured by the air quality index (AQI) reported by the Environmental Protection Agency. This index reports the pollution level and what associated health effects might be a concern. The index is calculated for five major air pollutants regulated by the Clean Air Act and takes values from 0 to 300, where a higher value indicates lower air quality. AQI was reported for a 356 days in 2022 in Durham, NC. The histogram below shows the distribution of the AQI values on these days.



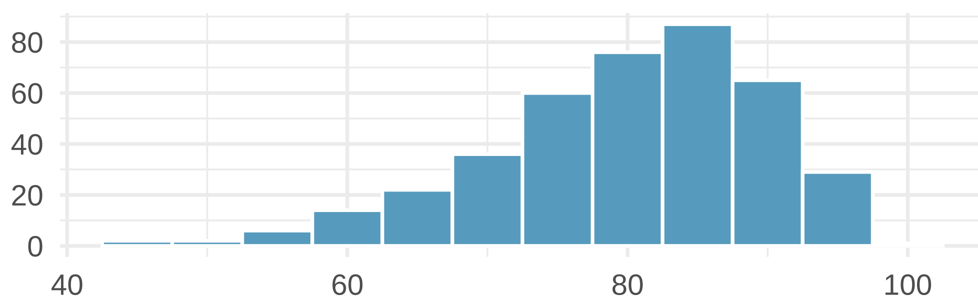
a. Estimate the median AQI value of this sample.

b. Would you expect the mean AQI value of this sample to be higher or lower than the median? Explain your reasoning.

c. Estimate Q1, Q3, and IQR for the distribution.

d. Would any of the days in this sample be considered to have an unusually low or high AQI? Explain your reasoning.

7. **Median vs. mean.** Estimate the median for the 400 observations shown in the histogram, and note whether you expect the mean to be higher or lower than the median.



8. **Facebook friends.** Facebook data indicate that 50% of Facebook users have 100 or more friends, and that the average friend count of users is 190. What do these findings suggest about the shape of the distribution of number of friends of Facebook users? (Backstrom 2011)

9. **Distributions and appropriate statistics.** For each of the following, state whether you expect the distribution to be symmetric, right skewed, or left skewed. Also specify whether the mean or median would best represent a typical observation in the data, and whether the variability of observations would be best represented using the standard deviation or IQR. Explain your reasoning.

a. Number of pets per household.

b. Distance to work, i.e., number of miles between work and home.

c. Heights of adult males.

d. Age at death.

e. Exam grade on an easy test.

10. **Distributions and appropriate statistics.** For each of the following, state whether you expect the distribution to be symmetric, right skewed, or left skewed. Also specify whether the mean or median would best represent a typical observation in the data, and whether the variability of observations would be best represented using the standard deviation or IQR. Explain your reasoning.

a. Housing prices in a country where 25% of the houses cost below \$350,000, 50% of the houses cost below \$450,000, 75% of the houses cost below \$1,000,000, and there are a meaningful number of houses that cost more than \$6,000,000.

b. Housing prices in a country where 25% of the houses cost below \$300,000, 50% of the houses cost below \$600,000, 75% of the houses cost below \$900,000, and very few houses that cost more than \$1,200,000.

c. Number of alcoholic drinks consumed by college students in a given week. Assume that most of these students don't drink since they are under 21 years old, and only a few drink excessively.

d. Annual salaries of the employees at a Fortune 500 company where only a few high level executives earn much higher salaries than all the other employees.

e. Gestation time in humans where 25% of the babies are born by 38 weeks of gestation, 50% of the babies are born by 39 weeks, 75% of the babies are born by 40 weeks, and the maximum gestation length is 46 weeks.

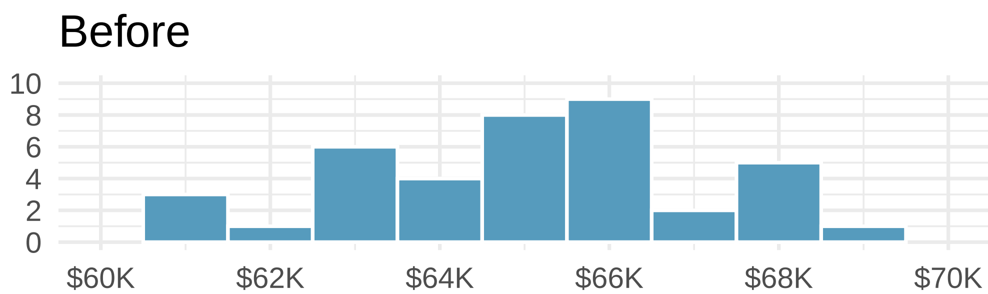
11. **TV watchers.** College students in a statistics class were asked how many hours of television they watch per week, including online streaming services. This sample yielded an average of 8.28 hours, with a standard deviation of 7.18 hours. Is the distribution of number of hours students watch television weekly symmetric? If not, what shape would you expect this distribution to have? Explain your reasoning.

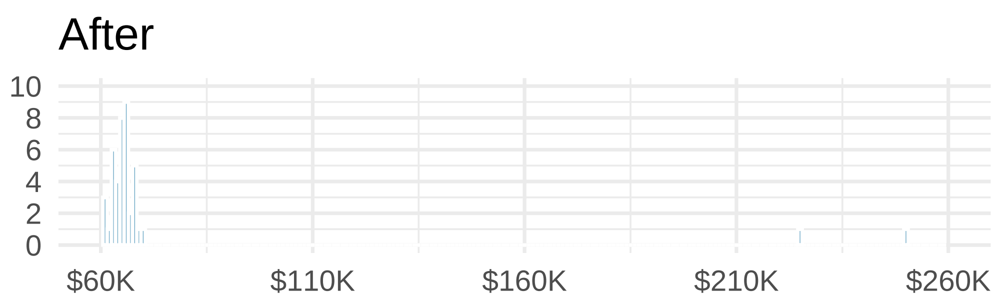
12. **Exam scores.** The average on a history exam (scored out of 100 points) was 85, with a standard deviation of 15. Is the distribution of the scores on this exam symmetric? If not, what shape would you expect this distribution to have? Explain your reasoning.

13. **Midrange.** The *midrange* of a distribution is defined as the average of the maximum and the minimum of that distribution. Is this statistic robust to outliers and extreme skew? Explain your reasoning.

14. **Stats scores.** The final exam scores of twenty introductory statistics students, arranged in ascending order, as as follows: 57, 66, 69, 71, 72, 73, 74, 77, 78, 78, 79, 79, 81, 81, 82, 83, 83, 88, 89, 94. Suppose students who score above the 75th percentile on the final exam get an A in the class. How many students will get an A in this class?

15. **Income at the coffee shop.** The first histogram below shows the distribution of the yearly incomes of 40 patrons at a college coffee shop. Suppose two new people walk into the coffee shop: one making $225,000 and the other $250,000. The second histogram shows the new income distribution. Summary statistics are also provided, rounded to the nearest whole number.





	n	Min	Q1	Median	Mean	Max	SD
Before	40	$60,679	$60,818	$65,238	$65,089	$69,885	$2,122
After	42	$60,679	$60,838	$65,352	$73,299	$250,000	$37,321

a. Would the mean or the median best represent what we might think of as a typical income for the 42 patrons at this coffee shop? What does this say about the robustness of the two measures?

b. Would the standard deviation or the IQR best represent the amount of variability in the incomes of the 42 patrons at this coffee shop? What does this say about the robustness of the two measures?

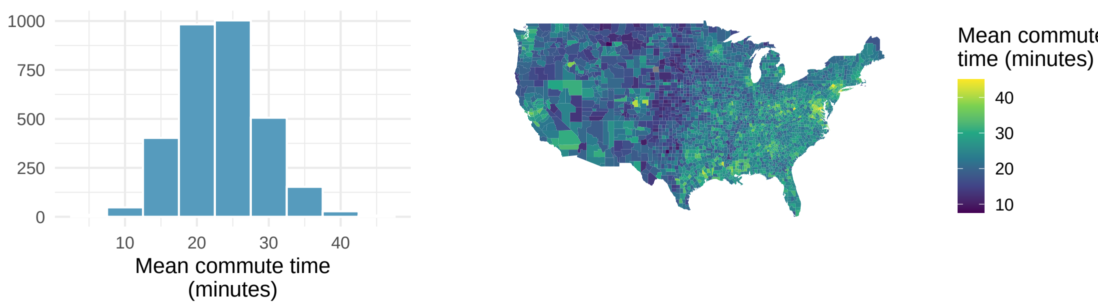
16. **A new statistic.** The statistic $\frac{\bar{x}}{median}$ can be used as a measure of skewness. Suppose we have a distribution where all observations are greater than 0, $x_i > 0$. What is the expected shape of the distribution under the following conditions? Explain your reasoning.

   a. $\frac{\bar{x}}{median} = 1$

   b. $\frac{\bar{x}}{median} < 1$

   c. $\frac{\bar{x}}{median} > 1$

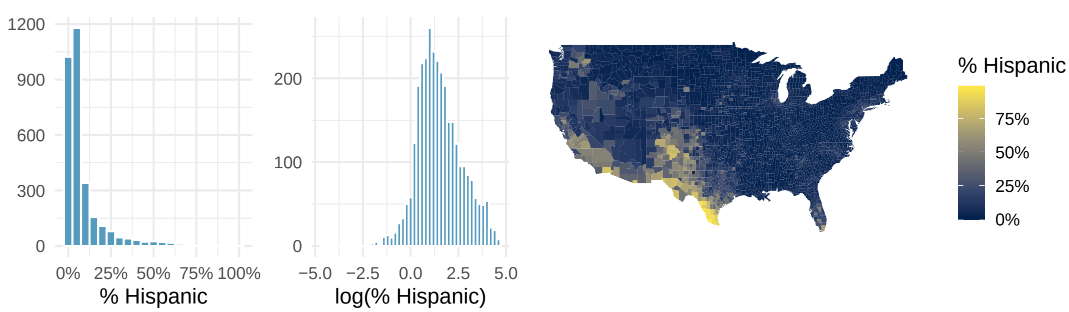
17. **Commute times.** The US census collects data on the time it takes Americans to commute to work, among many other variables. The histogram below shows the distribution of mean commute times in 3,142 US counties in 2017. Also shown below is a spatial intensity map of the same data.



   a. Describe the numerical distribution and comment on whether a log transformation may be advisable for these data.

   b. Describe the spatial distribution of commuting times using the map.

18. **Hispanic population.** The US census collects data on race and ethnicity of Americans, among many other variables. The histogram below shows the distribution of the percentage of the population that is Hispanic in 3,142 counties in the US in 2010. Also shown is a histogram of logs of these values.



   a. Describe the numerical distribution and comment on why we might want to use log-transformed values in analyzing or modeling these data.

   b. What features of the distribution of the Hispanic population in US counties are apparent in the map but not in the histogram? What features are apparent in the histogram but not the map?

   c. Is one visualization more appropriate or helpful than the other? Explain your reasoning.

Datasets: `cia_factbook` (openintro) | `pm25_2022_durham` (openintro) | `county_complete` (usdata)

Chapter 5

Relationships Between Variables

The preceding chapters focused on exploring one variable at a time. But the most interesting statistical questions involve **relationships between variables**. Does higher education lead to higher income? Is a drug more effective for one group than another? This chapter introduces tools for exploring relationships: side-by-side box plots and faceted histograms for comparing a quantitative variable across groups, and scatterplots for examining the association between two quantitative variables. We close with the critical distinction between association and causation.

StatLens tools for this chapter: Use the Grouped Statistics Explorer to compare distributions across groups with side-by-side box plots, histograms, and density overlays. Use the Regression Explorer for scatterplots of two quantitative variables.

5.1 Comparing groups

Some of the most interesting investigations involve comparing a quantitative variable across groups defined by a categorical variable.

5.1.1 Side-by-side box plots

Side-by-side box plots place a box plot for each group on the same scale, making it easy to compare centers, spreads, and shapes.

Consider comparing median household income for US counties, grouped by whether the county experienced population growth from 2010 to 2017:

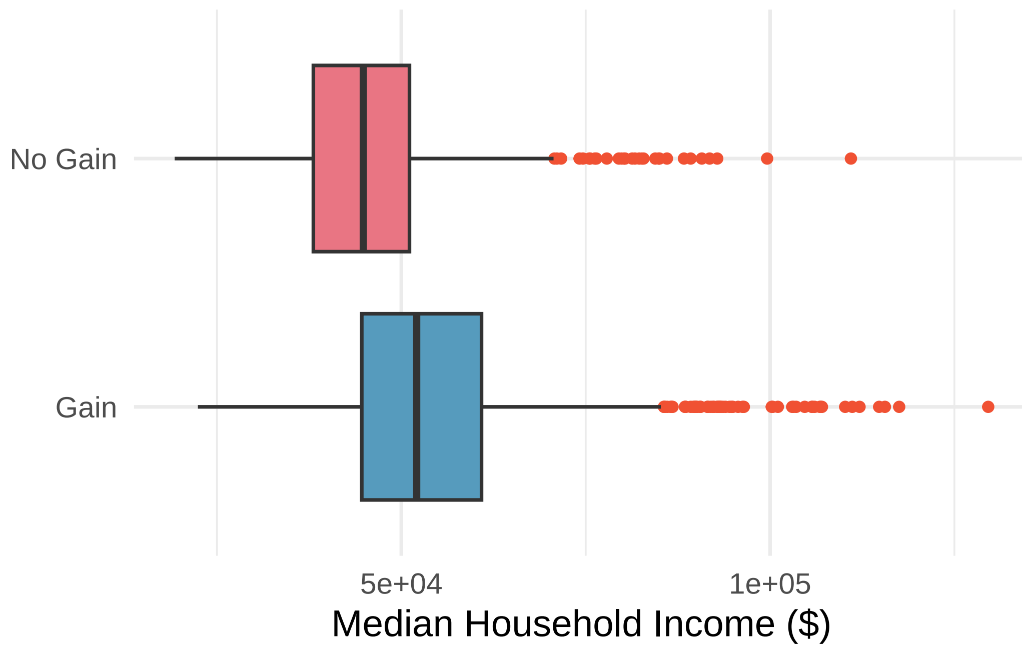


Figure 5.1: Side-by-side box plots of median household income by population change status (gain vs. no gain). Counties with population gain have a higher median income.

Using the box plots, compare the income distributions for the two groups. What do you notice about the centers, spreads, and shapes?

Show answer

Counties with population gains tend to have higher income (median about $45,000) compared to counties without gains (median about $40,000). The variability is also slightly larger for the gain group, as seen in the wider IQR. Both distributions show slight to moderate right skew, with outliers on the high end.

When are side-by-side box plots particularly useful? When might histograms be better?

Side-by-side box plots are excellent for comparing centers (medians) and spreads (IQRs) across groups in a compact display. However, they cannot reveal details about shape such as bimodality. Histograms or density plots are better when you want to see the full shape of each distribution, including the number of modes and the exact nature of the skew.

5.1.2 Faceted histograms

Faceting splits a graphical display into panels based on the levels of a categorical variable. Each panel shows the distribution for one group, using the same scale for easy comparison.

Figure 5.2: Faceted histograms of median household income, separated into panels for counties with population gain and those with no gain.

Faceting extends naturally to multiple categorical variables. For example, we could split the income data by both population change *and* whether the county is in a metropolitan area, creating a 2-by-2 grid of histograms.

What are the advantages of faceted histograms compared to side-by-side box plots?

Faceted histograms reveal the full shape of each distribution, including modality (unimodal vs. bimodal), the exact nature of any skewness, and gaps or clusters. Box plots, on the other hand, are more compact and make it easier to compare medians and IQRs at a glance, but they cannot show features like bimodality. The two displays complement each other well.

5.1.3 Ridge plots

A **ridge plot** (also called a joy plot) is a set of overlapping density plots, one for each group, drawn on the same horizontal scale but offset vertically. Ridge plots are useful for comparing the shapes of many distributions at once while maintaining a compact display.

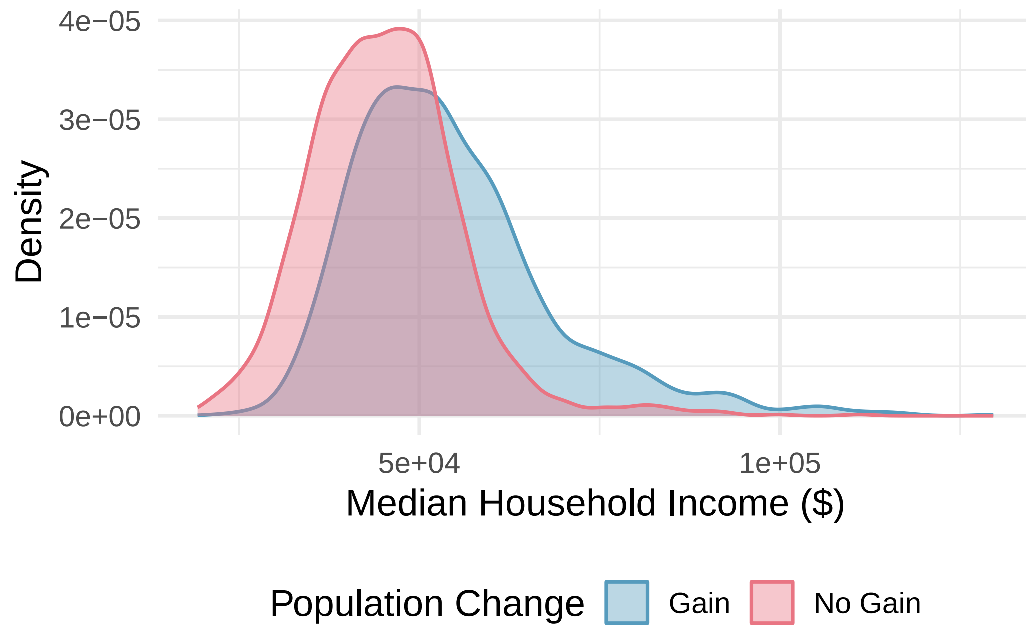


Figure 5.3: A ridge plot comparing median household income for counties with and without population gain. Each group gets its own smoothed density curve.

What does a ridge plot reveal that side-by-side box plots do not?

Show answer

Ridge plots show the full shape of each distribution, including modality and skewness, similar to histograms. Box plots compress all of this shape information into a five-number summary. Ridge plots are especially useful when you want to compare shapes across many groups in a compact space.

Try it yourself. Open the Grouped Statistics Explorer in StatLens to compare distributions across groups. Switch between box plots, dot plots, histograms, and density overlays to see how each visualization highlights different aspects of the comparison.

5.2 Scatterplots

5.2.1 Visualizing two quantitative variables

A **scatterplot** displays the relationship between two quantitative variables. Each observation is represented as a point, with one variable on the horizontal axis and the other on the vertical axis.

A **scatterplot** provides a case-by-case view of data for two quantitative variables. Each point represents one observation. The horizontal axis shows the values of one variable and the vertical axis shows the values of the other.

Consider a dataset of 50 loans, where we plot total income on the horizontal axis and loan amount on the vertical axis:

Figure 5.4: A scatterplot of loan amount versus total income for 50 loans. There is a moderate positive association.

What do scatterplots reveal about the data, and how are they useful?

Show answer

Scatterplots are helpful for quickly spotting associations between two quantitative variables — whether they tend to increase together, decrease together, or show no pattern. They can also reveal whether the relationship is linear or curved, how strong the association is, and whether there are any unusual points.

5.2.2 Describing associations

When examining a scatterplot, we describe the association using four characteristics:

Describing the association between two quantitative variables:

1. **Direction**: Is the association **positive** (both variables tend to increase together) or **negative** (one increases as the other decreases)?
2. **Form**: Is the relationship approximately **linear** (following a straight-line pattern) or **nonlinear** (curved)?
3. **Strength**: Is the association **strong** (points closely follow a pattern), **moderate**, or **weak** (points are loosely scattered)?
4. **Unusual observations**: Are there any **outliers** — points that deviate markedly from the overall pattern?

A scatterplot shows the poverty rate vs. median household income for over 3,000 US counties. How would you describe the association?

The association is **negative** (as poverty rate increases, median household income tends to decrease), **nonlinear** (the relationship curves, with income decreasing more steeply at lower poverty rates and leveling off at higher poverty rates), and **moderately strong** (the points follow the curving pattern fairly closely, though there is noticeable scatter). There do not appear to be any extreme outliers, though counties with very high poverty rates or very high incomes stand out somewhat.

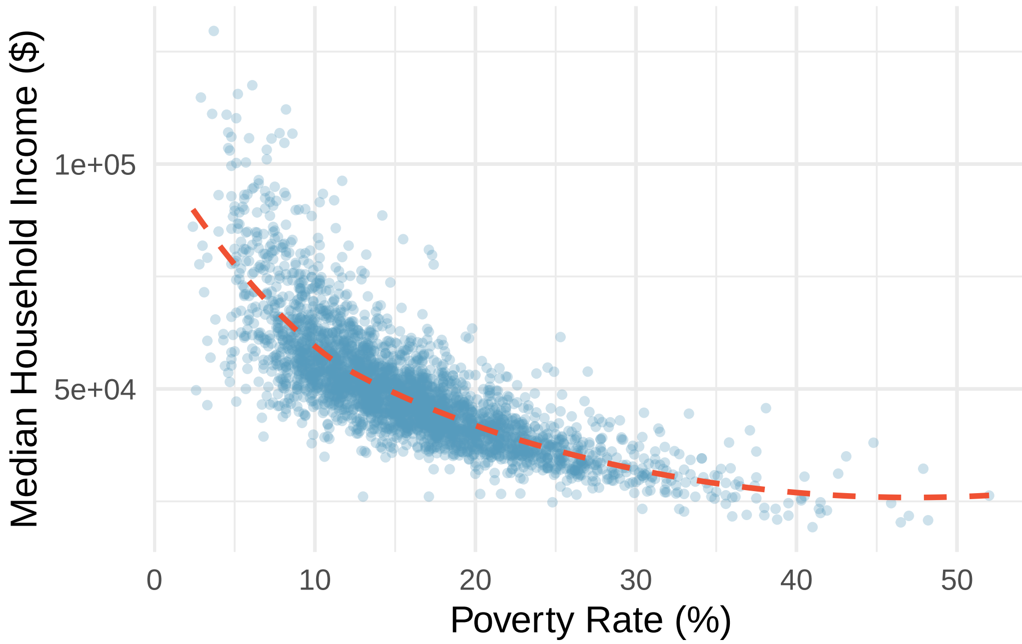


Figure 5.5: A scatterplot of median household income versus poverty rate for US counties, with a curved trend line. The relationship is negative and nonlinear.

Describe two variables that would have a horseshoe-shaped ($\cap$) association in a scatterplot.

Show answer

Consider a situation where the vertical axis represents something "good" and the horizontal axis represents something that is good only in moderation. For example, health and water consumption: we need some water to survive, but consuming too much becomes dangerous. Athletic performance and practice time is another example — some practice improves performance, but overtraining leads to injury and decline.

For the 50-loan dataset, the scatterplot of total income versus loan amount shows that borrowers with higher incomes tend to take out larger loans. Describe this association.

The association is **positive** (higher income is associated with higher loan amounts), approximately **linear** (the general trend follows a straight line, though the pattern is loose), and **weak to moderate** (there is considerable scatter — many borrowers with similar incomes take out very different loan amounts). There are a few borrowers with income above $250,000 who stand out on the right side of the plot.

5.2.3 Scatterplots with a third variable

We can add a third variable to a scatterplot by using color, shape, or size to represent a categorical or quantitative variable. For example, in a scatterplot of income vs. poverty rate, we could color the points by whether the county is in a metropolitan area. This technique helps reveal whether the relationship between two quantitative variables differs across groups.

5.3 Association vs. causation

Throughout this chapter, we have explored associations between variables. But association is not the same as causation.

Association does not imply causation.

When two variables are associated (correlated), it means they tend to vary together — but that does not necessarily mean one *causes* the other. There may be a **confounding variable** (a third variable) that is related to both, creating the illusion of a direct relationship.

A study finds a strong positive association between ice cream sales and drowning deaths. Does eating ice cream cause drowning?

Of course not. The association exists because both ice cream sales and drowning are related to a confounding variable: **temperature** (or season). In hot weather, people buy more ice cream *and* more people go swimming, which increases the risk of drowning. Temperature is the lurking variable that drives both.

A **confounding variable** (also called a **lurking variable**) is a variable that is related to both the explanatory and response variables. Confounding variables can create misleading associations between two variables that have no direct causal relationship.

A study finds that students who eat breakfast tend to have higher GPAs than students who skip breakfast. Can we conclude that eating breakfast causes higher GPAs?

Show answer

Not from this information alone. This is likely an observational study, and there could be confounding variables. For example, students who eat breakfast may come from more supportive home environments, may have better time management skills, or may generally take better care of their health — all of which could independently contribute to higher GPAs. To establish a causal connection, we would need a randomized experiment.

5.3.1 When can we infer causation?

Establishing a causal relationship between two variables generally requires:

1. **A randomized experiment**: Randomly assigning subjects to treatment and control groups helps eliminate confounding variables. If a difference is observed between groups, we can more confidently attribute it to the treatment.
2. **Temporal precedence**: The cause must precede the effect in time.
3. **Ruling out confounders**: We must be able to argue that no lurking variables explain the observed association.

In observational studies (where we observe but do not control the variables), we can identify associations but generally cannot draw causal conclusions. This is one of the most important distinctions in statistics.

Be especially skeptical of causal claims from observational data. Media reports often say "X causes Y" when the underlying study only found an association. Always ask: Was this a randomized experiment? Could there be confounding variables?

5.4 Chapter review

5.4.1 Summary

This chapter introduced tools for exploring relationships between variables. For **comparing a quantitative variable across groups** defined by a categorical variable, side-by-side box plots provide a compact comparison of centers and spreads, faceted histograms reveal full distributional shapes, and ridge plots overlay density curves for compact multi-group comparison. For **two quantitative variables**, scatterplots reveal the direction, form, strength, and unusual features of an association. We closed with the fundamental principle that **association does not imply causation** — confounding variables can create misleading associations, and establishing causation generally requires a randomized experiment.

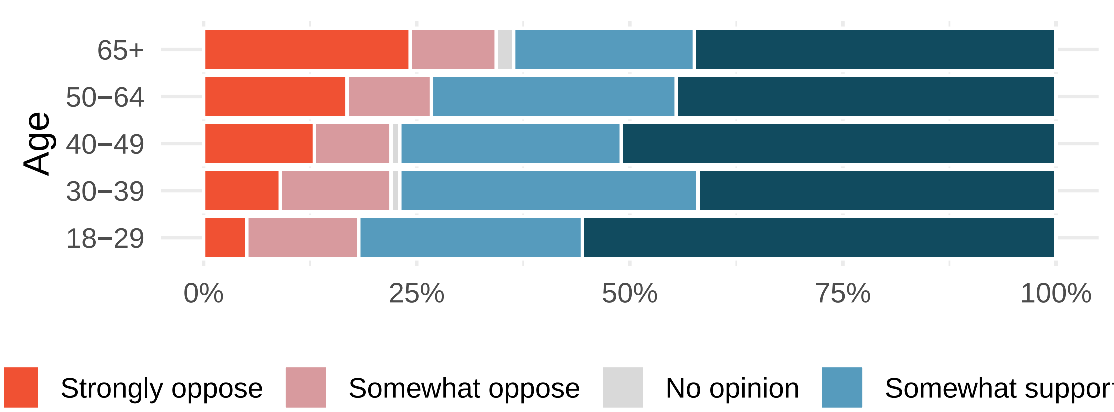
5.4.2 Key terms

- Side-by-side box plot, faceted plot, ridge plot
- Scatterplot
- Positive association, negative association
- Linear, nonlinear
- Direction, form, strength (of an association)
- Confounding variable (lurking variable)
- Association vs. causation

5.5 Exercises

Answers to odd-numbered exercises are provided inline.

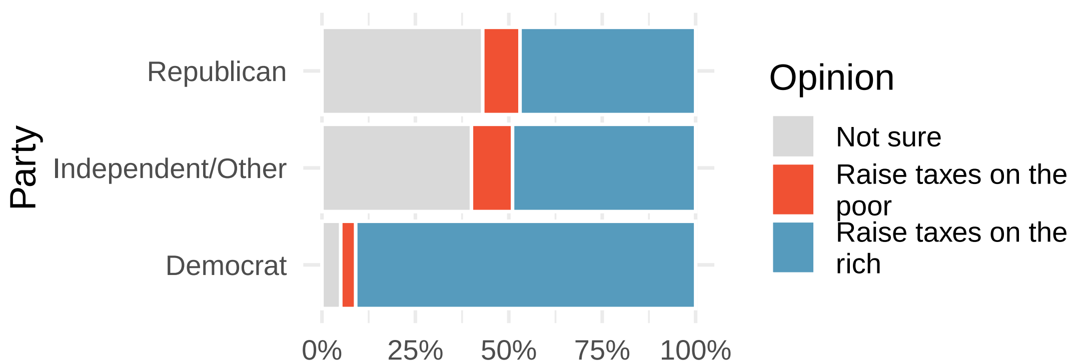
1. **Black Lives Matter.** A Washington Post-Schar School poll conducted in the United States in June 2020, among a random national sample of 1,006 adults, asked respondents whether they support or oppose protests following George Floyd's killing that have taken place in cities across the US. The survey also collected information on the age of the respondents. (Washington Post 2020) The results are summarized in the stacked bar plot below.



a. Based on the stacked bar plot, do views on the protests and age appear to be associated? Explain your reasoning.

b. Conjecture other possible variables that might explain the potential association between these two variables.

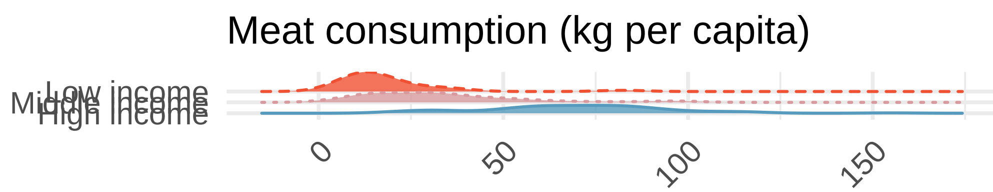
2. **Raise taxes.** A random sample of registered voters nationally were asked whether they think it's better to raise taxes on the rich or raise taxes on the poor. The survey also collected information on the political party affiliation of the respondents. (Polling 2015)

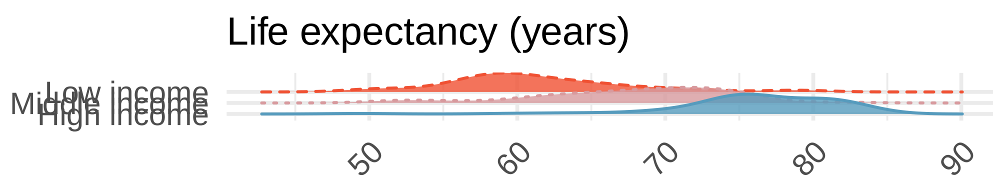


a. Based on the stacked bar plot shown above, do views on raising taxes and political affiliation appear to be associated? Explain your reasoning.

b. Conjecture other possible variables that might explain the potential association between these two variables.

3. **Meat consumption and life expectancy.** In data collected for (**You:2022?**), total meat intake is associated with life expectancy (at birth) in 175 countries. Meat intake is measured in kg per capita per year (averaged over 2011 to 2013). The two ridge plots show an association between income and meat consumption (higher income countries tend to eat more meat) and an association between income and life expectancy (higher income countries have higher life expectancy).





a. Do the graphs above demonstrate that meat consumption and life expectancy are associated? That is, can you tell if countries with low meat consumption have low life expectancy? Explain.

b. Let's assume that you had a plot comparing meat consumption and life expectancy, and they *do* seem associated. Your friend says that the plot shows that high meat consumption leads to a longer life. You correctly say, no, we can't tell if there is a causal realtionship because the relationship is confounded by income level. Explain what you mean.

c. How can you investigate the relationship between meat consumption and life expectancy in the presence of confounding variables (like income)?

4. **On-time arrivals.** Consider all of the flights out of New York City in 2013 that flew into Puerto Rico (BQN), Los Angeles (LAX), or San Francisco (SFO) on the following two airlines: JetBlue (B6) or United Airlines (UA). Below are the tabulated counts for the number of flights `delayed` and `on time` for each airline into each city.

dest	carrier	status	count
BQN	B6	delayed	271
BQN	B6	on time	322
BQN	UA	delayed	144
BQN	UA	on time	151
LAX	B6	delayed	670
LAX	B6	on time	999
LAX	UA	delayed	2368
LAX	UA	on time	3402
SFO	B6	delayed	405
SFO	B6	on time	615
SFO	UA	delayed	2694
SFO	UA	on time	4034

a. What percent of all JetBlue flights were delayed? What percent of all United Airlines flights were delayed? (Note, the overall delay proportions are typically what would be reported and associated with an airline.)

b. For each of the three airports, find the percent of delayed flights for each of JetBlue and United (you should have 6 numbers).

c. United has a higher proportion of delayed flights for each of the three cities, yet JetBlue has a higher proportion of delayed flights overall. Explain, using the data counts provided, how the seeming paradox could happen.

5. **US House of Representatives.** The US House of Representatives is dominated by two political parties: Democrats and Republicans. Democrats are thought to be the more liberal party and Republicans are considered to be the more conservative party. However, within each party there is an internal spectrum of liberal to conservative. For example, conservative Democrats and liberal Republicans would be labeled moderate. Consider an election where the only change in membership is that the most conservative Democrats are replaced by a set of liberal Republicans who are more liberal than the incumbent Republicans but more conservative than the Democrats they replaced.

a. After the election, is the Democratic wing of the House more conservative or more liberal? Explain.

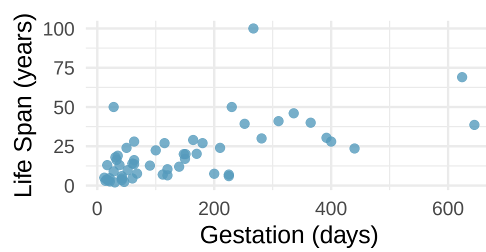
b. After the election, is the Republican wing of the House more conservative or more liberal? Explain.

c. After the election, is the overall House membership more conservative or more liberal? Explain.

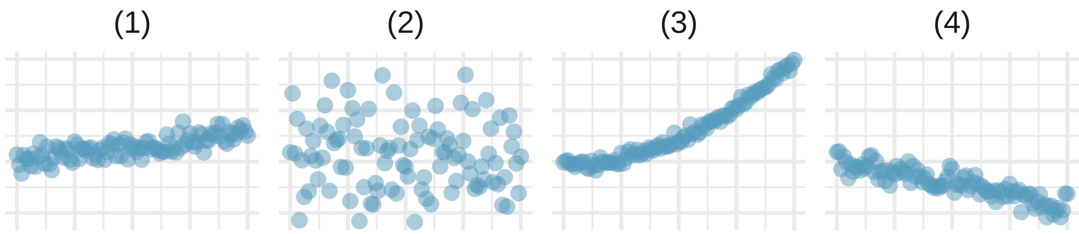
d. In what settings would you report the outcome of the change in House membership to be more conservative? And in what settings would you report the outcome of the change in House membership to be more liberal?

6. **Mammal life spans.** Data were collected on life spans (in years) and gestation lengths (in days) for 62 mammals. A scatterplot of life span versus length of gestation is shown below. (Allison and Cicchetti 1975)

a. What type of an association is apparent between life span and length of gestation?
b. What type of an association would you expect to see if the axes of the plot were reversed, i.e., if we plotted length of gestation versus life span?
c. Are life span and length of gestation independent? Explain your reasoning.



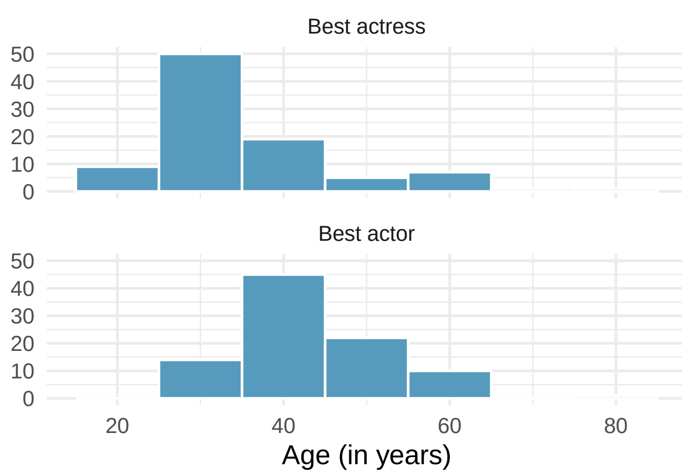
7. **Associations.** Indicate which of the plots show (a) a positive association, (b) a negative association, or (c) no association. Also determine if the positive and negative associations are linear or nonlinear. Each part may refer to more than one plot.



8. **Reproducing bacteria.** Suppose that there is only sufficient space and nutrients to support one million bacterial cells in a petri dish. You place a few bacterial cells in this petri dish, allow them to reproduce freely, and record the number of bacterial cells in the dish over time. Sketch a plot representing the relationship between number of bacterial cells and time.

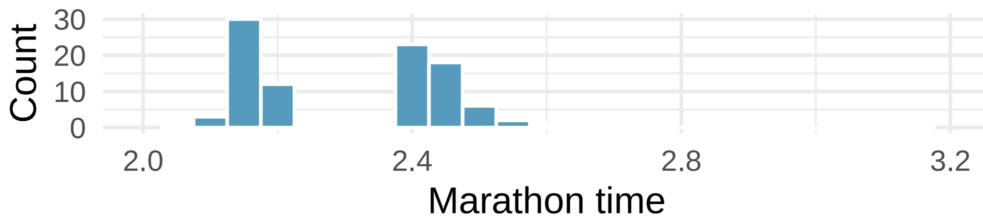
9. **Office productivity.** Office productivity is relatively low when the employees feel no stress about their work or job security. However, high levels of stress can also lead to reduced employee productivity. Sketch a plot to represent the relationship between stress and productivity.

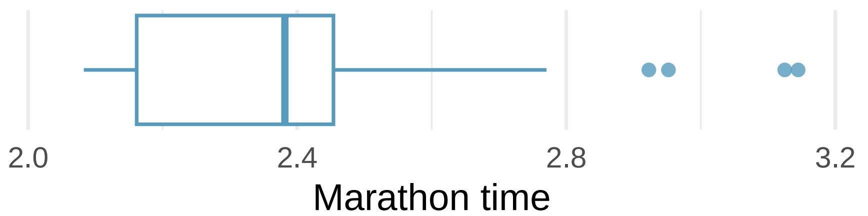
10. **Oscar winners.** The first Oscar awards for best actor and best actress were given out in 1929. The histograms below show the age distribution for all of the best actor and best actress winners from 1929 to 2019. Summary statistics for these distributions are also provided. Compare the distributions of ages of best actor and actress winners.



	Mean	SD	n
Best actor	43.8	8.8	92
Best actress	36.2	11.9	92

11. **NYC marathon winners.** The histogram and box plots below show the distribution of finishing times for male and female (combined) winners of the New York City Marathon between 1970 and 2023.

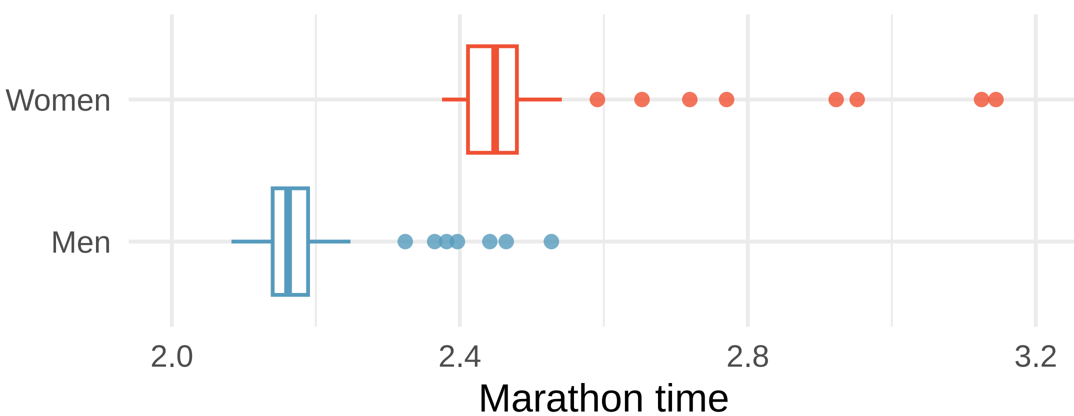




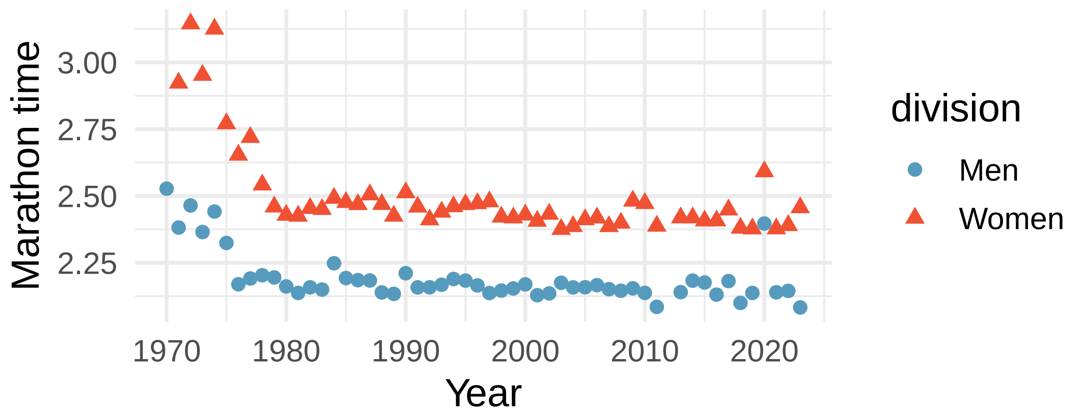
a. What features of the distribution are apparent in the histogram and not the box plot? What features are apparent in the box plot but not in the histogram?

b. What may be the reason for the bimodal distribution? Explain.

c. Compare the distribution of marathon times for men and women based on the box plot shown below.



d. The time series plot shown below is another way to look at these data. Describe what is visible in this plot but not in the others.



Datasets: `flights` (nycflights13) | `mammals` (openintro) | `oscars` (openintro) | `nyc_marathon` (openintro)

PART II

SIMULATE

How can we draw conclusions about a population from a single sample? We answer this question through *simulation*. By resampling data thousands of times, we build an intuitive understanding of sampling variability, hypothesis testing, and confidence intervals — without formulas or distribution tables.

This is the heart of the simulation-first approach: understand the *logic* of inference before learning the mathematical shortcuts.

Chapter 6

Sampling Variability

So far in this course, you have learned how to collect data, visualize it, and compute summary statistics. But statistics is about more than describing the data you have — it is about using data to learn about a larger population. The central challenge is this: **every sample is different**. If you took another sample from the same population, you would get different results. This chapter introduces the foundational ideas that make statistical inference possible: the distinction between parameters and statistics, the concept of a sampling distribution, and the standard error as a measure of how much sample statistics vary. These ideas are the bridge from describing data to drawing conclusions about the world.

6.1 Parameters vs. statistics

When we collect data, we typically have a larger group in mind. A political poll surveys 1,000 likely voters, but the real interest is in all voters. A quality engineer measures 50 widgets from an assembly line, but the real interest is in all widgets the line produces. The larger group is the **population**; the subset we actually observe is the **sample**.

A **population parameter** is a numerical summary of the entire population. It is a fixed number, but it is almost always unknown because we cannot observe the whole population.

A **sample statistic** (also called a **point estimate**) is a numerical summary computed from sample data. It is our best guess at the population parameter. Because the statistic is computed from a sample, it is known — but it changes from sample to sample.

We use specific notation to distinguish parameters from statistics:

Quantity	Parameter (population)	Statistic (sample)
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$
Difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$

The "hat" notation ($\hat{p}$) is a common convention: it signals that we are estimating the corresponding parameter.

A medical consultant has facilitated 62 liver donor surgeries, and 3 of her patients experienced complications. She uses this track record to attract new patients.

Let p represent the true complication rate for liver donors working with this consultant. What is the point estimate $\hat{p}$?

The sample proportion is $\hat{p} = 3/62 = 0.048$. That is, about 4.8% of her patients experienced complications. This is our point estimate of the true complication rate p.

A Stanford University graduate student recruited 120 tappers and 120 listeners for a study on communication. Only 3 out of 120 listeners were able to guess the tune being tapped. What is the point estimate $\hat{p}$ for the true proportion of listeners who can guess correctly?

Show answer

The point estimate is $\hat{p} = 3/120 = 0.025$, or 2.5%. This is our best single estimate of the true proportion p of listeners who can guess the tune.

In a study of sex discrimination, 48 male bank supervisors each reviewed a personnel file and decided whether the candidate should be promoted. The files were identical except that half listed the candidate as male and half as female. The results are summarized below:

	Promoted	Not promoted	Total
Male	21	3	24
Female	14	10	24
Total	35	13	48

What is the point estimate for the difference in promotion rates between male and female candidates?

The promotion rate for male candidates is $\hat{p}_M = 21/24 = 0.875$ and for female candidates is $\hat{p}_F = 14/24 = 0.583$. The point estimate of the difference is:

$$\hat{p}_M - \hat{p}_F = 0.875 - 0.583 = 0.292$$

The observed difference is 29.2 percentage points in favor of male candidates.

A point estimate is almost never exactly equal to the population parameter. This is not because we did anything wrong — it is an inherent consequence of working with a subset of the population. The critical question is: *how far off might the estimate be?*

6.2 Sampling distributions

To understand how far a statistic might be from its parameter, we need to understand what happens when we take **many samples** from the same population.

Imagine a large bowl containing thousands of beads, 60% red and 40% white. The proportion of red beads in the bowl is the population parameter: $p = 0.60$. Now suppose you reach in and scoop out a sample of $n = 50$ beads. You count 28 red beads, giving $\hat{p} = 28/50 = 0.56$. Your friend scoops a different sample of 50 and counts 32 red, giving $\hat{p} = 32/50 = 0.64$. Another friend gets 30 red for $\hat{p} = 0.60$.

Each person got a different sample proportion. No one did anything wrong — this is **sampling variability** in action.

Now imagine hundreds of people each scooping their own sample of 50 beads. If we recorded every $\hat{p}$ and made a histogram of all those values, we would see the **sampling distribution** of $\hat{p}$.

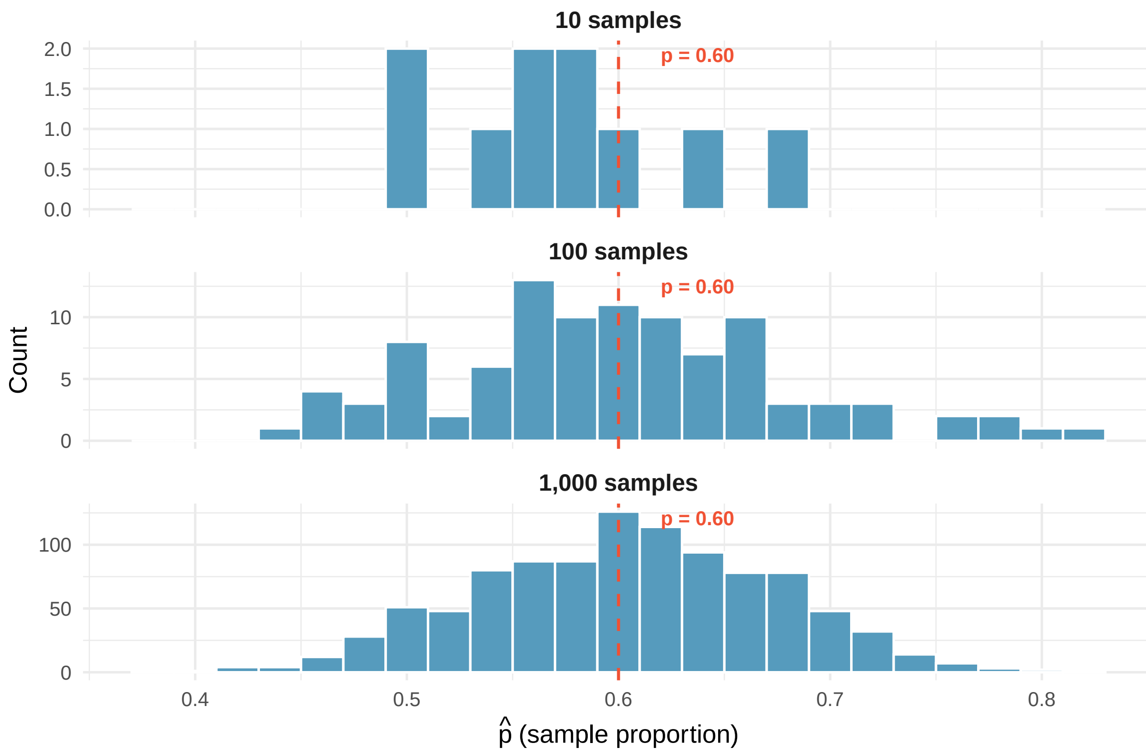


Figure 6.1: Building a sampling distribution. Each dot represents the proportion of red beads from one sample of n = 50. With more samples, the bell-shaped pattern centered at p = 0.60 becomes clear.

The **sampling distribution** of a statistic is the distribution of all possible values of that statistic computed from all possible samples of a given size from a given population.

A sampling distribution is *not* the distribution of individual data values. It is the distribution of a *summary statistic* (like $\hat{p}$ or $\bar{x}$) across many hypothetical samples.

Key properties of sampling distributions. For many common statistics, the sampling distribution has three important features:

1. **Center**: The sampling distribution is centered at the population parameter. On average, the sample statistic equals the parameter.
2. **Spread**: The sampling distribution has a measurable spread that decreases as sample size increases. Larger samples give more precise estimates.
3. **Shape**: For many statistics, the sampling distribution is approximately bell-shaped (normal) when the sample size is large enough.

Here is an analogy that may help. Imagine you are trying to determine the average depth of a lake. You drop a weighted line at a random spot and measure the depth — that is one sample. If you did this at many random spots and recorded all those measurements, you would get a distribution of individual depth measurements (the *data distribution*). But if instead, for each spot you visited, you dropped the line 30 times in a small cluster and computed the average of those 30 measurements, then moved to a new random spot and did it again, the collection of those averages would form a *sampling distribution* of the mean.

The two distributions are fundamentally different:

- The **data distribution** shows how individual observations vary.
- The **sampling distribution** shows how a *statistic* (like the mean) varies from sample to sample.

6.3 Variability of sample statistics

Different samples from the same population produce different statistics. This variability is not a flaw — it is a fundamental feature of sampling that we must understand and quantify.

Suppose the true proportion of all US adults who support a particular policy is $p = 0.52$. Three polling organizations each take a random sample of $n = 1{,}000$ adults and compute $\hat{p}$. Their results might look like:

Poll	Sample proportion $\hat{p}$
Poll A	0.49
Poll B	0.54
Poll C	0.51

All three polls were conducted correctly, yet they produced different results. Poll A even shows less than 50% support! This is sampling variability: the unavoidable variation that occurs when we estimate a population parameter from a sample.

If the three polls above each used a sample of $n = 10{,}000$ instead of $n = 1{,}000$, would you expect the three $\hat{p}$ values to be closer together or farther apart? Why?

Show answer

We would expect them to be closer together. Larger samples contain more information about the population, so the statistic is more precise — meaning less variability from sample to sample.

6.3.1 What determines how much a statistic varies?

Two factors control how much a sample statistic bounces around from sample to sample:

1. **Sample size** (n): Larger samples produce statistics that are closer to the parameter. This is intuitive — a sample of 1,000 tells you more about the population than a sample of 10.
2. **Population variability**: If the population is very homogeneous (everyone is similar), then any sample will look like any other sample, and the statistic won't vary much. If the population is very heterogeneous (lots of individual differences), then different samples can look quite different, and the statistic will vary more.

Of these two, sample size is the one we can control, and it turns out to be very powerful. Quadrupling the sample size cuts the variability of most statistics roughly in half.

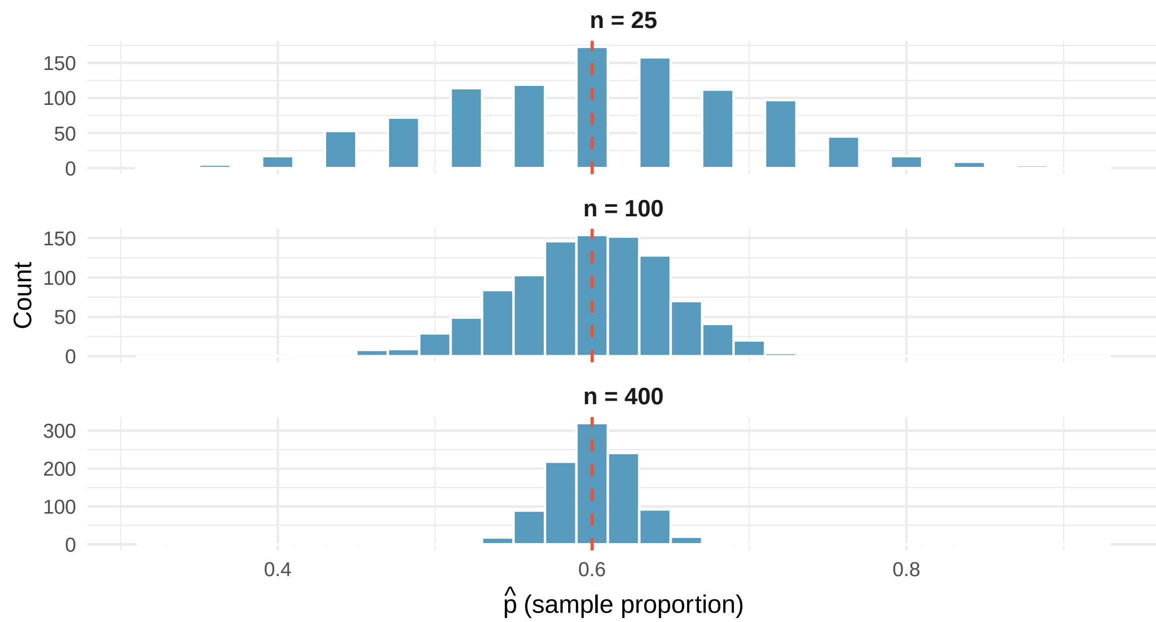


Figure 6.2: Sampling distributions of the sample proportion for three different sample sizes. All three are centered at p = 0.60, but the spread decreases dramatically as the sample size increases.

Figure 6.2 shows this dramatically: with $n = 25$, individual sample proportions range from about 0.35 to 0.85. With $n = 400$, they are tightly clustered between 0.55 and 0.65.

Explore sampling variability! Use StatLens's Sampling Distribution Simulator to see this in action:

Launch Sampling Distribution Simulator

1. Set the population proportion to $p = 0.60$ and the sample size to $n = 25$.
2. Draw 500 samples and observe the resulting sampling distribution.
3. Now change the sample size to $n = 100$ and draw 500 more samples.
4. Compare the two sampling distributions. How do their centers compare? Their spreads? Their shapes?

6.4 Standard error

We need a single number that quantifies how much a statistic varies from sample to sample. That number is the **standard error**.

The **standard error** (SE) of a statistic is the standard deviation of its sampling distribution. It measures the typical distance between a sample statistic and the population parameter.

A small standard error means the statistic tends to fall close to the parameter (precise estimate). A large standard error means the statistic could plausibly fall far from the parameter (imprecise estimate).

Do not confuse the standard error with the standard deviation of the data:

- The **standard deviation** (s or σ) measures how much *individual observations* vary around the mean.
- The **standard error** (SE) measures how much a *sample statistic* varies around the population parameter from sample to sample.

The standard error is always smaller than the standard deviation (for samples larger than $n = 1$) because averaging reduces variability.

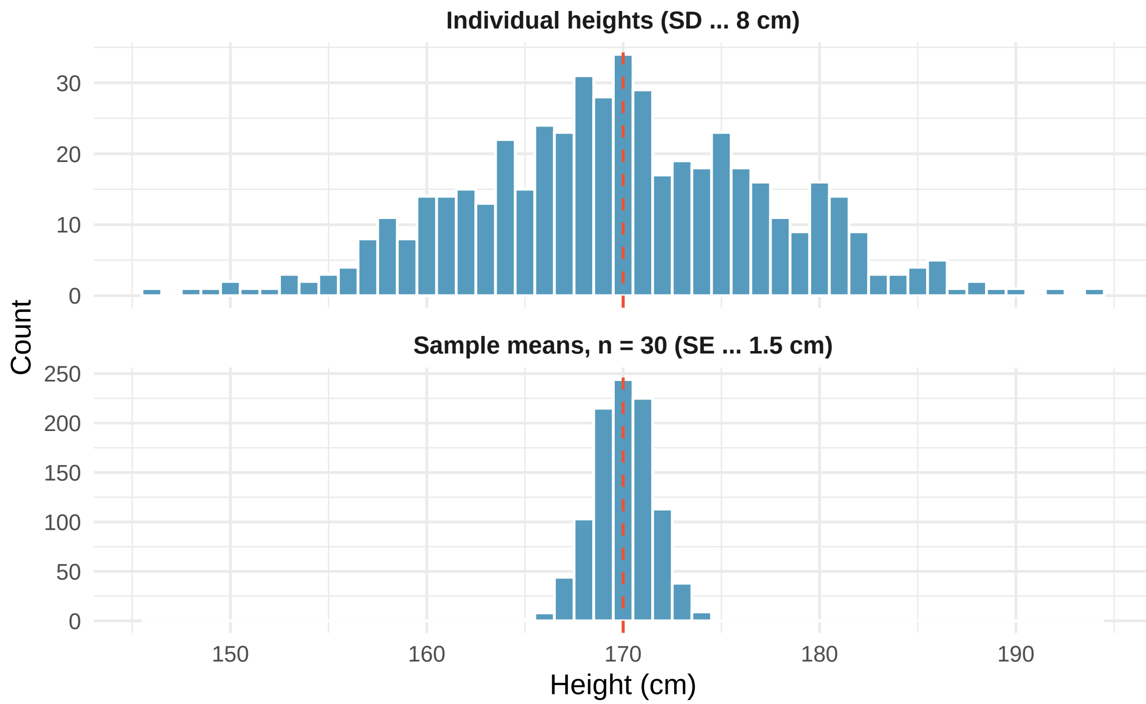


Figure 6.3: Comparing the data distribution (top) with the sampling distribution of the mean (bottom). Both are centered near the same value, but the sampling distribution is much narrower — its spread is the standard error, not the standard deviation.

In Figure 6.3, the top panel shows how individual heights vary (SD $\approx$ 8 cm), while the bottom panel shows how sample means vary (SE $\approx$ 1.5 cm). Both are centered near $\mu = 170$, but the sampling distribution of the mean is much narrower.

In the medical consultant example, $\hat{p} = 3/62 = 0.048$. If we could take many samples of 62 patients from this consultant's true patient population and compute $\hat{p}$ each time, the standard deviation of all those $\hat{p}$ values would be the standard error of $\hat{p}$.

While we usually cannot literally take repeated samples, simulation methods (which we will explore in Chapters 7 and 8) allow us to approximate the standard error. In this case, bootstrap simulations suggest that the standard error of $\hat{p}$ is approximately 0.027. This means that if we could repeat the study many times, the sample complication rate would typically differ from the true complication rate by about 2.7 percentage points.

6.4.1 Why the standard error matters

The standard error is the engine of statistical inference. Here is why:

- **Confidence intervals** are built by going a certain number of standard errors above and below the point estimate. A rough 95% confidence interval is:

$$\text{statistic} \pm 2 \times SE$$

- **Hypothesis tests** compare an observed statistic to what we would expect under the null hypothesis. The comparison is made in units of standard errors: "Is the observed result more than 2 standard errors from the null value?"
- **Sample size planning** uses the standard error to determine how many observations are needed to achieve a desired level of precision.

In the coming chapters, we will use simulation to *estimate* the standard error. In later chapters (Part III), we will use mathematical formulas to *calculate* it directly.

Two researchers both estimate the proportion of college students who prefer online textbooks. Researcher A surveys 100 students and finds $\hat{p}_A = 0.62$ with $SE_A = 0.049$. Researcher B surveys 400 students and finds $\hat{p}_B = 0.58$ with $SE_B = 0.025$.

(a) Whose estimate is more precise? How do you know?
(b) Why is Researcher B's standard error smaller?

Show answer

(a) Researcher B's estimate is more precise because the standard error is smaller (0.025 < 0.049), meaning the sample statistic is expected to be closer to the true parameter. (b) Researcher B surveyed four times as many students, and increasing the sample size reduces the standard error.

6.5 Looking ahead

This chapter has established the core vocabulary and ideas that drive all of statistical inference:

- We want to know about **parameters** but can only observe **statistics**.
- Statistics vary from sample to sample — this is **sampling variability**.
- The pattern of this variation is described by the **sampling distribution**.
- The **standard error** quantifies the typical amount of sampling variability.

The fundamental question of inference is: *Given the sampling variability we expect, is the pattern in our data real, or could it have happened by chance?*

In the next three chapters, we will answer this question using three different approaches — all rooted in simulation:

- **Chapter 7** (**Randomization Tests**): Shuffle the data to see what differences look like when there is no real effect. If the observed difference is extreme compared to the shuffled results, we have evidence of a real effect.
- **Chapter 8** (**Bootstrap Confidence Intervals**): Resample from the data to approximate the sampling distribution, and use it to construct a range of plausible values for the parameter.
- **Chapter 9** (**Sampling Distributions via Simulation**): Connect the simulation results from Chapters 7 and 8 to mathematical theory, setting the stage for the formula-based methods in Part III.

The goal of Part II is to build your *intuition* for inference. The formulas will come later; the ideas come first.

6.6 Chapter review

6.6.1 Summary

- A **population parameter** is a fixed but unknown numerical summary of the population. A **sample statistic** (point estimate) is computed from data and serves as our best guess of the parameter.
- Different samples from the same population yield different statistics. This is **sampling variability** — it is not a mistake, but a fundamental feature of working with samples.
- The **sampling distribution** of a statistic describes how the statistic varies across all possible samples of a given size. It is typically centered at the parameter, and its spread decreases as sample size increases.
- The **standard error** is the standard deviation of the sampling distribution. It measures the precision of the statistic as an estimate of the parameter.
- The standard error is not the same as the standard deviation of the data. The standard deviation measures variability of individual observations; the standard error measures variability of a statistic.
- Larger samples produce smaller standard errors, meaning more precise estimates.

6.6.2 Key terms

Term	Definition
Population parameter	A fixed numerical summary of the entire population (e.g., p, μ)
Sample statistic	A numerical summary computed from sample data (e.g., $\hat{p}$, $\bar{x}$)
Point estimate	Another name for sample statistic; our best guess at the parameter
Sampling variability	The fact that different samples produce different statistics
Sampling distribution	The distribution of a statistic over all possible samples of a given size
Standard error (SE)	The standard deviation of the sampling distribution

6.7 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Identify the parameter, I.** For each of the following situations, state whether the parameter of interest is a mean or a proportion. It may be helpful to examine whether individual responses are numerical or categorical.

a. In a survey, 100 college students are asked how many hours per week they spend on the Internet.

b. In a survey, 100 college students are asked: "What percentage of the time you spend on the Internet is part of your course work?"

c. In a survey, 100 college students are asked whether they cited information from Wikipedia in their papers.

d. In a survey, 100 college students are asked what percentage of their total weekly spending is on alcoholic beverages.

e. In a sample of 100 recent college graduates, it is found that 85 percent expect to get a job within one year of their graduation date.

2. **Identify the parameter, II.** For each of the following situations, state whether the parameter of interest is a mean or a proportion.

a. A poll shows that 64% of Americans personally worry a great deal about federal spending and the budget deficit.

b. A survey reports that local TV news has shown a 17% increase in revenue within a two year period while newspaper revenues decreased by 6.4% during this time period.

c. In a survey, high school and college students are asked whether they use geolocation services on their smart phones.

d. In a survey, smart phone users are asked whether they use a web-based taxi service.

e. In a survey, smart phone users are asked how many times they used a web-based taxi service over the last year.

Chapter 7

Randomization Tests

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Statistical inference is primarily concerned with understanding and quantifying the uncertainty of parameter estimates. While the equations and details change depending on the setting, the foundations for inference are the same throughout all of statistics. In this chapter, we introduce the **hypothesis testing framework** — a formal method for evaluating competing claims about a population using data. We learn how to build **randomization distributions** by shuffling data under the assumption that there is no effect, and we use these distributions to compute **p-values** that quantify the strength of the evidence.

7.1 The hypothesis testing framework

Throughout this course so far, you have worked with data in a variety of contexts. You have learned how to summarize and visualize data as well as how to describe relationships between variables. But more often than not, data have been collected to answer a research question about a larger group — and the data are a (hopefully) representative subset.

There is almost always variability in data — one dataset will not be identical to a second dataset even if they are both collected from the same population using the same methods. However, quantifying the variability in the data is neither obvious nor easy. Answering the question "*how* different is one dataset from another?" is not trivial.

Suppose your professor splits the students in your class into two groups: students who sit on the left side of the classroom and students who sit on the right side. If $\hat{p}_L$ represents the proportion of left-side students who prefer reading on a screen and $\hat{p}_R$ represents the proportion of right-side students who prefer reading on a screen, would you be surprised if $\hat{p}_L$ did not *exactly* equal $\hat{p}_R$?

While the proportions $\hat{p}_L$ and $\hat{p}_R$ would probably be close to each other, it would be unusual for them to be exactly the same. We would probably observe a small difference due to *chance*.

If we do not think the side of the room a person sits on in class is related to whether they prefer to read books on a screen, what assumption are we making about the relationship between these two variables?

Show answer

We would be assuming that these two variables are **independent** — knowing which side of the room someone sits on tells us nothing about their reading preference.

The question at the heart of hypothesis testing is: *Could the pattern we observe in the data have arisen from chance alone, or is there something real going on?*

Hypothesis testing.

A **hypothesis test** is a statistical technique used to evaluate competing claims using data. The process involves:

1. **State the hypotheses**: Define a null hypothesis (H_0) and an alternative hypothesis (H_A).
2. **Collect data**: Gather evidence that bears on the hypotheses.
3. **Assess the evidence**: Determine how likely the observed data would be if the null hypothesis were true.
4. **Draw a conclusion**: Decide whether the evidence is strong enough to reject the null hypothesis.

If the null hypothesis and the data notably disagree, then we reject the null hypothesis in favor of the alternative hypothesis. There are many nuances to hypothesis testing, and we will discuss these ideas throughout this chapter and those that follow.

7.2 Null and alternative hypotheses

The **null hypothesis** (H_0) often represents a skeptical perspective or a claim of "no difference" or "no effect." It assumes that any observed pattern in the data is due to chance alone.

The **alternative hypothesis** (H_A) represents the claim under investigation — typically that there *is* a real difference, a real effect, or a real relationship. The alternative hypothesis is usually the reason the research was conducted in the first place.

The US court system considers two possible claims about a defendant: they are either innocent or guilty. If we set these claims up in a hypothesis framework, which would be the null hypothesis and which the alternative?

The jury considers whether the evidence is so convincing that there is no reasonable doubt regarding the person's guilt. The skeptical perspective (null hypothesis) is that the person is innocent until evidence is presented that convinces the jury the person is guilty (alternative hypothesis).

$$H_0 : \text{The defendant is innocent.}$$
$$H_A : \text{The defendant is guilty.}$$

Notice that if a jury finds a defendant *not guilty*, this does not necessarily mean the jury is confident in the person's innocence. They are simply not convinced of the alternative. This is also the case with hypothesis testing: **even if we fail to reject the null hypothesis, we do not accept the null hypothesis as truth.** Failing to find evidence in favor of the alternative is not the same as finding evidence that the null hypothesis is true.

7.3 Case study: sex discrimination

We consider a study investigating sex discrimination in the 1970s, set in the context of personnel decisions within a bank. The research question: "Are individuals who identify as female discriminated against in promotion decisions made by their managers who identify as male?"

7.3.1 Observed data

The participants were 48 bank supervisors who identified as male, attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of personnel director of a bank and were given a personnel file to judge whether the person should be promoted to a branch manager position. The files were identical, except that half indicated the candidate identified as male and the other half indicated the candidate identified as female. These files were randomly assigned to the bank managers.

Is this an observational study or an experiment? How does the type of study impact what can be inferred from the results?

Show answer

The study is an experiment, as subjects were randomly assigned a "male" file or a "female" file (remember, all the files were actually identical in content). Since this is an experiment, the results can be used to evaluate a causal relationship between the sex of a candidate and the promotion decision.

The results are summarized in the table below:

	Promoted	Not promoted	Total
Male	21	3	24
Female	14	10	24
Total	35	13	48

When looking at the rates of promotion in this study, why might we be tempted to immediately conclude that individuals identifying as female are being discriminated against?

The large difference in promotion rates (58.3% for female personnel versus 87.5% for male personnel) suggests there might be discrimination against women in promotion decisions. However, we cannot yet be sure if the observed difference represents discrimination or is just due to random chance when there is no discrimination occurring. Since we wouldn't expect the sample proportions to be *exactly* equal, even if the truth was that the promotion decisions were independent of sex, we can't rule out random chance as a possible explanation.

The observed difference in promotion rates is:

$$\hat{p}_M - \hat{p}_F = \frac{21}{24} - \frac{14}{24} = 0.875 - 0.583 = 0.292$$

This is 29.2 percentage points. The point estimate is large, but the sample size is small, making it unclear if this represents real discrimination or simply chance variation. We label two competing claims:

- H_0: **Null hypothesis.** The variables `sex` and `decision` are independent. The difference in promotion rates of 29.2% was due to natural variability inherent in the random assignment.
- H_A: **Alternative hypothesis.** The variables `sex` and `decision` are *not* independent. The difference in promotion rates of 29.2% was not due to natural variability, and equally qualified female personnel are less likely to be promoted than male personnel.

7.4 Building a randomization distribution

What would it mean if the null hypothesis were true? It would mean each banker would decide whether to promote the candidate without regard to the sex indicated on the file. The 35 promotions and 13 non-promotions were determined by factors *other than sex*, and the difference we observed was just bad luck in how the files happened to be assigned.

We can test this by performing a **randomization** — simulating what would have happened if the bankers' decisions had been independent of sex but the files had been distributed differently.

7.4.1 The card-shuffling metaphor

Think of the data as 48 cards:

- 35 red cards (promoted)
- 13 white cards (not promoted)

In the actual study, 24 cards were dealt to the "male" pile and 24 to the "female" pile. Under the null hypothesis, the color of each card (the promotion decision) has nothing to do with which pile it lands in.

To simulate what the null hypothesis looks like, we:

1. **Thoroughly shuffle** all 48 cards together.
2. **Deal** 24 into a "male" pile and 24 into a "female" pile.
3. **Count** the number of red (promoted) cards in each pile and compute the difference in proportions.

This gives us one simulated difference *under the null hypothesis* — a difference that arose purely from the random shuffle, not from any real discrimination.

7.4.2 One simulation

Suppose after shuffling and dealing, we get the following result:

	Promoted	Not promoted	Total
Male	18	6	24
Female	17	7	24
Total	35	13	48

The simulated difference is $18/24-17/24 = 0.042$, or about 4.2% — much smaller than the observed 29.2%.

What is the difference in promotion rates in the simulated data above? How does this compare to the observed difference of 29.2% from the actual study?

Show answer

The simulated difference is $18/24 - 17/24 = 0.042$ or about 4.2% in favor of male personnel. This difference due to chance is much smaller than the 29.2% difference observed in the actual groups.

7.4.3 Many simulations

One simulation is not enough. We need to repeat the shuffle-and-deal process many times to build up a picture of what *chance alone* looks like. Using a computer, we can do this hundreds or thousands of times.

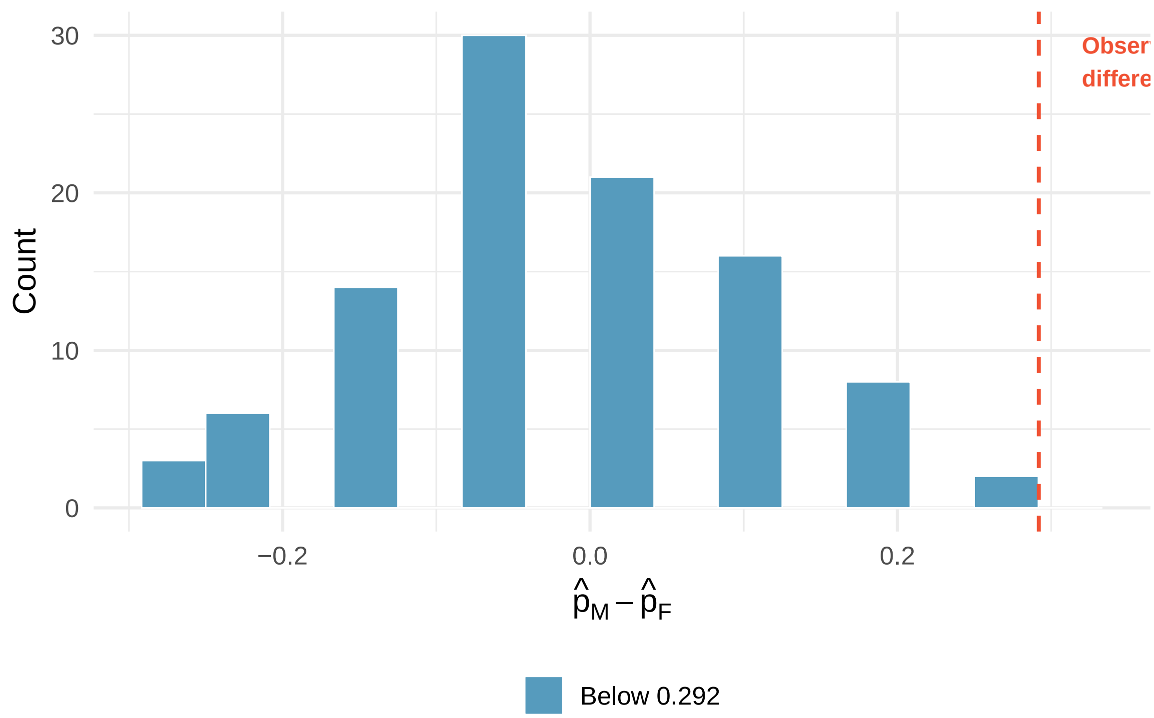


Figure 7.1: A histogram of differences from 100 simulations produced under the null hypothesis. The simulated sex and decision are independent. Differences at least as large as the observed 0.292 are highlighted in red.

The distribution of these simulated differences is centered around 0. This makes sense: under the null hypothesis, which makes no distinction between male and female files, we would expect the difference to be near zero with some random fluctuation.

How often would you observe a difference of at least 29.2% according to the randomization distribution? Often, sometimes, rarely, or never?

It appears that a difference of at least 29.2% under the null hypothesis would only happen about 2% of the time. Such a low probability indicates that observing such a large difference from chance alone is rare.

7.4.4 The conclusion

The difference of 29.2% is a rare event if there really is no effect of sex on promotion decisions. This provides us with two possible interpretations:

- If H_0 is true: Sex has no effect on promotion decisions, and we observed a difference that is so large it would only happen rarely by chance.
- If H_A is true: Sex has an effect on promotion decisions, and what we observed was actually due to equally qualified female candidates being discriminated against.

When we conduct formal studies, we reject a null position if the data strongly conflict with it. Since there was only about a 2% probability of obtaining a difference at least as large as 29.2% under the null hypothesis, we conclude that the data provide strong evidence of sex discrimination. We reject the null hypothesis in favor of the alternative.

Try it yourself! Use StatLens's Permutation Test Simulator to replicate the sex discrimination analysis:

Launch Randomization Test: Difference in Proportions

1. Enter the sex discrimination data (or use the built-in dataset).
2. Generate 1,000 shuffled differences.
3. Find how many simulated differences are at least as large as 0.292.
4. What proportion of simulations produced a difference this extreme or more?

7.5 Case study: opportunity cost

How rational and consistent is the behavior of the typical American college student? Here, we explore whether reminding students that money not spent now can be spent later causes them to be thriftier.

One hundred and fifty students were recruited for the study. All were given a scenario about purchasing a video on sale for $14.99. Half (the control group) were given these options:

(A) Buy this entertaining video.
(B) Not buy this entertaining video.

The other half (the treatment group) saw a slightly modified option (B):

(A) Buy this entertaining video.
(B) Not buy this entertaining video. Keep the $14.99 for other purchases.

The hypotheses are:

- H_0: Reminding students that they can save money for later purchases will not have any impact on their spending decisions.
- H_A: Reminding students that they can save money for later purchases will reduce the chance they will continue with a purchase.

7.5.1 Observed data

The results are summarized below:

	Buy video	Not buy video	Total
Control	56	19	75
Treatment	41	34	75
Total	97	53	150

We define a "success" as a student who chooses *not* to buy the video. The difference in the proportion choosing not to buy is:

$$\hat{p}_T - \hat{p}_C = \frac{34}{75} - \frac{19}{75} = 0.453 - 0.253 = 0.200$$

The proportion of students who chose not to buy was 20 percentage points higher in the treatment group. Is this 20% difference so prominent that it is unlikely to have occurred from chance alone?

7.5.2 Simulating under the null

Using the same randomization technique, we start with 150 index cards: 53 labeled "not buy video" and 97 labeled "buy video." We shuffle all cards together and deal them into two stacks of 75.

If we randomly assign the cards into simulated treatment and control groups, how many "not buy video" cards would we expect in each group? What would be the expected difference in proportions?

Since the groups are equal in size, we would expect about $53/2 \approx 26$ or 27 "not buy video" cards in each group, yielding a difference of roughly 0%. Due to random chance, we might sometimes observe a little more or fewer than 26–27.

After running 1,000 simulations, the results show that the distribution of simulated differences is centered at 0, as expected under the null hypothesis.

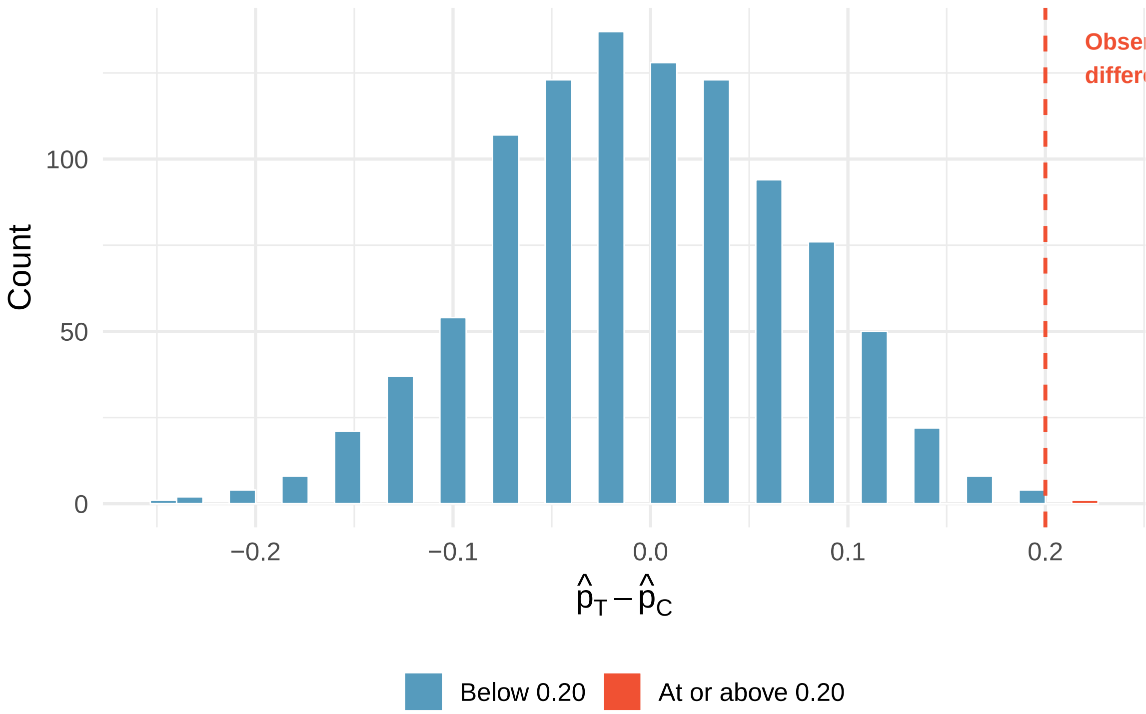


Figure 7.2: A histogram of 1,000 simulated differences produced under the null hypothesis. Differences at least as large as the observed 0.20 are highlighted in red.

Under the null hypothesis, we would observe a difference of at least +20% about 0.6% of the time. That is very rare! We conclude the data provide strong evidence there is a treatment effect: reminding students that they could spend money later lowers their chance of making a purchase. Because this is an experiment (with random assignment), we can make a causal claim.

7.6 P-values

The probability we computed in the previous sections — the chance of observing a result as extreme as or more extreme than the one actually observed, *if the null hypothesis were true* — has a name.

The **p-value** is the probability of observing data at least as favorable to the alternative hypothesis as our current dataset, if the null hypothesis were true. We typically use a summary statistic of the data, such as a difference in proportions, to compute the p-value. This summary value is called the **test statistic**.

In the sex discrimination study, the test statistic was the difference in promotion rates. What was the test statistic in the opportunity cost study?

The test statistic in the opportunity cost study was the difference in the proportion of students who decided against the video purchase in the treatment and control groups: $\hat{p}_T - \hat{p}_C = 0.200$.

Interpreting the p-value.

- A **small p-value** (e.g., below 0.05) means that the observed result would be rare if the null hypothesis were true. This is evidence *against* the null hypothesis.
- A **large p-value** (e.g., above 0.10) means that the observed result is not particularly unusual under the null hypothesis. The data do not provide strong evidence against the null hypothesis.
- The p-value is **not** the probability that the null hypothesis is true. It is the probability of the observed data (or more extreme data) *given that* the null hypothesis is true.

A common mistake is to interpret the p-value as the probability that the null hypothesis is true (e.g., "there is a 2% chance that sex has no effect on promotions"). This is incorrect. The p-value tells us how *surprising* the data would be *if* the null hypothesis were true, not how likely the null hypothesis is to be true.

7.7 Statistical significance

We say that the data provide **statistically significant** evidence against the null hypothesis if the p-value is less than a predetermined threshold called the **significance level**, denoted α (the Greek letter alpha).

The most commonly used significance level is $\alpha = 0.05$, meaning we reject the null hypothesis when the p-value is less than 5%. However, there is nothing magical about 0.05 — it is a convention, and in some fields or applications a stricter (e.g., 0.01) or more lenient (e.g., 0.10) threshold may be appropriate.

In the opportunity cost study, we found that a difference of at least 20% would occur only about 6-in-1,000 times if the reminder had no influence on student decisions. The p-value is approximately 0.006. Would you classify this as statistically significant?

Yes. Since p-value $= 0.006 < 0.05 = \alpha$, the data provide statistically significant evidence that US college students were actually influenced by the reminder.

What "statistically significant" does and does not mean.

"Statistically significant" means the p-value fell below the chosen significance level. It does **not** necessarily mean the effect is large or practically important. A very large sample can detect a tiny difference as "statistically significant" even if the difference is too small to matter in practice.

Conversely, "not statistically significant" does not mean there is no effect. It may simply mean the sample was too small to detect the effect.

7.7.1 Decision framework

The logic of hypothesis testing follows a consistent pattern:

If the p-value is...	Then we...	Because...
Less than α	**Reject** H_0	The data are unlikely under H_0; evidence supports H_A
Greater than or equal to α	**Fail to reject** H_0	The data are not sufficiently unusual under H_0

A researcher tests whether a new teaching method improves exam scores compared to the standard method. The p-value from the randomization test is 0.12. Using $\alpha = 0.05$, what conclusion should the researcher draw?

Show answer

Since p-value $= 0.12 > 0.05 = \alpha$, the researcher fails to reject the null hypothesis. The data do not provide sufficient evidence that the new teaching method improves exam scores. This does not mean the new method is ineffective — only that this study did not produce enough evidence to conclude it is effective.

7.8 Two-sided tests

In the sex discrimination study, we looked for differences in *one direction only* — whether female candidates were promoted at a lower rate. This is called a **one-sided** (or one-tailed) test. The alternative hypothesis specified a direction:

$$H_A : p_M - p_F > 0 \quad \text{(males promoted at a higher rate)}$$

Sometimes, however, we are interested in detecting a difference in *either direction*. For example, if we are studying whether a new drug changes blood pressure (it could increase it *or* decrease it), we would use a **two-sided** (or two-tailed) test:

$$H_A : p_1 \neq p_2 \quad \text{(the proportions are different, in either direction)}$$

In a **one-sided test**, the alternative hypothesis specifies a direction (greater than or less than). The p-value is computed as the proportion of the randomization distribution in one tail.

In a **two-sided test**, the alternative hypothesis does not specify a direction (just "different"). The p-value is computed as the proportion of the randomization distribution in *both* tails — values at least as extreme as the observed statistic in either direction.

Suppose we are testing whether a coin is fair. We flip it 100 times and get 60 heads.

- **One-sided** ($H_A : p > 0.5$): The p-value counts only simulations with 60 or more heads.
- **Two-sided** ($H_A : p \neq 0.5$): The p-value counts simulations with 60 or more heads *and* simulations with 40 or fewer heads (equally extreme in the other direction).

For a symmetric randomization distribution, the two-sided p-value is approximately twice the one-sided p-value. If the one-sided p-value is 0.03, the two-sided p-value is approximately 0.06.

A researcher wants to know if a new fertilizer changes plant growth compared to no fertilizer. Should this be a one-sided or two-sided test? Write the hypotheses using appropriate notation, where μ_T is the mean growth with fertilizer and μ_C is the mean growth without.

Show answer

This should be a two-sided test because the fertilizer could either increase or decrease growth (perhaps it has a toxic effect in some soils). The hypotheses are: $H_0 : \mu_T - \mu_C = 0$ (no difference in mean growth) and $H_A : \mu_T - \mu_C \neq 0$ (the fertilizer changes mean growth, in either direction).

Choosing one-sided vs. two-sided.

- Use a **one-sided test** when you have a specific directional hypothesis *before looking at the data*. For example, "we believe this drug *lowers* blood pressure."
- Use a **two-sided test** when you want to detect a difference in either direction, or when you are not sure which direction the difference might go.
- When in doubt, use a two-sided test. It is the more conservative choice and is less susceptible to bias from choosing the direction after seeing the data.

7.9 The randomization test procedure: a summary

We can summarize the randomization test procedure as follows:

1. **Frame the research question in terms of hypotheses.** The null hypothesis (H_0) usually represents no difference or no effect. The alternative hypothesis (H_A) represents the claim of interest.

2. **Collect data.** If the research question focuses on associations but not causation, use an observational study. If you want to establish a causal connection, use an experiment with random assignment.

3. **Model what would happen under the null hypothesis.** Shuffle (randomize) the data to break any association between the explanatory and response variables. Compute the test statistic for each shuffle. Repeat many times to build the **randomization distribution** (also called the null distribution).

4. **Compute the p-value.** Determine what proportion of the randomization distribution is at least as extreme as the observed test statistic.

5. **Draw a conclusion.** If the p-value is small (less than α), reject H_0 and conclude there is evidence for H_A. If the p-value is not small, fail to reject H_0. Write the conclusion in context, in plain language.

Practice the full procedure! Use StatLens's Permutation Test Simulator for different scenarios:

Launch Randomization Test: Difference in Proportions

Try entering your own data and running the full randomization test. Observe how the p-value changes when you adjust the sample size or the magnitude of the observed difference.

7.10 Chapter review

7.10.1 Summary

- A **hypothesis test** evaluates two competing claims: the null hypothesis (H_0), which represents no effect or no difference, and the alternative hypothesis (H_A), which represents the claim under investigation.
- A **randomization test** (or permutation test) assesses the null hypothesis by shuffling the data to simulate what would happen if there were truly no effect. The collection of simulated test statistics forms the **randomization distribution**.

- The **test statistic** is the summary value computed from the data (e.g., a difference in proportions) that we use to evaluate the hypotheses.
- The **p-value** is the probability of observing a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. It quantifies the strength of the evidence against H_0.
- We reject H_0 when the p-value is less than the **significance level** α (commonly 0.05). This is called a **statistically significant** result.
- A **one-sided test** looks for effects in a specified direction; a **two-sided test** looks for effects in either direction.
- Failing to reject H_0 is not the same as proving H_0 is true.
- Statistical significance does not imply practical importance.

7.10.2 Key terms

Term	Definition
Hypothesis test	A formal procedure for evaluating competing claims using data
Null hypothesis (H_0)	The claim of no effect or no difference; the skeptical position
Alternative hypothesis (H_A)	The claim that there is an effect or difference
Randomization test	A hypothesis test that builds a null distribution by shuffling data
Randomization distribution	The distribution of the test statistic under repeated randomizations
Test statistic	The summary value (e.g., difference in proportions) used to evaluate hypotheses
P-value	Probability of observing a result as extreme as or more extreme than the data, if H_0 is true
Significance level (α)	The threshold below which the p-value leads us to reject H_0
Statistically significant	A result whose p-value falls below α
One-sided test	Alternative hypothesis specifies a direction
Two-sided test	Alternative hypothesis does not specify a direction

7.11 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Hypotheses.** For each of the research statements below, note whether it represents a null hypothesis claim or an alternative hypothesis claim.

a. The number of hours that grade-school children spend doing homework predicts their future success on standardized tests.

b. King cheetahs on average run the same speed as standard spotted cheetahs.

c. For a particular student, the probability of correctly answering a 5-option multiple choice test is larger than 0.2 (i.e., better than guessing).

d. The mean length of African elephant tusks has changed over the last 100 years.

e. The risk of facial clefts is equal for babies born to mothers who take folic acid supplements compared with those from mothers who do not.

f. Caffeine intake during pregnancy affects mean birth weight.

g. The probability of getting in a car accident is the same if using a cell phone than if not using a cell phone.

2. **True null hypothesis.** Unbeknownst to you, let's say that the null hypothesis is actually true in the population. You plan to run a study anyway.

a. If the level of discernibility you choose (i.e., the cutoff for your p-value) is 0.05, how likely is it that you will mistakenly reject the null hypothesis?

b. If the level of discernibility you choose (i.e., the cutoff for your p-value) is 0.01, how likely is it that you will mistakenly reject the null hypothesis?

c. If the level of discernibility you choose (i.e., the cutoff for your p-value) is 0.10, how likely is it that you will mistakenly reject the null hypothesis?

3. **Identify hypotheses, I.** Write the null and alternative hypotheses in words and then symbols for each of the following situations.

a. New York is known as "the city that never sleeps". A random sample of 25 New Yorkers were asked how much sleep they get per night. Do these data provide convincing evidence that New Yorkers on average sleep less than 8 hours a night?

b. Employers at a firm are worried about the effect of March Madness, a basketball championship held each spring in the US, on employee productivity. They estimate that on a regular business day employees spend on average 15 minutes of company time checking personal email, making personal phone calls, etc. They also collect data on how much company time employees spend on such non- business activities during March Madness. They want to determine if these data provide convincing evidence that employee productivity decreases during March Madness.

4. **Identify hypotheses, II.** Write the null and alternative hypotheses in words and using symbols for each of the following situations.

a. Since 2008, chain restaurants in California have been required to display calorie counts of each menu item. Prior to menus displaying calorie counts, the average calorie intake of diners at a restaurant was 1100 calories. After calorie counts started to be displayed on menus, a nutritionist collected data on the number of calories consumed at this restaurant from a random sample of diners. Do these data provide convincing evidence of a difference in the average calorie intake of a diners at this restaurant?

b. Based on the performance of those who took the GRE exam between July 1, 2004 and June 30, 2007, the average Verbal Reasoning score was calculated to be 462. In 2021 the average verbal score was slightly higher. Do these data provide convincing evidence that the average GRE Verbal Reasoning score has changed since 2021?

5. **Side effects of Avandia.** Rosiglitazone is the active ingredient in the controversial type 2 diabetes medicine Avandia and has been linked to an increased risk of serious cardiovascular problems such as stroke, heart failure, and death. A common alternative treatment is Pioglitazone, the active ingredient in a diabetes medicine called Actos. In a nationwide retrospective observational study of 227,571 Medicare beneficiaries aged 65 years or older, it was found that 2,593 of the 67,593 patients using Rosiglitazone and 5,386 of the 159,978 using Pioglitazone had serious cardiovascular problems. These data are summarized in the contingency table below. (Graham et al. 2010)

Treatment	No	Yes	Total
Pioglitazone	154,592	5,386	159,978
Rosiglitazone	65,000	2,593	67,593
Total	219,592	7,979	227,571

See the next page for the questions.

a. Determine if each of the following statements is true or false. If false, explain why. *Be careful:* The reasoning may be wrong even if the statement's conclusion is correct. In such cases, the statement should be considered false.

   i. Since more patients on Pioglitazone had cardiovascular problems (5,386 vs. 2,593), we can conclude that the rate of cardiovascular problems for those on a Pioglitazone treatment is higher.

   ii. The data suggest that diabetic patients who are taking Rosiglitazone are more likely to have cardiovascular problems since the rate of incidence was (2,593 / 67,593 = 0.038) 3.8% for patients on this treatment, while it was only (5,386 / 159,978 = 0.034) 3.4% for patients on Pioglitazone.

   iii. The fact that the rate of incidence is higher for the Rosiglitazone group proves that Rosiglitazone causes serious cardiovascular problems.

   iv. Based on the information provided so far, we cannot tell if the difference between the rates of incidences is due to a relationship between the two variables or due to chance.

b. What proportion of all patients had cardiovascular problems?

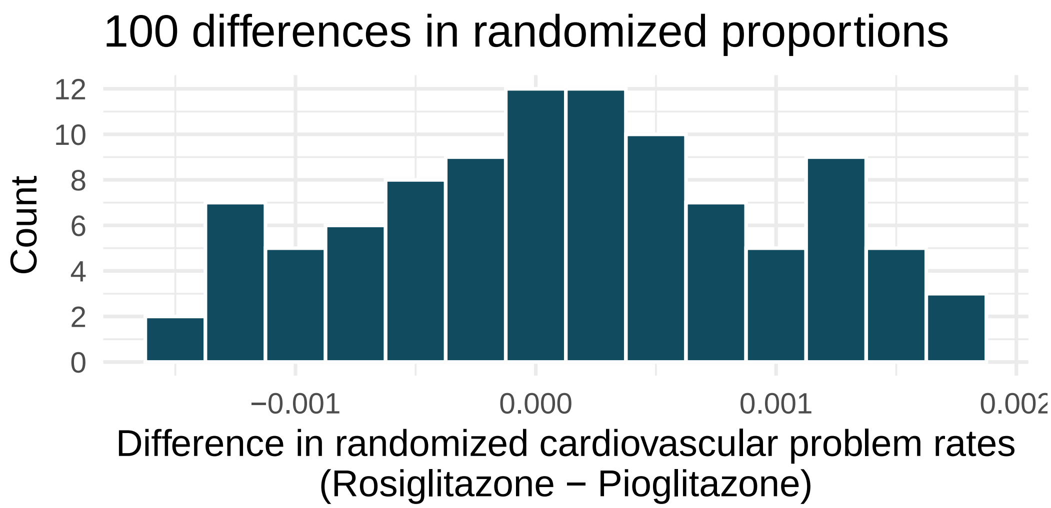
c. If the type of treatment and having cardiovascular problems were independent, how many patients in the Rosiglitazone group would we expect to have had cardiovascular problems?

d. We can investigate the relationship between outcome and treatment in this study using a randomization technique. While in reality we would carry out the simulations required for randomization using statistical software, suppose we actually simulate using index cards. In order to simulate from the independence model, which states that the outcomes were independent of the treatment, we write whether each patient had a cardiovascular problem on cards, shuffled all the cards together, then deal them into two groups of size 67,593 and 159,978. We repeat this simulation 100 times and each time record the difference between the proportions of cards that say "Yes" in the Rosiglitazone and Pioglitazone groups. Use the histogram of these differences in proportions to answer the following questions.

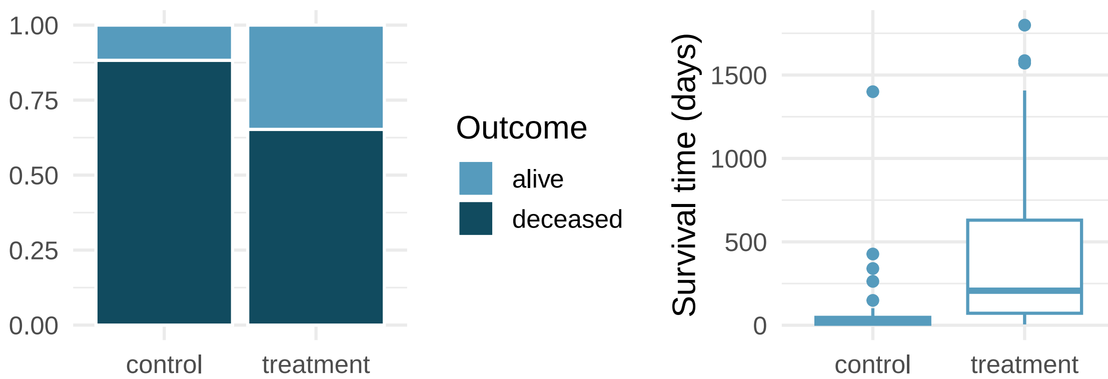
   i. What are the claims being tested?

   ii. Compared to the number calculated in part (b), which would provide more support for the alternative hypothesis, *higher* or *lower* proportion of patients with cardiovascular problems in the Rosiglitazone group?

   iii. What do the simulation results suggest about the relationship between taking Rosiglitazone and having cardiovascular problems in diabetic patients?


100 differences in randomized proportions
Count
12
10
8
6
4
2
0
−0.001
0.000
0.001
0.002
Difference in randomized cardiovascular problem rates
(Rosiglitazone − Pioglitazone)

6. **Heart transplants.** The Stanford University Heart Transplant Study was conducted to determine whether an experimental heart transplant program increased lifespan. Each patient entering the program was designated an official heart transplant candidate, meaning that they were gravely ill and would most likely benefit from a new heart. Some patients got a transplant and some did not. The variable `transplant` indicates which group the patients were in; patients in the treatment group got a transplant and those in the control group did not. Of the 34 patients in the control group, 30 died. Of the 69 people in the treatment group, 45 died. Another variable called `survived` was used to indicate whether the patient was alive at the end of the study. (Turnbull, Brown, and Hu 1974)



a. Does the stacked bar plot indicate that survival is independent of whether the patient got a transplant? Explain your reasoning.

b. What do the box plots suggest about the efficacy of heart transplants.

c. What proportions of patients in the treatment and control groups died?

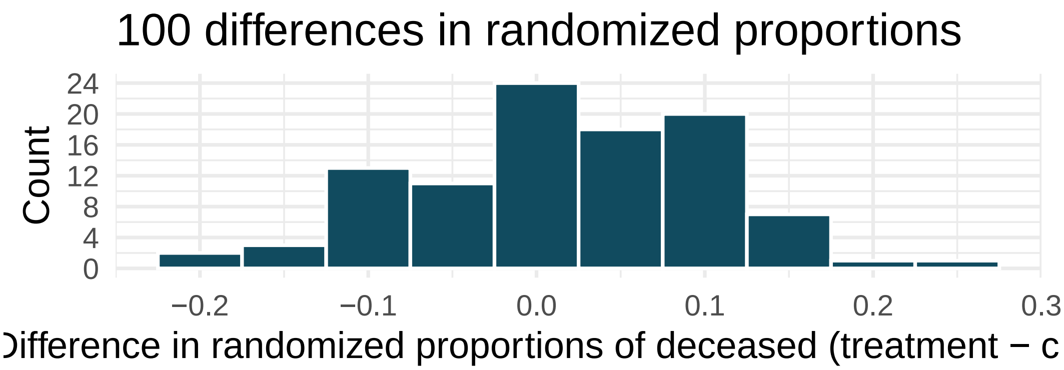
d. One approach for investigating whether the treatment is discernably effective is randomization testing.

   i. What are the claims being tested?

   ii. The paragraph below describes the set up for a randomization test, if we were to do it without using statistical software. Fill in the blanks with a number or phrase.

   We write *alive* on _______ cards representing patients who were alive at the end of the study, and *deceased* on _______ cards representing patients who were not. Then, we shuffle these cards and split them into two groups: one group of size _______ representing treatment, and another group of size _______ representing control. We calculate the difference between the proportion of *deceased* cards in the treatment and control groups (treatment - control) and record this value. We repeat this 100 times to build a distribution centered at _______. Lastly, we calculate the proportion of simulations where the simulated differences in proportions are _______. If this proportion is low, we conclude that it is unlikely to have observed such an outcome by chance and that the null hypothesis should be rejected in favor of the alternative.

   iii. What do the simulation results shown below suggest about the effectiveness of heart transplants?



Datasets: `avandia` (openintro) | `heart_transplant` (openintro)

Chapter 8

Bootstrap Confidence Intervals

In Chapter 6, we saw that sample statistics vary from sample to sample. This variability is not a nuisance — it's the key to quantifying uncertainty. In this chapter, we learn **bootstrapping**: a simulation-based method for constructing confidence intervals that requires no formulas and no assumptions about the shape of the population.

8.1 The big idea: resampling

In 2011, researchers collected a sample of 648 pennies in circulation and recorded how old each penny was (current year minus year minted). The question: what is the average age of *all* pennies currently in circulation in the United States?

The sample mean is $\bar{x} = 10.4$ years. This is our **point estimate** of the population mean μ. But how precise is this estimate? If we collected a different sample of 648 pennies, we'd get a different $\bar{x}$. We can't keep collecting new samples — we only have this one.

The bootstrap idea: Treat your sample as a stand-in for the population and resample *from it*. Each **bootstrap resample** is drawn with replacement from your original sample, at the same sample size n.

A **bootstrap resample** is a sample of size n drawn **with replacement** from the original sample. The statistic computed from a bootstrap resample is called a **bootstrap statistic**. The collection of many bootstrap statistics forms the **bootstrap distribution**.

Why does this work? If the original sample is representative of the population, then resampling from it mimics what would happen if we could repeatedly sample from the population itself. The variability we see in the bootstrap distribution approximates the true sampling variability of our statistic.

8.2 Bootstrapping a sample mean

Let's build a bootstrap confidence interval step by step.

8.2.1 Step 1: Collect data

Our sample of 648 pennies has a right-skewed distribution — many young pennies and a long tail of older ones:

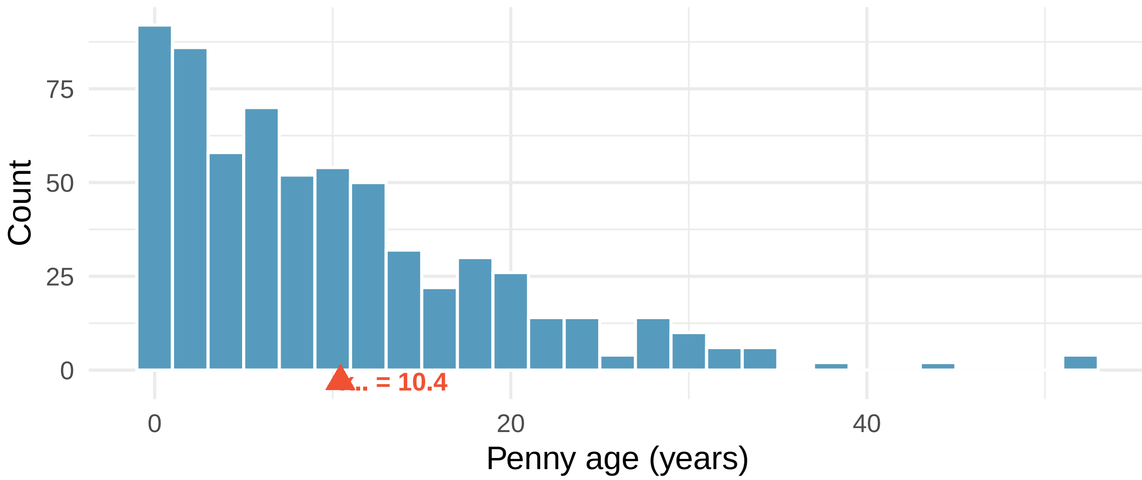


Figure 8.1: Distribution of ages for 648 pennies sampled from circulation. The distribution is right-skewed with a mean of about 10.4 years. The sample mean is marked with a red triangle. Open in StatLens

The sample mean is $\bar{x} = 10.4$ years. Notice the distribution is clearly not bell-shaped — it's skewed right. Bootstrapping doesn't care. It works regardless of the population shape.

8.2.2 Step 2: Resample with replacement

We draw a new sample of 648 values from the original 648, *with replacement*. Some pennies will appear more than once; others will be left out. Each such draw is one **bootstrap resample**, and the mean of that resample is a **bootstrap statistic** $\bar{x}^*$.

8.2.3 Step 3: Repeat many times

We repeat Step 2 a large number of times — typically $B = 1{,}000$ or more — and record the bootstrap mean $\bar{x}^*$ each time. This gives us a **bootstrap distribution**.

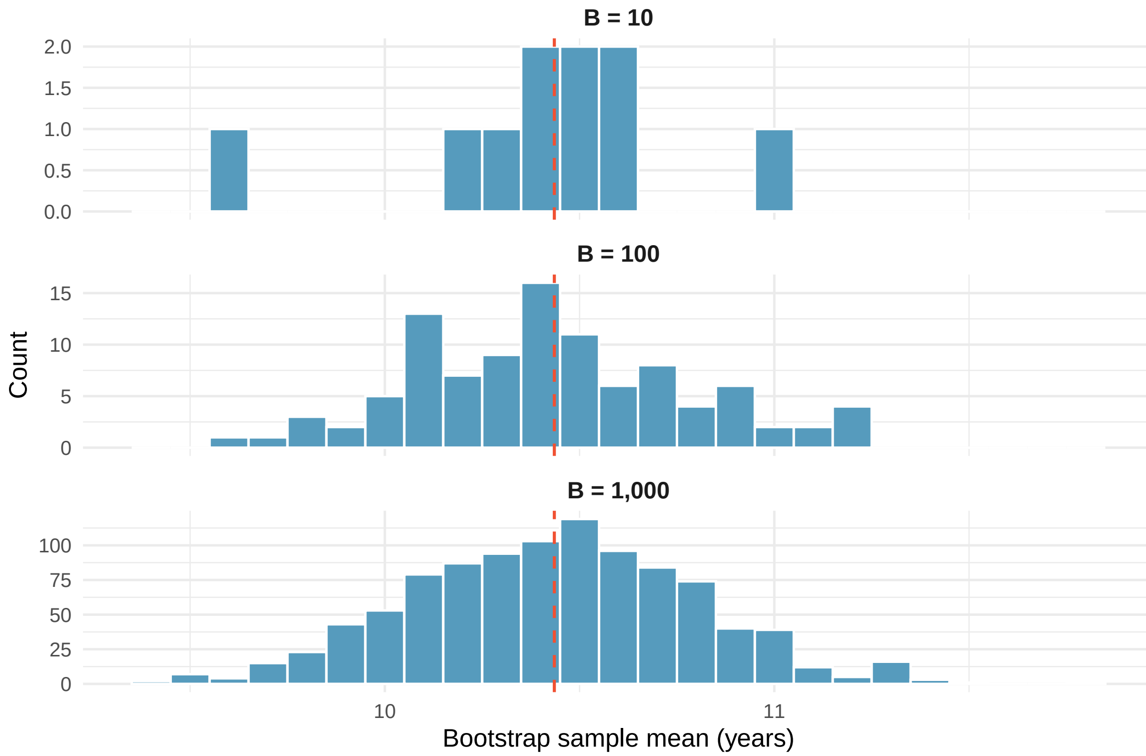


Figure 8.2: Building up the bootstrap distribution. With 10 resamples we see only a rough sketch; by 1,000 resamples the shape of the distribution is clear — and approximately bell-shaped (normal), even though the original data are skewed.

Notice two things: (1) the bootstrap distribution is centered near the original sample mean (dashed red line), and (2) even though the penny ages are right-skewed, the bootstrap distribution of the *mean* is approximately bell-shaped (what statisticians call a **normal** shape — recall this term from the empirical rule in Section 4.7). This is the Central Limit Theorem at work — we'll explore it more in the next chapter.

See it in action. Open the Bootstrap Resampling Activity to explore how bootstrap confidence intervals work step by step. The activity panel will guide you through generating resamples, watching the distribution build, and reading the confidence interval.

8.2.4 Step 4: Find the confidence interval

For a 95% confidence interval, we find the middle 95% of the bootstrap distribution by taking the **2.5th and 97.5th percentiles**.

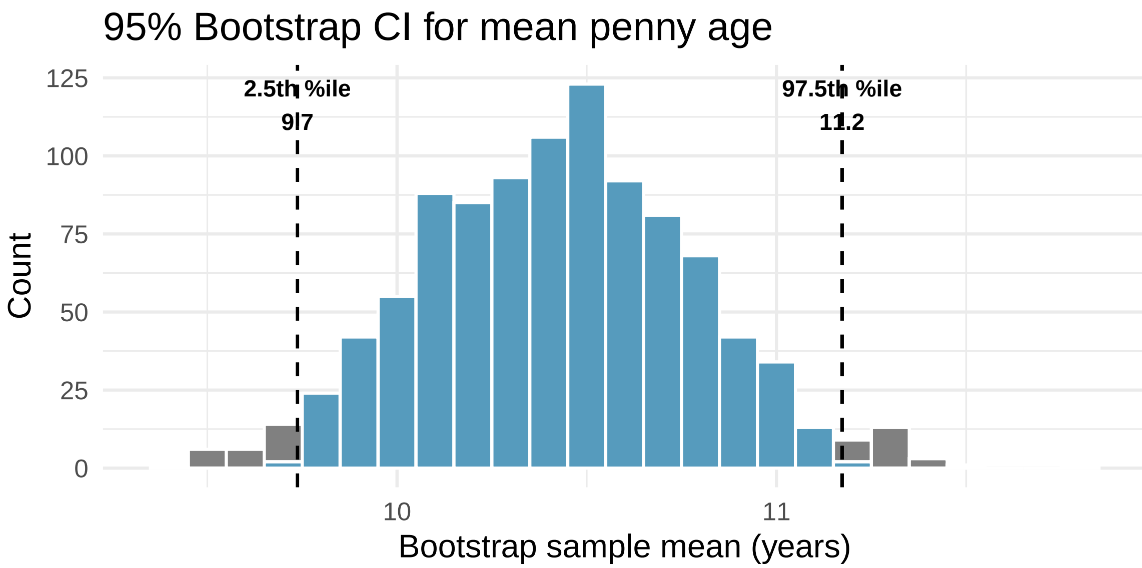


Figure 8.3: Bootstrap distribution of 1,000 sample means with the middle 95% shaded in blue. The 2.5th and 97.5th percentiles form the 95% confidence interval for the mean penny age.

The 95% bootstrap confidence interval for the mean penny age is approximately $(9.7, 11.2)$ years.

8.3 Interpreting a confidence interval

A **confidence interval** is a range of plausible values for the population parameter. A **95% confidence interval** is constructed using a procedure that, when applied repeatedly to many different samples, captures the true parameter in approximately 95% of all intervals produced.

What "95% confident" means: If we repeated the entire study many times — collecting a new sample each time and computing a new 95% confidence interval — approximately 95% of those intervals would contain the true population parameter. Any single interval either contains the parameter or it doesn't; the 95% refers to the long-run success rate of the *procedure*.

To illustrate the coverage concept, imagine we could treat our 648 pennies as if they were the entire population (with true mean $\mu \approx 10.4$ years). We then draw 25 smaller random samples of 50 pennies each and build a bootstrap CI from each one. This lets us check whether each interval captures the "true" mean — something we can never do with real data, because we don't know the true parameter.

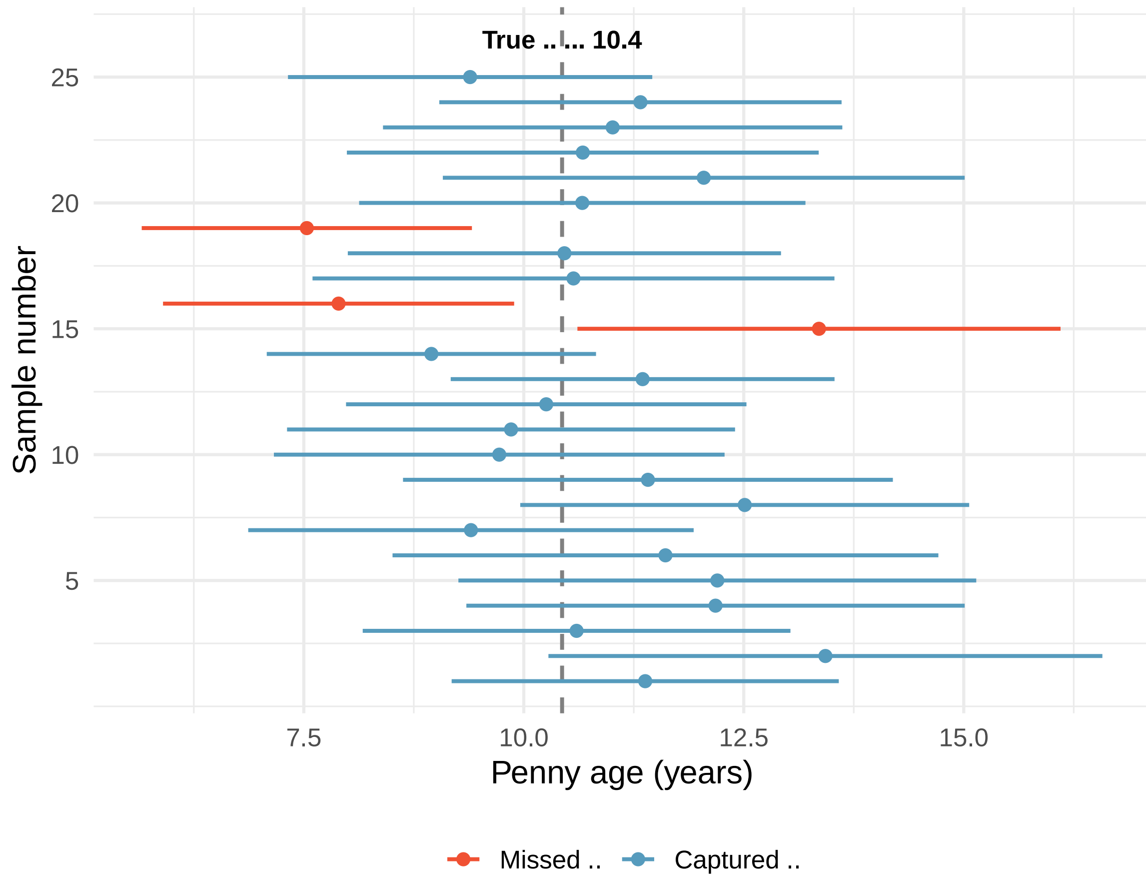


Figure 8.4: Twenty-five 95% confidence intervals from 25 different random samples of pennies. The true population mean is shown as a vertical dashed line. Approximately 95% of intervals capture the true mean; a few (shown in red) miss it. Open in StatLens

Each horizontal line in Figure 8.4 represents a 95% CI from a different sample. Most intervals (blue) capture the true mean, but a few (red) miss it entirely. Over many samples, roughly 95% of the intervals succeed. This is the meaning of "95% confidence."

Try it interactively. Open the CI Coverage demo and draw many confidence intervals from samples of 50 pennies. Watch how most intervals (blue) capture the true mean while a few (red) miss it. As you draw more samples, the coverage rate settles near 95%.

Common misinterpretation: "There is a 95% probability that the true mean is in this interval." This statement treats the population mean as random and the interval as fixed. In fact, the population mean is fixed (we just don't know it) and the interval is random (it depends on which sample we happened to collect).

A 95% bootstrap CI for the mean age of pennies in circulation is (9.7, 11.1). Which of the following interpretations is correct?

a. 95% of all pennies in circulation have ages between 9.7 and 11.1 years.
b. There is a 95% probability that the true mean penny age is between 9.7 and 11.1.
c. We are 95% confident that the true mean age of all pennies in circulation is between 9.7 and 11.1 years.

Show answer

Only (c) is correct. Statement (a) describes individual penny ages, not the mean. Statement (b) assigns a probability to the fixed parameter, which is incorrect. Statement (c) correctly describes confidence in the procedure.

8.4 The percentile method

The approach we've used is called the **percentile method** for bootstrap confidence intervals.

For a $(1-\alpha)\times 100\%$ confidence interval using the percentile method:

- **Lower bound** = the $(\alpha/2)\times 100$th percentile of the bootstrap distribution
- **Upper bound** = the $(1-\alpha/2)\times 100$th percentile of the bootstrap distribution

For a 95% CI ($\alpha = 0.05$): use the 2.5th and 97.5th percentiles. For a 90% CI ($\alpha = 0.10$): use the 5th and 95th percentiles. For a 99% CI ($\alpha = 0.01$): use the 0.5th and 99.5th percentiles.

If you increase the confidence level from 95% to 99%, what happens to the width of the confidence interval? Why?

Show answer

The interval gets wider. A 99% CI must capture the true parameter more often than a 95% CI, so it must cast a wider net. The 0.5th percentile is farther from the center than the 2.5th percentile.

8.5 Medical consultant case study

People providing an organ for donation sometimes seek the help of a special medical consultant. These consultants assist the patient in all aspects of the surgery, with the goal of reducing the possibility of complications during the medical procedure and recovery.

One consultant tried to attract patients by noting the average complication rate for liver donor surgeries in the US is about 10%, but her clients have had only 3 complications in the 62 liver donor surgeries she has facilitated. She claims this is strong evidence that her work reduces complications.

We estimate the true complication rate p using the sample proportion: $\hat{p} = 3/62 = 0.048$ (4.8%).

Is it possible to assess the consultant's claim — that the *reduction* in complications is *due to her work* — using these data?

No. The claim is causal, but the data are observational. Patients who can afford a medical consultant may also afford better medical care overall, which could lead to a lower complication rate. We cannot establish causation here. But we *can* estimate the consultant's true complication rate using a confidence interval.

To build a bootstrap CI for a proportion, we treat the 62 patients as marbles in a bag: 3 red (complication) and 59 white (no complication). We draw 62 marbles with replacement, compute $\hat{p}^*_{boot}$, and repeat 10,000 times.

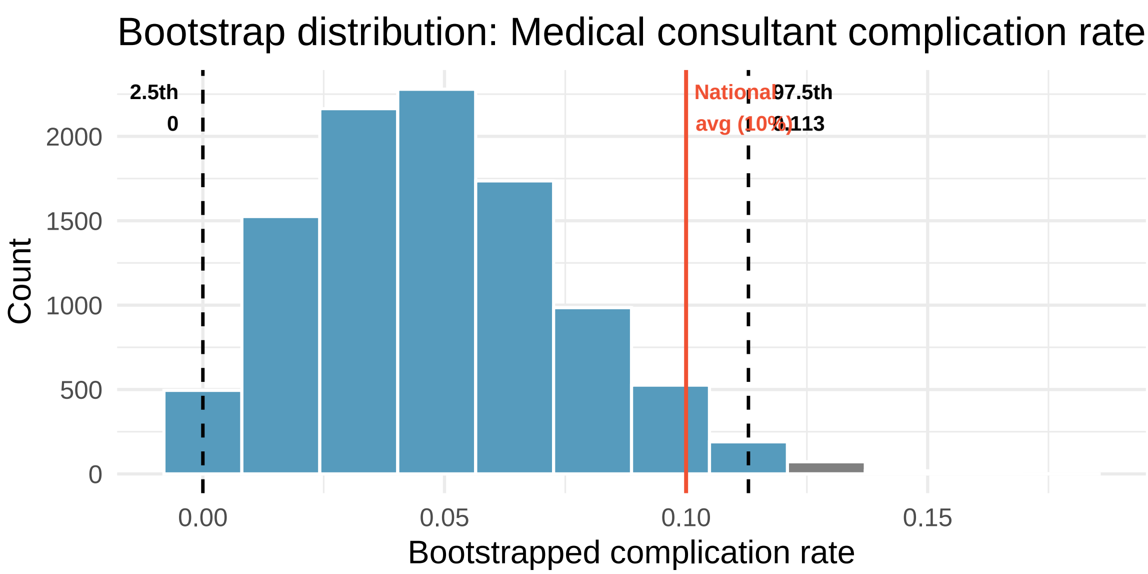


Figure 8.5: Bootstrap distribution of 10,000 resampled complication rates. The 95% CI extends from the 2.5th percentile to the 97.5th percentile. Because the interval includes 10%, we cannot conclude the consultant's rate is lower than the national average. Open in StatLens

The 95% bootstrap CI is approximately $(0, 0.113)$. The national average of 10% is right near the upper boundary of this interval — a borderline case. We certainly don't have strong evidence that the consultant's rate is lower than the national average. The sample is small $(n = 62)$ and the CI is wide, so complication rates anywhere from 0% up to around 10% are plausible.

Because there are only 62 patients and each is either "complication" or "no complication," the bootstrap proportions can only take values that are multiples of $1/62$ $(0, 1/62, 2/62, \ldots)$. This **discreteness** means the CI boundary can only land on one of these values — it can't fall smoothly between them. You may notice the CI upper bound jump between $6/62 \approx 0.097$ and $7/62 \approx 0.113$ depending on how many resamples you generate. This is normal with small-sample proportion data and is one reason why borderline conclusions should be stated cautiously. It also means the interval's actual confidence level may differ from the nominal 95% — here, the true coverage is closer to 99% because the discrete boundaries force the interval to be wider than intended.

Replicate this analysis. Open the Medical Consultant Activity to bootstrap the complication rate yourself and see whether the 95% CI includes the national average of 10%.

Suppose the consultant had facilitated 620 surgeries (instead of 62) with the same complication rate of 4.8% (about 30 complications). How would the bootstrap CI change? Would the conclusion change?

Show answer

With 10 times as much data, the bootstrap distribution would be much narrower. The 95% CI would likely fall entirely below 10%, providing strong evidence that the consultant's rate is lower. Larger samples give more precise estimates.

8.6 Tappers and listeners case study

A Stanford University graduate student named Elizabeth Newton conducted an experiment using a simple game: one person (the "tapper") taps out a well-known song on a desk, and another person (the "listener") tries to guess the song. About 50% of tappers expected the listener would guess correctly.

In Newton's study, only 3 out of 120 listeners ($\hat{p} = 3/120 = 0.025$) guessed the tune correctly. What is the true proportion of listeners who can guess the song?

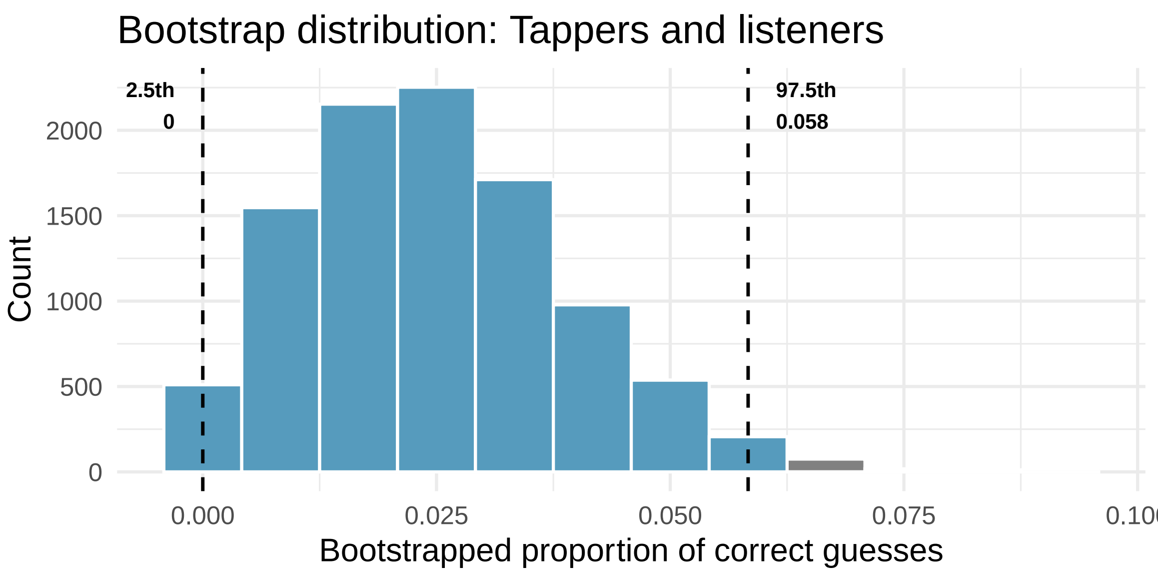


Figure 8.6: Bootstrap distribution for the proportion of listeners who guessed correctly. The 95% CI is far below the tappers' expected success rate of 50%, providing very convincing evidence against the tappers' expectation.

The 95% bootstrap CI is approximately $(0, 0.058)$. The tappers' expectation of 50% is nowhere near this interval — it provides **very convincing evidence** that listeners are far worse at guessing than tappers expect. This is an example of the "curse of knowledge": once you know something, it's hard to imagine not knowing it.

Do the data provide convincing evidence against the claim that 50% of listeners can guess the tapper's tune?

Show answer

Yes, very convincing evidence. The value 50% (0.50) is far outside the 95% CI, and even far from the largest bootstrapped proportion. The data strongly suggest that the true success rate is much lower than 50%.

8.7 What affects the width of a CI?

Three factors control how wide a confidence interval is:

1. **Sample size** (n): Larger samples → narrower intervals (more information about the population)
2. **Variability in the data**: More spread → wider intervals (less precision)
3. **Confidence level**: Higher confidence → wider intervals (more certainty requires a wider net)

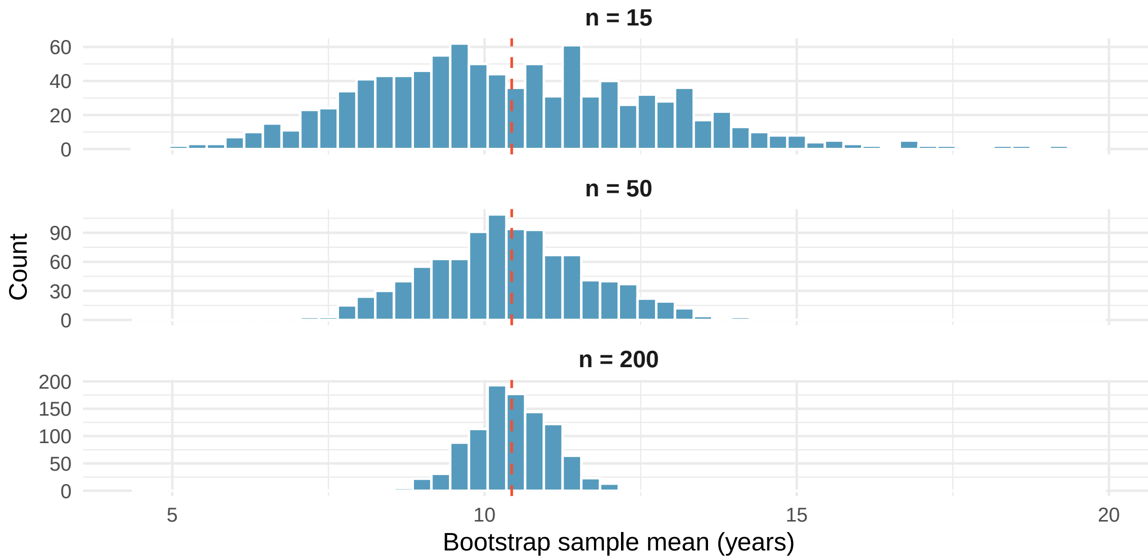


Figure 8.7: The effect of sample size on the bootstrap distribution. Larger samples produce narrower bootstrap distributions, leading to narrower confidence intervals. All three panels use the same population of penny ages.

As sample size increases, the bootstrap distribution becomes tighter around the sample mean. This is why large studies produce more precise estimates than small ones.

Explore the trade-off. Open the Confidence Level Activity to see firsthand how changing the confidence level from 90% to 95% to 99% affects the width of the interval — and why there's always a trade-off between confidence and precision.

8.8 When does bootstrapping work?

Bootstrapping is remarkably flexible, but it does require some conditions to produce reliable intervals:

1. **Independence**: The observations in the sample should be independent of one another. This is usually satisfied when data come from a random sample or a randomized experiment.
2. **Representative sample**: The sample should be reasonably representative of the population. If the sample is biased (e.g., only volunteers), the bootstrap CI will reflect that bias.
3. **Sufficient sample size**: Very small samples (e.g., $n < 10$) may not contain enough information about the population's shape. With small n, the bootstrap distribution can be unreliable.

Bootstrapping does **not** require the population to follow a bell-shaped (normal) distribution. This is one of its greatest strengths compared to traditional formula-based methods, which often do require normality. The penny ages data are clearly right-skewed (Figure 8.1), yet the bootstrap procedure works perfectly well. The bootstrap lets the data "speak for themselves."

A researcher surveys 8 of her friends about their study hours and computes a bootstrap CI. What condition is most clearly violated?

Show answer

Two conditions are violated: (1) the sample is a convenience sample of friends, not a random sample, so it may not be representative of the population, and (2) $n = 8$ is quite small, making the bootstrap distribution potentially unreliable.

8.9 Bootstrap CI for other statistics

The beauty of bootstrapping is its **generality**. The same resampling logic works for *any* statistic, not just the mean:

- **Median**: Resample → compute median → repeat → percentile CI
- **Proportion**: Resample → compute proportion → repeat → percentile CI
- **Standard deviation**: Resample → compute SD → repeat → percentile CI
- **Difference in means**: Resample from each group → compute difference → repeat → percentile CI

This is a powerful advantage. Many statistics (like the median or the difference in medians) don't have convenient formulas for their standard error. Bootstrapping handles them all the same way.

Try different statistics. Open the Bootstrap CI tool with penny ages and use the statistic dropdown to switch between the mean, median, standard deviation, Q1, and Q3. Generate 1,000 resamples for each and compare the bootstrap distributions and confidence intervals. Which statistics produce wider intervals? Which produce narrower ones?

8.10 Chapter review

8.10.1 Summary

- The **bootstrap** constructs a confidence interval by resampling with replacement from the original sample.
- The **bootstrap distribution** approximates the sampling distribution of the statistic.
- The **percentile method** uses the middle $(1 - \alpha) \times 100\%$ of the bootstrap distribution as the CI.
- A **95% confidence interval** means that if we repeated the study many times, about 95% of the resulting intervals would capture the true parameter.
- CI width depends on sample size, data variability, and confidence level.
- Bootstrap CIs work for any statistic — means, medians, proportions, differences, and more.
- Bootstrapping requires independent observations from a representative sample with sufficient sample size, but does **not** require the population to be bell-shaped (normal).

8.10.2 Key terms

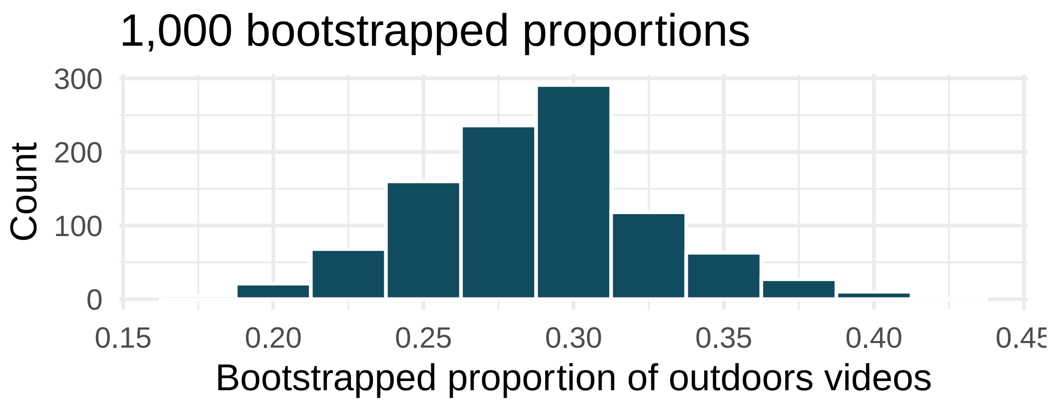
- Bootstrap resample
- Bootstrap statistic
- Bootstrap distribution
- Confidence interval
- Confidence level
- Percentile method
- Point estimate
- Parameter
- Sampling with replacement

8.11 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Outside YouTube videos.** Let's say that you want to estimate the proportion of YouTube videos which take place outside (define "outside" to be if any part of the video takes place outdoors). You take a random sample of 128 YouTube videos and determine that 37 of them take place

outside. You'd like to estimate the proportion of all YouTube videos which take place outside, so you decide to create a bootstrap interval from the original sample of 128 videos.



a. Describe in words the relevant statistic and parameter for this problem. If you know the numerical value for either one, provide it. If you don't know the numerical value, explain why the value is unknown.

b. What notation is used to describe, respectively, the statistic and the parameter?

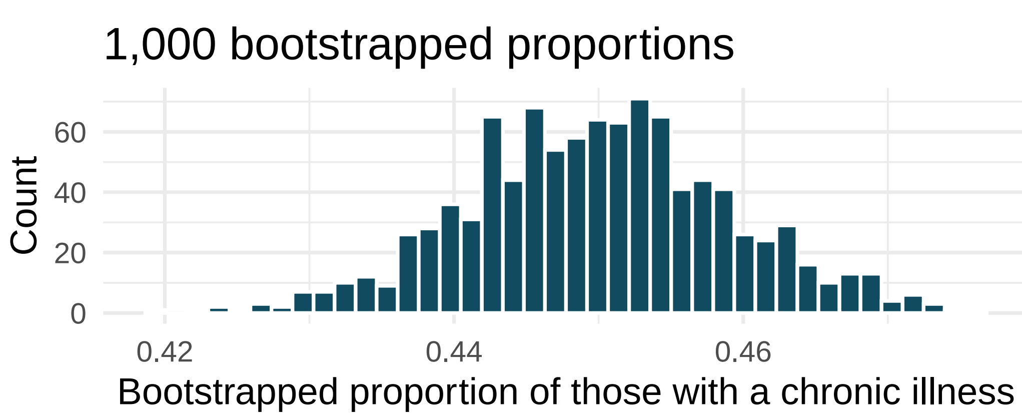
c. If using software to bootstrap the original dataset, what is the statistic calculated on each bootstrap sample?

d. When creating a bootstrap sampling distribution (histogram) of the bootstrapped sample proportions, where should the center of the histogram lie?

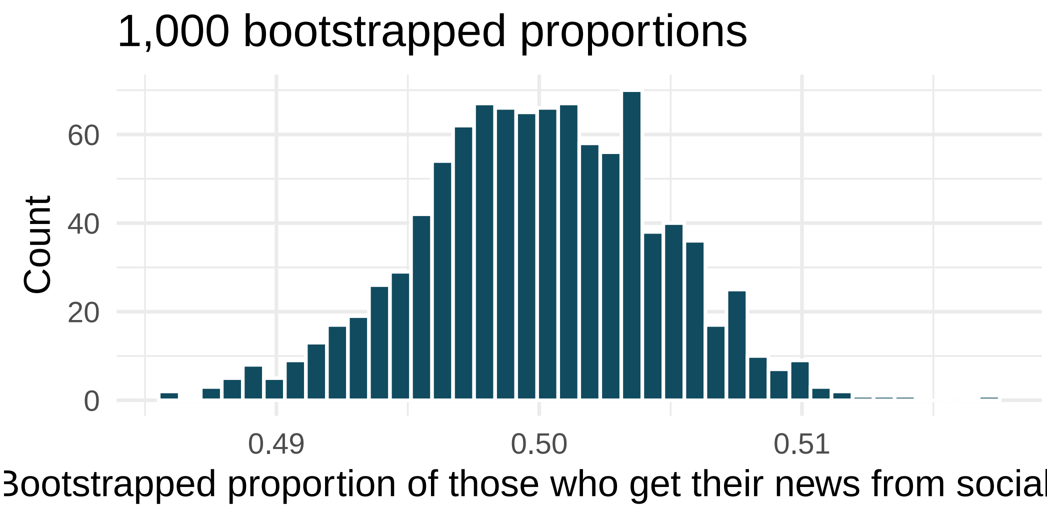
e. The histogram provides a bootstrap sampling distribution for the sample proportion (with 1000 bootstrap repetitions). Using the histogram, estimate a 90% confidence interval for the proportion of YouTube videos which take place outdoors.

f. Interpret the confidence interval in context of the data.

2. **Chronic illness.** In 2012 the Pew Research Foundation reported that "45% of US adults report that they live with one or more chronic conditions." However, this value was based on a sample, so it may not be a perfect estimate for the population parameter of interest on its own. The study was based on a sample of 3014 adults. Below is a distribution of 1000 bootstrapped sample proportions from the Pew dataset. (Pew Research Center 2013) Using the distribution of 1,000 bootstrapped proportions, approximate a 92% confidence interval for the true proportion of US adults who live with one or more chronic conditions and interpret it.



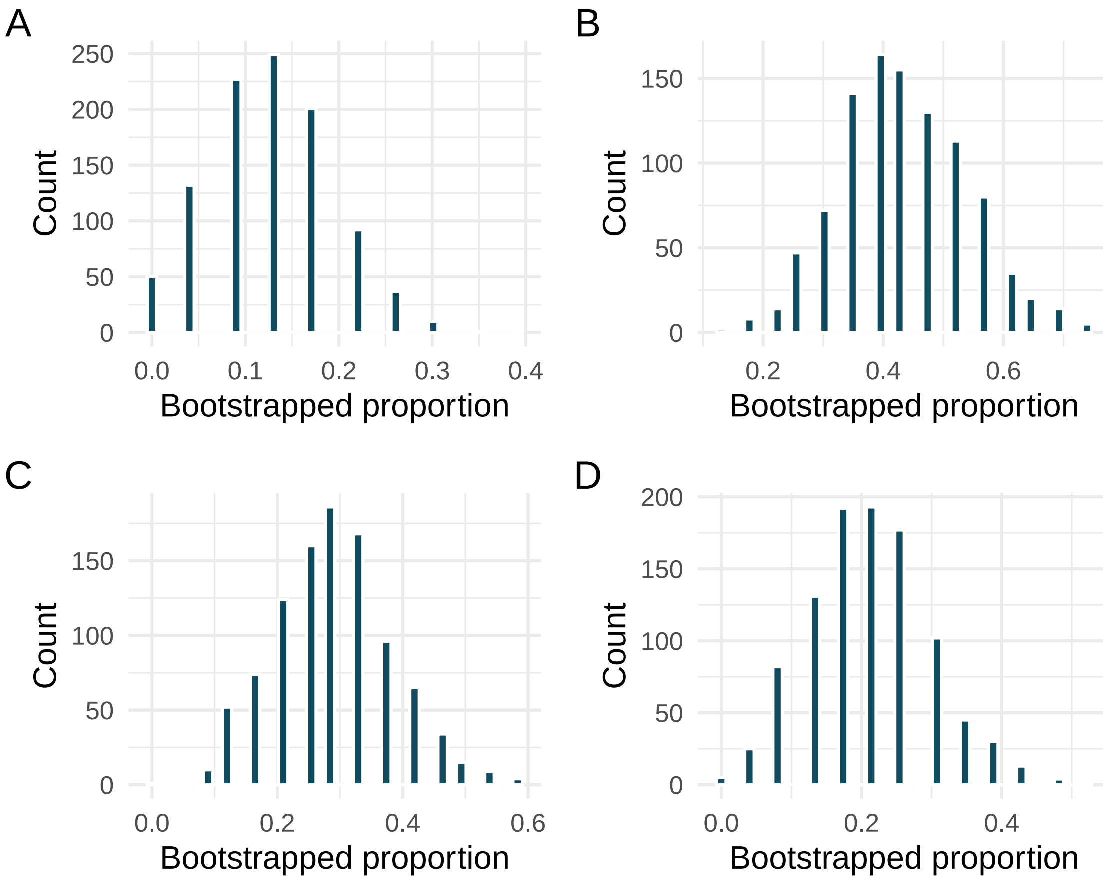
3. **Social media users and news, bootstrapping.** A poll conducted in 2022 found that 50% of U.S. adults get news from social media sometimes or often. However, the value was based on a sample, so it may not be a perfect estimate for the population parameter of interest on its own. The study was based on a sample of 12,147 adults. Below is a distribution of 1,000 bootstrapped sample proportions from the Pew dataset. (Pew Research Center 2022) Using the distribution of 1,000 bootstrapped proportions, approximate a 98% confidence interval for the true proportion of US adult social media users (in 2022) who get at least some of their news from Twitter. Interpret the interval in the context of the problem.



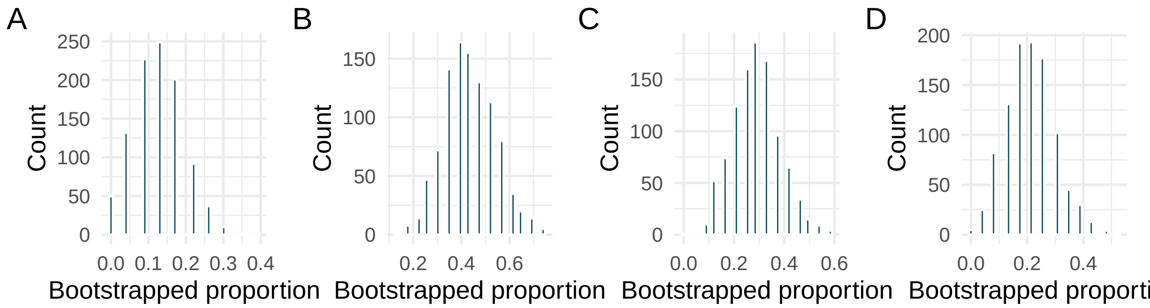
4. **Bootstrap distributions of $\hat{p}$, I.** Each of the following four distributions was created using a different dataset. Each dataset was based on $n = 23$ observations. The original datasets had the following proportions of successes:

$$\hat{p} = 0.13 \;\; \hat{p} = 0.22 \;\; \hat{p} = 0.30 \;\; \hat{p} = 0.43.$$

Match each histogram with the original data proportion of success.


A
B
C
D
Count
Bootstrapped proportion
0
50
100
150
200
250
0.0
0.1
0.2
0.3
0.4
0.6

5. **Bootstrap distributions of $\hat{p}$, II.** Each of the following four distributions was created using a different dataset. Each dataset was based on $n = 23$ observations.



Consider each of the following values for the true popluation p (proportion of success). Datasets A, B, C, D were bootstrapped 1000 times, with bootstrap proportions as given in the histograms provided. For each parameter value, list the datasets which could plausibly have come from that population. (Hint: there may be more than one dataset for each parameter value.)

a. $p = 0.05$

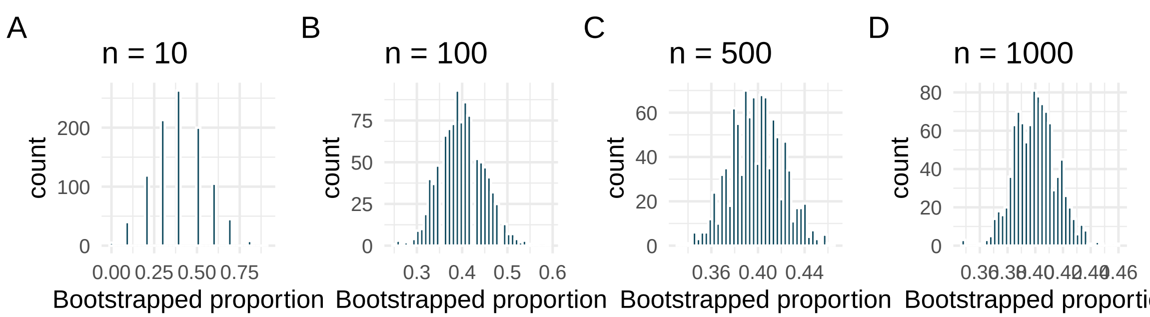
b. $p = 0.25$

c. $p = 0.45$

d. $p = 0.55$

e. $p = 0.75$

6. **Bootstrap distributions of $\hat{p}$, III.** Each of the following four distributions was created using a different dataset. Each dataset had the same proportion of successes ($\hat{p} = 0.4$) but a different sample size. The four datasets were given by $n = 10, 100, 500$, and 1000.



Consider each of the following values for the true popluation p (proportion of success). Datasets A, B, C, D were bootstrapped 1000 times, with bootstrap proportions as given in the histograms provided. For each parameter value, list the datasets which could plausibly have come from that population. (Hint: there may be more than one dataset for each parameter value.)

a. $p = 0.05$

b. $p = 0.25$

c. $p = 0.45$

d. $p = 0.55$

e. $p = 0.75$

7. **Cyberbullying rates.** Teens were surveyed about cyberbullying, and 54% to 64% reported experiencing cyberbullying (95% confidence interval). Answer the following questions based on this interval. (Pew Research Center 2018)

a. A newspaper claims that a majority of teens have experienced cyberbullying. Is this claim supported by the confidence interval? Explain your reasoning.

b. A researcher conjectured that 70% of teens have experienced cyberbullying. Is this claim supported by the confidence interval? Explain your reasoning.

c. Without actually calculating the interval, determine if the claim of the researcher from part (b) would be supported based on a 90% confidence interval?

8. **Waiting at an ER.** A 95% confidence interval for the mean waiting time at an emergency room (ER) of (128 minutes, 147 minutes). Answer the following questions based on this interval.

a. A local newspaper claims that the average waiting time at this ER exceeds 3 hours. Is this claim supported by the confidence interval? Explain your reasoning.

b. The Dean of Medicine at this hospital claims the average wait time is 2.2 hours. Is this claim supported by the confidence interval? Explain your reasoning.

c. Without actually calculating the interval, determine if the claim of the Dean from part (b) would be supported based on a 99% confidence interval?

Chapter 9

Sampling Distributions via Simulation

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In Chapters 7 and 8, we used simulation to answer two fundamental questions of inference: *Is there an effect?* (randomization tests) and *How big is the parameter?* (bootstrap confidence intervals). Both methods relied on the computer to generate distributions — randomization distributions and bootstrap distributions — that helped us quantify uncertainty. In this chapter, we step back to see the bigger picture. We examine the shape, center, and spread of these simulated distributions and discover a remarkable pattern: under certain conditions, they all look approximately **normal** (bell-shaped). This observation, formalized as the **Central Limit Theorem**, is the bridge from the simulation-based methods of Part II to the formula-based methods of Part III.

9.1 From simulation to theory

In Chapters 7 and 8, we encountered several case studies:

- **Sex discrimination** (Chapter 7): We shuffled promotion decisions and computed the difference in promotion rates 10,000 times.
- **Opportunity cost** (Chapter 7): We shuffled buying decisions and computed the difference in proportions 1,000 times.
- **Medical consultant** (Chapter 8): We bootstrapped the complication rate from 62 surgeries 10,000 times.
- **Tappers and listeners** (Chapter 8): We bootstrapped the proportion of correct guesses from 120 listeners 10,000 times.

Each of these simulations produced a distribution of a sample statistic. While they differed in their settings, outcomes, and the simulation technique used, they all shared something in common: the general shape of the distribution.

Think back to the randomization and bootstrap distributions you have seen. What shape did most of them have? Were they skewed, uniform, or something else?

Show answer

In general, the distributions were approximately symmetric and bell-shaped. The medical consultant example was the only one with noticeable skew (skewed right), which makes sense because the proportion was very close to zero.

This is no coincidence. The bell shape keeps appearing because of a deep mathematical result that underlies most of statistics. Whether we shuffle labels, resample with replacement, or imagine taking repeated samples from a population, the distribution of the resulting statistic tends to look the same — symmetric, unimodal, and bell-shaped.

In the simulation chapters, we let the computer show us this pattern directly. Now, we formalize it.

9.2 The Central Limit Theorem

The **Central Limit Theorem (CLT)** states that when we take sufficiently large random samples from a population, the sampling distribution of many common statistics (such as $\hat{p}$ or $\bar{x}$) is approximately **normal** (bell-shaped), regardless of the shape of the underlying population distribution.

This is a stunning result. Even if the population distribution is skewed, bimodal, or has some other non-normal shape, the distribution of the *sample statistic* will be approximately normal — provided the sample is large enough.

Central Limit Theorem: the key ideas.

1. The sampling distribution of a sample statistic is approximately normal when the sample size is large enough.
2. The center of the sampling distribution is at the population parameter.
3. The spread of the sampling distribution (the standard error) decreases as sample size increases.

These three facts together mean: with a large enough sample, the sample statistic will be close to the population parameter, and we can use the normal distribution to describe *how* close.

9.2.1 Conditions for the CLT

The normal approximation does not apply in all situations. Two conditions must hold:

1. **Independent observations.** The observations in the sample must be independent of each other. This is typically guaranteed by random sampling from a population, or by random assignment in an experiment.

2. **Large enough sample.** The sample size cannot be too small. What counts as "large enough" depends on the context:

   - For proportions: we generally need at least 10 expected successes and 10 expected failures.
   - For means: the rule of thumb depends on how skewed the population is. For roughly symmetric populations, $n \geq 30$ is often sufficient. For highly skewed populations, larger samples may be needed.

When these conditions are not met, the normal approximation may be poor and simulation-based methods (randomization tests, bootstrap CIs) are the better choice. This is one of the great advantages of simulation: it works even when mathematical conditions are not satisfied.

9.2.2 Visualizing the CLT

Imagine a population that is distinctly non-normal — say, a right-skewed distribution like household income. If we take a single random sample of $n = 5$ people and compute the mean income, that mean might be quite far from the population mean μ. But if we do this thousands of times (each time taking a new sample of 5 and computing the mean), the distribution of those thousands of sample means will already start to look somewhat bell-shaped, even though the underlying population is skewed.

Now increase the sample size to $n = 30$. The distribution of sample means looks even more bell-shaped. At $n = 100$, it is nearly indistinguishable from a perfect normal curve.

This progression — from skewed population, to somewhat normal at small n, to very normal at larger n — is the Central Limit Theorem in action.

Sample size	Shape of sampling distribution
$n = 5$	Retains some skew from the population
$n = 30$	Approximately normal for most populations
$n = 100$	Very close to normal for virtually all populations

The speed at which the sampling distribution approaches normality depends on how non-normal the population is. Symmetric populations converge quickly; highly skewed populations take longer.

See the CLT in action! Use StatLens's Sampling Distribution Simulator:

Launch Sampling Distribution Simulator

1. Select a skewed population (e.g., exponential or right-skewed).
2. Set the sample size to $n = 5$ and draw 1,000 samples. Observe the shape.
3. Increase to $n = 30$ and draw 1,000 more. How does the shape change?
4. Try $n = 100$. How close is the distribution to a normal curve?

9.3 Comparing randomization, bootstrap, and mathematical approaches

We now have three tools for statistical inference, each grounded in the same core ideas but using different mechanisms:

	Randomization	Bootstrap	Mathematical model
Question answered	Is there an effect? (hypothesis test)	How big is the parameter? (confidence interval)	Both
How variability is generated	Shuffle labels to simulate H_0	Resample with replacement from sample	Use formulas based on the normal distribution
What it models	Variability due to random assignment	Variability due to random sampling	Variability predicted by mathematical theory
Center of distribution	Null hypothesis value	Sample statistic	Depends on context (H_0 for tests, $\hat{p}$ for CIs)
Conditions required	Minimal	Minimal	Independence + large sample
Computational cost	Moderate (need many simulations)	Moderate (need many resamples)	Low (just plug into formula)

Key insight. All three methods answer the same fundamental question: *How much does the statistic vary?* They just measure that variability in different ways.

- **Randomization** asks: "How much would the statistic vary if the null hypothesis were true?"
- **Bootstrap** asks: "How much would the statistic vary across different samples from this population?"
- **Mathematical models** ask: "How much does theory predict the statistic would vary?"

When conditions are met, all three give similar answers. The mathematical approach is faster and requires no simulation, which is why it is widely used. But the simulation approaches are more flexible and work even when conditions for the mathematical model are not met.

9.3.1 Which method should I use?

Here is a practical guide:

1. **When conditions for the mathematical model are clearly met** (independent observations, large sample, not too skewed), the formula-based approach is convenient and widely accepted. This will be our primary tool in Part III.
2. **When conditions are questionable** (small sample, very skewed data, unusual statistic), simulation-based methods are safer. The bootstrap and randomization test make fewer assumptions.
3. **When you want to build intuition**, simulation is invaluable. Seeing 10,000 shuffles or resamples gives you a concrete feel for what "random variability" looks like — something a formula alone cannot provide.
4. **In practice**, many statisticians use simulation to check their formula-based results. If the two approaches agree, confidence in the conclusion is strengthened.

9.4 When is the normal approximation good enough?

The mathematical approach to inference (which we will develop in Part III) replaces simulation with the normal distribution. Instead of generating thousands of shuffles or resamples, we use a formula to compute the standard error and then rely on the bell curve to find p-values or construct confidence intervals.

But when can we trust this shortcut?

9.4.1 The success-failure condition for proportions

For a sample proportion $\hat{p}$, the sampling distribution is approximately normal when:

$$np \geq 10 \quad \text{and} \quad n(1-p) \geq 10$$

where n is the sample size and p is the proportion of interest. In practice, since p is unknown, we use $\hat{p}$ as a stand-in:

$$n\hat{p} \geq 10 \quad \text{and} \quad n(1-\hat{p}) \geq 10$$

This is called the **success-failure condition**.

In the medical consultant example, $n = 62$ and $\hat{p} = 3/62 = 0.048$. Check the success-failure condition.

- Expected successes: $n\hat{p} = 62 \times 0.048 = 3$ (less than 10)
- Expected failures: $n(1-\hat{p}) = 62 \times 0.952 = 59$ (at least 10)

The success-failure condition is **not met** because there are fewer than 10 expected successes. The normal approximation would be unreliable here. This is exactly the kind of situation where the bootstrap (as used in Chapter 8) is the better approach.

In the tappers and listeners study, $n = 120$ and $\hat{p} = 3/120 = 0.025$. Check the success-failure condition. Should we use the normal approximation?

Show answer

Expected successes: $120 \times 0.025 = 3$ (less than 10). Expected failures: $120 \times 0.975 = 117$ (at least 10). The condition is not met because there are too few successes. The normal approximation would not be appropriate. Bootstrap methods are preferred.

A poll of $n = 1{,}000$ voters finds that $\hat{p} = 0.52$ support a policy. Check the success-failure condition.

- Expected successes: $n\hat{p} = 1{,}000 \times 0.52 = 520$ (well above 10)
- Expected failures: $n(1 - \hat{p}) = 1{,}000 \times 0.48 = 480$ (well above 10)

The success-failure condition is comfortably met. The normal approximation is reliable here.

9.4.2 The sample size condition for means

For a sample mean $\bar{x}$, the CLT tells us the sampling distribution is approximately normal when:

- The sample size is "large enough," and
- The observations are independent.

A common guideline is $n \geq 30$, but this is only a rough rule of thumb. If the population distribution is nearly symmetric, the normal approximation can work well even for smaller samples. If the population is highly skewed or has extreme outliers, a larger sample may be needed.

Population shape	Minimum n for normal approximation
Symmetric, no outliers	$n \geq 15$ or even smaller
Slightly skewed	$n \geq 30$
Highly skewed or heavy-tailed	$n \geq 60$ or more

When in doubt, use simulation (bootstrap) to check whether the normal approximation is reasonable.

9.4.3 Simulation as a check on theory

One of the most powerful uses of simulation is as a **reality check** on mathematical results. Here is the workflow:

1. Compute a confidence interval or p-value using the formula-based (normal) approach.
2. Compute the same quantity using simulation (bootstrap CI or randomization test).
3. Compare. If the two results are similar, you can be confident in the mathematical approach. If they differ substantially, the conditions for the normal approximation may not be met, and you should rely on the simulation result.

This "trust but verify" approach is standard practice among professional statisticians and data scientists.

9.5 The normal distribution: a preview

Since the normal distribution will be our primary tool in Part III, let us briefly introduce its key features here.

9.5.1 The 68-95-99.7 rule

The normal distribution has a very useful property: the percentage of observations falling within 1, 2, and 3 standard deviations of the mean is always approximately the same.

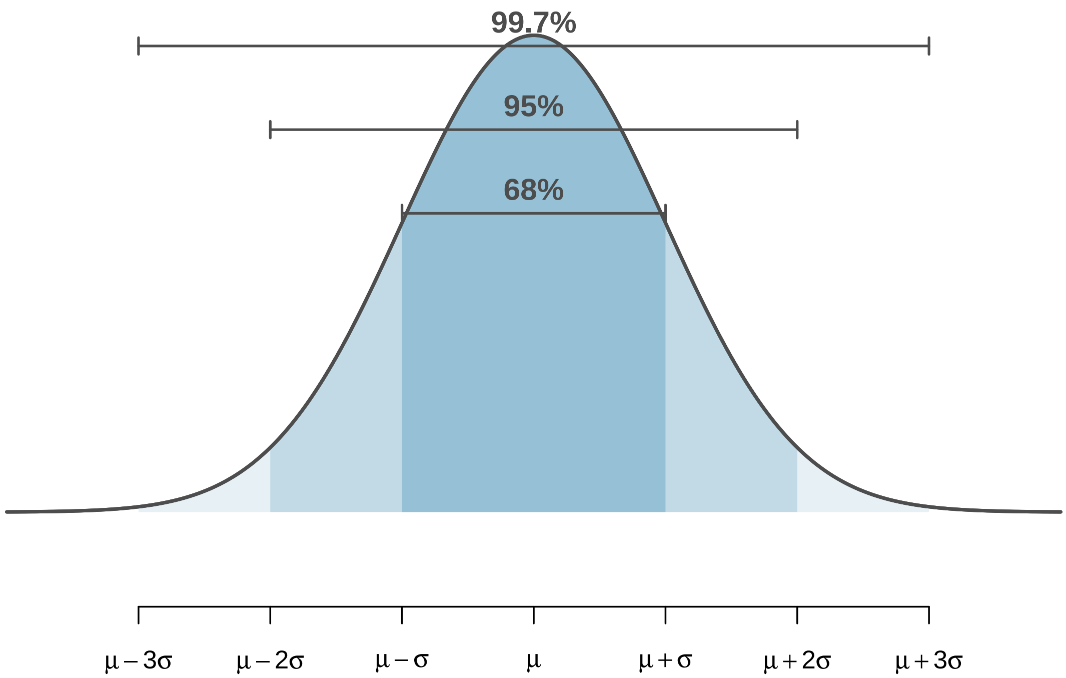


Figure 9.1: The 68-95-99.7 rule for normal distributions. Approximately 68% of values fall within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3.

The 68-95-99.7 rule.

For any normal distribution:

- About **68%** of values fall within 1 standard deviation of the mean: $\mu \pm \sigma$
- About **95%** of values fall within 2 standard deviations of the mean: $\mu \pm 2\sigma$
- About **99.7%** of values fall within 3 standard deviations of the mean: $\mu \pm 3\sigma$

This rule has an immediate connection to confidence intervals. In Chapter 8, we found that a 95% bootstrap confidence interval captures the middle 95% of the bootstrap distribution. If the bootstrap distribution is approximately normal, then the middle 95% corresponds to approximately ± 2 standard errors. This gives us the quick approximation:

$$95\% \text{ CI} \approx \text{statistic} \pm 2 \times SE$$

A poll of $n = 1{,}000$ voters finds $\hat{p} = 0.52$ with a standard error of $SE = 0.016$. Use the quick approximation to construct a 95% confidence interval.

$$\hat{p} \pm 2 \times SE = 0.52 \pm 2(0.016) = 0.52 \pm 0.032 = (0.488, 0.552)$$

We are 95% confident that the true proportion of voters who support the policy is between 48.8% and 55.2%.

Why is it "approximately 2" standard errors rather than exactly 2? What is the more precise multiplier for a 95% confidence interval based on the normal distribution?

Show answer

The more precise multiplier is 1.96, not 2. The value 1.96 is the Z-score that puts exactly 2.5% in each tail of the standard normal distribution, leaving 95% in the middle. The approximation of 2 is close enough for quick mental calculations.

9.5.2 Z scores

A **Z score** measures how many standard deviations an observation is above or below the mean:

$$Z = \frac{x - \mu}{\sigma}$$

In the context of sampling distributions, we can compute a Z score for an observed statistic:

$$Z = \frac{\text{statistic} - \text{parameter}}{SE}$$

This Z score tells us how unusual the observed statistic is, measured in standard-error units. In Part III, we will use Z scores (and their cousins, t-scores) extensively to compute p-values and confidence intervals without simulation.

In the sex discrimination study, the observed difference in promotion rates was 0.292. Suppose the standard error of the difference (computed under the null hypothesis) is 0.12. Compute the Z score.

Under H_0, the parameter (the true difference) is 0. So:

$$Z = \frac{0.292 - 0}{0.12} = 2.43$$

The observed difference is 2.43 standard errors above the null value of 0. Since values beyond 2 standard deviations are unusual for a normal distribution (the 95% rule), this is strong evidence against the null hypothesis — consistent with our earlier simulation result.

9.6 Recap: connecting the three approaches

Types of distributions (a reference guide).

You have now encountered several types of distributions in this course. Understanding the differences among them is essential:

- A **data distribution** (or sample distribution) describes the shape, center, and variability of the observed data.
- A **population distribution** describes the entire population — almost always unknown.
- A **sampling distribution** describes how a *sample statistic* varies from sample to sample. The CLT tells us this is approximately normal under certain conditions.
- A **randomization distribution** describes how a test statistic varies under random shuffling of the treatment variable. It is centered at the null hypothesis value.
- A **bootstrap distribution** describes how a statistic varies under resampling from the observed data. It is centered at the observed statistic.

9.6.1 The big picture of Part II

Part II has built your intuition for statistical inference using simulation:

Chapter	Method	Question	How variability is generated
6	Conceptual foundation	How much do statistics vary?	Thought experiments
7	Randomization test	Is there an effect?	Shuffle labels
8	Bootstrap CI	How big is the parameter?	Resample with replacement
9	CLT and normal model	Can we skip the simulation?	Mathematical formula

The key takeaway: **the simulation methods and the mathematical methods are answering the same questions**. The CLT tells us *when* the mathematical shortcuts are valid. When they are, we can replace thousands of computer simulations with a single formula.

In Part III, we will develop these formula-based methods for specific settings: one proportion, one mean, two proportions, two means, and more. But you now have the conceptual foundation to understand *why* those formulas work: they are mathematical descriptions of the same sampling variability that you observed through simulation.

9.7 Chapter review

9.7.1 Summary

- Randomization distributions, bootstrap distributions, and theoretical sampling distributions all tend to be approximately **bell-shaped** (normal) under common conditions. This is the **Central Limit Theorem** (CLT).
- The CLT states that for sufficiently large samples with independent observations, the sampling distribution of many common statistics is approximately normal, regardless of the shape of the population.
- The **success-failure condition** ($n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$) is the practical check for whether the normal approximation is appropriate for proportions.
- The **68-95-99.7 rule** describes the area under a normal curve within 1, 2, and 3 standard deviations of the mean.
- A **Z score** measures how many standard deviations (or standard errors) an observation or statistic falls from the mean: $Z = (x - \mu)/\sigma$.

- The three inference approaches — randomization, bootstrap, and mathematical models — answer the same fundamental questions. When conditions are met, they produce similar results. Mathematical models are faster; simulation methods are more flexible.
- This chapter bridges Part II (simulation-based inference) and Part III (formula-based inference). The CLT is the theoretical justification for the formulas we will use in Part III.

9.7.2 Key terms

Term	Definition
Central Limit Theorem (CLT)	The sampling distribution of a statistic is approximately normal for large samples
Normal distribution	A symmetric, bell-shaped distribution described by its mean and standard deviation
Success-failure condition	For proportions: need $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$
68-95-99.7 rule	About 68%, 95%, 99.7% of normal values fall within 1, 2, 3 SDs of the mean
Z score	Number of standard deviations an observation falls from the mean: $Z = (x - \mu)/\sigma$
Null distribution	The sampling distribution of the test statistic under H_0
Sampling distribution	Distribution of a statistic over all possible samples of a given size

9.8 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Repeated water samples.** A nonprofit wants to understand the fraction of households that have elevated levels of lead in their drinking water. They expect at least 5% of homes will have elevated levels of lead, but not more than about 30%. They randomly sample 800 homes and work with the owners to retrieve water samples, and they compute the fraction of these homes with elevated lead levels. They repeat this 1,000 times and build a distribution of sample proportions.

a. What is this distribution called?

b. Would you expect the shape of this distribution to be symmetric, right skewed, or left skewed? Explain your reasoning.

c. What is the name of the variability of this distribution.

d. Suppose the researchers' budget is reduced, and they are only able to collect 250 observations per sample, but they can still collect 1,000 samples. They build a new distribution of sample proportions. How will the variability of this new distribution compare to the variability of the distribution when each sample contained 800 observations?

2. **Repeated student samples.** Of all freshman at a large college, 16% made the dean's list in the current year. As part of a class project, students randomly sample 40 students and check if those students made the list. They repeat this 1,000 times and build a distribution of sample proportions.

a. What is this distribution called?

b. Would you expect the shape of this distribution to be symmetric, right skewed, or left skewed? Explain your reasoning.

c. What is the name of the variability of this distribution?

d. Suppose the students decide to sample again, this time collecting 90 students per sample, and they again collect 1,000 samples. They build a new distribution of sample proportions. How will the variability of this new distribution compare to the variability of the distribution when each sample contained 40 observations?

PART III

Calculate

You've built sampling distributions and computed confidence intervals through simulation. Now we introduce the mathematical tools — the normal distribution, the t-distribution — that approximate what simulation does. These theoretical methods are faster to compute and appear in published research, but they rest on the same logic you already understand.

Chapter 10

Normal Approximation

In Chapters 7–9, we used simulation-based methods – randomization tests and bootstrapping – to make inferences about population parameters. Those methods are powerful and flexible, but they share a limitation: every new problem requires running thousands of simulations. In this chapter, we introduce the **normal distribution**, a mathematical model that describes the bell-shaped pattern we keep seeing in our sampling distributions. When certain conditions are met, the normal distribution lets us calculate probabilities and construct confidence intervals using formulas instead of simulations. This is not a replacement for simulation-based thinking – it is a complement. The normal model provides the theoretical justification for *why* those simulations work the way they do.

StatLens tools for this chapter: **Normal Distribution Explorer** – Use it to visualize different normal curves, shade areas, compute probabilities, and see how changing μ and σ affects the distribution.

Launch Normal Distribution Explorer

10.1 The normal distribution

Among all the distributions encountered in statistics, one is overwhelmingly the most common. The symmetric, unimodal, bell-shaped curve is so ubiquitous that it goes by many names: the **normal curve**, the **normal model**, or the **normal distribution**.

The **normal distribution** is a symmetric, unimodal, bell-shaped distribution described by two **parameters**: the mean μ (which determines the center) and the standard deviation σ (which determines the spread). We write the distribution in shorthand as $N(\mu, \sigma)$.

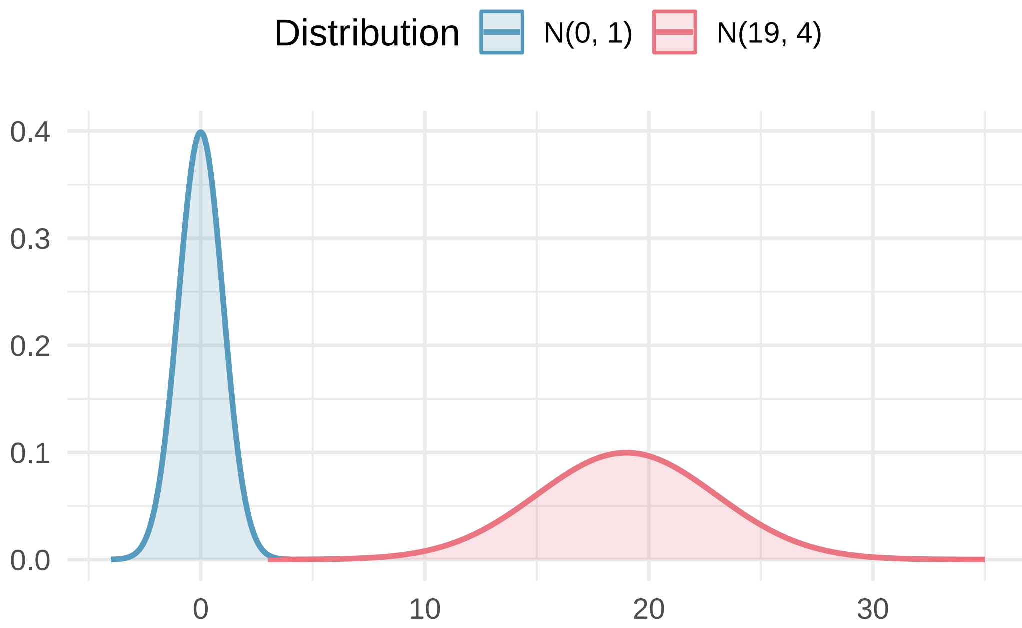


Figure 10.1: Two normal distributions with different parameters. The left curve has $\mu = 0$ and $\sigma = 1$; the right curve has $\mu = 19$ and $\sigma = 4$. Changing μ shifts the curve left or right; changing σ stretches or compresses it.

Key properties of the normal distribution:

- The curve is perfectly symmetric about the mean μ.
- The total area under the curve is exactly 1 (representing 100% of observations). We can interpret any area under the curve as a **proportion** — and, equivalently, as the **probability** of a randomly selected observation falling in that range.
- The tails extend infinitely in both directions but get very close to zero far from the center.
- Changing μ shifts the entire curve left or right without changing its shape.
- Increasing σ makes the curve wider and shorter; decreasing σ makes it narrower and taller.

Normal distribution facts. Many variables are *nearly* normal, but none are *exactly* normal. The normal distribution, while not perfect for any single problem, is extremely useful for a wide variety of problems. Variables like SAT scores, adult heights, and blood pressure measurements closely follow the normal distribution.

10.1.1 The standard normal distribution

The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is called the **standard normal distribution**, written $N(0, 1)$. It serves as a reference distribution that we will use throughout the course.

Write down the shorthand for a normal distribution with each set of parameters.

a. Mean 5 and standard deviation 3
b. Mean -100 and standard deviation 10
c. Mean 2 and standard deviation 9

---

a. $N(\mu = 5, \sigma = 3)$

b. $N(\mu = -100, \sigma = 10)$
c. $N(\mu = 2, \sigma = 9)$

10.2 The 68-95-99.7 rule

One of the most useful facts about the normal distribution is how observations spread out around the mean. This relationship is captured by the **68-95-99.7 rule** (sometimes called the **empirical rule**).

The 68-95-99.7 rule. For any normal distribution $N(\mu, \sigma)$:

- About **68%** of observations fall within **1 standard deviation** of the mean: between $\mu - \sigma$ and $\mu + \sigma$.
- About **95%** of observations fall within **2 standard deviations** of the mean: between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- About **99.7%** of observations fall within **3 standard deviations** of the mean: between $\mu - 3\sigma$ and $\mu + 3\sigma$.

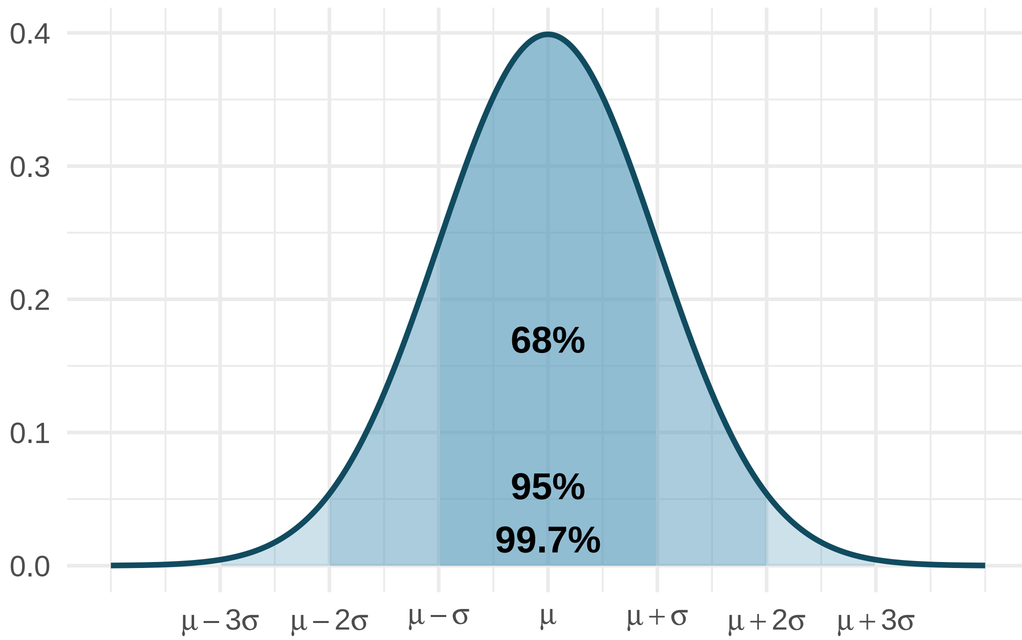


Figure 10.2: The 68-95-99.7 rule illustrated on a normal curve. The innermost shaded region (68%) extends from $\mu - \sigma$ to $\mu + \sigma$. The middle region (95%) extends from $\mu - 2\sigma$ to $\mu + 2\sigma$. The outermost region (99.7%) extends from $\mu - 3\sigma$ to $\mu + 3\sigma$.

This rule gives quick, approximate answers without needing a calculator or software.

SAT scores closely follow the normal distribution with mean $\mu = 1500$ and standard deviation $\sigma = 300$.

a. About what percent of test takers score between 1200 and 1800?
b. About what percent score between 1500 and 2100?

---

a. 1200 is one standard deviation below the mean $(1500 - 300 = 1200)$, and 1800 is one standard deviation above the mean $(1500 + 300 = 1800)$. By the 68-95-99.7 rule, about **68%** of test takers score between 1200 and 1800.

b. The range from 900 to 2100 covers $\mu \pm 2\sigma$, which contains about 95% of scores. Since the normal distribution is symmetric, the range from 1500 to 2100 (the upper half) contains about $\frac{95\%}{2} = 47.5\%$ of scores.

Adult male heights in the US follow approximately a normal distribution with mean 70 inches and standard deviation 3.3 inches. Using the 68-95-99.7 rule, approximately what percentage of adult males are between 63.4 and 76.6 inches tall?

Show answer

63.4 inches is $70 - 2(3.3) = 63.4$ and 76.6 inches is $70 + 2(3.3) = 76.6$. These are each 2 standard deviations from the mean, so by the 68-95-99.7 rule, approximately 95% of adult males have heights in this range.

Continuing with SAT scores ($\mu = 1500$, $\sigma = 300$), about what percent of test takers score above 2100?

Show answer

2100 is 2 standard deviations above the mean. By the 68-95-99.7 rule, about 95% of scores are within 2 standard deviations, leaving about 5% outside. Since the distribution is symmetric, about 2.5% are above 2100.

10.3 Z-scores and standardizing

When comparing observations from different distributions, raw values are not directly comparable. A score of 1800 on the SAT and a score of 24 on the ACT look very different, but we need a common scale to determine which performance is better relative to other test takers. The solution is **standardizing** the observations.

The **Z-score** of an observation x from a distribution with mean μ and standard deviation σ is the number of standard deviations it falls above or below the mean:

$$Z = \frac{x - \mu}{\sigma}$$

- Positive Z-scores indicate observations **above** the mean.
- Negative Z-scores indicate observations **below** the mean.
- A Z-score of 0 means the observation equals the mean.

If the observation x comes from a normal distribution $N(\mu, \sigma)$, then its Z-score follows the standard normal distribution $N(0, 1)$. Standardizing shifts the center to 0 and rescales the spread to 1, but preserves the normal shape.

SAT scores follow $N(\mu = 1500, \sigma = 300)$ and ACT scores follow $N(\mu = 21, \sigma = 5)$. Nel scored 1800 on the SAT and Sian scored 24 on the ACT. Who performed better relative to other test takers?

We compute Z-scores for each student.

Nel's Z-score: $Z_{\text{Nel}} = \frac{1800-1500}{300} = \frac{300}{300} = 1.00$

Sian's Z-score: $Z_{\text{Sian}} = \frac{24-21}{5} = \frac{3}{5} = 0.60$

Nel is 1 standard deviation above the SAT mean, while Sian is only 0.6 standard deviations above the ACT mean. Nel performed better relative to other test takers.

Head lengths of brushtail possums follow a nearly normal distribution with mean 92.6 mm and standard deviation 3.6 mm. Compute the Z-scores for possums with head lengths of 95.4 mm and 85.8 mm. Which possum has the more unusual head length?

Show answer

For $x_1 = 95.4$ mm: $Z_1 = \frac{95.4-92.6}{3.6} = \frac{2.8}{3.6} = 0.78$. For $x_2 = 85.8$ mm: $Z_2 = \frac{85.8-92.6}{3.6} = \frac{-6.8}{3.6} = -1.89$. The second possum has a more unusual head length because $|-1.89| > |0.78|$.

Let X represent a value drawn at random from $N(\mu = 3, \sigma = 2)$, and suppose we observe $x = 5.19$.

a. Find the Z-score.
b. How many standard deviations above or below the mean is this observation?

---

a. $Z = \frac{x-\mu}{\sigma} = \frac{5.19-3}{2} = \frac{2.19}{2} = 1.095$
b. The observation is 1.095 standard deviations *above* the mean. We know it is above because the Z-score is positive.

10.4 Finding probabilities from the normal distribution

One of the most important skills in statistics is finding the probability that a normally distributed variable falls in a specified range. The idea is straightforward: **area under the curve equals probability**. You've already been doing this intuitively — in Chapters 7 and 8, every time you counted "what fraction of simulated statistics were this extreme?" you were estimating a probability by looking at proportions in a distribution. The normal model just gives us a formula for the curve, so we can compute the area exactly instead of simulating it.

Strategy for normal probability problems. Always follow these steps:

1. **Draw a picture.** Sketch the normal curve, mark the mean, and shade the area you want to find.
2. **Find the Z-score.** Standardize the value(s) of interest using $Z = \frac{x-\mu}{\sigma}$.
3. **Look up the area.** Use a normal probability table, statistical software, or the StatLens Normal Distribution Explorer to find the area to the **left** of the Z-score.
4. **Adjust if needed.** If you need a right-tail area, subtract the left-tail area from 1. For an area between two values, subtract the smaller left-tail area from the larger.

10.4.1 Left-tail probabilities (percentiles)

The area to the left of a value on the normal curve gives the **percentile** – the proportion of observations that fall below that value.

The **percentile** of an observation is the percentage of values in the distribution that fall at or below that observation.

Nel scored 1800 on the SAT ($Z = 1.00$). What percentile is this?

We need the area to the left of $Z = 1.00$ on the standard normal curve.

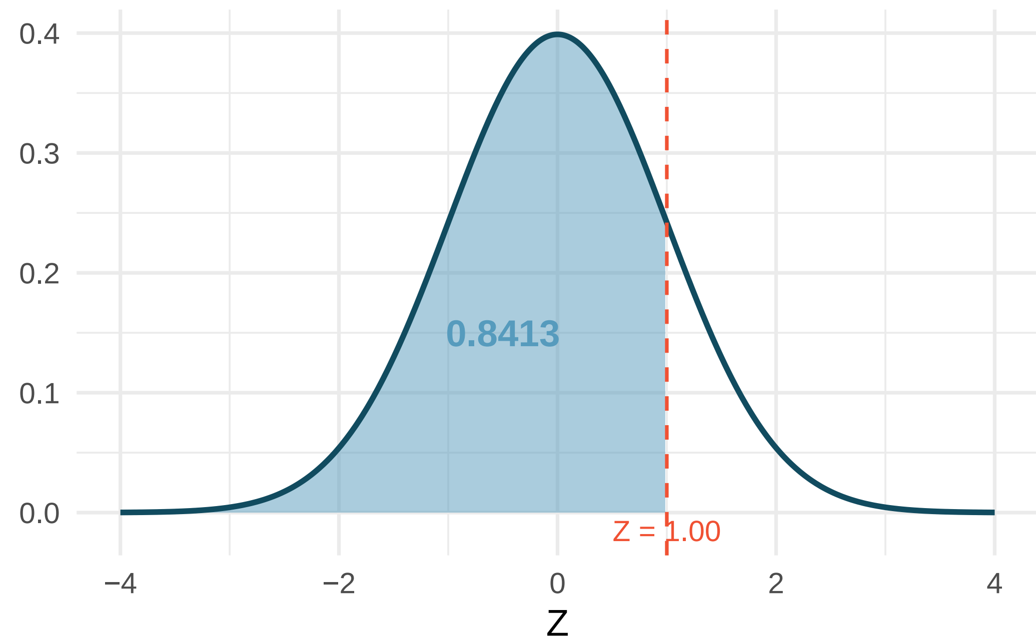


Figure 10.3: The standard normal curve with the area to the left of Z = 1.00 shaded. This area represents Nel's percentile.

Using a normal probability table or software, the area to the left of $Z = 1.00$ is **0.8413**. Nel is at the **84th percentile** – 84.13% of SAT takers scored lower.

10.4.2 Right-tail probabilities

Shannon is a randomly selected SAT taker. What is the probability Shannon scores at least 1630?

Step 1: Draw a picture. We shade the area above 1630 on the $N(1500, 300)$ curve.

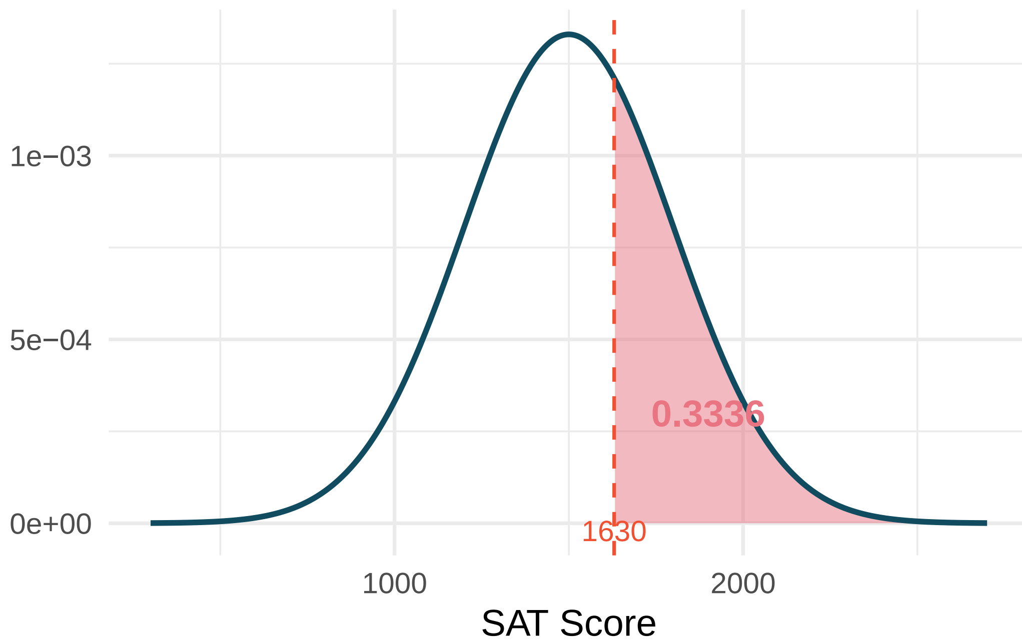


Figure 10.4: A normal curve with mean 1500 and standard deviation 300. The area to the right of 1630 is shaded.

Step 2: Find the Z-score.

$$Z = \frac{1630 - 1500}{300} = \frac{130}{300} = 0.43$$

Step 3: Look up the left-tail area. The area to the left of $Z = 0.43$ is 0.6664.

Step 4: Adjust. The area to the *right* is $1 - 0.6664 = 0.3336$.

The probability Shannon scores at least 1630 is **0.3336**, or about 33.4%.

If the probability of Shannon scoring at least 1630 is 0.3336, what is the probability Shannon scores *less than* 1630?

Show answer

The probability of scoring less than 1630 is the area to the left: 0.6664, or about 66.6%. The left-tail and right-tail areas always sum to 1.

10.4.3 Probabilities between two values

Adult male heights follow $N(\mu = 70, \sigma = 3.3)$ inches. What is the probability that a randomly selected adult male is between 5′9″ (69 inches) and 6′2″ (74 inches)?

Step 1: Draw a picture. Shade the area between 69 and 74 on the normal curve.

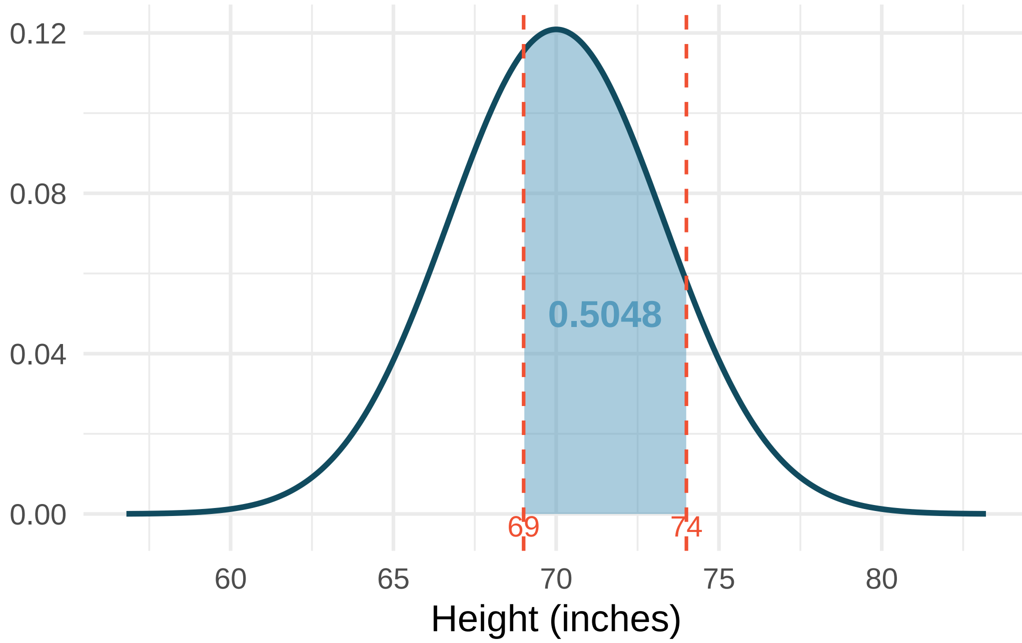


Figure 10.5: A normal curve centered at 70 with standard deviation 3.3. The area between 69 and 74 is shaded.

Step 2: Find both Z-scores.

$$Z_1 = \frac{69 - 70}{3.3} = -0.30 \qquad Z_2 = \frac{74 - 70}{3.3} = 1.21$$

Step 3: Look up both left-tail areas.

- Area to the left of $Z_1 = -0.30$: 0.3821
- Area to the left of $Z_2 = 1.21$: 0.8869

Step 4: Subtract.

$$P(69 < X < 74) = 0.8869 - 0.3821 = 0.5048$$

About **50.5%** of adult males are between 5′9″ and 6′2″.

What percent of SAT takers score between 1500 and 2000? (SAT scores follow $N(1500, 300)$.)

Show answer

First find the Z-scores: $Z_1 = \frac{1500-1500}{300} = 0.00$ and $Z_2 = \frac{2000-1500}{300} = 1.67$. The left-tail areas are 0.5000 and 0.9525. The area between them is $0.9525 - 0.5000 = 0.4525$, so about 45.25% score between 1500 and 2000.

10.4.4 Finding values from percentiles

Sometimes we want to work backwards: given a percentile, find the observation.

Adult male heights follow $N(70, 3.3)$. Yousef's height is at the 40th percentile. How tall is Yousef?

Step 1: Draw a picture. The 40th percentile means 40% of the area is to the left.

Step 2: Find the Z-score for the 40th percentile. Using a normal probability table (look for 0.40 in the body of the table) or software, $Z = -0.25$.

Step 3: Solve for x. Using the Z-score formula:

$$-0.25 = \frac{x - 70}{3.3}$$

$$x = 70 + (-0.25)(3.3) = 70 - 0.825 = 69.2 \text{ inches}$$

Yousef is approximately 69.2 inches (about 5′9″) tall.

What is the 95th percentile for SAT scores?

Show answer

The Z-score for the 95th percentile is $Z = 1.645$. Then $x = 1500 + 1.645 \times 300 = 1500 + 493.5 = 1993.5$. The 95th percentile for SAT scores is approximately 1994.

10.4.5 The normal probability table

A **normal probability table** (also called a Z-table) lists areas to the left of Z-scores for the standard normal distribution. The table is organized with the ones and tenths digits of the Z-score in the rows, and the hundredths digit in the columns.

Common Z-scores and areas worth remembering:

Table 10.1: Commonly used Z-scores and their corresponding left-tail areas.

Z-score	Left-tail area	Interpretation
-1.96	0.0250	2.5th percentile
-1.645	0.0500	5th percentile
-1.00	0.1587	15.87th percentile
0.00	0.5000	50th percentile (the mean)
1.00	0.8413	84.13th percentile
1.645	0.9500	95th percentile
1.96	0.9750	97.5th percentile
2.576	0.9950	99.5th percentile

Practice with the Normal Distribution Explorer. Use StatLens to verify the table entries above. Enter a Z-score and check that the shaded area matches. Then try working backwards: enter a percentile and find the corresponding Z-score.

Launch Normal Distribution Explorer

10.5 The normal approximation to sampling distributions

In Chapter 9, we used simulation to explore sampling distributions – the distributions of sample statistics computed from many repeated samples. A striking pattern emerged: regardless of whether we were looking at sample means, sample proportions, or differences in means or proportions, the sampling distributions tended to be **bell-shaped and symmetric**. This is not a coincidence.

10.5.1 The Central Limit Theorem

The Central Limit Theorem (CLT). When observations are independent and the sample size is sufficiently large, the sampling distribution of many common statistics (sample means, sample proportions, and their differences) is approximately normal, regardless of the shape of the population distribution.

Specifically:

- The sampling distribution of $\bar{x}$ is approximately $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$.
- The sampling distribution of $\hat{p}$ is approximately $N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$.

The CLT is one of the most remarkable results in all of mathematics. It explains why simulation-based null distributions and bootstrap distributions looked bell-shaped: they were approximating the normal distribution all along.

Two conditions for the CLT to apply:

1. **Independence.** The observations must be independent of one another. This is satisfied when data come from a random sample or a randomized experiment.
2. **Sample size.** The sample must be large enough. What counts as "large enough" depends on the context:
   - For **proportions**: at least 10 expected successes and 10 expected failures ($np \geq 10$ and $n(1-p) \geq 10$). This is called the **success-failure condition**.
   - For **means**: $n \geq 30$ is a common guideline, though smaller samples can work if the population is not strongly skewed.

10.5.2 Standard error: quantifying the variability of a statistic

The CLT tells us the sampling distribution is approximately normal. But how wide is that normal distribution? The answer is given by the **standard error**.

The **standard error (SE)** of a statistic is the standard deviation of its sampling distribution. It measures how much the statistic varies from sample to sample.

The standard error depends on the parameter of interest:

Table 10.2: Standard error formulas for common parameters.

Parameter	Point estimate	Standard error
Population mean μ	$\bar{x}$	$SE = \frac{s}{\sqrt{n}}$
Population proportion p	$\hat{p}$	$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Difference in means $\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Difference in proportions $p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

Margin of error. The distance $z^* \times SE$ is called the **margin of error**. For a 95% confidence interval, $z^* = 1.96$ (often rounded to 2), giving a margin of error of approximately $2 \times SE$. This is directly connected to the 68-95-99.7 rule: 95% of sample statistics fall within about 2 standard errors of the true parameter value.

10.5.3 Connecting simulation and normal approximation

Let's see the connection between simulation-based methods and the normal approximation in a concrete example.

In Chapter 7, we tested whether opportunity cost reminders reduce spending. We simulated 10,000 randomization samples and found a null distribution that was bell-shaped with a center of 0 and a spread described by $SE \approx 0.078$. The observed difference was 0.20.

Simulation approach: Count how many of the 10,000 simulated differences were at least as extreme as 0.20. The proportion gives the p-value: approximately 0.006.

Normal approximation approach: Model the null distribution as $N(0, 0.078)$. Compute the Z-score:

$$Z = \frac{0.20 - 0}{0.078} = 2.56$$

The area to the right of $Z = 2.56$ is 0.0052.

Both approaches give nearly the same p-value (0.006 vs. 0.0052) and the same conclusion: reject H_0. The normal approximation works because the conditions are met – independent observations and a sufficiently large sample.

10.5.4 Confidence intervals using the normal model

The normal approximation also lets us build confidence intervals without bootstrapping.

General confidence interval formula. When the sampling distribution of a statistic is approximately normal with standard error SE, a confidence interval takes the form:

$$\text{point estimate} \pm z^* \times SE$$

where z^* is chosen based on the desired confidence level:

Table 10.3: Critical values z^* for common confidence levels.

Confidence level	z^*
90%	1.645
95%	1.960
99%	2.576

A study examined whether stents reduce stroke risk. The observed difference in 30-day stroke rates (treatment minus control) was $\hat{p}_T - \hat{p}_C = 0.090$, with $SE = 0.028$. Construct a 95% confidence interval.

The conditions for the normal model have been verified. Using $z^* = 1.96$:

$$0.090 \pm 1.96 \times 0.028 = 0.090 \pm 0.055 = (0.035, 0.145)$$

We are 95% confident that implanting a stent increased the 30-day stroke rate by between 3.5 and 14.5 percentage points. Since the interval does not contain 0, we have evidence of a real difference.

10.6 When does the normal model apply – and when doesn't it?

The normal distribution is powerful but not universal. Here we summarize when it is and is not appropriate.

When the normal model is a good approximation:

- Sample proportions $\hat{p}$, when the success-failure condition is met ($np \geq 10$ and $n(1-p) \geq 10$).
- Sample means $\bar{x}$, when the sample size is at least 30 or the population is approximately normal (we will use the t-distribution for means – see Chapter 11).
- Differences in proportions or means, when both groups satisfy the above conditions.

When the normal model is NOT a good approximation:

- Small samples with a skewed population distribution.
- Proportions near 0 or 1 with small sample sizes (the success-failure condition fails).
- Data with strong dependence between observations.

A medical consultant claims that only 3 of their 62 liver transplant clients had complications, compared to the national rate of 10%. Is it appropriate to use the normal approximation to test whether this consultant's rate is lower?

Under the null hypothesis ($p_0 = 0.10$), we check the success-failure condition:

- Expected successes (complications): $np_0 = 62 \times 0.10 = 6.2$
- Expected failures (no complications): $n(1-p_0) = 62 \times 0.90 = 55.8$

Since the expected number of successes (6.2) is less than 10, the **success-failure condition is not met**. The normal approximation would underestimate the variability and give an inaccurate p-value. A simulation-based approach (as in Chapter 7) would be more appropriate here.

When conditions fail, use simulation. If the success-failure condition is not met or the sample size is too small for the normal approximation, we should fall back on the simulation-based methods from Chapters 7 and 8. The bootstrap and randomization approaches work well even when the normal model does not. This is one of the great advantages of simulation-based inference: it is more broadly applicable.

10.7 Chapter review

10.7.1 Summary

This chapter introduced the normal distribution, the most commonly encountered distribution in statistics. Key concepts include:

- The normal distribution is described by two parameters: the mean μ (center) and standard deviation σ (spread), written as $N(\mu, \sigma)$.
- The **68-95-99.7 rule** provides quick estimates: about 68%, 95%, and 99.7% of observations fall within 1, 2, and 3 standard deviations of the mean.
- A **Z-score** standardizes an observation by measuring how many standard deviations it lies from the mean: $Z = \frac{x-\mu}{\sigma}$.
- Normal probabilities are found by computing Z-scores and looking up areas under the standard normal curve using tables, software, or StatLens.
- The **Central Limit Theorem** guarantees that sampling distributions of means and proportions are approximately normal when the sample is large enough and observations are independent.

- The **standard error** measures how much a statistic varies from sample to sample and is the key ingredient in confidence intervals and hypothesis tests.
- The normal approximation and simulation-based methods give similar results when conditions are met, but simulation is more flexible when conditions fail.

10.7.2 Key terms

Normal distribution, standard normal distribution, parameters (μ and σ), 68-95-99.7 rule, Z-score, standardizing, percentile, Central Limit Theorem (CLT), standard error (SE), margin of error, success-failure condition.

10.8 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Chronic illness.** In 2013, the Pew Research Foundation reported that "45% of U.S. adults report that they live with one or more chronic conditions". However, this value was based on a sample, so it may not be a perfect estimate for the population parameter of interest on its own. The study reported a standard error of about 1.2%, and a normal model may reasonably be used in this setting.

a. Create a 95% confidence interval for the proportion of U.S. adults who live with one or more chronic conditions. Also interpret the confidence interval in the context of the study. (Pew Research Center 2013)

b. Identify each of the following statements as true or false. Provide an explanation to justify each of your answers.

   i. We can say with certainty that the confidence interval from part (a) contains the true percentage of U.S. adults who suffer from a chronic illness.

   ii. If we repeated this study 1,000 times and constructed a 95% confidence interval for each study, then approximately 950 of those confidence intervals would contain the true fraction of U.S. adults who suffer from chronic illnesses.

   iii. The poll provides statistically discernible evidence (at the $\alpha = 0.05$ level) that the percentage of U.S. adults who suffer from chronic illnesses is below 50%.

   iv. Since the standard error is 1.2%, only 1.2% of people in the study communicated uncertainty about their answer.

2. **Social media users and news, mathematical model.** A poll conducted in 2022 found that 50% of U.S. adults (i.e., a proportion of 0.5) get news from social media sometimes or often. The standard error for this estimate was 0.5% (i.e., 0.005), and a normal distribution may be used to model the sample proportion. (Pew Research Center 2022)

a. Construct a 99% confidence interval for the fraction of U.S. adults who get news on social media sometimes or often, and interpret the confidence interval in context.

b. Identify each of the following statements as true or false. Provide an explanation to justify each of your answers.

   i. The data provide statistically discernible evidence that more than half of U.S. adults users get news through social media sometimes or often. Use a discernibility level of $\alpha = 0.01$.

   ii. Since the standard error is 0.5%, we can conclude that 99.5% of all U.S. adults users were included in the study.

   iii. If we want to reduce the standard error of the estimate, we should collect less data.

   iv. If we construct a 90% confidence interval for the percentage of U.S. adults who get news through social media sometimes or often, the resulting confidence interval will be wider than a corresponding 99% confidence interval.

3. **Interpreting a Z score from a sample proportion.** Suppose that you conduct a hypothesis test about a population proportion and calculate the Z score to be 0.47. Which of the following is the best interpretation of this value? For the problems which are not a good interpretation, indicate the statistical idea being described.

a. The probability is 0.47 that the null hypothesis is true.

b. If the null hypothesis were true, the probability would be 0.47 of obtaining a sample proportion as far as observed from the hypothesized value of the population proportion.

c. The sample proportion is 0.47 standard errors greater than the hypothesized value of the population proportion.

d. The sample proportion is equal to 0.47 times the standard error.

e. The sample proportion is 0.47 away from the hypothesized value of the population.

f. The sample proportion is 0.47.

4. **Mental health.** The General Social Survey asked the question: "For how many days during the past 30 days was your mental health, which includes stress, depression, and problems with emotions, not good?" Based on responses from 1,151 US residents, the survey reported a 95% confidence interval of 3.40 to 4.24 days in 2010.

a. Interpret this interval in context of the data.

b. What does "95% confident" mean? Explain in the context of the application.

c. Suppose the researchers think a 99% confidence level would be more appropriate for this interval. Will this new interval be smaller or wider than the 95% confidence interval?

d. If a new survey were to be done with 500 Americans, do you think the standard error of the estimate be larger, smaller, or about the same.

5. **Area under the curve, Part I.** What percent of a standard normal distribution $N(\mu = 0, \sigma = 1)$ is found in each region? Be sure to draw a graph.

a. $Z < -1.35$

b. $Z > 1.48$

c. $-0.4 < Z < 1.5$

d. $|Z| > 2$

6. **Area under the curve, Part II.** What percent of a standard normal distribution $N(\mu = 0, \sigma = 1)$ is found in each region? Be sure to draw a graph.

a. $Z > -1.13$

b. $Z < 0.18$

c. $Z > 8$

d. $|Z| < 0.5$

7. **GRE scores, Part I.** Sophia who took the Graduate Record Examination (GRE) scored 160 on the Verbal Reasoning section and 157 on the Quantitative Reasoning section. The mean score for Verbal Reasoning section for all test takers was 151 with a standard deviation of 7, and the mean score for the Quantitative Reasoning was 153 with a standard deviation of 7.67. Suppose that both distributions are nearly normal.

a. Write down the short-hand for these two normal distributions.

b. What is Sophia's Z-score on the Verbal Reasoning section? On the Quantitative Reasoning section? Draw a standard normal distribution curve and mark these two Z-scores.

c. What do these Z-scores tell you?

d. Relative to others, which section did she do better on?

e. Find her percentile scores for the two exams.

f. What percent of the test takers did better than her on the Verbal Reasoning section? On the Quantitative Reasoning section?

g. Explain why simply comparing raw scores from the two sections could lead to an incorrect conclusion as to which section a student did better on.

h. If the distributions of the scores on these exams are not nearly normal, would your answers to parts (b) - (f) change? Explain your reasoning.

8. **Triathlon times, Part I.** In triathlons, it is common for racers to be placed into age and gender groups. Friends Leo and Mary both completed the Hermosa Beach Triathlon, where Leo competed in the *Men, Ages 30 - 34* group while Mary competed in the *Women, Ages 25 - 29* group. Leo completed the race in 1:22:28 (4948 seconds), while Mary completed the race in 1:31:53 (5513 seconds). Obviously Leo finished faster, but they are curious about how they did within their respective groups. Can you help them? Here is some information on the performance of their groups:

- The finishing times of the *Men, Ages 30 - 34* group has a mean of 4313 seconds with a standard deviation of 583 seconds.
- The finishing times of the *Women, Ages 25 - 29* group has a mean of 5261 seconds with a standard deviation of 807 seconds.
- The distributions of finishing times for both groups are approximately Normal.

Remember: a better performance corresponds to a faster finish.

a. Write down the short-hand for these two normal distributions.

b. What are the Z-scores for Leo's and Mary's finishing times? What do these Z-scores tell you?

c. Did Leo or Mary rank better in their respective groups? Explain your reasoning.

d. What percent of the triathletes did Leo finish faster than in his group?

e. What percent of the triathletes did Mary finish faster than in her group?

f. If the distributions of finishing times are not nearly normal, would your answers to parts (b) - (e) change? Explain your reasoning.

9. **GRE scores, Part II.** In a previous exercise we saw two distributions for GRE scores: $N(\mu = 151, \sigma = 7)$ for the verbal part of the exam and $N(\mu = 153, \sigma = 7.67)$ for the quantitative part. Use this information to compute each of the following:

a. The score of a student who scored in the 80^{th} percentile on the Quantitative Reasoning section.

b. The score of a student who scored worse than 70% of the test takers in the Verbal Reasoning section.

10. **Triathlon times, Part II.** In a previous exercise we saw two distributions for triathlon times: $N(\mu = 4313, \sigma = 583)$ for *Men, Ages 30 - 34* and $N(\mu = 5261, \sigma = 807)$ for the *Women, Ages 25 - 29* group. Times are listed in seconds. Use this information to compute each of the following:

a. The cutoff time for the fastest 5% of athletes in the men's group, i.e. those who took the shortest 5% of time to finish.

b. The cutoff time for the slowest 10% of athletes in the women's group.

11. **LA weather, Part I.** The average daily high temperature in June in LA is 77° F with a standard deviation of 5° F. Suppose that the temperatures in June closely follow a normal distribution.

a. What is the probability of observing an 83° F temperature or higher in LA during a randomly chosen day in June?

b. How cool are the coldest 10% of the days (days with lowest high temperature) during June in LA?

12. **CAPM.** The Capital Asset Pricing Model (CAPM) is a financial model that assumes returns on a portfolio are normally distributed. Suppose a portfolio has an average annual return of 14.7% (i.e. an average gain of 14.7%) with a standard deviation of 33%. A return of 0% means the value of the portfolio doesn't change, a negative return means that the portfolio loses money, and a positive return means that the portfolio gains money.

a. What percent of years does this portfolio lose money, i.e. have a return less than 0%?

b. What is the cutoff for the highest 15% of annual returns with this portfolio?

13. **LA weather, Part II.** A previous exercise states that average daily high temperature in June in LA is 77° F with a standard deviation of 5° F, and it can be assumed that they follow a normal distribution. We use the following equation to convert ° F (Fahrenheit) to ° C (Celsius):

$$C = (F - 32) \times \frac{5}{9}.$$

a. Write the probability model for the distribution of temperature in ° C in June in LA.

b. What is the probability of observing a 28° C (which roughly corresponds to 83° F) temperature or higher in June in LA? Calculate using the ° C model from part (a).

c. Did you get the same answer or different answers in part (b) of this question and part (a) of a previous exercise? Are you surprised? Explain.

d. Estimate the IQR of the temperatures (in ° C) in June in LA.

14. **Find the SD.** Cholesterol levels for women aged 20 to 34 follow an approximately normal distribution with mean 185 milligrams per deciliter (mg/dl). Women with cholesterol levels above 220 mg/dl are considered to have high cholesterol and about 18.5% of women fall into this category. What is the standard deviation of the distribution of cholesterol levels for women aged 20 to 34?

Chapter 11

Confidence Intervals for Means

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In Chapter 8, we used bootstrapping to construct confidence intervals – a flexible, simulation-based approach that works well in many settings. In Chapter 10, we saw that sampling distributions are often well approximated by the normal distribution. For means, however, we encounter a complication: the population standard deviation σ is almost never known, and estimating it with the sample standard deviation s introduces extra uncertainty. To account for this, we use a slightly different bell-shaped distribution called the t**-distribution**. In this chapter, we develop t-based confidence intervals for one mean, the difference between two independent means, and the mean of paired differences.

StatLens tools for this chapter: Use the t**-Distribution Explorer** to compare the t-distribution to the standard normal at various degrees of freedom.

Launch t-Distribution Explorer

11.1 The t-distribution

11.1.1 Why not the normal distribution?

In Chapter 10, we used the normal distribution to model sampling distributions. For proportions, this works well because the standard error formula $SE = \sqrt{\hat{p}(1-\hat{p})/n}$ depends only on the sample proportion and the sample size – both of which are known. For means, the standard error is $SE = \sigma/\sqrt{n}$, which depends on σ, the population standard deviation. Since we rarely know σ, we estimate it using the sample standard deviation s:

$$SE \approx \frac{s}{\sqrt{n}}$$

This substitution introduces additional uncertainty, especially in small samples. When n is small, s can be a poor estimate of σ, and the normal model underestimates how variable the statistic really is. The t-distribution corrects for this.

11.1.2 Properties of the t-distribution

The t**-distribution** is a bell-shaped, symmetric distribution that is similar to the normal distribution but has **heavier tails** (more probability in the extremes). The exact shape of the t-distribution depends on a parameter called the **degrees of freedom** (**df**).

- When df is small, the t-distribution has noticeably heavier tails than the normal – extreme values are more likely.
- As df increases, the t-distribution approaches the standard normal distribution.
- For $df \geq 30$, the t-distribution is nearly indistinguishable from the normal.

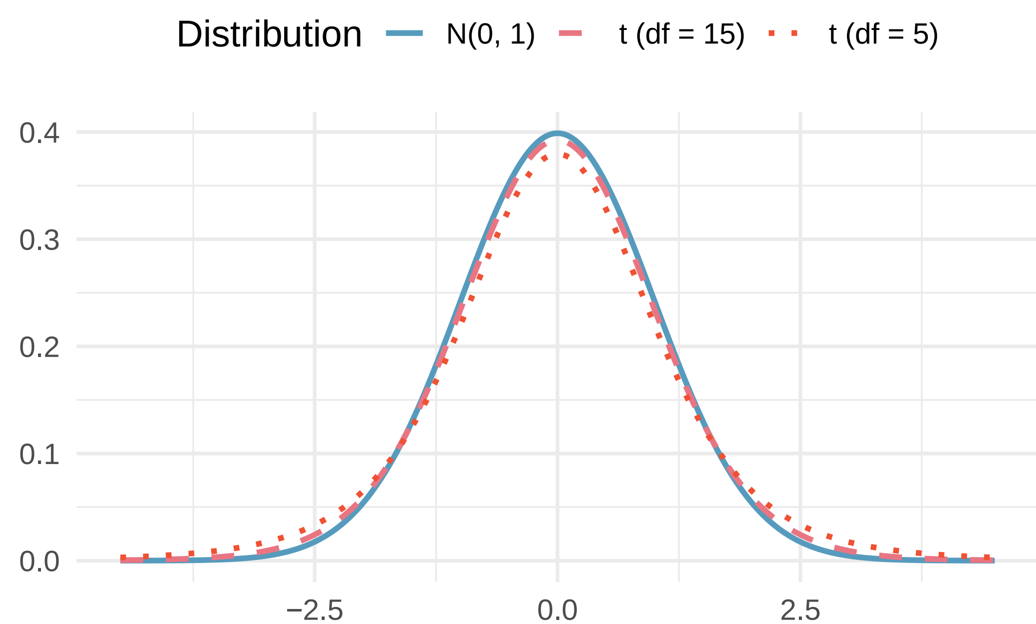


Figure 11.1: Three distributions plotted on the same axes: the standard normal distribution $N(0, 1)$, a t-distribution with $df = 5$, and a t-distribution with $df = 15$. All are centered at 0 and bell-shaped, but the t-distributions have heavier tails, especially when df is small.

Why heavier tails matter. Heavier tails mean that extreme values (large t-scores) are more probable under the t-distribution than under the normal distribution. In practice, this leads to **wider confidence intervals** when sample sizes are small – exactly the extra caution we need when σ is estimated by s.

How does the t-distribution compare to the normal distribution when $df = 2$? When $df = 100$?

Show answer

When $df = 2$, the t-distribution has very heavy tails – values far from 0 are much more likely than under the normal curve. When $df = 100$, the t-distribution is nearly identical to the standard normal. In general, the t-distribution becomes more like the normal as degrees of freedom increase.

See the t-distribution come alive. Open the t Distribution Explorer and drag the degrees of freedom slider. Start at $df = 2$ and watch the heavy tails — values far from zero are surprisingly common. Now slowly increase toward $df = 30$, then $df = 100$. The t-distribution gradually becomes indistinguishable from the normal curve. This is much harder to appreciate from a static figure — try it yourself.

11.1.3 Critical values from the t-distribution

Just as we use z^* for the normal distribution, we use t^*_{df} for the t-distribution. Because the t-distribution has heavier tails, t^* values are larger than the corresponding z^* values (especially for small df).

Selected critical values for 95% confidence intervals:

Table 11.1: Critical values t^*_{df} for a 95% confidence interval. As df increases, t^* approaches $z^* = 1.96$.

df	t^*_{df} (95% CI)	z^* (for comparison)
5	2.571	1.960
10	2.228	1.960
20	2.086	1.960
30	2.042	1.960
50	2.009	1.960
100	1.984	1.960
∞	1.960	1.960

Why is the critical value $t^*_5 = 2.571$ so much larger than $z^* = 1.96$?

Show answer

With only $df = 5$, there is substantial uncertainty in estimating σ with s. The t-distribution accounts for this by having heavier tails, which pushes the critical values further from zero. This results in a wider confidence interval, reflecting the greater uncertainty.

11.2 One-sample t confidence interval

11.2.1 The setup

We want to estimate the population mean μ based on a single sample. The point estimate is the sample mean $\bar{x}$, and the standard error is $SE = s/\sqrt{n}$.

One-sample t confidence interval for μ.

$$\bar{x} \pm t^*_{df} \times \frac{s}{\sqrt{n}}$$

where $df = n - 1$.

Conditions:

1. **Independence.** The observations are independent (e.g., from a random sample).
2. **Normality / sample size.** Either the population distribution is approximately normal, or the sample size is large enough ($n \geq 30$). For smaller samples, check for strong skewness or extreme outliers.

11.2.2 Checking conditions

The normality condition is about the shape of the **population**, not the sample. In practice:

- If $n < 30$: Check a histogram or dotplot of the sample data. If there is no strong skewness and no extreme outliers, the t-methods are reasonable.

- If $n \geq 30$: The Central Limit Theorem ensures that $\bar{x}$ is approximately normal, even if the population is moderately skewed. However, be cautious with extreme outliers, which can distort $\bar{x}$ and s.

Outliers and the t-test. The sample mean $\bar{x}$ and sample standard deviation s are sensitive to outliers. A single extreme observation can substantially shift $\bar{x}$ and inflate s, leading to misleading confidence intervals. When outliers are present, consider whether they represent data errors. If the outliers are genuine, a bootstrap confidence interval (Chapter 8) may be more reliable.

11.2.3 Worked example: Mercury levels in Risso's dolphins

Researchers measured mercury concentrations (in μg/g) in the muscle tissue of 19 Risso's dolphins from the Adriatic Sea. The sample mean was $\bar{x} = 4.4$ μg/g and the sample standard deviation was $s = 2.3$ μg/g. Construct a 95% confidence interval for the mean mercury level in the population of Risso's dolphins.

Check conditions.

1. *Independence.* The dolphins were sampled from different locations, so it is reasonable to treat the measurements as independent.
2. *Normality.* With $n = 19$, we need to check for strong skewness or outliers. Suppose a histogram of the data shows a roughly symmetric distribution with no extreme outliers. The condition is satisfied.

Compute the interval. With $df = 19 - 1 = 18$, the critical value for a 95% CI is $t^*_{18} = 2.101$.

$$SE = \frac{s}{\sqrt{n}} = \frac{2.3}{\sqrt{19}} = \frac{2.3}{4.359} = 0.528$$

$$\bar{x} \pm t^*_{18} \times SE = 4.4 \pm 2.101 \times 0.528 = 4.4 \pm 1.109 = (3.29, 5.51)$$

We are 95% confident that the mean mercury concentration in Risso's dolphins is between 3.29 and 5.51 μg/g.

A sample of 35 UW-La Crosse students reported their weekly study hours. The sample mean is $\bar{x} = 15.2$ hours and $s = 6.8$ hours. Construct a 90% confidence interval for the mean weekly study hours of all UW-La Crosse students. (Use $t^*_{34} = 1.691$ for a 90% CI.)

Show answer

First check conditions: the sample is a random sample (independence), and $n = 35 \geq 30$ (normality via CLT). Then $SE = 6.8/\sqrt{35} = 1.149$. The 90% CI is $15.2 \pm 1.691 \times 1.149 = 15.2 \pm 1.943 = (13.26, 17.14)$. We are 90% confident the mean weekly study hours for all UW-La Crosse students is between 13.26 and 17.14 hours.

11.3 Two-sample t confidence interval

11.3.1 The setup

When comparing the means of two independent groups, we estimate the difference $\mu_1 - \mu_2$ using $\bar{x}_1 - \bar{x}_2$.

Two-sample t confidence interval for $\mu_1 - \mu_2$.

$$(\bar{x}_1 - \bar{x}_2) \pm t^*_{df} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Degrees of freedom: The exact formula for df is complex and typically computed by software. A conservative (safe) approximation is:

$$df = \min(n_1 - 1,\ n_2 - 1)$$

Conditions:

1. **Independence (extended).** Observations are independent within each group *and* between groups (e.g., from two independent random samples or a randomized experiment).
2. **Normality / sample size.** Each group's data should be approximately normal or have $n \geq 30$. Check each group separately for strong skewness or extreme outliers.

11.3.2 Worked example: Embryonic stem cells and heart function

Researchers studied whether embryonic stem cell (ESC) treatment improves heart function in sheep after a heart attack. Nine sheep received the ESC treatment and nine were in the control group. The change in pumping capacity was measured:

Table 11.2: Summary statistics for the stem cell study.

Group	n	$\bar{x}$	s
ESC	9	3.50	5.17
Control	9	-4.33	2.76

Construct a 95% confidence interval for the difference in mean change in pumping capacity, $\mu_{ESC} - \mu_{Control}$.

Check conditions.

1. *Independence.* The sheep were randomly assigned to groups, so independence within and between groups is satisfied.
2. *Normality.* With $n = 9$ per group (small samples), we check histograms. Suppose they show no extreme outliers in either group. The condition is reasonably met.

Compute the interval.

$$SE = \sqrt{\frac{5.17^2}{9} + \frac{2.76^2}{9}} = \sqrt{\frac{26.73}{9} + \frac{7.62}{9}} = \sqrt{2.970 + 0.847} = \sqrt{3.817} = 1.954$$

Using $df = \min(9-1, 9-1) = 8$, the critical value for a 95% CI is $t^*_8 = 2.306$.

$$\text{Point estimate} = 3.50 - (-4.33) = 7.83$$

$$7.83 \pm 2.306 \times 1.954 = 7.83 \pm 4.506 = (3.32, 12.34)$$

We are 95% confident that the ESC treatment increases heart pumping capacity by between 3.32% and 12.34% compared to the control. Because the interval is entirely above zero, we have evidence that the ESC treatment improves heart function. Since this was a randomized experiment, we can attribute the improvement to the treatment.

Researchers collected birth weight data from a random sample of 1,000 US births. Nonsmoker mothers ($n = 867$) had babies with $\bar{x}_n = 7.27$ lbs ($s_n = 1.23$), and smoker mothers ($n = 114$) had babies with $\bar{x}_s = 6.68$ lbs ($s_s = 1.60$). Is there evidence that babies born to nonsmoker mothers weigh more on average? Compute a 95% confidence interval for the difference $\mu_n - \mu_s$.

Show answer

The point estimate is $7.27 - 6.68 = 0.59$ lbs. The SE is $\sqrt{1.23^2/867 + 1.60^2/114} = \sqrt{0.001745 + 0.02246} = \sqrt{0.02420} = 0.156$. Using $df = \min(866, 113) = 113$ and $t^*_{113} \approx 1.981$, the CI is $0.59 \pm 1.981 \times 0.156 = 0.59 \pm 0.309 = (0.28, 0.90)$. We are 95% confident that babies born to nonsmoker mothers weigh between 0.28 and 0.90 lbs more on average than babies born to smoker mothers.

11.4 Paired data confidence interval

11.4.1 What makes data "paired"?

Sometimes two measurements are made on the same individual (e.g., before and after a treatment) or on naturally matched pairs (e.g., the same textbook priced at two stores). In these cases, the two groups are **not independent** – they are **paired**.

Paired data. Two sets of observations are **paired** if each observation in one set has a special correspondence or connection with exactly one observation in the other set. Common examples include:

- Before/after measurements on the same individual
- Two treatments applied to the same subject (e.g., tires on the same car)
- Matched pairs in an experiment (e.g., twins assigned to different treatments)
- The same item measured under two conditions (e.g., textbook prices at two stores)

Table 11.3: Examples of paired designs.

Observational unit	Comparison groups	Measurement	Value of interest
Car	Smooth Turn vs. Quick Spin	Tire tread after 1,000 miles	Difference in tread
Textbook	UCLA Bookstore vs. Amazon	Price of new textbook	Difference in price
Individual	Pre-course vs. Post-course	Exam score	Difference in score

11.4.2 The key idea: analyze the differences

The key to working with paired data is simple: **compute the difference for each pair, then analyze those differences as a single sample.** This converts a two-sample problem into a one-sample problem.

Paired t confidence interval for μ_{diff}.

For paired data, compute the difference $d_i = x_{1,i} - x_{2,i}$ for each pair. Then the confidence interval for the mean difference μ_{diff} is:

$$\bar{x}_{diff} \pm t^*_{df} \times \frac{s_{diff}}{\sqrt{n_{diff}}}$$

where $\bar{x}_{diff}$ is the sample mean of the differences, s_{diff} is the sample standard deviation of the differences, n_{diff} is the number of pairs, and $df = n_{diff} - 1$.

Conditions:

1. **Independence of pairs.** The pairs themselves are independent (e.g., different cars, different textbooks).
2. **Normality of differences.** The distribution of the *differences* should be approximately normal, or the number of pairs should be large ($n_{diff} \geq 30$). Check for extreme outliers.

11.4.3 Worked example: Textbook prices

Researchers sampled 68 UCLA courses and compared textbook prices at the UCLA Bookstore to prices on Amazon. Summary statistics for the price differences (UCLA − Amazon) are:

Table 11.4: Summary statistics for the 68 price differences.

n_{diff}	$\bar{x}_{diff}$	s_{diff}
68	\$3.58	\$13.42

Construct a 95% confidence interval for the mean price difference.

Check conditions.

1. *Independence of pairs.* The textbooks were randomly sampled, so the pairs are independent.
2. *Normality of differences.* With $n_{diff} = 68$, the CLT ensures the sampling distribution of $\bar{x}_{diff}$ is approximately normal. Though the distribution of individual differences is right-skewed, the large sample size and absence of extreme outliers make the t-methods reasonable.

Compute the interval.

$$SE = \frac{s_{diff}}{\sqrt{n_{diff}}} = \frac{13.42}{\sqrt{68}} = \frac{13.42}{8.246} = 1.628$$

With $df = 68 - 1 = 67$, the critical value for a 95% CI is $t^*_{67} \approx 2.00$.

$$3.58 \pm 2.00 \times 1.628 = 3.58 \pm 3.256 = (0.32, 6.84)$$

We are 95% confident that UCLA Bookstore prices are, on average, between \$0.32 and \$6.84 higher than Amazon prices for UCLA course books.

A fitness program enrolled 25 participants and measured their resting heart rates before and after 8 weeks of training. The mean difference (before − after) was $\bar{x}_{diff} = 4.2$ beats per minute with $s_{diff} = 6.1$ bpm. Construct a 90% confidence interval for the mean decrease in resting heart rate. (Use $t^*_{24} = 1.711$.)

Show answer

First check conditions: pairs (individuals) were randomly selected (independence), and with $n = 25$ and no extreme outliers reported, the normality condition is reasonable. Then $SE = 6.1/\sqrt{25} = 6.1/5 = 1.22$. The 90% CI is $4.2 \pm 1.711 \times 1.22 = 4.2 \pm 2.09 = (2.11, 6.29)$. We are 90% confident the training program reduces resting heart rate by between 2.11 and 6.29 bpm on average.

11.4.4 Why not use a two-sample t interval for paired data?

Never use a two-sample procedure for paired data. A two-sample t interval treats the two groups as independent, which they are not. Ignoring the pairing throws away valuable information about the within-pair relationship and typically produces a wider (less precise) confidence interval. Always compute the differences first and then use the one-sample (paired) t procedure.

11.5 Connecting t-based and bootstrap confidence intervals

The t-based confidence intervals in this chapter and the bootstrap confidence intervals from Chapter 8 are answering the same question: "What is a plausible range for the population parameter?" When conditions are met, they give similar results.

When do they agree?

- Large samples with no extreme outliers: the t-interval and bootstrap interval will be nearly identical.
- Moderate samples from approximately normal populations: close agreement.

When might they differ?

- Small samples from skewed populations: the bootstrap can capture asymmetry in the sampling distribution, while the t-interval is always symmetric around $\bar{x}$.
- Presence of outliers: the bootstrap is somewhat more robust because it does not rely on s being a good estimate of σ.

Which should you use?

- If conditions are met, the t-interval is simpler to compute and has well-understood properties.
- If conditions are questionable (small sample, skewness, outliers), prefer the bootstrap.
- In practice, reporting both and noting any discrepancies is informative.

11.6 Chapter review

11.6.1 Summary

This chapter developed formula-based confidence intervals for means using the t-distribution.

- The **t-distribution** accounts for the extra uncertainty from estimating σ with s. It has heavier tails than the normal distribution, especially for small degrees of freedom.
- **One-sample t CI**: $\bar{x} \pm t^*_{n-1} \times \frac{s}{\sqrt{n}}$. Requires independent observations and approximate normality (or large n).
- **Two-sample t CI**: $(\bar{x}_1 - \bar{x}_2) \pm t^*_{df} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$. Requires independence within and between groups, and approximate normality or large samples in each group.
- **Paired t CI**: Compute differences for each pair, then apply the one-sample t CI to the differences: $\bar{x}_{diff} \pm t^*_{n_{diff}-1} \times \frac{s_{diff}}{\sqrt{n_{diff}}}$.
- Always check conditions before using t-methods: independence, normality/sample size, and absence of extreme outliers.

- When conditions are met, t-based and bootstrap confidence intervals give similar results. When conditions are questionable, the bootstrap may be more reliable.

11.6.2 Key terms

t-distribution, degrees of freedom (df), critical value (t^*), one-sample t confidence interval, two-sample t confidence interval, paired data, paired t confidence interval, standard error of the mean, independence condition, normality condition.

11.7 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Statistics vs. parameters: one mean.** Each of the following scenarios were set up to assess an average value. For each one, identify, in words: the statistic and the parameter.

a. A sample of 25 New Yorkers were asked how much sleep they get per night.

b. Researchers at two different universities in California collected information on undergraduates' heights.

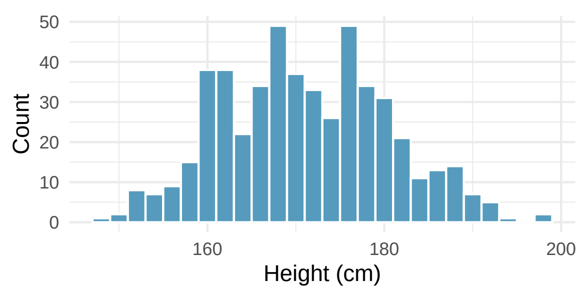
2. **Statistics vs. parameters: one mean.** Each of the following scenarios were set up to assess an average value. For each one, identify, in words: the statistic and the parameter.

a. Georgianna samples 20 children from a particular city and measures how many years they have each been playing piano.

b. Traffic police officers (who are regularly exposed to lead from automobile exhaust) had their lead levels measured in their blood.

3. **Heights of adults.** Researchers studying anthropometry collected body measurements, as well as age, weight, height and gender, for 507 physically active adults. Summary statistics for the distribution of heights (measured in centimeters, cm), along with a histogram, are provided below. (Heinz et al. 2003)

Min	147.2
Q1	163.8
Median	170.3
Mean	171.1
Q3	177.8
Max	198.1
SD	9.4
IQR	14.0



a. What are the point estimates for the average and median heights of active adults?

b. What are the point estimates for the standard deviation and IQR of heights of active adults?

c. Is a person who is 1m 80cm (180 cm) tall considered unusually tall? And is a person who is 1m 55cm (155cm) considered unusually short? Explain your reasoning.

d. The researchers take another random sample of physically active adults. Would you expect the mean and the standard deviation of this new sample to be the ones given above? Explain your reasoning.

e. The sample means obtained are point estimates for the mean height of all active individuals, if the sample of individuals is equivalent to a simple random sample. What measure do we use to quantify the variability of such an estimate? Compute this quantity using the data from the original sample under the condition that the data are a simple random sample.

4. **Heights of adults, standard error.** Heights of 507 physically active adults have a mean of 171 cm and a standard deviation of 9.4 cm. Provide an estimate for the standard error of the mean for samples of following sizes. (Heinz et al. 2003)

a. n = 10

b. n = 50

c. n = 100

d. n = 1000

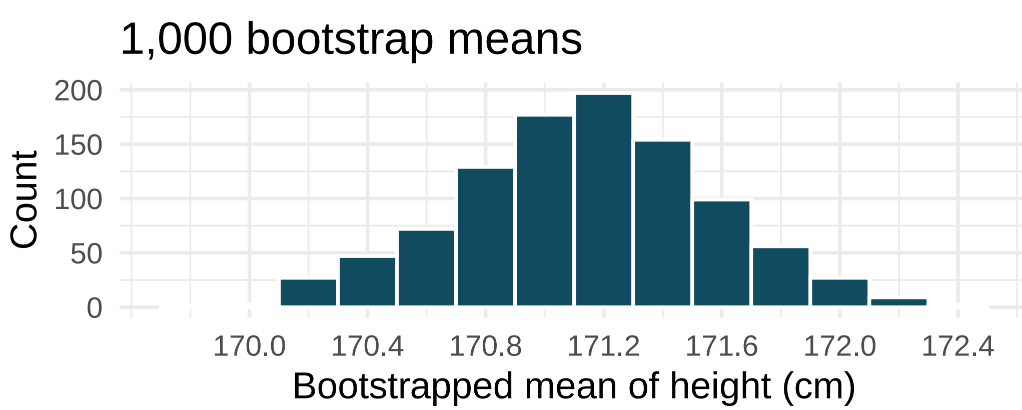
e. The standard error of the mean is a number which describes what?

5. **Heights of adults vs. kindergartners.** Heights of 507 physically active adults have a mean of 171 cm and a standard deviation of 9.4 cm. (Heinz et al. 2003)

a. Would you expect the standard deviation of the heights of a few hundred kindergartners to be higher or lower than 9.4 cm? Explain your reasoning.

b. Suppose many samples of size 100 adults is taken and, separately, many samples of size 100 kindergarteners are taken. For each of the many samples, the average height is computed. Which set of sample averages would have a larger standard error of the mean, the adult sample averages or the kindergartner sample averages?

6. **Heights of adults, bootstrap interval.** Researchers studying anthropometry collected body measurements, as well as age, weight, height and gender, for 507 physically active adults. The histogram below shows the sample distribution of bootstrapped means from 1,000 different bootstrap samples. (Heinz et al. 2003)



a. Given the bootstrap sampling distribution for the sample mean, find an approximate value for the standard error of the mean.

b. By looking at the bootstrap sampling distribution (1,000 bootstrap samples were taken), find an approximate 90% bootstrap percentile confidence interval for the true average adult height in the population from which the data were randomly sampled. Provide the interval as well as a one-sentence interpretation of the interval.

c. By looking at the bootstrap sampling distribution (1,000 bootstrap samples were taken), find an approximate 90% bootstrap SE confidence interval for the true average adult height in the population from which the data were randomly sampled. Provide the interval as well as a one-sentence interpretation of the interval.

7. **Identify the critical t.** A random sample is selected from an approximately normal population with unknown standard deviation. Find the degrees of freedom and the critical t-value ($t^\star$) for the given sample size and confidence level.

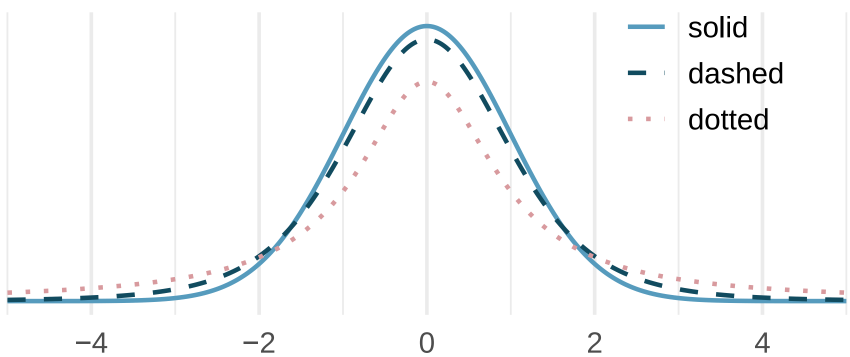
a. $n = 6$, CL = 90%

b. $n = 21$, CL = 98%

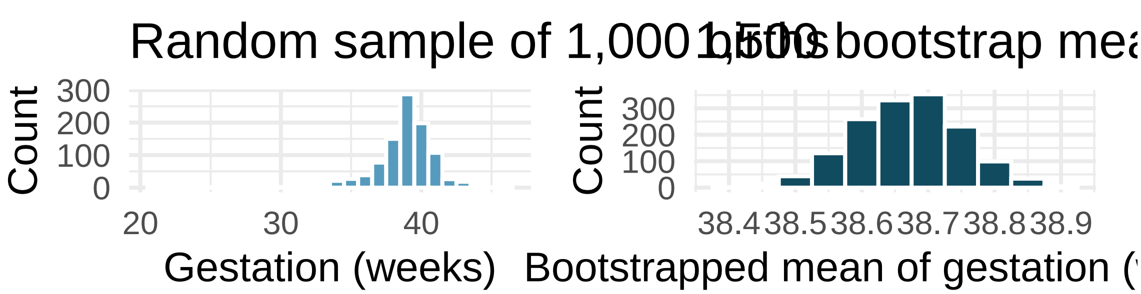
c. $n = 29$, CL = 95%

d. $n = 12$, CL = 99%

8. **t-distribution.** The figure below shows three unimodal and symmetric curves: the standard normal (z) distribution, the t-distribution with 5 degrees of freedom, and the t-distribution with 1 degree of freedom. Determine which is which, and explain your reasoning.



9. **Length of gestation, confidence interval.** Every year, the United States Department of Health and Human Services releases to the public a large dataset containing information on births recorded in the country. This dataset has been of interest to medical researchers who are studying the relation between habits and practices of expectant mothers and the birth of their children. In this exercise we work with a random sample of 1,000 cases from the dataset released in 2014. The length of pregnancy, measured in weeks, is commonly referred to as gestation. The histograms below show the distribution of lengths of gestation from the random sample of 1,000 births (on the left) and the distribution of bootstrapped means of gestation from 1,500 different bootstrap samples (on the right).



a. Given the bootstrap sampling distribution for the sample mean, find an approximate value for the standard error of the mean.

b. By looking at the bootstrap sampling distribution (1,500 bootstrap samples were taken), find an approximate 99% bootstrap percentile confidence interval for the true average gestation length in the population from which the data were randomly sampled. Provide the interval as well as a one-sentence interpretation of the interval.

c. By looking at the bootstrap sampling distribution (1,500 bootstrap samples were taken), find an approximate 99% bootstrap SE confidence interval for the true average gestation length in the population from which the data were randomly sampled. Provide the interval as well as a one-sentence interpretation of the interval.

10. **Interpreting confidence intervals for population mean.** For each of the following statements, indicate if they are a true or false interpretation of the confidence interval. If false, provide a reason or correction to the misinterpretation. You collect a large sample and calculate a 95% confidence interval for the average number of cans of sodas consumed annually per adult in the US to be (440 cans, 520 cans), i.e., on average, adults in the US consume just under two cans of soda per day.

a. 95% of adults in the US consume between 440 and 520 cans of soda per year.

b. There is a 95% probability that the true population average per adult yearly soda consumption is between 440 and 520 cans.

c. The true population average per adult yearly soda consumption is between 440 and 520 cans, with 95% confidence.

d. The average soda consumption of the people who were sampled is between 440 and 520 cans of soda per year, with 95% confidence.

11. **Working backwards, I.** A 95% confidence interval for a population mean, μ, is given as (18.985, 21.015). The population distribution is approximately normal and the population standard deviation is unknown. This confidence interval is based on a simple random sample of 36 observations. Assuming that all conditions necessary for inference are satisfied, and using the t-distribution, calculate the sample mean, the margin of error, and the sample standard deviation.

12. **Working backwards, II.** A 90% confidence interval for a population mean is (65, 77). The population distribution is approximately normal and the population standard deviation is unknown. This confidence interval is based on a simple random sample of 25 observations. Assuming that all conditions necessary for inference are satisfied, and using the t-distribution, calculate the sample mean, the margin of error, and the sample standard deviation.

13. $t^\star$ **for the correct confidence level.** As you've seen, the tails of a $t-$distribution are longer than the standard normal which results in $t^\star_{df}$ being larger than $z^\star$ for any given confidence level. When finding a CI for a population mean, explain how mistakenly using $z^\star$ (instead of the correct t^*_{df}) would affect the confidence level.

14. **Possible bootstrap samples.** Consider a simple random sample of the following observations: 47, 4, 92, 47, 12, 8. Which of the following could be a possible bootstrap samples from the observed data above? If the set of values could not be a bootstrap sample, indicate why not.

a. 47, 47, 47, 47, 47, 47

b. 92, 4, 13, 8, 47, 4

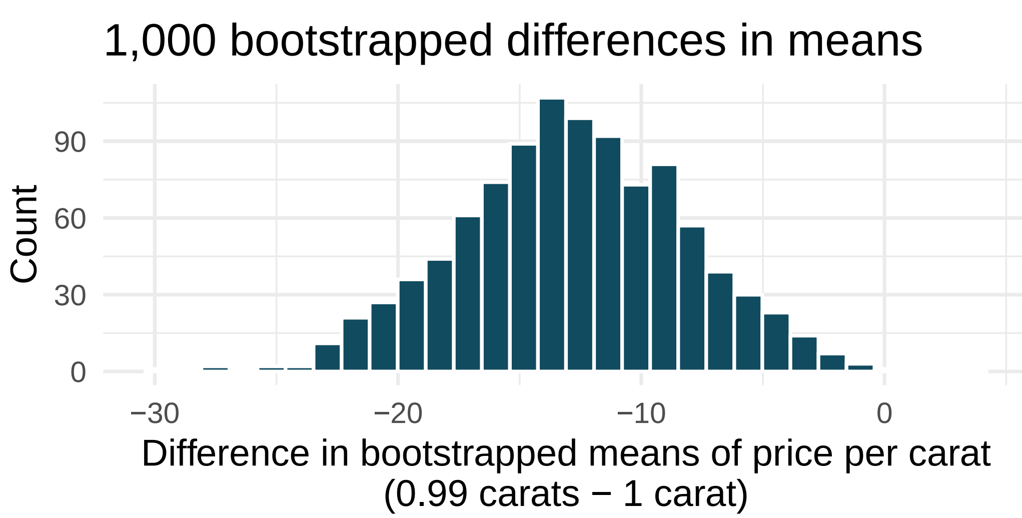
c. 92, 47, 12

d. 8, 47, 12, 12, 8, 4, 92

e. 12, 4, 8, 8, 92, 12

15. **Fill in the blanks.** We use a ___ to evaluate if data provide convincing evidence of a difference between two population means and we use a ___ to estimate this difference.

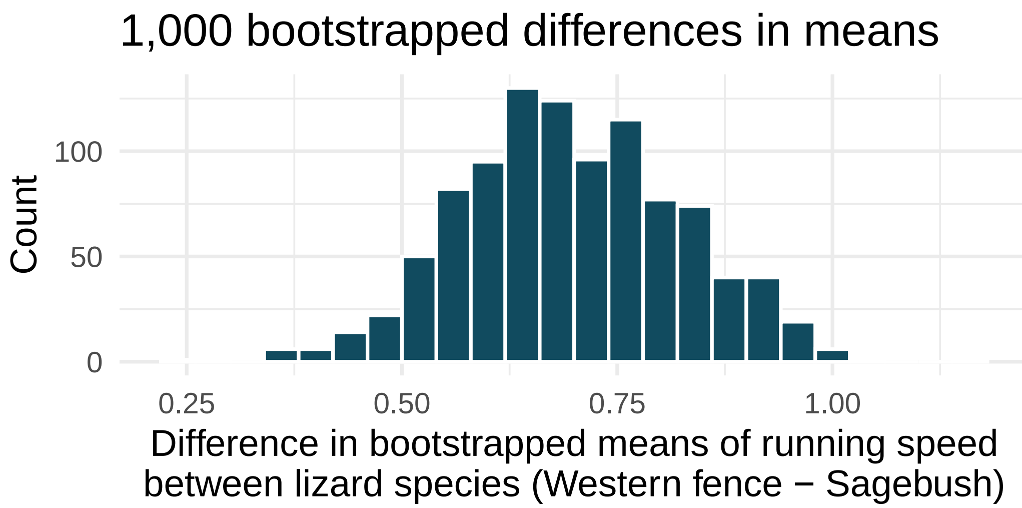
16. **Diamonds, bootstrap interval.** We have data on two random samples of diamonds: 23 0.99 carat diamonds and 23 1 carat diamonds. Provided below is a histogram of bootstrap differences in means of price per carat of diamonds that weigh 0.99 carats and diamonds that weigh 1 carat. (Wickham 2016)



a. Using the bootstrap distribution, create a (rough) 95% bootstrap percentile confidence interval for the true population difference in prices per carat of diamonds that weigh 0.99 carats and 1 carat.

b. Using the bootstrap distribution, create a (rough) 95% bootstrap SE confidence interval for the true population difference in prices per carat of diamonds that weigh 0.99 carats and 1 carat. Note that the standard error of the bootstrap distribution is 4.64.

17. **Lizards running, bootstrap interval.** We have data on top speeds (in m/sec) measured on a laboratory race track for two species of lizards: Western fence lizard (Sceloporus occidentalis) and Sagebrush lizard (Sceloporus graciosus). The bootstrap distribution below describes the variability of difference in means captured from 1,000 bootstrap samples of the lizard data. (Adolph 1987)



a. Using the bootstrap distribution, create a (rough) 90% percentile bootrap confidence interval for the true population difference in average speed of the Western fence lizard as compared with Sagebrush lizard.

b. Using the bootstrap distribution, create a (rough) 90% bootstrap SE confidence interval for the true population difference in average speed of the Western fence lizard as compared with Sagebrush lizard.

18. **Weight loss.** You are reading an article in which the researchers have created a 95% confidence interval for the difference in average weight loss for two diets. They are 95% confident that the true difference in average weight loss over 6 months for the two diets is somewhere between (1 lb, 25 lbs). The authors claim that, "therefore diet A ($\bar{x}_A$ = 20 lbs average loss) results in a much larger average weight loss as compared to diet B ($\bar{x}_B$ = 7 lbs average loss)." Comment on the authors' claim.

19. **Diamonds, mathematical interval.** We have data on two random samples of diamonds: one with diamonds that weigh 0.99 carats and one with diamonds that weigh 1 carat. Each sample has 23 diamonds. Sample statistics for the price per carat of diamonds in each sample are provided below. Assuming that the conditions for conducting inference using a mathematical model are satisfied, construct a 95% confidence interval for the true population difference in prices per carat of diamonds that weigh 0.99 carats and 1 carat. (Wickham 2016)

	Mean	SD	n
0.99 carats	\$44.51	\$13.32	23
1 carat	\$57.20	\$18.19	23

20. **True / False: comparing means.** Determine if the following statements are true or false, and explain your reasoning for statements you identify as false.

a. As the degrees of freedom increases, the t-distribution approaches normality.

b. If a 95% confidence interval for the difference between two population means contains 0, a 99% confidence interval calculated based on the same two samples will also contain 0.

c. If a 95% confidence interval for the difference between two population means contains 0, a 90% confidence interval calculated based on the same two samples will also contain 0.

21. **Difference of means.** We collect two random samples from two different populations In each part below, consider the sample means $\bar{x}_1$ and $\bar{x}_2$ that we might observe from these two samples.

	Mean	Standard deviation	Sample size
Population 1	15	20	50
Population 2	20	10	30

a. What is the associated mean and standard deviation of $\bar{x}_1$?

b. What is the associated mean and standard deviation of $\bar{x}_2$?

c. Calculate and interpret the mean and standard deviation associated with the difference in sample means for the two groups, $\bar{x}_2 - \bar{x}_1$.

d. How are the standard deviations from parts (a), (b), and (c) related?

22. **Air quality.** Air quality measurements were collected in a random sample of 25 country capitals in 2013, and then again in the same cities in 2014. We would like to use these data to compare average air quality between the two years. Should we use a paired or non-paired test? Explain your reasoning.

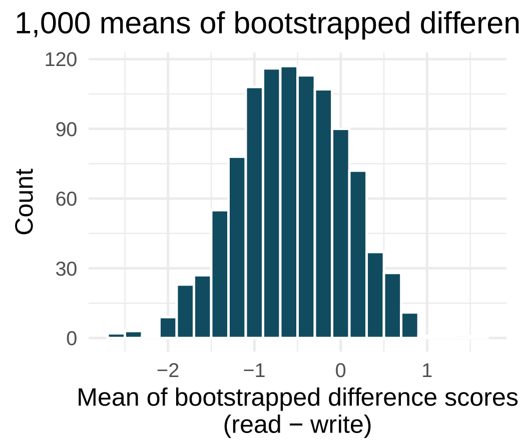
23. **True / False: paired.** Determine if the following statements are true or false. If false, explain.
  a. In a paired analysis we first take the difference of each pair of observations, and then we do inference on these differences.
  b. Two datasets of different sizes cannot be analyzed as paired data.
  c. Consider two sets of data that are paired with each other. Each observation in one dataset has a natural correspondence with exactly one observation from the other dataset.
  d. Consider two sets of data that are paired with each other. Each observation in one dataset is subtracted from the average of the other dataset's observations.

24. **Paired or not? I.** In each of the following scenarios, determine if the data are paired.
  a. Compare pre- (beginning of semester) and post-test (end of semester) scores of students.
  b. Assess gender-related salary gap by comparing salaries of randomly sampled men and women.
  c. Compare artery thicknesses at the beginning of a study and after 2 years of taking Vitamin E for the same group of patients.
  d. Assess effectiveness of a diet regimen by comparing the before and after weights of subjects.

25. **Paired or not? II.** In each of the following scenarios, determine if the data are paired.
  a. We would like to know if Intel's stock and Southwest Airlines' stock have similar rates of return. To find out, we take a random sample of 50 days, and record Intel's and Southwest's stock on those same days.
  b. We randomly sample 50 items from Target stores and note the price for each. Then we visit Walmart and collect the price for each of those same 50 items.
  c. A school board would like to determine whether there is a difference in average SAT scores for students at one high school versus another high school in the district. To check, they take a simple random sample of 100 students from each high school.

26. **Sample size and pairing.** Determine if the following statement is true or false, and if false, explain your reasoning: If comparing means of two groups with equal sample sizes, always use a paired test.

27. **High School and Beyond, bootstrap interval.** We considered the differences between the reading and writing scores of a random sample of 200 students who took the High School and Beyond Survey. The mean and standard deviation of the differences are $\bar{x}_{read-write} = -0.545$ and $s_{read-write}$ = 8.887 points. The bootstrap distribution below was produced by bootstrapping from the sample of differences in reading and writing scores 1,000 times.



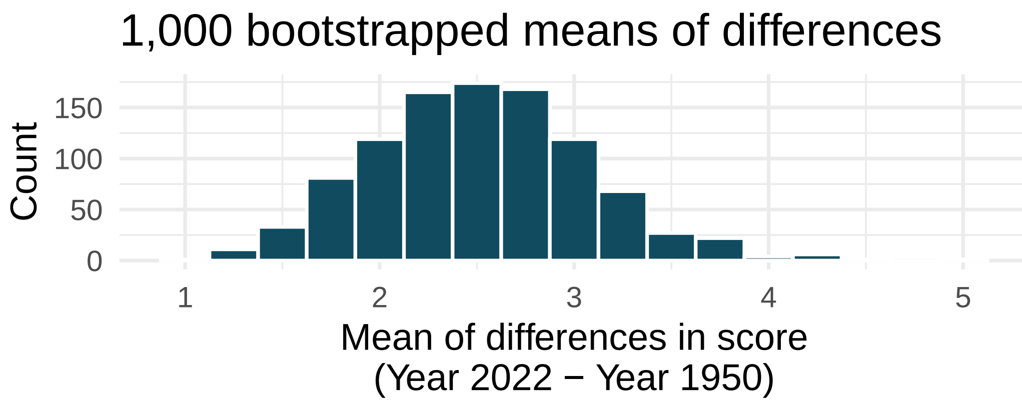
a. Find an approximate 95% bootstrap percentile confidence interval for the true average difference in scores (read - write).

b. Find an approximate 95% bootstrap SE confidence interval for the true average difference in scores (read - write).

c. Interpret both confidence intervals using words like "population" and "score".

d. From the confidence intervals calculated above, does it appear that there is a discernible difference in reading and writing scores, on average?

28. **Global warming, bootstrap interval.** We considered the change in the 90^{th} percentile high temperature in 1950 versus 2022 at 26 sampled locations from the NOAA database. (NOAA 2023) The mean and standard deviation of the reported differences are 2.53°F and 2.95°F.



a. Calculate a 90% bootstrap percentile confidence interval for the average difference of 90^{th} percentile high temperature between 1950 and 2022.

b. Calculate a 90% bootstrap SE confidence interval for the average difference of 90^{th} percentile high temperature between 1950 and 2022.

c. Interpret both intervals in context.

d. Do the confidence intervals provide convincing evidence that there were hotter high temperatures in 2022 than in 1950 at NOAA stations? Explain your reasoning.

29. **High school and beyond, mathematical interval.** We considered the differences between the reading and writing scores of a random sample of 200 students who took the High School and Beyond Survey. The mean and standard deviation of the differences are $\bar{x}_{read-write} = -0.545$ and $s_{read-write}$ = 8.887 points.

a. Calculate a 95% confidence interval for the average difference between the reading and writing scores of all students.

b. Interpret this interval in context.

c. Does the confidence interval provide convincing evidence that there is a real difference in the average scores? Explain.

30. **Global warming, mathematical interval.** We considered the change in the 90^{th} percentile high temperature in 1950 versus 2022 at 26 sampled locations from the NOAA database. (NOAA 2023) The mean and standard deviation of the reported differences are 2.53°F and 2.95°F.

a. Calculate a 90% confidence interval for the average difference of 90^{th} percentile high temperature between 1950 and 2022. We've already checked the conditions for you.

b. Interpret the interval in context.

c. Does the confidence interval provide convincing evidence that there were hotter high temperatures in 2022 than in 1950 at NOAA stations? Explain your reasoning.

31. **Forest management.** Forest rangers wanted to better understand the rate of growth for younger trees in the park. They took measurements of a random sample of 50 young trees in 2009 and again measured those same trees in 2019. The data below summarize their measurements, where the heights are in feet.

Year	Mean	SD	n
2009	12.0	3.5	50
2019	24.5	9.5	50
Difference	12.5	7.2	50

Construct a 99% confidence interval for the average growth of (what had been) younger trees in the park over 2009-2019.

Datasets: `bdims` (openintro) | `births14` (openintro)

Chapter 12

Hypothesis Tests for Means

In Chapter 7, we developed the logic of hypothesis testing using simulation: shuffle the data under the null hypothesis, build a null distribution, and measure how extreme the observed statistic is. In Chapter 11, we introduced the t-distribution as a mathematical model for the sampling distribution of means. Now we bring these ideas together. In this chapter, we use the t-distribution to conduct hypothesis tests for a single mean, the difference between two independent means, and the mean of paired differences. Along the way, we will see that the t-test and simulation-based tests generally agree when conditions are met, and we will discuss what to do when they disagree.

12.1 The t-test framework

Every hypothesis test follows the same four-step structure, whether we use simulation or a mathematical model:

1. **State hypotheses.** Write H_0 (null hypothesis) and H_A (alternative hypothesis) in both words and symbols.
2. **Check conditions.** Verify that the mathematical model is appropriate.
3. **Compute the test statistic and p-value.** Calculate how far the observed data are from what H_0 predicts, and determine how unusual that distance is.
4. **Draw a conclusion.** Compare the p-value to the significance level α and state the conclusion in context.

For means, the test statistic is a T **score** – the standardized distance from the observed sample mean to the hypothesized value, measured in units of standard errors.

12.2 One-sample t-test

12.2.1 Setup and hypotheses

The one-sample t-test is used when we have a single sample of quantitative data and want to test a claim about the population mean μ.

The test statistic for a single mean is a T.

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x}$ is the sample mean, μ_0 is the hypothesized value from H_0, s is the sample standard deviation, and n is the sample size.

When the null hypothesis is true and the conditions are met, T follows a t-distribution with $df = n - 1$.

Conditions:

- **Independent observations** (e.g., random sample).
- **Normality / sample size:** The population distribution should be approximately normal, or n should be large enough ($n \geq 30$). Check for strong skewness or extreme outliers.

12.2.2 Worked example: Cherry Blossom race times

The average finishing time for all runners in the Cherry Blossom 10-mile race in 2006 was $\mu = 93.29$ minutes. A random sample of 100 runners from a more recent year had a mean finishing time of $\bar{x} = 97.32$ minutes with $s = 16.98$ minutes. Is there evidence that the average finishing time has changed?

Step 1: State hypotheses.

- H_0: $\mu = 93.29$ (the average finishing time has not changed)
- H_A: $\mu \neq 93.29$ (the average finishing time has changed)

This is a two-sided test because we are looking for a change in either direction.

Step 2: Check conditions.

- *Independence.* The sample is a random sample of runners, so observations are independent.
- *Sample size.* With $n = 100 \geq 30$, the CLT guarantees the sampling distribution of $\bar{x}$ is approximately normal, even if the population distribution of finishing times is skewed.

Step 3: Compute the test statistic and p-value.

$$SE = \frac{s}{\sqrt{n}} = \frac{16.98}{\sqrt{100}} = 1.698$$

$$T = \frac{\bar{x} - \mu_0}{SE} = \frac{97.32 - 93.29}{1.698} = \frac{4.03}{1.698} = 2.37$$

With $df = 100 - 1 = 99$, we find the area to the right of $T = 2.37$ on the t-distribution: approximately 0.0099. Since this is a two-sided test, the p-value is $2 \times 0.0099 = 0.0198$.

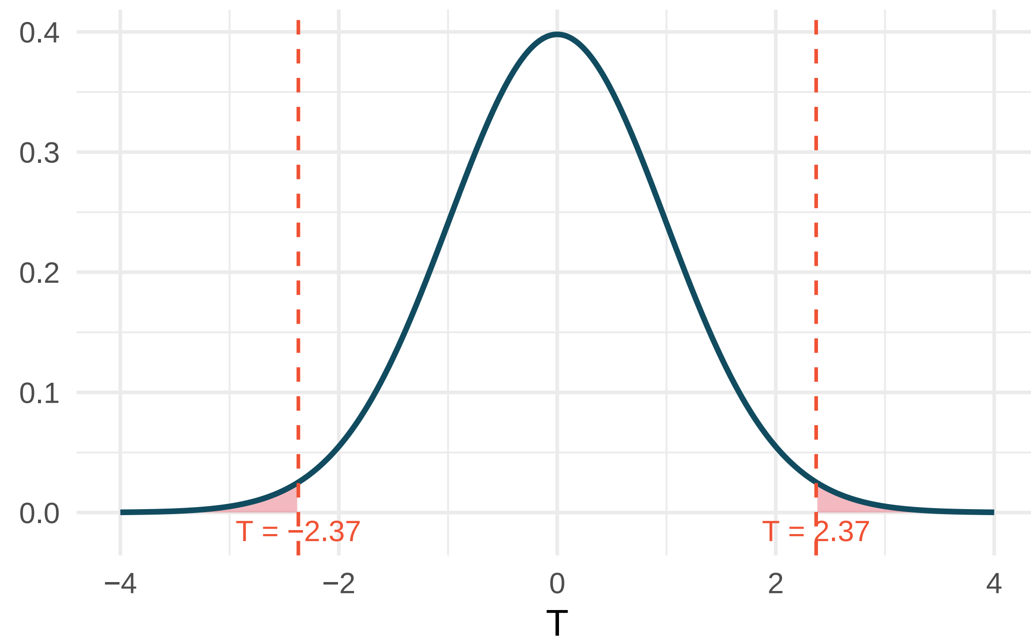


Figure 12.1: A t-distribution with $df = 99$ showing the two-sided p-value. The shaded areas in both tails represent the probability of observing a test statistic at least as extreme as 2.37 in either direction.

Step 4: Conclusion. The p-value of 0.0198 is less than $\alpha = 0.05$, so we reject H_0. The data provide statistically significant evidence that the average finishing time has changed from the 2006 average of 93.29 minutes. Based on the sample, runners appear to be finishing more slowly on average.

A nutrition study claims that the average American adult consumes 2,000 mg of sodium per day. A random sample of 45 adults shows $\bar{x} = 3,120$ mg and $s = 1,100$ mg. Test whether the data provide evidence that the true average sodium intake differs from 2,000 mg. Use $\alpha = 0.05$.

Show answer

Hypotheses: $H_0 : \mu = 2000$, $H_A : \mu \neq 2000$. Conditions: random sample (independence), $n = 45 \geq 30$ (normality via CLT). Test statistic: $SE = 1100/\sqrt{45} = 163.9, T = (3120 - 2000)/163.9 = 6.83$. With $df = 44$, the two-tailed p-value is essentially 0 (far less than 0.05). We reject H_0. There is very strong evidence that the average sodium intake is different from (and appears much higher than) 2,000 mg per day.

12.3 Two-sample t-test

12.3.1 Setup and hypotheses

The two-sample t-test compares the means of two independent groups. The parameter of interest is $\mu_1 - \mu_2$, and the null hypothesis typically claims no difference.

The test statistic for comparing two independent means is a T.

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

When the null hypothesis is true and conditions are met, T follows a t-distribution with $df = \min(n_1 - 1, n_2 - 1)$ (conservative approximation; software computes the exact df).

Conditions:

- **Independence (extended).** Observations are independent within and between the two groups.
- **Normality / sample size.** Each group should be approximately normal or have a large enough sample size. Check each group separately for strong skewness or extreme outliers.

12.3.2 Worked example: Birth weights and smoking

Is there evidence that newborns of mothers who smoke during pregnancy have a different average birth weight than newborns of non-smoking mothers? Data from a random sample of 1,000 US births in 2014:

Table 12.1: Summary statistics for birth weights by smoking status.

Group	n	Mean (lbs)	SD (lbs)
Nonsmoker	867	7.27	1.23
Smoker	114	6.68	1.60

Step 1: State hypotheses.

- H_0: $\mu_n - \mu_s = 0$ (no difference in average birth weight)
- H_A: $\mu_n - \mu_s \neq 0$ (there is a difference in average birth weight)

Step 2: Check conditions.

- *Independence.* The data come from a simple random sample, so observations are independent within and between groups.
- *Normality.* Both groups have large sample sizes ($n_n = 867$ and $n_s = 114$), both well above 30. Histograms of the data show no extreme outliers. Conditions are met.

Step 3: Compute the test statistic and p-value.

$$\bar{x}_n - \bar{x}_s = 7.27 - 6.68 = 0.59$$

$$SE = \sqrt{\frac{1.23^2}{867} + \frac{1.60^2}{114}} = \sqrt{\frac{1.5129}{867} + \frac{2.56}{114}} = \sqrt{0.001745 + 0.02246} = \sqrt{0.02420} = 0.156$$

$$T = \frac{0.59 - 0}{0.156} = 3.78$$

Using the conservative $df = \min(867 - 1, 114 - 1) = 113$, the one-tail area for $T = 3.78$ is approximately 0.00012. The two-tailed p-value is $2 \times 0.00012 = 0.00024$.

Step 4: Conclusion. With a p-value of 0.00024, which is far less than $\alpha = 0.05$, we reject H_0. The data provide strong evidence that there is a difference in average birth weight between babies born to smoking and non-smoking mothers. Specifically, babies born to non-smoking mothers weigh about 0.59 lbs more on average.

12.3.3 Worked example: Exam versions

An instructor created two versions of an exam and randomly distributed them. Summary statistics:

Table 12.2: Summary statistics for exam versions.

Version	n	Mean	SD	Min	Max
A	58	75.1	13.9	44	100
B	55	72.0	13.8	38	100

Test whether there is convincing evidence that one version is more difficult. Use $\alpha = 0.01$.

Step 1: State hypotheses.

- H_0: $\mu_A - \mu_B = 0$ (the exams are equally difficult)
- H_A: $\mu_A - \mu_B \neq 0$ (one exam is more difficult)

Step 2: Check conditions.

- *Independence.* Exams were randomly assigned, so independence is satisfied.
- *Normality.* Both groups have $n > 30$ and the summary statistics suggest roughly symmetric distributions. Conditions are met.

Step 3: Compute the test statistic and p-value.

$$\bar{x}_A - \bar{x}_B = 75.1 - 72.0 = 3.1$$

$$SE = \sqrt{\frac{13.9^2}{58} + \frac{13.8^2}{55}} = \sqrt{\frac{193.21}{58} + \frac{190.44}{55}} = \sqrt{3.331 + 3.463} = \sqrt{6.794} = 2.607$$

$$T = \frac{3.1}{2.607} = 1.19$$

Using $df = \min(57, 54) = 54$, the one-tail area for $T = 1.19$ is about 0.12. The two-tailed p-value is approximately 0.24.

Step 4: Conclusion. The p-value of 0.24 is much larger than $\alpha = 0.01$, so we fail to reject H_0. The data do not provide convincing evidence that one exam version is more difficult than the other.

Does the conclusion above mean that the two exams are equally difficult?

Show answer

No. Failing to reject H_0 does not prove that H_0 is true. The data are consistent with the exams being equally difficult, but they are also consistent with exam A being about 3.1 points easier. The test simply did not find strong enough evidence to distinguish these possibilities.

12.4 Paired t-test

12.4.1 Setup and hypotheses

When data are paired (see Chapter 11 for what makes data "paired"), we compute the difference within each pair and test whether the mean difference is zero.

The test statistic for a paired mean difference is a T.

$$T = \frac{\bar{x}_{diff} - 0}{s_{diff}/\sqrt{n_{diff}}}$$

When the null hypothesis is true and conditions are met, T follows a t-distribution with $df = n_{diff} - 1$.

Conditions:

- **Independently sampled pairs.**
- **Normality of differences.** The differences should be approximately normal, or n_{diff} should be large. Check for extreme outliers in the differences.

12.4.2 Worked example: Textbook prices

Is there evidence that UCLA Bookstore prices differ from Amazon prices, on average? Summary statistics for the 68 price differences (UCLA − Amazon):

Table 12.3: Summary statistics for the 68 price differences.

n_{diff}	$\bar{x}_{diff}$	s_{diff}
68	\$3.58	\$13.42

Step 1: State hypotheses.

- H_0: $\mu_{diff} = 0$ (no difference in average textbook price)
- H_A: $\mu_{diff} \neq 0$ (there is a difference in average price)

Step 2: Check conditions.

- *Independence.* The textbooks were a simple random sample of courses; pairs are independent.
- *Normality.* With $n_{diff} = 68$, the CLT applies. While some outliers exist (a few textbooks with large price differences), none are extreme enough to invalidate the approach given the large sample size.

Step 3: Compute the test statistic and p-value.

$$SE = \frac{s_{diff}}{\sqrt{n_{diff}}} = \frac{13.42}{\sqrt{68}} = 1.628$$

$$T = \frac{3.58 - 0}{1.628} = 2.20$$

With $df = 68 - 1 = 67$, the one-tail area for $T = 2.20$ is approximately 0.0156. The two-tailed p-value is $2 \times 0.0156 = 0.0312$.

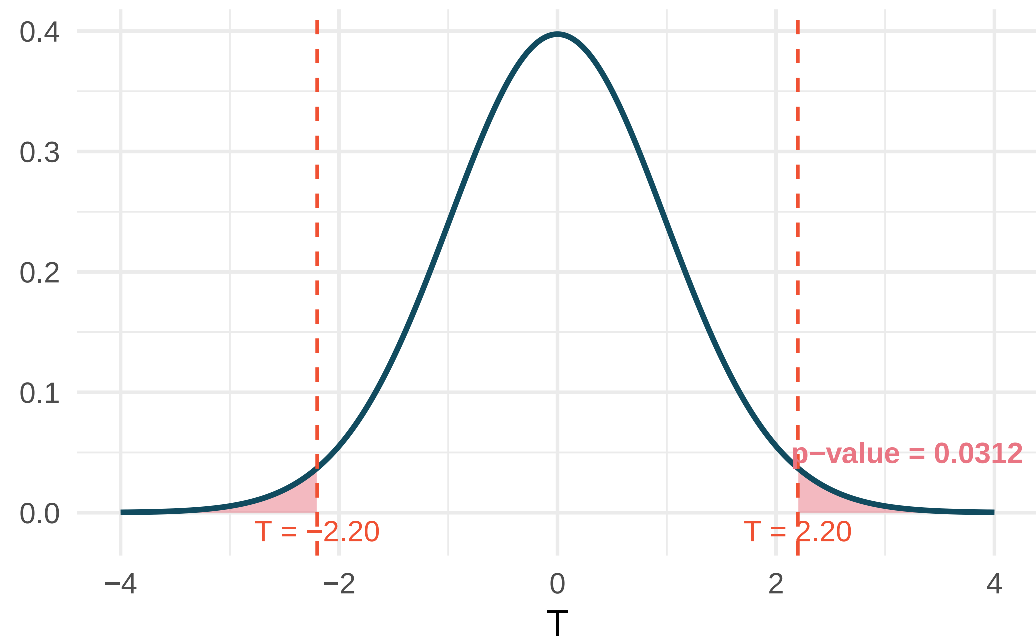


Figure 12.2: A t-distribution with $df = 67$ showing the two-sided p-value. The shaded areas in both tails represent the p-value of 0.0312.

Step 4: Conclusion. The p-value of 0.0312 is less than $\alpha = 0.05$, so we reject H_0. The data provide evidence that UCLA Bookstore prices are different from Amazon prices on average. Based on the positive mean difference, the UCLA Bookstore tends to charge more.

Researchers measured tire tread (in cm) for 25 cars, each equipped with one Smooth Turn and one Quick Spin tire. The mean tread difference (Smooth Turn − Quick Spin) was $\bar{x}_{diff} = 0.002$ cm with $s_{diff} = 0.004$ cm. Test whether there is a difference in average tread wear between the two brands. Use $\alpha = 0.05$.

Show answer

Hypotheses: $H_0 : \mu_{diff} = 0$, $H_A : \mu_{diff} \neq 0$. Conditions: cars are independent, $n = 25$ and no extreme outliers reported. $SE = 0.004/\sqrt{25} = 0.0008$. $T = 0.002/0.0008 = 2.50$. With $df = 24$, the two-tailed p-value is approximately 0.020. Since $0.020 < 0.05$, we reject H_0 and conclude there is evidence of a difference in tread wear between the two tire brands.

12.5 Connecting simulation-based and t-based tests

An important theme throughout this course is that simulation-based methods and mathematical models are complementary approaches to the same questions. When conditions are met, they should give similar answers. Here we compare the two approaches explicitly.

12.5.1 When they agree

For the birth weights example:

- **Randomization test:** Shuffle smoking/nonsmoking labels 10,000 times, compute the difference in means each time, and find the proportion of shuffled differences as extreme as 0.59 lbs. This gives a p-value around 0.001.

- **t-test:** The p-value was 0.00024.

Both approaches lead to the same conclusion (reject H_0) by a wide margin. This agreement is typical when sample sizes are moderate to large and conditions are met.

12.5.2 When they might disagree

Disagreement arises primarily in two situations:

1. **Small samples from skewed populations.** The t-test assumes the sampling distribution is symmetric, which may not hold for small samples from skewed distributions. A simulation-based approach can capture the asymmetry.
2. **Extreme outliers.** The t-test relies on $\bar{x}$ and s, both of which are sensitive to outliers. A randomization test, which works with ranks and permutations, can be more robust.

12.5.3 Practical guidance

- **When conditions are clearly met** (large samples, no extreme outliers), use the t-test – it is simpler and faster.
- **When conditions are borderline** (moderate samples, mild skewness), consider running both approaches. If they agree, report the t-test. If they disagree, investigate why and consider the simulation-based result more trustworthy.
- **When conditions are clearly violated** (small samples, strong skewness, extreme outliers), prefer the simulation-based approach.

12.6 Practical vs. statistical significance

Statistical significance $\neq$ practical significance.

A result is **statistically significant** if the p-value is below the chosen significance level α. But statistical significance only means the observed effect is unlikely to be due to chance alone. It says nothing about whether the effect is *large enough to matter in practice*.

With very large samples, even trivially small differences can be statistically significant. With very small samples, even large differences may not be statistically significant (due to low power).

A large study of 50,000 students found that students who eat breakfast score an average of 0.5 points higher on a 100-point exam than students who skip breakfast ($p < 0.001$). Is this result practically significant?

The result is statistically significant (the p-value is very small), but the effect size of 0.5 points on a 100-point exam is tiny. A teacher or student would likely not consider half a point to be a meaningful difference. This is an example where statistical significance does not imply practical significance.

12.6.1 Effect sizes

To bridge the gap between statistical and practical significance, researchers often report **effect sizes** – standardized measures of how large the observed effect is.

Cohen's d is a common measure of effect size for comparing two means. It expresses the difference in means as a multiple of the pooled standard deviation:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

where s_p is the pooled standard deviation. General guidelines for interpreting d:

Table 12.4: Guidelines for interpreting Cohen's d.

$\lvert d \rvert$	Interpretation
0.2	Small effect
0.5	Medium effect
0.8	Large effect

These guidelines are rough benchmarks. What constitutes a "meaningful" effect depends on the context. In medicine, even a small effect size might be important if it concerns a life-threatening condition. In education, a medium effect might be required for a costly intervention to be worthwhile.

In the birth weights study, the difference was 0.59 lbs with a pooled standard deviation of approximately 1.30 lbs. Compute Cohen's d and interpret it.

Show answer

$d = 0.59/1.30 = 0.45$. This is between a small and medium effect. While the difference is statistically significant, the practical impact is moderate – about half a standard deviation in birth weight.

12.7 Chapter review

12.7.1 Summary

This chapter presented hypothesis tests for means using the t-distribution:

- **One-sample t-test**: Tests whether a population mean μ equals a hypothesized value μ_0. Test statistic: $T = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ with $df = n - 1$.
- **Two-sample t-test**: Tests whether two population means are equal. Test statistic: $T = \frac{(\bar{x}_1-\bar{x}_2)}{\sqrt{s_1^2/n_1+s_2^2/n_2}}$ with $df = \min(n_1 - 1, n_2 - 1)$.
- **Paired t-test**: Tests whether the mean of paired differences equals zero. Compute differences first, then apply the one-sample t-test to the differences: $T = \frac{\bar{x}_{diff}}{s_{diff}/\sqrt{n_{diff}}}$ with $df = n_{diff} - 1$.
- All three tests require **independence** and **approximate normality or large sample size**.
- **Simulation-based and t-based tests** generally agree when conditions are met. When conditions are questionable, simulation is more reliable.
- **Statistical significance** (small p-value) does not imply **practical significance** (large enough to matter). Effect sizes like **Cohen's d** help assess practical importance.

12.7.2 Key terms

T test statistic, one-sample t-test, two-sample t-test, paired t-test, degrees of freedom, p-value, significance level (α), statistical significance, practical significance, effect size, Cohen's d.

12.8 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Find the p-value, I.** A random sample is selected from an approximately normal population with an unknown standard deviation. Find the p-value for the given sample size and test statistic. Also determine if the null hypothesis would be rejected at $\alpha = 0.05$.

a. $n = 11, T = 1.91$
b. $n = 17, T = -3.45$
c. $n = 7, T = 0.83$
d. $n = 28, T = 2.13$

2. **Find the p-value, II.** A random sample is selected from an approximately normal population with an unknown standard deviation. Find the p-value for the given sample size and test statistic. Also determine if the null hypothesis would be rejected at $\alpha = 0.01$.

a. $n = 26, T = 2.485$

b. $n = 18, T = 0.5$

3. **Length of gestation, hypothesis test.** In this exercise we work with a random sample of 1,000 cases from the dataset released by the United States Department of Health and Human Services in 2014. Provided below are sample statistics for gestation (length of pregnancy, measured in weeks) of births in this sample.

Min	Q1	Median	Mean	Q3	Max	SD	IQR
21	38	39	38.7	40	46	2.6	2

a. What is the point estimate for the average length of pregnancy for all women? What about the median?

b. You might have heard that human gestation is typically 40 weeks. Using the data, perform a complete hypothesis test, using mathematical models, to assess the 40 week claim. State the null and alternative hypotheses, find the T score, find the p-value, and provide a conclusion in context of the data.

c. A quick internet search validates the claim of "40 weeks gestation" for humans. A friend of yours claims that there are different ways to measure gestation (starting at first day of last period, ovulation, or conception) which will result in estimates that are a week or two different. Another friend mentions that recent increases in cesarean births is likely to have decreased length of gestation. Do the data provide a mechanism to distinguish between your two friends' claims?

4. **Interpreting p-values for population mean.** For each of the following statements, indicate if they are a true or false interpretation of the p-value. If false, provide a reason or correction to the misinterpretation. You are wondering if the average amount of cereal in a 10oz cereal box is greater than 10oz. You collect 50 boxes of cereal, weigh them carefully, find a T score, and a p-value of 0.23.

a. The probability that the average weight of all cereal boxes is 10 oz is 0.23.

b. The probability that the average weight of all cereal boxes is greater than 10 oz is 0.23.

c. Because the p-value is 0.23, the average weight of all cereal boxes is 10 oz.

d. Because the p-value is small, the population average must be just barely above 10 oz.

e. If H_0 is true, the probability of observing another sample with an average as or more extreme as the data is 0.23.

5. **Sleep habits of New Yorkers.** New York is known as "the city that never sleeps". A random sample of 25 New Yorkers were asked how much sleep they get per night. Statistical summaries of these data are shown below. The point estimate suggests New Yorkers sleep less than 8 hours a night on average. Evaluate the claim that New York is the city that never sleeps keeping in mind that, despite this claim, the true average number of hours New Yorkers sleep could be less than 8 hours or more than 8 hours.

n	Mean	SD	Min	Max
25	7.73	0.77	6.17	9.78

a. Write the hypotheses in symbols and in words.

b. Check conditions, then calculate the test statistic, T, and the associated degrees of freedom.

c. Find and interpret the p-value in this context. Drawing a picture may be helpful.

d. What is the conclusion of the hypothesis test?

e. If you were to construct a 90% confidence interval that corresponded to this hypothesis test, would you expect 8 hours to be in the interval?

6. **Find the mean.** You are given the hypotheses shown below. We know that the sample standard deviation is 8 and the sample size is 20. For what sample mean would the p-value be equal to 0.05? Assume that all conditions necessary for inference are satisfied.

$$H_0 : \mu = 60 \qquad H_A : \mu \neq 60$$

7. **Play the piano.** Georgianna claims that in a small city renowned for its music school, the average child takes less than 5 years of piano lessons. We have a random sample of 20 children from the city, with a mean of 4.6 years of piano lessons and a standard deviation of 2.2 years.

a. Evaluate Georgianna's claim (or that the opposite might be true) using a hypothesis test.

b. Construct a 95% confidence interval for the number of years students in this city take piano lessons, and interpret it in context of the data.

c. Do your results from the hypothesis test and the confidence interval agree? Explain your reasoning.

8. **Auto exhaust and lead exposure.** Researchers interested in lead exposure due to car exhaust sampled the blood of 52 police officers subjected to constant inhalation of automobile exhaust fumes while working traffic enforcement in a primarily urban environment. The blood samples of these officers had an average lead concentration of 124.32 μg/l and a SD of 37.74 μg/l; a previous study of individuals from a nearby suburb, with no history of exposure, found an average blood level concentration of 35 μg/l. (Mortada et al. 2000)

a. Write down the hypotheses that would be appropriate for testing if the police officers appear to have been exposed to a different concentration of lead.

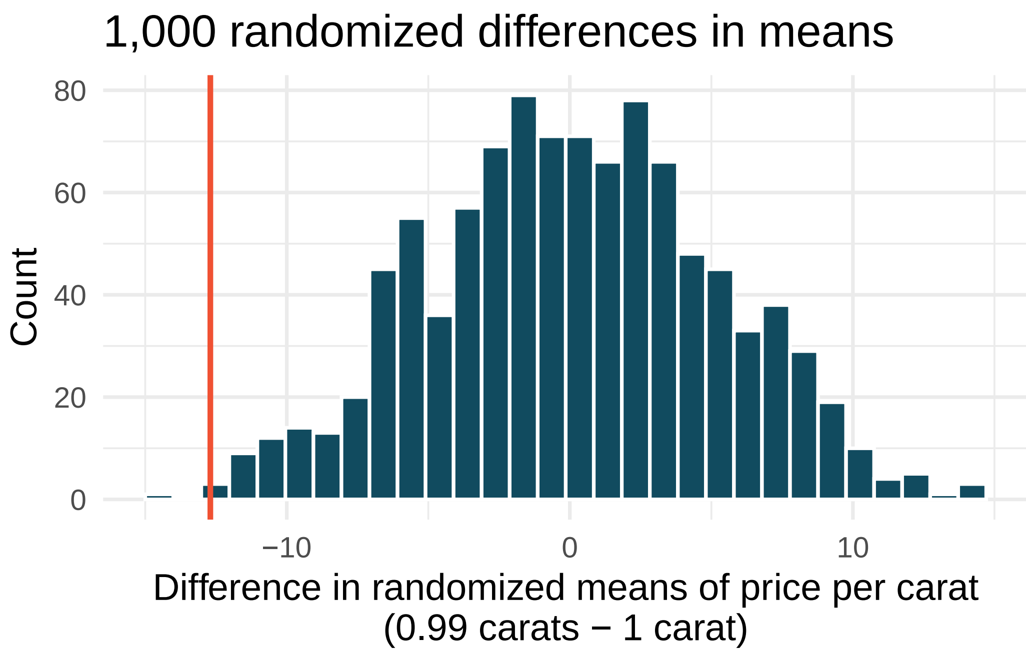
b. Explicitly state and check all conditions necessary for inference on these data.

c. Test the hypothesis that the downtown police officers have a higher lead exposure than the group in the previous study. Interpret your results in context.

9. **Experimental baker.** A baker working on perfecting their bagel recipe is experimenting with active dry (AD) and instant (I) yeast. They bake a dozen bagels with each type of yeast and score each bagel on a scale of 1 to 10 on how well the bagels rise. They come up with the following set of hypotheses for evaluating whether there is a difference in the average rise of bagels baked with active dry and instant yeast. What is wrong with the hypotheses as stated?

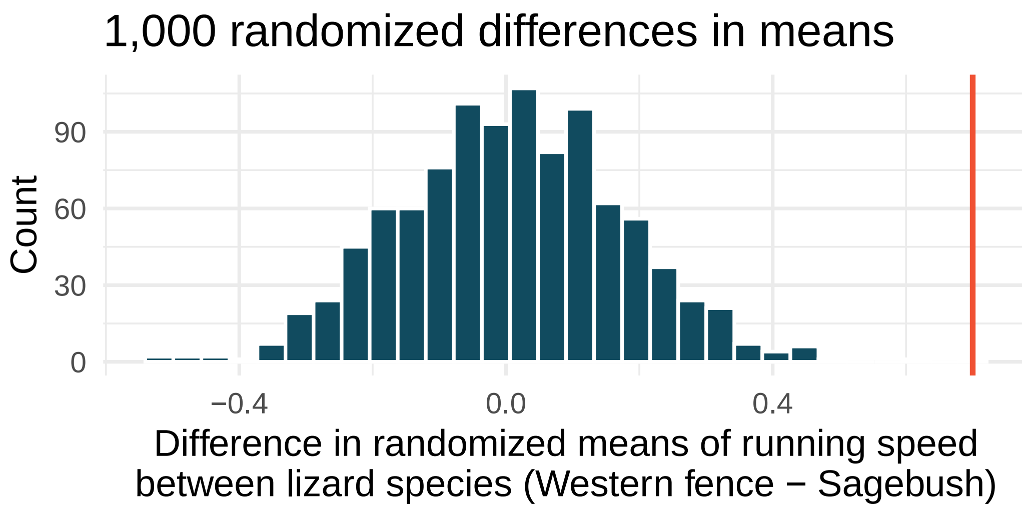
$$H_0 : \bar{x}_{AD} \leq \bar{x}_I \qquad H_A : \bar{x}_{AD} > \bar{x}_I$$

10. **Diamonds, randomization test.** The prices of diamonds go up as the carat weight increases, but the increase is not smooth. For example, the difference between the size of a 0.99 carat diamond and a 1 carat diamond is undetectable to the naked human eye, but the price of a 1 carat diamond tends to be much higher than the price of a 0.99 carat diamond. We have two random samples of diamonds: 23 0.99 carat diamonds and 23 1 carat diamonds. In order to be able to compare equivalent units, we first divide the price for each diamond by 100 times its weight in carats. That is, for a 0.99 carat diamond, we divide the price by 99 and for a 1 carat diamond, we divide it by 100. Then, we randomize the carat weight to the price values in order simulate the null distribution of differences in average prices of 0.99 carat and 1 carat diamonds. The null distribution (with 1,000 randomized differences) is shown below and depicts the distribution of differences in sample means (of price per carat) if there really was no difference in the population from which these diamonds came. (Wickham 2016)



Using the randomization distribution, conduct a hypothesis test to evaluate if there is a difference between the prices per carat of diamonds that weigh 0.99 carats and diamonds that weigh 1 carat. Make sure to state your hypotheses clearly and interpret your results in context of the data. (Wickham 2016)

11. **Lizards running, randomization test.** In order to assess physiological characteristics of common lizards, data on top speeds (in m/sec) measured on a laboratory race track for two species of lizards: Western fence lizard (Sceloporus occidentalis) and Sagebrush lizard (Sceloporus graciosus). The original observed difference in lizard speeds is $\bar{x}_{Western fence} - \bar{x}_{Sagebrush} = 0.7$m/sec. The histogram below shows the distribution of average differences when speed has been randomly allocated across lizard species 1,000 times. Using the randomization distribution, conduct a hypothesis test to evaluate if there is a difference between the average speed of the Western fence lizard as compared to the Sagebrush lizard. Make sure to state your hypotheses clearly and interpret your results in context of the data. (Adolph 1987)



12. **Possible randomized means.** Data were collected on data from two groups (A and B). There were 3 measurements taken on Group A and two measurements in Group B.

Group	Measurement 1	Measurement 2	Measurement 3
A	1	15	5
B	7	3	NA

If the data are (repeatedly) randomly allocated across the two conditions, provide the following: (1) the values which are assigned to group A, (2) the values which are assigned to group B, and (3) the difference in averages ($\bar{x}_A - \bar{x}_B$) for each of the following:

a. When the randomized difference in averages is as large as possible.

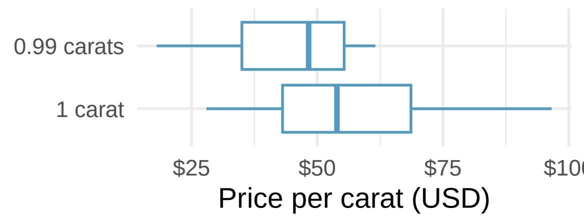
b. When the randomized difference in averages is as small as possible (a big in magnitude negative number).

c. When the randomized difference in averages is as close to zero as possible.

d. When the observed values are randomly assigned to the two groups, to which of the previous parts would you expect the difference in means to fall closest? Explain your reasoning.

13. **Diamonds, mathematical test.** We have data on two random samples of diamonds: one with diamonds that weigh 0.99 carats and one with diamonds that weigh 1 carat. Each sample has 23 diamonds. Sample statistics for the price per carat of diamonds in each sample are provided below. Conduct a hypothesis test using a mathematical model to evaluate if there is a difference between the prices per carat of diamonds that weigh 0.99 carats and diamonds that weigh 1 carat Make sure to state your hypotheses clearly, check relevant conditions, and interpret your results in context of the data. (Wickham 2016)

	Mean	SD	n
0.99 carats	\$44.51	\$13.32	23
1 carat	\$57.20	\$18.19	23



14. **A/B testing.** A/B testing is a user experience research methodology where two variants of a page are shown to users at random. A company wants to evaluate whether users will spend more time, on average, on Page A or Page B using an A/B test. Two user experience designers at the company, Lucie and Müge, are tasked with conducting the analysis of the data collected. They agree on how the null hypothesis should be set: on average, users spend the same amount of time on Page A and Page B. Lucie believes that Page B will provide a better experience for users and hence wants to use a one-tailed test, Müge believes that a two-tailed test would be a better choice. Which designer do you agree with, and why?

15. **Mindfulness intervention for nurses.** In order to address extremely challenging and stressful situations for intensive care unit nurses, researchers ran a mindfulness-based intervention (MBI) study on 60 nurses working in three hospitals in El-Beheira, Egypt. The participants were randomly allocated to one of the two groups: the treatment group (MBI) received 8 MBI sessions and the control group received no intervention. The nurses' emotional exhaustion was measured using 9 items from a questionnaire of the Maslach Burnout Inventory-Human Services Survey for Medical Personnel; the questions are recorded on a Likert scale where 0 indicated "Never" and 6 indicates "Every day". Nurses in the treatment group had an emotional exhaustion score of 15.47, with a standard deviation of 4.44, and nurses in the control group had an emotional exhaustion score of 32.43, with a standard deviation of 8.87. Do these data provide convincing evidence that the emotional exhaustion decrease is different for the patients in the treatment group compared to the control group? Assume that conditions for conducting inference using mathematical models are satisfied. (Othman, Hassan, and Mohamed 2023)

16. **Chicken diet: horsebean vs. linseed.** Chicken farming is a multi-billion dollar industry, and any methods that increase the growth rate of young chicks can reduce consumer costs while increasing company profits, possibly by millions of dollars. An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens. Newly hatched chicks were randomly allocated into six groups, and each group was given a different feed supplement. In this exercise we consider chicks that were fed horsebean and linseed. Below are some summary statistics from this dataset along with box plots showing the distribution of weights by feed type. (McNeil 1977)

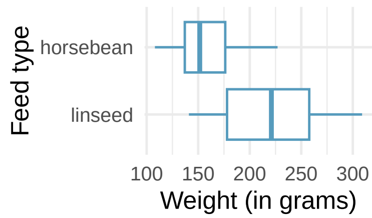
a. Describe the distributions of weights of chickens that were fed horsebean and linseed.

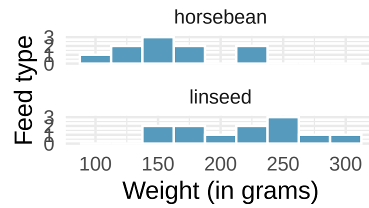
b. Do these data provide strong evidence that the average weights of chickens that were fed linseed and horsebean are different? Use a 5% discernibility level.

c. What type of error might we have committed? Explain.

d. Would your conclusion change if we used $\alpha = 0.01$?

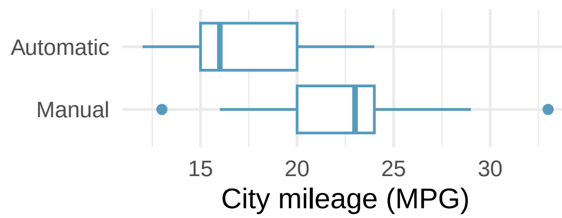
	Horsebean	Linseed
Mean	160.2	218.8
SD	38.6	52.2
n	10.0	12.0





17. **Fuel efficiency in the city.** Each year the US Environmental Protection Agency (EPA) releases fuel economy data on cars manufactured in that year. Below are summary statistics on fuel efficiency (in miles/gallon) from random samples of cars with manual and automatic transmissions manufactured in 2021. Do these data provide strong evidence of a difference between the average fuel efficiency of cars with manual and automatic transmissions in terms of their average city mileage? (US DOE EPA 2021)

CITY	Mean	SD	n
Automatic	17.4	3.44	25
Manual	22.7	4.58	25

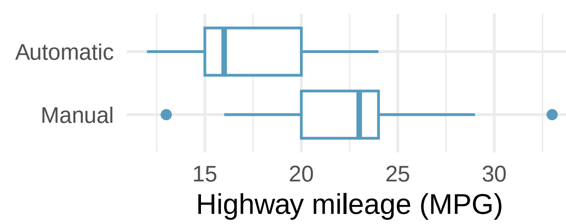


18. **Chicken diet: casein vs. soybean.** Casein is a common weight gain supplement for humans. Does it have an effect on chickens? An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens. Newly hatched chicks were randomly allocated into six groups, and each group was given a different feed supplement. In this exercise we consider chicks that were fed casein and soybean. Assume that the conditions for conducting inference using mathematical models are met, and using the data provided below, test the hypothesis that the average weight of chickens that were fed casein is different than the average weight of chickens that were fed soybean. If your hypothesis test yields a statistically discernible result, discuss whether the higher average weight of chickens can be attributed to the casein diet. (McNeil 1977)

Feed type	Mean	SD	n
casein	323.58	64.43	12
soybean	246.43	54.13	14

19. **Fuel efficiency on the highway.** Each year the US Environmental Protection Agency (EPA) releases fuel economy data on cars manufactured in that year. Below are summary statistics on fuel efficiency (in miles/gallon) from random samples of cars with manual and automatic transmissions manufactured in 2021. Do these data provide strong evidence of a difference between the average fuel efficiency of cars with manual and automatic transmissions in terms of their average highway mileage? (US DOE EPA 2021)

HIGHWAY	Mean	SD	n
Automatic	23.7	3.90	25
Manual	30.9	5.13	25

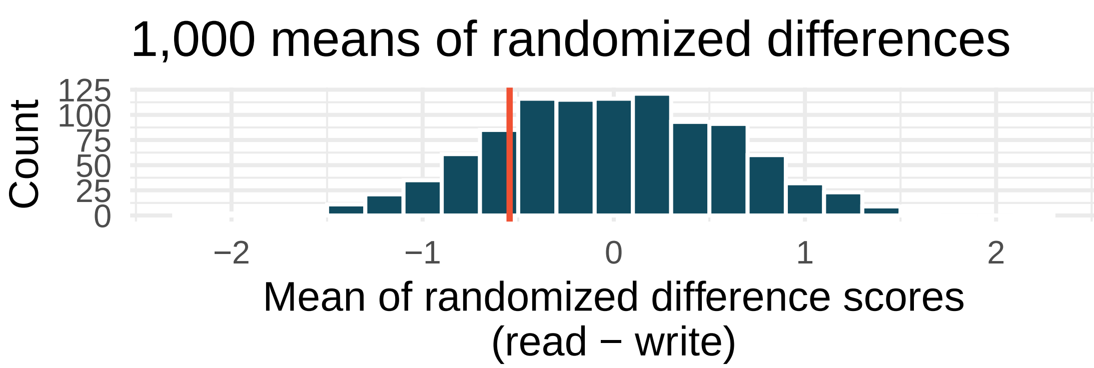


20. **Gaming, distracted eating, and intake.** A group of researchers who are interested in the possible effects of distracting stimuli during eating, such as an increase or decrease in the amount of food consumption, monitored food intake for a group of 44 patients who were randomized into two equal groups. The treatment group ate lunch while playing solitaire, and the control group ate lunch without any added distractions. Patients in the treatment group ate 52.1 grams of biscuits, with a standard deviation of 45.1 grams, and patients in the control group ate 27.1 grams of biscuits, with a standard deviation of 26.4 grams. Do these data provide convincing evidence that the average food intake (measured in amount of biscuits consumed) is different for the patients in the treatment group compared to the control group? Assume that conditions for conducting inference using mathematical models are satisfied. (Oldham-Cooper et al. 2011)

21. **Gaming, distracted eating, and recall.** A group of researchers who are interested in the possible effects of distracting stimuli during eating, such as an increase or decrease in the amount of food consumption, monitored food intake for a group of 44 patients who were randomized into two equal groups. The 22 patients in the treatment group who ate their lunch while playing solitaire were asked to do a serial-order recall of the food lunch items they ate. The average number of items recalled by the patients in this group was 4. 9, with a standard deviation of 1.8. The average number of items recalled by the patients in the control group (no distraction) was 6.1, with a standard deviation of 1.8. Do these data provide strong evidence that the average numbers of food items recalled by the patients in the treatment and control groups are different? Assume that conditions for conducting inference using mathematical models are satisfied. (Oldham-Cooper et al. 2011)

22. **High School and Beyond, randomization test.** The National Center of Education Statistics conducted a survey of high school seniors, collecting test data on reading, writing, and several other subjects. Here we examine a simple random sample of 200 students from this survey.

Side-by-side box plots of reading and writing scores as well as a histogram of the differences in scores are shown below. Also provided below is a histogram of randomized averages of paired differences of scores (read - write), with the observed difference ($\bar{x}_{read-write} = -0.545$) marked with a red vertical line. The randomization distribution was produced by doing the following 1000 times: for each student, the two scores were randomly assigned to either read or write, and the average was taken across all students in the sample.



a. Is there a clear difference in the average reading and writing scores?

b. Are the reading and writing scores of each student independent of each other?

c. Create hypotheses appropriate for the following research question: is there an evident difference in the average scores of students in the reading and writing exam?

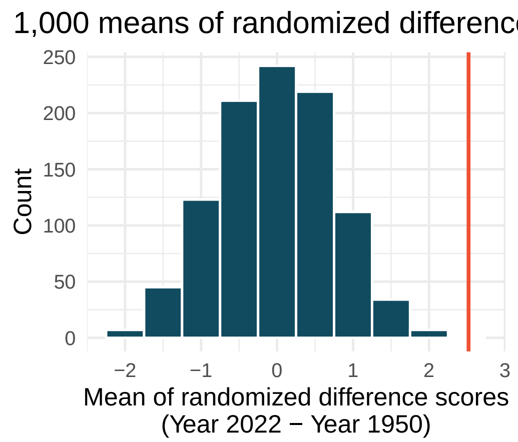
d. Is the average of the observed difference in scores ($\bar{x}_{read-write} = -0.545$) consistent with the distribution of randomized average differences? Explain.

e. Do these data provide convincing evidence of a difference between the average scores on the two exams? Estimate the p-value from the randomization test, and conclude the hypothesis test using words like "score on reading test" and "score on writing test."

23. **Global warming, randomization test.** Let's consider a limited set of climate data, examining temperature differences in 1950 vs 2022. We sampled 26 locations in the US from the National Oceanic and Atmospheric Administration's (NOAA) historical data, where the data was available for both years of interest. (NOAA 2023) The data are not a random sample, but they are selected to be a representative sample across the land area of the lower 48 United States. Using the hottest day of the year as a measure can make the results susceptible to outliers. Instead, to get a sense for how hot a year was, we calculate the 90^{th} percentile; that is, we find the maximum temperature on the day that was hotter than 90% of the days that year. We want to know: is the 90^{th} percentile high temperature greater in 2022 or in 1950? The difference in 90^{th} percentile high temperature (high temperature for 2022 - high temperature for 1950) was calculated for each of the 26 locations. The average of the 26 differences was 2.52°F with a standard deviation of 2.95°F. We are interested in determining whether these data provide strong evidence that the 90^{th} percentile high temperature is higher in 2022 than in 1950.

a. Create hypotheses appropriate for the following research question: is there an evident difference in the 90^{th} percentile high temp across the two years (1950 and 2022)?

b. Is the average of the observed difference in scores ($\bar{x}_{2022-1950} = 2.53^\circ F$) consistent with the distribution of randomized average differences? Explain.



c. Do these data provide convincing evidence of a difference between the 90^{th} percentile high temperature? Estimate the p-value from the randomization test, and conclude the hypothesis test using words like "90^{th} percentile high temperature in 1950" and "90^{th} percentile high temperature in 2022."

24. **High School and Beyond, mathematical test.** We considered the differences between the reading and writing scores of a random sample of 200 students who took the High School and Beyond Survey.

a. Create hypotheses appropriate for the following research question: is there an evident difference in the average scores of students in the reading and writing exam?

b. Check the conditions required to complete this test.

c. The average observed difference in scores is $\bar{x}_{read-write} = -0.545$, and the standard deviation of the differences is $s_{read-write} = 8.887$ points. Do these data provide convincing evidence of a difference between the average scores on the two exams?

d. What type of error might we have made? Explain what the error means in the context of the application.

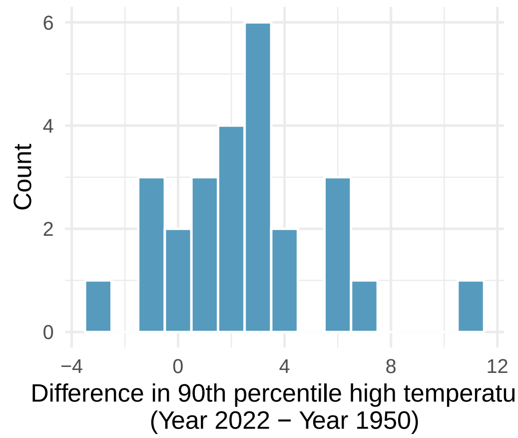
e. Based on the results of this hypothesis test, would you expect a confidence interval for the average difference between the reading and writing scores to include 0? Explain your reasoning.

25. **Global warming, mathematical test.** We considered the change in the 90^{th} percentile high temperature in 1950 versus 2022 at 26 sampled locations from the NOAA database. (NOAA 2023) The mean and standard deviation of the reported differences are 2.53°F and 2.95°F.

    a. Is there a relationship between the observations collected in 1950 and 2022? Or are the observations in the two groups independent? Explain your reasoning.

    b. Write hypotheses for this research in symbols and in words.

    c. Check the conditions required to complete this test.



    d. Calculate the test statistic and find the p-value.

    e. Use $\alpha = 0.05$ to evaluate the test, and interpret your conclusion in context.

    f. What type of error might we have made? Explain in context what the error means.

    g. Based on the results of this hypothesis test, would you expect a confidence interval for the average difference between the 90^{th} percentile high temperture from 1950 to 2022 to include 0? Explain your reasoning.

26. **Possible paired randomized differences.** Two observations were collected on each of five people. Which of the following could be a possible randomization of the paired differences given in the table below? If the set of values could not be a randomized set of differences, indicate why not.

    a. -2, 1, 1, 11, -2
    b. -4, 11, -2, 0, 1
    c. -2, 2, -11, 11, -2, 2, 0, 1, -1
    d. 0, -1, 2, -4, 11
    e. 4, -11, 2, 0, -1

	People				
	1	2	3	4	5
Observation 1	3	14	4	5	10
Observation 2	7	3	6	5	9
Difference	-4	11	-2	0	1

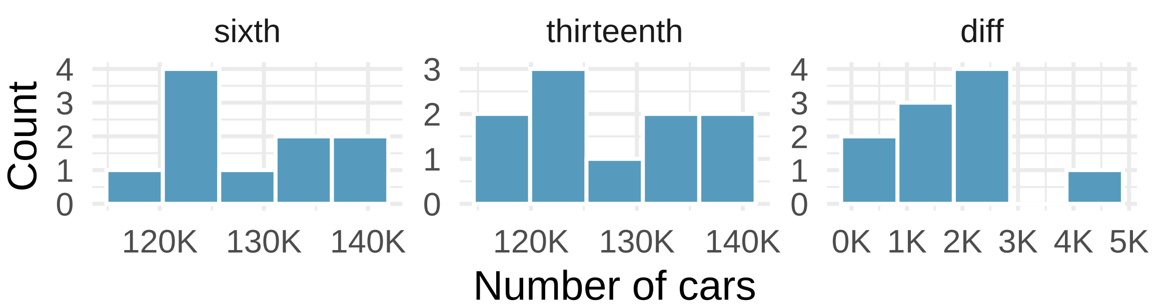
27. **Study environment.** In order to test the effects of listening to music while studying versus studying in silence, students agree to be randomized to two treatments (i.e., study with music or study in silence). There are two exams during the semester, so the researchers can either randomize the students to have one exam with music and one with silence (randomly selecting which exam corresponds to which study environment) or the researchers can randomize the students to one study habit for both exams.

The researchers are interested in estimating the true population difference of exam score for those who listen to music while studying as compared to those who study in silence.

a. Describe the experiment which is consistent with a paired designed experiment. How is the treatment assigned, and how are the data collected such that the observations are paired?

b. Describe the experiment which is consistent with an indpenedent samples experiment. How is the treatment assigned, and how are the data collected such that the observations are independent?

28. **Friday the 13th, traffic.** In the early 1990's, researchers in the UK collected data on traffic flow on Friday the 13th with the goal of addressing issues of how superstitions regarding Friday the 13th affect human behavior and and whether Friday the 13th is an unlucky day. The histograms below show the distributions of numbers of cars passing by a specific intersection on Friday the 6th and Friday the 13th for many such date pairs. Also provided are some sample statistics, where the difference is the number of cars on the 6th minus the number of cars on the 13th. (Scanlon et al. 1993)



	n	Mean	SD
sixth	10	128,385	7,259
thirteenth	10	126,550	7,664
diff	10	1,836	1,176

a. Are there any underlying structures in these data that should be considered in an analysis? Explain.

b. What are the hypotheses for evaluating whether the number of people out on Friday the 6^{th} is different than the number out on Friday the 13^{th}?

c. Check conditions to carry out the hypothesis test from part (b) using mathematical models.

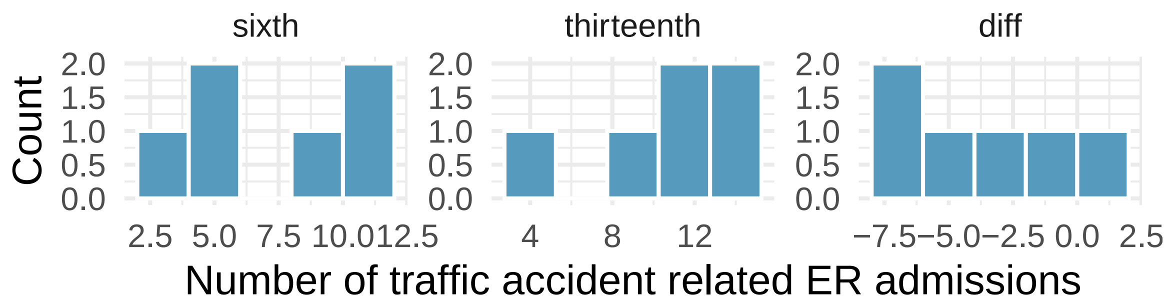
d. Calculate the test statistic and the p-value.

e. What is the conclusion of the hypothesis test?

f. Interpret the p-value in this context.

g. What type of error might have been made in the conclusion of your test? Explain.

29. **Friday the 13th, accidents.** In the early 1990's, researchers in the UK collected data the number of traffic accident related emergency room (ER) admissions on Friday the 13th with the goal of addressing issues of how superstitions regarding Friday the 13th affect human behavior and and whether Friday the 13th is an unlucky day. The histograms below show the distributions of numbers of ER admissions at specific emergency rooms on Friday the 6th and Friday the 13th for many such date pairs. Also provided are some sample statistics, where the difference is the ER admissions on the 6th minus the ER admissions on the 13th.(Scanlon et al. 1993)



	n	Mean	SD
sixth	6	8	3
thirteenth	6	11	4
diff	6	-3	3

a. Conduct a hypothesis test using mathematical models to evaluate if there is a difference between the average numbers of traffic accident related emergency room admissions between Friday the 6^{th} and Friday the 13^{th}.

b. Calculate a 95% confidence interval using mathematical models for the difference between the average numbers of traffic accident related emergency room admissions between Friday the 6^{th} and Friday the 13^{th}.

c. The conclusion of the original study states, "Friday 13th is unlucky for some. The risk of hospital admission as a result of a transport accident may be increased by as much as 52%. Staying at home is recommended." Do you agree with this statement? Explain your reasoning.

Datasets: `births14` (openintro) | `diamonds` (ggplot2) | `lizard_run` (openintro) | `chickwts` (datasets) | `epa2021` (openintro) | `hsb2` (openintro) | `us_temperature` (openintro) | `friday` (openintro)

Chapter 13

Inference for Proportions

> **Draft Chapter.** This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In earlier chapters, we used simulation-based methods to draw conclusions about population proportions. In Chapters 7 and 8, we built randomization distributions and bootstrap confidence intervals; in Chapter 10, we learned that the normal distribution provides a mathematical model for these sampling distributions when certain conditions are met. In this chapter, we bring everything together for proportions. We develop normal-based confidence intervals and hypothesis tests for a single proportion and for the difference between two proportions. Throughout, we compare these formula-based results to the simulation-based results from earlier chapters, reinforcing the idea that these are two routes to the same destination.

StatLens tools for this chapter: Use the **Bootstrap CI for One Proportion** and **Permutation Test Simulator** alongside the formula-based methods in this chapter to see both approaches side by side.

Launch Bootstrap CI Tool

13.1 Normal approximation for proportions

13.1.1 The sampling distribution of $\hat{p}$

In Chapter 10, we saw that the Central Limit Theorem guarantees that the sampling distribution of $\hat{p}$ is approximately normal when certain conditions are met. Let's state these conditions precisely.

Sampling distribution of $\hat{p}$.

The sampling distribution of the sample proportion $\hat{p}$, based on a sample of size n from a population with true proportion p, is approximately normal when:

1. **Independence.** The observations are independent (e.g., from a simple random sample).
2. **Success-failure condition.** We expect to see at least 10 successes and 10 failures: $np \geq 10$ and $n(1-p) \geq 10$.

When these conditions are met, the sampling distribution is approximately:

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

The **success-failure condition** requires that the expected number of successes (np) and the expected number of failures ($n(1-p)$) are both at least 10. This ensures the sampling distribution of $\hat{p}$ is not too skewed for the normal approximation to work well.

A medical consultant facilitated 62 liver transplant surgeries, with 3 complications. Under the null hypothesis that the true complication rate is $p_0 = 0.10$, is it appropriate to use the normal approximation?

Check the success-failure condition using $p_0 = 0.10$:

- Expected successes (complications): $np_0 = 62 \times 0.10 = 6.2$
- Expected failures: $n(1-p_0) = 62 \times 0.90 = 55.8$

The expected number of successes (6.2) is less than 10. The **success-failure condition is not met**, so the normal approximation should not be used here. A simulation-based approach (such as the parametric bootstrap from Chapter 8) would be more appropriate.

A poll surveys 826 randomly selected payday loan borrowers and finds that 51% support a new regulation. Check whether the normal approximation can be used to model $\hat{p}$ for a hypothesis test with $H_0 : p = 0.50$.

Show answer

Check the success-failure condition using $p_0 = 0.50$: $np_0 = 826 \times 0.50 = 413$ and $n(1-p_0) = 826 \times 0.50 = 413$. Both are well above 10, so the success-failure condition is met. Combined with the random sample (independence), the normal approximation is appropriate.

13.1.2 The standard error of $\hat{p}$

Standard error of $\hat{p}$.

When conditions are met, the variability of $\hat{p}$ is described by:

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Since we rarely know the true p:

- **For confidence intervals**, use $\hat{p}$ as the best guess: $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
- **For hypothesis tests**, use p_0 (the value from H_0): $SE = \sqrt{\frac{p_0(1-p_0)}{n}}$

Consider polls of size 300 where approximately 2/3 of voters support legalized marijuana. Calculate the standard error of $\hat{p}$.

Show answer

Using $p \approx 2/3$: $SE = \sqrt{\frac{(2/3)(1/3)}{300}} = \sqrt{\frac{2/9}{300}} = \sqrt{0.000741} = 0.027$.

13.2 Confidence interval for a single proportion

13.2.1 The formula

Confidence interval for a single proportion p.

$$\hat{p} \pm z^* \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where z^* is the critical value for the desired confidence level (e.g., $z^* = 1.96$ for 95%).

Steps:

1. Check conditions: independence and success-failure using $\hat{p}$.
2. Compute SE using $\hat{p}$.
3. Apply the confidence interval formula.

13.2.2 Worked example: Payday loan regulation

A simple random sample of 826 payday loan borrowers was surveyed, and 70% supported new regulations on payday lenders. Construct a 95% confidence interval for the true proportion who support the regulations.

Step 1: Check conditions.

- *Independence.* The data are a random sample.
- *Success-failure condition.* Using $\hat{p} = 0.70$: $n\hat{p} = 826 \times 0.70 = 578$ and $n(1-\hat{p}) = 826 \times 0.30 = 248$. Both are well above 10.

Step 2: Compute SE.

$$SE = \sqrt{\frac{0.70 \times 0.30}{826}} = \sqrt{\frac{0.21}{826}} = \sqrt{0.000254} = 0.016$$

Step 3: Construct the interval.

$$0.70 \pm 1.96 \times 0.016 = 0.70 \pm 0.031 = (0.669, 0.731)$$

We are 95% confident that between 66.9% and 73.1% of all payday loan borrowers support the new regulation.

13.2.3 Changing the confidence level

The confidence level is controlled by the choice of z^*:

Table 13.1: Critical values for common confidence levels.

Confidence level	z^*	Interval width
90%	1.645	Narrower
95%	1.960	Moderate
99%	2.576	Wider

Higher confidence requires a wider interval – we must cast a wider net to be more confident we have captured the true parameter.

Using the payday loan data ($\hat{p} = 0.70$, $n = 826$, $SE = 0.016$), construct a 99% confidence interval.

Show answer

Using $z^* = 2.576$: $0.70 \pm 2.576 \times 0.016 = 0.70 \pm 0.041 = (0.659, 0.741)$. The 99% interval is wider than the 95% interval, as expected.

A study on stent surgery found a point estimate of 0.090 (the difference in stroke rates) with $SE = 0.028$. Construct a 90% confidence interval.

Show answer

Using $z^* = 1.645$: $0.090 \pm 1.645 \times 0.028 = 0.090 \pm 0.046 = (0.044, 0.136)$. We are 90% confident that stent implantation increases the 30-day stroke rate by between 4.4% and 13.6%.

General confidence interval formula. If a point estimate follows a normal model with standard error SE, then a confidence interval for the population parameter is:

$$\text{point estimate} \pm z^* \times SE$$

where z^* corresponds to the chosen confidence level.

13.3 Hypothesis test for a single proportion

13.3.1 The Z test for a proportion

The test statistic for a single proportion is a Z.

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

where $\hat{p}$ is the sample proportion and p_0 is the hypothesized proportion from H_0.

When H_0 is true and conditions are met, Z follows a standard normal distribution $N(0, 1)$.

Conditions:

- Independent observations
- $np_0 \geq 10$ and $n(1 - p_0) \geq 10$

Note that for hypothesis tests, the standard error is computed using p_0 (not $\hat{p}$), because we are evaluating the data under the assumption that H_0 is true.

13.3.2 Worked example: Support for credit checks

From a random sample of 826 payday loan borrowers, 51% said they would support a regulation requiring lenders to pull credit reports. Is there convincing evidence that a majority (more than 50%) of borrowers support this regulation?

Step 1: State hypotheses.

- H_0: $p = 0.50$ (no majority support)
- H_A: $p > 0.50$ (majority support)

This is a one-sided test.

Step 2: Check conditions.

- *Independence.* Random sample.
- *Success-failure.* Using $p_0 = 0.50$: $np_0 = 826 \times 0.50 = 413$ and $n(1 - p_0) = 413$. Both ≥ 10.

Step 3: Compute the test statistic and p-value.

$$SE = \sqrt{\frac{0.50 \times 0.50}{826}} = \sqrt{\frac{0.25}{826}} = 0.017$$

$$Z = \frac{0.51 - 0.50}{0.017} = \frac{0.01}{0.017} = 0.59$$

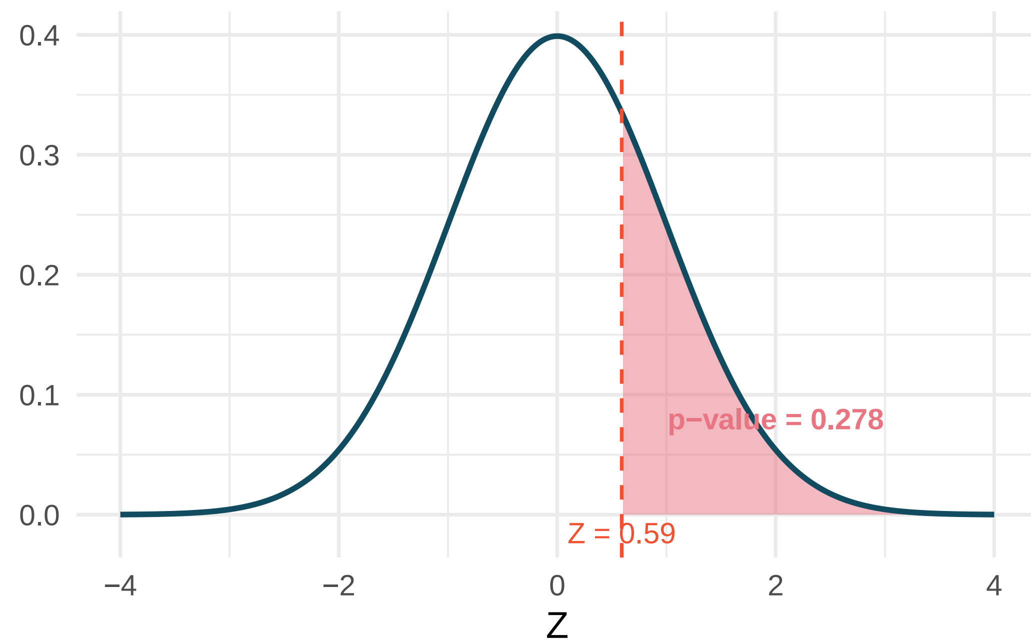


Figure 13.1: A standard normal curve with the area to the right of $Z = 0.59$ shaded, representing the one-sided p-value.

The area to the right of $Z = 0.59$ is 0.278.

Step 4: Conclusion. The p-value of 0.278 is much larger than $\alpha = 0.05$, so we fail to reject H_0. The poll does not provide convincing evidence that a majority of payday loan borrowers support the regulation requiring credit checks.

A candidate claims that more than 60% of voters in a district support her. A random sample of 500 voters finds $\hat{p} = 0.64$. Test the claim at $\alpha = 0.05$.

Show answer

Hypotheses: $H_0 : p = 0.60$, $H_A : p > 0.60$. Success-failure: $500(0.60) = 300$ and $500(0.40) = 200$, both ≥ 10. $SE = \sqrt{0.60 \times 0.40/500} = \sqrt{0.000480} = 0.0219$. $Z = (0.64 - 0.60)/0.0219 = 1.83$. The right-tail area is 0.034. Since $0.034 < 0.05$, we reject H_0. There is sufficient evidence that more than 60% of voters support the candidate.

13.3.3 When conditions fail

When the success-failure condition is not met, the normal model can underestimate the variability of $\hat{p}$, leading to inaccurate p-values and confidence intervals. In these cases, use simulation-based methods:

- **Bootstrap hypothesis test**: Simulate samples under H_0 by drawing from a population with proportion p_0 (the parametric bootstrap from Chapter 8).
- **Bootstrap confidence interval**: Resample from the original data to estimate the variability of $\hat{p}$.

13.4 Difference of two proportions

13.4.1 The parameter and point estimate

When comparing two groups, the parameter of interest is the difference in population proportions: $p_1 - p_2$. The point estimate is $\hat{p}_1 - \hat{p}_2$.

13.4.2 Confidence interval for a difference in proportions

Confidence interval for $p_1 - p_2$.

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Conditions:

1. **Independence (extended).** Observations are independent within and between groups.
2. **Success-failure condition (each group separately).** $n_1\hat{p}_1 \geq 10$, $n_1(1-\hat{p}_1) \geq 10$, $n_2\hat{p}_2 \geq 10$, and $n_2(1-\hat{p}_2) \geq 10$.

In a CPR study, patients who had a heart attack were randomly assigned to receive a blood thinner (treatment) or not (control). The survival rates after 24 hours:

Table 13.2: CPR study results.

Group	n	Survived	$\hat{p}$
Treatment	50	14	0.280
Control	40	11	0.275

Construct a 90% confidence interval for the difference in survival rates, $p_T - p_C$.

Check conditions.

- *Independence.* Patients were randomly assigned, so groups are independent.
- *Success-failure.* Treatment: $50(0.28) = 14 \geq 10$ and $50(0.72) = 36 \geq 10$. Control: $40(0.275) = 11 \geq 10$ and $40(0.725) = 29 \geq 10$. All conditions met.

Compute the interval.

$$\hat{p}_T - \hat{p}_C = 0.280 - 0.275 = 0.005$$

$$SE = \sqrt{\frac{0.280(0.720)}{50} + \frac{0.275(0.725)}{40}} = \sqrt{\frac{0.2016}{50} + \frac{0.1994}{40}} = \sqrt{0.004032 + 0.004985} = \sqrt{0.009017} = 0.095$$

Using $z^* = 1.645$ for 90% confidence:

$$0.005 \pm 1.645 \times 0.095 = 0.005 \pm 0.156 = (-0.151, 0.161)$$

We are 90% confident that the difference in 24-hour survival rates is between -15.1% and $+16.1\%$. Because the interval contains 0, we cannot conclude that the blood thinner affects survival rates.

13.4.3 Hypothesis test for a difference in proportions

The test statistic for comparing two proportions is a Z.

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where the **pooled proportion** is:

$$\hat{p}_{pool} = \frac{\text{total successes in both groups}}{\text{total sample size}} = \frac{x_1 + x_2}{n_1 + n_2}$$

When H_0 is true ($p_1 = p_2$), both groups share the same population proportion, so we pool the data to get a single best estimate.

Conditions:

- Independence within and between groups.
- $n_1\hat{p}_{pool} \geq 10$, $n_1(1 - \hat{p}_{pool}) \geq 10$, $n_2\hat{p}_{pool} \geq 10$, $n_2(1 - \hat{p}_{pool}) \geq 10$.

Does the blood thinner improve 24-hour survival after CPR? Using the data from the previous example (treatment: 14/50 survived; control: 11/40 survived), test at $\alpha = 0.05$.

Step 1: State hypotheses.

- H_0: $p_T - p_C = 0$ (the blood thinner has no effect)
- H_A: $p_T - p_C \neq 0$ (the blood thinner has some effect)

Step 2: Check conditions.

The pooled proportion is $\hat{p}_{pool} = \frac{14+11}{50+40} = \frac{25}{90} = 0.278$. Check:

- $50(0.278) = 13.9 \geq 10$, $50(0.722) = 36.1 \geq 10$
- $40(0.278) = 11.1 \geq 10$, $40(0.722) = 28.9 \geq 10$

Conditions are met.

Step 3: Compute the test statistic and p-value.

$$SE = \sqrt{0.278(0.722)\left(\frac{1}{50} + \frac{1}{40}\right)} = \sqrt{0.2007 \times 0.045} = \sqrt{0.009032} = 0.095$$

$$Z = \frac{0.005 - 0}{0.095} = 0.05$$

The two-sided p-value for $Z = 0.05$ is $2 \times P(Z > 0.05) = 2 \times 0.480 = 0.960$.

Step 4: Conclusion. The p-value of 0.960 is far larger than 0.05. We fail to reject H_0. There is no evidence that the blood thinner affects 24-hour survival rates.

13.4.4 Why pool for hypothesis tests but not for confidence intervals?

You may have noticed a subtle difference:

- **Hypothesis tests** use the **pooled proportion** $\hat{p}_{pool}$ in the standard error. This is because under H_0, we assume $p_1 = p_2$, so the best estimate of this common proportion comes from combining both samples.
- **Confidence intervals** use the **individual sample proportions** $\hat{p}_1$ and $\hat{p}_2$ separately. We are not assuming the proportions are equal – we are estimating how different they are.

This distinction applies only to proportions. For means, we use separate sample standard deviations in both tests and confidence intervals.

13.5 Comparing simulation-based and normal-based methods

Throughout this course, we have developed two parallel approaches to inference:

Table 13.3: Comparison of simulation-based and normal approximation approaches.

Feature	Simulation-based	Normal approximation
Requires conditions?	Minimal (mainly independence)	Independence + success-failure
Small samples?	Works well	May be inaccurate
Computational cost	Higher (needs many simulations)	Lower (one formula)
Conceptual clarity	High (directly simulates the process)	Moderate (requires distribution theory)
Precision	Approximate (varies by number of simulations)	Approximate (varies by how well conditions hold)

13.5.1 An example comparison

Consider the opportunity cost study from Chapter 7. The observed difference in proportions was 0.20 ($\hat{p}_T - \hat{p}_C$), and we wanted to test $H_0 : p_T - p_C = 0$.

Randomization test (Chapter 7): We shuffled the group labels 10,000 times. The p-value was approximately 0.006.

Normal approximation (this chapter): Using $SE = 0.078$ and $Z = 0.20/0.078 = 2.56$, the p-value was 0.005.

Both methods give essentially the same answer and the same conclusion. This is reassuring and typical when conditions are met.

13.5.2 When do they disagree?

The methods can give different answers when:

1. **The success-failure condition fails.** The normal approximation may undercount the probability in the tails, leading to p-values that are too small. The simulation approach remains valid.
2. **The sampling distribution is noticeably skewed.** The normal model is symmetric by definition, while the true sampling distribution of $\hat{p}$ near 0 or 1 is skewed.

The medical consultant example illustrates this: with only 6.2 expected complications, the normal model gave a p-value of 0.085, while the simulation gave 0.122 – nearly 45% higher. The simulation-based p-value is more trustworthy in this case.

13.5.3 Practical recommendation

- **When conditions are met**: Both methods work. Use the normal approximation for its simplicity, or use simulation for its conceptual directness.
- **When conditions are borderline or not met**: Use simulation. The bootstrap and randomization approaches make fewer assumptions and are more reliable in these settings.
- **Always check conditions first.** The success-failure condition is a simple, quick check that helps you decide which approach to use.

13.6 Chapter review

13.6.1 Summary

This chapter developed normal-based inference methods for proportions:

- **Success-failure condition.** The normal approximation for $\hat{p}$ requires at least 10 expected successes and 10 expected failures. For confidence intervals, check with $\hat{p}$; for hypothesis tests, check with p_0.
- **CI for one proportion**: $\hat{p} \pm z^* \times \sqrt{\hat{p}(1-\hat{p})/n}$. Use $\hat{p}$ in the SE.
- **Hypothesis test for one proportion**: $Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$. Use p_0 in the SE.
- **CI for difference of proportions**: $(\hat{p}_1 - \hat{p}_2) \pm z^* \times SE$, where SE uses separate sample proportions.
- **Hypothesis test for difference of proportions**: $Z = \frac{\hat{p}_1-\hat{p}_2}{SE}$, where SE uses the **pooled proportion** $\hat{p}_{pool}$.
- **Confidence level** is controlled by z^*: higher confidence means wider intervals.
- **When conditions fail** (especially the success-failure condition), use simulation-based methods from Chapters 7 and 8 instead.
- **Normal-based and simulation-based methods agree** when conditions are met and can give different answers when conditions are violated.

13.6.2 Key terms

Success-failure condition, standard error of $\hat{p}$, Z test for a proportion, pooled proportion, confidence interval for a proportion, confidence interval for a difference of proportions, margin of error, critical value (z^*), confidence level.

13.7 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Do aliens exist?** In May 2021, YouGov asked 4,839 adult Great Britain residents whether they think aliens exist, and if so, if they have or have not visited Earth. You want to evaluate if more than a quarter (25%) of Great Britain adults think aliens don't exist. In the survey 22% responded "I think they exist, and have visited Earth", 28% responded "I think they exist, but have not visited Earth", 29% responded "I don't think they exist", and 22% responded "Don't know". A friend of yours offers to help you with setting up the hypothesis test and comes up with the following hypotheses. Indicate any errors you see.

$H_0 : \hat{p} = 0.29 \qquad H_A : \hat{p} > 0.29$

2. **Married at 25.** A study suggests that the 25% of 25 year-olds have gotten married. You believe that this is incorrect and decide to collect your own sample for a hypothesis test. From a random sample of 776 25 year-olds, you find that 24% of them are married. A friend of yours offers to help you with setting up the hypothesis test and comes up with the following hypotheses. Indicate any errors you see.

$H_0 : \hat{p} = 0.24 \qquad H_A : \hat{p} \neq 0.24$

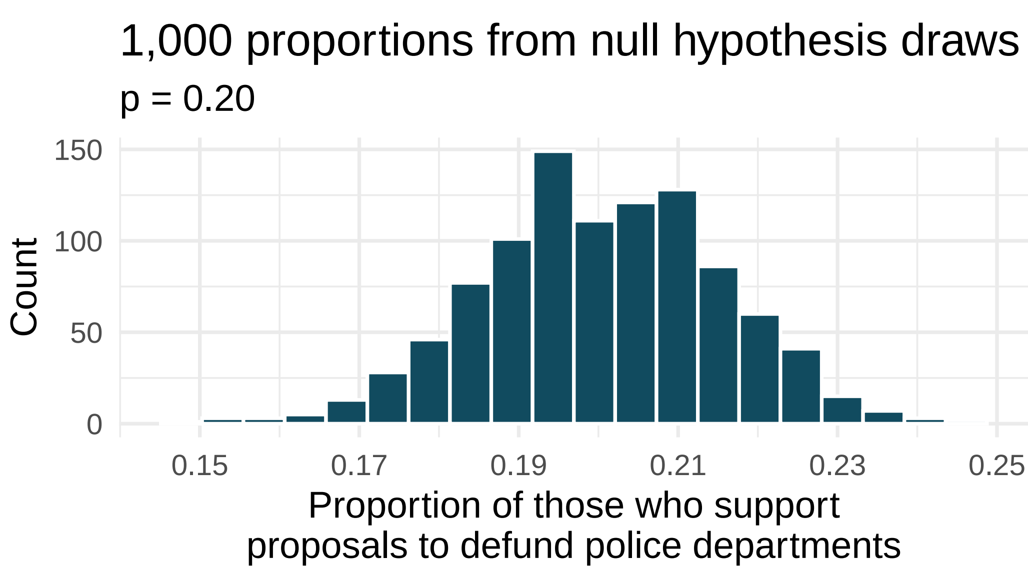
3. **Defund the police.** A Survey USA poll conducted in Seattle, WA in May 2021 reports that of the 650 respondents (adults living in this area), 159 support proposals to defund police departments. (Survey USA 2021)

a. A journalist writing a news story on the poll results wants to use the headline "More than 1 in 5 adults living in Seattle support proposals to defund police departments." You caution the journalist that they should first conduct a hypothesis test to see if the poll data provide convincing evidence for this claim Write the hypotheses for this test.

b. Calculate the proportion of Seattle adults in the sample who support proposals to defund police departments.

c. Describe a setup for a simulation that would be appropriate in this situation and how the p-value can be calculated using the simulation results.

d. The histogram below shows the distribution of 1,000 $\hat{p}_{sim}$s under the null hypothesis. Estimate the p-value using the plot and use it to evaluate the hypotheses.

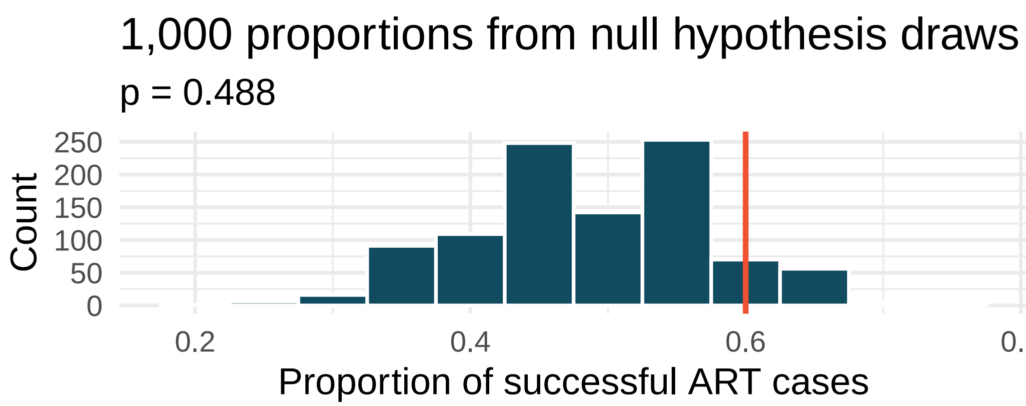


4. **Assisted reproduction.** Assisted Reproductive Technology (ART) is a collection of techniques that help facilitate pregnancy (e.g., in vitro fertilization). The 2018 ART Fertility Clinic Success Rates Report published by the Centers for Disease Control and Prevention reports that ART has been successful in leading to a live birth in 48.8% of cases where the patient is under 35 years old. (CDC 2018) A new fertility clinic claims that their success rate is higher than average for this age group. A random sample of 30 of their patients yielded a success rate of 60%. A consumer watchdog group would like to determine if the data provides strong evidence to support the company's claim.

a. Write the hypotheses to test if the success rate for ART at this clinic is discernibly higher than the success rate reported by the CDC.

b. Describe a setup for a simulation that would be appropriate in this situation and how the p-value can be calculated using the simulation results.

c. The histogram below shows the distribution of 1,000 $\hat{p}_{sim}$s under the null hypothesis. Estimate the p-value using the plot and use it to evaluate the hypotheses.

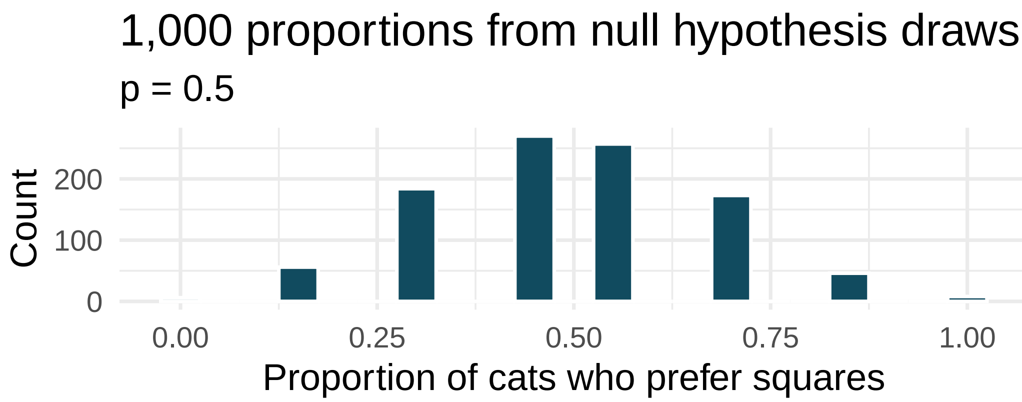


d. After performing this analysis, the consumer group releases the following news headline: "Infertility clinic falsely advertises better success rates". Comment on the appropriateness of this statement.

5. **If I fits, I sits, simulated null hypothesis.** A citizen science project on which type of enclosed spaces cats are most likely to sit in compared (among other options) two different spaces taped to the ground. The first was a square, and the second was a shape known as Kanizsa square illusion. When comparing the two options given to 7 cats, 5 chose the square, and 2 chose the Kanizsa square illusion. We are interested to know whether these data provide convincing evidence that cats prefer one of the shapes over the other. (Smith, Chouinard, and Byosiere 2021)

a. What are the null and alternative hypotheses for evaluating whether these data provide convincing evidence that cats have preference for one of the shapes

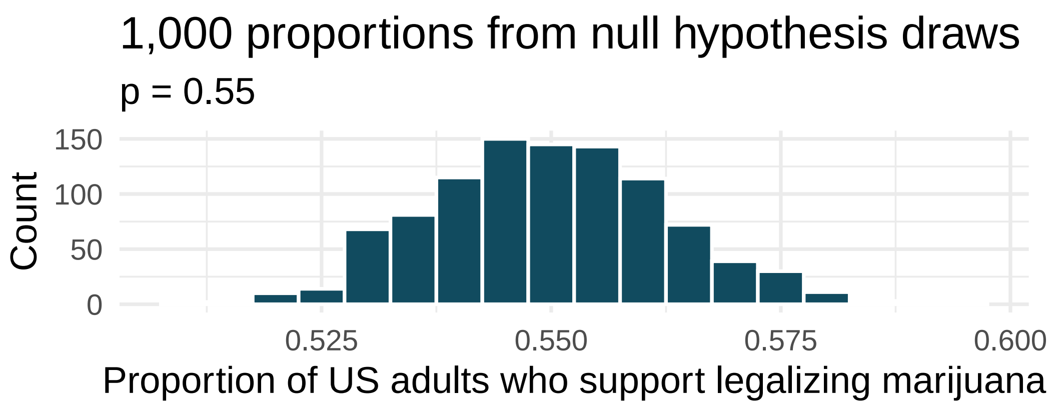
b. A null hypothesis simulation (with 1,000 draws) was run, and the resulting null distribution is displayed in the histogram below. Find the p-value using this distribution and conclude the hypothesis test in the context of the problem.


1,000 proportions from null hypothesis draws
p = 0.5
Count
200
100
0
0.00
0.25
0.50
0.75
1.00
Proportion of cats who prefer squares

6. **Legalization of marijuana, simulated null hypothesis.** The 2022 General Social Survey asked a random sample of 1,207 US adults: "Do you think the use of marijuana should be made legal, or not?" 65.3% of the respondents said it should be made legal. (NORC 2022) Consider a scenario where, in order to become legal, 55% (or more) of voters must approve.

a. What are the null and alternative hypotheses for evaluating whether these data provide convincing evidence that, if voted on, marijuana would be legalized in the US.

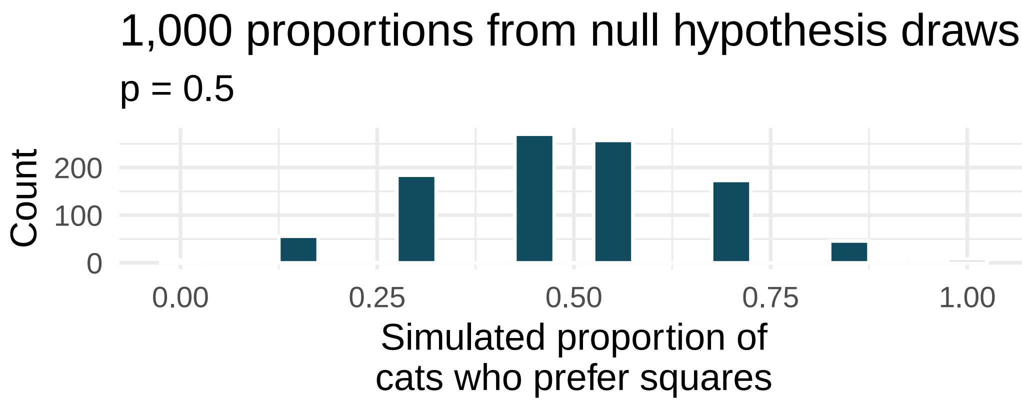
b. A null hypothesis simulation (with 1,000 draws) was run, and the resulting null distribution is displayed in the histogram below. Find the p-value using this distribution and conclude the hypothesis test in the context of the problem.



7. **If I fits, I sits, standard errors.** The results of a study on the type of enclosed spaces cats are most likely to sit in show that 5 out of 7 cats chose a square taped to the ground over a shape known as Kanizsa square illusion, which was preferred by the remaining 2 cats. To evaluate whether these data provide convincing evidence that cats prefer one of the shapes over the other, we set $H_0 : p = 0.5$, where p is the population proportion of cats who prefer square over the Kanizsa square illusion and $H_A : p \neq 0.5$, which suggests some preference, without specifying which shape is more preferred. (Smith, Chouinard, and Byosiere 2021)

a. Using the mathematical model, calculate the standard error of the sample proportion in repeated samples of size 7.

b. A null hypothesis simulation (with 1,000 draws) was run, and the resulting null distribution is displayed in the histogram below. This distribution shows the variability of the sample proportion in samples of size 7 when 50% of cats prefer the square shape over the Kanizsa square illusion. What is the approximate standard error of the sample proportion based on this distribution?



c. Do the mathematical model and simulated draws yield similar standard errors?

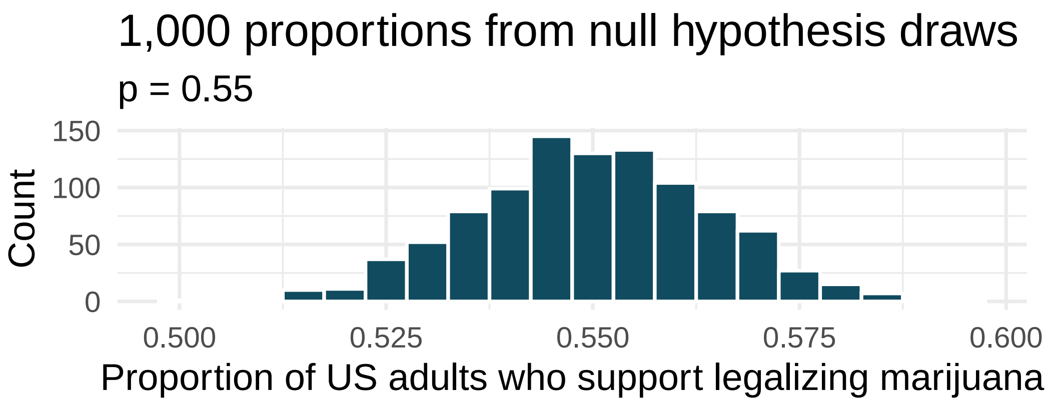
d. In order to approach the problem using the mathematical model, is the success-failure condition met for this study? Explain.

e. What features of the null distribution shown above tells us that the mathematical model should probably not be used?

8. **Legalization of marijuana, standard errors.** According to the 2022 General Social Survey, in a random sample of 1,207 US adults, 65.3% think marijuana should be made legal. (NORC 2022) Consider a scenario where, in order to become legal, 55% (or more) of voters must approve.

a. Calculate the standard error of the sample proportion using the mathematical model.

b. 1,000 sample proportions from samples of size 1,207 were drawn from a null distribution where 55% of voters approve legalizing marijuana. The distribution of these proportions is shown in the histogram below. Approximate the standard error of the sample proportion based on this distribution.

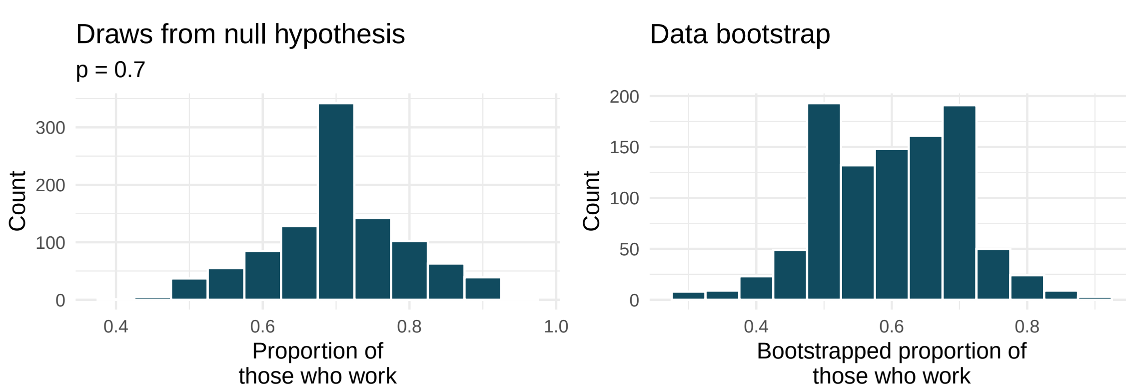


c. Do the mathematical model and simulated draws yield similar standard errors?

d. In this setting (to test whether the true underlying population proportion is greater than 0.55), would there be a strong reason to choose the mathematical model over the simulated null hypothesis (or vice versa)?

9. **Statistics and employment, describe the bootstrap.** A large university knows that about 70% of the full-time students are employed at least 5 hours per week. The members of the Statistics Department wonder if the same proportion of their students work at least 5 hours per week. They randomly sample 25 majors and find that 15 of the students work 5 or more hours each week.

Two sampling distributions are created to describe the variability in the proportion of statistics majors who work at least 5 hours per week. The null hypothesis simulation imposes a true population proportion of $p = 0.7$ while the data bootstrap resamples from the actual data (which has 60% of the observations who work at least 5 hours per week).



a. The sampling was done under two different settings to generate each of the distributions shown above. Describe the two different settings.

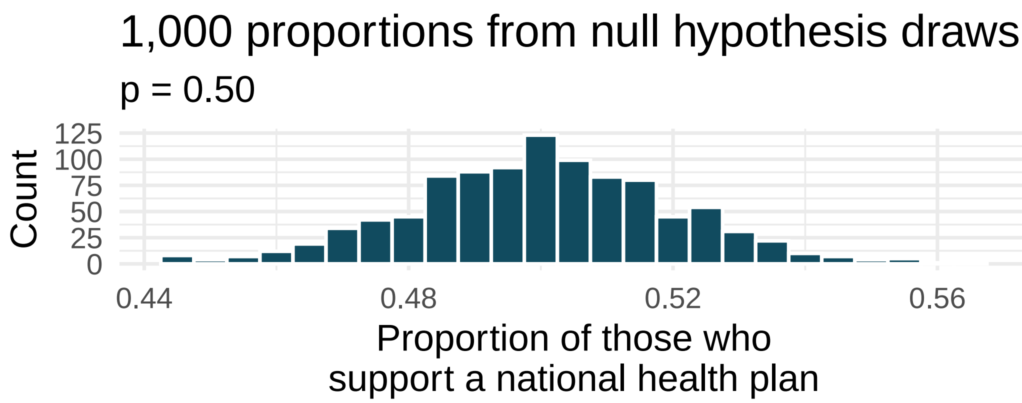
b. Where are each of the two distributions centered? How do their centers compare?

c. Estimate the standard error of the simulated proportions based on each distribution. Are the two standard errors you estimate roughly equal?

d. Describe the shapes of the two distributions. Are they roughly the same?

10. **National Health Plan, simulated null hypothesis.** A Kaiser Family Foundation poll for a random sample of US adults in 2019 found that 79% of Democrats, 55% of Independents, and 24% of Republicans supported a generic "National Health Plan". There were 347 Democrats, 298 Republicans, and 617 Independents surveyed. (Kaiser Family Foundation 2019)

A political pundit on TV claims that a majority of Independents support a National Health Plan. Do these data provide strong evidence to support this type of statement? One approach to assessing the question of whether a majority of Independents support a National Health Plan is to simulate 1,000 draws from a null hypothesis with $p = 0.5$ as the proportion of Independents in support.



a. The histogram above displays 1000 values of what?

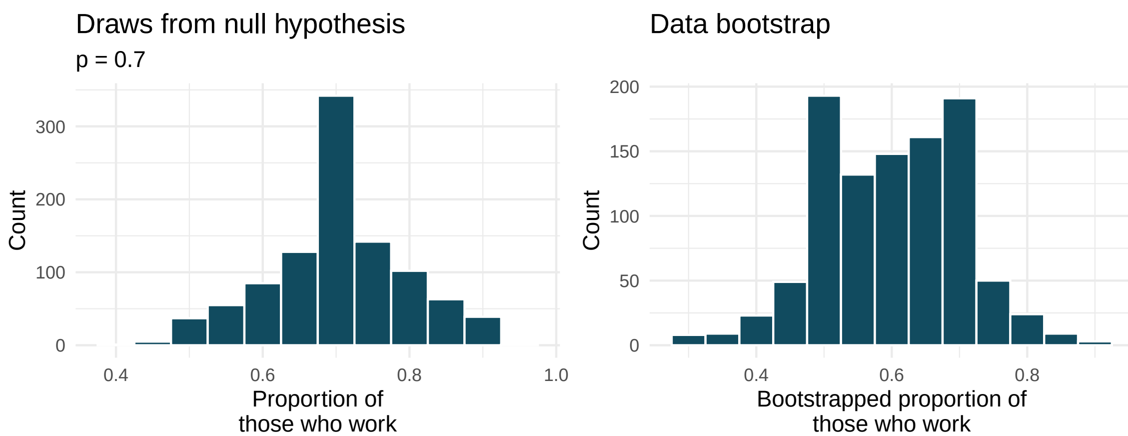
b. Is the observed proportion of Independents consistent with the null hypothesis simulated proportions under the setting where $p = 0.5$?

c. In order to test the claim that "a majority of Independents support a National Health Plan" what are the null and alternative hypotheses?

d. Using the simulated null hypothesis distribution, find the p-value and conclude the hypothesis test in the context of the problem.

11. **Statistics and employment, use the bootstrap.** In a large university where 70% of the full-time students are employed at least 5 hours per week, the members of the Statistics Department wonder if the same proportion of their students work at least 5 hours per week. They randomly sample 25 majors and find that 15 of the students work 5 or more hours each week.

Two sampling distributions are created to describe the variability in the proportion of statistics majors who work at least 5 hours per week. The null hypothesis distribution imposes a true population proportion of $p = 0.7$ while the data bootstrap resamples from the actual data (which has 60% of the observations who work at least 5 hours per week).



See the next page for the questions.

a. Which distribution should be used to test whether the proportion of all statistics majors who work at least 5 hours per week is 70%? And which distribution should be used to find a confidence interval for the true poportion of statistics majors who work at least 5 hours per week?

b. Using the appropriate histogram, test the claim that 70% of statistics majors, like their peers, work at least 5 hours per week. State the null and alternative hypotheses, find the p-value, and conclude the test in the context of the problem.

c. Using the appropriate histogram, find a 98% bootstrap percentile confidence interval for the true proportion of statistics majors who work at least 5 hours per week. Interpret the confidence interval in the context of the problem.

d. Using the appropriate historgram, find a 98% bootstrap SE confidence interval for the true proportion of statistics majors who work at least 5 hours per week. Interpret the confidence interval in the context of the problem.

12. **CLT for proportions.** Define the term "sampling distribution" of the sample proportion, and describe how the shape, center, and spread of the sampling distribution change as the sample size increases when $p = 0.1$.

13. **Vegetarian college students.** Suppose that 8% of college students are vegetarians. Determine if the following statements are true or false, and explain your reasoning.

a. The distribution of the sample proportions of vegetarians in random samples of size 60 is approximately normal since $n \geq 30$.

b. The distribution of the sample proportions of vegetarian college students in random samples of size 50 is right skewed.

c. A random sample of 125 college students where 12% are vegetarians would be considered unusual.

d. A random sample of 250 college students where 12% are vegetarians would be considered unusual.

e. The standard error would be reduced by one-half if we increased the sample size from 125 to 250.

14. **Young Americans, American dream.** About 77% of young adults think they can achieve the American dream. Determine if the following statements are true or false, and explain your reasoning. (Vaughn 2011)

a. The distribution of sample proportions of young Americans who think they can achieve the American dream in random samples of size 20 is left skewed.

b. The distribution of sample proportions of young Americans who think they can achieve the American dream in random samples of size 40 is approximately normal since $n \geq 30$.

c. A random sample of 60 young Americans where 85% think they can achieve the American dream would be considered unusual.

d. A random sample of 120 young Americans where 85% think they can achieve the American dream would be considered unusual.

15. **Orange tabbies.** Suppose that 90% of orange tabby cats are male. Determine if the following statements are true or false, and explain your reasoning.

a. The distribution of sample proportions of random samples of size 30 is left skewed.

b. Using a sample size that is 4 times as large will reduce the standard error of the sample proportion by one-half.

c. The distribution of sample proportions of random samples of size 140 is approximately normal.

d. The distribution of sample proportions of random samples of size 280 is approximately normal.

16. **Young Americans, starting a family.** About 25% of young Americans have delayed starting a family due to the continued economic slump. Determine if the following statements are true or false, and explain your reasoning. (Demos 2011)

a. The distribution of sample proportions of young Americans who have delayed starting a family due to the continued economic slump in random samples of size 12 is right skewed.

b. In order for the distribution of sample proportions of young Americans who have delayed starting a family due to the continued economic slump to be approximately normal, we need random samples where the sample size is at least 40.

c. A random sample of 50 young Americans where 20% have delayed starting a family due to the continued economic slump would be considered unusual.

d. A random sample of 150 young Americans where 20% have delayed starting a family due to the continued economic slump would be considered unusual.

e. Tripling the sample size will reduce the standard error of the sample proportion by one-third.

17. **Sex equality.** The General Social Survey asked a random sample of 1,390 Americans the following question: "On the whole, do you think it should or should not be the government's responsibility to promote equality between men and women?" 82% of the respondents said it "should be". At a 95% confidence level, this sample has 2% margin of error. Based on this information, determine if the following statements are true or false, and explain your reasoning. (NORC 2016)

a. We are 95% confident that 80% to 84% of Americans in this sample think it's the government's responsibility to promote equality between men and women.

b. We are 95% confident that 80% to 84% of all Americans think it's the government's responsibility to promote equality between men and women.

c. If we considered many random samples of 1,390 Americans, and we calculated 95% confidence intervals for each, 95% of these intervals would include the true population proportion of Americans who think it's the government's responsibility to promote equality between men and women.

d. In order to decrease the margin of error to 1%, we would need to quadruple (multiply by 4) the sample size.

e. Based on this confidence interval, there is sufficient evidence to conclude that a majority of Americans think it's the government's responsibility to promote equality between men and women.

18. **Elderly drivers.** A Marist Poll report states that 66% of American adults think licensed drivers should be required to retake their road test once they reach 65 years of age, based on a random sample of 1,018 American adults. They also report a margin of error was 3% at the 95% confidence level. (Poll 2011)

a. Verify the margin of error reported by The Marist Poll using a mathematical model.

b. Based on a 95% confidence interval, does the poll provide convincing evidence that *more than* two thirds of the population think that licensed drivers should be required to retake their road test once they turn 65?

19. **Fireworks on July 4$^{\text{th}}$.** A local news outlet reported that 56% of 600 randomly sampled Kansas residents planned to set off fireworks on July 4^{th}. Determine the margin of error for the 56% point estimate using a 95% confidence level using a mathematical model. (Survey USA 2012)

20. **Proof of COVID-19 vaccination.** In the US, businesses and schools shut down due to the COVID-19 pandemic in March 2020, and a vaccine became publicly available for the first time in April 2021. That month, a Gallup poll surveyed a random sample of 3,731 US adults, asking how they felt about the COVID-19 vaccine requirement for air travel. The poll found that 57% said they would favor it. (Gallup 2021b)

  a. Describe the population parameter of interest. What is the value of the point estimate of this parameter?

  b. Check if the conditions required for constructing a confidence interval using a mathematical model based on these data are met.

  c. Construct a 95% confidence interval for the proportion of US adults who favor requiring proof of COVID-19 vaccination for travel by airplane.

  d. Without doing any calculations, describe what would happen to the confidence interval if we decided to use a higher confidence level.

  e. Without doing any calculations, describe what would happen to the confidence interval if we used a larger sample.

21. **Study abroad.** A survey on 1,509 high school seniors who took the SAT and who completed an optional web survey shows that 55% of high school seniors are fairly certain that they will participate in a study abroad program in college. (American Council on Education 2008)

  a. Is this sample a representative sample from the population of all high school seniors in the US? Explain your reasoning.

  b. Suppose the conditions for inference are met, regardless of your answer to part (a). Using a mathematical model, construct a 90% confidence interval for the proportion of high school seniors (of those who took the SAT) who are fairly certain they will participate in a study abroad program in college, and interpret this interval in context.

  c. What does "90% confidence" mean?

  d. Based on this interval, would it be appropriate to claim that the majority of high school seniors are fairly certain that they will participate in a study abroad program in college?

22. **Legalization of marijuana, mathematical interval.** The General Social Survey asked a random sample of 1,563 US adults: "Do you think the use of marijuana should be made legal, or not?" 60% of the respondents said it should be made legal. (NORC 2022)

  a. Is 60% a sample statistic or a population parameter? Explain.

  b. Using a mathematical model, construct a 95% confidence interval for the proportion of US adults who think marijuana should be made legal, and interpret it in the context of the data.

  c. A critic points out that this 95% confidence interval is only accurate if the statistic follows a normal distribution, or if the normal model is a good approximation. Do the technical conditions hold for these data? Explain.

  d. A news piece on this survey's findings states, "Majority of US adults think marijuana should be legalized." Based on your confidence interval, is the news piece's statement justified?

23. **National Health Plan, mathematical inference.** A Kaiser Family Foundation poll for a random sample of US adults in 2019 found that 79% of Democrats, 55% of Independents, and 24% of Republicans supported a generic "National Health Plan". There were 347 Democrats, 298 Republicans, and 617 Independents surveyed. (Kaiser Family Foundation 2019)

   a. A political pundit on TV claims that a majority of Independents support a National Health Plan. Do these data provide strong evidence to support this type of statement? Your response should use a mathematical model.

   b. Would you expect a confidence interval for the proportion of Independents who oppose the public option plan to include 0.5? Explain.

24. **Is college worth it?** Among a simple random sample of 331 American adults who do not have a four-year college degree and are not currently enrolled in school, 48% said they decided not to go to college because they could not afford school. (Pew Research Center 2011)

   a. A newspaper article states that only a minority of the Americans who decide not to go to college do so because they cannot afford it and uses the point estimate from this survey as evidence. Conduct a hypothesis test to determine if these data provide strong evidence supporting this statement.

   b. Would you expect a confidence interval for the proportion of American adults who decide not to go to college because they cannot afford it to include 0.5? Explain.

25. **Taste test.** Some people claim that they can tell the difference between a diet soda and a regular soda in the first sip. A researcher wanting to test this claim randomly sampled 80 such people. He then filled 80 plain white cups with soda, half diet and half regular through random assignment, and asked each person to take one sip from their cup and identify the soda as diet or regular. 53 participants correctly identified the soda.

   a. Do these data provide strong evidence that these people are able to detect the difference between diet and regular soda, in other words, are the results discernibly better than just random guessing? Your response should use a mathematical model.

   b. Interpret the p-value in this context.

26. **Will the coronavirus bring the world closer together?** In early 2020 the COVID-19 pandemic arrived in the US; by December 2020 the first COVID-19 vaccine was available. An April 2021 YouGov poll asked 4,265 UK adults whether they think the coronavirus bring the world closer together or leave us further apart. 12% of the respondents said it will bring the world closer together. 37% said it would leave us further apart, 39% said it won't make a difference and the remainder didn't have an opinion on the matter. (YouGov 2021)

   a. Calculate, using a mathematical model, a 90% confidence interval for the proportion of UK adults who think the coronavirus will bring the world closer together, and interpret the interval in context.

   b. Suppose we wanted the margin of error for the 90% confidence level to be about 0.5%. How large of a sample size would you recommend for the poll?

27. **Quality control.** As part of a quality control process for computer chips, an engineer at a factory randomly samples 212 chips during a week of production to test the current rate of chips with severe defects. She finds that 27 of the chips are defective.

a. What population is under consideration in the dataset?

b. What parameter is being estimated?

c. What is the point estimate for the parameter?

d. What is the name of the statistic that can be used to measure the uncertainty of the point estimate?

e. Compute the value of the statistic from part (d) using a mathematical model.

f. The historical rate of defects is 10%. Should the engineer be surprised by the observed rate of defects during the current week?

g. Suppose the true population value was found to be 10%. If we use this proportion to recompute the value in part (d) using $p = 0.1$ instead of $\hat{p}$, how much does the resulting value of the statistic change?

28. **Nearsighted children.** Nearsightedness (myopia) is a common vision condition in which you can see near objects clearly, but farther away objects blurry. It is believed that nearsightedness affects about 8% of all children. In a random sample of 194 children, 21 are nearsighted. Using a mathematical model, conduct a hypothesis test for the following question: do these data provide evidence that the 8% value is inaccurate?

29. **Website registration.** A website is trying to increase registration for first-time visitors, exposing 1% of these visitors to a new site design. Of 752 randomly sampled visitors over a month who saw the new design, 64 registered.

a. Check the conditions for constructing a confidence interval for the proportion of first-time visitors of the site who would register under the new design using a mathematical model.

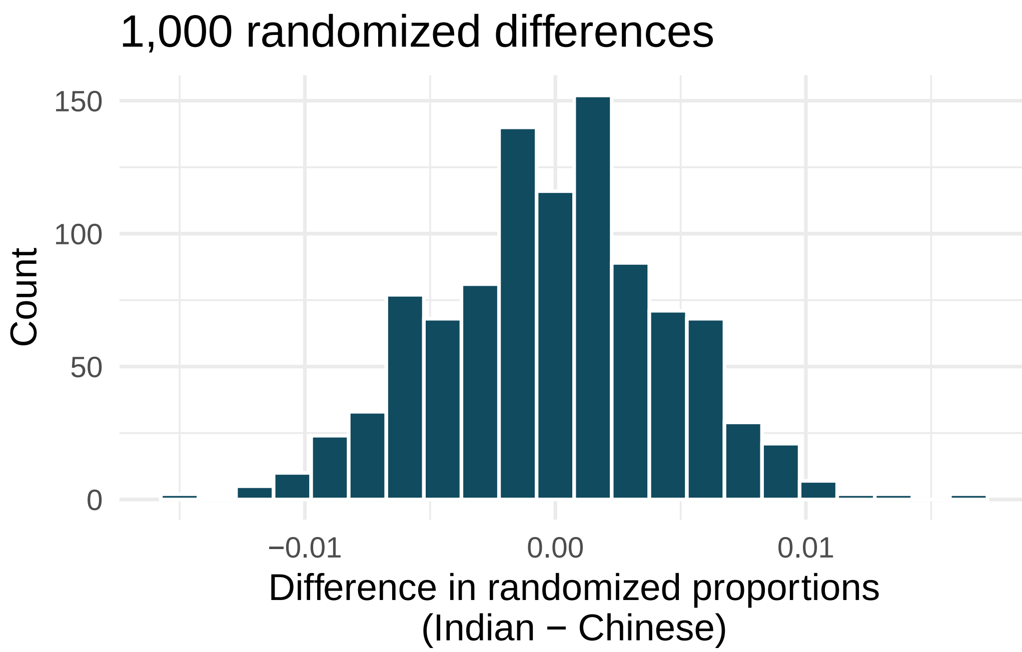
b. Compute the standard error which would describe the variability os the point estimate associated with repeated samples of size 752.

c. Construct and interpret a 90% confidence interval for the fraction of first-time visitors of the site who would register under the new design (assuming stable behaviors by new visitors over time).

30. **Coupons driving visits.** A store randomly samples 603 shoppers over the course of a year and finds that 142 of them made their visit because of a coupon they'd received in the mail. Using a mathematical model, construct a 95% confidence interval for the fraction of all shoppers during the year whose visit was because of a coupon they'd received in the mail.

31. **Disaggregating Asian American tobacco use, hypothesis testing.** Understanding cultural differences in tobacco use across different demographic groups can lead to improved health care education and treatment. A recent study disaggregated tobacco use across Asian American ethnic groups including Asian-Indian (n = 4,373), Chinese (n = 4,736), and Filipino (n = 4,912), in comparison to non-Hispanic Whites (n = 275,025). The number of current smokers in each group was reported as Asian-Indian (n = 223), Chinese (n = 279), Filipino (n = 609), and non-Hispanic Whites (n = 50,880). (Rao et al. 2021)

To determine whether the proportion of Asian-Indian Americans who are current smokers is different from the proportion of Chinese Americans who are smokers, a randomization simulation was performed.

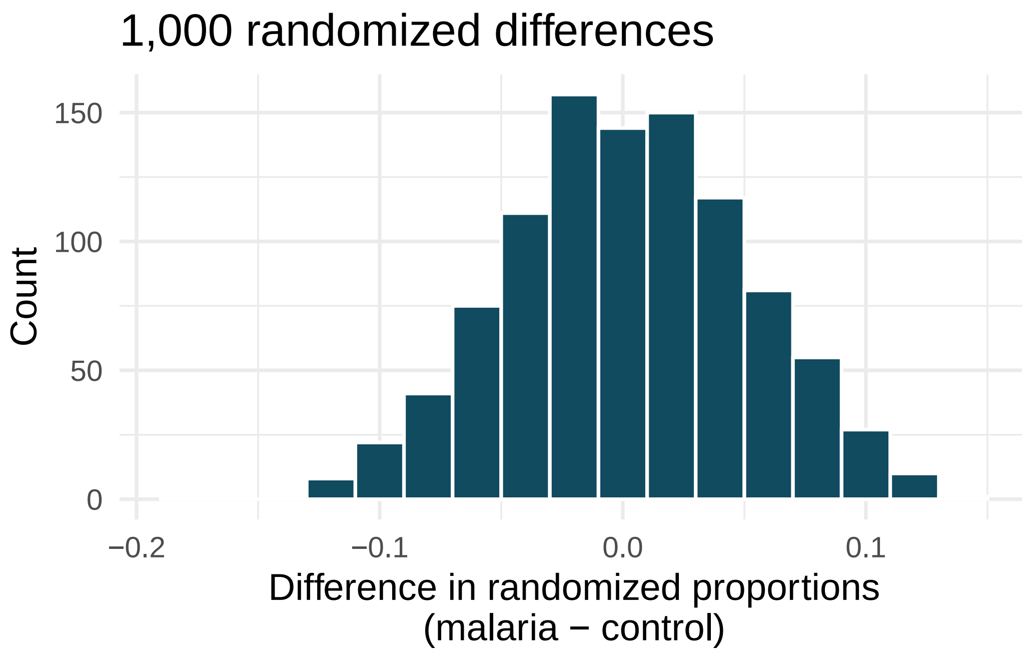


a. In both words and symbols provide the parameter and statistic of interest for this study. Do you know the numerical value of either the parameter or statistic of interest? If so, provide the numerical value.

b. The histogram above provides the sampling distribution (under randomization) for $\hat{p}_{Asian-Indian} - \hat{p}_{Chinese}$ under repeated null randomizations ($\hat{p}$ is the proportion in the sample who are current smokers). Estimate the standard error of $\hat{p}_{Asian-Indian} - \hat{p}_{Chinese}$ based on the randomization histogram.

c. Consider the hypothesis test to determine if there is a difference in proportion of Asian-Indian Americans as compared to Chinese Americans who are current smokers. Write out the null and alternative hypotheses, estimate a p-value using the randomization histogram, and conclude the test in the context of the problem.

32. **Malaria vaccine effectiveness, hypothesis test.** With no currently licensed vaccines to inhibit malaria, good news was welcomed with a recent study reporting long-awaited vaccine success for children in Burkina Faso. With 450 children randomized to either one of two different doses of the malaria vaccine or a control vaccine, 89 of 292 malaria vaccine and 106 out of 147 control vaccine children contracted malaria within 12 months after the treatment. (Datoo et al. 2021)

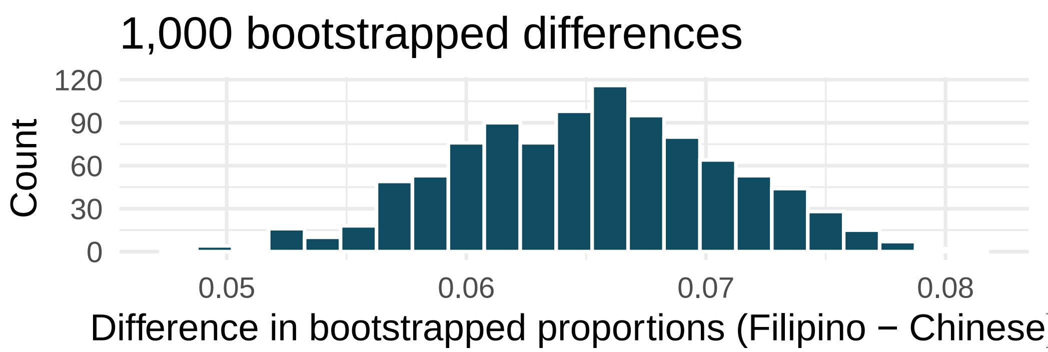


a. In both words and symbols provide the parameter and statistic of interest for this study. Do you know the numerical value of either the parameter or statisic of interest? If so, provide the numerical value.

b. The histogram above provides the sampling distribution (under randomization) for $\hat{p}_{malaria} - \hat{p}_{control}$ under repeated null randomizations ($\hat{p}$ is the proportion of children in the sample who contracted malaria). Estimate the standard error of $\hat{p}_{malaria} - \hat{p}_{control}$ based on the randomization histogram.

c. Consider the hypothesis test constructed to show a lower proportion of children contracting malaria on the malaria vaccine as compared to the control vaccine. Write out the null and alternative hypotheses, estimate a p-value using the randomization histogram, and conclude the test in the context of the problem.

33. **Disaggregating Asian American tobacco use, confidence interval.** Based on a study on the degree to which smoking practices differ across ethnic groups. a confidence interval for the difference in current smoking status for Filipino versus Chinese Americans is desired. (Rao et al. 2021)

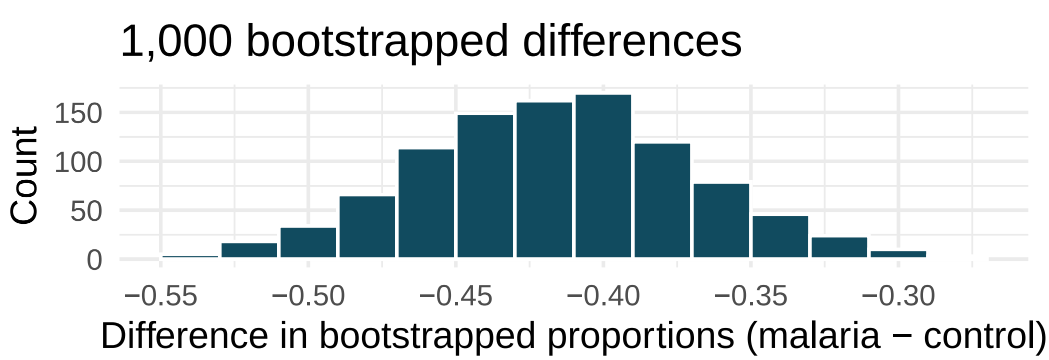


a. Consider the bootstrap distribution of difference in sample proportions of current smokers (Filipino Americans minus Chinese Americans) in 1,000 bootstrap repetitions as above. Estimate the standard error of the difference in sample proportions, as seen in the histogram.

b. Using the standard error from the bootstrap distribution, find a 95% bootstrap SE confidence interval for the true difference in proportion of current smokers (Filipino Americans minus Chinese Americans) in the population. Interpret the interval in the context of the problem.

c. Using the entire bootstrap distribution, find a 95% bootstrap percentile confidence interval for the true difference in proportion of current smokers (Filipino Americans minus Chinese Americans) in the population. Interpret the interval in the context of the problem.

34. **Malaria vaccine effectiveness, confidence interval.** With no currently licensed vaccines to inhibit malaria, good news was welcomed with a recent study reporting long-awaited vaccine success for children in Burkina Faso. With 450 children randomized to either one of two different doses of the malaria vaccine or a control vaccine, 89 of 292 malaria vaccine and 106 out of 147 control vaccine children contracted malaria within 12 months after the treatment. (Datoo et al. 2021)



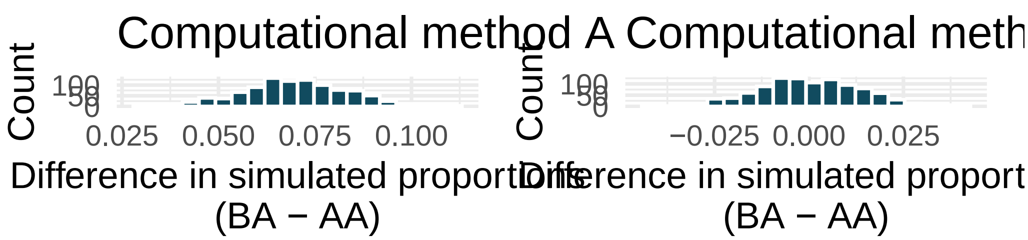
a. Consider the bootstrap distribution of difference in sample proportions of children who contracted malaria (malaria vaccine minus control vaccine) in 1000 bootstrap repetitions as above. Estimate the standard error of the difference in sample proportions, as seen in the histogram.

b. Using the standard error from the bootstrap distribution, find a 95% bootstrap SE confidence interval for the true difference in proportion of children who contract malaria (malaria vaccine minus control vaccine) in the population. Interpret the interval in the context of the problem.

c. Using the entire bootstrap distribution, find a 95% bootstrap percentile confidence interval for the true difference in proportion of children who contract malaria (malaria vaccine minus control vaccine) in the population. Interpret the interval in the context of the problem.

35. **COVID-19 and degree completion.** A 2021 Gallup poll surveyed 3,941 students pursuing a bachelor's degree and 2,064 students pursuing an associate degree (students were not randomly selected but were weighted so as to represent a random selection of currently enrolled US college students). The poll found that 51% of the bachelor's degree students and 44% of associate degree students said that the COVID-19 pandemic will negatively impact their ability to complete the degree. (Gallup 2021a)

Below are two histograms generated with different computational approaches (both use 1,000 repetitions) to research questions which could be asked of these data. One of the histograms can be used to do a randomization test on whether the proportions of bachelor's and associate students who think the COVID-19 pandemic will negatively impact their ability to complete the degree. The other histogram is a bootstrap distribution used to quantify the difference in the proportions of bachelor's and associate's students who feel this way.



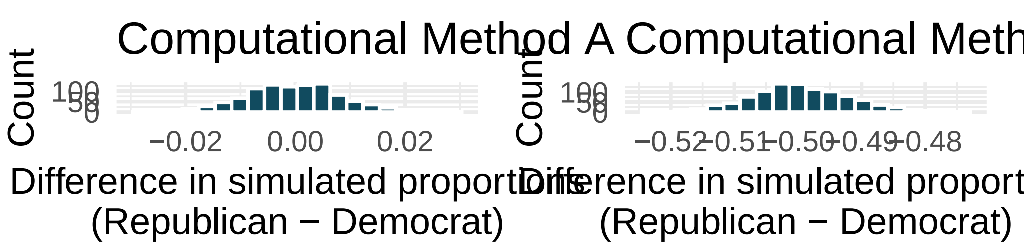
a. Are the center and standard error of the two graphs approximately the same? Explain.

b. Write a research question that can be addressed using the histogram generated with computational method A.

c. Write a research question that can addressed using the histogram generated with computational method B.

36. **Renewable energy.** A 2021 Gallup poll surveyed 5,447 randomly sampled US adults who are Republican (or Republican leaning) and 7,962 who are Democrats (or Democrat leaning). 31% of Republicans and 81% of Democrats said "government regulations are necessary to encourage businesses and consumers to rely more on renewable energy sources". (Gallup 2021a)

Below are two histograms generated with different computational approaches (both use 1,000 repetitions) to research questions which could be asked of these data. One of the histograms can be used to do a randomization test on whether the proportions of Republicans and Democrats who think government regulations are necessary to encourage businesses and consumers to rely more on renewable energy sources are different. The other histogram is a bootstrap distribution used to quantify the difference in the proportions of Republicans and Democrats who agree with this statement.



a. Are the center and standard error of the two graphs approximately the same? Explain.

b. Write a research question that can addressed using the histogram generated with computational method A.

c. Write a research question that can addressed using the histogram generated with computational method B.

37. **HIV in sub-Saharan Africa.** In July 2008 the US National Institutes of Health announced that it was stopping a clinical study early because of unexpected results. The study population consisted of HIV-infected women in sub-Saharan Africa who had been given single dose Nevaripine (a treatment for HIV) while giving birth, to prevent transmission of HIV to the infant. The study was a randomized comparison of continued treatment of a woman (after successful childbirth) with Nevaripine vs Lopinavir, a second drug used to treat HIV. 240 women participated in the study; 120 were randomized to each of the two treatments. Twenty-four weeks after starting the study treatment, each woman was tested to determine if the HIV infection was becoming worse (an outcome called *virologic failure*). Twenty-six of the 120 women treated with Nevaripine experienced virologic failure, while 10 of the 120 women treated with the other drug experienced virologic failure. (Lockman et al. 2007)

a. Create a two-way table presenting the results of this study.

b. State appropriate hypotheses to test for difference in virologic failure rates between treatment groups.

c. Complete the hypothesis test and state an appropriate conclusion. (Reminder: Verify any necessary conditions for the test.)

38. **Supercommuters.** The fraction of workers who are considered "supercommuters", because they commute more than 90 minutes to get to work, varies by state. Suppose the 1% of Nebraska residents and 6% of New York residents are supercommuters. Now suppose that we plan a study to survey 1000 people from each state, and we will compute the sample proportions $\hat{p}_{NE}$ for Nebraska and $\hat{p}_{NY}$ for New York.

a. What is the associated mean and standard deviation of $\hat{p}_{NE}$ in repeated samples of size 1000?

b. What is the associated mean and standard deviation of $\hat{p}_{NY}$ in repeated samples of size 1000?

c. Calculate and interpret the mean and standard deviation associated with the difference in sample proportions for the two groups, $\hat{p}_{NY} - \hat{p}_{NE}$ in repeated samples of 1000 in each group.

d. How are the standard deviations from parts (a), (b), and (c) related?

39. **National Health Plan.** A Kaiser Family Foundation poll for US adults in 2019 found that 79% of Democrats, 55% of Independents, and 24% of Republicans supported a generic "National Health Plan". There were 347 Democrats, 298 Republicans, and 617 Independents surveyed. 79% of 347 Democrats and 55% of 617 Independents support a National Health Plan. (Kaiser Family Foundation 2019)

a. Calculate a 95% confidence interval for the difference between the proportion of Democrats and Independents who support a National Health Plan $(p_D - p_I)$, and interpret it in this context. We have already checked conditions for you.

b. True or false: If we had picked a random Democrat and a random Independent at the time of this poll, it is more likely that the Democrat would support the National Health Plan than the Independent.

40. **Sleep deprivation, CA vs. OR, confidence interval.** According to a report on sleep deprivation by the Centers for Disease Control and Prevention, the proportion of California residents who reported insufficient rest or sleep during each of the preceding 30 days is 8.0%, while this proportion is 8.8% for Oregon residents. These data are based on simple random samples of 11,545 California and 4,691 Oregon residents. Calculate a 95% confidence interval for the difference between the proportions of Californians and Oregonians who are sleep deprived and interpret it in context of the data. (CDC 2008)

41. **Gender pay gap in medicine.** A study examined the average pay for men and women entering the workforce as doctors for 21 different positions. (Lo Sasso et al. 2011)

  a. If each gender was equally paid, then we would expect about half of those positions to have men paid more than women and women would be paid more than men in the other half of positions. Write appropriate hypotheses to test this scenario.

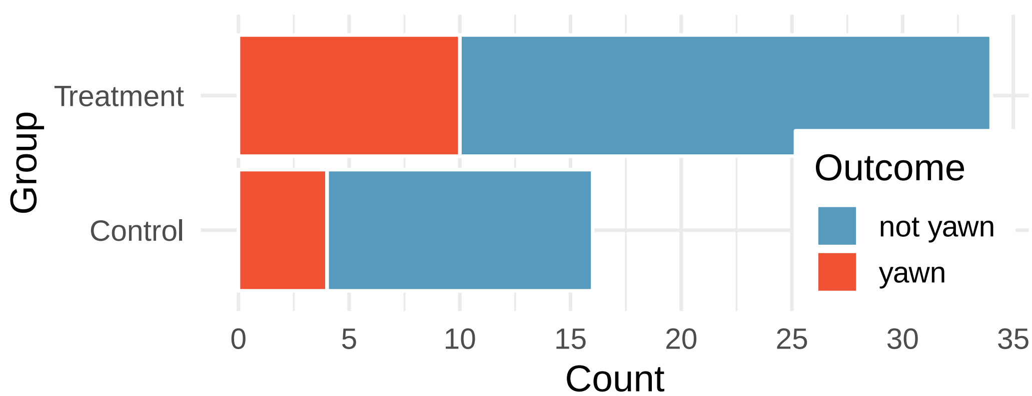
  b. Men were, on average, paid more in 19 of those 21 positions. Complete a hypothesis test using your hypotheses from part (a).

42. **Sleep deprivation, CA vs. OR, hypothesis test.** A CDC report on sleep deprivation rates shows that the proportion of California residents who reported insufficient rest or sleep during each of the preceding 30 days is 8.0%, while this proportion is 8.8% for Oregon residents. These data are based on simple random samples of 11,545 California and 4,691 Oregon residents.

  a. Conduct a hypothesis test to determine if these data provide strong evidence that the rate of sleep deprivation is different for the two states. (Reminder: Check conditions)

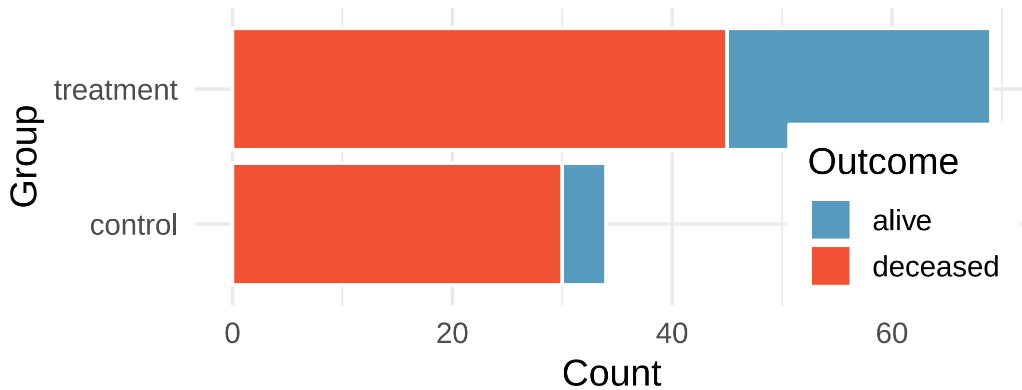
  b. It is possible the conclusion of the test in part (a) is incorrect. If this is the case, what type of error was made?

43. **Is yawning contagious?** An experiment conducted by the MythBusters, a science entertainment TV program on the Discovery Channel, tested if a person can be subconsciously influenced into yawning if another person near them yawns. 50 people were randomly assigned to two groups: 34 to a group where a person near them yawned (treatment) and 16 to a group where there wasn't a person yawning near them (control). The visualization below displays how many participants yawned in each group.



Suppose we are interested in estimating the difference in yawning rates between the control and treatment groups using a confidence interval. Explain why we cannot construct such an interval using the normal approximation. What might go wrong if we constructed the confidence interval despite this problem?

44. **Heart transplant success.** The Stanford University Heart Transplant Study was conducted to determine whether an experimental heart transplant program increased lifespan. Each patient entering the program was officially designated a heart transplant candidate, meaning that he was gravely ill and might benefit from a new heart. Patients were randomly assigned into treatment and control groups. Patients in the treatment group received a transplant, and those in the control group did not. The visualization below displays how many patients survived and died in each group. (Turnbull, Brown, and Hu 1974)



Suppose we are interested in estimating the difference in survival rate between the control and treatment groups using a confidence interval. Explain why we cannot construct such an interval using the normal approximation. What might go wrong if we constructed the confidence interval despite this problem?

45. **Government shutdown.** The United States federal government shutdown of 2018–2019 occurred from December 22, 2018 until January 25, 2019, a span of 35 days. A Survey USA poll of 614 randomly sampled Americans during this time period reported that 48% of those who make less than $40,000 per year and 55% of those who make $40,000 or more per year said the government shutdown has not at all affected them personally. A 95% confidence interval for $(p_{<40\text{K}} - p_{\geq 40\text{K}})$, where p is the proportion of those who said the government shutdown has not at all affected them personally, is (-0.16, 0.02). Based on this information, determine if the following statements are true or false, and explain your reasoning if you identify the statement as false. (Survey USA 2019)

a. At the 5% discernibility level, the data provide convincing evidence of a real difference in the proportion who are not affected personally between Americans who make less than $40,000 annually and Americans who make $40,000 annually.

b. We are 95% confident that 16% more to 2% fewer Americans who make less than $40,000 per year are not at all personally affected by the government shutdown compared to those who make $40,000 or more per year.

c. A 90% confidence interval for $(p_{<40\text{K}} - p_{\geq 40\text{K}})$ would be wider than the $(-0.16, 0.02)$ interval.

d. A 95% confidence interval for $(p_{\geq 40\text{K}} - p_{<40\text{K}})$ is (-0.02, 0.16).

46. **Online harassment.** A Pew Research poll asked US adults aged 18-29 and 30-49 whether they have personally experienced harassment online. A 95% confidence interval for the difference between the proportions of 18-29 year-olds and 30-49 year-olds who have personally experienced harassment online $(p_{18-29} - p_{30-49})$ was calculated to be (0.115, 0.185). Based on this information, determine if the following statements are true or false, and explain your reasoning for each statement you identify as false. (Pew Research Center 2021b)

a. We are 95% confident that the true proportion of 18-29 year-olds who have personally experienced harassment online is 11.5% to 18.5% lower than the true proportion of 30-49 year-olds who have personally experienced harassment online.

b. We are 95% confident that the true proportion of 18-29 year-olds who have personally experienced harassment online is 11.5% to 18.5% higher than the true proportion of 30-49 year-olds who have personally experienced harassment online.

c. 95% of random samples will produce 95% confidence intervals that include the true difference between the population proportions of 18-29 year-olds and 30-49 year-olds who have personally experienced harassment online.

d. We can conclude that there is a discernible difference between the proportions of 18-29 year-olds and 30-49 year-olds who have personally experienced harassment online is too large to plausibly be due to chance, if in fact there is no difference between the two proportions.

e. The 90% confidence interval for $(p_{18-29} - p_{30-49})$ cannot be calculated with only the information given in this exercise.

47. **Decision errors and comparing proportions I.** In the following research studies, conclusions were made based on the data provided. It is always possible that the analysis conclusion could be wrong, although we will almost never actually know if an error has been made or not. For each study conclusion, specify which of a Type I or Type II error could have been made, and state the error in the context of the problem.

a. The malaria vaccine was seen to be effective at lowering the rate of contracting malaria (when compared to the control vaccine).

b. In the US population, Asian-Indian Americans and Chinese Americans are not observed to have different proportions of current smokers.

c. There is no evidence to claim a difference in the proportion of Americans who are not affected personally by a government shutdown when comparing Americans who make less than $40,000 annually and Americans who make $40,000 annually.

48. **Decision errors and comparing proportions II.** In the following research studies, conclusions were made based on the data provided. It is always possible that the analysis conclusion could be wrong, although we will almost never actually know if an error has been made or not. For each study conclusion, specify which of a Type I or Type II error could have been made, and state the error in the context of the problem.

a. Of registered voters in California, the proportion who report not knowing enough to voice an opinion on whether they support off shore drilling is different across those who have a college degree and those who do not.

b. In comparing Californians and Oregonians, there is no evidence to support a difference in the proportion of each who are sleep deprived.

49. **Active learning.** A teacher wanting to increase the active learning component of her course is concerned about student reactions to changes she is planning to make. She conducts a survey in her class, asking students whether they believe more active learning in the classroom (hands on exercises) instead of traditional lecture will helps improve their learning. She does this at the beginning and end of the semester and wants to evaluate whether students' opinions have changed over the semester. Can she used the methods we learned in this chapter for this analysis? Explain your reasoning.

50. **An apple a day keeps the doctor away.** A physical education teacher at a high school wanting to increase awareness on issues of nutrition and health asked her students at the beginning of the semester whether they believed the expression "an apple a day keeps the doctor away". 40% of the students responded yes. Throughout the semester she started each class with a discussion of a study highlighting positive effects of eating more fruits and vegetables. She conducted the same apple-a-day survey at the end of the semester, and this time 60% of the students responded yes. Can she used a two-proportion method from this section for this analysis? Explain your reasoning.

51. **Malaria vaccine effectiveness, effect size.** A randomized controlled trial on malaria vaccine effectiveness randomly assigned 450 children intro either one of two different doses of the malaria vaccine or a control vaccine. 89 of 292 malaria vaccine and 106 out of 147 control vaccine children contracted malaria within 12 months after the treatment. (Datoo et al. 2021)

Recall that in order to reject the null hypothesis that the two vaccines (malaria and control) are equivalent, we'd need the sample proportion to be about 2 standard errors below the hypothesized value of zero.

Say that the true difference (in the population) is given as δ, the sample sizes are the same in both groups ($n_{malaria} = n_{control}$), and the true proportion who contract malaria on the control vaccine is $p_{control} = 0.7$. If you ran your own study (in the future), how likely is it that you would get a difference in sample proportions that was sufficiently far from zero that you could reject under each of the conditions below. (*Hint:* Use the mathematical model.)

a. $\delta = -0.1$ and $n_{malaria} = n_{control} = 20$

b. $\delta = -0.4$ and $n_{malaria} = n_{control} = 20$

c. $\delta = -0.1$ and $n_{malaria} = n_{control} = 100$

d. $\delta = -0.4$ and $n_{malaria} = n_{control} = 100$

e. What can you conclude about values of δ and the sample size?

52. **Diabetes and unemployment.** A Gallup poll surveyed Americans about their employment status and whether they have diabetes. The survey results indicate that 1.5% of the 47,774 employed (full or part time) and 2.5% of the 5,855 unemployed 18-29 year-olds have diabetes. (Gallup 2012)

a. Create a two-way table presenting the results of this study.

b. State appropriate hypotheses to test for difference in proportions of diabetes between employed and unemployed Americans.

c. The sample difference is about 1%. If we completed the hypothesis test, we would find that the p-value is very small (about 0), meaning the difference is statistically discernible. Use this result to explain the difference between statistically discernible and practically important findings.

Datasets: `yawn` (openintro) | `heart_transplant` (openintro)

PART IV

Apply

You now have the full inference toolkit — simulation-based and formula-based. In this part, we apply that toolkit to new data types and richer questions: testing categorical distributions with chi-square, comparing multiple groups with ANOVA, and modeling relationships with correlation and linear regression.

Chapter 14

Chi-Square Goodness of Fit

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In previous chapters, we have developed inferential methods for comparing proportions across two groups. But what if we want to assess whether a single categorical variable with multiple levels follows a particular distribution? For instance, we might want to know if a jury pool is racially representative of the community, or whether the days of the week on which accidents occur are equally likely. The **chi-square goodness of fit test** is designed for exactly this purpose. It compares a set of observed counts to a set of expected counts derived from a hypothesized distribution, using a single test statistic that summarizes all of the differences at once.

14.1 Observed versus expected counts

Consider data from a random sample of 275 jurors in a small county. Jurors identified their racial group, and we would like to determine if these jurors are racially representative of the county's population of registered voters. The observed counts and the population proportions are shown below.

	White	Black	Hispanic	Other	Total
Observed jurors	205	26	25	19	275
Population proportion	0.72	0.07	0.12	0.09	1.00

While the proportions in the jury sample do not precisely match the population proportions, it is unclear whether these data provide convincing evidence that the sample is not representative. If the jurors really were randomly sampled from the registered voters, we might expect small differences due to chance. However, unusually large differences may provide convincing evidence that the juries were not representative.

14.1.1 Computing expected counts

If the jurors were randomly sampled from the population (i.e., if the null hypothesis is true), we would expect the sample proportions to roughly match the population proportions. The **expected count** for each category is found by multiplying the sample size by the hypothesized proportion for that category.

Of the 275 people who served on a jury, about how many would we expect to be White if jurors are randomly selected from the population? How many would we expect to be Black?

About 72% of the population is White, so we would expect about 72% of the jurors to be White: $0.72 \times 275 = 198$. Similarly, we would expect about 7% of the jurors to be Black, which would correspond to about $0.07 \times 275 = 19.25$ Black jurors.

Twelve percent of the population is Hispanic and 9% represent other races. How many of the 275 jurors would we expect to be Hispanic? How many from other races?

Show answer

We would expect $0.12 \times 275 = 33$ Hispanic jurors and $0.09 \times 275 = 24.75$ jurors from other races.

The complete table of observed and expected counts is:

	White	Black	Hispanic	Other	Total
Observed counts	205	26	25	19	275
Expected counts	198	19.25	33	24.75	275

Computing expected counts for a goodness of fit test.

If the null hypothesis specifies proportions $p_1, p_2, \ldots, p_k$ for k categories, and the total sample size is n, then the expected count for category i is:

$$E_i = n \times p_i$$

14.1.2 Setting up hypotheses

The sample proportions from each racial group were not a precise match for the population proportions. We need to test whether the differences are strong enough to provide convincing evidence that the jurors are not a random sample. These ideas can be organized into hypotheses:

- H_0: The jurors are a random sample from the population. The observed counts reflect natural sampling fluctuation.
- H_A: The jurors are not randomly sampled from the population. There is bias in juror selection.

To evaluate these hypotheses, we need to quantify how different the observed counts are from the expected counts. Strong evidence for the alternative hypothesis would come in the form of unusually large deviations from what would be expected based on sampling variation alone.

14.2 The chi-square statistic

To build a test statistic, we first compute the standardized difference between each observed and expected count. For each category i:

$$Z_i = \frac{O_i - E_i}{\sqrt{E_i}}$$

where O_i is the observed count and E_i is the expected count. The denominator $\sqrt{E_i}$ serves as the standard error for the count.

For the jury data:

$$\begin{aligned}
Z_{\text{White}} &= \frac{205 - 198}{\sqrt{198}} = 0.50 \\
Z_{\text{Black}} &= \frac{26 - 19.25}{\sqrt{19.25}} = 1.54 \\
Z_{\text{Hispanic}} &= \frac{25 - 33}{\sqrt{33}} = -1.39 \\
Z_{\text{Other}} &= \frac{19 - 24.75}{\sqrt{24.75}} = -1.16
\end{aligned}$$

We would like to use a single test statistic to determine if these four standardized differences are unusually far from zero as a group. We do this by squaring each standardized difference and adding them up:

$$X^2 = Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 = 0.50^2 + 1.54^2 + (-1.39)^2 + (-1.16)^2 = 5.89$$

Squaring each standardized difference before adding them together does two things:

1. Any standardized difference that is squared will be positive, so negative and positive deviations both contribute to the test statistic.
2. Differences that are already unusual (e.g., a standardized difference of 2.5) become much larger after squaring, which gives them more weight.

Chi-square test statistic.

The **chi-square test statistic** is the sum of the squared standardized differences between observed and expected counts:

$$X^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

where the sum is over all k categories, O_i is the observed count in category i, and E_i is the expected count in category i under the null hypothesis.

14.3 Simulation-based goodness of fit test

Before turning to the mathematical chi-square distribution, it is instructive to see how a simulation-based approach works for goodness of fit tests.

The idea is the same as in other randomization tests: if the null hypothesis is true (the data come from the hypothesized distribution), we can simulate many datasets that would arise under the null hypothesis and compute the chi-square statistic for each. This gives us a null distribution of the chi-square statistic, against which we can compare our observed statistic.

How the simulation works:

1. Under H_0, each observation has probability p_i of falling into category i.
2. For each simulated sample of size n, randomly assign each observation to a category according to the null probabilities.
3. Compute the chi-square statistic X^2 for the simulated data.
4. Repeat many times (e.g., 1,000 or 10,000 times) to build a null distribution.

5. Compare the observed X^2 to the null distribution. The p-value is the proportion of simulated statistics that are as large or larger than the observed statistic.

For the jury example, each simulated dataset would randomly assign 275 jurors to the four racial categories with probabilities 0.72, 0.07, 0.12, and 0.09. The resulting chi-square statistics from the simulated datasets form the null distribution.

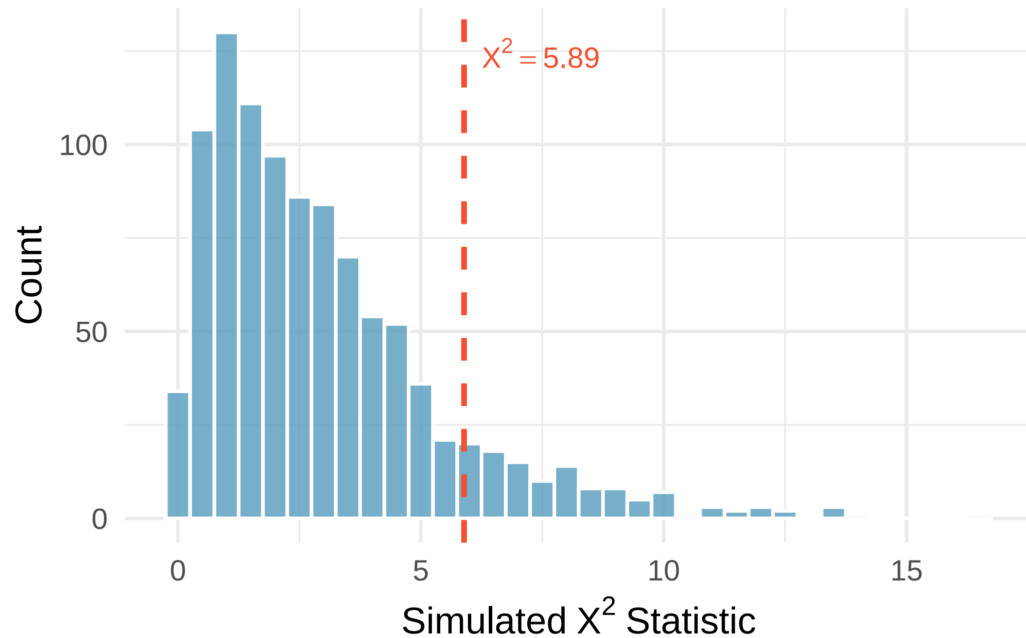


Figure 14.1: Histogram of chi-square statistics from 1,000 simulations under the null hypothesis for the jury example. The observed statistic of 5.89 is marked. Most simulated values fall below 5.89, but a meaningful fraction exceed it.

The simulation-based approach confirms what we will find with the mathematical model: the observed chi-square statistic of 5.89 is not unusual enough to reject the null hypothesis at a 0.05 significance level.

StatLens: Categorical Data Explorer. Use the Categorical Data Explorer to enter observed and expected counts and run a simulation-based chi-square goodness of fit test. The tool will generate a null distribution and compute a p-value.

14.4 The chi-square distribution

The chi-square test statistic X^2 follows a **chi-square distribution** when the null hypothesis is true and the sample size conditions are met. The chi-square distribution is characterized by a single parameter called the **degrees of freedom** (df), which influences the shape, center, and spread of the distribution.

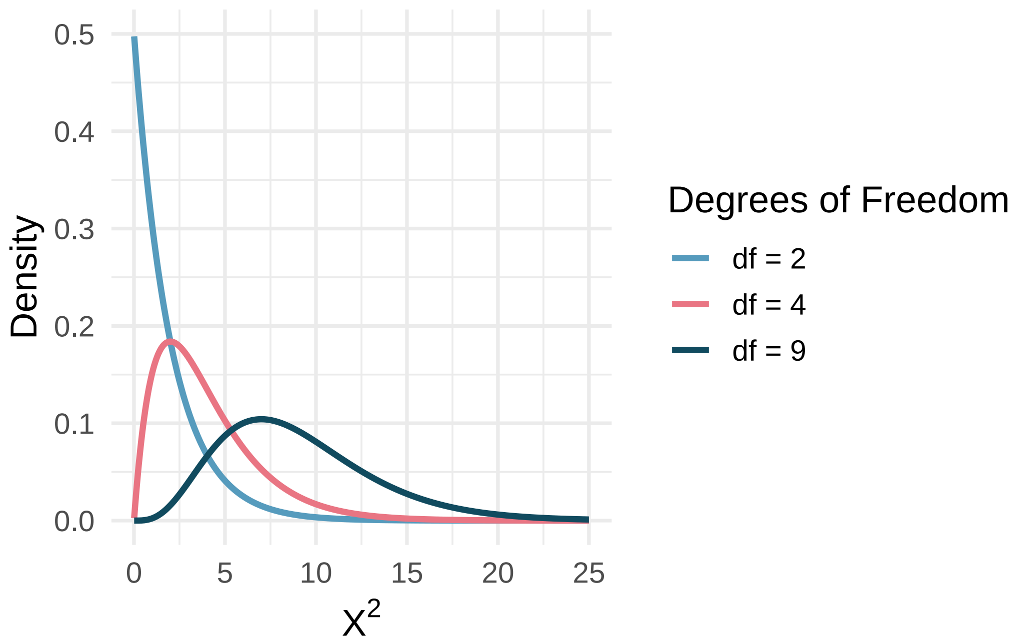


Figure 14.2: Three chi-square distributions with degrees of freedom 2, 4, and 9. As the degrees of freedom increase, the distribution becomes more symmetric, the center moves to the right, and the variability increases.

Key properties of the chi-square distribution:

- It is always non-negative (the chi-square statistic is a sum of squared values).
- It is right-skewed, especially for small degrees of freedom.
- As the degrees of freedom increase, the distribution becomes more symmetric and shifts to the right.
- The mean of the distribution equals the degrees of freedom.

14.4.1 Degrees of freedom for goodness of fit

For a goodness of fit test with k categories, the degrees of freedom are:

$$df = k - 1$$

How many categories were there in the juror example? How many degrees of freedom should be associated with the chi-square distribution?

There were $k = 4$ categories: White, Black, Hispanic, and Other. The degrees of freedom are $df = k - 1 = 3$.

14.5 Finding a p-value

If the null hypothesis is true, X^2 follows a chi-square distribution with $k - 1$ degrees of freedom. Because larger values of X^2 correspond to greater deviations from the null hypothesis, the p-value is always found from the **upper tail** of the distribution.

For the jury data, the test statistic is $X^2 = 5.89$ and the degrees of freedom are $df = 3$. Find the p-value.

Using a chi-square distribution with 3 degrees of freedom, the area to the right of 5.89 is approximately 0.117.

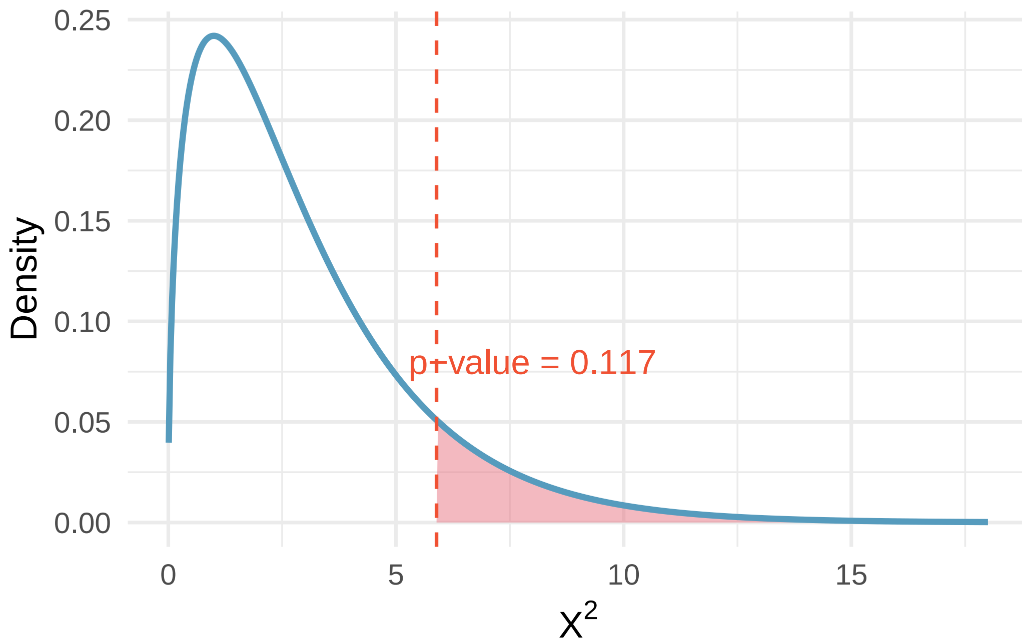


Figure 14.3: Chi-square distribution with 3 degrees of freedom, with the area above 5.89 shaded to represent the p-value of approximately 0.117.

Since the p-value of 0.117 is larger than $\alpha = 0.05$, we do not reject the null hypothesis. The data do not provide convincing evidence of racial bias in the juror selection process.

Consider a chi-square distribution with 5 degrees of freedom and a test statistic of 5.1. Find the p-value.

Using statistical software or a chi-square table, the area to the right of 5.1 on a chi-square distribution with 5 degrees of freedom is approximately 0.404. This is a very large p-value, so we would not reject the null hypothesis.

A chi-square distribution with 4 degrees of freedom has a test statistic of 10. Find the approximate p-value.

Show answer

Using statistical software, the area to the right of 10 on a chi-square distribution with 4 degrees of freedom is approximately 0.040. Since this is less than 0.05, we would reject the null hypothesis.

14.6 Chi-square goodness of fit test

We can now summarize the complete procedure for a chi-square goodness of fit test.

Chi-square test for goodness of fit.

Suppose we want to evaluate whether the observed counts $O_1, O_2, \ldots, O_k$ in k categories are consistent with a hypothesized distribution that specifies expected counts $E_1, E_2, \ldots, E_k$.

Hypotheses:

- H_0: The data follow the hypothesized distribution (the observed counts are consistent with the expected counts, up to sampling variability).
- H_A: The data do not follow the hypothesized distribution.

Test statistic:

$$X^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

When the null hypothesis is true and the conditions are met, X^2 follows a chi-square distribution with $df = k - 1$.

The p-value is found from the upper tail of the chi-square distribution.

Conditions:

- **Independence.** Each case that contributes a count must be independent of all other cases.
- **Sample size.** Each expected count must be at least 5.

14.7 Evaluating goodness of fit for a distribution

The goodness of fit test can also be used to evaluate whether data follow a particular theoretical distribution.

Suppose we observe the daily change in a stock index over 1,362 trading days. We classify each day according to how many consecutive days of increase preceded it, and compare the observed frequencies to what we would expect if each day's movement were independent of the previous day's (a geometric distribution model). The observed and expected counts are shown below.

Days since last increase	1	2	3	4	5	6	7+
Observed	717	369	155	69	28	14	10
Expected	743	338	154	70	32	14	12

Are these data consistent with the geometric distribution model?

We set up the hypotheses:

- H_0: Stock returns are independent from one day to the next and follow a geometric model.
- H_A: Stock returns show a pattern inconsistent with the geometric model.

The chi-square statistic is:

$$X^2 = \frac{(717-743)^2}{743} + \frac{(369-338)^2}{338} + \frac{(155-154)^2}{154} + \frac{(69-70)^2}{70} + \frac{(28-32)^2}{32} + \frac{(14-14)^2}{14} + \frac{(10-12)^2}{12} = 4.61$$

With $k = 7$ categories, $df = 6$. The p-value for $X^2 = 4.61$ with 6 degrees of freedom is approximately 0.595.

Since the p-value is very large, we do not reject the null hypothesis. The data are consistent with the geometric distribution model, suggesting that stock movements may be independent from day to day over this period.

14.8 Chapter review

14.8.1 Summary

In this chapter, we developed the chi-square goodness of fit test, which assesses whether a sample of categorical data is consistent with a hypothesized distribution. The test uses the chi-square statistic, which measures how far the observed counts deviate from the expected counts. When the null hypothesis is true and the conditions are met, this statistic follows a chi-square distribution with $k - 1$ degrees of freedom, where k is the number of categories. The p-value is always computed from the upper tail of the chi-square distribution, since larger values indicate greater discrepancy between the observed and expected counts.

14.8.2 Key terms

- Expected count
- Chi-square statistic (X^2)
- Chi-square distribution
- Degrees of freedom (goodness of fit)
- Goodness of fit test

14.9 Exercises

1. **True or false, Part I.** Determine if the statements below are true or false. For each false statement, suggest an alternative wording to make it a true statement.

a. The chi-square distribution, just like the normal distribution, has two parameters, mean and standard deviation.

b. The chi-square distribution is always right skewed, regardless of the value of the degrees of freedom parameter.

c. The chi-square statistic is always positive.

d. As the degrees of freedom increases, the shape of the chi-square distribution becomes more skewed.

2. **True or false, Part II.** Determine if the statements below are true or false. For each false statement, suggest an alternative wording to make it a true statement.

a. As the degrees of freedom increases, the mean of the chi-square distribution increases.

b. If you found $\chi^2 = 10$ with $df = 5$ you would fail to reject H_0 at the 5% significance level.

c. When finding the p-value of a chi-square test, we always shade the tail areas in both tails.

d. As the degrees of freedom increases, the variability of the chi-square distribution decreases.

3. **Open source textbook.** A professor using an open source introductory statistics book predicts that 60% of the students will purchase a hard copy of the book, 25% will print it out from the web, and 15% will read it online. At the end of the semester he asks his students to complete a survey where they indicate what format of the book they used. Of the 126 students, 71 said they bought a hard copy of the book, 30 said they printed it out from the web, and 25 said they read it online.

a. State the hypotheses for testing if the professor's predictions were inaccurate.

b. How many students did the professor expect to buy the book, print the book, and read the book exclusively online?

c. This is an appropriate setting for a chi-square test. List the conditions required for a test and verify they are satisfied.

d. Calculate the chi-squared statistic, the degrees of freedom associated with it, and the p-value.

e. Based on the p-value calculated in part (d), what is the conclusion of the hypothesis test? Interpret your conclusion in this context.

4. **Barking deer.** Microhabitat factors associated with forage and bed sites of barking deer in Hainan Island, China were examined. In this region woods make up 4.8% of the land, cultivated grass plot makes up 14.7%, and deciduous forests make up 39.6%. Of the 426 sites where the deer forage, 4 were categorized as woods, 16 as cultivated grassplot, and 61 as deciduous forests. The table below summarizes these data.

Woods | Cultivated grassplot | Deciduous forests | Other | Total |

a. Write the hypotheses for testing if barking deer prefer to forage in certain habitats over others.

b. What type of test can we use to answer this research question?

c. Check if the assumptions and conditions required for this test are satisfied.

d. Do these data provide convincing evidence that barking deer prefer to forage in certain habitats over others? Conduct an appropriate hypothesis test to answer this research question.

Chapter 15

Chi-Square Test of Independence

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In the previous chapter, we used the chi-square statistic to test whether a single categorical variable follows a particular distribution. In this chapter, we extend the chi-square framework to two-way tables, where we have two categorical variables and want to assess whether they are associated. The key question becomes: is the distribution of one variable the same across all levels of the other variable, or does the distribution shift depending on the group? As with our other inference methods, we begin with a simulation-based approach (randomization) and then introduce the mathematical chi-square approximation.

Note that with two-way tables, there is not an obvious single parameter of interest. Instead, research questions focus on how the proportions of the response variable change (or not) across the different levels of the explanatory variable. Because there is no single population parameter to estimate, we focus on hypothesis testing (and not confidence intervals).

15.1 Randomization test of independence

15.1.1 Case study: The iPod experiment

We all buy used products – cars, computers, textbooks – and we sometimes assume the sellers will be forthright about any underlying problems. Researchers recruited 219 participants in a study where they would sell a used iPod that was known to have frozen twice in the past. The participants were incentivized to get as much money as they could. The researchers wanted to understand what types of questions would elicit the seller to disclose the freezing issue.

The buyers (who were collaborating with the researchers) asked one of three scripted questions:

- **General**: "What can you tell me about it?"
- **Positive Assumption**: "It does not have any problems, does it?"
- **Negative Assumption**: "What problems does it have?"

The results are shown below:

Question	Disclose problem	Hide problem	Total
General	2	71	73

Question	Disclose problem	Hide problem	Total
Positive assumption	23	50	73
Negative assumption	36	37	73
Total	**61**	**158**	**219**

The data suggest that asking "What problems does it have?" was the most effective at getting the seller to disclose the past freezing issues. But could we see these results due to chance alone, or is this evidence that some questions are more effective at getting to the truth?

15.1.2 Expected counts in two-way tables

The hypothesis test is about assessing whether there is convincing evidence that the question asked was related to whether the seller disclosed the problem. In other words, we want to check whether the buyer's question was **independent** of whether the seller disclosed a problem.

From the experiment, the overall proportion of sellers who disclosed the freezing problem was $61/219 = 0.2785$. If there really is no difference among the questions and 27.85% of sellers were going to disclose regardless, how many of the 73 people in the General group would we have expected to disclose?

We would predict that $0.2785 \times 73 = 20.33$ sellers would disclose the problem. We observed far fewer than this, though it is not yet clear if that is due to chance variation or because the questions truly differ in effectiveness.

If the questions were actually equally effective, meaning about 27.85% of respondents would disclose the freezing issue regardless of what question they were asked, about how many sellers in the Positive Assumption group would we expect to *hide* the freezing problem?

Show answer

We would expect $(1 - 0.2785) \times 73 = 52.67$.

We can compute the expected number for all cells using the row totals, column totals, and table total. For instance, the proportion who disclosed overall was $61/219$. Multiplying by each row total:

$$\frac{61}{219} \times 73 = 20.33 \quad \text{(for each question group)}$$

More generally, this can be rewritten as:

$$\text{Expected Count}_{\text{row } i, \text{ col } j} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{\text{table total}}$$

Computing expected counts in a two-way table.

To calculate the expected count for the i^{th} row and j^{th} column, compute:

$$\text{Expected Count}_{\text{row } i, \text{ col } j} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{\text{table total}}$$

Using this formula, the observed counts with expected counts in parentheses are:

Question	Disclose problem	Hide problem	Total
General	2 (20.33)	71 (52.67)	73
Positive assumption	23 (20.33)	50 (52.67)	73
Negative assumption	36 (20.33)	37 (52.67)	73
Total	**61**	**158**	**219**

15.1.3 The observed chi-square statistic

The chi-square test statistic for a two-way table is found by computing the squared standardized difference for every cell in the table and summing them:

$$\begin{array}{ll} \text{General formula} & \dfrac{(\text{observed count} - \text{expected count})^2}{\text{expected count}} \\ \text{Row 1, Col 1} & \dfrac{(2 - 20.33)^2}{20.33} = 16.53 \\ \text{Row 2, Col 1} & \dfrac{(23 - 20.33)^2}{20.33} = 0.35 \\ \vdots & \qquad \vdots \\ \text{Row 3, Col 2} & \dfrac{(37 - 52.67)^2}{52.67} = 4.66 \end{array}$$

Adding the computed value for each cell gives the chi-square test statistic:

$$X^2 = 16.53 + 0.35 + \cdots + 4.66 = 40.13$$

15.1.4 Variability of the statistic

Is 40.13 a big number? Does it indicate that the observed and expected values are really different, or is it a value we might expect to see just due to natural variability?

Assuming that the individuals would disclose or hide the problems **regardless** of the question they were given (i.e., that the null hypothesis is true), we can randomize the data by reassigning the 61 disclosed problems and 158 hidden problems to the three question groups at random. This is exactly the same type of randomization we used in previous chapters.

One such randomization might produce:

Question	Disclose problem	Hide problem	Total
General	29	44	73
Positive assumption	15	58	73
Negative assumption	17	56	73
Total	**61**	**158**	**219**

Computing the chi-square statistic for this randomized table:

$$X^2 = \frac{(29 - 20.33)^2}{20.33} + \frac{(15 - 20.33)^2}{20.33} + \cdots + \frac{(56 - 52.67)^2}{52.67} = 8$$

This is much smaller than 40.13.

15.1.5 Observed statistic vs. null chi-square statistics

One randomization is not sufficient. We generate many chi-square statistics under the null hypothesis to build a null distribution.

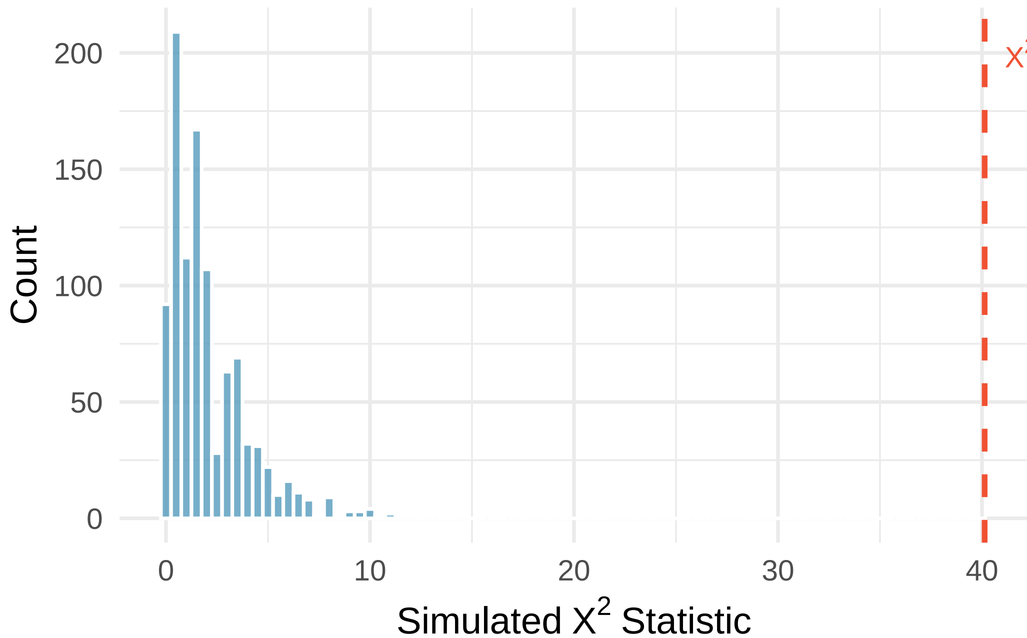


Figure 15.1: A histogram of chi-square statistics from 1,000 simulations produced under the null hypothesis, where the question is independent of the response. The observed statistic of 40.13 is marked by a red line. None of the 1,000 simulations had a chi-square value anywhere close to 40.13.

The observed value is so far from the null statistics that the simulated p-value is essentially zero. We conclude that the decision of whether to disclose the iPod's problem **is changed** by the question asked. We use the causal language of "changed" because the study was a randomized experiment.

Note that with a chi-square test, we only know that the two variables (question type and response) are related (i.e., not independent). We are not able to claim which specific type of question causes which specific type of response.

15.2 Mathematical model for test of independence

15.2.1 The chi-square test of independence

The chi-square test statistic follows a **chi-square distribution** when the null hypothesis is true and the conditions are met. For two-way tables, the degrees of freedom are:

$$df = (R - 1) \times (C - 1)$$

where R is the number of rows and C is the number of columns.

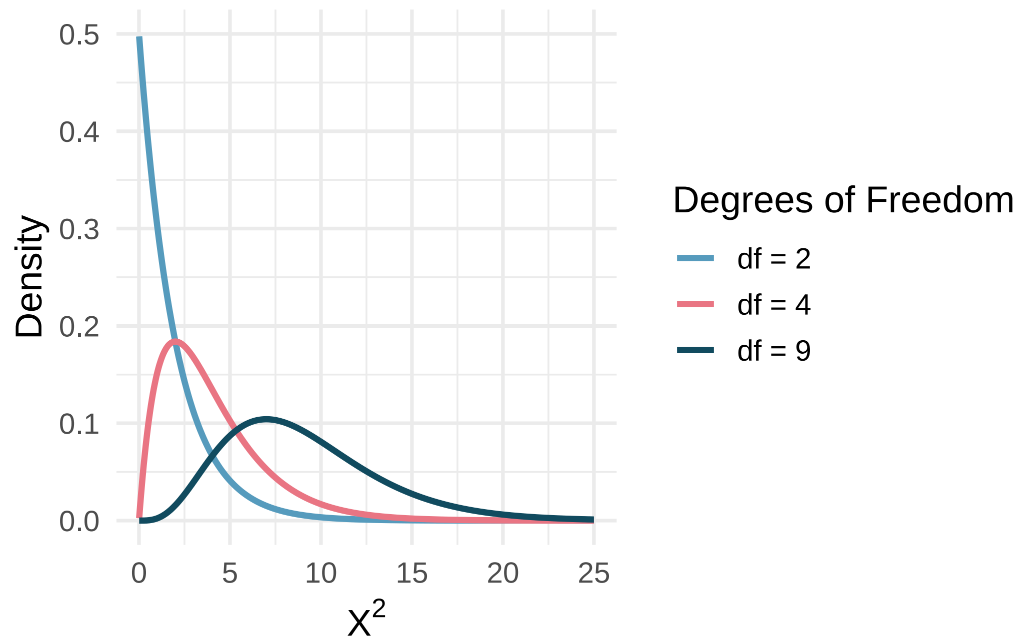


Figure 15.2: The chi-square distribution for different degrees of freedom. The larger the degrees of freedom, the longer the right tail extends. The smaller the degrees of freedom, the more peaked the mode on the left becomes.

15.2.2 Variability of the chi-square statistic

As it turns out, the chi-square test statistic follows a chi-square distribution when the null hypothesis is true. For two-way tables, the degrees of freedom equal $(R-1)\times(C-1)$. In the iPod example, this is $(3-1)\times(2-1)=2$.

The test statistic for assessing independence between two categorical variables is X^2.

The X^2 statistic is a ratio of how the observed counts differ from the expected counts, standardized by the expected counts:

$$X^2 = \sum_{i,j} \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

When the null hypothesis is true and the conditions are met, X^2 has a chi-square distribution with $df=(R-1)\times(C-1)$.

Conditions:

- Independent observations
- Large samples: at least 5 expected counts in each cell

Computing degrees of freedom for a two-way table.

When applying the chi-square test to a two-way table, use $df=(R-1)\times(C-1)$ where R is the number of rows and C is the number of columns.

15.2.3 Finding the p-value

We can safely assume that the observations in the iPod experiment are independent (the question groups were randomly assigned). Each expected count exceeds 5, so the conditions for using the chi-square distribution are met. The test statistic $X^2 = 40.13$ follows a chi-square distribution with 2 degrees of freedom.

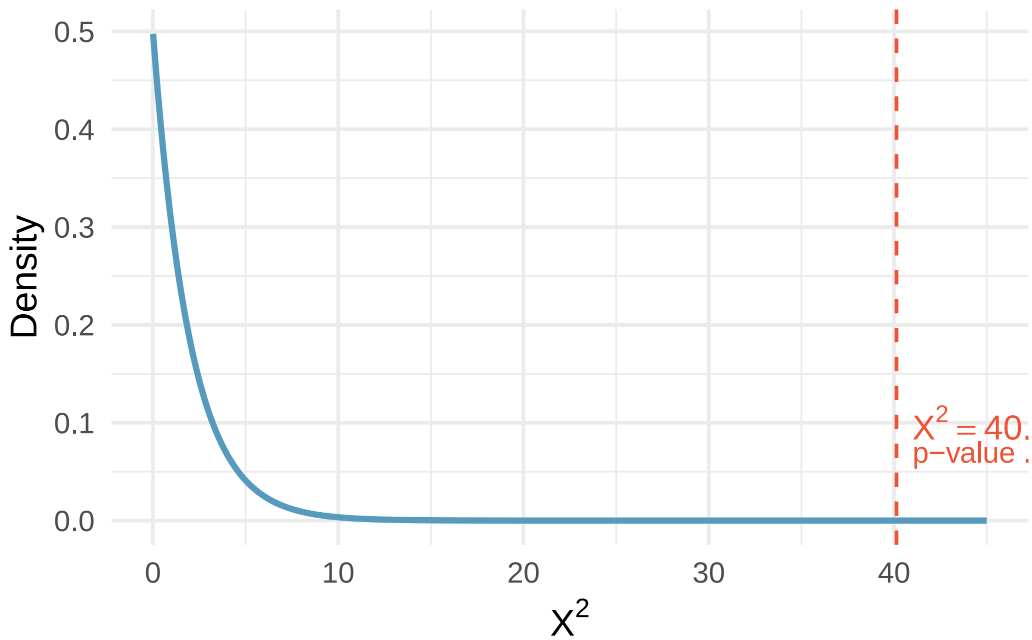


Figure 15.3: The chi-square distribution with $df = 2$, showing the area to the right of $X^2 = 40.13$. The p-value is so small it is not visible on the plot.

Find the p-value and draw a conclusion about whether the question affects the seller's likelihood of reporting the freezing problem.

Using a chi-square distribution with 2 degrees of freedom, the tail area above $X^2 = 40.13$ is approximately 0.000000002.

Using a significance level of $\alpha = 0.05$, the null hypothesis is rejected since the p-value is much smaller. The data provide convincing evidence that the question asked did affect a seller's likelihood to tell the truth about problems with the iPod.

15.2.4 Another example: Diabetes treatment

An experiment evaluated three treatments for Type 2 Diabetes in patients aged 10–17 who were being treated with metformin. The three treatments were: continued metformin alone (met), metformin combined with rosiglitazone (rosi), or a lifestyle intervention program. Each patient had a primary outcome of either failure (lacked glycemic control) or success (maintained glycemic control). The results are shown below.

Treatment	Failure	Success	Total
met	120	114	234

Treatment	Failure	Success	Total
rosi	90	143	233
lifestyle	109	123	232
Total	**319**	**380**	**699**

What are appropriate hypotheses for this test?

- H_0: There is no difference in the effectiveness of the three treatments.
- H_A: There is some difference in effectiveness between the three treatments, e.g., perhaps the rosiglitazone treatment performed better than lifestyle intervention.

Compute the expected counts for each of the six cells in the diabetes treatment table.

Show answer

Row 1, Col 1: $\frac{234\times319}{699} = 106.8$. Row 1, Col 2: $\frac{234\times380}{699} = 127.2$. Row 2, Col 1: $\frac{233\times319}{699} = 106.3$. Row 2, Col 2: $\frac{233\times380}{699} = 126.7$. Row 3, Col 1: $\frac{232\times319}{699} = 105.9$. Row 3, Col 2: $\frac{232\times380}{699} = 126.1$.

Note: when analyzing 2-by-2 contingency tables (where both variables have only two categories), it is generally preferable to use the two-proportion methods from earlier chapters.

15.3 Comparing the simulation and chi-square approaches

Both the randomization test and the mathematical chi-square test address the same question: are two categorical variables independent? The two approaches generally give very similar results, especially when the sample size is large and the expected counts are well above 5.

The **randomization test** is more flexible because:

- It does not require any minimum expected count.
- It works well even for small samples.
- It requires no distributional assumptions beyond independence.

The **mathematical chi-square test** is more convenient because:

- It does not require a computer to simulate thousands of randomizations.
- It provides a simple formula and distribution for computing p-values.
- It is widely implemented in statistical software.

In practice, the chi-square test is the standard approach when conditions are met, while the randomization test provides a useful check or alternative when conditions are questionable.

15.4 Chapter review

15.4.1 Summary

In this chapter, we extended the chi-square framework to two-way tables, testing whether two categorical variables are independent. The randomization approach shuffles the data under the null hypothesis to build a distribution of the chi-square statistic, while the mathematical approach uses the chi-square distribution with $(R-1)\times(C-1)$ degrees of freedom. Both approaches require that observations are independent and that expected counts are sufficiently large (at least 5 in each cell).

The chi-square test of independence tells us *whether* two variables are related, but does not tell us *how* they are related or *which* groups differ. To understand the nature of the relationship, we must examine the individual cells and their contributions to the chi-square statistic.

15.4.2 Key terms

- Independence
- Expected counts (two-way table)
- Chi-square statistic
- Chi-square distribution
- Degrees of freedom (two-way table)

15.5 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Quitters.** Does being part of a support group affect the ability of people to quit smoking? A county health department enrolled 300 smokers in a randomized experiment. 150 participants were randomly assigned to a group that used a nicotine patch and met weekly with a support group; the other 150 received the patch and did not meet with a support group. At the end of the study, 40 of the participants in the patch plus support group had quit smoking while only 30 smokers had quit in the other group.

a. Create a two-way table presenting the results of this study.

b. Answer each of the following questions under the null hypothesis that being part of a support group does not affect the ability of people to quit smoking, and indicate whether the expected values are higher or lower than the observed values.

   i. How many subjects in the "patch + support" group would you expect to quit?

   ii. How many subjects in the "patch only" group would you expect to not quit?

2. **Act on climate change.** The table below summarizes results from a Pew Research poll which asked respondents whether they have personally taken action to help address climate change within the last year and their generation. The differences in each generational group may be due to chance. Complete the following computations under the null hypothesis of independence between an individual's generation and whether they have personally taken action to help address climate change within the last year. (Pew Research Center 2021a)

	Response		
Generation	Took action	Didn't take action	Total
Gen Z	292	620	912
Millenial	885	2,275	3,160
Gen X	809	2,709	3,518
Boomer & older	1,276	4,798	6,074
Total	3,262	10,402	13,664

If there is no relationship between age and action,

a. how many Gen Z'ers would you expect to have personally taken action to help address climate change within the last year?

b. how many Millenials would you expect to have personally taken action to help address climate change within the last year?

c. how many Gen X'ers would you expect to have personally taken action to help address climate change within the last year?

d. how many Boomers and older would you expect to have personally taken action to help address climate change within the last year?

3. **Lizard habitats, data.** In order to assess whether habitat conditions are related to the sunlight choices a lizard makes for resting, Western fence lizard (*Sceloporus occidentalis*) were observed across three different microhabitats. (Adolph 1990; Asbury and Adolph 2007)

	sunlight			
site	sun	partial	shade	Total
desert	16	32	71	119
mountain	56	36	15	107
valley	42	40	24	106
Total	114	108	110	332

a. If the variables describing the habitat and the amount of sunlight are independent, what proporiton of lizards (total) would be expected in each of the three sunlight categories?

b. Given the proportions of each sunlight condition, how many lizards of each type would you expect to see in the sun? in the partial sun? in the shade?

c. Compare the observed (original data) and expected (part b.) tables. From a first glance, does it seem as though the habitat and choice of sunlight may be associated?

d. Regardless of your answer to part (c), is it possible to tell from looking only at the expected and observed counts whether the two variables are associated?

4. **Disaggregating Asian American tobacco use, data.** Understanding cultural differences in tobacco use across different demographic groups can lead to improved health care education and treatment. A recent study disaggregated tobacco use across Asian American ethnic groups including Asian-Indian (n = 4,373), Chinese (n = 4,736), and Filipino (n = 4,912), in comparison to non-Hispanic Whites (n = 275,025). The number of current smokers in each group was reported as Asian-Indian (n = 223), Chinese (n = 279), Filipino (n = 609), and non-Hispanic Whites (n = 50,880). (Rao et al. 2021)

In order to assess whether there is a difference in current smoking rates across three Asian American ethnic groups, the observed data is compared to the data that would be expected if there were no association between the variables.

	Smoking		
ethnicity	don't smoke	smoke	Total
Asian-Indian	4,150	223	4,373
Chinese	4,457	279	4,736
Filipino	4,303	609	4,912
Total	12,910	1,111	14,021

a. If the variables on ethnicity and smoking status are independent, estimate the proporiton of individuals (total) who smoke?

b. Given the overall proportion who smoke, how many of each Asian American ethnicity would you expect to smoke?

c. Compare the observed and expected counts. From a first glance, does it seem as though the Asian American ethnicity and choice of smoking may be associated?

d. Regardless of your answer to part (c), is it possible to tell from looking only at the expected and observed counts whether the two variables are associated?

5. **Lizard habitats, randomize once.** In order to assess whether habitat conditions are related to the sunlight choices a lizard makes for resting, Western fence lizard (*Sceloporus occidentalis*) were observed across three different microhabitats. (Adolph 1990; Asbury and Adolph 2007) Then, the data were randomized once, where sunlight preference was randomly assigned to the lizards across different sites. The original data are shown on the left and the results of the randomization is shown on the right.

Original data

	sunlight			
site	sun	partial	shade	Total
desert	16	32	71	119
mountain	56	36	15	107
valley	42	40	24	106
Total	114	108	110	332

Randomized data

	sunlight			
site	sun	partial	shade	Total
desert	44	42	33	119
mountain	39	31	37	107
valley	31	35	40	106
Total	114	108	110	332

Recall that the Chi-squared statistic (X^2) measures the difference between the expected and observed counts. Without calculating the actual statistic, report on whether the original data or the randomized data will have a larger Chi-squared statistic. Explain your choice.

6. **Disaggregating Asian American tobacco use, randomize once.** In a study that aims to disaggregate tobacco use across Asian American ethnic groups (Asian-Indian, Chinese, and Filipino, in comparison to non-Hispanic Whites), respondents were asked whether they smoke tobacco or not. (Rao et al. 2021) Then, the data were randomized once, where smoking status was randomly assigned to the participants across different ethnicities. The original data are shown on the left and the results of the randomization is shown on the right.

Original data

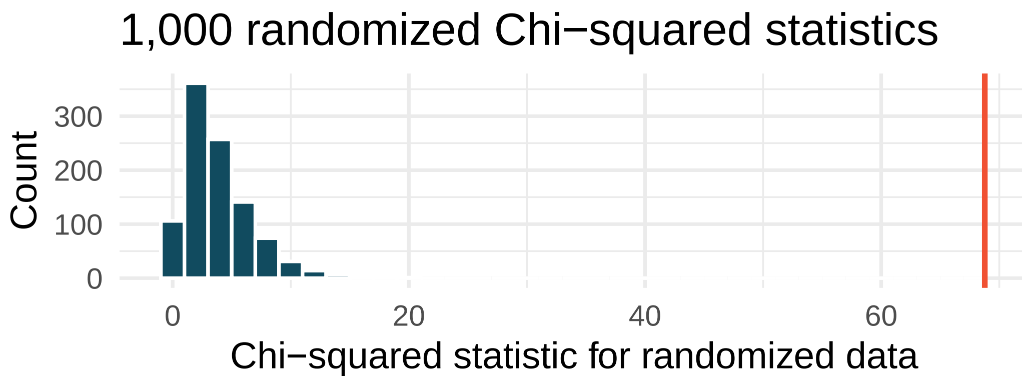
	Smoking		
ethnicity	don't smoke	smoke	Total
Asian-Indian	4,150	223	4,373
Chinese	4,457	279	4,736
Filipino	4,303	609	4,912
Total	12,910	1,111	14,021

Randomized data

	Smoking		
ethnicity	don't smoke	smoke	Total
Asian-Indian	4,015	358	4,373
Chinese	4,385	351	4,736
Filipino	4,510	402	4,912
Total	12,910	1,111	14,021

Recall that the Chi-squared statistic (X^2) measures the difference between the expected and observed counts. Without calculating the actual statistic, report on whether the original data or the randomized data will have a larger Chi-squared statistic. Explain your choice.

7. **Lizard habitats, randomization test.** In order to assess whether habitat conditions are related to the sunlight choices a lizard makes for resting, Western fence lizard (*Sceloporus occidentalis*) were observed across three different microhabitats. (Adolph 1990; Asbury and Adolph 2007) The original data were randomized 1,000 times (sunlight variable randomly assigned to the observations across different habitats), and the histogram of the Chi-squared statistic on each randomization is displayed.

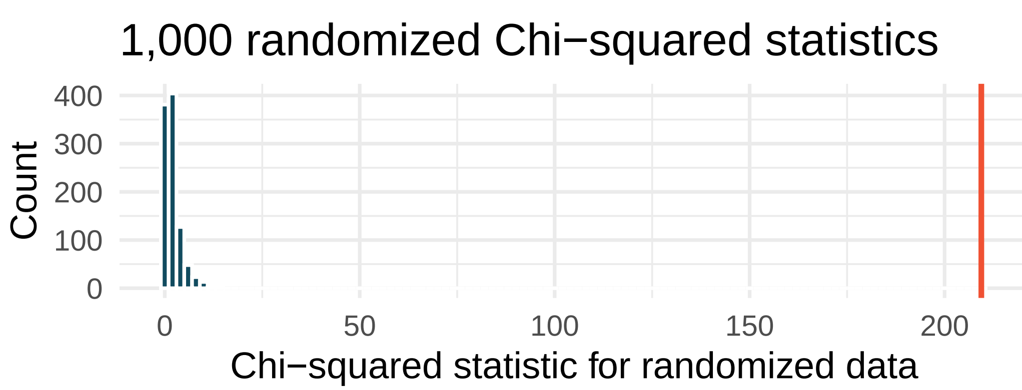


a. The histogram above describes the Chi-squared statistics for 1,000 different randomization datasets. When randomizing the data, is the imposed structure that the variables are independent or that the variables are associated? Explain.

b. What is the range of plausible values for the randomized Chi-squared statistic?

c. The observed Chi-squared statistic is 68.8 (marked in red on plot). Does the observed value provide evidence against the null hypothesis? To answer the question, state the null and alternative hypotheses, approximate the p-value, and conclude the test in the context of the problem.

8. **Disaggregating Asian American tobacco use, randomization test.** Understanding cultural differences in tobacco use across different demographic groups can lead to improved health care education and treatment. A recent study disaggregated tobacco use across Asian American ethnic groups including Asian-Indian (n = 4373), Chinese (n = 4736), and Filipino (n = 4912), in comparison to non-Hispanic Whites (n = 275,025). The number of current smokers in each group was reported as Asian-Indian (n = 223), Chinese (n = 279), Filipino (n = 609), and non-Hispanic Whites (n = 50,880). (Rao et al. 2021) The original data were randomized 1000 times (smoking status randomly assigned to the observations across ethnicities), and the histogram of the Chi-squared statistic on each randomization is displayed.



a. The histogram above describes the Chi-squared statistics for 1000 different randomization datasets. When randomizing the data, is the imposed structure that the variables are independent or that the variables are associated? Explain.

b. What is the range of plausible values for the randomized Chi-squared statistic?

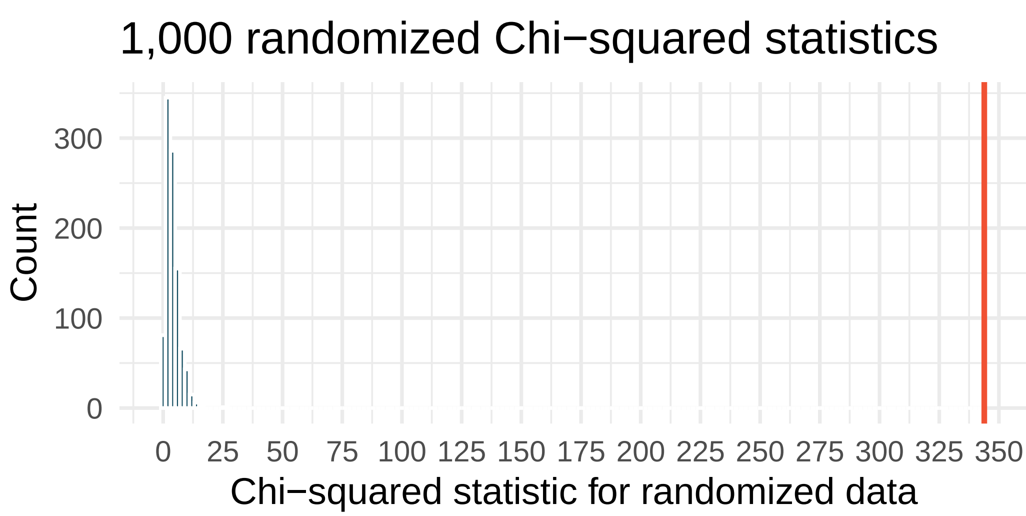
c. The observed Chi-squared statistic is 209.42 (marked in red on plot). Does the observed value provide evidence against the null hypothesis? To answer the question, state the null and alternative hypotheses, approximate the p-value, and conclude the test in the context of the problem.

9. **Lizard habitats, larger data.** In order to assess whether habitat conditions are related to the sunlight choices a lizard makes for resting, Western fence lizard (*Sceloporus occidentalis*) were observed across three different microhabitats. (Adolph 1990; Asbury and Adolph 2007)

Consider the situation where the dataset is 5 times *larger* than the original data (but have the same proportional representation in each category). The distribution of lizards in each of the sites resting in the sun, partial sun, and shade are as follows.

Larger data				
	sunlight			
site	sun	partial	shade	Total
desert	80	160	355	595
mountain	280	180	75	535
valley	210	200	120	530
Total	570	540	550	1,660

The larger dataset was randomized 1,000 times (sunlight preference randomly assigned to the observations across sites), and the histogram of the Chi-squared statistic on each randomization is displayed.



a. The histogram above describes the Chi-squared statistics for 1,000 different randomization of the larger dataset. When randomizing the data, is the imposed structure that the variables are independent or that the variables are associated? Explain.

b. What is the (approximate) range of plausible values for the randomized Chi-squared statistic?

c. The observed Chi-squared statistic is 343.865 (and seen in red on the graph). Does the observed value provide evidence against the null hypothesis? To answer the question, state the null and alternative hypotheses, approximate the p-value, and conclude the test in the context of the problem.

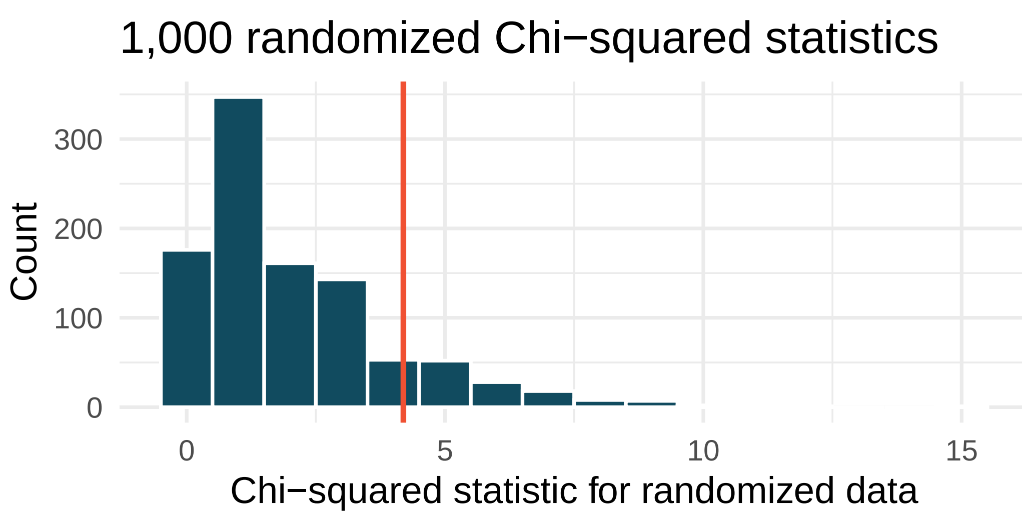
d. If the alternative hypothesis is true, how does the sample size effect the ability to reject the null hypothesis? (*Hint:* Consider the original data as compared with the larger dataset that have the same proportional values.)

10. **Disaggregating Asian American tobacco use, smaller data.** Understanding cultural differences in tobacco use across different demographic groups can lead to improved health care education and treatment. A recent study disaggregated tobacco use across Asian American ethnic groups. (Rao et al. 2021)

Consider the situation where the dataset is 50 times *smaller* than the original data (but have the same proportional representation in each category). The distribution of smokers in each of the ethnicity groups in the smaller data are as follows.

Smaller data			
	Smoking		
ethnicity	don't smoke	smoke	Total
Asian-Indian	83	4	87
Chinese	89	6	95
Filipino	86	12	98
Total	258	22	280

The smaller dataset was randomized 1,000 times (smoking status randomly assigned to the observations across ethnicities), and the histogram of the Chi-squared statistic on each randomization is displayed.



a. The histogram above describes the Chi-squared statistics for 1,000 different randomization of the smaller dataset. When randomizing the data, is the imposed structure that the variables are independent or that the variables are associated? Explain.

b. What is the (approximate) range of plausible values for the randomized Chi-squared statistic?

c. The observed Chi-squared statistic is 4.19 (and seen in red on the graph). Does the observed value provide evidence against the null hypothesis? To answer the question, state the null and alternative hypotheses, approximate the p-value, and conclude the test in the context of the problem.

d. If the alternative hypothesis is true, how does the sample size effect the ability to reject the null hypothesis? (*Hint:* Consider the original data as compared with the smaller dataset that have the same proportional values.)

11. **True / False, I.** Determine if the statements below are true or false. For each false statement, suggest an alternative wording to make it a true statement.

a. The Chi-square distribution, just like the normal distribution, has two parameters, mean and standard deviation.

b. The Chi-square distribution is always right skewed, regardless of the value of the degrees of freedom parameter.

c. The Chi-square statistic is always greater than or equal to 0.

d. As the degrees of freedom increases, the shape of the Chi-square distribution becomes more skewed.

12. **True / False, II.** Determine if the statements below are true or false. For each false statement, suggest an alternative wording to make it a true statement.

a. As the degrees of freedom increases, the mean of the Chi-square distribution increases.

b. If you found $\chi^2 = 10$ with $df = 5$ you would fail to reject H_0 at the 5% discernibility level.

c. When finding the p-value of a Chi-square test, we always shade the tail areas in both tails.

d. As the degrees of freedom increases, the variability of the Chi-square distribution decreases.

13. **Sleep deprived transportation workers.** The National Sleep Foundation conducted a survey on the sleep habits of randomly sampled transportation workers and randomly sampled non-transportation workers that serve as a "control" for comparison. (National Sleep Foundation 2012) The results of the survey are shown below. Conduct a hypothesis test to evaluate if these data provide evidence of an association between sleep levels and profession.

Profession	Less than 6 hours	6 to 8 hours	More than 8 hours	Total
Non-transportation workers	35	193	64	292
Transportation workers	104	499	192	795
Total	139	692	256	1,087

14. **Parasitic worm.** Lymphatic filariasis is a disease caused by a parasitic worm. Complications of the disease can lead to extreme swelling and other complications. Here we consider results from a randomized experiment that compared three different drug treatment options to clear people of the this parasite, which people are working to eliminate entirely. The results for the second year of the study are given below: (King et al. 2018)

	Outcome		
group	Clear at Year 2	Not Clear at Year 2	Total
Three drugs	52	2	54
Two drugs	31	24	55
Two drugs annually	42	14	56
Total	125	40	165

a. Set up hypotheses for evaluating whether there is any difference in the performance of the treatments, and also check conditions.

b. Statistical software was used to run a Chi-square test, which output: $X^2 = 23.7 \quad df = 2 \quad$ p-value < 0.0001. Use these results to evaluate the hypotheses from part (a), and provide a conclusion in the context of the problem.

15. **Shipping holiday gifts.** A local news survey asked 500 randomly sampled Los Angeles residents which shipping carrier they prefer to use for shipping holiday gifts. The table below shows the distribution of responses by age group as well as the expected counts for each cell (shown in italics).

	Age						
Shipping method	18-34		35-54		55+		Total
USPS	72	*81*	97	*102*	76	*62*	245
UPS	52	*53*	76	*68*	34	*41*	162
FedEx	31	*21*	24	*27*	9	*16*	64
Something else	7	*5*	6	*7*	3	*4*	16
Not sure	3	*5*	6	*5*	4	*3*	13
Total	165		209		126		500

a. State the null and alternative hypotheses for testing for independence of age and preferred shipping method for holiday gifts among Los Angeles residents.

b. Are the conditions for inference using a Chi-square test satisfied?

16. **Coffee and depression.** Researchers conducted a study investigating the relationship between caffeinated coffee consumption and risk of depression in women. They collected data on 50,739 women free of depression symptoms at the start of the study in the year 1996, and these women were followed through 2006. The researchers used questionnaires to collect data on caffeinated coffee consumption, asked each individual about physician- diagnosed depression, and also asked about the use of antidepressants. The table below shows the distribution of incidences of depression by amount of caffeinated coffee consumption. (Lucas et al. 2011)

	Caffeinated coffee consumption					
Clinical depression	1 cup / week or fewer	2-6 cups / week	1 cups / day	2-3 cups / day	4 cups / day or more	Total
Yes	670	___	905	564	95	2,607
No	11,545	6,244	16,329	11,726	2,288	48,132
Total	12,215	6,617	17,234	12,290	2,383	50,739

a. What type of test is appropriate for evaluating if there is an association between coffee intake and depression?

b. Write the hypotheses for the test you identified in part (a).

c. Calculate the overall proportion of women who do and do not suffer from depression.

d. Identify the expected count for the empty cell, and calculate the contribution of this cell to the test statistic.

e. The test statistic is $\chi^2 = 20.93$. What is the p-value?

f. What is the conclusion of the hypothesis test?

g. One of the authors of this study was quoted on the New York Times as saying it was "too early to recommend that women load up on extra coffee" based on just this study. (O'Connor 2011) Do you agree with this statement? Explain your reasoning.

Datasets: `lizard_habitat` (openintro)

Chapter 16

Analysis of Variance

> **Draft Chapter.** This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In earlier chapters, we developed methods to compare the mean of a quantitative outcome across two groups. An important aspect of those analyses was to look at the difference in sample means as an estimate for the difference in population means. When comparing more than two groups, simple subtraction will not fully capture the variation across three or more groups. As with two groups, the research question focuses on whether group membership is independent of the quantitative response variable. In this chapter, we focus on a new statistic that incorporates differences in means across more than two groups. Although the ideas are similar to the t-test, they have earned their own name: **AN**alysis **O**f **VA**riance, or **ANOVA**.

Sometimes we want to compare means across many groups. We might initially think to do pairwise comparisons. For example, if there were three groups, we might compare the first mean with the second, then with the third, and finally compare the second and third means – a total of three comparisons. However, this strategy can be treacherous. If we have many groups and do many comparisons, it is likely that we will eventually find a difference just by chance, even if there is no difference in the populations. Instead, we should apply a holistic test to check whether there is evidence that at least one pair of groups is different, which is where ANOVA saves the day.

ANOVA uses a single hypothesis test to check whether the means across many groups are equal:

- H_0: The mean outcome is the same across all groups: $\mu_1 = \mu_2 = \cdots = \mu_k$ where μ_j represents the mean of the outcome for observations in category j.
- H_A: At least one mean is different.

Generally, we must check three conditions before performing ANOVA:

- The observations are independent within and between groups.
- The responses within each group are nearly normal.
- The variability across the groups is about equal.

When these conditions are met, we may perform ANOVA to determine whether the data provide convincing evidence against the null hypothesis.

College departments commonly run multiple sections of the same introductory course. Consider a statistics department that runs three sections of an intro stats course. We might like to determine whether there are substantial differences in first exam scores across sections A, B, and C. Describe appropriate hypotheses.

- H_0: The average score is identical in all sections, $\mu_A = \mu_B = \mu_C$. The observed differences are due to chance.
- H_A: The average score varies by class. We would reject the null hypothesis if there were larger differences among the class averages than what we might expect from chance alone.

Strong evidence favoring the alternative hypothesis in ANOVA is described by unusually large differences among the group means. Assessing the variability of the group means relative to the variability among individual observations within each group is key to ANOVA's success.

Consider the figure below showing two sets of three groups.

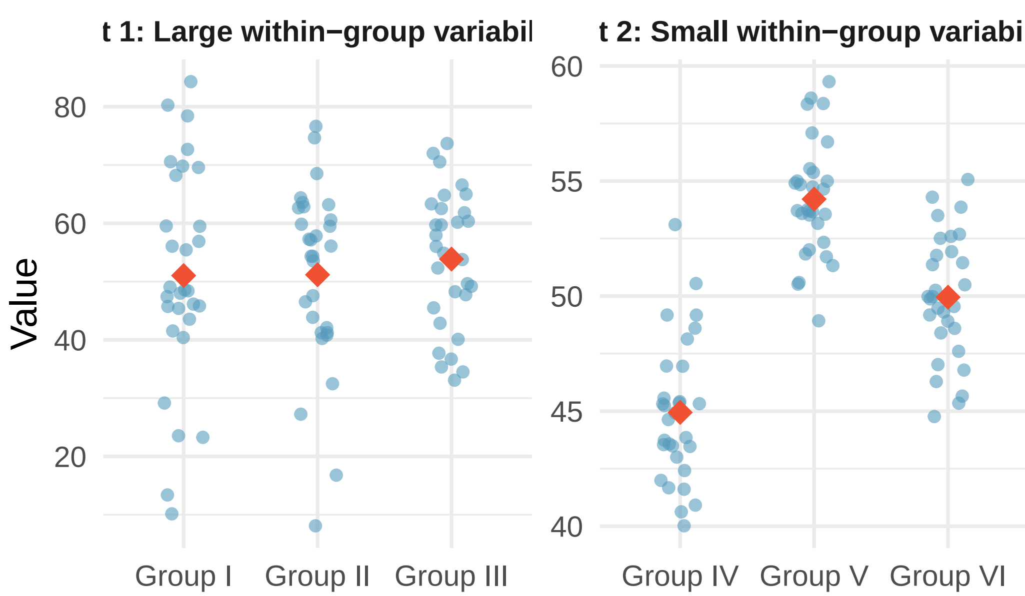


Figure 16.1: Side-by-side dot plots for two sets of three groups. In the first set (Groups I, II, III), the within-group variability is large relative to the between-group differences. In the second set (Groups IV, V, VI), the within-group variability is small and the between-group differences are clearly visible.

Compare Groups I, II, and III. Can you visually determine if the group differences are real? Now compare Groups IV, V, and VI.

Any real difference in Groups I, II, and III is difficult to discern because the data within each group are very volatile relative to differences in the group averages. On the other hand, Groups IV, V, and VI show noticeable differences because those differences are large *relative to the variability in individual observations within each group*.

16.1 Case study: MLB batting

We would like to determine whether there are real differences in batting performance among baseball players according to their position: outfielder (OF), infielder (IF), and catcher (C). We use data from 429 Major League Baseball players from the 2018 season who had at least 100 at-bats. The measure of batting performance is the on-base percentage (OBP), which roughly represents the fraction of the time a player successfully gets on base or hits a home run.

Summary statistics for each group:

Position	n	Mean	SD
OF	160	0.320	0.044
IF	205	0.318	0.037
C	64	0.302	0.038

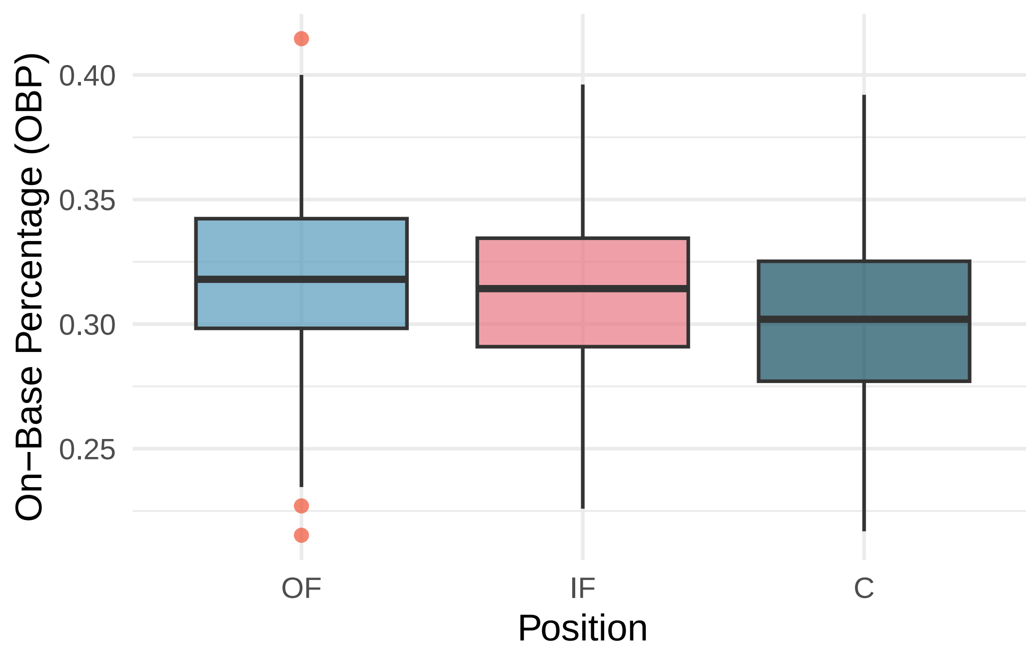


Figure 16.2: Side-by-side box plots of on-base percentage for 429 players across three positions. The variability appears approximately constant across groups. Catchers seem to have a slightly lower on-base percentage.

The null hypothesis under consideration is $\mu_{OF} = \mu_{IF} = \mu_{C}$. Write the null and alternative hypotheses in plain language.

Show answer

H_0: The average on-base percentage is equal across the three positions. H_A: The average on-base percentage varies across some (or all) groups.

What would be an appropriate point estimate of the on-base percentage by outfielders, μ_{OF}?

A good estimate of the on-base percentage by outfielders would be the sample average of OBP for just those players who are outfielders: $\bar{x}_{OF} = 0.320$.

16.1.1 Why not just compare the most different groups?

The largest difference between sample means is between catchers and outfielders. Why might it be inappropriate to simply test whether μ_C and μ_{OF} are different?

The primary issue is **data snooping** (also called **data fishing**). We are inspecting the data before picking the groups to compare. It is inappropriate to examine all data informally and only afterward decide which parts to formally test. Naturally, we would pick the groups with the largest differences, which inflates the Type I error rate.

For example, suppose we measure aptitude for students in 20 classrooms where all students are randomly assigned. With so many groups, we will probably observe a few that look different just by chance. If we select only those classes and perform a formal test, we will probably reach the wrong conclusion.

16.2 Randomization test for comparing many means

16.2.1 The F-statistic

The method of analysis of variance focuses on answering one question: is the variability in the sample means so large that it seems unlikely to be from chance alone? We quantify the between-group variability using the **mean square between groups** (**MSG**), which has degrees of freedom $df_G = k - 1$ where k is the number of groups. The MSG can be thought of as a scaled variance formula for the group means.

We also need a benchmark for how much variability to expect if the null hypothesis is true. For this, we compute the **mean square error** (**MSE**), which measures the variability *within* groups and has degrees of freedom $df_E = n - k$.

The **F-statistic** is the ratio of these two quantities:

$$F = \frac{MSG}{MSE}$$

The MSG represents between-group variability, and MSE measures within-group variability. When the null hypothesis is true, any differences among sample means are due to chance, and the MSG and MSE should be about equal, making F close to 1. Large values of F suggest that the between-group differences are larger than what we would expect from random variation.

The test statistic for three or more means is an F.

The F statistic is a ratio of how the groups differ (MSG) as compared to how the observations within a group vary (MSE).

$$F = \frac{MSG}{MSE}$$

When the null hypothesis is true and the conditions are met, F has an F-distribution with $df_1 = k-1$ and $df_2 = n - k$.

Conditions:

- Independent observations, both within and across groups
- Large samples and no extreme outliers

16.2.2 Variability of the statistic

Suppose we have three exams (A, B, and C) and want to investigate whether they differ in difficulty:

- H_0: $\mu_A = \mu_B = \mu_C$. The exams are equally difficult.
- H_A: At least one exam is more (or less) difficult than the others.

Exam	n	Mean	SD	Min	Max
A	58	75.1	13.9	44	100
B	55	72.0	13.8	38	100
C	51	79.4	14.1	45	100

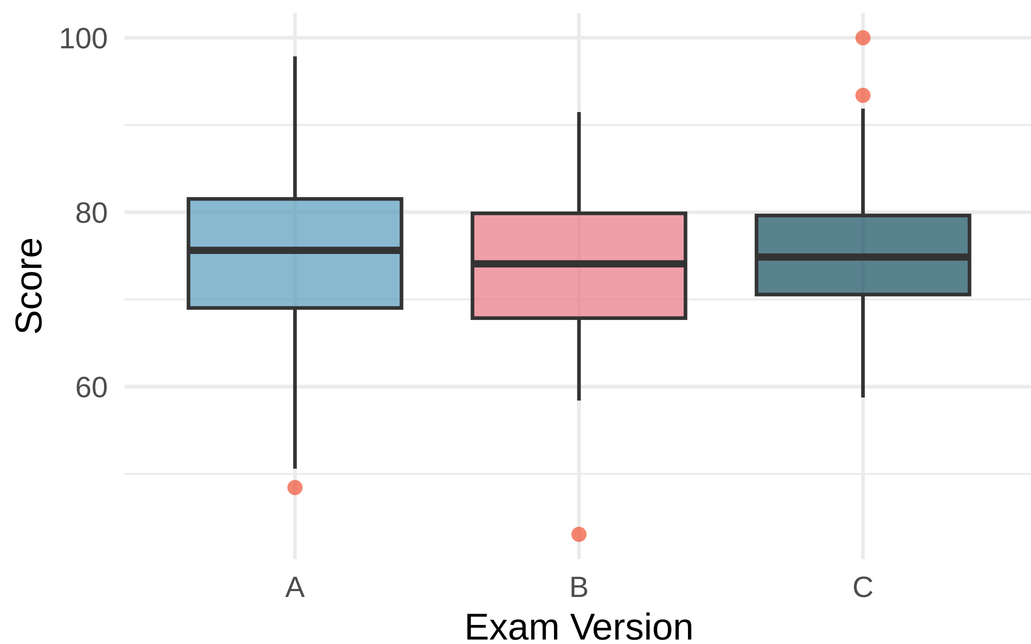


Figure 16.3: Box plots of exam scores for three exam versions. Exam C appears to have a higher median, but the distributions overlap.

To perform a randomization test, we randomly reassign the exam labels to the observed scores. If the null hypothesis is true, the exam version should not matter, so any score could belong to any exam version.

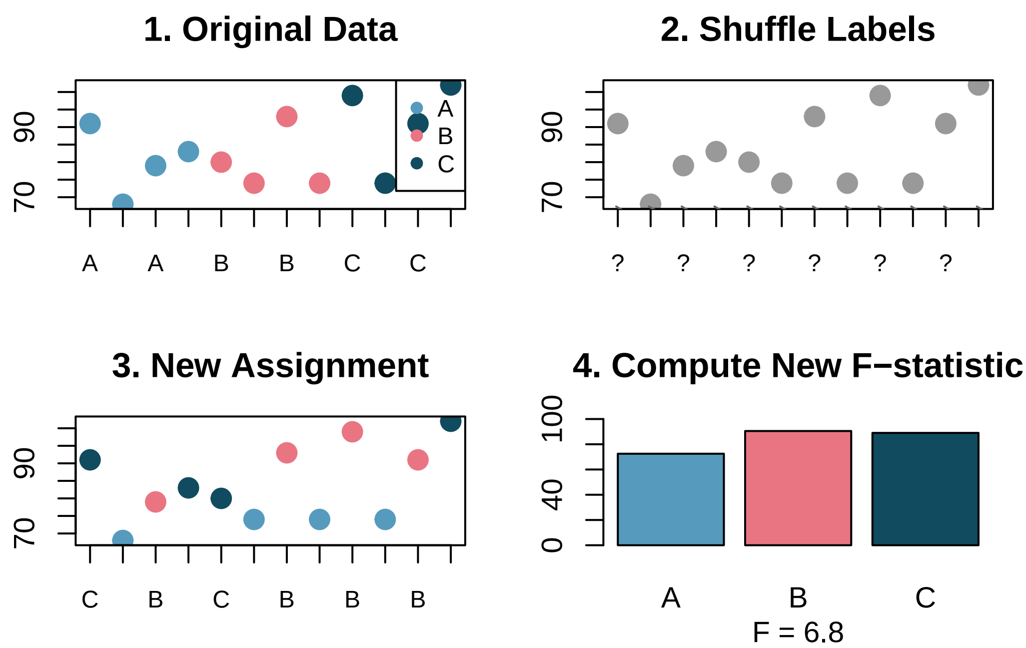


Figure 16.4: The randomization process: exam version labels are randomly shuffled across the observed scores to generate a new assignment, from which a new F-statistic is computed.

By repeating this process 1,000 times, we build a null distribution of F-statistics.

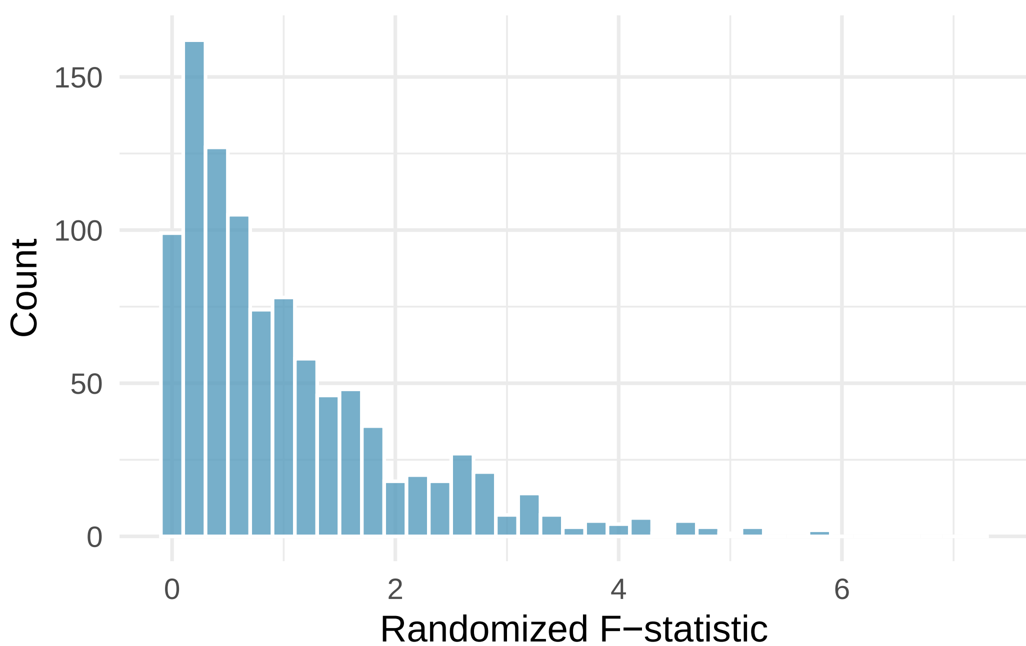


Figure 16.5: Histogram of F-statistics from 1,000 randomizations. The distribution is right-skewed with most values below 2. The tail extends to about 6.

16.2.3 Observed statistic vs. null statistics

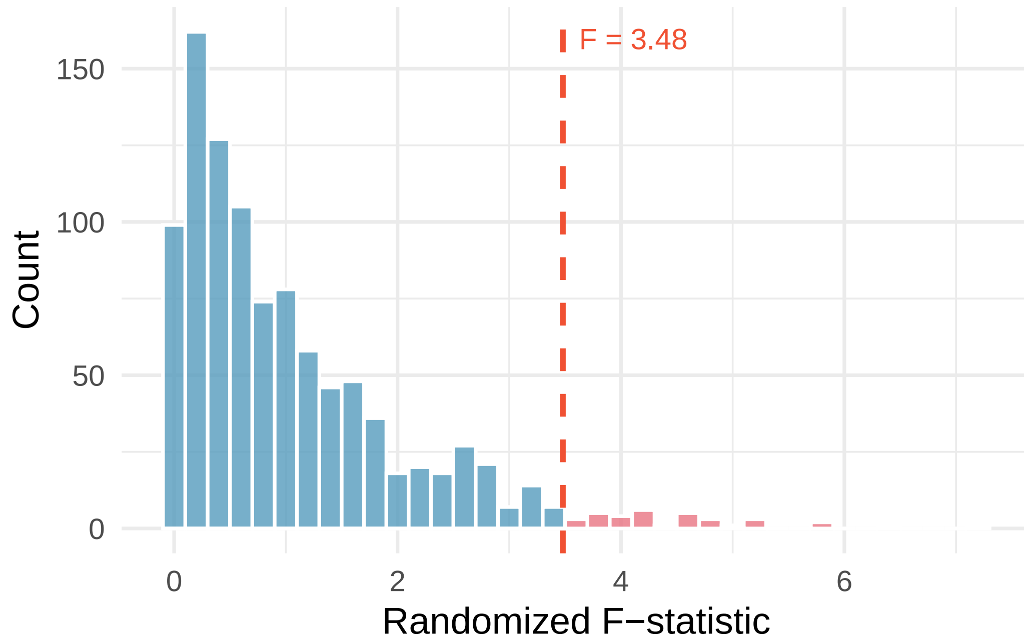


Figure 16.6: Histogram of F-statistics from 1,000 randomizations with the observed F-statistic of 3.48 marked by a red line. The area to the right represents the p-value.

Using statistical software, we find that 3.6% of the randomized F statistics were at or above the observed test statistic of $F = 3.48$. That is, the p-value is 0.036. Assuming a significance level of $\alpha = 0.05$, the p-value is smaller than the significance level, which leads us to reject the null hypothesis. We conclude that the difficulty level is different for at least one of the exams.

16.3 Mathematical model for ANOVA

The randomization test on the F-statistic has a corresponding mathematical theory: the **F-distribution**.

16.3.1 Variability of the statistic

The larger the observed variability in the sample means (MSG) relative to the within-group variability (MSE), the larger the F-statistic and the stronger the evidence against the null hypothesis. We use the upper tail of the F-distribution to compute the p-value.

The F-statistic and the F-test.

Analysis of variance (ANOVA) uses a test statistic, the F-statistic, which represents a standardized ratio of variability in the sample means relative to the variability within the groups. If H_0 is true and the model conditions are satisfied, the F-statistic follows an F-distribution with parameters $df_1 = k - 1$ and $df_2 = n - k$. The upper tail of the F-distribution is used to compute the p-value.

For the MLB data, $MSG = 0.00803$ and $MSE = 0.00158$. Identify the degrees of freedom associated with MSG and MSE and verify the F-statistic is approximately 5.077.

Show answer

There are $k = 3$ groups, so $df_G = k - 1 = 2$. There are $n = 429$ total observations, so $df_E = n - k = 426$. The F-statistic is $F = \frac{MSG}{MSE} = \frac{0.00803}{0.00158} = 5.08 \approx 5.077$.

16.3.2 The p-value from the F-distribution

A p-value can be computed from the F-statistic using an F-distribution with $df_1 = df_G$ and $df_2 = df_E$.

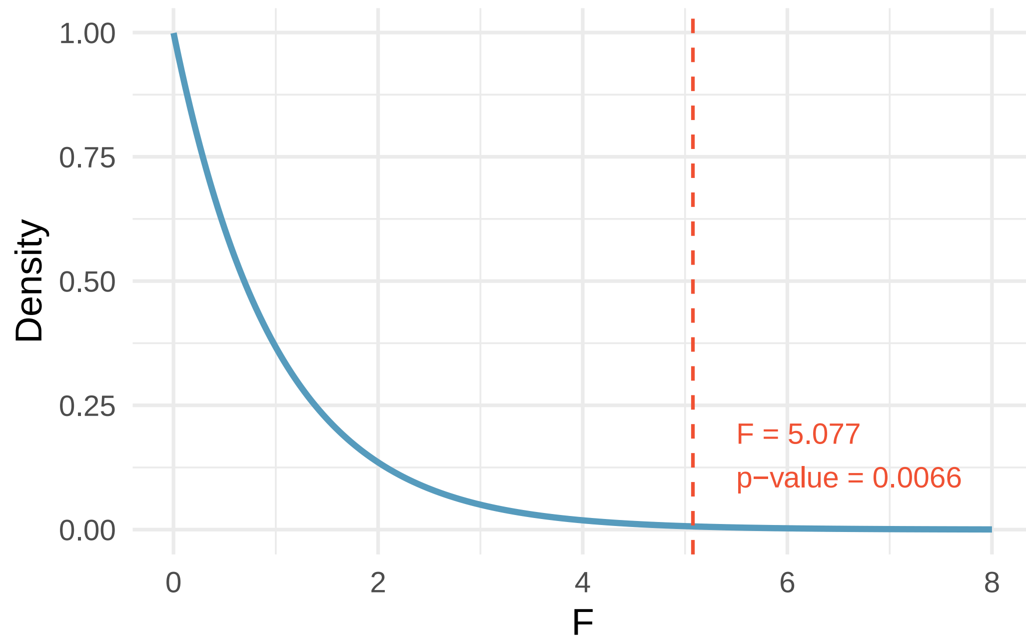


Figure 16.7: An F-distribution with $df_1 = 2$ and $df_2 = 426$. The area to the right of $F = 5.077$ is shaded, representing a p-value of approximately 0.0066.

The p-value corresponding to $F = 5.077$ with $df_1 = 2$ and $df_2 = 426$ is approximately 0.0066. Does this provide strong evidence against the null hypothesis?

The p-value is smaller than 0.05, indicating the evidence is strong enough to reject the null hypothesis at a significance level of 0.05. The data provide strong evidence that the average on-base percentage varies by player's primary field position.

Note that the small p-value tells us there is a notable difference somewhere, but ANOVA does **not** tell us *which* group is different. Follow-up analyses (discussed in the next chapter on multiple comparisons) are needed.

16.3.3 Reading an ANOVA table

ANOVA results are commonly presented in a table. Here is the ANOVA table for the MLB data:

Source	df	Sum Sq	Mean Sq	F value	p-value
position	2	0.0161	0.00803	5.077	0.0066
Residuals	426	0.6740	0.00158		

The key information is in the last two columns: the F-statistic (5.077) and the p-value (0.0066).

Components of an ANOVA table.

- **Sum of Squares Between Groups (SSG):** Measures the total variability of the group means around the overall mean. $SSG = \sum_{j=1}^{k} n_j(\bar{x}_j - \bar{x})^2$.
- **Sum of Squared Errors (SSE):** Measures the total variability of observations around their group means. $SSE = \sum_{j=1}^{k}(n_j - 1)s_j^2$.
- **Sum of Squares Total (SST):** The total variability in all observations. $SST = SSG + SSE$.
- **Mean Square Between Groups (MSG):** $MSG = SSG/(k-1)$.
- **Mean Square Error (MSE):** $MSE = SSE/(n-k)$.
- **F-statistic:** $F = MSG/MSE$.

16.4 Conditions for ANOVA

There are three conditions to check before performing ANOVA:

- **Independence.** If the data are a simple random sample, this condition is generally satisfied. For experiments, carefully consider whether the data may be independent (e.g., no pairing).
- **Approximately normal.** As with one- and two-sample testing for means, the normality assumption is especially important when sample sizes are small. With large sample sizes within each group, we look mainly for extreme outliers.
- **Constant variance.** The variability in the groups should be about equal. This can be checked by examining side-by-side box plots or comparing the standard deviations across groups. A common rule of thumb is that the ratio of the largest to smallest group standard deviation should be less than 2.

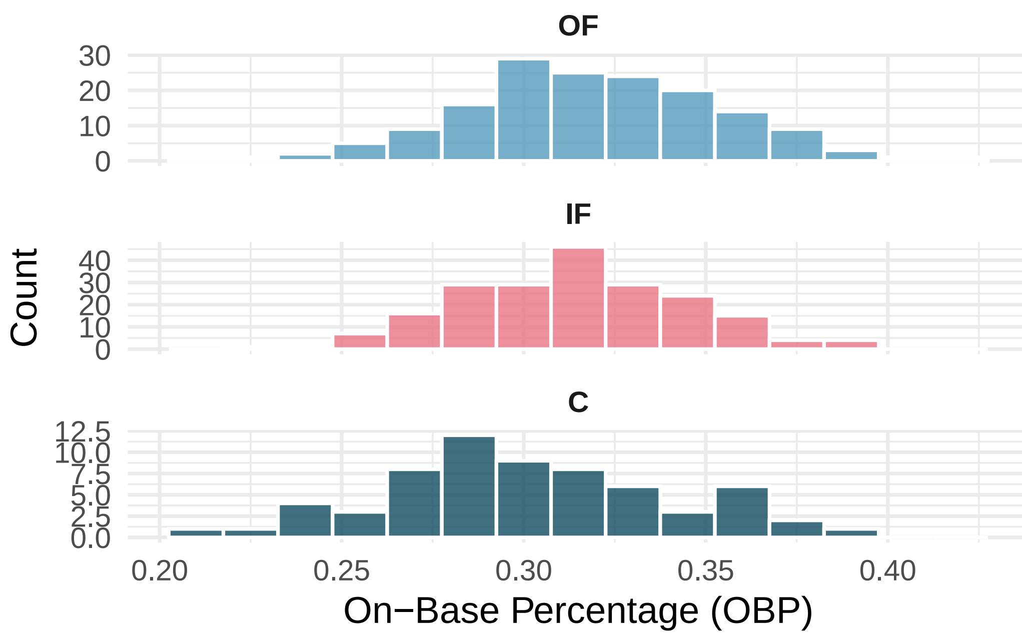


Figure 16.8: Histograms of OBP for each field position, showing that the distributions are reasonably bell-shaped and symmetric.

Diagnostics for an ANOVA analysis.

Independence is always important. The normality condition matters most when group sizes are small (with large groups, look only for extreme outliers). The constant variance condition is especially important when group sizes differ.

16.5 Chapter review

16.5.1 Summary

In this chapter, we introduced both the randomization test and the mathematical model for comparing means across two or more groups using ANOVA. The F-statistic measures the ratio of between-group variability (MSG) to within-group variability (MSE). When the null hypothesis is true and the conditions are satisfied, the F-statistic follows an F-distribution. A significant result tells us that at least one group mean differs from the others, but does not identify *which* groups differ. For that, we need multiple comparison procedures, which are covered in the next chapter.

16.5.2 Key terms

- ANOVA (analysis of variance)
- F-statistic
- F-distribution
- F-test
- Mean square between groups (MSG)
- Mean square error (MSE)
- Sum of squares between groups (SSG)
- Sum of squared errors (SSE)
- Sum of squares total (SST)
- Degrees of freedom (ANOVA)
- Data snooping / data fishing

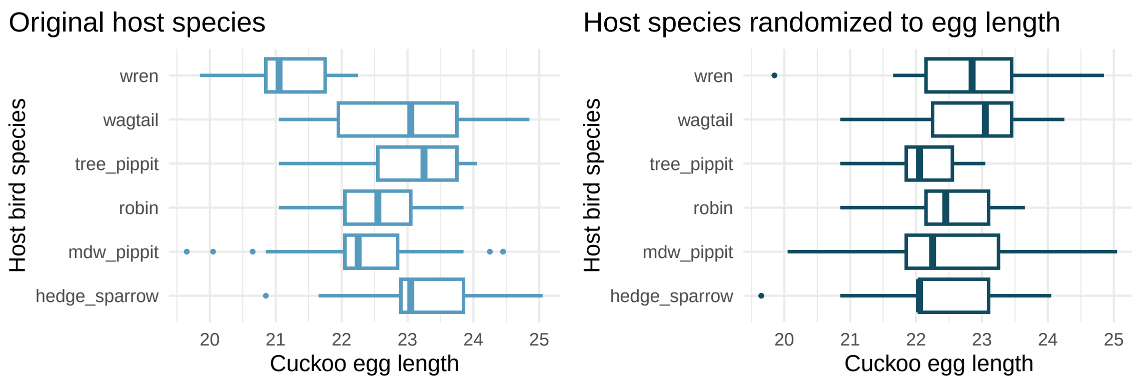
16.6 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Fill in the blank.** When doing an ANOVA, you observe large differences in means between groups. Within the ANOVA framework, this would most likely be interpreted as evidence strongly favoring the ____________ hypothesis.

2. **Which test?** We would like to test if students who are in the social sciences, natural sciences, arts and humanities, and other fields spend the same amount of time, on average, studying for a course. What type of test should we use? Explain your reasoning.

3. **Cuckoo bird egg lengths, randomize once.** Cuckoo birds lay their eggs in other birds' nests, making them known as brood parasites. One question relates to whether the size of the cuckoo egg differs depending on the species of the host bird. (Latter 1902) Consider the following plots, one represents the original data, the second represents data where the host species has been randomly assigned to the egg length.

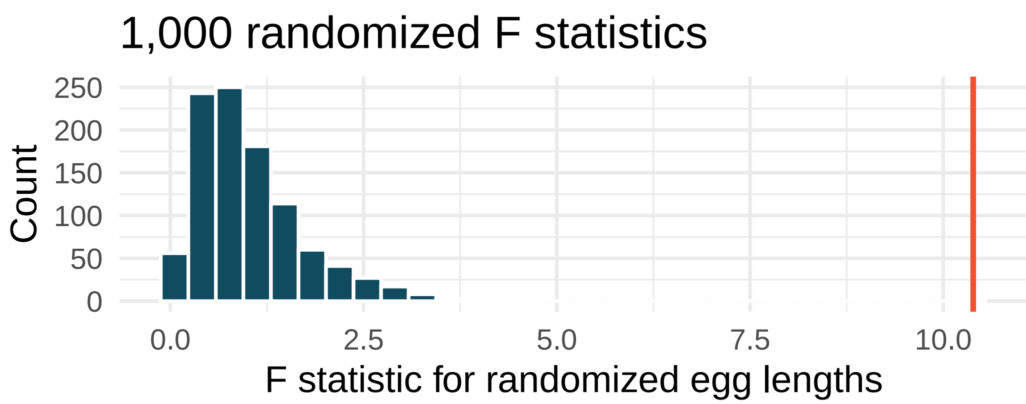


a. Consider the average length of the eggs for each species. Is the average length for the original data: more variable, less variable, or about the same as the randomized species? Describe what you see in the plots.

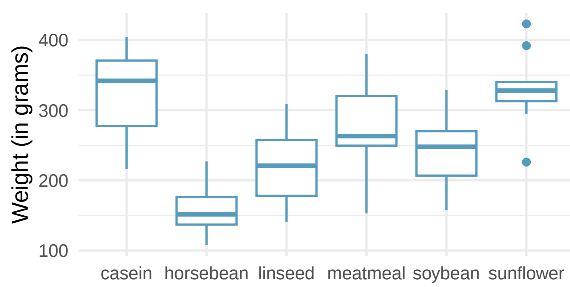
b. Consider the standard deviation of the lengths of the eggs within each species. Is the within species standard deviation of the length for the original data: bigger, smaller, or about the same as the randomized species?

c. Recall that the F statistic's numerator measures how much the groups vary (MSG) with the denominator measuring how much the within species values vary (MSE), which of the plots above would have a larger F statistic, the original data or the randomized data? Explain.

4. **Cuckoo bird egg lengths, randomization test.** Cuckoo birds lay their eggs in other birds' nests, making them known as brood parasites. One question relates to whether the size of the cuckoo egg differs depending on the species of the host bird. (Latter 1902) Using the randomization distribution of the F statistic (host species randomized to egg length), conduct a hypothesis test to evaluate if there is a difference, in the population, between the average egg lengths for different host bird species. Make sure to state your hypotheses clearly and interpret your results in context of the data.



5. **Chicken diet and weight, many groups.** An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens. Newly hatched chicks were randomly allocated into six groups, and each group was given a different feed supplement. Sample statistics and a visualization of the observed data are shown below. (McNeil 1977)



Feed type	Mean	SD	n
casein	323.58	64.43	12
horsebean	160.20	38.63	10
linseed	218.75	52.24	12
meatmeal	276.91	64.90	11
soybean	246.43	54.13	14
sunflower	328.92	48.84	12

Using the ANOVA output below, conduct a hypothesis test to determine if these data provide convincing evidence that the average weight of chicks varies across some (or all) groups. Make sure to check relevant conditions.

term	df	sumsq	meansq	statistic	p.value
`feed`	5	231,129	46,226	15.4	<0.0001
`Residuals`	65	195,556	3,009	NA	NA

6. **Teaching descriptive statistics.** A study compared five different methods for teaching descriptive statistics. The five methods were traditional lecture and discussion, programmed textbook instruction, programmed text with lectures, computer instruction, and computer instruction with lectures. 45 students were randomly assigned, 9 to each method. After completing the course, students took a 1-hour exam.

a. What are the hypotheses for evaluating if the average test scores are different for the different teaching methods?

b. What are the degrees of freedom associated with the F-test for evaluating these hypotheses?

c. Suppose the p-value for this test is 0.0168. What is the conclusion?

7. **Coffee, depression, and physical activity.** Caffeine is the world's most widely used stimulant, with approximately 80% consumed in the form of coffee. Participants in a study investigating the relationship between coffee consumption and exercise were asked to report the number of hours they spent per week on moderate (e.g., brisk walking) and vigorous (e.g., strenuous sports and jogging) exercise. Based on these data the researchers estimated the total hours of metabolic equivalent tasks (MET) per week, a value always greater than 0. The table below gives summary statistics of MET for women in this study based on the amount of coffee consumed. (Lucas et al. 2011)

	Caffeinated coffee consumption				
	1 cup / week or fewer	2-6 cups / week	1 cups / day	2-3 cups / day	4 cups / day or more
Mean	18.7	19.6	19.3	18.9	17.5
SD	21.1	25.5	22.5	22.0	22.0
n	12,215.0	6,617.0	17,234.0	12,290.0	2,383.0

a. Write the hypotheses for evaluating if the average physical activity level varies among the different levels of coffee consumption.

b. Check conditions and describe any assumptions you must make to proceed with the test.

c. Below is the output associated with this test. What is the conclusion of the test?

	df	sumsq	meansq	statistic	p.value
`cofee`	4	10,508	2,627	5.2	0
`Residuals`	50,734	25,564,819	504	NA	NA
`Total`	50,738	25,575,327	NA	NA	NA

8. **Student performance across discussion sections.** A professor who teaches a large introductory statistics class (197 students) with eight discussion sections would like to test if student performance differs by discussion section, where each discussion section has a different teaching assistant. The summary table below shows the average final exam score for each discussion section as well as the standard deviation of scores and the number of students in each section.

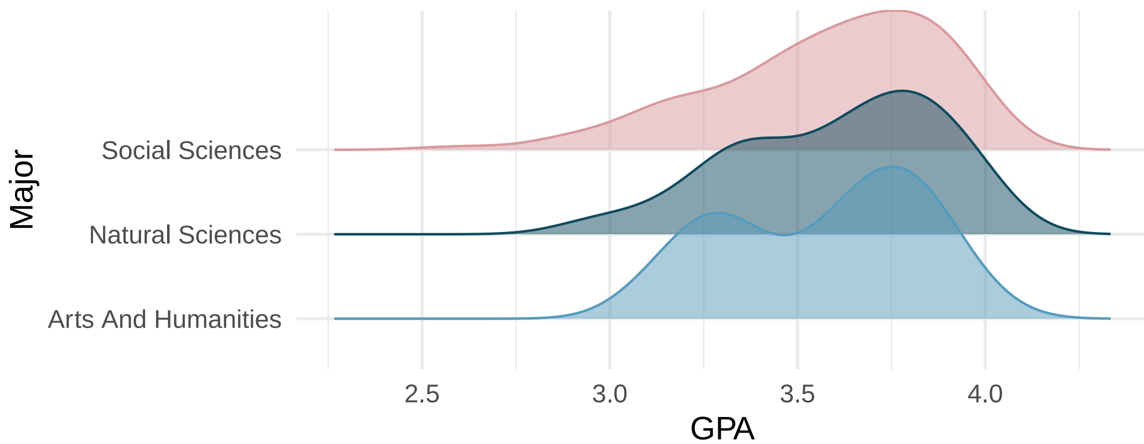
	Sec 1	Sec 2	Sec 3	Sec 4	Sec 5	Sec 6	Sec 7	Sec 8
Mean	92.94	91.11	91.80	92.45	89.30	88.30	90.12	93.35
SD	4.21	5.58	3.43	5.92	9.32	7.27	6.93	4.57
n	33.00	19.00	10.00	29.00	33.00	10.00	32.00	31.00

The ANOVA output below can be used to test for differences between the average scores from the different discussion sections.

	df	sumsq	meansq	statistic	p.value
`section`	7	525	75.0	1.87	0.077
`Residuals`	189	7,584	40.1	NA	NA
`Total`	196	8,109	NA	NA	NA

Conduct a hypothesis test to determine if these data provide convincing evidence that the average score varies across some (or all) groups. Check conditions and describe any assumptions you must make to proceed with the test.

9. **GPA and major.** Undergraduate students in an introductory statistics course at Duke University conducted a survey about GPA and major. The density plots show the distributions of GPA among three groups of majors. ANOVA output is also provided.



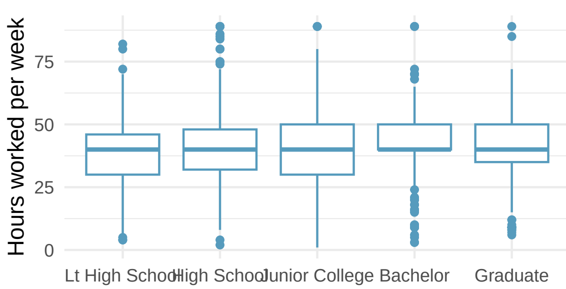
term	df	sumsq	meansq	statistic	p.value
`major`	2	0.03	0.02	0.21	0.81
`Residuals`	195	15.77	0.08	NA	NA

a. Write the hypotheses for testing for a difference between average GPA across majors.

b. What is the conclusion of the hypothesis test?

c. How many students answered the questions on the survey, i.e., what is the sample size?

10. **Work hours and education.** The General Social Survey collects data on demographics, education, and work, among many other characteristics of US residents. (NORC 2010) Using ANOVA, we can consider educational attainment levels for all 1,172 respondents at once. Below are the distributions of hours worked by educational attainment and relevant summary statistics that will be helpful in carrying out this analysis.



Educational attainment	Mean	SD	n
Lt High School	38.7	15.8	121
High School	39.6	15.0	546
Junior College	41.4	18.1	97
Bachelor	42.5	13.6	253
Graduate	40.8	15.5	155

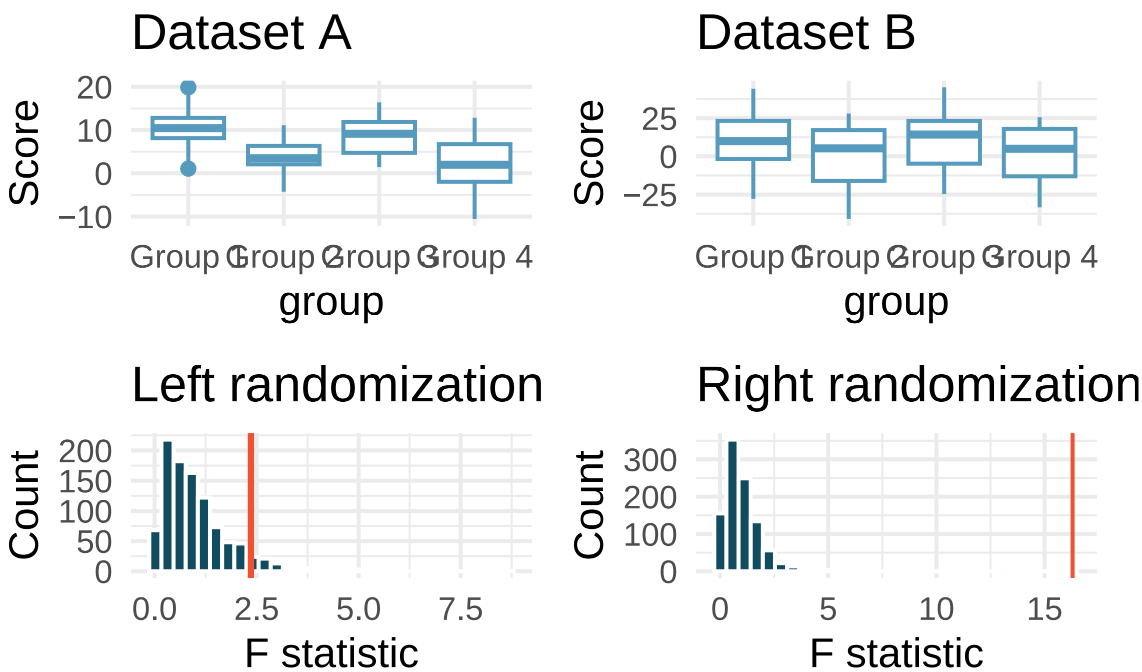
a. Write hypotheses for evaluating whether the average number of hours worked varies across the five groups.

b. Check conditions and describe any assumptions you must make to proceed with the test.

c. Below is the output associated with this test. What is the conclusion of the test?

term	df	sumsq	meansq	statistic	p.value
`degree`	4	2,006	502	2.19	0.07
`Residuals`	1,167	267,382	229	NA	NA

11. **Matching observed data with randomized F statistics.** Consider the following two datasets. The response variable is the `score` and the explanatory variable is whether the individual is in one of four groups.

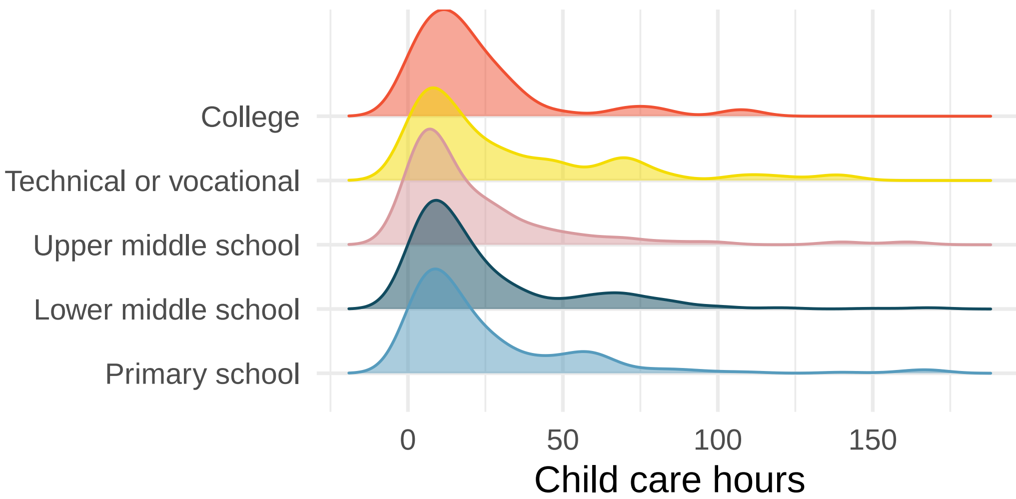


The randomizations (randomly assigning group to the score, calculating a randomization F statistic) were done 1000 times for each of Dataset A and B. The red line on each plot indicates the observed F statistic for the original (unrandomized) data.

a. Does the randomization distribution on the left correspond to Dataset A or B? Explain.

b. Does the randomization distribution on the right correspond to Dataset A or B? Explain.

12. **Child care hours.** The China Health and Nutrition Survey aims to examine the effects of the health, nutrition, and family planning policies and programs implemented by national and local governments. (UNC Carolina Population Center 2006) It, for example, collects information on number of hours Chinese parents spend taking care of their children under age 6. The side-by-side box plots below show the distribution of this variable by educational attainment of the parent. Also provided below is the ANOVA output for comparing average hours across educational attainment categories.



term	df	sumsq	meansq	statistic	p.value
`edu`	4	4,142	1,036	1.26	0.28
`Residuals`	794	653,048	822	NA	NA

a. Write the hypotheses for testing for a difference between the average number of hours spent on child care across educational attainment levels.

b. What is the conclusion of the hypothesis test?

Datasets: `Cuckoo` (Stat2Data)

Chapter 17

Multiple Comparisons

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

When an ANOVA test rejects the null hypothesis, we know that at least one group mean differs from the others – but we do not know *which* groups differ. This chapter addresses the natural follow-up question: which specific pairs of groups have different means? Attempting many pairwise comparisons inflates the overall Type I error rate, so we need specialized methods that control for the fact that we are conducting multiple tests simultaneously. We introduce the multiple testing problem, the Bonferroni correction, and Tukey's Honest Significant Difference (HSD) method.

17.1 The multiple testing problem

When we conduct a single hypothesis test at significance level $\alpha = 0.05$, there is a 5% chance of incorrectly rejecting a true null hypothesis (a Type I error). But what happens when we conduct many tests?

Suppose we compare $k = 4$ groups. How many pairwise comparisons are possible? If we conduct each test at $\alpha = 0.05$ and all null hypotheses are actually true, what is the probability that we make at least one Type I error?

With 4 groups, there are $\binom{4}{2} = 6$ pairwise comparisons. If we conduct each at $\alpha = 0.05$, the probability that any *single* test does not produce a Type I error is $1 - 0.05 = 0.95$. If the tests were independent, the probability that *none* of the 6 tests produces a Type I error would be:

$$P(\text{no errors}) = 0.95^6 = 0.735$$

So the probability of making **at least one** Type I error is:

$$P(\text{at least one error}) = 1 - 0.735 = 0.265$$

That is about a 27% chance of at least one false positive – much higher than the 5% we intended.

17.1.1 Family-wise error rate

The probability of making at least one Type I error across all tests in a family of comparisons is called the **family-wise error rate** (FWER). When conducting m independent tests, each at significance level α, the FWER is:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

The table below shows how the FWER grows with the number of comparisons when $\alpha = 0.05$:

Number of groups (k)	Number of comparisons (m)	FWER
3	3	0.143
4	6	0.265
5	10	0.401
6	15	0.537
10	45	0.901

With 10 groups, there is a 90% chance of making at least one Type I error if we use unadjusted pairwise tests. This is the core of the multiple comparisons problem: we need procedures that control the FWER at an acceptable level (typically 0.05).

Family-wise error rate (FWER).

The **family-wise error rate** is the probability of making one or more Type I errors among all the hypothesis tests in a family of comparisons. Methods like the Bonferroni correction and Tukey's HSD are designed to keep the FWER at or below a specified level α.

17.2 The Bonferroni correction

The simplest method for controlling the family-wise error rate is the **Bonferroni correction**. The idea is straightforward: if we are conducting m tests and want the overall FWER to be at most α, we use a stricter significance level for each individual test.

Bonferroni correction.

When conducting m hypothesis tests and we want the family-wise error rate to be at most α, the Bonferroni correction uses an adjusted significance level for each individual test:

$$\alpha^* = \frac{\alpha}{m}$$

Equivalently, we can multiply each p-value by m (the adjusted p-value) and compare the result to the original α.

After an ANOVA with $k = 4$ groups rejects H_0, we want to conduct all pairwise comparisons at an overall $\alpha = 0.05$. How many comparisons are there, and what adjusted significance level should we use for each?

There are $\binom{4}{2} = 6$ pairwise comparisons. The Bonferroni-adjusted significance level is:

$$\alpha^* = \frac{0.05}{6} = 0.0083$$

So each individual two-sample t-test must have a p-value below 0.0083 to be considered significant.

The Bonferroni correction is simple and broadly applicable – it can be used with any type of test, not just pairwise comparisons of means. However, it is **conservative**, meaning it may fail to detect real differences when they exist (higher Type II error). When the number of comparisons is very large, the Bonferroni correction can be overly strict.

17.2.1 Example: Exam scores

Suppose an ANOVA comparing three exam versions (A, B, C) gave $F = 3.48$ with $p = 0.036$. We rejected H_0 and now want to know which exams differ.

With $k = 3$ groups, there are $\binom{3}{2} = 3$ pairwise comparisons. The Bonferroni-adjusted significance level is $\alpha^* = 0.05/3 = 0.0167$.

Suppose the three pairwise t-test p-values are:

Comparison	p-value	Significant at $\alpha^* = 0.0167$?
A vs. B	0.245	No
A vs. C	0.068	No
B vs. C	0.011	Yes

Only the comparison between exams B and C is significant after the Bonferroni correction. We conclude that there is evidence of a difference in difficulty between exams B and C, but we cannot say the other pairs differ.

17.3 Tukey's Honest Significant Difference

A more powerful alternative to the Bonferroni correction for pairwise comparisons of means is **Tukey's Honest Significant Difference (HSD)** method. While the Bonferroni correction is general-purpose, Tukey's method is specifically designed for comparing all pairs of group means after ANOVA, and it is less conservative.

17.3.1 How Tukey's HSD works

Tukey's HSD computes a single critical difference – the minimum difference between two group means that would be considered statistically significant. The procedure uses the **Studentized range distribution** to determine this critical value.

For each pair of groups j and j', the test statistic is:

$$q = \frac{\bar{x}_j - \bar{x}_{j'}}{\sqrt{MSE/n^*}}$$

where MSE is the mean square error from the ANOVA, and n^* is a measure of the group sample size (when group sizes are equal, $n^* = n_j$; when unequal, a harmonic mean is often used). The statistic q is compared to critical values from the Studentized range distribution.

Tukey's HSD.

Tukey's Honest Significant Difference method controls the family-wise error rate at α when making all pairwise comparisons of group means. It computes a confidence interval for each pairwise difference:

$$(\bar{x}_j - \bar{x}_{j'}) \pm q^* \cdot \sqrt{\frac{MSE}{n^*}}$$

where q^* is the critical value from the Studentized range distribution with k groups and $n-k$ degrees of freedom. If the confidence interval does not contain zero, the difference is significant.

17.3.2 Example: MLB batting positions

After the ANOVA for MLB batting data ($F = 5.077, p = 0.0066$), we rejected H_0 and want to identify which position pairs differ in on-base percentage. Tukey's HSD produces the following pairwise comparisons:

Comparison	Difference	95% CI	Significant?
IF - OF	-0.002	(-0.016, 0.012)	No
C - OF	-0.018	(-0.038, 0.002)	No
C - IF	-0.016	(-0.035, 0.003)	No

Interestingly, although the overall ANOVA was significant, none of the individual Tukey pairwise comparisons reaches significance at the 0.05 level. This can happen because the ANOVA considers all groups simultaneously, while the pairwise comparisons are more conservative. In this case, the overall pattern of catchers having lower OBP than the other two positions drives the ANOVA result, but no single pairwise difference is large enough to be declared significant on its own.

17.4 Interpreting post-hoc results

Post-hoc tests (like Bonferroni or Tukey's HSD) are conducted *after* a significant ANOVA result. Several important principles guide their interpretation:

17.4.1 When to use post-hoc tests

Post-hoc pairwise comparisons should only be conducted after ANOVA rejects the null hypothesis. If the ANOVA p-value is not significant, there is no justification for exploring pairwise differences – doing so would be data snooping.

17.4.2 Reporting results

When reporting results of multiple comparisons, it is common to use **compact letter displays**. Groups that share a letter are not significantly different from each other:

Group	Mean	Grouping
A	75.1	ab
B	72.0	a
C	79.4	b

In this notation, Groups A and B share the letter "a" (not significantly different), and Groups A and C share the letter "b" is not shared by B and C (they are significantly different). Group A, sharing a letter with both B and C, is not significantly different from either.

17.4.3 Confidence intervals for differences

An especially useful way to present multiple comparison results is through confidence intervals for pairwise differences. If a confidence interval contains zero, the two groups are not significantly different. If the interval is entirely positive or entirely negative, the groups differ.

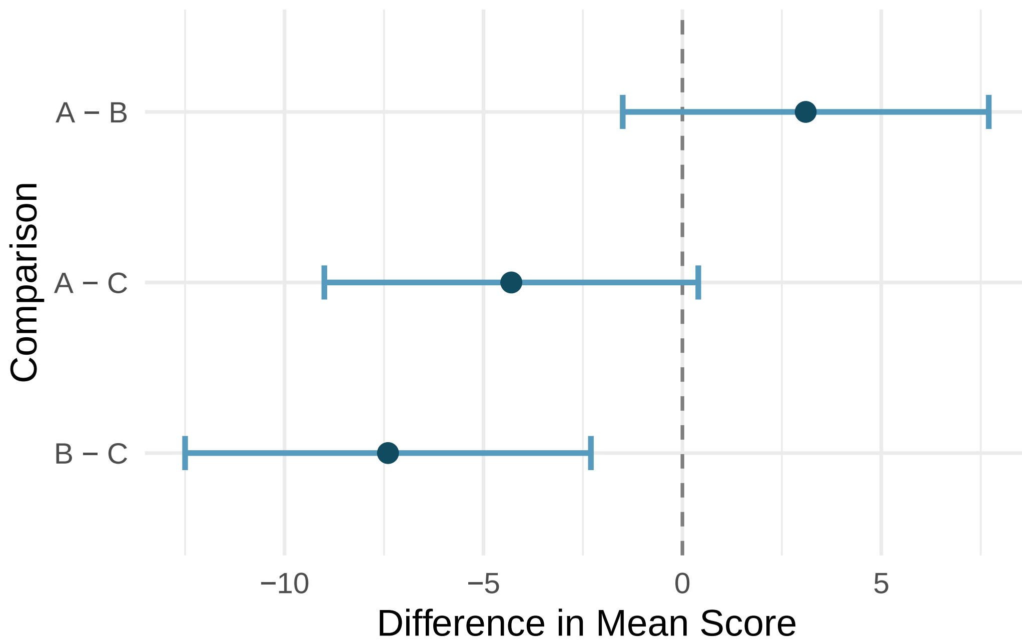


Figure 17.1: Confidence intervals for pairwise differences in mean exam scores. The interval for B vs. C does not contain zero, indicating a significant difference.

17.4.4 Common mistakes

1. **Running multiple pairwise tests without correction.** This inflates the Type I error rate. Always use a correction method when conducting multiple comparisons.
2. **Conducting post-hoc tests after a non-significant ANOVA.** If the overall F-test does not reject H_0, there is no justification for exploring pairwise differences.
3. **Ignoring practical significance.** A statistically significant difference may be too small to matter in practice. Always consider the magnitude of the differences, not just the p-values.
4. **Over-interpreting non-significant pairwise comparisons.** A non-significant pairwise comparison after a significant ANOVA does not mean the groups are equal – it means we do not have enough evidence to declare them different individually.

17.5 Bonferroni vs. Tukey: When to use which

Both methods control the family-wise error rate, but they have different strengths:

Feature	Bonferroni	Tukey's HSD
Generality	Any set of tests	Pairwise comparisons of means only

Feature	Bonferroni	Tukey's HSD
Power	More conservative (less powerful)	Less conservative (more powerful)
Assumptions	None beyond the individual tests	Equal group variances (same as ANOVA)
Best for	Small number of planned comparisons	All pairwise comparisons after ANOVA

Rule of thumb: If you are comparing all pairs of group means after a significant ANOVA, use Tukey's HSD. If you have a small number of pre-planned comparisons (not necessarily all pairs), the Bonferroni correction is appropriate.

17.6 Chapter review

17.6.1 Summary

When ANOVA rejects the null hypothesis, the natural follow-up question is: which groups differ? Performing many pairwise tests without correction inflates the family-wise error rate, making false positives likely. The Bonferroni correction divides the significance level by the number of comparisons, which is simple but conservative. Tukey's HSD method is specifically designed for all pairwise comparisons of means after ANOVA and is typically more powerful. Both methods control the family-wise error rate at the desired level.

17.6.2 Key terms

- Multiple comparisons problem
- Family-wise error rate (FWER)
- Bonferroni correction
- Tukey's Honest Significant Difference (HSD)
- Post-hoc test
- Compact letter display

17.7 Exercises

Answers to odd-numbered exercises are provided inline.

1. **True / False: ANOVA, I.** Determine if the following statements are true or false in ANOVA, and explain your reasoning for statements you identify as false.

a. As the number of groups increases, the modified discernibility level for pairwise tests increases as well.

b. As the total sample size increases, the degrees of freedom for the residuals increases as well.

c. The constant variance condition can be somewhat relaxed when the sample sizes are relatively consistent across groups.

d. The independence assumption can be relaxed when the total sample size is large.

2. **True / False: ANOVA, II.** Determine if the following statements are true or false, and explain your reasoning for statements you identify as false.

If the null hypothesis that the means of four groups are all the same is rejected using ANOVA at a 5% discernibility level, then...

a. we can then conclude that all the means are different from one another.

b. the standardized variability between groups is higher than the standardized variability within groups.

c. the pairwise analysis will identify at least one pair of means that are discernibly different.

d. the appropriate α to be used in pairwise comparisons is 0.05 / 4 = 0.0125 since there are four groups.

Chapter 18

Correlation

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

When we have two quantitative variables, we often want to understand the relationship between them. Does one variable increase as the other increases? Does it decrease? Or is there no apparent pattern? In this chapter, we introduce tools for visualizing and quantifying the strength and direction of the linear relationship between two quantitative variables. We begin with scatterplots, then introduce the correlation coefficient r, discuss its properties and limitations, and address the ever-important distinction between correlation and causation.

18.1 Scatterplots

A **scatterplot** displays the relationship between two quantitative variables. Each point represents a single observation, with one variable on the horizontal axis and the other on the vertical axis. Scatterplots are the most useful tool for visualizing the relationship between two quantitative variables.

When examining a scatterplot, we look for four key features:

1. **Direction**: Is the relationship positive (upward trend), negative (downward trend), or neither?
2. **Form**: Is the relationship linear (points cluster around a line) or nonlinear (points follow a curve)?
3. **Strength**: How tightly do the points cluster around the trend? A strong relationship has little scatter; a weak relationship has lots of scatter.
4. **Unusual observations**: Are there any outliers that deviate from the overall pattern?

Researchers captured 104 brushtail possums in Australia and took body measurements before releasing them. Consider the relationship between total body length (cm) and head length (mm).

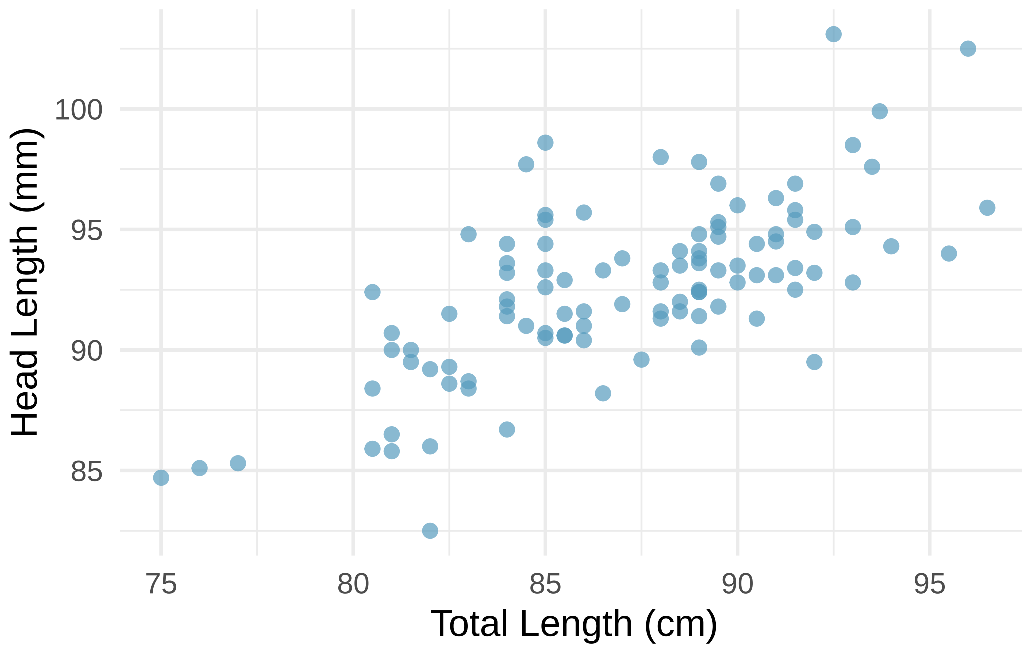


Figure 18.1: A scatterplot showing head length against total length for 104 brushtail possums. The relationship is moderately strong, positive, and linear.

Describe the relationship between total length and head length.

The relationship is **positive** (possums with longer total lengths tend to have longer heads), **linear** (the points cluster around a straight line, not a curve), and **moderate in strength** (the points are somewhat scattered around the trend, but the pattern is clear). There do not appear to be any extreme outliers.

18.1.1 When linear models are not appropriate

There are cases where fitting a straight line to the data is not helpful, even if there is a clear relationship between the variables.

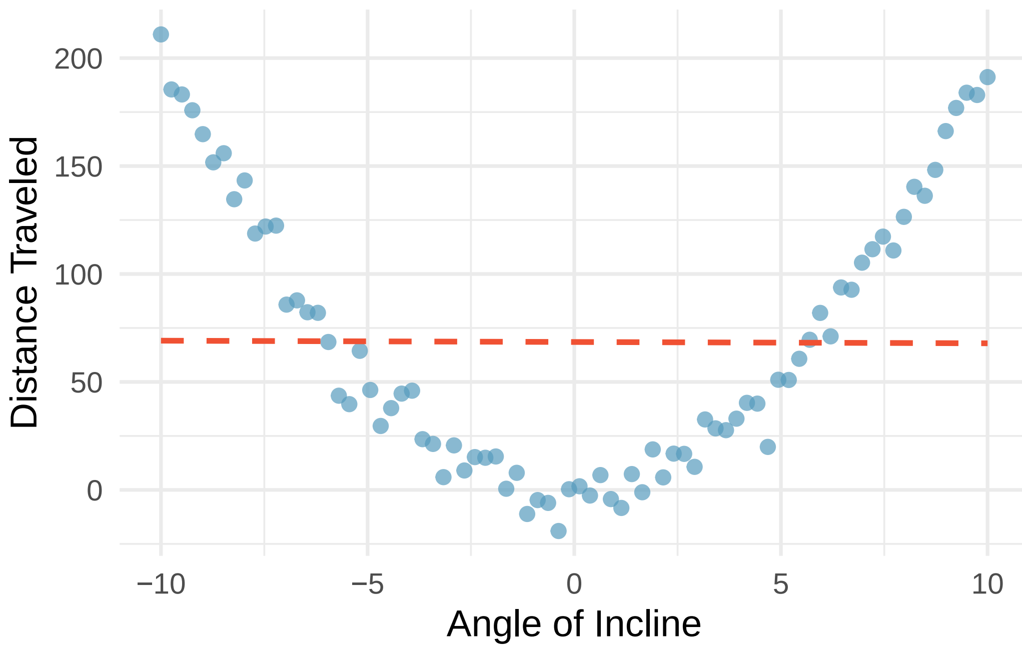


Figure 18.2: A scatterplot showing a clear quadratic (U-shaped) relationship between angle of incline and distance traveled. The best-fitting straight line is flat and does not capture the pattern at all.

When the relationship is nonlinear, fitting a straight line can be misleading. We must always examine scatterplots before computing correlations or fitting linear models.

18.2 The correlation coefficient

We need a way to quantify the strength and direction of a linear relationship. The **correlation coefficient** r does exactly this.

Correlation: strength of a linear relationship.

Correlation, denoted r, always takes values between -1 and 1 and describes the strength and direction of the linear relationship between two variables.

The correlation is unitless and is not affected by linear changes in the units of measurement (e.g., converting from inches to centimeters).

The formula for the correlation coefficient is:

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \frac{x_i - \bar{x}}{s_x} \cdot \frac{y_i - \bar{y}}{s_y}$$

where $\bar{x}$ and $\bar{y}$ are the sample means, and s_x and s_y are the sample standard deviations for each variable. In practice, we always use a computer or calculator to compute r.

18.2.1 Interpreting correlation values

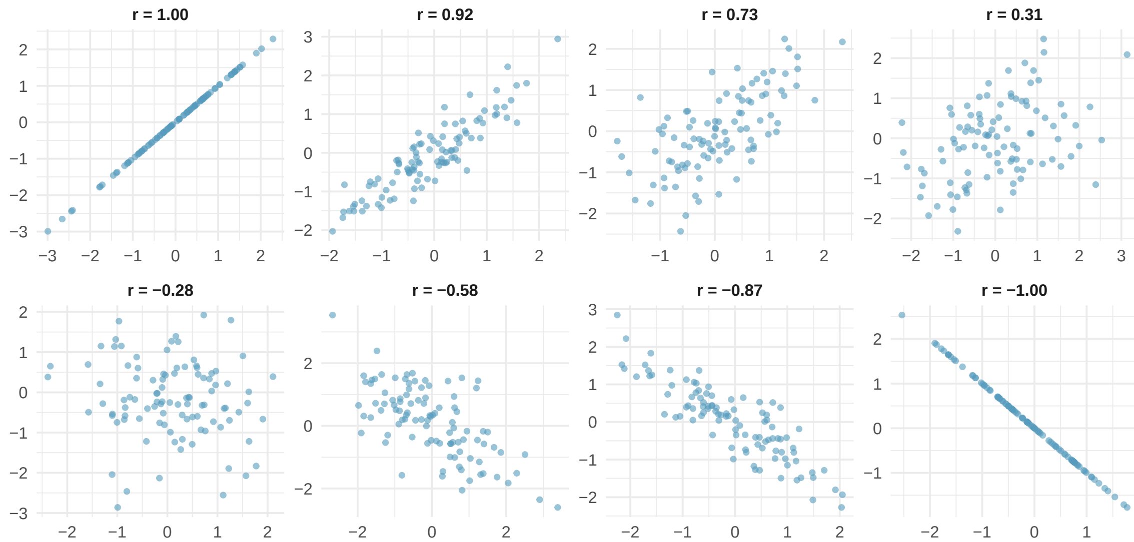


Figure 18.3: Eight scatterplots showing different correlation values. The first row shows positive relationships with r values of 1.00, 0.92, 0.73, and 0.31. The second row shows negative relationships with r values of -0.28, -0.58, -0.87, and -1.00.

Key patterns to notice:

- When $r = 1$ or $r = -1$, all points fall exactly on a straight line (a perfect linear relationship).
- When r is close to $+1$ or -1, the relationship is strong and linear.
- When r is close to 0, there is no linear relationship (but there could still be a nonlinear relationship!).
- Positive r means both variables tend to increase together; negative r means one tends to decrease as the other increases.

18.2.2 Properties of correlation

The correlation coefficient has several important properties:

1. **Range:** $-1 \leq r \leq 1$.
2. **Unitless:** Correlation has no units. Changing the units of x or y (e.g., from cm to inches) does not change r.
3. **Symmetric:** The correlation of x with y is the same as the correlation of y with x.
4. **Not affected by linear transformations:** Adding a constant to or multiplying a variable by a positive constant does not change r.
5. **Only measures linear relationships:** Correlation can be misleading for nonlinear relationships.

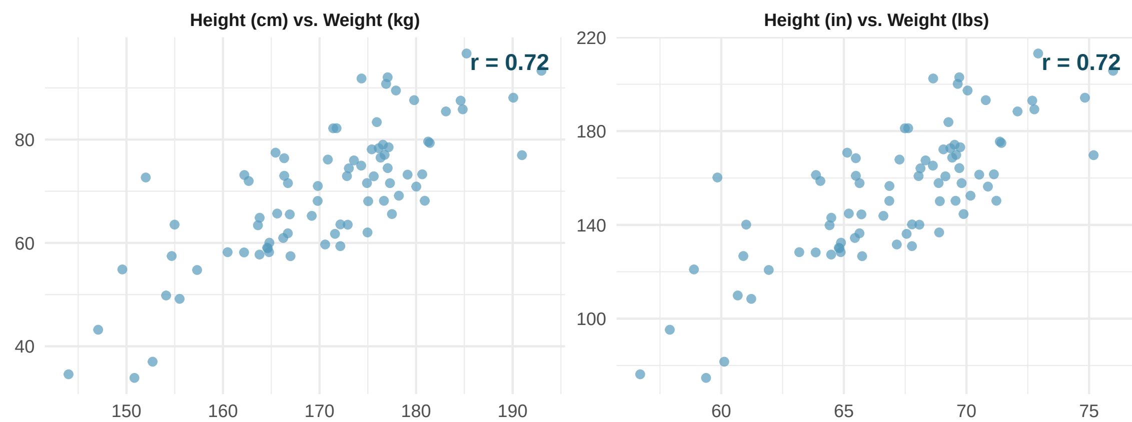


Figure 18.4: Two scatterplots showing the same relationship between weight and height, measured in different units (kg/cm vs. lbs/in). The correlation $r = 0.72$ is the same in both plots.

18.2.3 Correlation does not capture nonlinear relationships

The correlation coefficient is designed to measure the strength of *linear* trends. Nonlinear trends, even when strong, can produce misleading correlations.

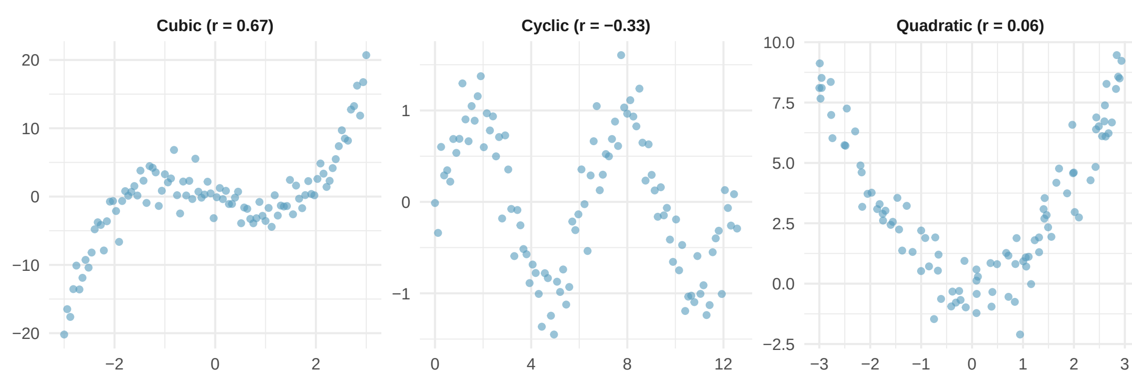


Figure 18.5: Three scatterplots showing strong nonlinear relationships (quadratic, cyclic, and a complex curve) that have weak correlations.

No straight line is a good fit for any of the datasets in the figure above. If you drew nonlinear curves on each plot, what general features would be important in your fit?

Show answer

The curves should be close to most points and reflect the overall nonlinear trend in the data.

Correlation only measures linear relationships. Always examine a scatterplot before computing or interpreting a correlation. A correlation near zero does not necessarily mean there is no relationship – it means there is no *linear* relationship. There may still be a strong nonlinear association.

Examine the scatterplots below showing the relationship between various crop yields across countries. Order the plots from strongest negative to strongest positive linear relationship.

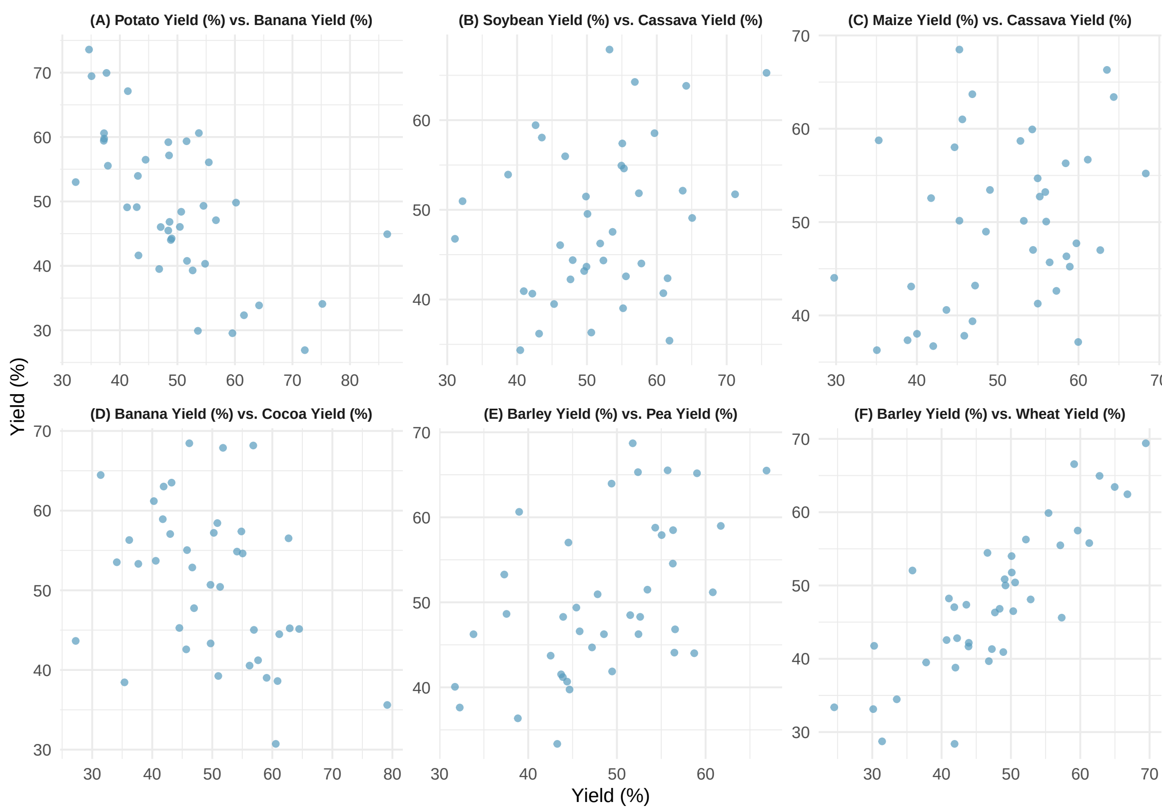


Figure 18.6: Six scatterplots of crop yield percentages across countries. Each shows a different pair of crop types with varying degrees of correlation.

Ordering from most negative to most positive correlation: the plots showing (A) bananas vs. potatoes, (D) cocoa vs. bananas, (B) cassava vs. soybeans, (C) cassava vs. maize, (E) peas vs. barley, and (F) wheat vs. barley.

18.3 Correlation and causation

Correlation does not imply causation.

Just because two variables are correlated does not mean that one causes the other. There may be a **confounding variable** (also called a lurking variable) that is associated with both x and y, creating a correlation between them even though there is no direct causal link.

Studies have found a positive correlation between the number of firefighters sent to a fire and the amount of damage the fire causes. Does this mean sending more firefighters causes more damage?

No. The confounding variable is the **severity of the fire**. Larger fires cause more damage *and* require more firefighters. The number of firefighters does not cause the damage; both are driven by the severity of the fire.

To establish causation, we need one of the following:

1. A **randomized experiment** where the explanatory variable is randomly assigned.

2. Strong theoretical justification combined with careful control of confounding variables.

Observational studies can reveal associations (correlations), but they cannot establish causation on their own.

18.4 Sampling variability of correlation

Like any sample statistic, the correlation coefficient r computed from a sample is just an estimate of the true population correlation ρ (the Greek letter "rho"). Different samples from the same population will produce different values of r.

When the sample size is small, r can vary substantially from sample to sample, even when the true population correlation is zero. This means we should be cautious about interpreting correlations from small samples.

Suppose the true population correlation between two variables is $\rho = 0$ (no linear relationship). If we draw random samples of size $n = 10$, the sample correlation r could reasonably fall anywhere between about -0.55 and $+0.55$ just by chance.

With larger samples ($n = 100$), the sampling variability shrinks, and r would typically fall between about -0.20 and $+0.20$.

This illustrates why statistical tests of significance are important: we need to determine whether an observed correlation is large enough to be unlikely under the null hypothesis that $\rho = 0$.

We will return to inference for correlation when we discuss inference for regression in a later chapter. In particular, testing whether the slope of a regression line is zero is equivalent to testing whether the population correlation is zero.

StatLens: Correlation Simulator. Use the Correlation Simulator to explore how the correlation coefficient changes as you add or move points in a scatterplot. Experiment with different sample sizes to see how sampling variability affects the observed correlation.

18.5 Chapter review

18.5.1 Summary

In this chapter, we introduced scatterplots as the primary tool for visualizing the relationship between two quantitative variables. We then defined the correlation coefficient r as a measure of the strength and direction of the linear relationship. Key properties of r include: it ranges from -1 to $+1$, it is unitless, it is symmetric, and it measures only linear relationships. We emphasized that correlation does not imply causation – confounding variables can create spurious associations. Finally, we noted that the sample correlation is subject to sampling variability, especially in small samples.

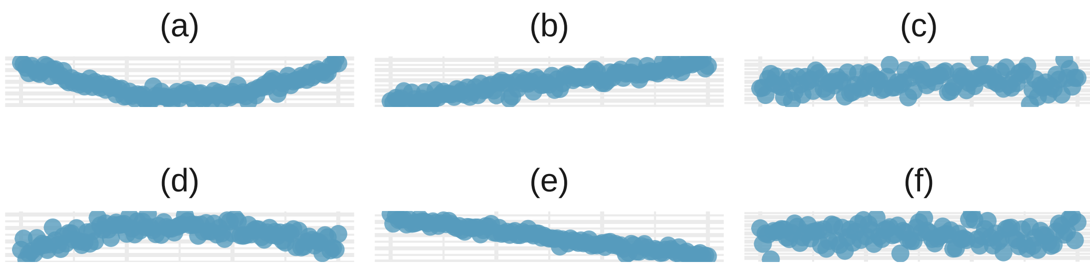
18.5.2 Key terms

- Scatterplot
- Correlation (r)
- Positive association / negative association
- Nonlinear relationship
- Confounding variable
- Correlation does not imply causation
- Population correlation (ρ)

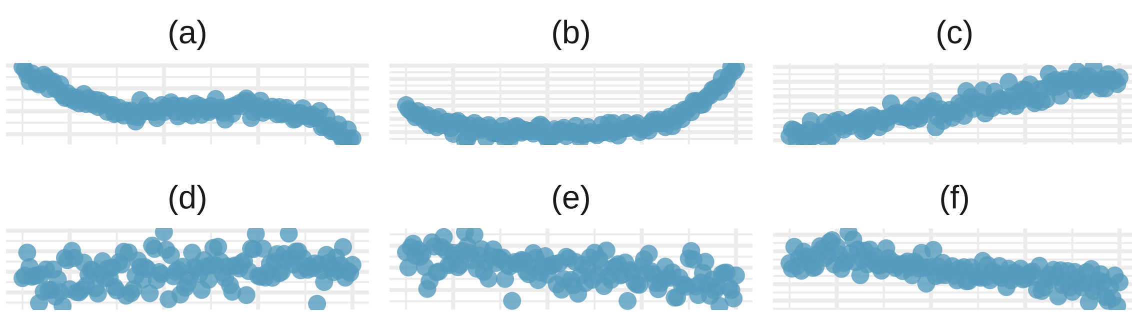
18.6 Exercises

Answers to odd-numbered exercises are provided inline.

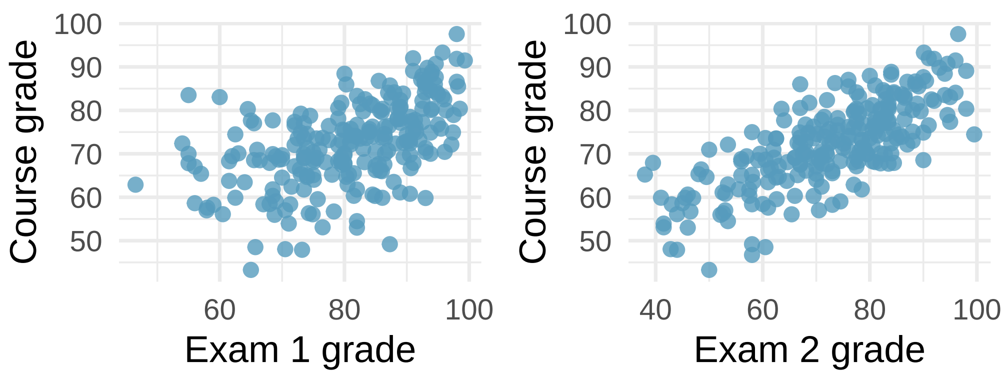
1. **Identify relationships, I.** For each of the six plots, identify the strength of the relationship (e.g., weak, moderate, or strong) in the data and whether fitting a linear model would be reasonable.



2. **Identify relationships, II.** For each of the six plots, identify the strength of the relationship (e.g., weak, moderate, or strong) in the data and whether fitting a linear model would be reasonable.



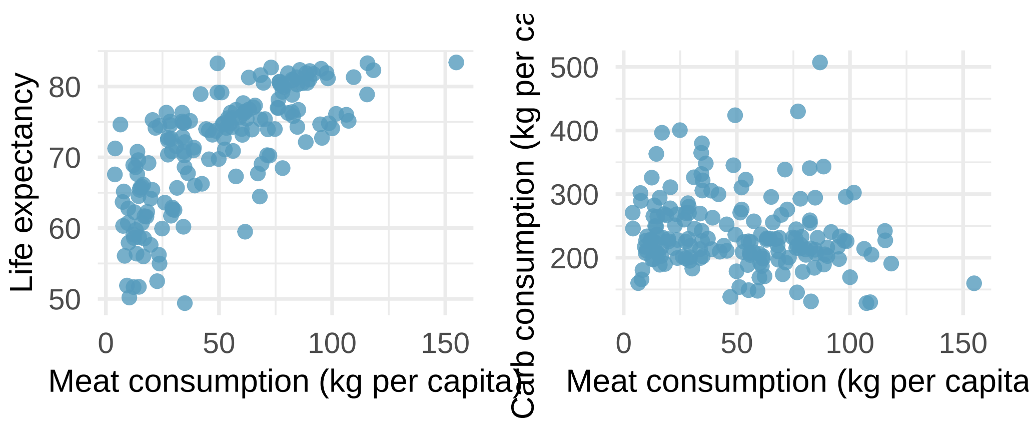
3. **Midterms and final.** The two scatterplots below show the relationship between the overall course average and two midterm exams (Exam 1 and Exam 2) recorded for 233 students during several years for a statistics course at a university.



a. Based on these graphs, which of the two exams has the strongest correlation with the course grade? Explain.

b. Can you think of a reason why the correlation between the exam you chose in part (a) and the course grade is higher?

4. **Meat consumption and life expectancy.** In data collected for (**You:2022?**), total meat intake is associated with life expectancy (at birth) in 175 countries. Meat intake is measured in kg per capita per year (averaged over 2011 to 2013). Additionally, the authors collected data on carbohydrate crops (e.g., cereals, root, sugar, etc.) in kg per capita per year (averaged over 2011 and 2013). The scatterplot on the left shows the life expectancy at birth plotted against the per capita meat consumption. The scatterplot on the right shows the amount of carbohydrate consumption plotted against the per capita meat consumption.



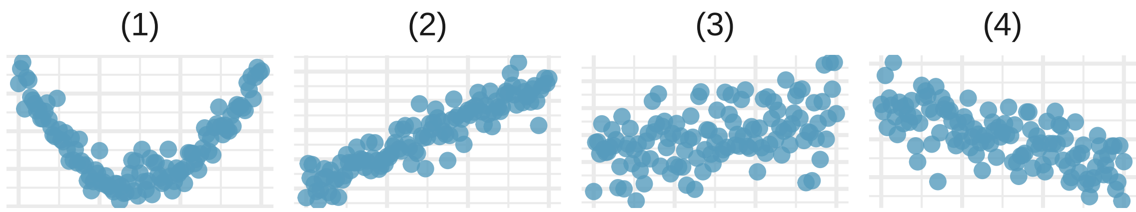
a. Describe the relationship between meat consumption and life expectancy.

b. Describe the relationship between meat consumption and carbohydrate consumption.

c. Which plot shows a stronger correlation? Explain your reasoning.

d. Data on consumption were originally collected in kilograms. How would the plots above and the correlation values change if the consumption variables were converted to pounds?

5. **Match the correlation, I.** Match each correlation to the corresponding scatterplot.



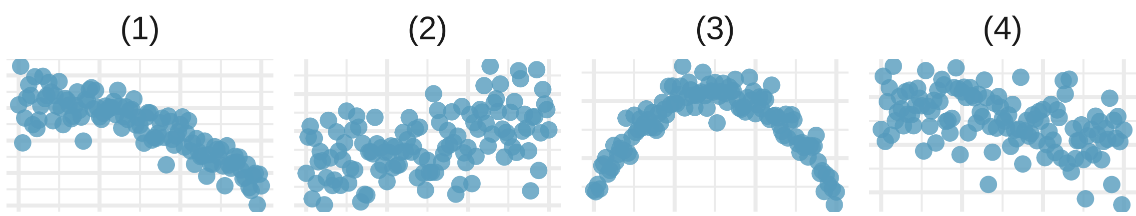
a. $r = -0.7$

b. $r = 0.45$

c. $r = 0.06$

d. $r = 0.92$

6. **Match the correlation, II.** Match each correlation to the corresponding scatterplot.



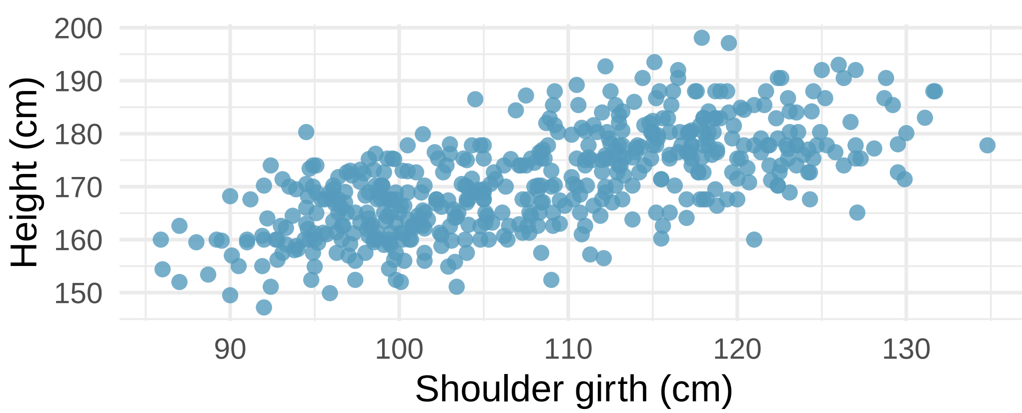
a. $r = 0.49$

b. $r = -0.48$

c. $r = -0.03$

d. $r = -0.85$

7. **Body measurements, correlation.** Researchers studying anthropometry collected body and skeletal diameter measurements, as well as age, weight, height and sex for 507 physically active individuals. The scatterplot below shows the relationship between height and shoulder girth (circumference of shoulders measured over deltoid muscles), both measured in centimeters. (Heinz et al. 2003)

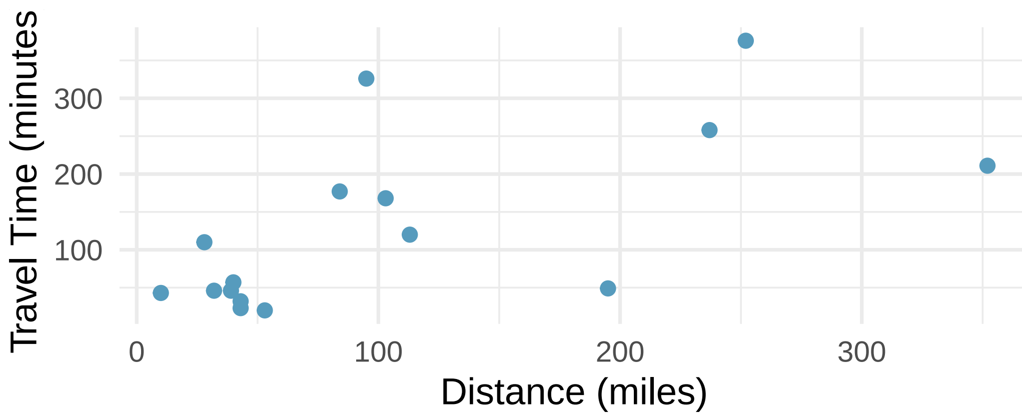


a. Describe the relationship between shoulder girth and height.

b. How would the relationship change if shoulder girth was measured in inches while the units of height remained in centimeters?

8. **Compare correlations.** Eduardo and Rosie are both collecting data on number of rainy days in a year and the total rainfall for the year. Eduardo records rainfall in inches and Rosie in centimeters. How will their correlation coefficients compare?

9. **The Coast Starlight, correlation.** The Coast Starlight Amtrak train runs from Seattle to Los Angeles. The scatterplot below displays the distance between each stop (in miles) and the amount of time it takes to travel from one stop to another (in minutes).

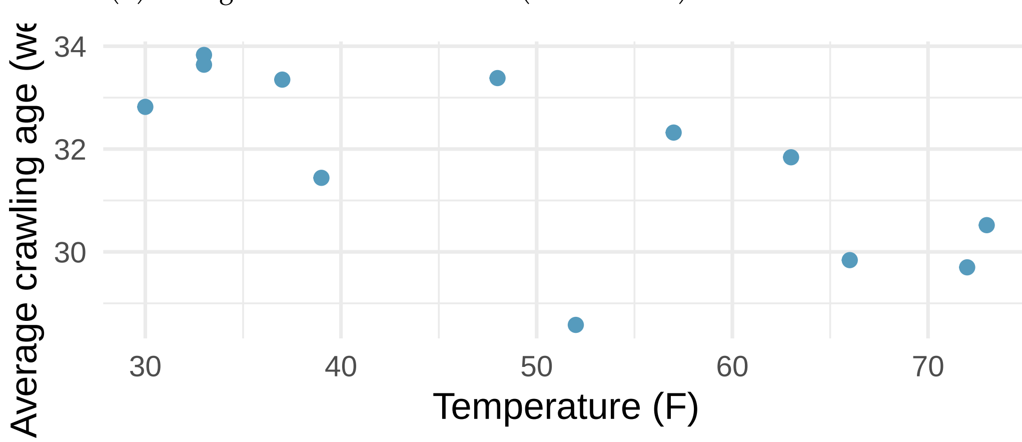


a. Describe the relationship between distance and travel time.

b. How would the relationship change if travel time was instead measured in hours, and distance was instead measured in kilometers?

c. Correlation between travel time (in miles) and distance (in minutes) is $r = 0.636$. What is the correlation between travel time (in kilometers) and distance (in hours)?

10. **Crawling babies, correlation.** A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. Infants born during the study year were split into twelve groups, one for each birth month. We consider the average crawling age of babies in each group against the average temperature when the babies are six months old (that's when babies often begin trying to crawl). Temperature is measured in degrees Fahrenheit (F) and age is measured in weeks. (Benson 1993)



a. Describe the relationship between temperature and crawling age.

b. How would the relationship change if temperature was measured in degrees Celsius (C) and age was measured in months?

c. The correlation between temperature in F and age in weeks was $r = -0.70$. If we converted the temperature to C and age to months, what would the correlation be?

11. **Meat and carbohydrate consumption.** What would be the correlation between the per capita meat consumption and per capita carbohydrate consumption if, for each country, people always consumed

12. 3 kg more of meat than of carbohydrates each year?

13. 2 kg less of meat than of carbohydrates each year?

14. half as much meat as carbohydrates each year?

12. **Graduate degrees and salaries.** What would be the correlation between the annual salaries of people with and without a graduate degree at a company if, for a certain type of position, someone with a graduate degree always made

a. $5,000 more than those without a graduate degree?

b. 25% more than those without a graduate degree?

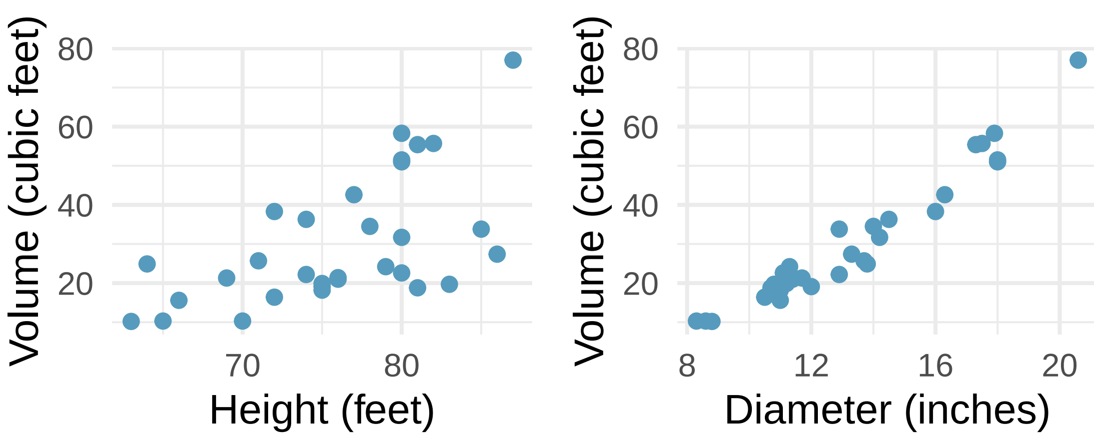
c. 15% less than those without a graduate degree?

13. **True / False.** Determine if the following statements are true or false. If false, explain why.

a. A correlation coefficient of -0.90 indicates a stronger linear relationship than a correlation of 0.5.

b. Correlation is a measure of the association between any two variables.

14. **Cherry trees.** The scatterplots below show the relationship between height, diameter, and volume of timber in 31 felled black cherry trees. The diameter of the tree is measured 4.5 feet above the ground.

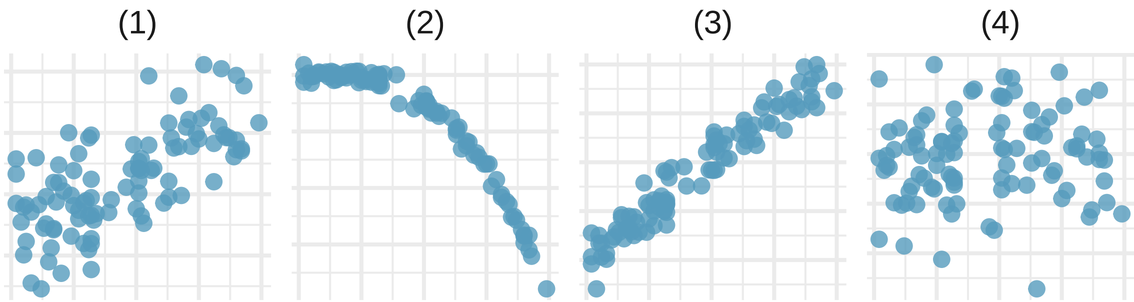


a. Describe the relationship between volume and height of these trees.

b. Describe the relationship between volume and diameter of these trees.

c. Suppose you have height and diameter measurements for another black cherry tree. Which of these variables would be preferable to use to predict the volume of timber in this tree using a simple linear regression model? Explain your reasoning.

15. **Match the correlation, III.** Match each correlation to the corresponding scatterplot.



a. r = 0.69

b. r = 0.09

c. r = -0.91

d. r = 0.97

Chapter 19

Linear Regression

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Linear regression is a powerful statistical technique for describing the relationship between two quantitative variables. Many people have some familiarity with regression just from reading the news, where straight lines are overlaid on scatterplots. In this chapter, we build on the ideas of correlation from the previous chapter to develop the least squares regression line. We learn how to interpret the slope and intercept, use residuals to assess model fit, quantify the strength of the model with R^2, and perform inference on the slope to test whether a linear relationship exists in the population.

19.1 Fitting a line to data

Linear regression is the statistical method for fitting a line to data where the relationship between two variables, x and y, can be modeled by a straight line with some error:

$$y = b_0 + b_1 x + e$$

The values b_0 and b_1 represent the model's **intercept** and **slope**, respectively, and the error is represented by e. These values are calculated from the data – they are sample statistics. If the observed data are a random sample from a target population, these values are point estimates for the population parameters β_0 and β_1.

When we use x to predict y, we call x the **predictor** variable (or explanatory variable) and y the **outcome** variable (or response variable). We often write the model without the error term when focusing on the predicted average outcome:

$$\hat{y} = b_0 + b_1 x$$

The "hat" on y signifies that this is a predicted (estimated) value.

It is rare for all data to fall perfectly on a straight line. Instead, data more commonly appear as a cloud of points, where the trend may be strong or weak.

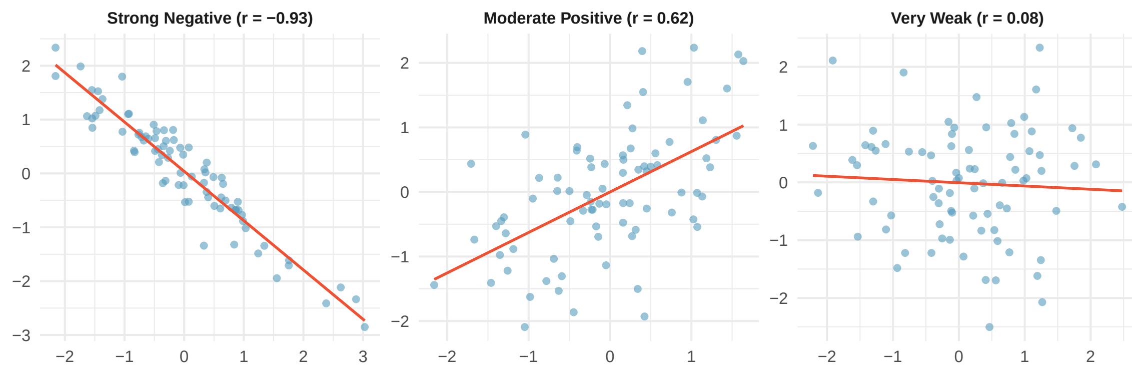


Figure 19.1: Three scatterplots with fabricated data showing a strong negative linear trend, a moderate positive linear trend, and a very weak trend.

There are also cases where fitting a straight line is inappropriate, even if there is a clear relationship between the variables. When the relationship is nonlinear, a linear model can be misleading.

19.1.1 Using linear regression to predict possum head lengths

Researchers captured 104 brushtail possums in Australia and took body measurements. We consider total body length (cm) as the predictor to predict head length (mm).

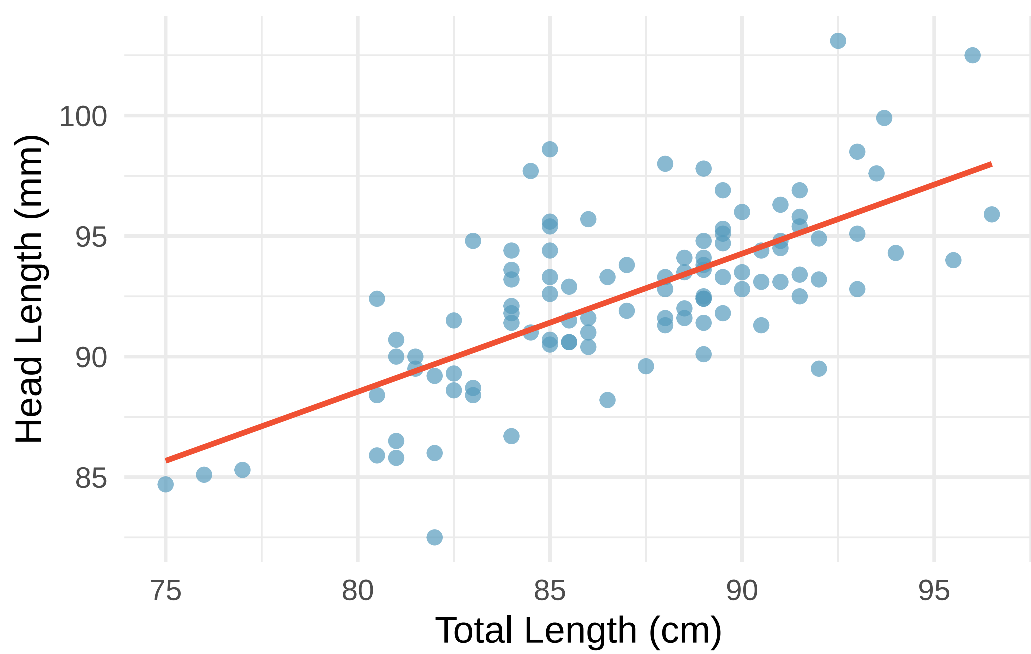


Figure 19.2: A scatterplot showing head length against total length for 104 brushtail possums, with a linear model fit to the data.

The equation for this line is:

$$\hat{y} = 41 + 0.59x$$

We can use this line to make predictions. For instance, the equation predicts a possum with a total length of 80 cm will have a head length of:

$$\hat{y} = 41 + 0.59 \times 80 = 88.2 \text{ mm}$$

This estimate may be viewed as an average: the equation predicts that possums with a total length of 80 cm will have an average head length of 88.2 mm.

19.2 Residuals

Residuals are the leftover variation in the data after accounting for the model fit:

$$\text{Data} = \text{Fit} + \text{Residual}$$

Each observation has a residual. If an observation is above the regression line, its residual is positive. If it is below the line, the residual is negative.

Residual: Difference between observed and expected.

The residual of the i^{th} observation (x_i, y_i) is the difference between the observed outcome and the predicted outcome:

$$e_i = y_i - \hat{y}_i$$

We identify $\hat{y}_i$ by plugging x_i into the model.

The linear fit is $\hat{y} = 41 + 0.59x$. Compute the residual for the observation $(76.0, 85.1)$.

First compute the predicted value: $\hat{y} = 41 + 0.59 \times 76.0 = 85.84$. The residual is $e = 85.1 - 85.84 = -0.74$ mm. The negative residual indicates that the model overpredicted head length for this possum.

If a model underestimates an observation, will the residual be positive or negative? What about if it overestimates?

Show answer

If a model underestimates an observation, the residual is positive (the actual value exceeds the prediction). If it overestimates, the residual is negative.

19.2.1 Residual plots

Residuals are helpful in evaluating how well a linear model fits a dataset. We display them in a **residual plot**, where the horizontal axis shows the predicted values and the vertical axis shows the residuals.

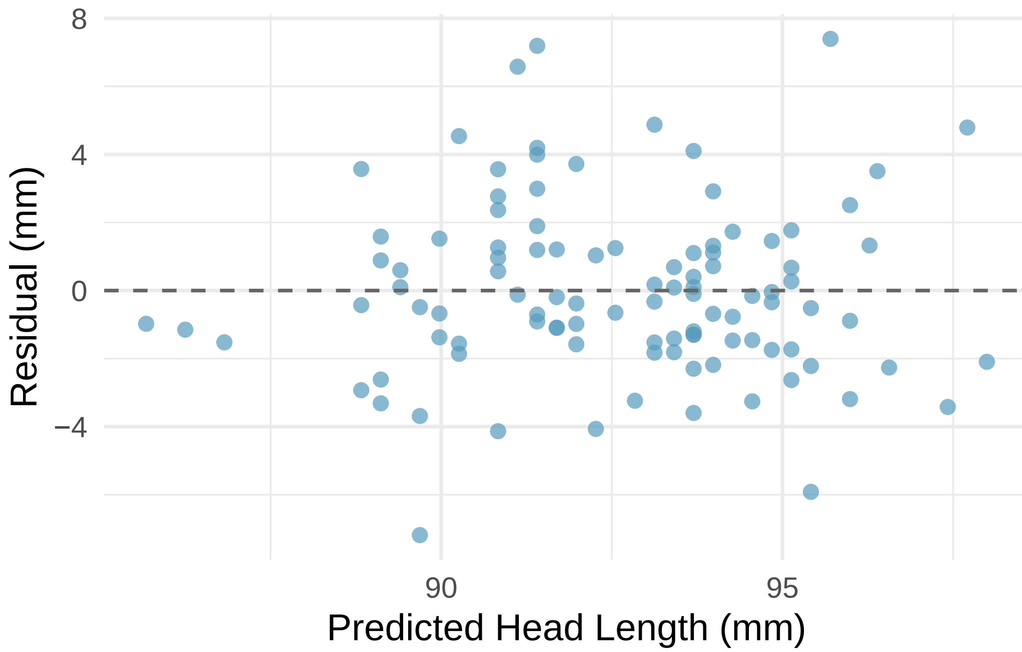


Figure 19.3: A residual plot for the possum model. The residuals appear randomly scattered around zero with no obvious pattern.

What patterns should we look for in a residual plot?

- **No pattern (good):** Residuals scattered randomly around zero suggest the linear model is appropriate.
- **Curved pattern (bad):** A U-shape or other curve in the residuals indicates a nonlinear relationship that a straight line fails to capture.
- **Fan shape (bad):** Increasing or decreasing spread in the residuals indicates non-constant variability.

19.3 Least squares regression

Fitting linear models by eye is subjective. **Least squares regression** provides an objective, rigorous approach.

19.3.1 The objective: minimize squared residuals

We want a line with small residuals. The **least squares line** minimizes the sum of the squared residuals:

$$e_1^2 + e_2^2 + \cdots + e_n^2$$

Why square the residuals rather than just sum their absolute values?

1. It is the most commonly used method.
2. It is widely supported in statistical software.
3. A residual twice as large is usually more than twice as bad. Squaring accounts for this.

4. The inference theory connecting the model to population parameters is most straightforward with least squares.

19.3.2 Interpreting the slope and intercept

For the Elmhurst College data, the least squares line predicting gift aid (in $1,000s) from family income (in $1,000s) is:

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times \text{family\_income}$$

Term	Estimate	Std. Error	T statistic	p-value
(Intercept)	24.32	1.29	18.83	<0.0001
family_income	-0.0431	0.0108	-3.98	0.0002

What do the intercept and slope mean?

Slope ($b_1 = -0.0431$)**:** For each additional $1,000 of family income, we would expect a student to receive $43.10 less in gift aid, on average. The negative coefficient means higher income is associated with less aid. We must be cautious: this is an observational study, so we cannot interpret a causal connection.

Intercept ($b_0 = 24.32$)**:** The model predicts that a student whose family has no income would receive $24,320 in gift aid, on average. This is meaningful because the family income for some students is near zero. In other applications, the intercept may have no practical meaning if $x = 0$ is outside the range of the data.

Interpreting parameters estimated by least squares.

The **slope** describes the estimated difference in the predicted average outcome of y if the predictor variable x were one unit larger.

The **intercept** describes the average outcome of y if $x = 0$, provided the linear model is valid at $x = 0$ (which may not be the case if $x = 0$ is far outside the observed data range).

19.3.3 Computing the slope and intercept from summary statistics

The slope can be computed using the correlation and the standard deviations of the two variables:

$$b_1 = \frac{s_y}{s_x} \cdot r$$

The least squares line always passes through the point $(\bar{x}, \bar{y})$. Using the point-slope form:

$$b_0 = \bar{y} - b_1 \cdot \bar{x}$$

Identifying the least squares line from summary statistics.

- Estimate the slope: $b_1 = (s_y/s_x) \cdot r$.
- Find the intercept using the fact that $(\bar{x}, \bar{y})$ is on the line: $b_0 = \bar{y} - b_1\bar{x}$.

19.4 R-squared

We evaluated the strength of a linear relationship using r. More commonly, we describe the strength of a linear fit using R^2, called **R-squared** or the **coefficient of determination**.

Coefficient of determination: proportion of variability explained by the model.

R^2 measures the proportion of variation in the outcome variable y that is explained by the linear model with predictor x. Since r is always between -1 and 1, R^2 is always between 0 and 1.

$$R^2 = 1 - \frac{SSE}{SST}$$

where:

- $SST = \sum(y_i - \bar{y})^2$ is the **total sum of squares**, measuring total variability in y.
- $SSE = \sum(y_i - \hat{y}_i)^2 = \sum e_i^2$ is the **sum of squared errors**, measuring leftover variability after using the model.

For simple linear regression, $R^2 = r^2$.

For the Elmhurst data, the correlation is $r = -0.499$. What proportion of the variability in gift aid is explained by family income?

$R^2 = (-0.499)^2 = 0.249$, or about 25%. Family income explains about 25% of the variability in gift aid among these students.

If a linear model has a very strong negative relationship with $r = -0.97$, how much of the variation in the outcome is explained by the predictor?

Show answer

$R^2 = (-0.97)^2 = 0.94$, or about 94% of the variation is explained.

19.5 Inference for the slope

We have been computing sample statistics b_0 and b_1, but these are estimates of population parameters β_0 and β_1. We now turn to formal inference: testing whether a linear relationship exists in the population.

19.5.1 Hypotheses

- H_0: $\beta_1 = 0$. There is no linear relationship between x and y.
- H_A: $\beta_1 \neq 0$. There is a linear relationship between x and y.

19.5.2 Randomization test for the slope

If the null hypothesis is true ($\beta_1 = 0$), then x and y are not linearly related. We can simulate this by **permuting** the response variable: randomly shuffling the y values while keeping the x values fixed. This breaks any relationship between x and y.

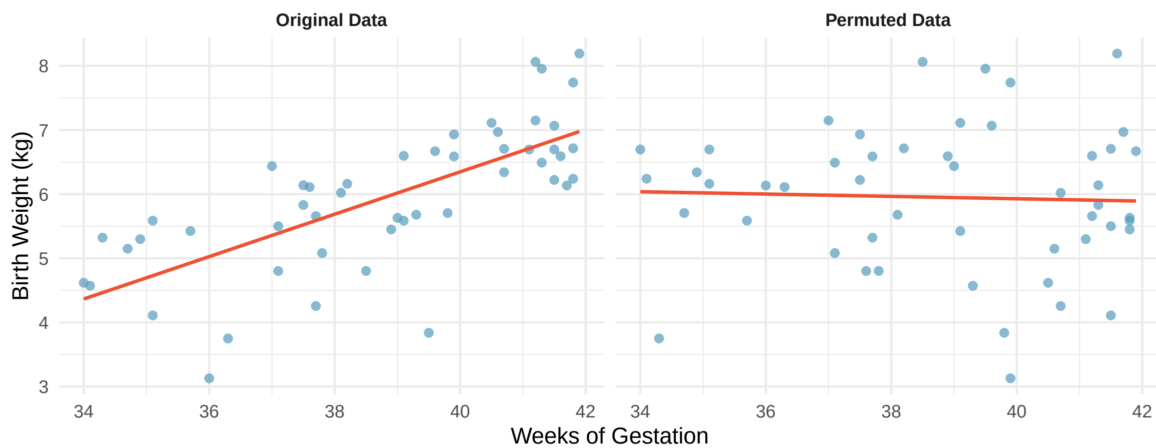


Figure 19.4: Original data showing a positive linear relationship between weeks of gestation and birth weight, alongside the same data after permuting the weight variable, which destroys the relationship.

By repeating the permutation many times and computing the slope each time, we build a null distribution of slopes.

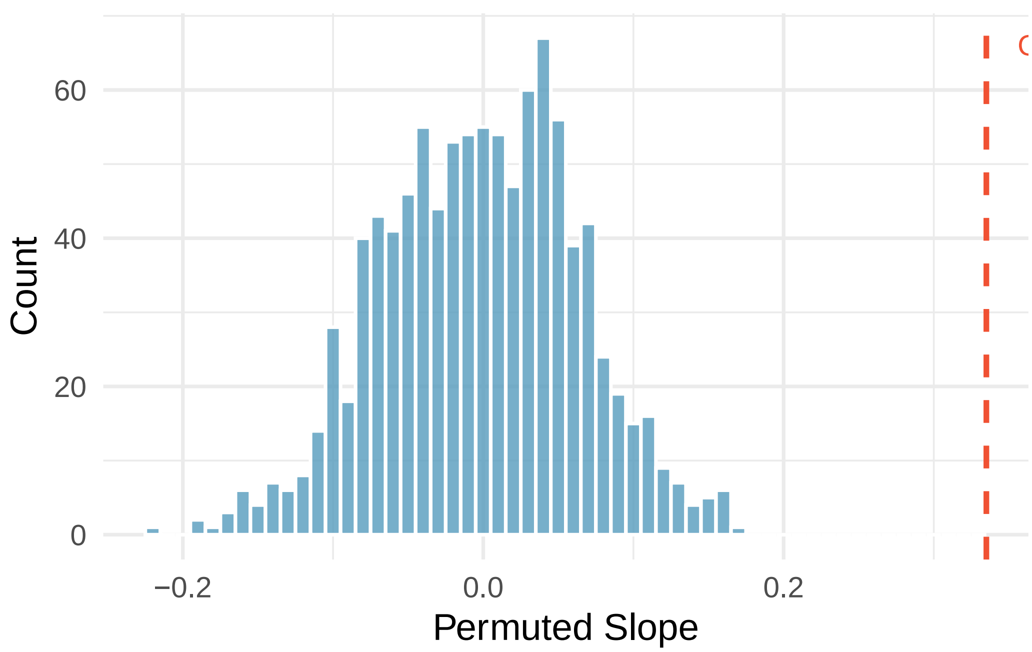


Figure 19.5: Histogram of slopes from 1,000 permutations of birth weight. The permuted slopes range from about -0.15 to $+0.15$, centered at zero. The observed slope of 0.335 is far from this distribution.

The observed slope of 0.335 is far from any value in the null distribution. We reject H_0 and conclude there is a linear relationship between weeks of gestation and birth weight.

19.5.3 Mathematical model: the t-test for slope

When the technical conditions are met, we can use the t-distribution to test the slope:

$$T = \frac{b_1 - 0}{SE_{b_1}}$$

This T-statistic follows a t-distribution with $df = n - 2$.

The regression output for predicting the change in House seats for the President's party from the unemployment rate is:

Term	Estimate	Std. Error	T statistic	p-value
(Intercept)	-7.36	5.16	-1.43	0.1649
unemp	-0.89	0.835	-1.07	0.2961

The p-value of 0.2961 is not significant. The data do not provide convincing evidence that the unemployment rate is a useful predictor of midterm election outcomes.

19.5.4 Confidence intervals for the slope

Confidence intervals for model coefficients.

Confidence intervals for model coefficients can be computed using the t-distribution:

$$b_1 \pm t^*_{df} \times SE_{b_1}$$

where t^*_{df} is the critical value corresponding to the confidence level, with $df = n - 2$.

Using the Elmhurst College regression output ($b_1 = -0.0431$, $SE = 0.0108$, $df = 48$), compute a 95% confidence interval for the slope.

The critical value is $t^*_{48} = 2.01$. The confidence interval is:

$$-0.0431 \pm 2.01 \times 0.0108 = (-0.0648, -0.0214)$$

We are 95% confident that for each additional $1,000 in family income, the university's gift aid decreases by between $21.40 and $64.80 on average.

19.6 Conditions for linear regression

For the mathematical inference to be valid, we check four conditions, often remembered by the **LINE** mnemonic:

- **L – Linear model:** The data should show a linear trend. Check the scatterplot and residual plot.
- **I – Independent observations:** Be cautious with time series data or clustered observations.
- **N – Nearly normal residuals:** Residuals should be approximately normally distributed. This is less critical for large samples (Central Limit Theorem), but outliers are always a concern.
- **E – Equal variability:** The variability of points around the line should be roughly constant across all values of x. A "fan shape" in the residual plot indicates a violation.

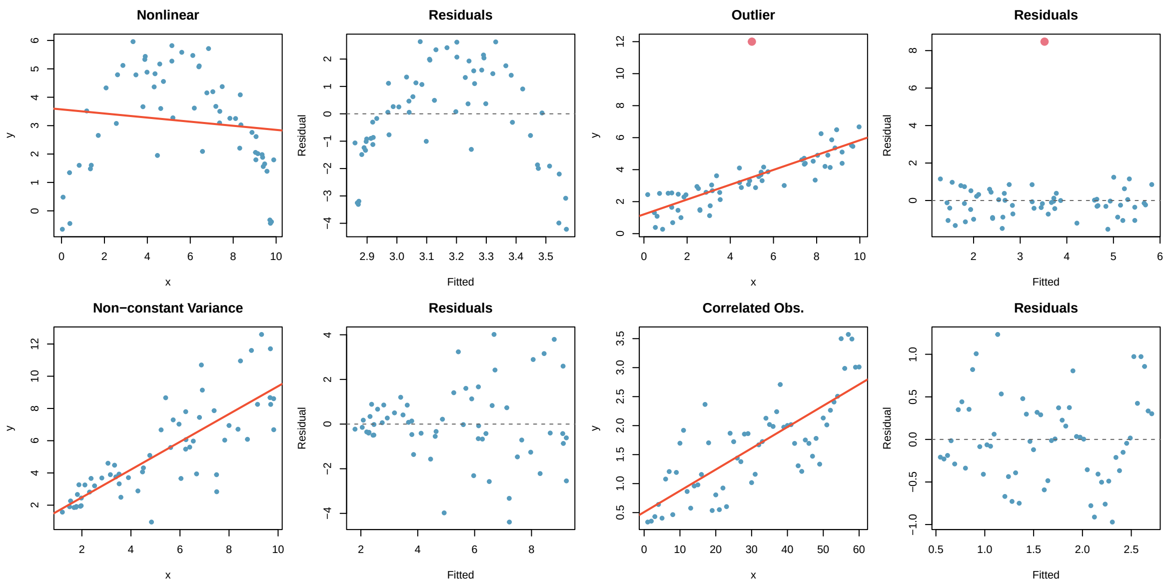


Figure 19.6: Four scatterplots and their residual plots showing common violations: nonlinearity, an outlier, increasing variability, and correlated observations.

Diagnostics for linear regression.

Independence is always important. The normality condition matters most for small samples. The constant variance condition is especially important for inference. The linearity condition is the most fundamental – if the true relationship is not linear, the slope estimate has no meaningful interpretation.

19.7 Chapter review

19.7.1 Summary

In this chapter, we developed the least squares regression line as a tool for modeling the linear relationship between two quantitative variables. The slope describes the predicted change in y for a one-unit increase in x, and the intercept describes the predicted value of y when $x = 0$. Residuals measure the difference between observed and predicted values, and residual plots help assess whether the linear model is appropriate. The coefficient of determination R^2 quantifies the proportion of variability in y explained by the model. For inference, we use either a randomization test or the t-distribution to test whether the slope is significantly different from zero, and we can construct confidence intervals for the slope. The LINE conditions (linearity, independence, normality, equal variance) must be checked for the mathematical model to be valid.

19.7.2 Key terms

- Predictor / outcome variable
- Least squares line
- Slope (b_1) and intercept (b_0)
- Residual
- Residual plot
- R-squared (R^2) / coefficient of determination
- Total sum of squares (SST) / sum of squared errors (SSE)

- T-test for slope
- LINE conditions
- Extrapolation

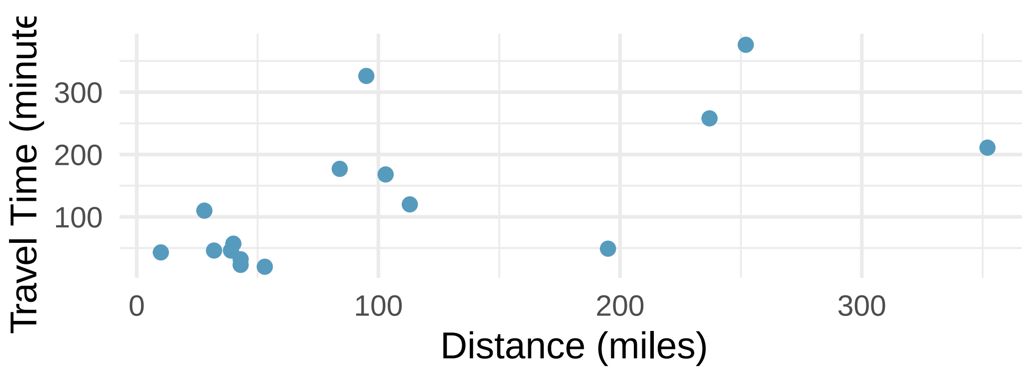
19.8 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Units of regression.** Consider a regression predicting the number of calories (cal) from width (cm) for a sample of square shaped chocolate brownies. What are the units of the correlation coefficient, the intercept, and the slope?

2. **Which is higher?** Determine if (I) or (II) is higher or if they are equal: "*For a regression line, the uncertainty associated with the slope estimate, b_1, is higher when (I) there is a lot of scatter around the regression line or (II) there is very little scatter around the regression line.*" Explain your reasoning.

3. **The Coast Starlight, regression.** The Coast Starlight Amtrak train runs from Seattle to Los Angeles. The scatterplot below displays the distance between each stop (in miles) and the amount of time it takes to travel from one stop to another (in minutes). The mean travel time from one stop to the next on the Coast Starlight is 129 mins, with a standard deviation of 113 minutes. The mean distance traveled from one stop to the next is 108 miles with a standard deviation of 99 miles. The correlation between travel time and distance is 0.636.



a. Write the equation of the regression line for predicting travel time.

b. Interpret the slope and the intercept in this context.

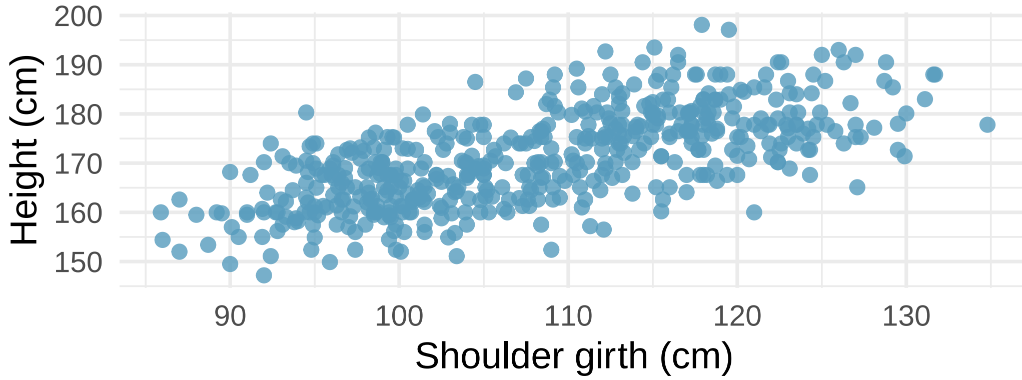
c. Calculate R^2 of the regression line for predicting travel time from distance traveled for the Coast Starlight, and interpret R^2 in the context of the application.

d. The distance between Santa Barbara and Los Angeles is 103 miles. Use the model to estimate the time it takes for the Starlight to travel between these two cities.

e. It takes the Coast Starlight about 168 mins to travel from Santa Barbara to Los Angeles. Calculate the residual and explain the meaning of this residual value.

f. Suppose Amtrak is considering adding a stop to the Coast Starlight 500 miles away from Los Angeles. Would it be appropriate to use this linear model to predict the travel time from Los Angeles to this point?

4. **Body measurements, regression.** Researchers studying anthropometry collected body and skeletal diameter measurements, as well as age, weight, height and sex for 507 physically active individuals. The scatterplot below shows the relationship between height and shoulder girth (circumference of shoulders measured over deltoid muscles), both measured in centimeters. The mean shoulder girth is 107.20 cm with a standard deviation of 10.37 cm. The mean height is 171.14 cm with a standard deviation of 9.41 cm. The correlation between height and shoulder girth is 0.67. (Heinz et al. 2003)



a. Write the equation of the regression line for predicting height.

b. Interpret the slope and the intercept in this context.

c. Calculate R^2 of the regression line for predicting height from shoulder girth, and interpret it in the context of the application.

d. A randomly selected student from your class has a shoulder girth of 100 cm. Predict the height of this student using the model.

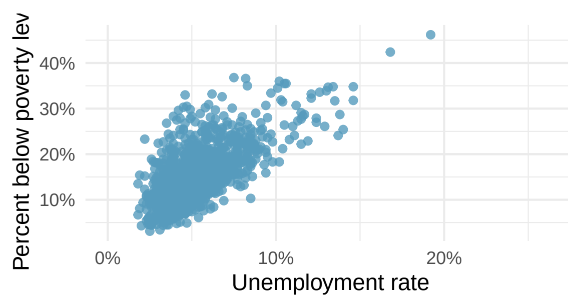
e. The student from part (d) is 160 cm tall. Calculate the residual, and explain what this residual means.

f. A one year old has a shoulder girth of 56 cm. Would it be appropriate to use this linear model to predict the height of this child?

5. **Poverty and unemployment.** The following scatterplot shows the relationship between percent of population below the poverty level (`poverty`) from unemployment rate among those ages 20-64 (`unemployment_rate`) in counties in the US, as provided by data from the 2019 American Community Survey. The regression output for the model for predicting `poverty` from `unemployment_rate` is also provided.

term	estimate	std.error	statistic	p.value
`(Intercept)`	4.60	0.349	13.2	<0.0001
`unemployment_rate`	2.05	0.062	33.1	<0.0001

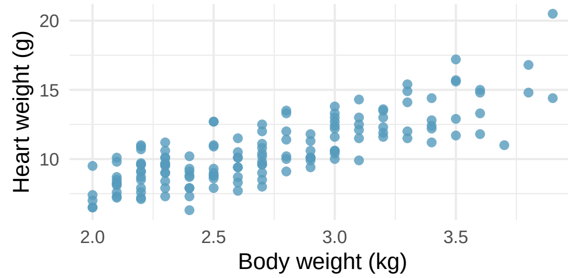
a. Write out the linear model.
b. Interpret the intercept.
c. Interpret the slope.
d. The R^2 of this model is 46%. Interpret this value.
e. Calculate the correlation coefficient.



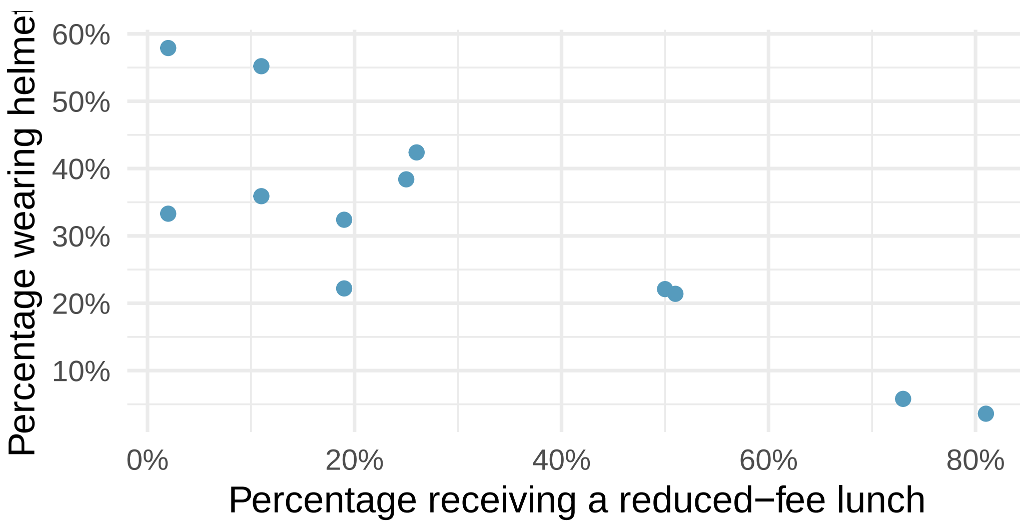
6. **Cat weights.** The following regression output is for predicting the heart weight (`Hwt`, in g) of cats from their body weight (`Bwt`, in kg). The coefficients are estimated using a dataset of 144 domestic cats.

term	estimate	std.error	statistic	p.value
`(Intercept)`	-0.357	0.692	-0.515	0.6072
`Bwt`	4.034	0.250	16.119	<0.0001

a. Write out the linear model.
b. Interpret the intercept.
c. Interpret the slope.
d. The R^2 of this model is 65%. Interpret R^2.
e. Calculate the correlation coefficient.



7. **Helmets and lunches.** The scatterplot shows the relationship between socioeconomic status measured as the percentage of children in a neighborhood receiving reduced-fee lunches at school (`lunch`) and the percentage of bike riders in the neighborhood wearing helmets (`helmet`). The average percentage of children receiving reduced-fee lunches is 30.83% with a standard deviation of 26.72% and the average percentage of bike riders wearing helmets is 30.88% with a standard deviation of 16.95%.



a. If the R^2 for the least-squares regression line for these data is 72%, what is the correlation between `lunch` and `helmet`?

b. Calculate the slope and intercept for the least-squares regression line for these data.

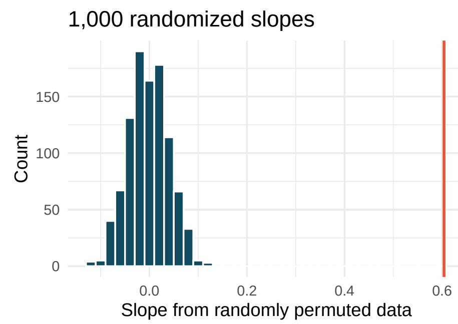
c. Interpret the intercept of the least-squares regression line in the context of the application.

d. Interpret the slope of the least-squares regression line in the context of the application.

e. What would the value of the residual be for a neighborhood where 40% of the children receive reduced-fee lunches and 40% of the bike riders wear helmets? Interpret the meaning of this residual in the context of the application.

8. **Body measurements, randomization test.** Researchers studying anthropometry collected body and skeletal diameter measurements, as well as age, weight, height and sex for 507 physically active individuals. A linear model is built to predict height based on shoulder girth (circumference of shoulders measured over deltoid muscles), both measured in centimeters. (Heinz et al. 2003) Shown below are the linear model output for predicting height from shoulder girth and the histogram of slopes from 1,000 randomized datasets (1,000 times, `hgt` was permuted and regressed against `sho_gi`). The red vertical line is drawn at the observed slope value which was produced in the linear model output.

term	estimate	std.error	statistic	p.value
`(Intercept)`	105.832	3.27	32.3	<0.0001
`sho_gi`	0.604	0.03	20.0	<0.0001



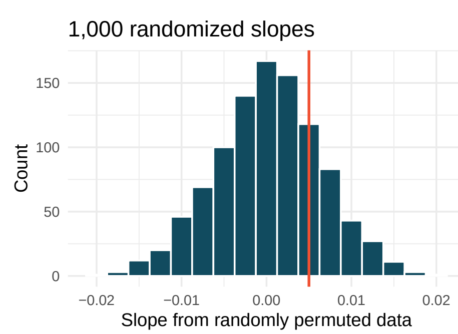
a. What are the null and alternative hypotheses for evaluating whether the slope of the model predicting height from shoulder girth is differen than 0.

b. Using the histogram which describes the distribution of slopes when the null hypothesis is true, find the p-value and conclude the hypothesis test in the context of the problem (use words like shoulder girth and height).

c. Is the conclusion based on the histogram of randomized slopes consistent with the conclusion from the mathematical model? Explain your reasoning.

9. **Baby's weight and father's age, randomization test.** US Department of Health and Human Services, Centers for Disease Control and Prevention collect information on births recorded in the country. The data used here are a random sample of 1000 births from 2014. Here, we study the relationship between the father's age and the weight of the baby. (ICPSR 2014) Shown below are the linear model output for predicting baby's weight (in pounds) from father's age (in years) and the histogram of slopes from 1000 randomized datasets (1000 times, `weight` was permuted and regressed against `fage`). The red vertical line is drawn at the observed slope value which was produced in the linear model output.

term	estimate	std.error	statistic	p.value
`(Intercept)`	7.101	0.199	35.674	<0.0001
`fage`	0.005	0.006	0.757	0.4495

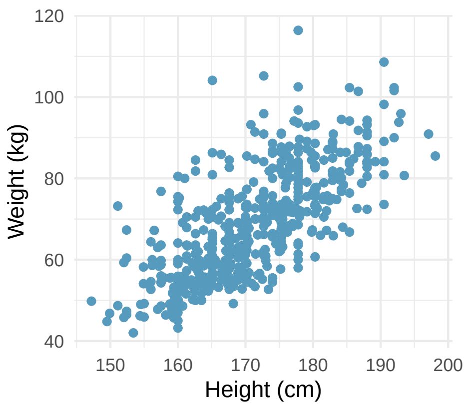


a. What are the null and alternative hypotheses for evaluating whether the slope of the model for predicting baby's weight from father's age is different than 0?

b. Using the histogram which describes the distribution of slopes when the null hypothesis is true, find the p-value and conclude the hypothesis test in the context of the problem (use words like father's age and weight of baby). What does the conclusion of your test say about whether the father's age is a useful predictor of baby's weight?

c. Is the conclusion based on the histogram of randomized slopes consistent with the conclusion from the mathematical model? Explain your reasoning.

10. **Body measurements, mathematical test.** The scatterplot and least squares summary below show the relationship between weight measured in kilograms and height measured in centimeters of 507 physically active individuals. (Heinz et al. 2003)



term	estimate	std.error	statistic	p.value
`(Intercept)`	-105.01	7.54	-13.9	<0.0001
`hgt`	1.02	0.04	23.1	<0.0001

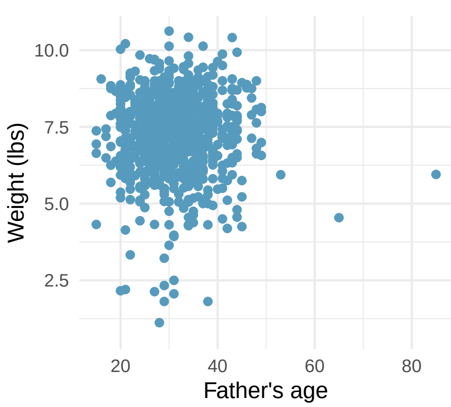
a. Describe the relationship between height and weight.

b. Write the equation of the regression line. Interpret the slope and intercept in context.

c. Do the data provide convincing evidence that the true slope parameter is different than 0? State the null and alternative hypotheses, report the p-value (using a mathematical model), and state your conclusion.

d. The correlation coefficient for height and weight is 0.72. Calculate R^2 and interpret it in context.

11. **Baby's weight and father's age, mathematical test.** Is the father's age useful in predicting the baby's weight? The scatterplot and least squares summary below show the relationship between baby's weight (measured in pounds) and father's age for a random sample of babies. (ICPSR 2014)



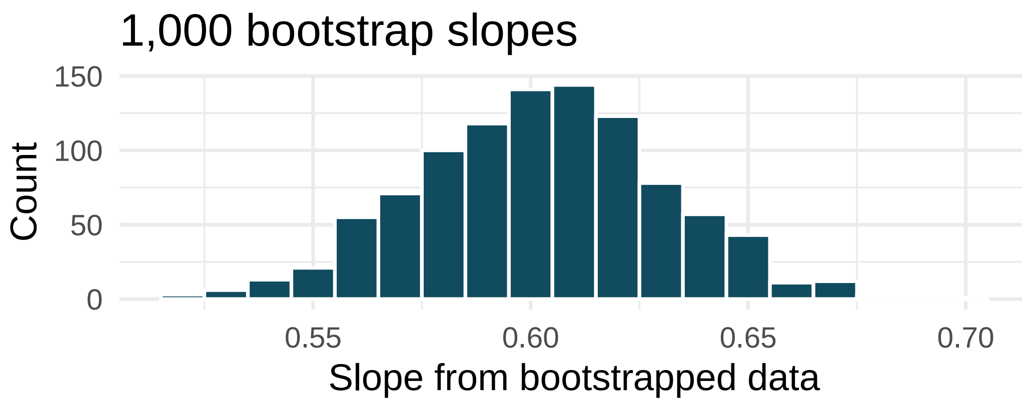
term	estimate	std.error	statistic	p.value
`(Intercept)`	7.1042	0.1936	36.698	<0.0001
`fage`	0.0047	0.0061	0.779	0.4359

a. What is the predicted weight of a baby whose father is 30 years old?

b. Do the data provide convincing evidence that the model for predicting baby weights from father's age has a slope different than 0? State the null and alternative hypotheses, report the p-value (using a mathematical model), and state your conclusion.

c. Based on your conclusion, is father's age a useful predictor of baby's weight?

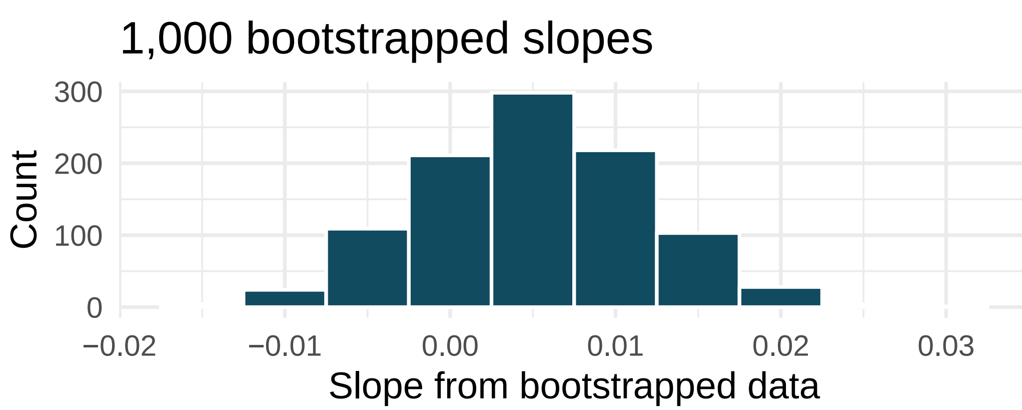
12. **Body measurements, bootstrap percentile interval.** In order to estimate the slope of the model predicting height based on shoulder girth (circumference of shoulders measured over deltoid muscles), 1,000 bootstrap samples are taken from a dataset of body measurements from 507 people. A linear model predicting height based on shoulder girth is fit to each bootstrap sample, and the slope is estimated. A histogram of these slopes is shown below. (Heinz et al. 2003)



a. Using the bootstrap percentile method and the histogram above, find a 98% confidence interval for the slope parameter.

b. Interpret the confidence interval in the context of the problem.

13. **Baby's weight and father's age, bootstrap percentile interval.** US Department of Health and Human Services, Centers for Disease Control and Prevention collect information on births recorded in the country. The data used here are a random sample of 1000 births from 2014. Here, we study the relationship between the father's age and the weight of the baby. Below is the bootstrap distribution of the slope statistic from 1,000 different bootstrap samples of the data. (ICPSR 2014)

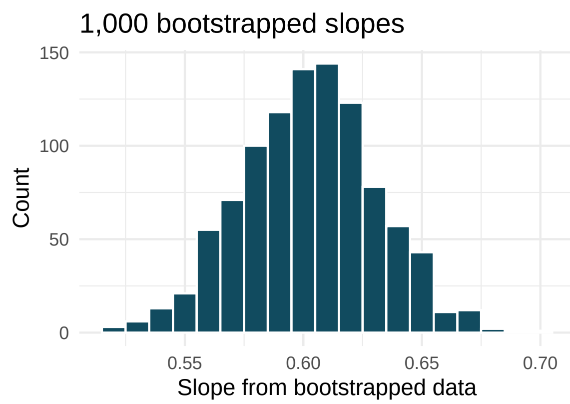


a. Using the bootstrap percentile method and the histogram above, find a 95% confidence interval for the slope parameter.

b. Interpret the confidence interval in the context of the problem.

14. **Body measurements, standard error bootstrap interval.** A linear model is built to predict height based on shoulder girth (circumference of shoulders measured over deltoid muscles), both measured in centimeters. (Heinz et al. 2003) Shown below are the linear model output for predicting height from shoulder girth and the bootstrap distribution of the slope statistic from 1,000 different bootstrap samples of the data.

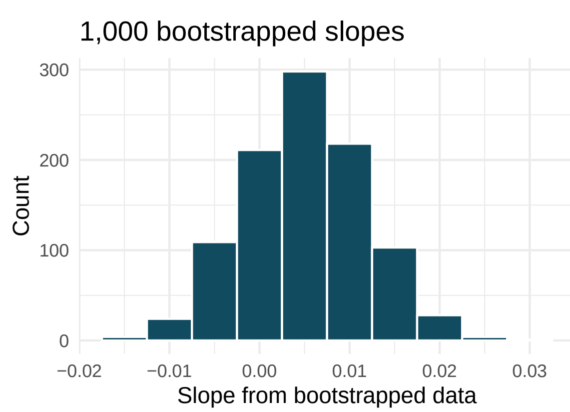
term	estimate	std.error	statistic	p.value
`(Intercept)`	105.832	3.27	32.3	<0.0001
`sho_gi`	0.604	0.03	20.0	<0.0001



a. Using the histogram, approximate the standard error of the slope statistic (that is, quantify the variability of the slope statistic from sample to sample).
b. Find a 98% bootstrap SE confidence interval for the slope parameter.
c. Interpret the confidence interval in the context of the problem.

15. **Baby's weight and father's age, standard error bootstrap interval.** US Department of Health and Human Services, Centers for Disease Control and Prevention collect information on births recorded in the country. The data used here are a random sample of 1000 births from 2014. Here, we study the relationship between the father's age and the weight of the baby. (ICPSR 2014) Shown below are the linear model output for predicting baby's weight (in pounds) from father's age (in years) and the the bootstrap distribution of the slope statistic from 1000 different bootstrap samples of the data.

term	estimate	std.error	statistic	p.value
`(Intercept)`	7.101	0.199	35.674	<0.0001
`fage`	0.005	0.006	0.757	0.4495

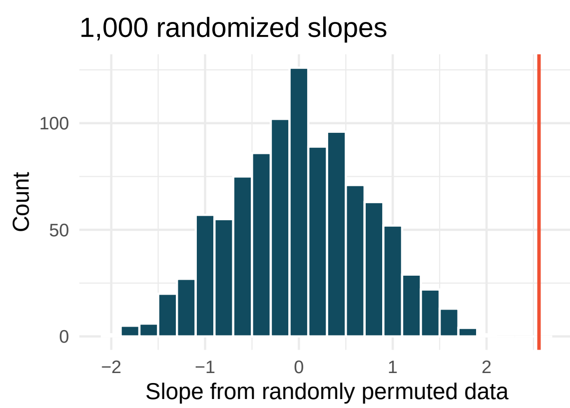


a. Using the histogram, approximate the standard error of the slope statistic (that is, quantify the variability of the slope statistic from sample to sample).
b. Find a 95% bootstrap SE confidence interval for the slope parameter.
c. Interpret the confidence interval in the context of the problem.

16. **Murders and poverty, randomization test.** The following regression output is for predicting annual murders per million (`annual_murders_per_mil`) from percentage living in poverty (`perc_pov`) in a random sample of 20 metropolitan areas. Shown below are the linear model output for predicting annual murders per million from percentage living in poverty for metropolitan areas and the histogram of slopes from 1000 randomized datasets (1000 times, `annual_murders_per_mil` was permuted and regressed against `perc_pov`). The red vertical line is drawn at the observed slope value which was produced in the linear model output.

term	estimate	std.error	statistic	p.value
`(Intercept)`	-29.90	7.79	-3.84	0.0012
`perc_pov`	2.56	0.39	6.56	<0.0001

a. What are the null and alternative hypotheses for evaluating whether the slope of the model for predicting annual murder rate from poverty percentage is different than 0?
b. Using the histogram which describes the distribution of slopes when the null hypothesis is true, find the p-value and conclude the hypothesis test in the context of the problem (use words like murder rate and poverty).
c. Is the conclusion based on the histogram of randomized slopes consistent with the conclusion which would have been obtained using the mathematical model? Explain your reasoning.



17. **Murders and poverty, mathematical test.** The table below shows the output of a linear model annual murders per million (`annual_murders_per_mil`) from percentage living in poverty (`perc_pov`) in a random sample of 20 metropolitan areas.

term	estimate	std.error	statistic	p.value
`(Intercept)`	-29.90	7.79	-3.84	0.0012
`perc_pov`	2.56	0.39	6.56	<0.0001

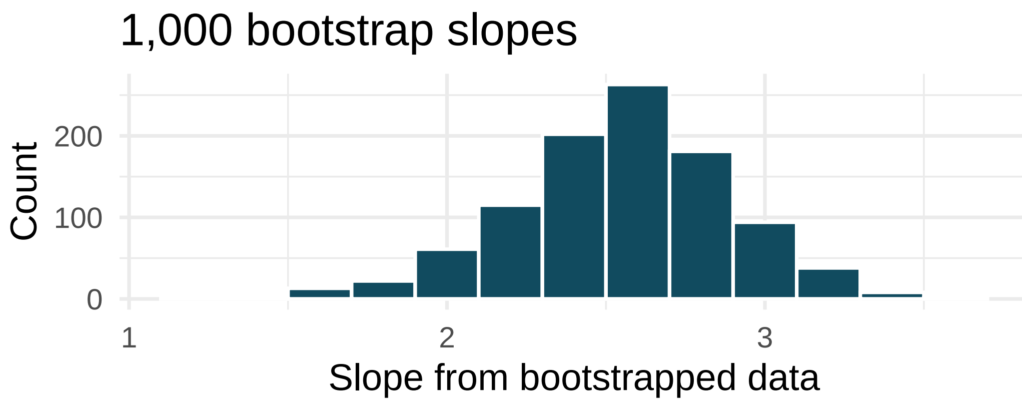
a. What are the hypotheses for evaluating whether the slope of the model predicting annual murder rate from poverty percentage is different than 0?

b. State the conclusion of the hypothesis test from part (a) in context. What does this say about whether poverty percentage is a useful predictor of annual murder rate?

c. Calculate a 95% confidence interval for the slope of poverty percentage, and interpret it in context.

d. Do your results from the hypothesis test and the confidence interval agree? Explain your reasoning.

18. **Murders and poverty, bootstrap percentile interval.** Data on annual murders per million (`annual_murders_per_mil`) and percentage living in poverty (`perc_pov`) is collected from a random sample of 20 metropolitan areas. Using these data we want to estimate the slope of the model predicting `annual_murders_per_mil` from `perc_pov`. We take 1,000 bootstrap samples of the data and fit a linear model predicting `annual_murders_per_mil` from `perc_pov` to each bootstrap sample. A histogram of these slopes is shown below.

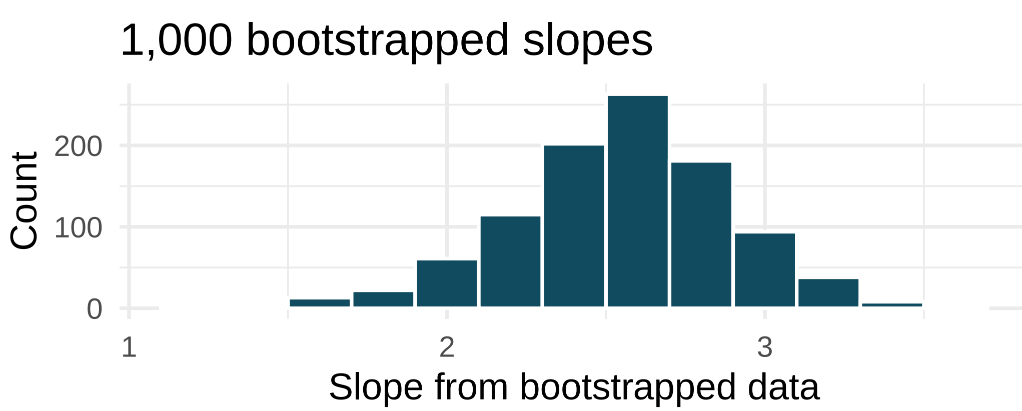


a. Using the percentile bootstrap method and the histogram above, find a 90% confidence interval for the slope parameter.

b. Interpret the confidence interval in the context of the problem.

19. **Murders and poverty, standard error bootstrap interval.** A linear model is built to predict annual murders per million (`annual_murders_per_mil`) from percentage living in poverty (`perc_pov`) in a random sample of 20 metropolitan areas. Shown below are the standard linear model output for predicting annual murders per million from percentage living in poverty for metropolitan areas and the bootstrap distribution of the slope statistic from 1000 different bootstrap samples of the data.

term	estimate	std.error	statistic	p.value
`(Intercept)`	-29.90	7.79	-3.84	0.0012
`perc_pov`	2.56	0.39	6.56	<0.0001



a. Using the histogram, approximate the standard error of the slope statistic (that is, quantify the variability of the slope statistic from sample to sample).

b. Find a 90% bootstrap SE confidence interval for the slope parameter.

c. Interpret the confidence interval in the context of the problem.

20. **I heart cats.** Researchers collected data on heart and body weights of 144 domestic adult cats. The table below shows the output of a linear model predicting heart weight (measured in grams) from body weight (measured in kilograms) of these cats.

term	estimate	std.error	statistic	p.value
`(Intercept)`	-0.357	0.692	-0.515	0.6072
`Bwt`	4.034	0.250	16.119	<0.0001

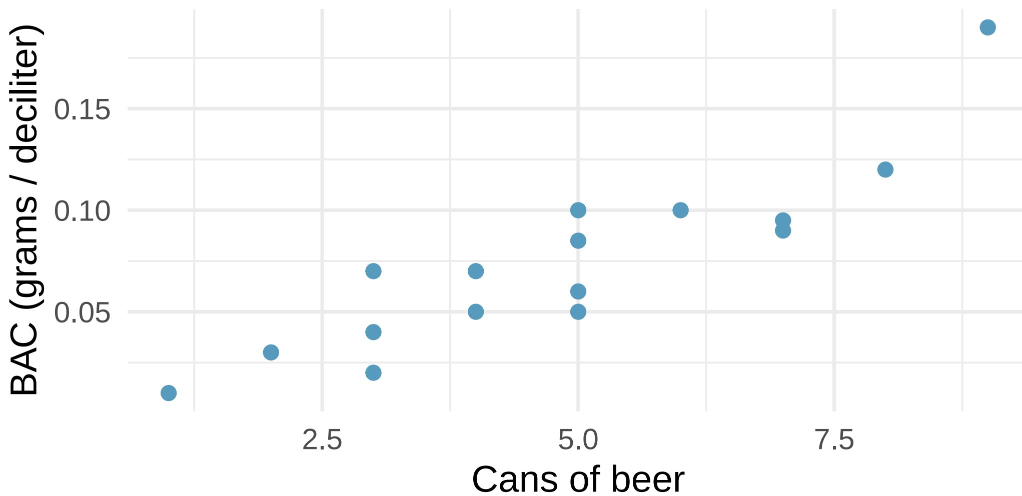
a. What are the hypotheses for evaluating whether body weight is positively associated with heart weight in cats?

b. State the conclusion of the hypothesis test from part (a) in context.

c. Calculate a 95% confidence interval for the slope of body weight, and interpret it in context.

d. Do your results from the hypothesis test and the confidence interval agree? Explain your reasoning.

21. **Beer and blood alcohol content.** Many people believe that weight, drinking habits, and many other factors are much more important in predicting blood alcohol content (BAC) than simply considering the number of drinks a person consumed. Here we examine data from sixteen student volunteers at Ohio State University who each drank a randomly assigned number of cans of beer. These students were evenly divided between men and women, and they differed in weight and drinking habits. Thirty minutes later, a police officer measured their blood alcohol content (BAC) in grams of alcohol per deciliter of blood. The scatterplot and regression table summarize the findings. (Malkevitch and Lesser 2008)



term	estimate	std.error	statistic	p.value
`(Intercept)`	-0.0127	0.0126	-1.00	0.332
`beers`	0.0180	0.0024	7.48	<0.0001

a. Describe the relationship between the number of cans of beer and BAC.

b. Write the equation of the regression line. Interpret the slope and intercept in context.

c. Do the data provide convincing evidence that drinking more cans of beer is associated with an increase in blood alcohol? State the null and alternative hypotheses, report the p-value, and state your conclusion.

d. The correlation coefficient for number of cans of beer and BAC is 0.89. Calculate R^2 and interpret it in context.

e. Suppose we visit a bar in our own town, ask people how many drinks they have had, and also measure their BAC. Would the relationship between number of drinks and BAC would be as strong as the relationship found in the Ohio State study? Why?

Datasets: `coast_starlight` (openintro) | `bdims` (openintro) | `county_2019` (usdata) | `cats` (MASS) | `births14` (openintro) | `cats` (MASS) | `bac` (openintro)

Chapter 20

Prediction and Model Fit

> **Draft Chapter.** This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In the previous chapter, we developed the least squares regression line, interpreted its slope and intercept, and performed inference to test whether a linear relationship exists. In this chapter, we focus on using the regression model for prediction and assessing how well it fits the data. We discuss the dangers of extrapolation, introduce residual diagnostics for identifying problems with the model, examine leverage and influential points, and briefly introduce confidence intervals and prediction intervals for forecasts.

20.1 Making predictions

One of the primary uses of a regression model is to predict the value of the response variable y for a given value of the explanatory variable x. To make a prediction, we simply plug the value of x into the regression equation.

The least squares line for predicting gift aid from family income at Elmhurst College is:

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times \text{family\_income}$$

Predict the gift aid for a student whose family income is \$80,000 (i.e., $x = 80$ in \$1,000s).

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times 80 = 24.3 - 3.45 = 20.85$$

The model predicts this student will receive approximately \$20,850 in gift aid.

Suppose a high school senior is considering Elmhurst College. Can they simply use the linear equation to calculate their financial aid?

They may use it as an estimate, but with important qualifiers. First, the data come from a single year, and aid policies may change. Second, the equation provides an imperfect estimate – while the linear model captures the overall trend, individual students' aid will vary around the predicted value.

20.2 Extrapolation

Use the Elmhurst model to estimate the aid for a student whose family has an income of \$1 million (i.e., $x = 1000$).

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times 1000 = -18.8$$

The model predicts $-\$18,800$ in aid! Elmhurst College does not charge extra tuition to wealthy students. This nonsensical prediction illustrates the danger of **extrapolation**.

Extrapolation is treacherous.

Extrapolation means applying a model to make predictions outside the range of the observed data. A linear model is only an approximation of the true relationship, and that approximation may break down badly outside the range of the data used to build it.

Generally, predictions should only be made within the range of the observed predictor values. Predictions far outside this range are unreliable and should be treated with great skepticism.

Before making a prediction, always check whether the value of x falls within the range of the data used to build the model. If it does not, acknowledge that you are extrapolating and that the prediction may be unreliable.

20.3 Residual diagnostics

In the previous chapter, we introduced residual plots as a way to assess whether a linear model is appropriate. In this section, we examine residual diagnostics in more detail.

20.3.1 Patterns in residual plots

A well-fitting linear model produces residuals that appear randomly scattered around zero. Systematic patterns in the residuals indicate problems:

Consider the three pairs of scatterplots and residual plots below.

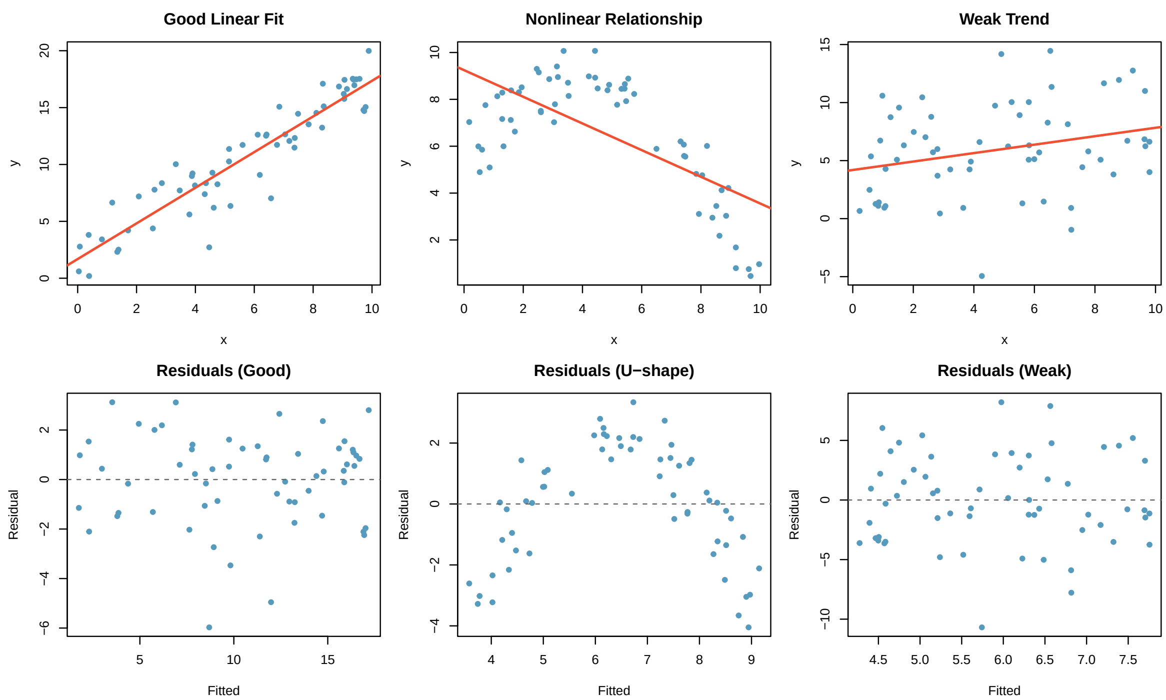


Figure 20.1: Three scatterplots with linear models in the top row and their corresponding residual plots in the bottom row. The first shows a good linear fit with random residuals. The second shows a curved relationship with a U-shaped residual pattern. The third shows a weak trend with random residuals.

Describe any patterns in the residuals.

Dataset 1: The residuals show no obvious patterns. They are scattered randomly around zero. The linear model is appropriate.

Dataset 2: There is a curved pattern in the residuals – they are negative at the edges and positive in the middle. This indicates a nonlinear relationship. A straight line is not a good model for these data.

Dataset 3: The residuals show no patterns, but the trend is very weak. The linear model is technically appropriate, but the relationship may not be practically meaningful.

20.3.2 Non-constant variability (heteroscedasticity)

When the spread of the residuals changes as the predicted value increases, we have **non-constant variability** (also called **heteroscedasticity**). This often appears as a "fan shape" in the residual plot – residuals spread out (or narrow) as the predicted values increase.

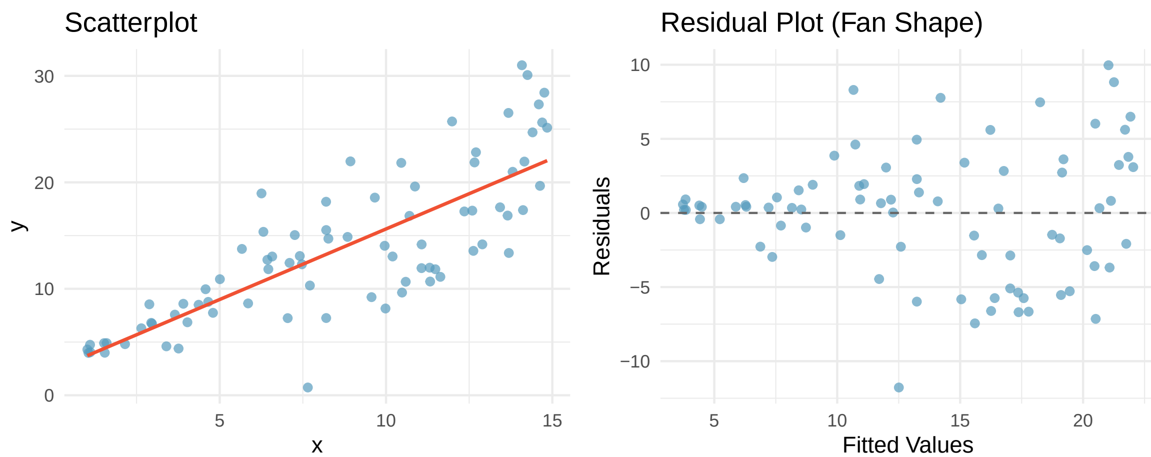


Figure 20.2: A scatterplot and residual plot showing increasing variability as x increases. The residual plot shows a fan shape.

Non-constant variability does not bias the regression line itself, but it does affect inference: standard errors, confidence intervals, and p-values may be unreliable.

20.4 Leverage and influential points

Not all observations have the same impact on the regression line. Some points, by virtue of their position, can exert a disproportionate influence on the model.

20.4.1 Leverage

Leverage.

A point has **high leverage** if its x-value is far from $\bar{x}$ (the mean of the predictor variable). High-leverage points have the *potential* to strongly influence the slope and intercept of the regression line.

High leverage does not automatically mean the point is problematic. If a high-leverage point falls close to the trend established by the other data, it may actually strengthen the model. A high-leverage point is only problematic if its y-value does not follow the pattern of the rest of the data.

20.4.2 Influential points

Influential point.

A point is **influential** if removing it would substantially change the regression line (i.e., the estimated slope or intercept). Influential points are typically high-leverage points that do not follow the pattern of the rest of the data.

Consider three scenarios:

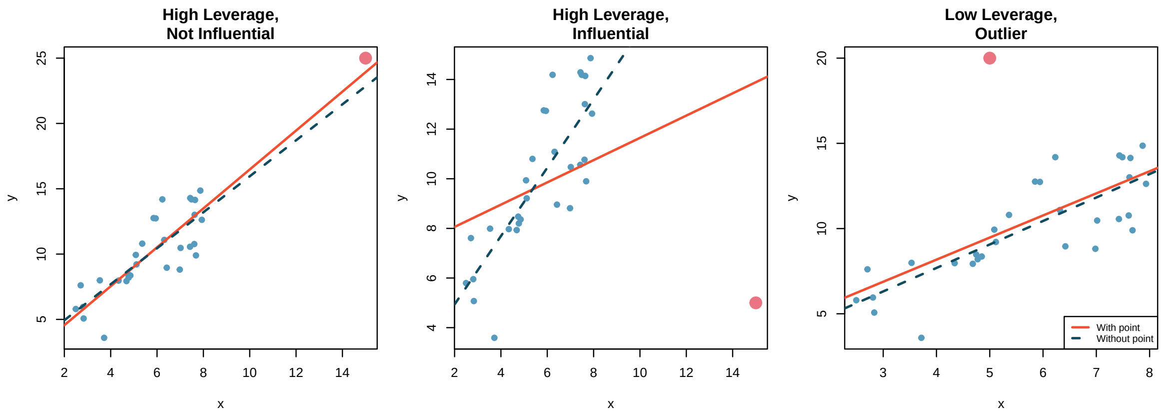


Figure 20.3: Three scatterplots with regression lines. In the first, a point in the corner has high leverage but follows the trend (not influential). In the second, a point far from the trend has high leverage and is influential – removing it changes the slope substantially. In the third, a point near the mean of x is an outlier in the y-direction but has low leverage and is not very influential.

Scenario 1: The point in the far right corner has high leverage (it is far from $\bar{x}$), but it falls along the trend of the other data. Removing it would barely change the regression line. It is *not* influential.

Scenario 2: The point in the far right corner has high leverage *and* falls far from the trend. Removing it would substantially change the slope. It *is* influential.

Scenario 3: The point near the center of the x-values is an outlier in the y-direction. It has low leverage because its x-value is near $\bar{x}$. Removing it would change the intercept slightly but not the slope very much. It has limited influence.

Dealing with influential points.

When you identify an influential point:

1. Verify that it is not a data entry error.
2. Run the analysis with and without the point and report both results.
3. Do not automatically remove it – if it is a legitimate observation, it contains real information.
4. Be transparent about its effect on the conclusions.

20.5 Confidence and prediction intervals

When we use a regression model to make predictions, there are two types of uncertainty to consider:

20.5.1 Confidence intervals for the mean response

A **confidence interval for the mean response** estimates the average value of y for all individuals with a particular value of x. This interval captures the uncertainty in estimating the population regression line.

For example, we might ask: "What is the average gift aid for all Elmhurst students with a family income of $80,000?" The confidence interval for the mean response at $x = 80$ would give a range of plausible values for this population average.

The formula for a $(1 - \alpha) \times 100\%$ confidence interval for the mean response at $x = x^*$ is:

$$\hat{y} \pm t^*_{n-2} \cdot SE_{\hat{y}}$$

where the standard error depends on the distance of x^* from $\bar{x}$:

$$SE_{\hat{y}} = s_e \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$$

Here $s_e = \sqrt{MSE}$ is the residual standard error.

20.5.2 Prediction intervals for individual observations

A **prediction interval** estimates the value of y for a *single* new individual with a particular value of x. This interval is always wider than the confidence interval for the mean response because it must account for both:

1. The uncertainty in estimating the regression line (same as the confidence interval).
2. The natural variability of individual observations around the line.

The prediction interval at $x = x^*$ is:

$$\hat{y} \pm t^*_{n-2} \cdot s_e \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$$

Notice the extra "1+" under the square root compared to the confidence interval formula. This is what makes the prediction interval wider.

20.5.3 Comparing the two intervals

For the possum data, suppose we want to predict the head length for possums with a total length of 85 cm.

The point prediction is $\hat{y} = 41 + 0.59 \times 85 = 91.15$ mm.

- A 95% **confidence interval for the mean** head length of all possums with total length 85 cm might be (89.8, 92.5) mm.
- A 95% **prediction interval** for the head length of a single possum with total length 85 cm might be (83.2, 99.1) mm.

The confidence interval is relatively narrow because it estimates an average, which is estimated with moderate precision. The prediction interval is much wider because it must also account for the natural variation among individual possums – even possums with the same total length have different head lengths.

Confidence interval vs. prediction interval.

- A **confidence interval for the mean response** estimates the average y for a given x. It accounts for uncertainty in the regression line.
- A **prediction interval** estimates the y for a single new observation at a given x. It accounts for both the uncertainty in the regression line and the variability of individual observations.
- Prediction intervals are always wider than confidence intervals.
- Both intervals are narrowest at $x = \bar{x}$ and widen as x moves away from $\bar{x}$.

The fact that both intervals widen as x moves away from $\bar{x}$ connects back to the danger of extrapolation: predictions become increasingly uncertain farther from the center of the data.

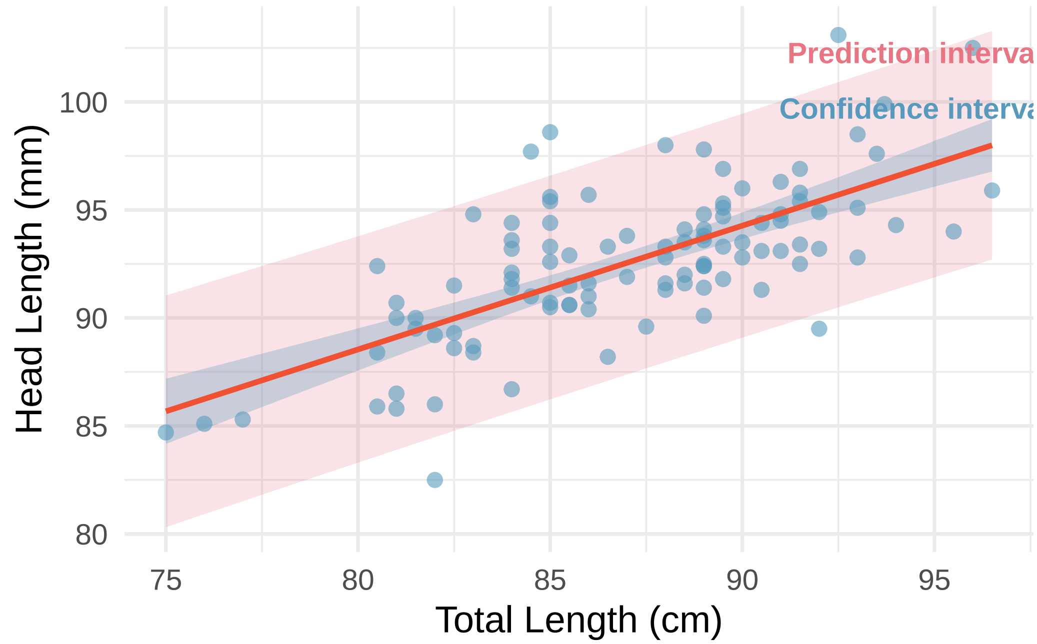


Figure 20.4: A scatterplot with a regression line and both 95% confidence bands (narrow) and 95% prediction bands (wide). Both bands are narrowest near the mean of x and flare outward.

20.6 Chapter review

20.6.1 Summary

In this chapter, we focused on using regression models for prediction and evaluating their fit. We learned that extrapolation – predicting outside the range of the observed data – is dangerous and can lead to nonsensical predictions. Residual diagnostics help us identify violations of the model assumptions, including nonlinearity, non-constant variance, and the influence of unusual points. Leverage measures how far a point's x-value is from the center of the data; influential points are high-leverage observations that do not follow the overall trend and substantially affect the regression line when removed. Finally, we distinguished between confidence intervals for the mean response and prediction intervals for individual observations, noting that prediction intervals are always wider because they account for individual variability.

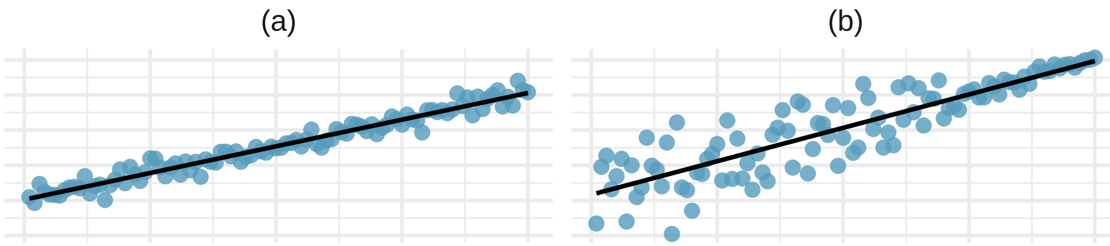
20.6.2 Key terms

- Extrapolation
- Residual diagnostics
- Non-constant variability (heteroscedasticity)
- Leverage
- Influential point
- Confidence interval for the mean response
- Prediction interval

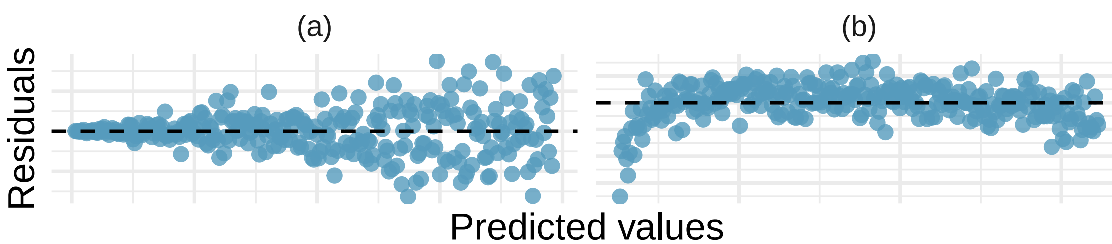
20.7 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Visualizing residuals.** The scatterplots shown below each have a superimposed regression line. If we were to construct a residual plot (residuals versus x) for each, describe in words what those plots would look like.



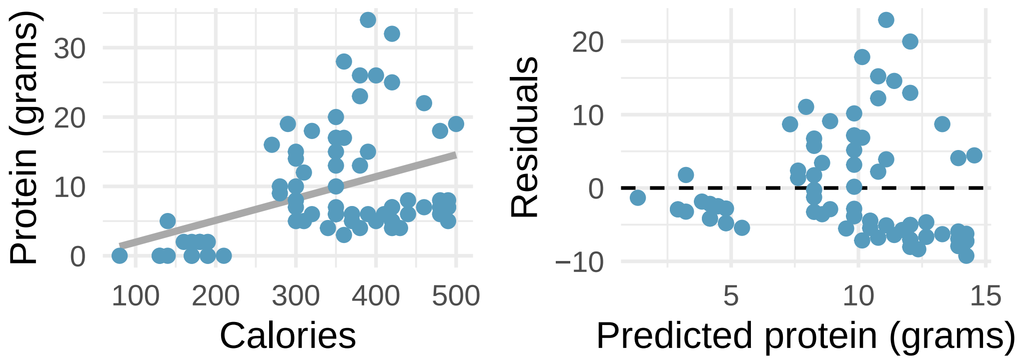
2. **Trends in residuals.** Shown below are two plots of residuals remaining after fitting a linear model to two different sets of data. For each plot, describe important features and determine if a linear model would be appropriate for these data. Explain your reasoning.



3. **Over-under, I.** Suppose we fit a regression line to predict the shelf life of an apple based on its weight. For a particular apple, we predict the shelf life to be 4.6 days. The apple's residual is -0.6 days. Did we over or under estimate the shelf-life of the apple? Explain your reasoning.

4. **Over-under, II.** Suppose we fit a regression line to predict the number of incidents of skin cancer per 1,000 people from the number of sunny days in a year. For a particular year, we predict the incidence of skin cancer to be 1.5 per 1,000 people, and the residual for this year is 0.5. Did we over or under estimate the incidence of skin cancer? Explain your reasoning.

5. **Starbucks, calories, and protein.** The scatterplot below shows the relationship between the number of calories and amount of protein (in grams) Starbucks food menu items contain. Since Starbucks only lists the number of calories on the display items, we might be interested in predicting the amount of protein a menu item has based on its calorie content.



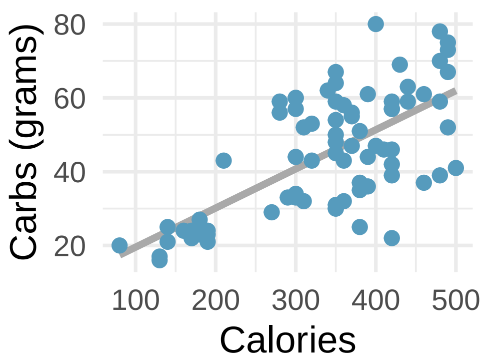
a. Describe the relationship between number of calories and amount of protein (in grams) that Starbucks food menu items contain.

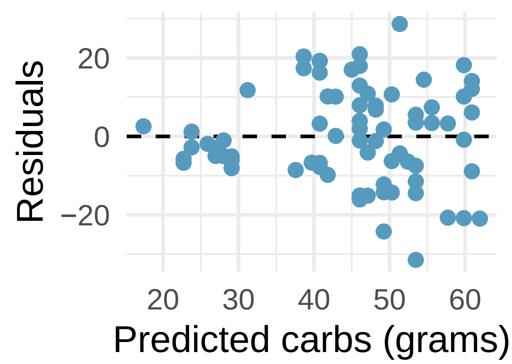
b. In this scenario, what are the predictor and outcome variables?

c. Why might we want to fit a regression line to these data?

d. What does the residuals vs. predicted plot tell us about the variability in prediction errors based on this model for items with lower vs. higher predicted protein?

6. **Starbucks, calories, and carbs.** The scatterplot below shows the relationship between the number of calories and amount of carbohydrates (in grams) Starbucks food menu items contain. Since Starbucks only lists the number of calories on the display items, we might be interested in predicting the amount of carbs a menu item has based on its calorie content.





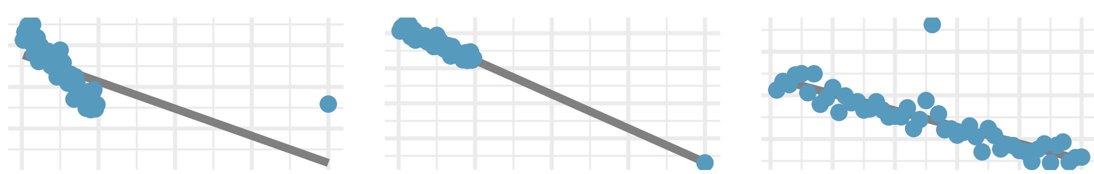
a. Describe the relationship between number of calories and amount of carbohydrates (in grams) that Starbucks food menu items contain.

b. In this scenario, what are the predictor and outcome variables?

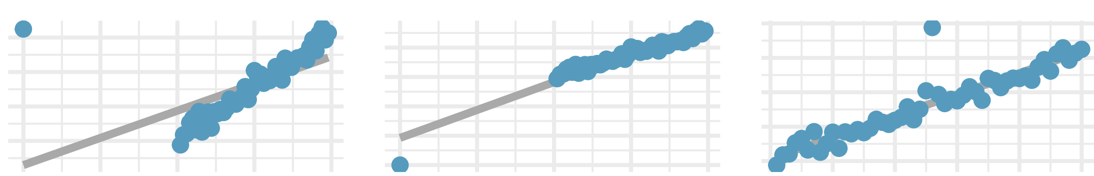
c. Why might we want to fit a regression line to these data?

d. What does the residuals vs. predicted plot tell us about the variability in prediction errors based on this model for items with lower vs. higher predicted carbs?

7. **Outliers, I.** Identify the outliers in the scatterplots shown below, and determine what type of outliers they are. Explain your reasoning.

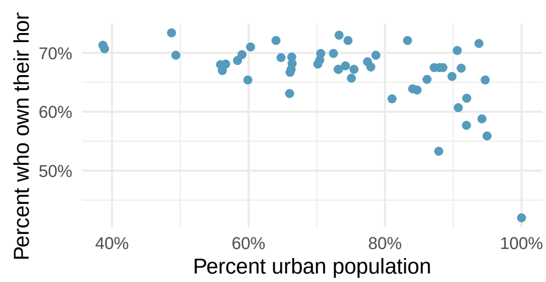


8. **Outliers, II.** Identify the outliers in the scatterplots shown below and determine what type of outliers they are. Explain your reasoning.

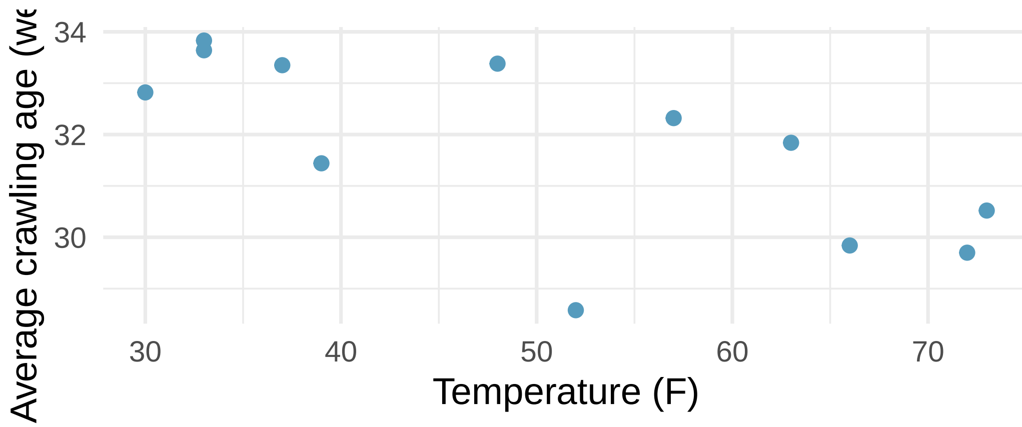


9. **Urban homeowners, outliers.** The scatterplot below shows the percent of families who own their home vs. the percent of the population living in urban areas. There are 52 observations, each corresponding to a state in the US. Puerto Rico and District of Columbia are also included.

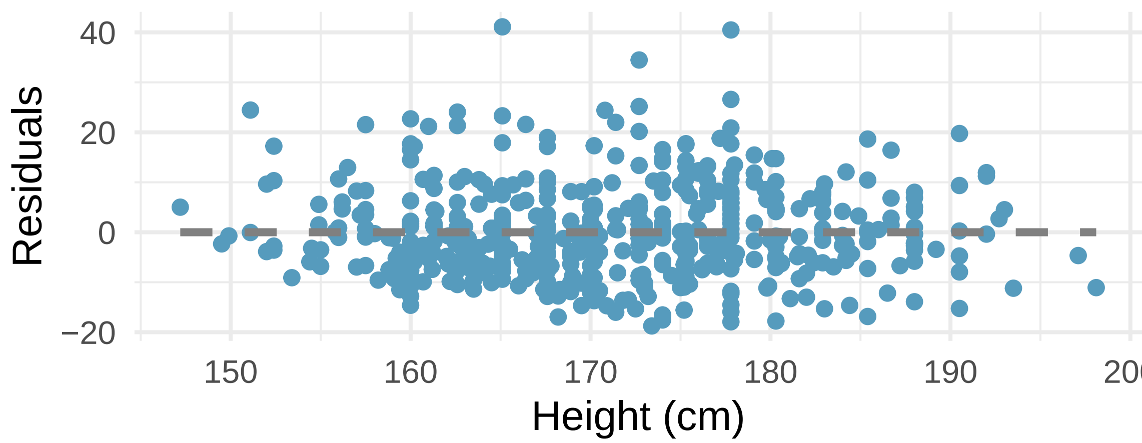
a. Describe the relationship between the percent of families who own their home and the percent of the population living in urban areas.
b. The outlier at the bottom right corner is District of Columbia, where 100% of the population is considered urban. What type of an outlier is this observation?



10. **Crawling babies, outliers.** A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. The plot below shows the relationship between average crawling age of babies born in each month and the average temperature in the month when the babies are six months old. The plot reveals a potential outlying month when the average temperature is about 53F and average crawling age is about 28.5 weeks. Does this point have high leverage? Is it an influential point? (Benson 1993)



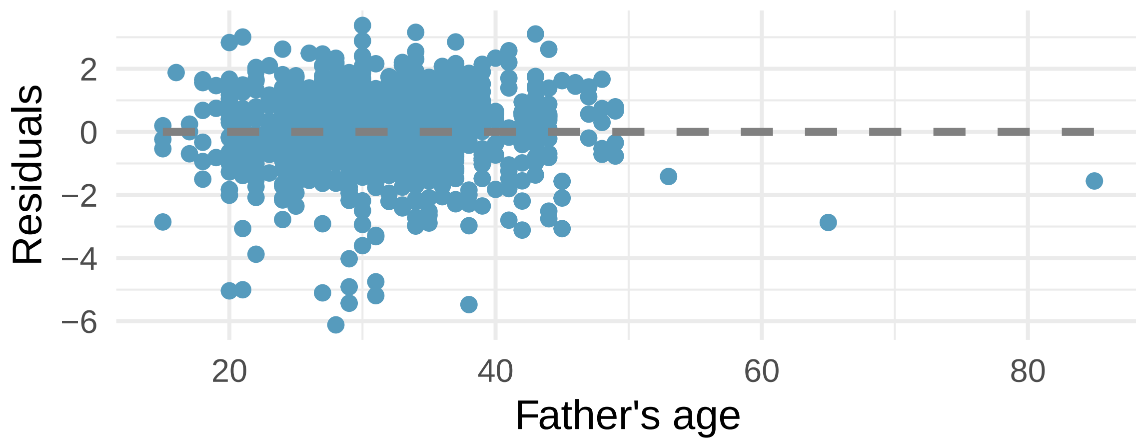
11. **Body measurements, conditions.** The scatterplot below shows the residuals (on the y-axis) from the linear model of weight vs. height from a dataset of body measurements from 507 physically active individuals. The x-axis is the height of the individuals, in cm. (Heinz et al. 2003)



a. For these data, R^2 is 51.84%. What is the value of the correlation coefficient? How can you tell if it is positive or negative? (Hint: you may need to look at a previous exercise.)

b. Examine the residual plot. What do you observe? Is a simple least squares fit appropriate for these data? Which of the LINE conditions are met or not met?

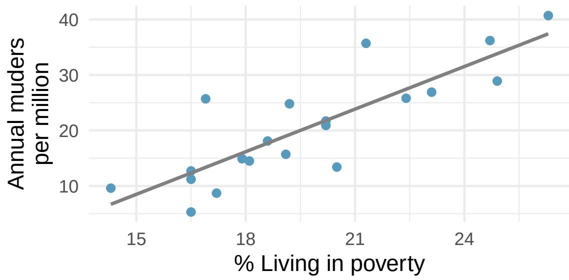
12. **Baby's weight and father's age, conditions.** The scatterplot below shows the residuals (on the y-axis) from the linear model of baby's weight (measured in pounds) vs. father's age for a random sample of babies. Father's age is on the x-axis. (ICPSR 2014)

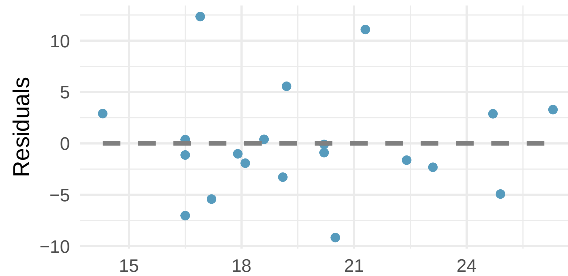


a. For these data, R^2 is 0.09%. What is the value of the correlation coefficient? How can you tell if it is positive or negative? (Hint: you may need to look at a previous exercise.)

b. Examine the residual plot. What do you observe? Is a simple least squares fit appropriate for these data? Which of the LINE conditions are met or not met?

13. **Murders and poverty, conditions.** The scatterplot below shows the annual murders per million vs. percentage living in poverty in a random sample of 20 metropolitan areas. The second figure plots residuals on the y-axis and percent living in poverty on the x-axis.

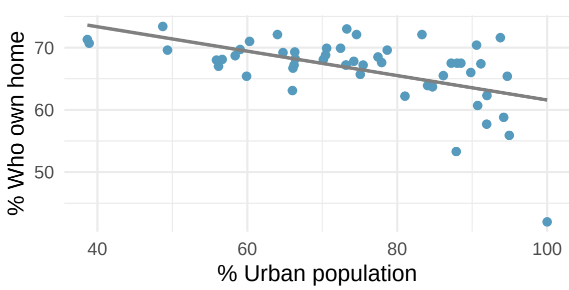


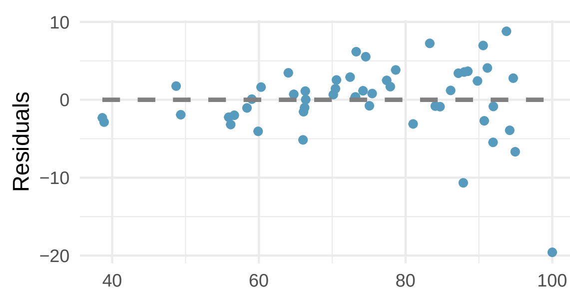


a. For these data, R^2 is 70.56%. What is the value of the correlation coefficient? How can you tell if it is positive or negative?

b. Examine the residual plot. What do you observe? Is a simple least squares fit appropriate for the data? Which of the LINE conditions are met or not met?

14. **Urban homeowners, conditions.** The scatterplot below shows the percent of families who own their home vs. the percent of the population living in urban areas. (US Census Bureau 2010) There are 52 observations, each corresponding to a state in the US. Puerto Rico and District of Columbia are also included. The second figure plots residuals on the y-axis and percent of the population living in urban areas on the x-axis.

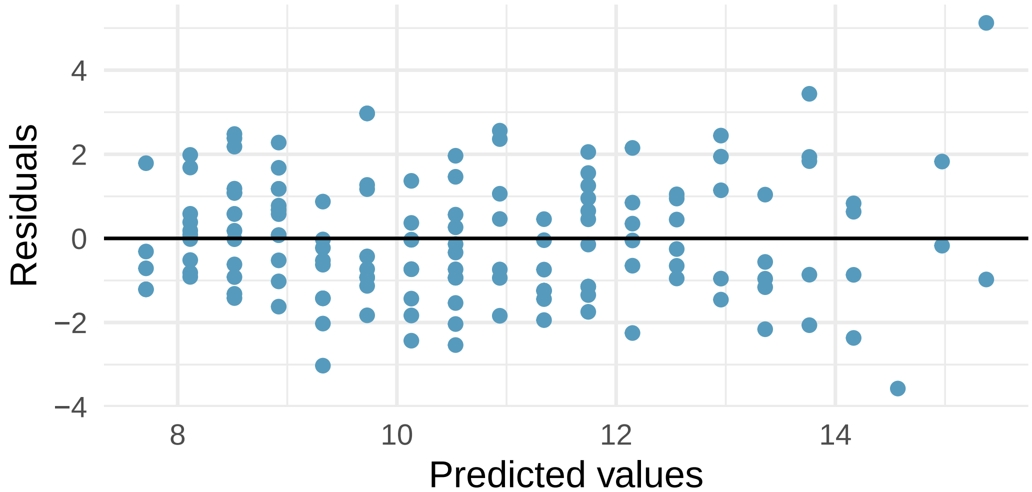




a. For these data, R^2 is 29.16%. What is the value of the correlation coefficient? How can you tell if it is positive or negative?

b. Examine the residual plot. What do you observe? Is a simple least squares fit appropriate for the data? Which of the LINE conditions are met or not met?

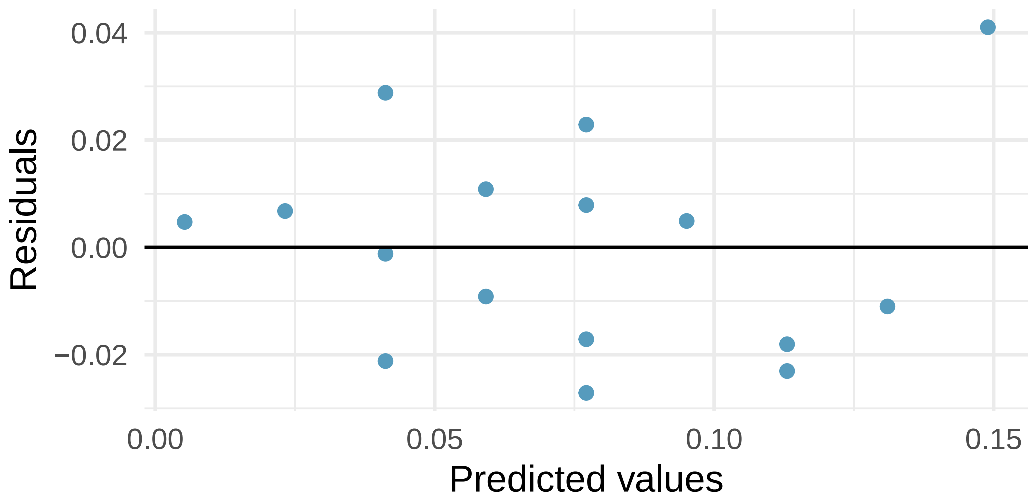
15. **I heart cats, LINE conditions.** Researchers collected data on heart and body weights of 144 domestic adult cats. The figure below shows the output of the predicted values and residuals generated from a linear model predicting heart weight (measured in grams) from body weight (measured in kilograms) of these cats.



a. Examine the residual plot. Notice that for the small predicted values the residuals have a smaller magnitude than the larger residuals seen with the larger predicted values. The change in magnitude of the residuals across the predicted values is an indication of violation of which LINE technical condition?

b. If the LINE condtion described in part (a) is violated, might it lead to an incorrect conclusion about the model (i.e., the least squares regression line itself), the inference of the model (i.e., the p-value associated with the least squares regression line), neither, or both? Explain your reasoning.

16. **Beer and blood alcohol content, LINE conditions.** The figure below shows the output of the predicted values and residuals generated from a linear model predicting the blood alcohol content (BAC) from number of cans of beer drunk by sixteen student volunteers at Ohio State University. (Malkevitch and Lesser 2008)



a. Examine the residual plot. Notice that it is difficult to identify any convincing patterns for or against violation of the LINE technical conditions. What is it about the residual plot that makes it difficult to assess the LINE technical conditions?

b. Is there anything about the residual plot which would make you hesitate about using the linear model for inference about all students? Is there anything about the experimental design of the study which would make you hesitate about using the linear model for inference about all students?

Datasets: `starbucks` (openintro) | `urban_owner` (usdata) | `babies_crawl` (openintro) | `bac` (openintro)

PART V

FOUNDATIONS

You have already been *doing* probability — every simulation you ran was an experiment in randomness. This part formalizes the rules governing random events, introduces random variables, and develops the binomial and normal distributions as mathematical models. These foundations explain *why* the methods from earlier parts work.

The final chapters explore what happens when inference goes wrong: Type II errors, statistical power, and how to plan studies large enough to detect meaningful effects. These topics are essential for designing good research and critically evaluating published results.

Chapter 21

Probability Rules

> **Draft Chapter.** This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Probability forms the foundation of statistics, and you are probably already aware of many of the ideas we will discuss in this chapter. However, formalizing these concepts is likely new. While this chapter provides the theoretical foundation for the inference methods you have already been using, it also offers practical tools for reasoning about uncertainty. We will define probability, develop rules for combining probabilities, and explore conditional probability and independence — all concepts that underpin the inference methods from Parts II and III.

StatLens tools for this chapter: The probability concepts in this chapter provide the theoretical foundation for the distribution calculators and inference tools you used in earlier parts of the book. Try the **Binomial Distribution Calculator** to see probability in action — set $n = 1$ for a single Bernoulli trial.

Launch Binomial Distribution Calculator

21.1 Defining probability

Statistics is built on probability, and probability gives us the language to describe uncertainty. We begin with a few simple examples that may feel familiar.

A "die," the singular of "dice," is a cube with six faces numbered 1, 2, 3, 4, 5, and 6. What is the chance of getting a 1 when rolling a fair die?

If the die is fair, then the chance of a 1 is as good as the chance of any other number. Since there are six equally likely outcomes, the chance must be 1-in-6 or, equivalently, $1/6$.

What is the chance of getting a 1 or 2 on the next roll?

The outcomes 1 and 2 constitute two of the six equally likely outcomes, so the chance of getting one of these two is $2/6 = 1/3$.

What is the chance of not rolling a 2?

Since the chance of rolling a 2 is $1/6$, the chance of not rolling a 2 must be $1 - 1/6 = 5/6$. Alternatively, not rolling a 2 means getting a 1, 3, 4, 5, or 6, which makes up five of the six equally likely outcomes.

We use probability to describe and understand apparent randomness. We frame probability in terms of a **random process** giving rise to an **outcome**.

Random process	Possible outcomes
Roll a die	1, 2, 3, 4, 5, or 6
Flip a coin	H or T

The **probability** of an outcome is the proportion of times the outcome would occur if we observed the random process an infinite number of times. Probability always takes values between 0 and 1 (inclusive) and may also be expressed as a percentage between 0% and 100%.

Probability can be illustrated by rolling a die many times. Let $\hat{p}_n$ be the proportion of outcomes that are 1 after the first n rolls. As the number of rolls increases, $\hat{p}_n$ converges to the probability $p = 1/6$.

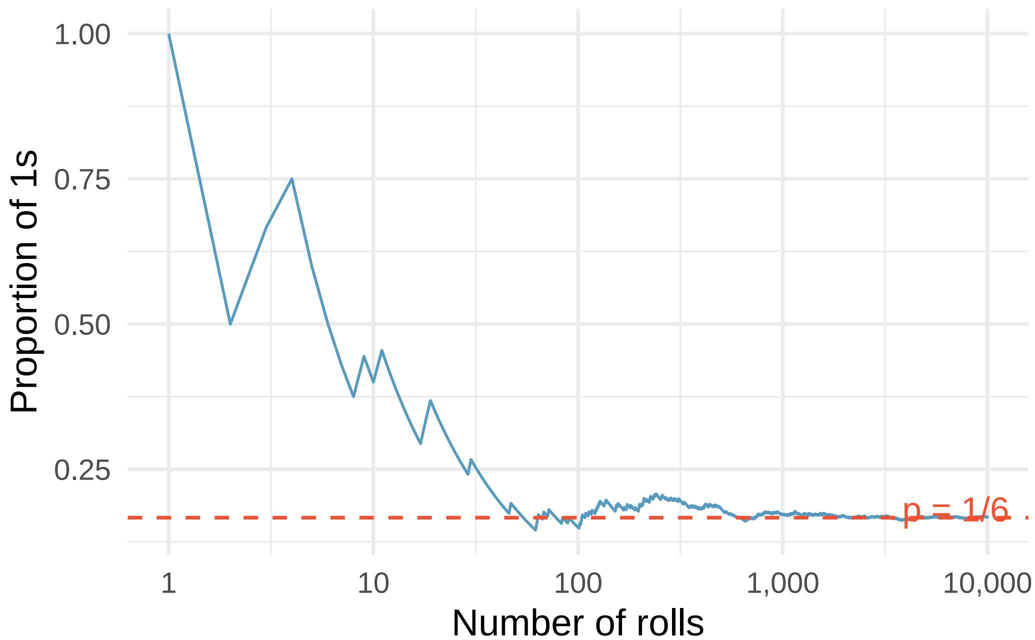


Figure 21.1: The fraction of die rolls that are 1 at each stage in a simulation. The proportion tends to get closer to 1/6 as the number of rolls increases.

Law of Large Numbers. As more observations are collected, the proportion $\hat{p}_n$ of occurrences with a particular outcome converges to the probability p of that outcome.

Occasionally the proportion will veer away from the true probability, but these deviations become smaller as the number of trials increases.

We write the probability of rolling a 1 as $P(\text{rolling a }1)$. When the context is clear, we abbreviate this as $P(1)$.

Random processes include rolling a die and flipping a coin. (a) Think of another random process. (b) Describe all the possible outcomes of that process.

Show answer

(a) and (b) Here are several examples. (i) Whether someone gets sick next month: outcomes are *sick* and *not sick*. (ii) Randomly selecting a person and measuring their height: the outcome is a positive number. (iii) Whether the stock market goes up or down next week: outcomes are *up*, *down*, and *no change*. (iv) The number of emails you receive tomorrow: the outcome is a non-negative integer.

21.1.1 Simulation: The law of large numbers in action

One of the most powerful ways to build intuition about probability is through simulation. Imagine writing a computer program that simulates flipping a fair coin 10,000 times and tracking the running proportion of heads after each flip.

- After 10 flips, the proportion of heads might be 0.30 or 0.70 — quite far from 0.50.
- After 100 flips, the proportion is likely between 0.40 and 0.60.
- After 1,000 flips, the proportion is usually between 0.47 and 0.53.
- After 10,000 flips, it is almost certainly within 0.01 of 0.50.

This is the Law of Large Numbers at work. The key insight is that probability is a *long-run* concept. In the short run, anything can happen. In the long run, patterns emerge with remarkable stability.

This simulation perspective will recur throughout our course. In Parts II and III, we used simulation to build sampling distributions and test hypotheses. Here in Part IV, we develop the *mathematical* rules that explain *why* those simulations work.

21.2 Sample spaces and events

The **sample space** S of a random process is the set of all possible outcomes. An **event** is a subset of the sample space — that is, a collection of one or more outcomes.

For rolling a single die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.

Let A represent the event that the die roll results in 1 or 2, so $A = \{1, 2\}$. Let B represent the event that the die roll is 4 or 6, so $B = \{4, 6\}$. These events are shown visually in the diagram below.

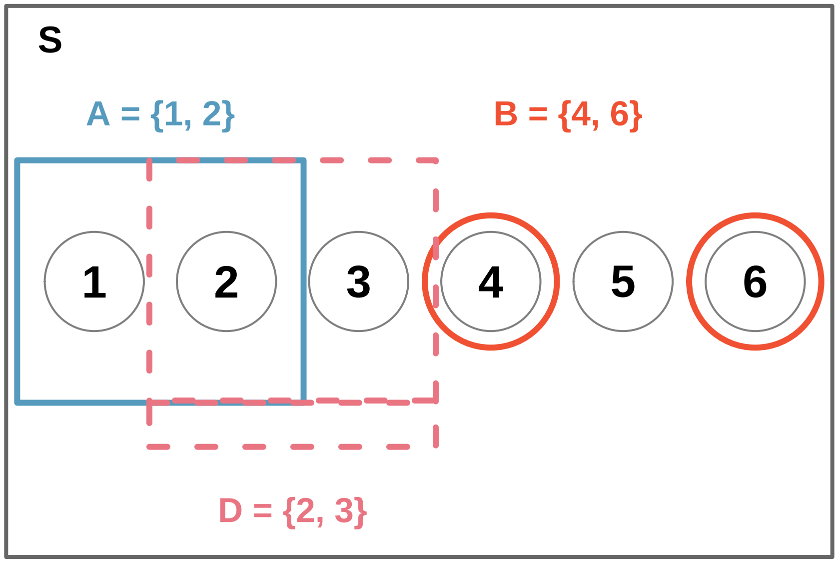


Figure 21.2: Three events for die outcomes. A and B are disjoint since they share no outcomes.

21.3 The complement rule

Let $D = \{2, 3\}$ represent the event that the outcome of a die roll is 2 or 3. Then the **complement** of D, denoted D^c, represents all outcomes in the sample space that are *not* in D:

$$D^c = \{1, 4, 5, 6\}$$

(a) Compute $P(D^c) = P(\text{rolling a } 1, 4, 5, \text{ or } 6)$. (b) What is $P(D) + P(D^c)$?

Show answer

(a) The outcomes are disjoint and each has probability $1/6$, so $P(D^c) = 4/6 = 2/3$. (b) $P(D) = 1/6 + 1/6 = 1/3$. Since D and D^c are disjoint, $P(D) + P(D^c) = 1/3 + 2/3 = 1$.

A complement A^c is constructed so that (i) every possible outcome not in A is in A^c, and (ii) A and A^c are disjoint. This gives us:

Complement rule. The complement of event A is denoted A^c and represents all outcomes not in A. The probabilities of A and A^c are related by:

$$P(A) + P(A^c) = 1 \qquad \text{equivalently,} \qquad P(A) = 1 - P(A^c)$$

The complement rule is especially useful when computing a probability directly would be tedious but the complement is simple.

Let A represent the event where the sum of two dice is less than 12. What is $P(A)$?

The complement A^c is the event that the sum equals 12. The only way to roll a sum of 12 is to get two 6s, so $P(A^c) = 1/36$. Using the complement rule:

$$P(A) = 1 - P(A^c) = 1 - \frac{1}{36} = \frac{35}{36}$$

Find the following probabilities for rolling two dice: (a) The sum is *not* 6. (b) The sum is at least 4. (c) The sum is no more than 10.

Show answer

(a) $P(\text{sum} = 6) = 5/36$, so $P(\text{not } 6) = 1{-}5/36 = 31/36$. (b) $P(\text{sum} = 2 \text{ or } 3) = 1/36{+}2/36 = 1/12$, so $P(\text{at least } 4) = 1 - 1/12 = 11/12$. (c) $P(\text{sum} = 11 \text{ or } 12) = 2/36 + 1/36 = 1/12$, so $P(\text{no more than } 10) = 1 - 1/12 = 11/12$.

21.4 The addition rule

21.4.1 Disjoint (mutually exclusive) events

Two outcomes or events are called **disjoint** (or **mutually exclusive**) if they cannot both happen at the same time. For instance, when rolling a die, the outcomes 1 and 2 are disjoint since they cannot both occur on a single roll.

Addition rule for disjoint events. If A_1 and A_2 are disjoint events, then:

$$P(A_1 \text{ or } A_2) = P(A_1) + P(A_2)$$

More generally, if $A_1, A_2, \ldots, A_k$ are all mutually disjoint, then:

$$P(A_1 \text{ or } A_2 \text{ or } \cdots \text{ or } A_k) = P(A_1) + P(A_2) + \cdots + P(A_k)$$

We are interested in the probability of rolling a 1, 4, or 5. (a) Explain why the outcomes 1, 4, and 5 are disjoint. (b) Apply the addition rule to determine $P(1 \text{ or } 4 \text{ or } 5)$.

Show answer

(a) At most one of these outcomes can occur on a single die roll, so they are disjoint. (b) $P(1 \text{ or } 4 \text{ or } 5) = P(1) + P(4) + P(5) = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$.

21.4.2 Probability distributions

A **probability distribution** is a table of all disjoint outcomes and their associated probabilities. For example, the distribution for the sum of two dice:

Dice sum	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Rules for probability distributions. A valid probability distribution must satisfy three rules:

1. The outcomes listed must be disjoint.
2. Each probability must be between 0 and 1.
3. The probabilities must total 1.

21.4.3 The general addition rule

When events are *not* disjoint, we cannot simply add their probabilities — doing so would double-count the outcomes they share.

Consider a standard deck of 52 cards. Let A be the event "the card is a diamond" and B be the event "the card is a face card." These events overlap: the jack, queen, and king of diamonds are both diamonds and face cards.

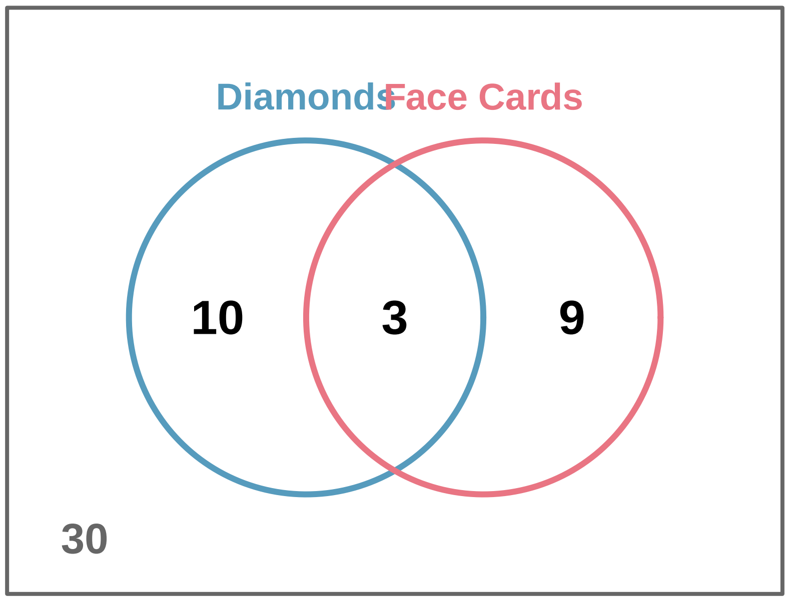


Figure 21.3: A Venn diagram for diamonds and face cards showing 10 diamonds that are not face cards, 9 face cards that are not diamonds, and 3 cards that are both.

If we compute $P(A) + P(B) = 13/52 + 12/52 = 25/52$, we have counted the 3 cards in the overlap twice. We correct this:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26}$$

General addition rule. For any two events A and B (disjoint or not):

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

"Or" is inclusive in statistics. When we write "A or B," we mean "A, B, or both" unless explicitly stated otherwise.

(a) If A and B are disjoint, explain why $P(A \text{ and } B) = 0$. (b) Show that the general addition rule simplifies to the simpler addition rule when A and B are disjoint.

Show answer

(a) If A and B are disjoint, they can never occur simultaneously, so $P(A \text{ and } B) = 0$. (b) Substituting 0 for $P(A \text{ and } B)$ in the general addition rule gives $P(A \text{ or } B) = P(A) + P(B) - 0 = P(A) + P(B)$, which is the simpler addition rule.

21.5 Conditional probability

21.5.1 Exploring conditional probability with data

There can be rich relationships between two or more variables, and conditional probability provides the framework for exploring them. Suppose we have data from a study classifying 1,822 photos using both a machine learning (ML) algorithm and human judgment:

	Truth: fashion	Truth: not	Total
ML: pred_fashion	197	22	219
ML: pred_not	112	1491	1603
Total	309	1513	1822

If a photo is actually about fashion, what is the chance the ML classifier correctly identified it?

Of the 309 fashion photos, the ML algorithm correctly classified 197:

$$P(\text{ML predicts fashion} \mid \text{truth is fashion}) = \frac{197}{309} = 0.638$$

We call this a **conditional probability** because we computed the probability *under a condition* — that the photo was actually about fashion.

21.5.2 Marginal and joint probabilities

A **marginal probability** is a probability based on a single variable, without regard to other variables. A **joint probability** is the probability of two or more outcomes occurring together.

From the photo classification data: $P(\text{ML predicts fashion}) = 219/1822 = 0.120$ is a marginal probability, while $P(\text{ML predicts fashion and truth is fashion}) = 197/1822 = 0.108$ is a joint probability.

21.5.3 The conditional probability formula

Conditional probability. The conditional probability of outcome A given condition B is:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

The vertical bar "|" is read as "given."

Using the photo classification data, we can verify our earlier calculation using proportions alone:

$$P(\text{truth is fashion} \mid \text{ML predicts fashion}) = \frac{P(\text{truth is fashion and ML predicts fashion})}{P(\text{ML predicts fashion})} = \frac{0.108}{0.120} = 0.900$$

If the ML classifier says a photo is about fashion, there is a 90% probability it actually is.

The smallpox dataset provides data on 6,224 individuals in Boston in 1721. Of these, 3.92% were inoculated. Among those not inoculated, 14.11% died. Among those inoculated, only 2.46% died. Write out, in formal notation, the probability a randomly selected person who was not inoculated died from smallpox.

Show answer

$P(\text{result} = \text{died} \mid \text{inoculated} = \text{no}) = \frac{P(\text{died and not inoculated})}{P(\text{not inoculated})} = \frac{0.1356}{0.9608} = 0.1411.$

21.6 The multiplication rule

21.6.1 Multiplication rule for independent processes

Consider rolling two dice. If $1/6$ of the time the first die is a 1, and $1/6$ of *those* times the second die is also a 1, then:

$$P(\text{both are } 1) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Multiplication rule for independent processes. If A and B represent events from two different and independent processes, then:

$$P(A \text{ and } B) = P(A) \times P(B)$$

If there are k events $A_1, \ldots, A_k$ from k independent processes, then:

$$P(A_1 \text{ and } A_2 \text{ and } \cdots \text{ and } A_k) = P(A_1) \times P(A_2) \times \cdots \times P(A_k)$$

About 9% of people are left-handed. Suppose 2 people are selected at random from the US population. (a) What is the probability that both are left-handed? (b) What is the probability that both are right-handed?

Show answer

(a) $0.09 \times 0.09 = 0.0081$. (b) $P(\text{right-handed}) = 1 - 0.09 = 0.91$, so $0.91 \times 0.91 = 0.8281$.

Suppose 5 people are selected at random. (a) What is the probability that all are right-handed? (b) What is the probability that not all are right-handed?

Show answer

(a) $0.91^5 = 0.624$. (b) Using the complement: $P(\text{not all right-handed}) = 1 - 0.624 = 0.376$.

21.6.2 The general multiplication rule

The general multiplication rule handles events that may not be independent:

General multiplication rule. If A and B represent two outcomes or events, then:

$$P(A \text{ and } B) = P(A \mid B) \times P(B)$$

This is simply a rearrangement of the conditional probability formula.

In the smallpox data, 96.08% of residents were not inoculated, and 85.88% of those who were not inoculated survived. What is the probability that a resident was not inoculated and lived?

$P(\text{lived and not inoculated}) = P(\text{lived} \mid \text{not inoculated}) \times P(\text{not inoculated}) = 0.8588 \times 0.9608 = 0.8251$

21.6.3 Tree diagrams

Tree diagrams are a tool to organize outcomes and probabilities when two or more processes occur in sequence and each process is conditioned on its predecessors.

Suppose 13% of students earned an A on the midterm. Of those who earned an A on the midterm, 47% received an A on the final. Of those who earned lower than an A on the midterm, 11% received an A on the final. You randomly pick up a final exam and notice the student received an A. What is the probability the student also earned an A on the midterm?

We organize the information into a tree diagram:

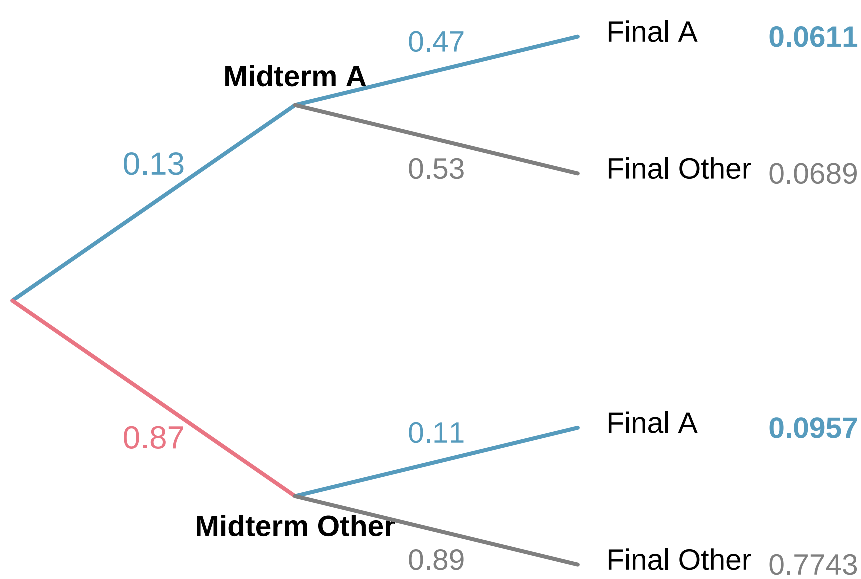


Figure 21.4: Tree diagram for midterm and final exam grades.

From the tree:

- $P(\text{midterm A and final A}) = 0.13 \times 0.47 = 0.0611$
- $P(\text{final A}) = 0.0611 + 0.0957 = 0.1568$
- $P(\text{midterm A} \mid \text{final A}) = \frac{0.0611}{0.1568} = 0.390$

The probability the student also earned an A on the midterm is about 0.39.

Sum of conditional probabilities. If $A_1, \dots, A_k$ represent all disjoint outcomes of a variable, and B is an event, then:

$$P(A_1 \mid B) + P(A_2 \mid B) + \cdots + P(A_k \mid B) = 1$$

The complement rule also holds for conditional probabilities: $P(A \mid B) = 1 - P(A^c \mid B)$.

21.7 Independence

Two processes are **independent** if knowing the outcome of one provides no useful information about the outcome of the other. Mathematically, events A and B are independent if:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Equivalently, A and B are independent if $P(A \mid B) = P(A)$.

If we draw one card from a shuffled deck, is the event that the card is a heart independent of the event that the card is an ace?

$P(\heartsuit) = 1/4$, $P(\text{ace}) = 1/13$, and $P(\heartsuit \text{ and ace}) = 1/52$. We check:

$$P(\heartsuit) \times P(\text{ace}) = \frac{1}{4} \times \frac{1}{13} = \frac{1}{52} = P(\heartsuit \text{ and ace})$$

Because the equation holds, "heart" and "ace" are independent events.

Let X and Y represent the outcomes of rolling two dice. (a) What is $P(X = 1)$? (b) What is $P(Y = 1 \text{ and } X = 1)$? (c) Compute $P(Y = 1 \mid X = 1)$. (d) Compare $P(Y = 1)$ with the result from (c).

Show answer

(a) $1/6$. (b) $1/36$. (c) $\frac{1/36}{1/6} = 1/6$. (d) $P(Y = 1) = 1/6$, the same as part (c). Knowledge about X does not change the probability for Y, confirming the dice are independent.

The gambler's fallacy. Ron is watching a roulette table and notices the last five spins were all black. He bets on red, reasoning that "black six times in a row" is unlikely. This reasoning is flawed: each spin is independent of the previous spins. Past outcomes provide no information about future ones. Casinos often post recent outcomes precisely to encourage this fallacy.

21.8 Bayes' theorem (optional)

In many situations, we are given $P(B \mid A)$ but need $P(A \mid B)$ — the "inverted" conditional probability. Bayes' theorem provides a systematic way to perform this inversion.

In Canada, about 0.35% of women over 40 develop breast cancer in any given year. Mammograms produce false negatives 11% of the time (indicating no cancer when cancer is present) and false positives 7% of the time (indicating cancer when none is present). If a randomly selected woman over 40 tests positive, what is the probability she actually has breast cancer?

We organize the information into a tree diagram:

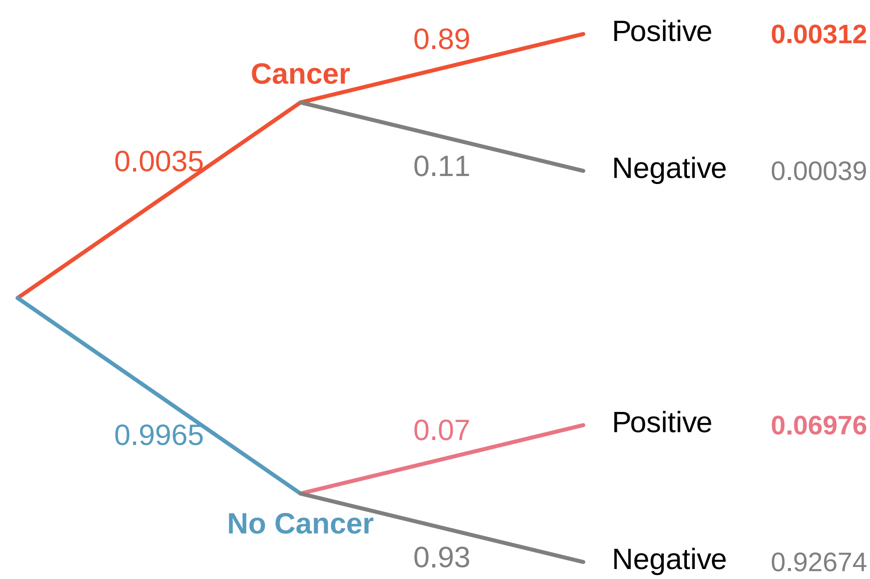


Figure 21.5: Tree diagram for breast cancer screening.

From the tree:

- $P(\text{cancer and positive}) = 0.0035 \times 0.89 = 0.00312$
- $P(\text{positive}) = 0.00312 + 0.06976 = 0.07288$
- $P(\text{cancer} \mid \text{positive}) = \frac{0.00312}{0.07288} \approx 0.043$

Even with a positive mammogram, there is only about a 4% chance the patient has breast cancer. This is why doctors often run additional tests.

Bayes' theorem. If $A_1, A_2, \dots, A_k$ represent all possible disjoint outcomes of a first variable, and B is an observed outcome of a second variable, then:

$$P(A_1 \mid B) = \frac{P(B \mid A_1)\, P(A_1)}{P(B \mid A_1)\, P(A_1) + P(B \mid A_2)\, P(A_2) + \cdots + P(B \mid A_k)\, P(A_k)}$$

The numerator is the probability of getting both A_1 and B. The denominator is the total (marginal) probability of B.

To apply Bayes' theorem:

1. Identify the marginal probabilities: $P(A_1), P(A_2), \dots, P(A_k)$.
2. Identify the conditional probabilities: $P(B \mid A_1), P(B \mid A_2), \dots, P(B \mid A_k)$.
3. Substitute into the formula.

Jose visits campus on Thursday evenings. Academic events occur 35% of evenings, sporting events 20%, and no events 45%. The parking garage fills up 25% of the time during academic events, 70% during sporting events, and 5% when there are no events. If Jose finds the garage full, what is the probability there is a sporting event?

Show answer

Using Bayes' theorem with A_1 = sporting, A_2 = academic, A_3 = none, and B = garage full: $P(\text{sporting} \mid \text{full}) = \frac{0.7\times0.2}{0.7\times0.2+0.25\times0.35+0.05\times0.45} = \frac{0.14}{0.25} = 0.56$. There is a 56% probability of a sporting event.

The strategy of updating beliefs based on new evidence is the foundation of **Bayesian statistics**, an important branch of statistics that we will not have time to explore further in this course.

21.9 Chapter review

21.9.1 Summary

In this chapter, we developed the mathematical rules of probability:

- **Probability** is the long-run proportion of times an outcome occurs, governed by the **Law of Large Numbers**.
- The **sample space** S is the set of all possible outcomes; an **event** is a subset of S.
- **Complement rule**: $P(A) = 1 - P(A^c)$.
- **Addition rule**: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. For disjoint events, $P(A \text{ and } B) = 0$.
- **Conditional probability**: $P(A \mid B) = P(A \text{ and } B)/P(B)$.
- **Multiplication rule**: $P(A \text{ and } B) = P(A \mid B) \times P(B)$. For independent events, $P(A \text{ and } B) = P(A) \times P(B)$.
- Events are **independent** if knowing one occurred does not change the probability of the other.
- **Bayes' theorem** inverts conditional probabilities.

21.10 Exercises

1. **True or false.** Determine if the statements below are true or false, and explain your reasoning.

a. If a fair coin is tossed many times and the last eight tosses are all heads, then the chance that the next toss will be heads is somewhat less than 50%.

b. Drawing a face card (jack, queen, or king) and drawing a red card from a full deck of playing cards are mutually exclusive events.

c. Drawing a face card and drawing an ace from a full deck of playing cards are mutually exclusive events.

2. **Roulette wheel.** The game of roulette involves spinning a wheel with 38 slots: 18 red, 18 black, and 2 green. A ball is spun onto the wheel and will eventually land in a slot, where each slot has an equal chance of capturing the ball.

a. You watch a roulette wheel spin 3 consecutive times and the ball lands on a red slot each time. What is the probability that the ball will land on a red slot on the next spin?

b. You watch a roulette wheel spin 300 consecutive times and the ball lands on a red slot each time. What is the probability that the ball will land on a red slot on the next spin?

c. Are you equally confident of your answers to parts (a) and (b)? Why or why not?

3. **Four games, one winner.** Below are four versions of the same game. Your archnemesis gets to pick the version of the game, and then you get to choose how many times to flip a coin: 10 times or 100 times. Identify how many coin flips you should choose for each version of the game. It costs $1 to play each game. Explain your reasoning.

a. If the proportion of heads is larger than 0.60, you win $1.

b. If the proportion of heads is larger than 0.40, you win \$1.

c. If the proportion of heads is between 0.40 and 0.60, you win \$1.

d. If the proportion of heads is smaller than 0.30, you win \$1.

4. **Backgammon.** Backgammon is a board game for two players in which the playing pieces are moved according to the roll of two dice. Players win by removing all of their pieces from the board, so it is usually good to roll high numbers. You are playing backgammon with a friend and you roll two 6s in your first roll and two 6s in your second roll. Your friend rolls two 3s in his first roll and again in his second row. Your friend claims that you are cheating, because rolling double 6s twice in a row is very unlikely. Using probability, show that your rolls were just as likely as his.

5. **Coin flips.** If you flip a fair coin 10 times, what is the probability of

a. getting all tails?

b. getting all heads?

c. getting at least one tails?

6. **Dice rolls.** If you roll a pair of fair dice, what is the probability of

a. getting a sum of 1?

b. getting a sum of 5?

c. getting a sum of 12?

7. **Swing voters.** A Pew Research survey asked 2,373 randomly sampled registered voters their political affiliation (Republican, Democrat, or Independent) and whether or not they identify as swing voters. 35% of respondents identified as Independent, 23% identified as swing voters, and 11% identified as both.

a. Are being Independent and being a swing voter disjoint, i.e. mutually exclusive?

b. Draw a Venn diagram summarizing the variables and their associated probabilities.

c. What percent of voters are Independent but not swing voters?

d. What percent of voters are Independent or swing voters?

e. What percent of voters are neither Independent nor swing voters?

f. Is the event that someone is a swing voter independent of the event that someone is a political Independent?

8. **Poverty and language.** The American Community Survey is an ongoing survey that provides data every year to give communities the current information they need to plan investments and services. The 2010 American Community Survey estimates that 14.6% of Americans live below the poverty line, 20.7% speak a language other than English (foreign language) at home, and 4.2% fall into both categories.

a. Are living below the poverty line and speaking a foreign language at home disjoint?

b. Draw a Venn diagram summarizing the variables and their associated probabilities.

c. What percent of Americans live below the poverty line and only speak English at home?

d. What percent of Americans live below the poverty line or speak a foreign language at home?

e. What percent of Americans live above the poverty line and only speak English at home?

f. Is the event that someone lives below the poverty line independent of the event that the person speaks a foreign language at home?

9. **Disjoint vs. independent.** In parts (a) and (b), identify whether the events are disjoint, independent, or neither (events cannot be both disjoint and independent).

a. You and a randomly selected student from your class both earn A's in this course.

b. You and your class study partner both earn A's in this course.

c. If two events can occur at the same time, must they be dependent?

10. **Guessing on an exam.** In a multiple choice exam, there are 5 questions and 4 choices for each question (a, b, c, d). Nancy has not studied for the exam at all and decides to randomly guess the answers. What is the probability that:

a. the first question she gets right is the 5^{th} question?

b. she gets all of the questions right?

c. she gets at least one question right?

11. **Educational attainment of couples.** The table below shows the distribution of education level attained by US residents by gender based on data collected in the 2010 American Community Survey.

c c }		*Gender*
	9th to 12th grade, no diploma	0.10
Highest	HS graduate (or equivalent)	0.30
education	Some college, no degree	0.22
attained	Associate's degree	0.06
	Bachelor's degree	0.16
	Graduate or professional degree	0.09

a. What is the probability that a randomly chosen man has at least a Bachelor's degree?

b. What is the probability that a randomly chosen woman has at least a Bachelor's degree?

c. What is the probability that a man and a woman getting married both have at least a Bachelor's degree? Note any assumptions you must make to answer this question.

d. If you made an assumption in part (c), do you think it was reasonable? If you didn't make an assumption, double check your earlier answer and then return to this part.

12. **School absences.** Data collected at elementary schools in DeKalb County, GA suggest that each year roughly 25% of students miss exactly one day of school, 15% miss 2 days, and 28% miss 3 or more days due to sickness.

a. What is the probability that a student chosen at random doesn't miss any days of school due to sickness this year?

b. What is the probability that a student chosen at random misses no more than one day?

c. What is the probability that a student chosen at random misses at least one day?

d. If a parent has two kids at a DeKalb County elementary school, what is the probability that neither kid will miss any school? Note any assumption you must make to answer this question.

e. If a parent has two kids at a DeKalb County elementary school, what is the probability that both kids will miss some school, i.e. at least one day? Note any assumption you make.

f. If you made an assumption in part (d) or (e), do you think it was reasonable? If you didn't make any assumptions, double check your earlier answers.

13. **Joint and conditional probabilities.** P(A) = 0.3, P(B) = 0.7

a. Can you compute P(A and B) if you only know P(A) and P(B)?

b. Assuming that events A and B arise from independent random processes,

c. what is P(A and B)?

d. what is P(A or B)?

e. what is P(A|B)?

f. If we are given that P(A and B) = 0.1, are the random variables giving rise to events A and B independent?

g. If we are given that P(A and B) = 0.1, what is P(A|B)?

14. **PB & J.** Suppose 80% of people like peanut butter, 89% like jelly, and 78% like both. Given that a randomly sampled person likes peanut butter, what's the probability that he also likes jelly?

15. **Global warming.** A Pew Research poll asked 1,306 Americans "From what you've read and heard, is there solid evidence that the average temperature on earth has been getting warmer over the past few decades, or not?". The table below shows the distribution of responses by party and ideology, where the counts have been replaced with relative frequencies.

		Response
		warming
Party and	Mod/Lib Republican	0.06
Ideology	Mod/Cons Democrat	0.25
	Liberal Democrat	0.18

a. Are believing that the earth is warming and being a liberal Democrat mutually exclusive?

b. What is the probability that a randomly chosen respondent believes the earth is warming or is a liberal Democrat?

c. What is the probability that a randomly chosen respondent believes the earth is warming given that he is a liberal Democrat?

d. What is the probability that a randomly chosen respondent believes the earth is warming given that he is a conservative Republican?

e. Does it appear that whether or not a respondent believes the earth is warming is independent of their party and ideology? Explain your reasoning.

f. What is the probability that a randomly chosen respondent is a moderate/liberal Republican given that he does not believe that the earth is warming?

16. **Health coverage, relative frequencies.** The Behavioral Risk Factor Surveillance System (BRFSS) is an annual telephone survey designed to identify risk factors in the adult population and report emerging health trends. The following table displays the distribution of health status of respondents to this survey (excellent, very good, good, fair, poor) and whether or not they have health insurance.

		Health Status	
Coverage	Yes	0.2099	0.3123

a. Are being in excellent health and having health coverage mutually exclusive?

b. What is the probability that a randomly chosen individual has excellent health?

c. What is the probability that a randomly chosen individual has excellent health given that he has health coverage?

d. What is the probability that a randomly chosen individual has excellent health given that he doesn't have health coverage?

e. Do having excellent health and having health coverage appear to be independent?

17. **Burger preferences.** A 2010 SurveyUSA poll asked 500 Los Angeles residents, "What is the best hamburger place in Southern California? Five Guys Burgers? In-N-Out Burger? Fat Burger? Tommy's Hamburgers? Umami Burger? Or somewhere else?" The distribution of responses by gender is shown below.

r r r }		*Gender*
	In-N-Out Burger	162
Best	Fat Burger	10
hamburger	Tommy's Hamburgers	27
place	Umami Burger	5
	Other	26
	Not Sure	13

a. Are being female and liking Five Guys Burgers mutually exclusive?

b. What is the probability that a randomly chosen male likes In-N-Out the best?

c. What is the probability that a randomly chosen female likes In-N-Out the best?

d. What is the probability that a man and a woman who are dating both like In-N-Out the best? Note any assumption you make and evaluate whether you think that assumption is reasonable.

e. What is the probability that a randomly chosen person likes Umami best or that person is female?

18. **Assortative mating.** Assortative mating is a nonrandom mating pattern where individuals with similar genotypes and/or phenotypes mate with one another more frequently than what would be expected under a random mating pattern. Researchers studying this topic collected data on eye colors of 204 Scandinavian men and their female partners. The table below summarizes the results.

		Partner (*female*)
Self (*male*)	Brown	19
	Green	11

a. What is the probability that a randomly chosen male respondent or his partner has blue eyes?

b. What is the probability that a randomly chosen male respondent with blue eyes has a partner with blue eyes?

c. What is the probability that a randomly chosen male respondent with brown eyes has a partner with blue eyes? What about the probability of a randomly chosen male respondent with green eyes having a partner with blue eyes?

d. Does it appear that the eye colors of male respondents and their partners are independent? Explain your reasoning.

19. **Drawing box plots.** After an introductory statistics course, 80% of students can successfully construct box plots. Of those who can construct box plots, 86% passed, while only 65% of those students who could not construct box plots passed.

a. Construct a tree diagram of this scenario.

b. Calculate the probability that a student is able to construct a box plot if it is known that he passed.

20. **Predisposition for thrombosis.** A genetic test is used to determine if people have a predisposition for *thrombosis*, which is the formation of a blood clot inside a blood vessel that obstructs the flow of blood through the circulatory system. It is believed that 3% of people actually have this predisposition. The genetic test is 99% accurate if a person actually has the predisposition, meaning that the probability of a positive test result when a person actually has the predisposition is 0.99. The test is 98% accurate if a person does not have the predisposition. What is the probability that a randomly selected person who tests positive for the predisposition by the test actually has the predisposition?

21. **It's never lupus.** Lupus is a medical phenomenon where antibodies that are supposed to attack foreign cells to prevent infections instead see plasma proteins as foreign bodies, leading to a high risk of blood clotting. It is believed that 2% of the population suffer from this disease. The test is 98% accurate if a person actually has the disease. The test is 74% accurate if a person does not have the disease. There is a line from the Fox television show *House* that is often used after a patient tests positive for lupus: "It's never lupus." Do you think there is truth to this statement? Use appropriate probabilities to support your answer.

22. **Exit poll.** Edison Research gathered exit poll results from several sources for the Wisconsin recall election of Scott Walker. They found that 53% of the respondents voted in favor of Scott Walker. Additionally, they estimated that of those who did vote in favor for Scott Walker, 37% had a college degree, while 44% of those who voted against Scott Walker had a college degree. Suppose we randomly sampled a person who participated in the exit poll and found that he had a college degree. What is the probability that he voted in favor of Scott Walker?

23. **Marbles in an urn.** Imagine you have an urn containing 5 red, 3 blue, and 2 orange marbles in it.

a. What is the probability that the first marble you draw is blue?

b. Suppose you drew a blue marble in the first draw. If drawing with replacement, what is the probability of drawing a blue marble in the second draw?

c. Suppose you instead drew an orange marble in the first draw. If drawing with replacement, what is the probability of drawing a blue marble in the second draw?

d. If drawing with replacement, what is the probability of drawing two blue marbles in a row?

e. When drawing with replacement, are the draws independent? Explain.

24. **Socks in a drawer.** In your sock drawer you have 4 blue, 5 gray, and 3 black socks. Half asleep one morning you grab 2 socks at random and put them on. Find the probability you end up wearing

a. 2 blue socks

b. no gray socks

c. at least 1 black sock

d. a green sock

e. matching socks

25. **Chips in a bag.** Imagine you have a bag containing 5 red, 3 blue, and 2 orange chips.

a. Suppose you draw a chip and it is blue. If drawing without replacement, what is the probability the next is also blue?

b. Suppose you draw a chip and it is orange, and then you draw a second chip without replacement. What is the probability this second chip is blue?

c. If drawing without replacement, what is the probability of drawing two blue chips in a row?

d. When drawing without replacement, are the draws independent? Explain.

26. **Books on a bookshelf.** The table below shows the distribution of books on a bookcase based on whether they are nonfiction or fiction and hardcover or paperback.

	Format
Nonfiction	15

a. Find the probability of drawing a hardcover book first then a paperback fiction book second when drawing without replacement.

b. Determine the probability of drawing a fiction book first and then a hardcover book second, when drawing without replacement.

c. Calculate the probability of the scenario in part (b), except this time complete the calculations under the scenario where the first book is placed back on the bookcase before randomly drawing the second book.

d. The final answers to parts (b) and (c) are very similar. Explain why this is the case.

27. **Student outfits.** In a classroom with 24 students, 7 students are wearing jeans, 4 are wearing shorts, 8 are wearing skirts, and the rest are wearing leggings. If we randomly select 3 students without replacement, what is the probability that one of the selected students is wearing leggings and the other two are wearing jeans? Note that these are mutually exclusive clothing options.

28. **The birthday problem.** Suppose we pick three people at random. For each of the following questions, ignore the special case where someone might be born on February 29th, and assume that births are evenly distributed throughout the year.

a. What is the probability that the first two people share a birthday?

b. What is the probability that at least two people share a birthday?

29. **Health coverage, frequencies.** The Behavioral Risk Factor Surveillance System (BRFSS) is an annual telephone survey designed to identify risk factors in the adult population and report emerging health trends. The following table summarizes two variables for the respondents: health status and health coverage, which describes whether each respondent had health insurance.

		Health Status	
Coverage	Yes	4,198	6,245

a. If we draw one individual at random, what is the probability that the respondent has excellent health and doesn't have health coverage?

b. If we draw one individual at random, what is the probability that the respondent has excellent health or doesn't have health coverage?

30. **HIV in Swaziland.** Swaziland has the highest HIV prevalence in the world: 25.9% of this country's population is infected with HIV. The ELISA test is one of the first and most accurate tests for HIV. For those who carry HIV, the ELISA test is 99.7% accurate. For those who do not carry HIV, the test is 92.6% accurate. If an individual from Swaziland has tested positive, what is the probability that he carries HIV?

31. **Twins.** About 30% of human twins are identical, and the rest are fraternal. Identical twins are necessarily the same sex – half are males and the other half are females. One-quarter of fraternal twins are both male, one-quarter both female, and one-half are mixes: one male, one female. You have just become a parent of twins and are told they are both girls. Given this information, what is the probability that they are identical?

Chapter 22

Random Variables

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In the previous chapter, we developed rules for computing probabilities of events. Now we shift our focus to **random variables** — numerical summaries of random processes. A random variable assigns a number to each outcome of a random process, which allows us to apply mathematical tools for prediction and analysis. We distinguish between discrete and continuous random variables, introduce probability distributions, and build the foundation for the specific distributions (binomial and normal) that follow in later chapters.

22.1 Random variables

It is often useful to model a random process by assigning a number to each outcome. This numerical model is called a **random variable**.

A **random variable** is a numerical outcome of a random process. We typically denote random variables with capital letters such as X, Y, or Z, and their observed values with the corresponding lowercase letters x, y, or z.

Two books are assigned for a statistics class: a textbook (\$137) and a study guide (\$33). The university bookstore has determined that 20% of enrolled students buy neither book, 55% buy the textbook only, and 25% buy both books. If X represents the amount a single student spends on books, then X is a random variable with possible values \$0, \$137, and \$170.

Random variables come in two types — discrete and continuous — and the distinction matters because they require different mathematical tools.

22.2 Discrete random variables

A **discrete random variable** takes on a finite or countably infinite number of distinct values. These values can be listed (even if the list is infinitely long).

Examples of discrete random variables:

- The number showing when you roll a die: $X \in \{1, 2, 3, 4, 5, 6\}$

- The number of heads in 10 coin flips: $X \in \{0, 1, 2, \dots, 10\}$
- The number of customers arriving at a store in an hour: $X \in \{0, 1, 2, 3, \dots\}$

22.2.1 Why the distinction matters

In Parts II and III, we worked extensively with sample statistics like the sample mean $\bar{x}$ and the sample proportion $\hat{p}$. These are actually *observed values* of random variables. The sampling distribution of $\bar{x}$ that we studied in Chapter 9 is really the probability distribution of a random variable. Formalizing the language of random variables helps us state results more precisely and connect our earlier simulation-based reasoning to mathematical theory.

22.2.2 Probability mass functions

The complete description of a discrete random variable is given by its **probability mass function** (PMF), which specifies the probability of each possible value.

The **probability mass function** (PMF) of a discrete random variable X is the function $P(X = x)$ that gives the probability that X takes on each possible value x. A valid PMF must satisfy:

1. $P(X = x) \geq 0$ for every value x.
2. The sum of all probabilities equals 1: $\sum_{\text{all } x} P(X = x) = 1$.

The bookstore example from the previous section has the following PMF:

x_i	\$0	\$137	\$170
$P(X = x_i)$	0.20	0.55	0.25

We can verify this is a valid PMF: all probabilities are between 0 and 1, and $0.20 + 0.55 + 0.25 = 1.00$.

Suppose a campus shuttle arrives every 10 minutes. The number of students who board the shuttle at a particular stop can be modeled with the following distribution:

Number of students (x)	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10

(a) Verify this is a valid probability distribution. (b) What is the probability that 2 or fewer students board? (c) What is the probability that more than 2 students board?

Show answer

(a) All probabilities are between 0 and 1, and $0.10 + 0.25 + 0.35 + 0.20 + 0.10 = 1.00$. (b) $P(X \leq 2) = 0.10 + 0.25 + 0.35 = 0.70$. (c) $P(X > 2) = 1 - P(X \leq 2) = 1 - 0.70 = 0.30$, or equivalently $0.20 + 0.10 = 0.30$.

22.2.3 Visualizing discrete distributions

Discrete probability distributions are often displayed as bar plots where the height of each bar represents the probability of that outcome. When the outcomes are numerical, we place the bars at their numerical locations, creating a plot that resembles a histogram.

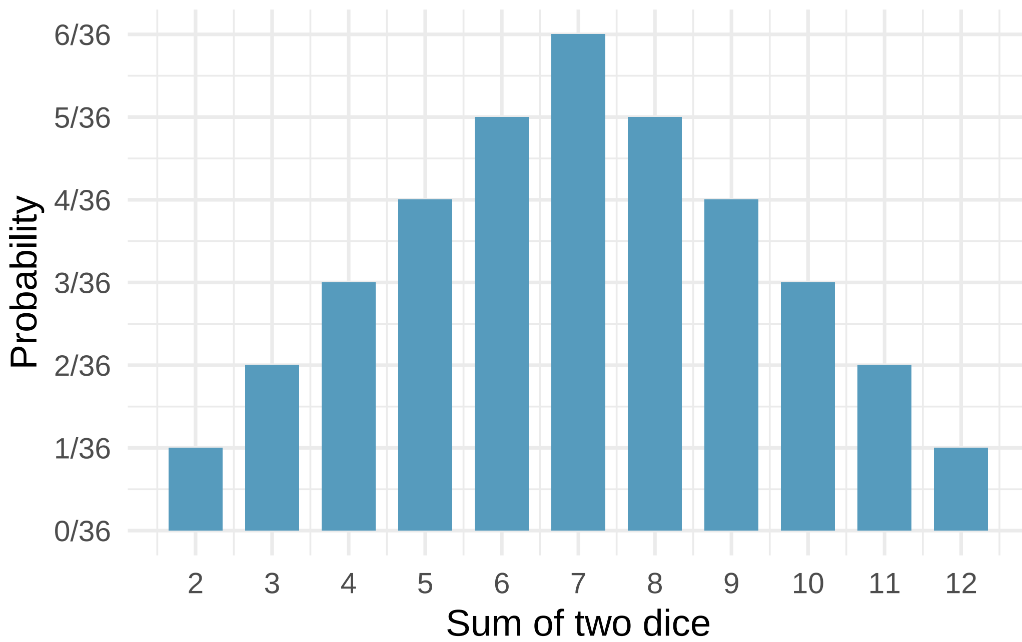


Figure 22.1: Probability distribution for the sum of two dice displayed as a bar plot.

22.2.4 Building intuition with simulation

We can also *estimate* a probability distribution by simulating the random process many times and recording the relative frequencies of each outcome. For example, if we simulate rolling two dice 10,000 times and record the sum each time, the histogram of those 10,000 sums will closely approximate the theoretical probability distribution. As the number of simulations increases, the approximation improves — this is the Law of Large Numbers in action.

This connection between simulation and theory is one of the most important ideas in the course. Every time we used simulation in Parts II and III to build a null distribution or a bootstrap distribution, we were approximating a theoretical probability distribution.

22.3 Cumulative distribution functions

The **cumulative distribution function** (CDF) of a random variable X is the function:

$$F(x) = P(X \leq x)$$

The CDF gives the probability that X takes a value less than or equal to x.

The CDF is a useful alternative to the PMF for answering "at most" or "at least" types of questions. For a discrete random variable, the CDF is a step function that jumps at each possible value of X.

Consider the sum of two dice. The CDF at a few key values:

x	2	3	4	5	6	7	8	9	10	11	12
$F(x) = P(X \leq x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$

Key properties of any CDF:

- $F(x)$ is non-decreasing: if $a < b$, then $F(a) \leq F(b)$.
- $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$.
- $P(a < X \leq b) = F(b) - F(a)$.

The CDF becomes especially important for continuous random variables, where individual values have zero probability and we must always work with intervals.

Using the CDF for the sum of two dice above, find: (a) $P(X \leq 5)$. (b) $P(X > 9)$. (c) $P(4 < X \leq 8)$.

Show answer

(a) $F(5) = 10/36$. (b) $P(X > 9) = 1 - F(9) = 1 - 30/36 = 6/36 = 1/6$. (c) $P(4 < X \leq 8) = F(8) - F(4) = 26/36 - 6/36 = 20/36 = 5/9$.

22.4 Continuous random variables

A **continuous random variable** can take any value in some interval of real numbers. Its probability distribution is described by a smooth curve called a **probability density function** (PDF).

Examples of continuous random variables:

- The height of a randomly selected adult
- The time until the next bus arrives
- The temperature at noon tomorrow

22.4.1 From histograms to density curves

When we have a very large sample and use very narrow bins in a histogram, the outline of the histogram starts to look like a smooth curve. This smooth curve is a probability density function.

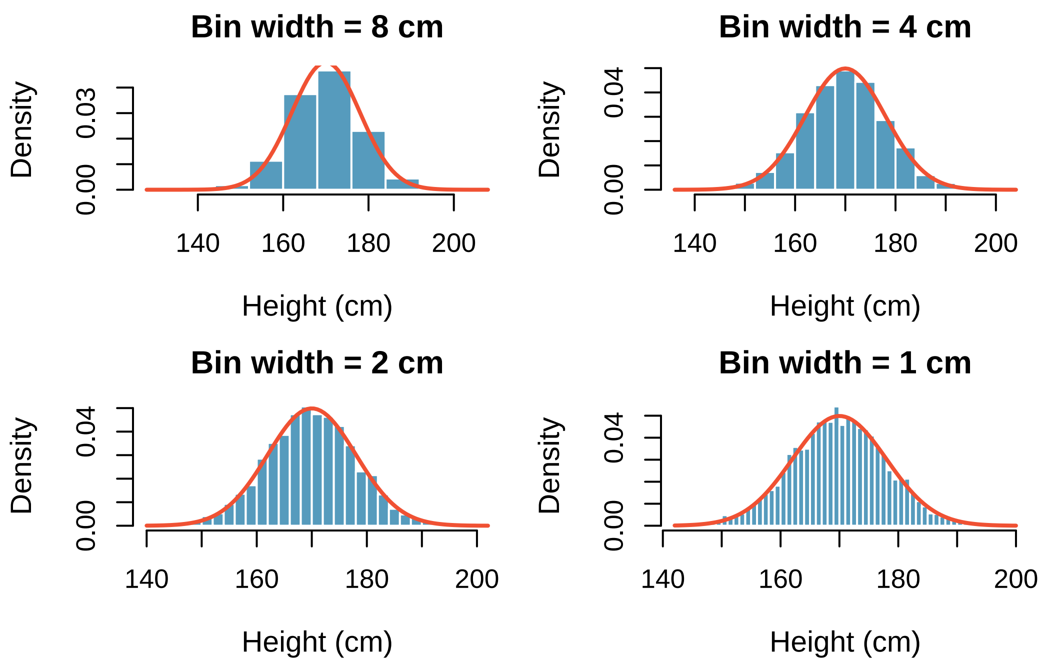


Figure 22.2: The transition from a histogram to a smooth density curve as the bin width decreases and sample size increases.

Properties of a probability density function (PDF):

1. The density curve is always on or above the horizontal axis: $f(x) \geq 0$ for all x.
2. The total area under the curve equals 1.
3. The probability that X falls between a and b equals the area under the curve between a and b: $P(a \leq X \leq b) =$ area between a and b.

22.4.2 Probabilities from continuous distributions

For a continuous random variable, probabilities are *areas* under the density curve.

The heights of US adults follow a continuous distribution. What proportion of the population has heights between 180 cm and 185 cm?

Using the density curve, we find the area under the curve between 180 and 185:

$$P(180 \leq X \leq 185) = \text{area between 180 and 185} \approx 0.1157$$

About 11.6% of US adults are between 180 and 185 cm tall.

A key difference: the probability of an exact value is zero for continuous random variables.

For a continuous random variable, $P(X = c) = 0$ for any specific value c. There is no area under the curve at a single point. This means:

$$P(X \leq a) = P(X < a)$$

This is *not* true for discrete random variables, where individual outcomes carry positive probability.

If $P(180 \leq \text{height} \leq 185) = 0.1157$ for a randomly selected US adult, and three adults are selected at random: (a) What is the probability all three have heights between 180 and 185 cm? (b) What is the probability none of them do?

Show answer

(a) Since the selections are independent: $0.1157^3 \approx 0.0015$. (b) $(1-0.1157)^3 = 0.8843^3 \approx 0.692$.

22.4.3 Discrete vs. continuous: a summary

Feature	Discrete	Continuous
Values	Finite or countable list	Any value in an interval
Described by	Probability mass function (PMF)	Probability density function (PDF)
$P(X = c)$	Can be positive	Always equals 0
Probabilities	Sum probabilities of individual values	Find areas under the curve
Visualization	Bar plot	Smooth curve
Example	Number of coin flips until heads	Height of a randomly selected person

In this course, we will work primarily with two specific distributions:

- The **binomial distribution** (Chapter 24) — a discrete distribution
- The **normal distribution** (Chapter 25) — a continuous distribution

Understanding the general concepts from this chapter will make those specific distributions much easier to work with.

22.5 Chapter review

22.5.1 Summary

- A **random variable** assigns a numerical value to each outcome of a random process.
- **Discrete random variables** take on a finite or countable set of values. Their distributions are described by probability mass functions (PMFs).
- **Continuous random variables** can take any value in an interval. Their distributions are described by probability density functions (PDFs), and probabilities are calculated as areas under the curve.
- The **cumulative distribution function** (CDF) gives $P(X \leq x)$ and works for both discrete and continuous random variables.
- For continuous random variables, $P(X = c) = 0$ for any specific value c.
- A valid probability distribution has non-negative probabilities that sum (or integrate) to 1.

22.6 Exercises

1. **College smokers.** At a university, 13% of students smoke.

a. Calculate the expected number of smokers in a random sample of 100 students from this university.

b. The university gym opens at 9 am on Saturday mornings. One Saturday morning at 8:55 am there are 27 students outside the gym waiting for it to open. Should you use the same approach from part (a) to calculate the expected number of smokers among these 27 students?

2. **Ace of clubs wins.** Consider the following card game with a well-shuffled deck of cards. If you draw a red card, you win nothing. If you get a spade, you win \$5. For any club, you win \$10 plus an extra \$20 for the ace of clubs.

a. Create a probability model for the amount you win at this game. Also, find the expected winnings for a single game and the standard deviation of the winnings.

b. What is the maximum amount you would be willing to pay to play this game? Explain your reasoning.

3. **Hearts win.** In a new card game, you start with a well-shuffled full deck and draw 3 cards without replacement. If you draw 3 hearts, you win \$50. If you draw 3 black cards, you win \$25. For any other draws, you win nothing.

a. Create a probability model for the amount you win at this game, and find the expected winnings. Also compute the standard deviation of this distribution.

b. If the game costs \$5 to play, what would be the expected value and standard deviation of the net profit (or loss)? (*Hint: profit = winnings − cost;* $X - 5$)

c. If the game costs \$5 to play, should you play this game? Explain.

4. **Is it worth it.** Andy is always looking for ways to make money fast. Lately, he has been trying to make money by gambling. Here is the game he is considering playing: The game costs \$2 to play. He draws a card from a deck. If he gets a number card (2-10), he wins nothing. For any face card (jack, queen or king), he wins \$3. For any ace, he wins \$5, and he wins an *extra* \$20 if he draws the ace of clubs.

a. Create a probability model and find Andy's expected profit per game.

b. Would you recommend this game to Andy as a good way to make money? Explain.

5. **Cat weights.** The histogram shown below represents the weights (in kg) of 47 female and 97 male cats.

a. What fraction of these cats weigh less than 2.5 kg?

b. What fraction of these cats weigh between 2.5 and 2.75 kg?

c. What fraction of these cats weigh between 2.75 and 3.5 kg?

6. **Income and gender.** The relative frequency table below displays the distribution of annual total personal income (in 2009 inflation-adjusted dollars) for a representative sample of 96,420,486 Americans. These data come from the American Community Survey for 2005-2009. This sample is comprised of 59% males and 41% females.

a. Describe the distribution of total personal income.

b. What is the probability that a randomly chosen US resident makes less than \$50,000 per year?

c. What is the probability that a randomly chosen US resident makes less than \$50,000 per year and is female? Note any assumptions you make.

d. The same data source indicates that 71.8% of females make less than \$50,000 per year. Use this value to determine whether or not the assumption you made in part (c) is valid.

{

\$10,000 to \$14,999	4.7%
\$15,000 to \$24,999	15.8%
\$25,000 to \$34,999	18.3%
\$35,000 to \$49,999	21.2%
\$50,000 to \$64,999	13.9%
\$65,000 to \$74,999	5.8%
\$75,000 to \$99,999	8.4%
\$100,000 or more	9.7%

}

7. **Grade distributions.** Each row in the table below is a proposed grade distribution for a class. Identify each as a valid or invalid probability distribution, and explain your reasoning.

	Grades
(b)	0
(c)	0.3
(d)	0.3
(e)	0.2
(f)	0

8. **Roulette winnings.** In the game of roulette, a wheel is spun and you place bets on where it will stop. One popular bet is that it will stop on a red slot; such a bet has an 18/38 chance of winning. If it stops on red, you double the money you bet. If not, you lose the money you bet. Suppose you play 3 times, each time with a \$1 bet. Let Y represent the total amount won or lost. Write a probability model for Y.

Chapter 23

Expected Value and Variance

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

How much revenue should a bookstore expect per student? How much variability should a stock investor anticipate in their portfolio? These questions require us to summarize a random variable's distribution with just two numbers: its **expected value** (center) and **variance** (spread). In this chapter, we define these quantities, develop useful computational properties, and extend them to linear combinations of random variables. These tools will be essential when we study the binomial and normal distributions in the chapters that follow.

23.1 Expected value of a random variable

We introduced random variables in Chapter 22. Now we develop the tools to summarize them numerically, starting with the expected value — the long-run average of a random variable.

Two books are assigned for a statistics class: a textbook (\$137) and a study guide (\$33). From past experience, the bookstore knows that 20% of enrolled students buy neither book, 55% buy the textbook only, and 25% buy both books. If there are 100 students enrolled, how many books should the bookstore expect to sell?

About 20 students buy 0 books, 55 buy 1 book (55 books total), and 25 buy 2 books (50 books total). The bookstore should expect to sell about $0 + 55 + 50 = 105$ books.

What is the average revenue per student?

The expected total revenue is $\$0 \times 20 + \$137 \times 55 + \$170 \times 25 = \$11{,}785$, and there are 100 students. The expected revenue per student is $\$11{,}785/100 = \$117.85$.

Notice that the "average revenue per student" was computed by multiplying each outcome by its probability and summing:

$$0 \times 0.20 + 137 \times 0.55 + 170 \times 0.25 = 117.85$$

This is the **expected value** of the random variable X.

Expected value of a discrete random variable. If X takes outcomes $x_1, x_2, \dots, x_k$ with probabilities $P(X = x_1), P(X = x_2), \dots, P(X = x_k)$, the **expected value** of X is:

$$E(X) = x_1 \cdot P(X = x_1) + x_2 \cdot P(X = x_2) + \cdots + x_k \cdot P(X = x_k) = \sum_{i=1}^{k} x_i \, P(X = x_i)$$

The Greek letter μ may be used in place of $E(X)$.

The expected value represents the **long-run average** of the random variable. If we were to observe the random process many, many times, the average of all observed values would converge to $E(X)$.

23.1.1 The expected value as a center of gravity

In physics, the expected value corresponds to the **center of gravity** (or balance point) of a distribution. Imagine cutting out the shape of a probability distribution from a piece of cardboard. The expected value is the point where you could balance the cutout on the tip of your finger. This is why we sometimes call $E(X)$ the "mean" — it is exactly analogous to the sample mean $\bar{x}$, but for a probability distribution rather than a data set.

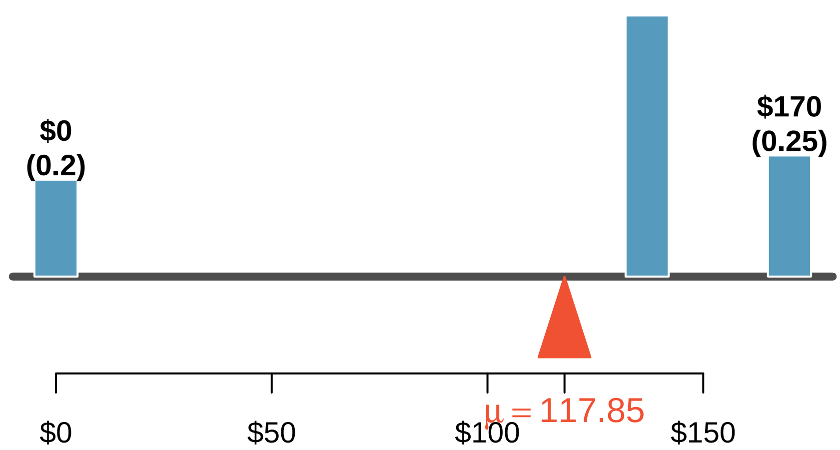


Figure 23.1: A weight system representing the bookstore distribution balanced at the mean of $117.85.

A campus shuttle arrives every 10 minutes. The number of students boarding follows this distribution:

x	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10

What is the expected number of students who board?

Show answer

$E(X) = 0(0.10) + 1(0.25) + 2(0.35) + 3(0.20) + 4(0.10) = 0 + 0.25 + 0.70 + 0.60 + 0.40 = 1.95$. On average, about 1.95 students board the shuttle at this stop.

The expected value does not have to be a value that the random variable can actually take. In the shuttle example, $E(X) = 1.95$, but you can never have 1.95 students board. The expected value is a long-run average, not a prediction of any single outcome.

23.2 Variance of a random variable

The expected value tells us where the center of a distribution lies, but it says nothing about the *spread*. The bookstore might want to know not just the average revenue per student, but how much that revenue varies from student to student.

Variance of a discrete random variable. If X takes outcomes $x_1, \dots, x_k$ with probabilities $P(X = x_1), \dots, P(X = x_k)$ and expected value $\mu = E(X)$, then the **variance** of X is:

$$\text{Var}(X) = \sigma^2 = \sum_{i=1}^{k} (x_i - \mu)^2 \, P(X = x_i)$$

The **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.

The variance is the average squared deviation from the mean, weighted by the probabilities. The standard deviation has the same units as X, which makes it more interpretable.

Compute the variance and standard deviation of the bookstore revenue per student.

We organize the calculation in a table with $\mu = 117.85$:

	$x_1 = 0$	$x_2 = 137$	$x_3 = 170$	Total
$P(X = x_i)$	0.20	0.55	0.25	
$x_i - \mu$	-117.85	19.15	52.15	
$(x_i - \mu)^2$	13,888.6	366.7	2,719.6	
$(x_i - \mu)^2 \cdot P(X = x_i)$	2,777.7	201.7	679.9	3,659.3

The variance is $\sigma^2 = 3{,}659.3$ and the standard deviation is $\sigma = \sqrt{3{,}659.3} = \$60.49$.

The bookstore also sells a chemistry textbook ($159) and supplement ($41). From past experience, 15% of chemistry students buy neither, 25% buy the textbook only, and 60% buy both. Let Y be the revenue from a single chemistry student. (a) Write out the probability distribution of Y. (b) Compute $E(Y)$. (c) Compute SD(Y).

Show answer

(a) Y takes values $0, $159, $200 with probabilities 0.15, 0.25, 0.60. (b) $E(Y) = 0(0.15) + 159(0.25) + 200(0.60) = 0 + 39.75 + 120 = \$159.75$. (c) $\text{Var}(Y) = (0 - 159.75)^2(0.15) + (159 - 159.75)^2(0.25) + (200 - 159.75)^2(0.60) = 3828.0 + 0.1 + 972.0 = 4800.1$, so SD$(Y) = \sqrt{4800.1} \approx \$69.28$.

23.3 Properties of expected value

The expected value has several important algebraic properties that make it easy to work with. These properties follow from the definition but are worth stating explicitly because we will use them frequently.

Properties of expected value. For a random variable X and constants a and b:

1. **Adding a constant shifts the mean**: $E(X + b) = E(X) + b$
2. **Multiplying by a constant scales the mean**: $E(aX) = a \cdot E(X)$
3. **General linear transformation**: $E(aX + b) = a \cdot E(X) + b$

Why these make sense: If every student's textbook cost increases by $10 (adding a constant $b = 10$), then the average cost also increases by $10. If the bookstore marks up all prices by 20% (multiplying by $a = 1.20$), then the average revenue also increases by 20%.

Suppose the bookstore offers a 10% discount to students who buy both books. The discounted price is $0.90 \times X$, where X is the original price. What is the expected revenue per student under the discount?

Using the property $E(aX) = aE(X)$:

$$E(0.90X) = 0.90 \times E(X) = 0.90 \times 117.85 = \$106.07$$

The expected revenue per student drops from \$117.85 to \$106.07.

The bookstore charges a flat \$5 processing fee in addition to book costs. If X is the cost of books and the total charge is $Y = X + 5$, what is $E(Y)$?

Show answer

$E(Y) = E(X + 5) = E(X) + 5 = 117.85 + 5 = \$122.85$.

23.4 Properties of variance

Variance has different algebraic properties than expected value. The most important distinction: adding a constant does not change the variance.

Properties of variance. For a random variable X and constants a and b:

1. **Adding a constant does not change variance**: $\text{Var}(X + b) = \text{Var}(X)$
2. **Multiplying by a constant scales variance by the square**: $\text{Var}(aX) = a^2 \cdot \text{Var}(X)$
3. **General linear transformation**: $\text{Var}(aX + b) = a^2 \cdot \text{Var}(X)$

For standard deviations:

$$\text{SD}(aX + b) = |a| \cdot \text{SD}(X)$$

Why adding a constant does not change variance: If every student pays \$5 more, the *spread* of payments does not change — everyone's payment shifts by the same amount. The center moves, but the variability stays the same.

Why multiplying scales by the square: If every price is doubled ($a = 2$), then deviations from the mean are also doubled, so *squared* deviations are multiplied by $2^2 = 4$.

For the bookstore revenue with $\text{Var}(X) = 3{,}659.3$ and $\sigma_X = \$60.49$:

- If a \$5 fee is added: $\text{Var}(X + 5) = \text{Var}(X) = 3{,}659.3$ and $\text{SD}(X + 5) = \$60.49$.
- If a 10% discount is applied: $\text{Var}(0.90X) = 0.90^2 \times 3{,}659.3 = 0.81 \times 3{,}659.3 = 2{,}964.0$ and $\text{SD}(0.90X) = 0.90 \times 60.49 = \$54.44$.

Suppose temperatures in Celsius are converted to Fahrenheit using $F = 1.8C + 32$. If the standard deviation of daily high temperatures is $\sigma_C = 5^\circ$C, what is the standard deviation in Fahrenheit?

Show answer

$\text{SD}(1.8C + 32) = 1.8 \times \text{SD}(C) = 1.8 \times 5 = 9^\circ$F. The additive constant 32 does not affect the spread.

23.5 Linear combinations of random variables

Sometimes the total outcome is best described as a sum of several random variables. For instance, a person's total weekly commute time is the sum of five daily commute times.

A **linear combination** of random variables X and Y is an expression of the form $aX + bY$, where a and b are fixed constants.

John travels to work five days a week. Let X_i represent his travel time on day i. His total weekly commute time is:

$$W = X_1 + X_2 + X_3 + X_4 + X_5$$

If the average daily commute is 18 minutes, the expected weekly total is:

$$E(W) = E(X_1) + E(X_2) + \cdots + E(X_5) = 18 + 18 + 18 + 18 + 18 = 90 \text{ minutes}$$

23.5.1 Expected value of a linear combination

Expected value of a linear combination. For random variables X and Y with constants a and b:

$$E(aX + bY) = a \cdot E(X) + b \cdot E(Y)$$

This generalizes to any number of random variables:

$$E(a_1X_1 + a_2X_2 + \cdots + a_kX_k) = a_1E(X_1) + a_2E(X_2) + \cdots + a_kE(X_k)$$

This property holds whether or not the random variables are independent.

Elena is selling a TV at auction (expected profit X: \$175) and buying a toaster oven (expected cost Y: \$23). Her net gain is $X - Y$. What is her expected net gain?

Show answer

$E(X - Y) = E(X) - E(Y) = 175 - 23 = \$152$.

Leonard has invested \$6,000 in Caterpillar stock (CAT) and \$2,000 in Exxon Mobil stock (XOM). If X represents CAT's monthly percentage return (in decimal form) and Y represents XOM's return, his portfolio change is $6000X + 2000Y$. If $E(X) = 0.020$ and $E(Y) = 0.002$:

$$E(6000X + 2000Y) = 6000 \times 0.020 + 2000 \times 0.002 = \$124$$

Leonard expects to gain \$124 per month on average.

23.5.2 Variance of a linear combination

Variance of a linear combination (independent case). If X and Y are **independent** random variables, then:

$$\text{Var}(aX + bY) = a^2 \cdot \text{Var}(X) + b^2 \cdot \text{Var}(Y)$$

The standard deviation is: $\text{SD}(aX + bY) = \sqrt{a^2\text{Var}(X) + b^2\text{Var}(Y)}$.

Independence is required for the variance formula. The formula $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$ is only valid when X and Y are independent. If the variables are not independent, an additional covariance term is needed.

However, the expected value formula $E(aX+bY) = aE(X)+bE(Y)$ always holds, regardless of independence.

John's daily commute has a standard deviation of 4 minutes, and the commute times for different days are independent. What is the standard deviation of his weekly commute time $W = X_1 + X_2 + X_3 + X_4 + X_5$?

Each coefficient is 1, and $\text{Var}(X_i) = 4^2 = 16$:

$$\text{Var}(W) = 1^2(16) + 1^2(16) + 1^2(16) + 1^2(16) + 1^2(16) = 5 \times 16 = 80$$

$$\text{SD}(W) = \sqrt{80} \approx 8.94 \text{ minutes}$$

Notice that the standard deviation of the sum (8.94 minutes) is *not* five times the daily standard deviation (which would be 20 minutes). Variabilities do not add up as simply as means do — we must add *variances*, not standard deviations.

Leonard's portfolio has $\text{Var}(X) = 0.0057$ and $\text{Var}(Y) = 0.0021$ (where the stocks are approximately independent). What is the standard deviation of his monthly return?

$$\text{Var}(6000X + 2000Y) = 6000^2 \times 0.0057 + 2000^2 \times 0.0021 = 205{,}200 + 8{,}400 = 213{,}600$$

$$\text{SD} = \sqrt{213{,}600} \approx \$462$$

While an average monthly return of \$124 sounds nice, a standard deviation of \$462 means the returns are highly volatile.

Elena's TV sale has $\text{SD}(X) = \$25$ and her toaster oven purchase has $\text{SD}(Y) = \$8$. Assuming the auctions are independent, compute the standard deviation of her net gain $X - Y$.

Show answer

Note that $X - Y = 1 \cdot X + (-1) \cdot Y$. So $\text{Var}(X - Y) = 1^2 \times 25^2 + (-1)^2 \times 8^2 = 625 + 64 = 689$. Thus $\text{SD}(X-Y) = \sqrt{689} \approx \$26.25$. The negative coefficient is eliminated when we square it — subtracting a random variable introduces *more* variability, not less.

23.5.3 Why variances add even when we subtract

A common source of confusion: the variance of a difference is the *sum* of the variances (not the difference). This makes sense intuitively. If you buy a bag of apples and a bag of oranges, the uncertainty in the total weight is a combination of both uncertainties — regardless of whether you are adding or subtracting the weights. Two sources of randomness always create more total uncertainty.

23.6 Chapter review

23.6.1 Summary

- The **expected value** $E(X) = \sum x_i P(X = x_i)$ is the long-run average of a random variable.
- The **variance** $\text{Var}(X) = \sum(x_i - \mu)^2 P(X = x_i)$ measures spread; the **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.
- **Expected value properties**: $E(aX + b) = aE(X) + b$ (always holds).
- **Variance properties**: $\text{Var}(aX + b) = a^2\text{Var}(X)$ (adding a constant does not change variance).
- **Linear combinations**: $E(aX+bY) = aE(X)+bE(Y)$ (always). $\text{Var}(aX+bY) = a^2\text{Var}(X)+b^2\text{Var}(Y)$ (requires independence).
- Variances always add, even when random variables are subtracted.

23.7 Exercises

1. **Portfolio return.** A portfolio's value increases by 18% during a financial boom and by 9% during normal times. It decreases by 12% during a recession. What is the expected return on this portfolio if each scenario is equally likely?

2. **Baggage fees.** An airline charges the following baggage fees: \$25 for the first bag and \$35 for the second. Suppose 54% of passengers have no checked luggage, 34% have one piece of checked luggage and 12% have two pieces. We suppose a negligible portion of people check more than two bags.

a. Build a probability model, compute the average revenue per passenger, and compute the corresponding standard deviation.

b. About how much revenue should the airline expect for a flight of 120 passengers? With what standard deviation? Note any assumptions you make and if you think they are justified.

3. **American roulette.** The game of American roulette involves spinning a wheel with 38 slots: 18 red, 18 black, and 2 green. A ball is spun onto the wheel and will eventually land in a slot, where each slot has an equal chance of capturing the ball. Gamblers can place bets on red or black. If the ball lands on their color, they double their money. If it lands on another color, they lose their money. Suppose you bet $1 on red. What's the expected value and standard deviation of your winnings?

4. **European roulette.** The game of European roulette involves spinning a wheel with 37 slots: 18 red, 18 black, and 1 green. A ball is spun onto the wheel and will eventually land in a slot, where each slot has an equal chance of capturing the ball. Gamblers can place bets on red or black. If the ball lands on their color, they double their money. If it lands on another color, they lose their money.

a. Suppose you play roulette and bet $3 on a single round. What is the expected value and standard deviation of your total winnings?

b. Suppose you bet $1 in three different rounds. What is the expected value and standard deviation of your total winnings?

c. How do your answers to parts (a) and (b) compare? What does this say about the riskiness of the two games?

5. **Cost of breakfast.** Sally gets a cup of coffee and a muffin every day for breakfast from one of the many coffee shops in her neighborhood. She picks a coffee shop each morning at random and independently of previous days. The average price of a cup of coffee is $1.40 with a standard deviation of 30¢ ($0.30), the average price of a muffin is $2.50 with a standard deviation of 15¢, and the two prices are independent of each other.

a. What is the mean and standard deviation of the amount she spends on breakfast daily?

b. What is the mean and standard deviation of the amount she spends on breakfast weekly (7 days)?

6. **Scooping ice cream.** Ice cream usually comes in 1.5 quart boxes (48 fluid ounces), and ice cream scoops hold about 2 ounces. However, there is some variability in the amount of ice cream in a box as well as the amount of ice cream scooped out. We represent the amount of ice cream in the box as X and the amount scooped out as Y. Suppose these random variables have the following means, standard deviations, and variances:

Y | 2 | 0.25 | 0.0625 |

a. An entire box of ice cream, plus 3 scoops from a second box is served at a party. How much ice cream do you expect to have been served at this party? What is the standard deviation of the amount of ice cream served?

b. How much ice cream would you expect to be left in the box after scooping out one scoop of ice cream? That is, find the expected value of $X - Y$. What is the standard deviation of the amount left in the box?

c. Using the context of this exercise, explain why we add variances when we subtract one random variable from another.

7. **Variance of a mean, Part I.** Suppose we have independent observations X_1 and X_2 from a distribution with mean μ and standard deviation σ. What is the variance of the mean of the two values: $\frac{X_1+X_2}{2}$?

8. **Variance of a mean, Part II.** Suppose we have 3 independent observations X_1, X_2, X_3 from a distribution with mean μ and standard deviation σ. What is the variance of the mean of these 3 values: $\frac{X_1+X_2+X_3}{3}$?

9. **Variance of a mean, Part III.** Suppose we have n independent observations X_1, X_2, ..., X_n from a distribution with mean μ and standard deviation σ. What is the variance of the mean of these n values: $\frac{X_1+X_2+\cdots+X_n}{n}$?

Chapter 24

Binomial Distribution

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

How many of 400 randomly sampled adults are left-handed? How many of 20 randomly selected parts from an assembly line are defective? These "count of successes" questions arise constantly in statistics and are answered by the **binomial distribution**. In this chapter, we identify the four conditions that define a binomial setting, develop the formula for computing binomial probabilities, derive the mean and standard deviation, and show how the normal distribution can approximate binomial probabilities when the sample size is large.

StatLens tools for this chapter: Use the **Binomial Distribution Calculator** to compute binomial probabilities, visualize the distribution, and explore how changing n and p affects the shape.

Launch Binomial Distribution Calculator

24.1 The binomial setting

The binomial distribution describes the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes.

The four binomial conditions. A random variable X follows a binomial distribution if:

1. **Fixed number of trials**: The number of trials n is fixed in advance.
2. **Binary outcomes**: Each trial results in one of two outcomes, which we label "success" and "failure."
3. **Independence**: The trials are independent — the outcome of one trial does not affect the outcome of any other trial.
4. **Constant probability**: The probability of success p is the same for every trial.

If all four conditions are met, we say X follows a binomial distribution with parameters n and p, written $X \sim \text{Binomial}(n, p)$.

A health insurance company found that 70% of the people they insure stay below their deductible in any given year. Consider a random sample of four individuals. Does the number who stay below their deductible follow a binomial distribution?

1. **Fixed** n: Yes, $n = 4$ individuals.
2. **Binary**: Yes, each person either stays below the deductible (success) or exceeds it (failure).
3. **Independence**: Yes, because the sample is random from a large population.
4. **Constant** p: Yes, each person has a 70% chance of staying below the deductible.

All four conditions are met, so $X \sim \text{Binomial}(n = 4, p = 0.70)$.

The labels "success" and "failure" are just conventions — a "success" does not have to be a good outcome. If we are counting defective parts, a "success" might be finding a defective part. The mathematical framework does not depend on which outcome we call a success, as long as we are consistent.

The probability that a random smoker will develop a severe lung condition is about 0.3. If you have 4 friends who smoke, are the conditions for the binomial model satisfied?

Show answer

This is questionable. If the friends know each other, the independence condition may be violated. Acquaintances may have similar smoking habits or might influence each other. If, however, we could treat them as a random sample of smokers, the conditions would be approximately met.

24.1.1 The Bernoulli distribution: a single trial

A single trial with two outcomes is called a **Bernoulli trial**, and the resulting random variable is a **Bernoulli random variable**.

If X is a random variable that takes value 1 with probability p (success) and 0 with probability $1 - p$ (failure), then X is a **Bernoulli random variable** with:

$$\mu = p \qquad \sigma = \sqrt{p(1-p)}$$

A binomial random variable is simply the sum of n independent Bernoulli random variables, each with the same probability p.

24.2 Computing binomial probabilities

24.2.1 Building the formula

Let's find the probability that exactly 1 of 4 randomly selected insured individuals exceeds the deductible (so 3 "successes" — staying below — and 1 "failure").

Consider the specific scenario where person A exceeds and persons B, C, D do not:

$$P(A = \text{exceed}, B = \text{not}, C = \text{not}, D = \text{not}) = (0.3)(0.7)(0.7)(0.7) = (0.7)^3(0.3)^1 = 0.103$$

But there are four such scenarios — any one of the four people could be the one who exceeds. Each scenario has the same probability. So:

$$P(\text{exactly 3 successes in 4 trials}) = 4 \times (0.7)^3(0.3)^1 = 0.412$$

The general pattern is:

$$P(X = k) = [\text{number of arrangements}] \times [\text{probability of one arrangement}]$$

24.2.2 The binomial coefficient

The number of ways to arrange k successes in n trials is given by the **binomial coefficient**:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

read as "n choose k," where $n!$ (read "n factorial") means $n \times (n-1) \times \cdots \times 2 \times 1$, with the special case $0! = 1$.

For our example: $\binom{4}{3} = \frac{4!}{3!\cdot 1!} = \frac{24}{6\times 1} = 4$, confirming the four arrangements we identified.

24.2.3 The binomial probability formula

Binomial probability formula. If $X \sim \text{Binomial}(n, p)$, the probability of exactly k successes is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

for $k = 0, 1, 2, \ldots, n$.

What is the probability that 5 of 8 randomly selected insured individuals will not exceed the deductible?

With $n = 8$, $k = 5$, and $p = 0.70$:

$$P(X = 5) = \binom{8}{5} (0.70)^5 (0.30)^3$$

Computing the binomial coefficient:

$$\binom{8}{5} = \frac{8!}{5!\cdot 3!} = \frac{8\times 7\times 6}{3\times 2\times 1} = 56$$

And the probability of a single arrangement: $(0.70)^5(0.30)^3 \approx 0.00454$.

$$P(X = 5) = 56 \times 0.00454 \approx 0.254$$

Suppose 4 friends who smoke can be treated as a random sample, with $P(\text{severe lung condition}) = 0.3$ for each. (a) What is the probability that none develop a severe lung condition? (b) What is the probability that exactly one does? (c) What is the probability that at most one does?

Show answer

(a) $P(X = 0) = \binom{4}{0}(0.3)^0(0.7)^4 = 1 \times 1 \times 0.2401 = 0.2401$. (b) $P(X = 1) = \binom{4}{1}(0.3)^1(0.7)^3 = 4 \times 0.3 \times 0.343 = 0.4116$. (c) $P(X \le 1) = P(X = 0) + P(X = 1) = 0.2401 + 0.4116 = 0.6517$.

What is the probability that at least 2 of the 4 smoking friends will develop a severe lung condition?

Show answer

Using the complement: $P(X \ge 2) = 1 - P(X \le 1) = 1 - 0.6517 = 0.3483$.

24.2.4 Computing binomial probabilities in practice

For small values of n and k, the binomial formula can be computed by hand. For larger problems, use technology:

- **StatLens**: The Binomial Distribution calculator computes probabilities and visualizes the distribution.
- **Calculator**: Most graphing calculators have a `binompdf` and `binomcdf` function.
- **Software**: Any statistical software can compute binomial probabilities. Look for functions named something like "binomial CDF" or "binomial probability."

For example, to find $P(X \le 42)$ when $X \sim \text{Binomial}(400, 0.15)$, computing 43 individual probabilities by hand would be impractical. Using the StatLens Binomial calculator with $n = 400$, $p = 0.15$, and $P(X \le 42)$, we get approximately 0.0054.

24.3 Mean and standard deviation

Mean and standard deviation of a binomial distribution. If $X \sim \text{Binomial}(n, p)$, then:

$$\mu = np \qquad \sigma = \sqrt{np(1-p)}$$

These formulas follow from the fact that a binomial random variable is the sum of n independent Bernoulli random variables, each with mean p and variance $p(1-p)$.

If 40 individuals are randomly sampled and each has a 70% chance of staying below their insurance deductible, how many would you expect to stay below? What is the standard deviation?

$$\mu = np = 40 \times 0.70 = 28$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{40 \times 0.70 \times 0.30} = \sqrt{8.4} \approx 2.9$$

We expect about 28 individuals to stay below their deductible, give or take about 2.9. Using the rough "95% within 2 standard deviations" guideline, we would expect between about 22 and 34 individuals in most samples.

Suppose 7 friends who smoke can be treated as a random sample, with $p = 0.3$ for developing a severe lung condition. How many would you expect to develop the condition? What is the standard deviation?

Show answer

$\mu = np = 7 \times 0.3 = 2.1$. $\sigma = \sqrt{7 \times 0.3 \times 0.7} = \sqrt{1.47} \approx 1.21$. You would expect about 2 of the 7 friends to develop a severe lung condition.

24.3.1 Simulation perspective: Does the formula match reality?

We can verify the binomial formulas using simulation. Consider $X \sim \text{Binomial}(n = 20, p = 0.3)$:

- **Theory**: $\mu = 20 \times 0.3 = 6$ and $\sigma = \sqrt{20 \times 0.3 \times 0.7} = \sqrt{4.2} \approx 2.05$
- **Simulation**: Generate 10,000 binomial random samples, each with $n = 20$ and $p = 0.3$. The sample mean of the 10,000 counts will be very close to 6, and the sample standard deviation will be very close to 2.05.

This is another instance of the Law of Large Numbers: the simulated distribution converges to the theoretical distribution as we increase the number of simulations.

Try it in StatLens: Use the Binomial Distribution Explorer to set $n = 20$ and $p = 0.3$. Simulate 1,000 binomial experiments and compare the histogram of results to the theoretical formula. As you increase the number of simulations, the histogram converges to the theoretical distribution.

24.4 Normal approximation to the binomial

The binomial formula can be cumbersome for large n, especially when computing cumulative probabilities like $P(X \leq 42)$ for $n = 400$. Fortunately, when n is large enough, the binomial distribution is well approximated by the normal distribution.

Approximately 15% of the US population smokes cigarettes. A local government surveyed 400 randomly selected individuals and found only 42 smokers. If the true proportion is 0.15, what is the probability of observing 42 or fewer smokers?

The exact answer requires summing 43 binomial probabilities, which gives $P(X \leq 42) = 0.0054$.

As the sample size increases, the binomial distribution becomes increasingly bell-shaped and symmetric. This is illustrated in the figure below.

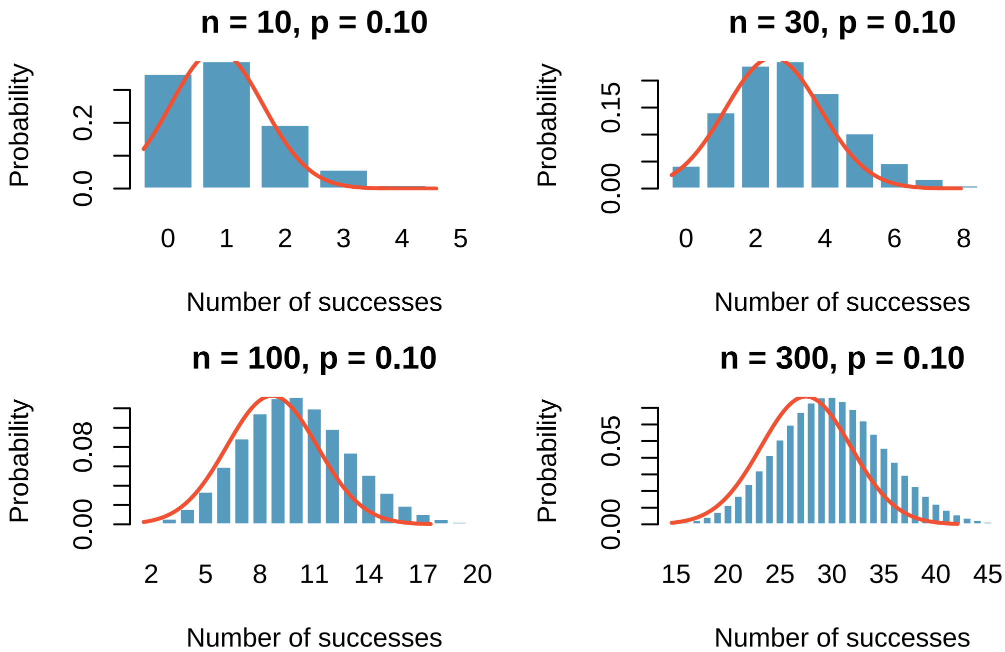


Figure 24.1: Binomial distributions with $p = 0.10$ and sample sizes $n = 10, 30, 100$, and 300. As n increases, the distribution approaches a normal curve.

Normal approximation to the binomial. The binomial distribution with parameters n and p is approximately normal when both $np \geq 10$ and $n(1-p) \geq 10$. The approximating normal distribution has:

$$\mu = np \qquad \sigma = \sqrt{np(1-p)}$$

Use the normal approximation to estimate $P(X \leq 42)$ for $X \sim \text{Binomial}(400, 0.15)$.

First, check the conditions: $np = 400 \times 0.15 = 60 \geq 10$ and $n(1-p) = 400 \times 0.85 = 340 \geq 10$. The normal approximation is appropriate.

The approximating distribution is $N(\mu = 60, \sigma = 7.14)$, where $\sigma = \sqrt{400 \times 0.15 \times 0.85} = 7.14$.

Compute the Z-score:

$$Z = \frac{42 - 60}{7.14} = \frac{-18}{7.14} = -2.52$$

Using the standard normal distribution: $P(Z \leq -2.52) = 0.0059$.

This is very close to the exact binomial probability of 0.0054.

A coin is flipped 100 times. Use the normal approximation to find the probability of getting 60 or more heads.

Show answer

Check conditions: $np = 100 \times 0.5 = 50 \geq 10$ and $n(1-p) = 50 \geq 10$. The approximation is appropriate. $\mu = 50$, $\sigma = \sqrt{100 \times 0.5 \times 0.5} = 5$. $Z = (60 - 50)/5 = 2$. $P(Z \geq 2) = 1 - 0.9772 = 0.0228$. There is about a 2.3% chance of getting 60 or more heads in 100 flips of a fair coin.

24.4.1 Connecting to inference

The normal approximation to the binomial is not just a computational shortcut — it is the theoretical basis for the inference methods we developed in Parts II and III. When we studied the sampling distribution of the sample proportion $\hat{p}$ in Chapter 9, we were implicitly using the fact that the count of successes X in a sample of size n follows a binomial distribution, and that this binomial distribution is approximately normal for large n.

Specifically, if $X \sim$ Binomial(n, p), then $\hat{p} = X/n$, and the normal approximation tells us:

$$\hat{p} \approx N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

This is exactly the result we used when constructing confidence intervals and performing hypothesis tests for proportions in Chapter 13. The theory developed in this chapter *explains* why those methods work.

24.5 Chapter review

24.5.1 Summary

- The **binomial distribution** counts the number of successes in n independent trials, each with the same probability of success p.
- **Four conditions**: fixed n, binary outcomes, independence, constant p.
- **Binomial probability formula**: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
- **Mean and standard deviation**: $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.
- When $np \geq 10$ and $n(1-p) \geq 10$, the binomial distribution is well approximated by the normal distribution $N(np, \sqrt{np(1-p)})$.
- The normal approximation to the binomial provides the theoretical justification for the inference methods on proportions from Chapter 13.

24.6 Exercises

1. **Underage drinking, Part I.** Data collected by the Substance Abuse and Mental Health Services Administration (SAMSHA) suggests that 69.7% of 18-20 year olds consumed alcoholic beverages in any given year.

a. Suppose a random sample of ten 18-20 year olds is taken. Is the use of the binomial distribution appropriate for calculating the probability that exactly six consumed alcoholic beverages? Explain.
b. Calculate the probability that exactly 6 out of 10 randomly sampled 18- 20 year olds consumed an alcoholic drink.
c. What is the probability that exactly four out of ten 18-20 year olds have *not* consumed an alcoholic beverage?
d. What is the probability that at most 2 out of 5 randomly sampled 18-20 year olds have consumed alcoholic beverages?
e. What is the probability that at least 1 out of 5 randomly sampled 18-20 year olds have consumed alcoholic beverages?

2. **Chickenpox, Part I.** Boston Children's Hospital estimates that 90% of Americans have had chickenpox by the time they reach adulthood.

a. Suppose we take a random sample of 100 American adults. Is the use of the binomial distribution appropriate for calculating the probability that exactly 97 out of 100 randomly sampled American adults had chickenpox during childhood? Explain.
b. Calculate the probability that exactly 97 out of 100 randomly sampled American adults had chickenpox during childhood.
c. What is the probability that exactly 3 out of a new sample of 100 American adults have *not* had chickenpox in their childhood?

d. What is the probability that at least 1 out of 10 randomly sampled American adults have had chickenpox?

e. What is the probability that at most 3 out of 10 randomly sampled American adults have *not* had chickenpox?

3. **Underage drinking, Part II.** We learned in a previous exercise that about 70% of 18-20 year olds consumed alcoholic beverages in any given year. We now consider a random sample of fifty 18-20 year olds.

a. How many people would you expect to have consumed alcoholic beverages? And with what standard deviation?

b. Would you be surprised if there were 45 or more people who have consumed alcoholic beverages?

c. What is the probability that 45 or more people in this sample have consumed alcoholic beverages? How does this probability relate to your answer to part (b)?

4. **Chickenpox, Part II.** We learned in a previous exercise that about 90% of American adults had chickenpox before adulthood. We now consider a random sample of 120 American adults.

a. How many people in this sample would you expect to have had chickenpox in their childhood? And with what standard deviation?

b. Would you be surprised if there were 105 people who have had chickenpox in their childhood?

c. What is the probability that 105 or fewer people in this sample have had chickenpox in their childhood? How does this probability relate to your answer to part (b)?

5. **Game of dreidel.** A dreidel is a four-sided spinning top with the Hebrew letters *nun*, *gimel*, *hei*, and *shin*, one on each side. Each side is equally likely to come up in a single spin of the dreidel. Suppose you spin a dreidel three times. Calculate the probability of getting

a. at least one *nun*?

b. exactly 2 *nun*s?

c. exactly 1 *hei*?

d. at most 2 *gimel*s?

6. **Arachnophobia.** A Gallup Poll found that 7% of teenagers (ages 13 to 17) suffer from arachnophobia and are extremely afraid of spiders. At a summer camp there are 10 teenagers sleeping in each tent. Assume that these 10 teenagers are independent of each other.%

a. Calculate the probability that at least one of them suffers from arachnophobia.

b. Calculate the probability that exactly 2 of them suffer from arachnophobia.

c. Calculate the probability that at most 1 of them suffers from arachnophobia.

d. If the camp counselor wants to make sure no more than 1 teenager in each tent is afraid of spiders, does it seem reasonable for him to randomly assign teenagers to tents?

7. **Eye color, Part II.** a previous exercise introduces a husband and wife with brown eyes who have 0.75 probability of having children with brown eyes, 0.125 probability of having children with blue eyes, and 0.125 probability of having children with green eyes.

a. What is the probability that their first child will have green eyes and the second will not?

b. What is the probability that exactly one of their two children will have green eyes?

c. If they have six children, what is the probability that exactly two will have green eyes?

d. If they have six children, what is the probability that at least one will have green eyes?

e. What is the probability that the first green eyed child will be the 4^{th} child?

f. Would it be considered unusual if only 2 out of their 6 children had brown eyes?

8. **Sickle cell anemia.** Sickle cell anemia is a genetic blood disorder where red blood cells lose their flexibility and assume an abnormal, rigid, "sickle" shape, which results in a risk of various complications. If both parents are carriers of the disease, then a child has a 25% chance of having the disease, 50% chance of being a carrier, and 25% chance of neither having the disease nor being a carrier. If two parents who are carriers of the disease have 3 children, what is the probability that

a. two will have the disease?

b. none will have the disease?

c. at least one will neither have the disease nor be a carrier?

d. the first child with the disease will the be 3^{rd} child?

9. **Exploring permutations.** The formula for the number of ways to arrange n objects is $n! = n \times (n-1) \times \cdots \times 2 \times 1$. This exercise walks you through the derivation of this formula for a couple of special cases.

A small company has five employees: Anna, Ben, Carl, Damian, and Eddy. There are five parking spots in a row at the company, none of which are assigned, and each day the employees pull into a random parking spot. That is, all possible orderings of the cars in the row of spots are equally likely.

a. On a given day, what is the probability that the employees park in alphabetical order?

b. If the alphabetical order has an equal chance of occurring relative to all other possible orderings, how many ways must there be to arrange the five cars?

c. Now consider a sample of 8 employees instead. How many possible ways are there to order these 8 employees' cars?

10. **Male children.** While it is often assumed that the probabilities of having a boy or a girl are the same, the actual probability of having a boy is slightly higher at 0.51. Suppose a couple plans to have 3 kids.

a. Use the binomial model to calculate the probability that two of them will be boys.

b. Write out all possible orderings of 3 children, 2 of whom are boys. Use these scenarios to calculate the same probability from part (a) but using the addition rule for disjoint outcomes. Confirm that your answers from parts (a) and (b) match.

c. If we wanted to calculate the probability that a couple who plans to have 8 kids will have 3 boys, briefly describe why the approach from part (b) would be more tedious than the approach from part (a).

11. **University admissions.** Suppose a university announced that it admitted 2,500 students for the following year's freshman class. However, the university has dorm room spots for only 1,786 freshman students. If there is a 70% chance that an admitted student will decide to accept the offer and attend this university, what is the approximate probability that the university will not have enough dormitory room spots for the freshman class?

12. **Survey response rate.** Pew Research reported that the typical response rate to their surveys is only 9%. If for a particular survey 15,000 households are contacted, what is the probability that at least 1,500 will agree to respond?

Chapter 25

Normal Distribution

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

The normal distribution is the most important probability model in statistics. Its familiar bell shape appears in an extraordinary range of settings — from SAT scores and adult heights to measurement errors and stock returns. In this chapter, we study the normal distribution as a **probability model** for individual observations: we learn to compute probabilities, find percentiles, and assess whether data plausibly come from a normal distribution. This complements Chapter 10, which uses the normal distribution as an approximation for **sampling distributions** in inference. Here, the focus is squarely on probability.

StatLens tools for this chapter: Use the **Normal Distribution Calculator** to visualize normal curves, shade areas, compute probabilities, and find percentiles. Adjust μ and σ to see how the distribution changes.

Launch Normal Distribution Calculator

25.1 The normal distribution as a probability model

Among all distributions we encounter in practice, one is overwhelmingly the most common. The symmetric, unimodal, bell-shaped curve known as the **normal distribution** (also called the **Gaussian distribution**) arises in a remarkable variety of settings.

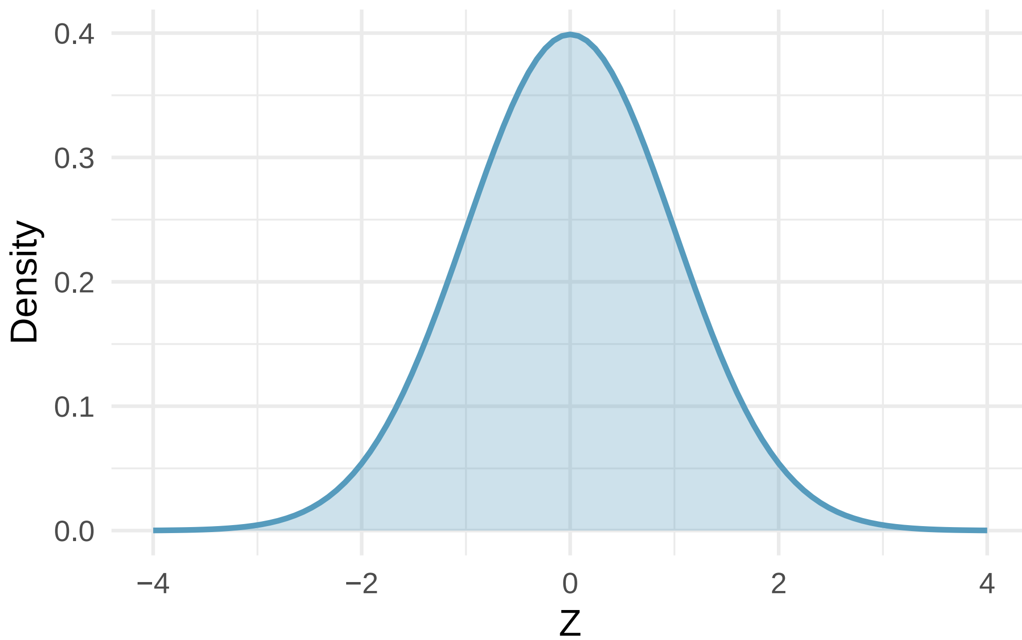


Figure 25.1: A standard normal curve — symmetric, unimodal, and bell-shaped.

Many variables are *nearly* normal, but none are *exactly* normal. The normal distribution, while not a perfect description of any single real-world variable, is an extraordinarily useful model for a wide variety of problems.

25.1.1 Parameters: mean and standard deviation

The normal distribution is completely specified by two **parameters**:

- The **mean** μ determines the center (location) of the bell curve.
- The **standard deviation** σ determines the spread (width) of the bell curve.

If a random variable X follows a normal distribution with mean μ and standard deviation σ, we write:

$$X \sim N(\mu, \sigma)$$

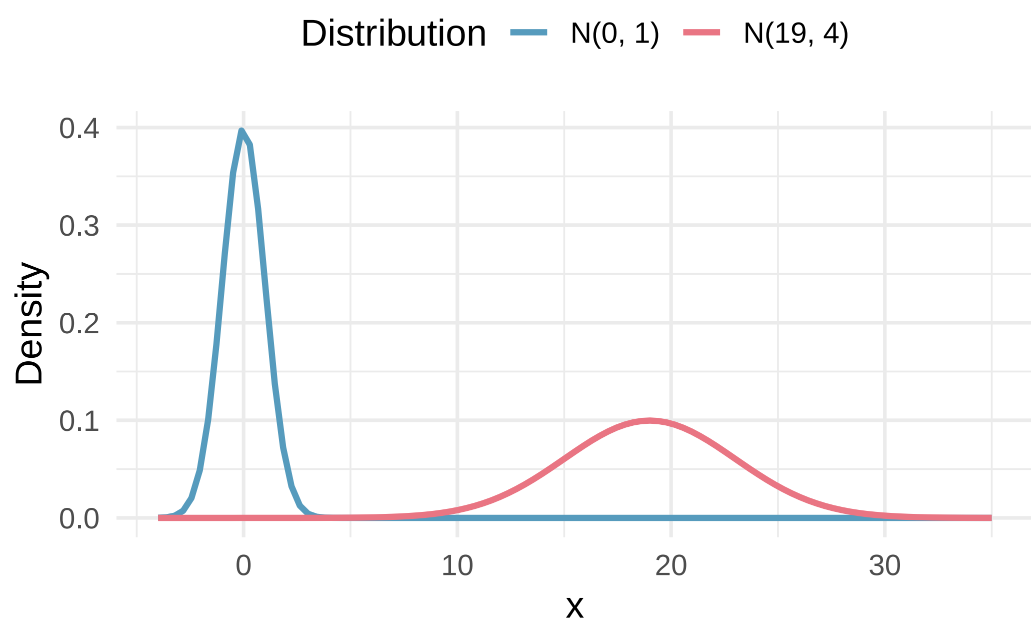


Figure 25.2: Two normal distributions: $N(0, 1)$ and $N(19, 4)$. Changing μ shifts the curve; changing σ stretches or compresses it.

The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is called the **standard normal distribution**. It is often denoted $Z \sim N(0, 1)$.

Write the shorthand notation for normal distributions with: (a) mean 5 and standard deviation 3, (b) mean -100 and standard deviation 10, (c) mean 2 and standard deviation 9.

Show answer

(a) $N(\mu = 5, \sigma = 3)$. (b) $N(\mu = -100, \sigma = 10)$. (c) $N(\mu = 2, \sigma = 9)$.

25.1.2 The 68-95-99.7 rule

A useful rule of thumb describes how probability is distributed within a normal distribution:

The 68-95-99.7 rule. For a normal distribution:

- About **68%** of observations fall within 1 standard deviation of the mean ($\mu \pm \sigma$).
- About **95%** of observations fall within 2 standard deviations of the mean ($\mu \pm 2\sigma$).
- About **99.7%** of observations fall within 3 standard deviations of the mean ($\mu \pm 3\sigma$).

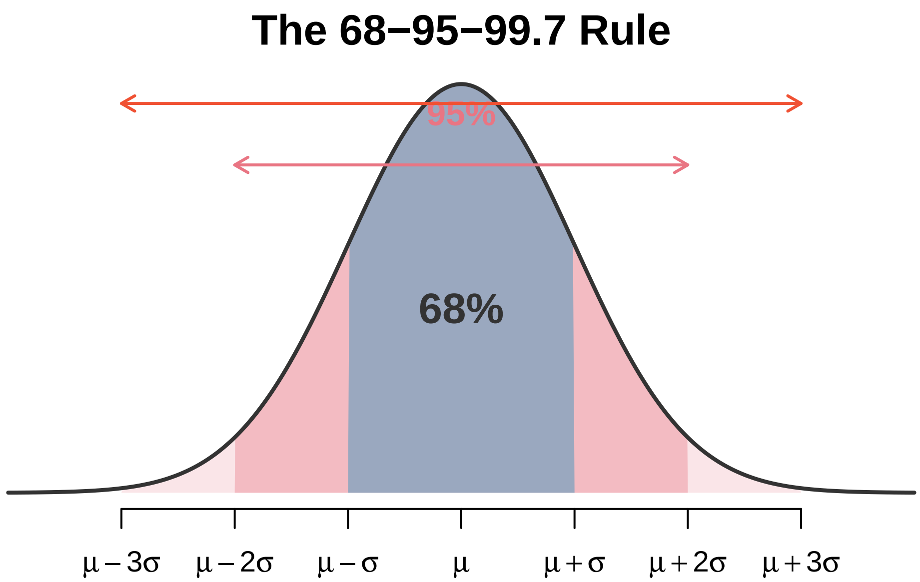


Figure 25.3: The 68-95-99.7 rule illustrated on a normal distribution.

It is possible for a normal random variable to fall 4, 5, or even more standard deviations from the mean, but such events are extremely rare. The probability of being more than 4 standard deviations from the mean is about 1 in 15,000.

SAT scores closely follow $N(\mu = 1100, \sigma = 200)$. (a) About what percent of test takers score between 700 and 1500? (b) What percent score between 1100 and 1500?

Show answer

(a) 700 and 1500 are each 2 standard deviations from the mean, so about 95% of test takers score in this range. (b) By symmetry, half of the 95% falls above the mean: $95\%/2 = 47.5\%$.

25.1.3 Scope note: Chapter 25 vs. Chapter 10

In Chapter 10, you used the normal distribution as an approximation for **sampling distributions** — describing the behavior of statistics like $\bar{x}$ and $\hat{p}$ across many samples. Here, we study the normal distribution in a different role: as a **probability model for individual observations**. You use it to answer questions like:

- What proportion of adults are taller than 6 feet?
- What SAT score puts a student in the 95th percentile?
- Is it plausible that this data set came from a normal distribution?

The same mathematical object — the normal distribution — serves both purposes, but the *interpretation* is different. In Chapter 10, we modeled the behavior of sample statistics; here, we are modeling data.

25.2 Finding probabilities: $P(X < a)$

25.2.1 Z-scores: standardizing observations

To find probabilities from a normal distribution, we first convert the observation to a **Z-score**, which measures how many standard deviations it falls above or below the mean.

The **Z-score** of an observation x from a distribution with mean μ and standard deviation σ is:

$$Z = \frac{x - \mu}{\sigma}$$

Observations above the mean have positive Z-scores; observations below the mean have negative Z-scores; an observation equal to the mean has a Z-score of 0.

SAT scores follow $N(1100, 200)$ and ACT scores follow $N(21, 6)$. Ann scored 1300 on the SAT and Tom scored 24 on the ACT. Who performed better relative to their peers?

$$Z_{\text{Ann}} = \frac{1300 - 1100}{200} = 1.0 \qquad Z_{\text{Tom}} = \frac{24 - 21}{6} = 0.5$$

Ann is 1 standard deviation above the SAT mean, while Tom is only 0.5 standard deviations above the ACT mean. Ann performed relatively better.

Head lengths of brushtail possums follow $N(92.6, 3.6)$ mm. Compute the Z-scores for possums with head lengths of 95.4 mm and 85.8 mm. Which is more unusual?

Show answer

$Z_1 = (95.4 - 92.6)/3.6 = 0.78$ and $Z_2 = (85.8 - 92.6)/3.6 = -1.89$. Since $|-1.89| > |0.78|$, the 85.8 mm head length is more unusual.

25.2.2 Computing tail areas

Once we have a Z-score, we can find the probability of observing a value at least that extreme. This is called a **tail area** or **tail probability**.

Always draw a picture first. For any normal probability problem, draw and label the normal curve, shade the area of interest, and then find the Z-score. The picture provides a visual estimate and helps avoid errors.

There are several ways to find the area under the normal curve:

1. **StatLens**: The Normal Distribution calculator lets you enter a mean, standard deviation, and boundary value, and it shades and computes the area for you.
2. **Calculator**: Most graphing calculators have a `normalcdf` function.
3. **Probability tables**: Tables of standard normal areas are provided in many textbooks (see Appendix).
4. **Statistical software**: Any statistical software can compute normal probabilities. Look for a function or menu option related to "normal CDF" or "normal probability."

Shannon is a randomly selected SAT taker. What is the probability she scores at least 1190?

Step 1: Draw the picture — shade the area above 1190 on $N(1100, 200)$.

Step 2: Compute the Z-score.

$$Z = \frac{1190 - 1100}{200} = \frac{90}{200} = 0.45$$

Step 3: Find the area. Using software, the area to the *left* of $Z = 0.45$ is 0.6736. Since we want the area to the *right*:

$$P(X \geq 1190) = 1 - 0.6736 = 0.3264$$

The probability Shannon scores at least 1190 is about 32.6%.

Finding areas to the right. Most software returns the area to the *left* of a Z-score. To find the area to the *right*, subtract from 1:

$$P(X > a) = 1 - P(X \leq a)$$

Edward scored 1030 on his SAT. What is his percentile?

The **percentile** is the proportion of scores *below* his. Compute the Z-score:

$$Z = \frac{1030 - 1100}{200} = -0.35$$

Using software: $P(Z \leq -0.35) = 0.3632$. Edward is at the 36th percentile.

Stuart scored 1500 on the SAT. (a) What is his percentile? (b) What percent of test takers scored higher?

Show answer

(a) $Z = (1500 - 1100)/200 = 2.0$. $P(Z \leq 2) = 0.9772$. Stuart is at the 97.7th percentile. (b) $1 - 0.9772 = 0.0228$, or about 2.3%.

25.2.3 Finding probabilities between two values

To find the probability between two values, compute the area between them by subtracting tail areas.

Adult male heights in the US follow $N(70.0'', 3.3'')$. What is the probability a randomly selected adult male is between 5′9′ ′ (69′ ′) and 6′2′ ′ (74′ ′)?

Z-scores:

$$Z_{69} = \frac{69 - 70}{3.3} = -0.30 \qquad Z_{74} = \frac{74 - 70}{3.3} = 1.21$$

Areas:

- $P(Z \leq -0.30) = 0.3821$ (area to the left of 69′ ′)
- $P(Z \leq 1.21) = 0.8869$ (area to the left of 74′ ′)

Area between:

$$P(69 \leq X \leq 74) = 0.8869 - 0.3821 = 0.5048$$

About 50.5% of adult males are between 5′9′ ′ and 6′2′ ′.

SAT scores follow $N(1100, 200)$. What percent of test takers score between 1100 and 1400?

Show answer

$Z_{1100} = 0 \to P(Z \leq 0) = 0.5000$. $Z_{1400} = 1.5 \to P(Z \leq 1.5) = 0.9332$. Area between: $0.9332 - 0.5000 = 0.4332$, or about 43.3%.

25.3 Finding percentiles (inverse normal)

Sometimes we know a tail probability (or percentile) and need to find the corresponding observation value. This is the *inverse* of the normal probability problem.

Strategy:

1. Draw the picture and identify the tail area.
2. Find the Z-score corresponding to that tail area (using software, a calculator, or a table).
3. Use the Z-score formula to solve for x: $x = \mu + Z \cdot \sigma$.

Erik's height is at the 40th percentile of the US adult male height distribution $N(70.0, 3.3)$. How tall is he?

Step 1: The lower tail has area 0.40.

Step 2: Using software, the Z-score corresponding to the 40th percentile is $Z = -0.25$.

Step 3: Solve for Erik's height:

$$x = \mu + Z \cdot \sigma = 70.0 + (-0.25)(3.3) = 70.0 - 0.825 = 69.18 \text{ inches}$$

Erik is about 5'9''.

What is the height at the 82nd percentile for adult males?

Using software, the Z-score at the 82nd percentile is $Z = 0.92$. Then:

$$x = 70.0 + 0.92 \times 3.3 = 70.0 + 3.04 = 73.04 \text{ inches}$$

The 82nd percentile corresponds to about 6'1''.

SAT scores follow $N(1100, 200)$. (a) What is the 95th percentile for SAT scores? (b) What is the 97.5th percentile?

Show answer

(a) $Z_{0.95} = 1.645$, so $x = 1100 + 1.645 \times 200 = 1429$. (b) $Z_{0.975} = 1.96$, so $x = 1100 + 1.96 \times 200 = 1492$.

25.3.1 Using StatLens for normal probabilities and percentiles

The StatLens Normal Distribution Explorer provides an interactive way to work with the normal distribution. You can:

- Set μ and σ to define the distribution
- Shade regions to find probabilities (left tail, right tail, or between two values)
- Enter a probability to find the corresponding cutoff value (inverse normal)
- Toggle between showing the Z-score and the original scale

Try it in StatLens: Open the Normal Distribution Explorer and set $\mu = 1100$, $\sigma = 200$ for SAT scores. Shade the region above 1300 to find the probability of scoring above 1300. Then switch to "find value" mode and enter 0.95 to find the 95th percentile.

25.4 QQ plots and checking normality

Before using the normal distribution as a model, we should check whether the assumption of normality is reasonable. Two visual methods are commonly used.

25.4.1 Method 1: Histogram with normal curve overlay

The simplest approach is to plot a histogram of the data and overlay the best-fitting normal curve (using $\bar{x}$ for μ and s for σ). If the histogram closely matches the curve, the normal model is reasonable.

25.4.2 Method 2: Normal probability plots (QQ plots)

A **normal probability plot** (also called a **QQ plot**, short for "quantile-quantile plot") plots the observed data values against the values we would expect if the data were perfectly normal. If the data are approximately normal, the points will fall close to a straight line.

How to read a QQ plot:

- **Points on or near the line**: The data are consistent with a normal distribution.

- **Systematic curve (S-shape or other)**: The data deviate from normality in a specific way.
- **Points curving up at both ends**: The distribution has heavier tails than normal (more extreme values).
- **Points curving down at both ends**: The distribution has lighter tails than normal.
- **Points curving up at the right**: The distribution is right-skewed.
- **Points curving down at the left**: The distribution is left-skewed.

25.4.3 Interpreting QQ plots: Examples

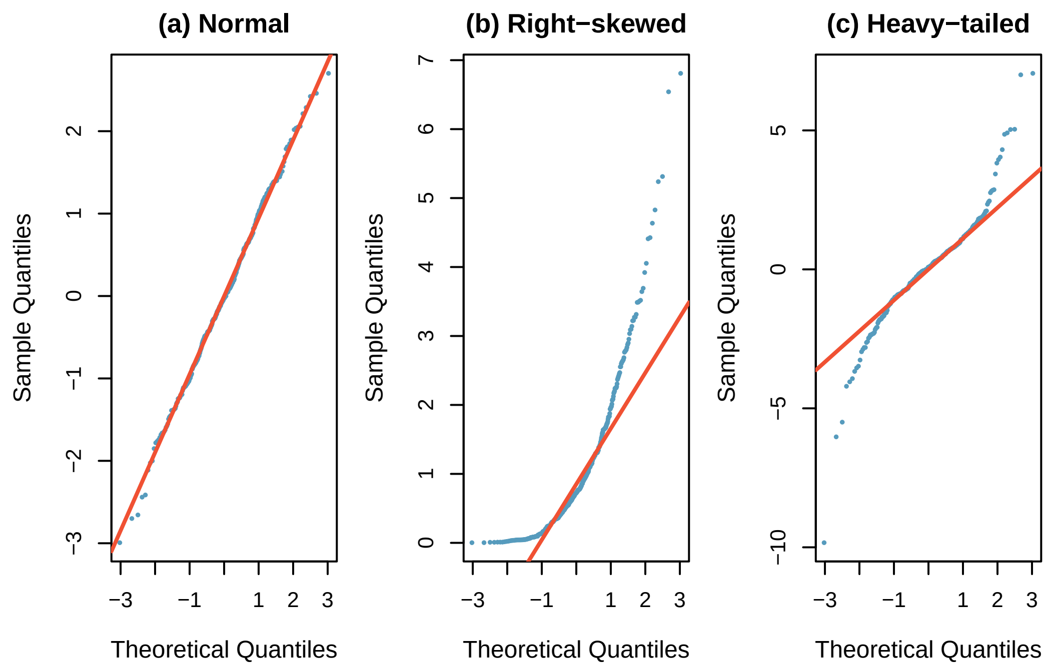


Figure 25.4: Three QQ plots: (a) approximately normal data with $n = 400$ — points fall close to the line; (b) right-skewed data — points curve upward at the right; (c) heavy-tailed data — points curve away from the line at both ends.

Sample size matters. With small sample sizes ($n < 30$), we should expect more variability in the QQ plot even when the data truly come from a normal distribution. Deviations must be more severe to be considered evidence against normality when n is small.

A researcher collects the following data and creates a QQ plot. The points follow the diagonal line closely in the middle of the distribution but curve upward at the right end. What does this suggest about the shape of the data?

Show answer

The QQ plot suggests the data are right-skewed. The bulk of the data are approximately normal, but there are larger values in the upper tail than the normal distribution would predict.

25.4.4 When is the normal model appropriate?

The normal distribution is a good model for data that are:

- **Symmetric**: The histogram is roughly mirror-image around the center.
- **Unimodal**: There is a single peak.
- **Bell-shaped**: The histogram has the characteristic tapered shape of the normal curve.
- **No extreme outliers**: Outliers can distort both the mean and standard deviation, making the normal model a poor fit.

Many variables are approximately normal:

- Standardized test scores (SAT, ACT, GRE)
- Heights of adults (within a sex)
- Measurement errors in scientific instruments
- Many biological measurements (blood pressure, cholesterol levels)

Many variables are **not** normal:

- Income distributions (right-skewed)
- Waiting times (often right-skewed)
- Count data (often Poisson or binomial)
- Variables with natural boundaries (proportions, ages)

When the normal model is not appropriate, other distributions may be used, or nonparametric methods may be applied.

25.5 Chapter review

25.5.1 Summary

- The **normal distribution** $N(\mu, \sigma)$ is a continuous probability model defined by its mean μ (center) and standard deviation σ (spread).
- The **68-95-99.7 rule** provides quick estimates: 68% of values fall within 1 SD, 95% within 2 SDs, and 99.7% within 3 SDs of the mean.
- The **Z-score** $Z = (x - \mu)/\sigma$ standardizes an observation by measuring how many standard deviations it falls from the mean.
- **Finding probabilities**: Convert to a Z-score, then use software, a calculator, or a table to find the area under the standard normal curve.
- **Finding percentiles** (inverse normal): Look up the Z-score corresponding to the desired area, then solve $x = \mu + Z\sigma$.
- **QQ plots** assess whether data are approximately normal by plotting observed values against expected normal values. Points near a straight line indicate normality.
- Recall that Chapter 10 used the normal distribution for **sampling distributions** in inference. This chapter treats it as a **probability model** for individual observations — the theoretical foundation for that earlier work.

25.6 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Area under the curve, Part I.** What percent of a standard normal distribution $N(\mu = 0, \sigma = 1)$ is found in each region denoted by a Z inequality below? Be sure to draw a graph. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

a. $Z < -1.35$

b. $Z > 1.48$

c. $-0.4 < Z < 1.5$

d. $|Z| > 2$

2. **Area under the curve, Part II.** What percent of a standard normal distribution $N(\mu = 0, \sigma = 1)$ is found in each region denoted by a Z inequality below? Be sure to draw a graph. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

a. $Z > -1.13$
b. $Z < 0.18$
c. $Z > 8$
d. $|Z| < 0.5$

3. **GRE scores, Z scores.** Sophia who took the Graduate Record Examination (GRE) scored 160 on the Verbal Reasoning section and 157 on the Quantitative Reasoning section. The mean score for Verbal Reasoning section for all test takers was 151 with a standard deviation of 7, and the mean score for the Quantitative Reasoning was 153 with a standard deviation of 7.67. Suppose that both distributions are nearly normal. Use the information to compute each of the following. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

a. Write down the short-hand for each of the two normal distributions.
b. What is Sophia's Z score on the Verbal Reasoning section? On the Quantitative Reasoning section? Draw a standard normal distribution curve and mark the two Z scores.
c. What do the Z scores tell you?
d. Relative to others, which section did Sophia do better on?
e. Find her percentile scores for each of the two exams.
f. What percent of the test takers did better than her on the Verbal Reasoning section? On the Quantitative Reasoning section?
g. Explain why simply comparing raw scores from the two sections could lead to an incorrect conclusion as to which section a student did better on.
h. If the distributions of the scores on these exams are not nearly normal, would your answers to parts (b) - (f) change? Explain your reasoning.

4. **Triathlon times, Z scores.** In triathlons, it is common for racers to be placed into age and gender groups. Two friends, Leo and Mary, both completed the Hermosa Beach Triathlon, where Leo competed in the "Men, Ages 30 - 34" group and Mary competed in the "Women, Ages 25 - 29" group. Leo completed the race in 1:22:28 (4948 seconds), while Mary completed the race in 1:31:53 (5513 seconds). We can see that Leo finished faster, but they are curious about how they did within their respective groups. Can you help them? Below is some information on the performance of their groups. Use the information to compute each of the following. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

- The finishing times of the "Men, Ages 30 - 34" group has a mean of 4313 seconds with a standard deviation of 583 seconds.
- The finishing times of the "Women, Ages 25 - 29" group has a mean of 5261 seconds with a standard deviation of 807 seconds.
- The distributions of finishing times for both groups are approximately Normal.
  Remember: a better performance corresponds to a faster finish.

a. Write down the short-hand for the two normal distributions.
b. What are the Z scores for each of Leo's and Mary's finishing times? What do the Z scores tell you?
c. Did Leo or Mary rank better in their respective group? Explain your reasoning.
d. What percent of the triathletes did Leo finish faster than in his group?
e. What percent of the triathletes did Mary finish faster than in her group?
f. If the distributions of finishing times are not nearly normal, would your answers to parts (b) – (e) change? Explain your reasoning.

5. **GRE scores, cutoffs.** Consider the previous two distributions for GRE scores: $N(\mu = 151, \sigma = 7)$ for the Verbal Reasoning part of the exam and $N(\mu = 153, \sigma = 7.67)$ for the Quantitative Reasoning part. Use the information to compute each of the following. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

a. The score of a student who scored in the 80^{th} percentile on the Quantitative Reasoning section.
b. The score of a student who scored worse than 70% of test takers in the Verbal Reasoning section.

6. **Triathlon times, cutoffs.** Recall the two different distributions for triathlon times: $N(\mu = 4313, \sigma = 583)$ for "Men, Ages 30 - 34" and $N(\mu = 5261, \sigma = 807)$ for the "Women, Ages 25 - 29" group. Times are listed in seconds. Use this information to compute each of the following. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

a. The cutoff time for the fastest 5% of athletes in the men's group, i.e., those who took the shortest 5% of time to finish.
b. The cutoff time for the slowest 10% of athletes in the women's group.

7. **LA weather, Fahrenheit.** The average daily high temperature in June in LA is 77° F with a standard deviation of 5° F. Suppose that the temperatures in June closely follow a normal distribution. Use the information to compute each of the following. In the text above, we used R to calculate normal probabilities. You might choose to use a different source, such as a Shiny App or a normal table.

a. What is the probability of observing an 83° F temperature or higher in LA during a randomly chosen day in June?
b. How cool are the coldest 10% of the days (days with lowest high temperature) during June in LA?

8. **CAPM.** The Capital Asset Pricing Model (CAPM) is a financial model that assumes returns on a portfolio are normally distributed. Suppose a portfolio has an average annual return of 14.7% (i.e., an average gain of 14.7%) with a standard deviation of 33%. A return of 0% means the value of the portfolio doesn't change, a negative return means that the portfolio loses money, and a positive return means that the portfolio gains money.

a. What percent of years does this portfolio lose money, i.e., have a return less than 0%?

b. What is the cutoff for the highest 15% of annual returns with this portfolio?

9. **LA weather, Celsius.** Recall the set-up that average daily high temperature in June in LA is 77° F with a standard deviation of 5° F, and it can be assumed that the high temperatures follow a normal distribution. We use the following equation to convert °F (Fahrenheit) to °C (Celsius): $C = (F - 32) \times \frac{5}{9}$.

a. Write the probability model for the distribution of temperature in °C in June in LA.

b. What is the probability of observing a 28° C (which roughly corresponds to 83° F) temperature or higher in June in LA? Calculate using the °C model from part (a).

c. Did you get the same answer or different answers in part (b) of this question and part (a) of the previous question on the LA weather? Are you surprised? Explain.

d. Estimate the IQR of the temperatures (in °C) in June in LA.

10. **Find the SD.** Find the standard deviation of the distribution in the following situations.

a. MENSA is an organization whose members have IQs in the top 2% of the population. IQs are normally distributed with mean 100, and the minimum IQ score required for admission to MENSA is 132.

b. Cholesterol levels for women aged 20 to 34 follow an approximately normal distribution with mean 185 milligrams per deciliter (mg/dl). Women with cholesterol levels above 220 mg/dl are considered to have high cholesterol and about 18.5% of women fall into this category.

Chapter 26

Type II Error

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In Chapter 7, we introduced the hypothesis testing framework and saw how data can provide evidence against a null hypothesis. We focused on the risk of incorrectly rejecting a true null hypothesis — a Type I error. But there is a second, equally important kind of mistake: failing to detect a real effect. This chapter examines **Type II error**, explores the tradeoff between the two error types, and develops intuition for why some studies miss real effects.

StatLens tools for this chapter: Use the randomization test simulators from Chapter 7 to explore Type II error. Run tests with small sample sizes or small effect sizes and notice how often the test fails to reject H_0 — those are Type II errors in action.

Launch Randomization Test: Difference in Means

26.1 Review: hypothesis testing and Type I error

Recall the basic structure of a hypothesis test. We begin with two competing claims:

- H_0 (the **null hypothesis**): There is no effect, no difference, or no relationship.
- H_A (the **alternative hypothesis**): There *is* an effect, difference, or relationship.

We collect data, compute a test statistic, and calculate a p-value — the probability of observing results at least as extreme as ours if H_0 were true. If the p-value is small enough (below our chosen significance level α), we reject H_0.

But rejecting H_0 when it is actually true is a mistake. This mistake has a name.

A **Type I error** occurs when we reject the null hypothesis H_0 even though H_0 is actually true. In other words, we conclude there is an effect when in reality there is none.

The probability of a Type I error is controlled by the significance level α. When we set $\alpha = 0.05$, we are accepting a 5% chance of making this kind of mistake — declaring a "discovery" that is really just noise.

Type I error is the kind of mistake that gets the most attention in introductory statistics, and for good reason: it is the error rate we directly control when we choose α. But controlling one kind of error does not mean we have eliminated all risk. There is a second kind of mistake lurking in every hypothesis test, and it can be just as costly — sometimes more so.

26.2 Type II error: missing a real effect

Suppose a new medication truly does lower blood pressure compared to a placebo. A clinical trial is conducted, data are collected, and the researchers perform a hypothesis test. But the p-value comes back at 0.14 — not small enough to reject H_0 at the $\alpha = 0.05$ level. The researchers conclude: "We did not find sufficient evidence that the medication works."

But the medication *does* work. The study simply failed to detect the real effect. This is a **Type II error**.

A **Type II error** occurs when we fail to reject the null hypothesis H_0 even though the alternative hypothesis H_A is actually true. In other words, there is a real effect, but our test misses it.

Type II errors are sometimes called "false negatives" — the test comes back negative (fail to reject H_0) when the truth is positive (the effect exists). In medical screening, this is analogous to a test that says "no disease" when the patient actually has the disease. In a court system, it is analogous to acquitting a guilty person.

26.3 The decision table

Every hypothesis test ends with one of two decisions: reject H_0 or fail to reject H_0. And the truth is one of two realities: H_0 is true or H_A is true. Combining these gives four scenarios:

Table 26.1: Four possible outcomes of a hypothesis test.

	H_0 is true	H_A is true
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

Two of the four outcomes are correct decisions:

- Rejecting H_0 when H_A is true — we correctly detected the real effect.
- Failing to reject H_0 when H_0 is true — we correctly avoided a false alarm.

The other two outcomes are errors:

- **Type I error** (upper-left): We rejected H_0 but it was actually true. False alarm.
- **Type II error** (lower-right): We failed to reject H_0 but H_A was actually true. Missed detection.

In a US court, the defendant is either innocent (H_0) or guilty (H_A). What does a Type I error represent in this context? What does a Type II error represent? Table 26.1 may be useful.

If the court makes a Type I error, this means the defendant is innocent (H_0 true) but wrongly convicted. A Type II error means the court failed to reject H_0 (i.e., failed to convict the person) when they were in fact guilty (H_A true).

A pharmaceutical company tests whether a new drug reduces cholesterol more than a placebo. They set up H_0: the drug has no effect, and H_A: the drug reduces cholesterol. Describe what a Type I error and a Type II error would mean in this context.

Show answer

A Type I error would mean concluding the drug reduces cholesterol when it actually does not — the company might invest millions in producing an ineffective drug. A Type II error would mean failing to detect a real cholesterol-lowering effect — the company might shelve a drug that actually works.

A quality control inspector tests whether a batch of light bulbs has a defect rate above 2%. She sets up H_0: defect rate ≤ 0.02 and H_A: defect rate > 0.02. Which error is worse in this context: Type I or Type II? Why?

Show answer

A Type II error is arguably worse here: the inspector would fail to detect that the defect rate is too high, and defective bulbs would be shipped to customers. A Type I error means rejecting a good batch, which wastes product but doesn't harm customers. The relative costs depend on the specific situation, but in quality control, missing a real problem (Type II) often has more serious consequences.

26.4 Error rates: α and β

We give the probabilities of the two error types their own Greek-letter names:

The **Type I error rate**, denoted α, is the probability of rejecting H_0 when H_0 is true:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

The **Type II error rate**, denoted β, is the probability of failing to reject H_0 when H_A is true:

$$\beta = P(\text{fail to reject } H_0 \mid H_A \text{ is true})$$

The significance level α is something we choose before the test — it is under our direct control. Common choices are $\alpha = 0.05$, $\alpha = 0.01$, and $\alpha = 0.10$.

The Type II error rate β, on the other hand, is **not** something we directly choose. It depends on several factors that we will explore shortly: the true effect size, the sample size, the variability in the data, and the chosen significance level α. Unlike α, β is usually unknown because we don't know the true state of the world — if we knew H_A were true and knew the exact effect size, we wouldn't need to run the test.

This asymmetry between α and β is a fundamental feature of the hypothesis testing framework. We control one error rate directly (α) and try to manage the other (β) through careful study design.

26.5 The tradeoff between α and β

How could we reduce the Type I error rate in US courts? What influence would this have on the Type II error rate?

To lower the Type I error rate, we might raise our standard for conviction from "beyond a reasonable doubt" to "beyond a conceivable doubt" so fewer people would be wrongly convicted. However, this would also make it more difficult to convict the people who are actually guilty, so we would make more Type II errors.

How could we reduce the Type II error rate in US courts? What influence would this have on the Type I error rate?

Show answer

To lower the Type II error rate, we want to convict more guilty people. We could lower the standards for conviction from "beyond a reasonable doubt" to "beyond a little doubt". Lowering the bar for guilt will also result in more wrongful convictions, raising the Type I error rate.

The example and guided practice above provide an important lesson: if we reduce how often we make one type of error, we generally make more of the other type.

This tradeoff is not just a courtroom analogy — it is baked into the mathematics of hypothesis testing. Here is the core intuition:

- **Lowering** α means we require stronger evidence (a smaller p-value) before rejecting H_0. This makes us less likely to reject H_0 when it is true (good), but also less likely to reject H_0 when it is false (bad). So β increases.

- **Raising** α means we accept weaker evidence for rejecting H_0. This makes us more likely to detect real effects (β decreases), but also more likely to make false discoveries (α increases).

The α-β tradeoff. For a fixed sample size and effect size, decreasing α increases β, and vice versa. You cannot reduce both error rates simultaneously without changing something else — typically the sample size.

To see why this makes sense, imagine a number line representing all possible values of a test statistic. Under H_0, the test statistic is centered at some null value. The rejection region is the set of values far enough from the null to trigger rejection. Making α smaller means shrinking the rejection region — fewer values lead to rejection. But if H_A is true, the test statistic is centered somewhere else. A smaller rejection region means less of the alternative distribution falls inside it, so we are less likely to reject — β goes up.

The only way to reduce *both* α and β at the same time is to collect more data. More data makes the test statistic more precise (smaller standard error), which makes it easier to distinguish between H_0 and H_A — allowing us to keep α small while also keeping β small. This is why sample size planning (Chapter 28) is so important.

26.6 Choosing a significance level

The **significance level** (also called the **discernibility level**) provides the cutoff for the p-value which will lead to a decision of "reject the null hypothesis." The traditional level is 0.05. However, it is sometimes helpful to adjust the significance level based on the application. We may select a level that is smaller or larger than 0.05 depending on the consequences of any conclusions reached from the test.

If making a Type I error is dangerous or especially costly, we should choose a small significance level (e.g., 0.01 or 0.001). If we want to be very cautious about rejecting the null hypothesis, we demand very strong evidence favoring the alternative H_A before we would reject H_0.

If a Type II error is relatively more dangerous or much more costly than a Type I error, then we should choose a higher significance level (e.g., 0.10). Here we want to be cautious about failing to reject H_0 when the null is actually false.

Significance levels should reflect consequences of errors. The significance level selected for a test should reflect the real-world consequences associated with making a Type I or Type II error.

A city water department tests whether the lead content in drinking water exceeds the safe limit. They set H_0: lead level $\leq$ safe limit, and H_A: lead level $>$ safe limit. Should they use $\alpha = 0.01$ or $\alpha = 0.10$?

A Type II error here — failing to detect dangerous lead levels — could have serious public health consequences. A Type I error — declaring the water unsafe when it is actually fine — would cause inconvenience and expense but is far less harmful than allowing people to drink contaminated water. Therefore, the water department should use a larger α (like 0.10) to make it easier to detect real contamination, accepting a slightly higher chance of false alarms.

A social media company wants to test whether a new recommendation algorithm increases average time spent on the platform. They set H_0: no difference, H_A: the new algorithm increases engagement. Should they use $\alpha = 0.01$ or $\alpha = 0.10$?

The stakes here are relatively low. A Type I error means rolling out an algorithm that doesn't actually help — some wasted development effort. A Type II error means keeping the old algorithm when the new one would have been slightly better — a missed opportunity but not a disaster. A standard $\alpha = 0.05$ or even a conservative $\alpha = 0.01$ is reasonable, since the company can easily run another test with more data.

26.7 Visualizing Type II error

The most effective way to build intuition about Type II error is to think about what happens when there *is* a real effect but our test fails to detect it. Simulation makes this concrete.

26.7.1 A simulation thought experiment

Imagine the following scenario. A tutoring program truly raises students' exam scores by an average of 5 points (on a 100-point scale). The standard deviation of exam scores in the population is 15 points. A researcher recruits $n = 20$ students, has them go through the tutoring program, and compares their scores to the known population mean using a one-sample t-test at $\alpha = 0.05$.

Here is what might happen:

- **The truth**: $\mu = 75$ (population mean with tutoring), while $\mu_0 = 70$ (the null hypothesis value without tutoring). The real effect is $\mu - \mu_0 = 5$ points.
- **One simulated study**: The researcher collects 20 scores. Due to natural variability, the sample mean might be $\bar{x} = 73.2$ with $s = 14.8$. The test statistic is:

$$t = \frac{73.2 - 70}{14.8/\sqrt{20}} = \frac{3.2}{3.31} = 0.97$$

The p-value for this test is about 0.17 — not significant at $\alpha = 0.05$. The researcher fails to reject H_0, even though the tutoring program really does work. This is a Type II error.

- **Another simulated study**: A different random sample of 20 students happens to include several who benefited greatly from tutoring. The sample mean is $\bar{x} = 76.8$ with $s = 13.5$. Now:

$$t = \frac{76.8 - 70}{13.5/\sqrt{20}} = \frac{6.8}{3.02} = 2.25$$

The p-value is about 0.018 — significant at $\alpha = 0.05$. This time the researcher correctly rejects H_0. This is a correct decision.

26.7.2 What happens across many studies?

If we repeated this experiment 1,000 times — each time drawing a fresh sample of 20 students from a population where the true effect is 5 points — we would find that roughly 340 of those experiments reject H_0 and roughly 660 do not. That means:

- About 34% of the time, we correctly detect the real effect.
- About 66% of the time, we commit a Type II error.

The Type II error rate is $\beta \approx 0.66$ — alarmingly high! Nearly two-thirds of studies with this design would miss the real effect. This is not because the effect doesn't exist; it's because the sample size is too small relative to the effect size and variability.

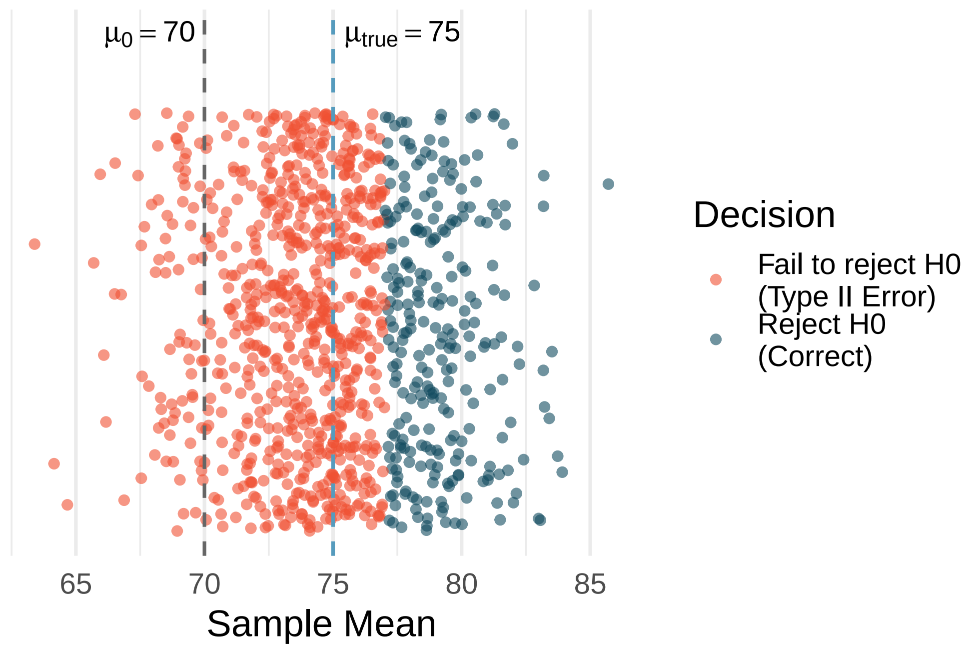


Figure 26.1: Visualization of 1,000 simulated hypothesis tests when the true effect is 5 points. Each dot represents one simulated study. The green dots are studies that correctly rejected H_0 (correct decisions), and the red dots are studies that failed to reject H_0 (Type II errors).

In the tutoring program simulation above, what would happen to the Type II error rate if the researcher increased the sample size from 20 to 100 students? Would β increase, decrease, or stay the same?

Show answer

The Type II error rate would decrease substantially. With 100 students instead of 20, the standard error drops from $15/\sqrt{20} = 3.35$ to $15/\sqrt{100} = 1.50$. The test becomes much more precise, making it easier to distinguish the true mean of 75 from the null value of 70. In fact, with $n = 100$, the power would be about 94%, meaning $\beta \approx 0.06$ — the test would miss the real effect only about 6% of the time.

StatLens: Power Simulation Tool. Use the Power Simulation Tool to explore Type II error visually. Set the true effect size, sample size, significance level, and population variability, then run 1,000 simulated experiments. Watch how the proportion of studies that reject H_0 changes as you adjust each parameter. The simulations that fail to reject H_0 represent Type II errors.

26.8 Factors that affect the Type II error rate

Four key factors determine how likely a study is to commit a Type II error:

26.8.1 1. Effect size

The **effect size** is the magnitude of the difference between the true parameter value and the null hypothesis value. Larger effects are easier to detect.

- If a tutoring program raises scores by 20 points, almost any reasonable study will detect it.
- If a tutoring program raises scores by 2 points, you need a very large or very precise study to detect it.

Intuitively, a large effect "sticks out" from the noise, while a small effect can easily be hidden by random variability.

26.8.2 2. Sample size

Larger samples provide more precise estimates and smaller standard errors. The standard error of the sample mean is $SE = \sigma/\sqrt{n}$, which decreases as n increases. A smaller standard error means the sampling distribution is narrower, making it easier to distinguish the true effect from zero.

More data, fewer missed effects. Increasing the sample size is the most common way to reduce the Type II error rate. This is why large clinical trials with thousands of participants are more likely to detect treatment effects than small pilot studies.

26.8.3 3. Population variability (σ)

When individual observations are highly variable, it is harder to estimate the population mean precisely. The standard error depends directly on σ: high variability means a wide sampling distribution, which makes it harder to distinguish H_A from H_0.

For example, if exam scores range from 20 to 100 with $\sigma = 25$, it is much harder to detect a 5-point effect than if scores range from 60 to 80 with $\sigma = 5$.

A researcher is studying whether a dietary supplement improves reaction time. She can either recruit participants from the general population (where reaction times are highly variable) or from a group of trained athletes (where reaction times are more consistent). Which group would give her study a lower Type II error rate, assuming the supplement has the same true effect in both populations?

Show answer

The group of trained athletes would give a lower Type II error rate because their reaction times are less variable (smaller σ). A smaller σ leads to a smaller standard error, making it easier to detect the true effect of the supplement.

26.8.4 4. Significance level (α)

As we discussed in Section 26.5, a smaller α makes the test more conservative — it demands stronger evidence before rejecting H_0. This directly increases β because the rejection region shrinks.

The table below summarizes these four factors:

Table 26.2: How each factor affects the Type II error rate.

Factor	Change	Effect on β
Effect size	Larger effect	β decreases
Sample size (n)	Larger n	β decreases
Population variability (σ)	Larger σ	β increases
Significance level (α)	Smaller α	β increases

A researcher plans a two-sample t-test to compare a new therapy to a standard therapy. She knows from pilot data that the standard deviation of patient outcomes is about $\sigma = 12$. She is hoping to detect a difference of at least 4 points between the two therapies. Her initial plan calls for 30 patients per group with $\alpha = 0.05$.

The standard error for the difference in means is approximately:

$$SE = \sqrt{\frac{12^2}{30} + \frac{12^2}{30}} = \sqrt{\frac{144}{30} + \frac{144}{30}} = \sqrt{9.6} = 3.10$$

The rejection boundary for a two-sided test is approximately $1.96 \times SE = 1.96 \times 3.10 = 6.07$. Since the true effect is only 4 points — smaller than the rejection boundary of 6.07 — the study will more often than not fail to detect the effect. The Type II error rate will be high.

What could she change to reduce the Type II error rate?

She has several options: (1) increase the sample size to reduce the standard error, (2) try to reduce variability through better measurement or a more homogeneous study population, or (3) increase α to 0.10 (though this increases the Type I error rate). The most practical option is usually to increase the sample size.

26.9 Practical vs. statistical significance revisited

The concepts of Type I and Type II error are closely connected to the distinction between **statistical significance** and **practical significance** that we first encountered in earlier chapters.

Statistical significance means the data provide enough evidence to reject H_0 at our chosen α level. It says: "The effect is probably not zero."

Practical significance means the effect is large enough to matter in the real world. It says: "The effect is big enough to be useful."

These are different questions, and they can give different answers:

Scenario	Statistically significant?	Practically significant?	What happened?
Large effect, large n	Yes	Yes	Ideal outcome
Large effect, small n	Possibly not	Yes	Type II error risk — effect exists but may be missed
Small effect, large n	Yes	No	"Significant but trivial" — very large studies can detect tiny, unimportant effects
Small effect, small n	No	No	No news

The lower-left cell is the Type II error danger zone: a real, important effect exists, but the study is too small to detect it. The upper-right cell represents a different problem: a large study that detects a trivially small effect that has no practical importance.

A study with 50,000 participants finds that a new teaching method raises test scores by 0.3 points on a 100-point scale, with $p = 0.002$. Is this result practically significant?

The result is statistically significant ($p = 0.002 < 0.05$), meaning we have strong evidence that the effect is not zero. However, a 0.3-point improvement on a 100-point scale is almost certainly too small to matter in practice. No school would adopt a new teaching method for a gain of one-third of a point. This is a case of statistical significance without practical significance — the study was so large that it detected a real but trivially small effect.

Statistical significance is not the same as practical significance. A statistically significant result tells you the effect is probably real, but it does not tell you the effect is important. Always consider the *size* of the effect, not just whether it is statistically significant.

A small pilot study with 12 participants finds that a new exercise program reduces resting heart rate by 8 beats per minute, but the result is not statistically significant ($p = 0.09$). Does this mean the exercise program doesn't work?

Show answer

Not necessarily. An 8 bpm reduction in resting heart rate is a clinically meaningful effect. The study may simply have been too small to detect it — this is a potential Type II error. The non-significant result means we don't have enough evidence to be confident the effect is real, but it doesn't prove the program is ineffective. A larger study with more participants would be warranted.

26.10 Chapter review

26.10.1 Summary

A hypothesis test can go wrong in two ways. A **Type I error** occurs when we reject H_0 even though it is true — a false alarm. A **Type II error** occurs when we fail to reject H_0 even though H_A is true — a missed detection. The probability of a Type I error is α (the significance level), and the probability of a Type II error is β.

There is a fundamental tradeoff: for a fixed sample size and effect size, decreasing α increases β, and vice versa. The only way to reduce both error rates simultaneously is to increase the sample size, which makes the test more precise.

Four factors affect the Type II error rate: the effect size (larger effects are easier to detect), the sample size (more data reduces β), the population variability (more variability increases β), and the significance level (smaller α increases β).

Finally, statistical significance and practical significance are not the same thing. A large study can detect trivially small effects, and a small study can miss important effects. Researchers should always consider both the statistical evidence and the practical importance of their findings.

26.11 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Testing for Fibromyalgia.** A patient named Diana was diagnosed with Fibromyalgia, a long-term syndrome of body pain, and was prescribed anti-depressants. Being the skeptic that she is, Diana didn't initially believe that anti-depressants would help her symptoms. However after a couple months of being on the medication she decides that the anti-depressants are working, because she feels like her symptoms are in fact getting better.

a. Write the hypotheses in words for Diana's skeptical position when she started taking the anti-depressants.
b. What is a Type I error in this context?
c. What is a Type II error in this context?

2. **Testing for food safety.** A food safety inspector is called upon to investigate a restaurant with a few customer reports of poor sanitation practices. The food safety inspector uses a hypothesis testing framework to evaluate whether regulations are not being met. If he decides the restaurant is in gross violation, its license to serve food will be revoked.

a. Write the hypotheses in words.
b. What is a Type I error in this context?
c. What is a Type II error in this context?
d. Which error is more problematic for the restaurant owner? Why?
e. Which error is more problematic for the diners? Why?
f. As a diner, would you prefer that the food safety inspector requires strong evidence or very strong evidence of health concerns before revoking a restaurant's license? Explain your reasoning.

3. **True / False.** Determine if the following statements are true or false, and explain your reasoning. If false, state how it could be corrected.

a. If a given value (for example, the null hypothesized value of a parameter) is within a 95% confidence interval, it will also be within a 99% confidence interval.

b. Decreasing the discernibility level (α) will increase the probability of making a Type I error.

c. Suppose the null hypothesis is $p = 0.5$ and we fail to reject H_0. Under this scenario, the true population proportion is 0.5.

d. With large sample sizes, even small differences between the null value and the observed point estimate, a difference often called the effect size, will be identified as statistically discernible.

4. **Online communication.** A study suggests that 60% of college student spend 10 or more hours per week communicating with others online. You believe that this is incorrect and decide to collect your own sample for a hypothesis test. You randomly sample 160 students from your dorm and find that 70% spent 10 or more hours a week communicating with others online. A friend of yours, who offers to help you with the hypothesis test, comes up with the following set of hypotheses. Indicate any errors you see.

$$H_0 : \hat{p} < 0.6 \qquad H_A : \hat{p} > 0.7$$

5. **Estimating π.** In a class activity, each of 100 students experimentally estimates the value of π, 10 separate times. Using the 10 measurements for π (10 values of $\hat{\pi}$), each student calculates a confidence interval for π. In grading the 100 student assignments, the professor marks 7 of the assignments wrong, indicating that the 7 students must have done their experiments or analysis incorrectly because each of the 7 students reported confidence intervals that did not capture the known true value of π, roughly 3.14159. Was the professor correct to mark the assignments wrong for having CIs that did not capture the value of 3.14159? Explain.

6. **Fermenting yeast.** Twenty students work individually in a biology lab to test whether using raw sucrose versus refined sugar will lead to the same yeast fermentation rate. Each student runs a full experiment independently of the other students in the lab. Of the twenty students, twelve are able to reject the null hypothesis and to claim that the fermentation rates are different.

a. Explain what type of error was likely to have occurred in this situation.
b. What change would you suggest that would lower the error rate?

7. **Hypothesis statements.** For each of the research claims below, fill in the value and the direction of the null and alternative hypotheses. That is, complete all aspects of the following hypothesis statements. Additionally, for each item, describe p in words.

$$H_0 : p\_\_\_\_\_\_\_\_ \qquad H_A : p\_\_\_\_\_\_\_\_$$

a. On a pre-test to assess knowledge of the upcoming material, a professor wants to determine if their students know, on average, more than if they were just randomly guessing. The pre-test is 30 multiple choice questions, where each question has 5 possible responses.
b. A standard treatment is known to reduce blood pressure in 32% of patients. A clinical trial is conducted to assess whether a new medical intervention will produce results which are different than the standard treatment, in terms of the percent of patients who will have reduced blood pressure.
c. In the last presidential election 67% of registered voters turned out to vote. Will the next presidential election have a higher turn-out of voters?

Chapter 27

Statistical Power

> **Extended Content.** This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In Chapter 26, we saw that a hypothesis test can miss a real effect — a Type II error. The flip side of this coin is **power**: the probability that a test *correctly* detects a real effect. Power is not just a theoretical concept; it is the foundation of good study design. Before collecting data, researchers need to ask: "Is my study large enough to find the effect I'm looking for?" This chapter defines power, explores the factors that influence it, and shows how simulation can estimate power when formulas are unavailable.

StatLens tools for this chapter: Use the randomization test simulators to explore power empirically. Try running tests with increasing sample sizes and observe how the rejection rate changes — that's power. A dedicated Power Simulation Tool is planned for a future release.

Launch Randomization Test: Difference in Means

27.1 What is power?

In Chapter 26, we defined the Type II error rate β as the probability of failing to reject H_0 when H_A is true. Power is simply the complement of β.

The **power** of a hypothesis test is the probability of rejecting the null hypothesis when the alternative hypothesis is true:

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_A \text{ is true})$$

Power answers the question: "If there is a real effect, how likely is my study to detect it?"

The four possible outcomes of a hypothesis test, with their probabilities, are:

Table 27.1: Probabilities of each outcome in a hypothesis test.

	H_0 is true	H_A is true
Reject H_0	α (Type I error)	$1 - \beta$ (Power)
Fail to reject H_0	$1 - \alpha$ (Correct)	β (Type II error)

Notice the two columns must each sum to 1:

- When H_0 is true: $\alpha + (1 - \alpha) = 1$
- When H_A is true: $(1 - \beta) + \beta = 1$

Power = 1 − β**.** A test with high power (close to 1) is very likely to detect a real effect. A test with low power (close to 0) will often miss real effects, even when they exist.

27.2 Why power matters

Power matters for both scientific and ethical reasons.

27.2.1 Designing studies that can actually detect effects

Imagine spending a year and $500,000 on a clinical trial, only to discover at the end that the study was too small to detect the treatment effect. The trial produces an inconclusive result — a non-significant p-value — even though the treatment actually works. The money is gone, the patients' time was wasted, and the scientific question remains unanswered.

This scenario is unfortunately common. A study with low power is essentially a gamble: you might detect the effect (if you're lucky with your sample), or you might not (if natural variability obscures the signal). Low-powered studies are wasteful because they frequently produce null results that leave everyone uncertain.

27.2.2 Ethical considerations

In medical research, every patient enrolled in a clinical trial is exposed to some degree of risk — side effects, time away from proven treatments, or the inconvenience of participation. If a study is underpowered and unlikely to produce a clear answer, then patients are being exposed to risk with little prospect of scientific benefit. Many institutional review boards (IRBs) now require a power analysis before approving research involving human subjects.

27.2.3 The scientific literature

Low-powered studies distort the scientific literature. When only a small fraction of underpowered studies happen to produce significant results, the published literature overestimates effect sizes (because only the "lucky" studies with unusually large observed effects made it past the significance threshold). This phenomenon is sometimes called the **winner's curse** — the studies that "win" (reject H_0) in a low-power setting tend to have inflated effect sizes.

A researcher wants to study whether a new fertilizer increases crop yield. Based on previous research, she expects the fertilizer to increase yield by about 8%. She plans to compare treated and untreated fields. If her study has power of 0.50, what does this mean in practical terms?

Power of 0.50 means that even if the fertilizer truly increases yield by 8%, there is only a 50% chance that her study will detect it — essentially a coin flip. She has a 50% chance of committing a Type II error and concluding there is no evidence the fertilizer works. This is generally considered unacceptable. She should increase her sample size or consider other design improvements before proceeding.

27.3 Factors affecting power

The same four factors that affect β (Chapter 26) determine power, since power = $1 - \beta$. Here we frame each factor in terms of power rather than Type II error rate.

27.3.1 Sample size: $\uparrow n \rightarrow \uparrow$ power

This is the most important lever researchers have. Larger samples produce smaller standard errors, which make the sampling distribution narrower. A narrower distribution makes it easier to distinguish the alternative from the null.

Power.

The power of the test is the probability of rejecting the null claim when the alternative claim is true.

How easy it is to detect the effect depends on both how big the effect is (e.g., how good the medical treatment is) as well as the sample size.

Consider the blood pressure medication example from Chapter 26. A new medication is expected to lower blood pressure by 3 mmHg compared to a standard medication. The standard deviation of blood pressure change is $\sigma = 12$ mmHg, and we use a two-sided test with $\alpha = 0.05$.

What is the approximate power with $n = 100$ patients per group? With $n = 500$ per group?

With $n = 100$ per group:

The standard error for the difference in means is:

$$SE = \sqrt{\frac{12^2}{100} + \frac{12^2}{100}} = \sqrt{2.88} \approx 1.70$$

The rejection boundary for a two-sided test at $\alpha = 0.05$ is approximately $\pm 1.96 \times 1.70 = \pm 3.33$ mmHg.

If the true effect is -3 mmHg, then the alternative distribution is centered at -3 with standard deviation 1.70. We need to find the probability that a value from this distribution falls below -3.33:

$$Z = \frac{-3.33 - (-3)}{1.70} = \frac{-0.33}{1.70} = -0.20 \quad \rightarrow \quad P(Z < -0.20) \approx 0.42$$

The power is approximately **42%** — far too low.

With $n = 500$ per group:

$$SE = \sqrt{\frac{12^2}{500} + \frac{12^2}{500}} = \sqrt{0.576} \approx 0.76$$

The rejection boundary is $\pm 1.96 \times 0.76 = \pm 1.49$ mmHg.

$$Z = \frac{-1.49 - (-3)}{0.76} = \frac{1.51}{0.76} = 1.99 \quad \rightarrow \quad P(Z < 1.99) \approx 0.977$$

The power is approximately **97.7%** — excellent. With 500 patients per group, we would almost certainly detect a 3 mmHg effect.

27.3.2 Effect size: $\uparrow$ effect $\rightarrow \uparrow$ power

A larger effect is easier to detect. If a drug lowers blood pressure by 10 mmHg instead of 3 mmHg, the signal is much stronger relative to the noise, and even a modest sample size will detect it.

Using the same setup as the worked example above ($\sigma = 12$, $\alpha = 0.05$, $n = 100$ per group), estimate the power to detect a true difference of -8 mmHg instead of -3 mmHg.

Show answer

With $SE = 1.70$ and rejection boundary at -3.33, the alternative is centered at -8. We compute $Z = (-3.33 - (-8))/1.70 = 4.67/1.70 = 2.75$, giving $P(Z < 2.75) \approx 0.997$. The power is about 99.7%. An 8 mmHg effect is easy to detect even with only 100 patients per group.

27.3.3 Significance level: $\uparrow \alpha \rightarrow \uparrow$ power

A larger α means a wider rejection region. With more "room" to reject H_0, the test is more sensitive — but also more prone to false alarms (Type I errors).

Using the blood pressure setup ($\sigma = 12$, $n = 100$, true effect $= -3$ mmHg), compare the power at $\alpha = 0.05$ versus $\alpha = 0.10$.

Show answer

At $\alpha = 0.05$: rejection boundary $= \pm 1.96 \times 1.70 = \pm 3.33$. $Z = (-3.33 - (-3))/1.70 = -0.20$, power ≈ 0.42.
At $\alpha = 0.10$: rejection boundary $= \pm 1.645 \times 1.70 = \pm 2.80$. $Z = (-2.80 - (-3))/1.70 = 0.12$, power ≈ 0.55.
Power increases from 42% to 55% when we relax α from 0.05 to 0.10, but at the cost of a higher false alarm rate.

27.3.4 Variability: $\uparrow \sigma \rightarrow \downarrow$ power

Greater variability in the data means a larger standard error, which spreads out the sampling distribution and makes it harder to distinguish the alternative from the null.

Table 27.2: Factors that affect power. All relationships assume the other factors are held constant.

Factor	Change	Effect on Power
Sample size (n)	Increase n	Power increases
Effect size	Larger effect	Power increases
Significance level (α)	Increase α	Power increases
Population variability (σ)	Decrease σ	Power increases

The researcher's primary tool for increasing power is sample size. Effect size is determined by nature (we don't control how effective a treatment is). Variability can sometimes be reduced through better measurement or study design, but it is largely a feature of the population. The significance level α is usually fixed by convention. That leaves sample size as the main adjustable dial.

27.4 Simulation-based power estimation

When exact power formulas are available (as for t-tests and z-tests), we can compute power mathematically. But many real-world situations involve test procedures where exact formulas are complicated or unavailable — for example, bootstrap tests, permutation tests, or tests involving non-normal data. In these cases, **simulation** offers a straightforward alternative.

27.4.1 The simulation recipe

The idea is simple: if we know (or hypothesize) the true state of the world under H_A, we can simulate the data-collection process many times, run the hypothesis test each time, and count how often we reject H_0.

Here is the step-by-step recipe:

1. **Specify the alternative hypothesis**: Choose the effect size you want to detect. For example: "The true mean difference is 5 points."
2. **Specify the data-generating process**: Define the population(s) from which data will be drawn. For example: "Each observation comes from a normal distribution with mean 75 and standard deviation 15."
3. **Choose the sample size** and **significance level**.

4. **Simulate one experiment**: Draw a random sample of size n from the specified population, compute the test statistic, and determine whether you reject H_0.
5. **Repeat many times**: Run step 4 a total of B times (e.g., $B = 1{,}000$ or $B = 10{,}000$).
6. **Estimate power**: Count the proportion of simulations that rejected H_0. This proportion is the estimated power.

$$\widehat{\text{Power}} = \frac{\text{Number of simulations that reject } H_0}{B}$$

A biologist plans to test whether a new habitat restoration technique increases the average number of bird species observed at a site. She expects the technique to increase the count from a baseline of 12 species to about 15 species, with a standard deviation of 6 species. She will survey $n = 25$ sites using a one-sample t-test at $\alpha = 0.05$.

Describe how she could use simulation to estimate the power of her study.

---

1. **Alternative**: true mean $= 15$, null value $= 12$.
2. **Data-generating process**: draw 25 observations from a normal distribution with mean 15 and standard deviation 6.
3. **Sample size** $n = 25$, **significance level** $\alpha = 0.05$.
4. **One simulated experiment**: generate 25 random values from $N(15, 6)$, compute $\bar{x}$ and s, calculate $t = (\bar{x} - 12)/(s/\sqrt{25})$, find the p-value, and check if $p < 0.05$.
5. **Repeat** 10,000 times.
6. **Estimate power**: suppose 8,247 of the 10,000 simulations rejected H_0. Then $\widehat{\text{Power}} = 8{,}247/10{,}000 = 0.825$.

She would estimate the power at about 82.5%, which exceeds the conventional 80% target.

The beauty of simulation-based power analysis is its flexibility. It works for any test procedure — not just t-tests — and can accommodate complex scenarios like non-normal data, unequal group sizes, missing data, or multi-stage designs.

StatLens: Power Simulation Tool. The Power Simulation Tool implements exactly this recipe. Set the true effect size, sample size, variability, and significance level. Click “Run Simulation” to generate 1,000 simulated experiments. The tool displays the proportion that reject H_0 (the estimated power) and shows a histogram of simulated test statistics with the rejection region shaded.

27.5 Power curves

A single power calculation tells us the power for one specific combination of sample size, effect size, α, and σ. But researchers often want to see how power changes across a range of values. **Power curves** show this relationship graphically.

27.5.1 Power vs. sample size

The most common power curve plots power on the y-axis against sample size on the x-axis, for a fixed effect size, α, and σ. This curve answers: “How many observations do I need to achieve a desired level of power?”

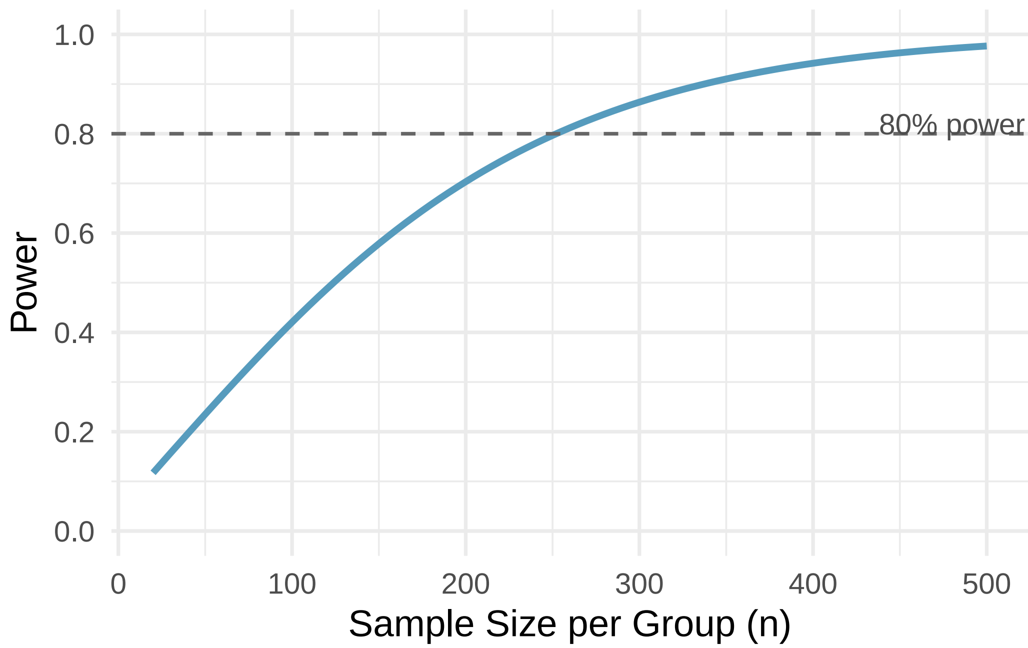


Figure 27.1: A power curve showing power (y-axis) versus sample size per group (x-axis) for the blood pressure example with true effect = 3 mmHg, $\sigma = 12$, $\alpha = 0.05$. The dashed horizontal line marks 80% power; the curve crosses it at approximately $n = 250$ per group.

Key features of power curves:

- **Power starts low** for small sample sizes and **increases toward 1** as n grows.
- The curve has a characteristic **S-shape**: power increases slowly at first, then more rapidly in the middle range, then levels off as it approaches 1.
- The curve **never reaches exactly 1** (there is always some chance of a Type II error, though it becomes negligibly small for very large n).
- The steepest part of the curve — where adding more observations gives the biggest payoff in power — is typically in the middle range.

27.5.2 Power vs. effect size

We can also plot power against effect size, holding n, α, and σ fixed. This curve shows how the detectable effect depends on its magnitude.

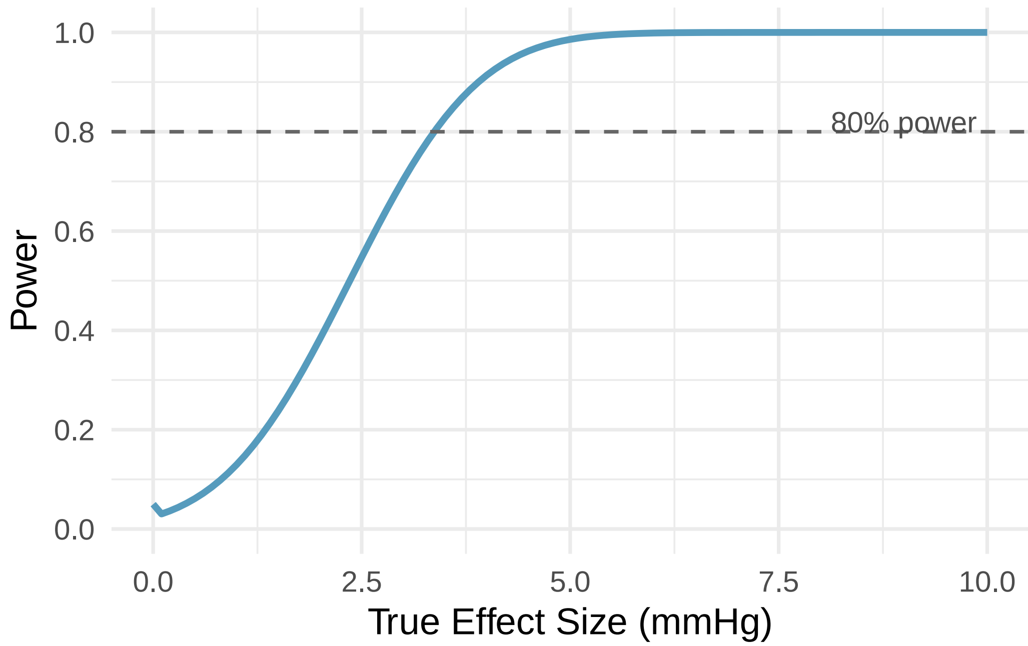


Figure 27.2: A power curve showing power (y-axis) versus true effect size (x-axis) for the blood pressure example with $n = 200$ per group, $\sigma = 12$, $\alpha = 0.05$. Larger effects are easier to detect.

At an effect size of 0, the "power" is just α — this is the Type I error rate, the probability of rejecting H_0 when there is no effect. As the effect size grows, power increases.

Look at the power-vs-sample-size curve description above. If a researcher can only afford to recruit 100 patients per group, approximately what power would she have? Is this adequate?

Show answer

From the curve, at $n = 100$ per group the power is approximately 42%. This is far below the conventional minimum of 80%. The researcher should either find a way to increase the sample size, or reconsider whether the study is worth conducting at this scale.

27.6 Typical power targets

How much power is "enough"? There is no universal answer, but strong conventions have emerged in statistical practice.

The 80% power convention. The most widely used target is **power = 0.80** (80%). This means the study has an 80% chance of detecting the specified effect. Equivalently, $\beta = 0.20$ — a 20% chance of a Type II error.

Some fields or funding agencies require higher power:

- **80% power** is the conventional minimum for most research.
- **90% power** is sometimes required for clinical trials, especially when the consequences of a Type II error are severe.
- **95% power** is occasionally used in high-stakes contexts but requires substantially larger sample sizes.

Why is 80% a reasonable target? Why not 95% or 99%?

The choice of 80% reflects a practical tradeoff. Increasing power beyond 80% requires increasingly large sample sizes, and the gains become smaller. Going from 80% to 90% power might require 30-50% more observations. Going from 90% to 95% might require another 30% or more. At some point, the cost and logistical difficulty of recruiting additional participants outweigh the marginal reduction in Type II error risk.

The 80% target also reflects the observation that in most settings, a Type II error (missing a real effect) is considered less serious than a Type I error (claiming a false effect), so we are comfortable with $\beta = 0.20$ even when $\alpha = 0.05$. If the consequences of missing an effect are severe — for example, in a drug safety trial — a higher power target is appropriate.

Power analysis must be done before data collection. A power analysis conducted *after* data have been collected and analyzed is called **post hoc power analysis** (or "observed power"), and it is widely regarded as misleading. Post hoc power is mathematically determined by the p-value and adds no new information. The time to think about power is during the study design phase.

27.7 Computing power for a two-sample test

While a full derivation of power formulas is beyond the scope of this course, it is useful to see the mechanics of a power calculation for the common case of comparing two group means. This section walks through the logic step by step using the blood pressure medication example.

27.7.1 Setting up the null and alternative distributions

Suppose a pharmaceutical company has developed a new drug to lower blood pressure. They plan a randomized experiment comparing the new drug to a standard medication. The hypotheses are:

- H_0: $\mu_{\text{treatment}} - \mu_{\text{control}} = 0$ (no difference)
- H_A: $\mu_{\text{treatment}} - \mu_{\text{control}} \neq 0$ (there is a difference)

From pilot studies, the standard deviation of blood pressure measurements within each group is about $\sigma = 12$ mmHg. The test will use $\alpha = 0.05$.

Step 1: Determine the null distribution. Under H_0, the difference $\bar{x}_{\text{treatment}} - \bar{x}_{\text{control}}$ has mean 0 and standard error:

$$SE = \sqrt{\frac{\sigma^2}{n} + \frac{\sigma^2}{n}} = \sigma\sqrt{\frac{2}{n}}$$

With $n = 100$ per group: $SE = 12\sqrt{2/100} = 12 \times 0.1414 = 1.70$.

So under H_0, the test statistic approximately follows a normal distribution centered at 0 with standard deviation 1.70.

Step 2: Find the rejection regions. At $\alpha = 0.05$ (two-sided), we reject H_0 when the observed difference falls more than $1.96 \times SE = 1.96 \times 1.70 = 3.33$ units from zero. That is, we reject when the observed difference is below -3.33 or above $+3.33$.

Step 3: Specify the alternative. Suppose the new drug truly lowers blood pressure by 3 mmHg more than the standard medication, so the true difference is $\mu_{\text{treatment}} - \mu_{\text{control}} = -3$.

Under this alternative, the observed difference follows a normal distribution centered at -3 with standard deviation 1.70 (the standard error doesn't change — it depends on σ and n, not the true mean).

Step 4: Compute the power. Power is the probability that the observed difference falls in the rejection region when the alternative is true. The rejection boundary closest to the true effect of -3 is at -3.33. So we need:

$$P(\bar{x}_{\text{treatment}} - \bar{x}_{\text{control}} < -3.33 \mid \text{true difference} = -3)$$

Standardizing:

$$Z = \frac{-3.33 - (-3)}{1.70} = \frac{-0.33}{1.70} = -0.20$$

$$P(Z < -0.20) \approx 0.42$$

The power is approximately **42%**. This is quite low — the study has less than a coin-flip chance of detecting the 3 mmHg effect.

Using the same setup ($\sigma = 12$, $\alpha = 0.05$, true effect $= -3$ mmHg), compute the power when $n = 250$ per group.

Show answer

With $n = 250$: $SE = 12\sqrt{2/250} = 12 \times 0.0894 = 1.073$. Rejection boundary: $1.96 \times 1.073 = 2.10$. Under H_A centered at -3: $Z = (-2.10 - (-3))/1.073 = 0.90/1.073 = 0.84$. $P(Z < 0.84) \approx 0.80$. The power is approximately 80% — right at the conventional target. So about 250 patients per group would be needed to achieve 80% power for this effect size.

27.8 Chapter review

27.8.1 Summary

Power is the probability of correctly rejecting a false null hypothesis: Power $= 1 - \beta$. A well-designed study should have power of at least 0.80 (80%), meaning it has an 80% chance of detecting the effect of interest.

Power depends on four factors: sample size (more data gives more power), effect size (larger effects are easier to detect), significance level (higher α gives more power but more false alarms), and population variability (less variability gives more power). Of these, sample size is the factor most under the researcher's control.

When exact power formulas are unavailable, simulation provides a flexible alternative: generate many datasets under the assumed alternative, run the test on each, and count the proportion that reject H_0. Power curves visualize how power changes with sample size or effect size, helping researchers identify the "sweet spot" where power is adequate without wasting resources.

Power analysis should always be done *before* data collection. Post hoc power analysis — computing power after the study is over — is misleading and widely discouraged.

27.9 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Which is higher?** In each part below, there is a value of interest and two scenarios: (i) and (ii). For each part, report if the value of interest is larger under scenario (i), scenario (ii), or whether the value is equal under the scenarios.

a. The standard error of $\hat{p}$ when (i) $n = 125$ or (ii) $n = 500$.
b. The margin of error of a confidence interval when the confidence level is (i) 90% or (ii) 80%.
c. The p-value for a Z-statistic of 2.5 calculated based on a (i) sample with $n = 500$ or based on a (ii) sample with $n = 1000$.
d. The probability of making a Type II error when the alternative hypothesis is true and the discernibility level is (i) 0.05 or (ii) 0.10.

2. **Same observation, different sample size.** Suppose you conduct a hypothesis test based on a sample where the sample size is $n = 50$, and arrive at a p-value of 0.08. You then refer back to your notes and discover that you made a careless mistake, the sample size should have been $n = 500$. Will your p-value increase, decrease, or stay the same? Explain.

3. **Practical importance vs. statistical discernibility.** Determine whether the following statement is true or false, and explain your reasoning: "With large sample sizes, even small differences between the null value and the observed point estimate can be statistically discernible."

Chapter 28

Sample Size Planning

> **Extended Content.** This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

How many observations do you need? This deceptively simple question is one of the most important in all of applied statistics. Too few observations and the study lacks the power to detect real effects (Chapter 27). Too many and the study wastes time, money, and — in medical contexts — exposes unnecessary participants to risk. This chapter develops formulas and strategies for determining the right sample size before a study begins.

StatLens tools for this chapter: Power Simulation Tool (sample size slider)

28.1 Why plan sample size?

Sample size planning — also called **sample size determination** or **power analysis** — is the process of deciding how many observations to collect before beginning a study. There are three main reasons to plan carefully.

28.1.1 Statistical reasons

As we saw in Chapter 27, the power of a hypothesis test depends critically on the sample size. A study that is too small has low power and a high probability of committing a Type II error — missing a real effect. A study that is too large may have power close to 100%, but the excess observations provide diminishing returns.

Similarly, for estimation, the width of a confidence interval depends on the sample size through the standard error. More observations produce narrower intervals and more precise estimates. Sample size planning for confidence intervals asks: "How many observations do I need to achieve a desired margin of error?"

28.1.2 Ethical reasons

In medical research, participants are exposed to potential risks: side effects, discomfort, time away from proven treatments. If a study is underpowered and unlikely to produce useful results, participants bear risk without commensurate benefit. Many institutional review boards (IRBs) and research ethics committees require a sample size justification as part of the study approval process.

At the other extreme, an unnecessarily large study exposes more people to risk than needed. If a treatment has a large, harmful side effect, ethical principles demand that we detect it with as few participants as possible.

28.1.3 Practical reasons

Research costs money. Each additional participant means more time, more supplies, more lab work, more data entry. A well-planned sample size ensures that resources are used efficiently — enough to answer the research question, but not so many that the budget is strained or the timeline is unrealistic.

Plan your sample size before collecting data. Determining the sample size after data have been collected defeats the purpose. The goal is to ensure that the study *will have* adequate power or precision, not to verify that it *did*.

28.2 Sample size for estimating a mean

One of the most common sample size questions is: "How many observations do I need to estimate a population mean with a given margin of error?"

28.2.1 The margin of error formula

Recall that a confidence interval for a population mean μ takes the form:

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

where z^* is the critical value for the desired confidence level (e.g., $z^* = 1.96$ for 95% confidence) and σ is the population standard deviation. The **margin of error** (ME) is the "plus-or-minus" part:

$$ME = z^* \cdot \frac{\sigma}{\sqrt{n}}$$

If we want the margin of error to be no larger than some target value, we can solve for n:

$$ME = z^* \cdot \frac{\sigma}{\sqrt{n}} \quad \Longrightarrow \quad \sqrt{n} = \frac{z^* \cdot \sigma}{ME} \quad \Longrightarrow \quad n = \left(\frac{z^* \cdot \sigma}{ME}\right)^2$$

The **sample size for estimating a mean** with margin of error ME at confidence level C is:

$$n = \left(\frac{z^* \cdot \sigma}{ME}\right)^2$$

where z^* is the critical value corresponding to confidence level C (e.g., $z^* = 1.96$ for 95% confidence) and σ is the population standard deviation.

Always round up. When the formula gives a non-integer answer, always round *up* to the next whole number. Rounding down would give a margin of error slightly larger than desired.

A university wants to estimate the average amount of student loan debt among its graduates to within \$1,500 with 95% confidence. Previous studies suggest the standard deviation of student loan debt is about \$12,000. How many graduates should they survey?

We have:

- $ME = 1{,}500$
- $z^* = 1.96$ (for 95% confidence)
- $\sigma = 12{,}000$

$$n = \left(\frac{1.96 \times 12{,}000}{1{,}500}\right)^2 = \left(\frac{23{,}520}{1{,}500}\right)^2 = (15.68)^2 = 245.86$$

Rounding up, the university should survey $n = 246$ **graduates**.

A health researcher wants to estimate the average daily calorie intake of adults in a city to within 50 calories with 99% confidence. From national data, the standard deviation of daily calorie intake is approximately 500 calories. How large a sample does she need?

Show answer

With $ME = 50$, $z^* = 2.576$ (for 99% confidence), and $\sigma = 500$: $n = (2.576 \times 500/50)^2 = (25.76)^2 = 663.6$. Rounding up, she needs $n = 664$ participants.

28.2.2 Where does σ come from?

There is a practical catch in the formula: we need to know the population standard deviation σ before we collect data. But if we already knew σ, we would know more about the population than we actually do. In practice, σ is estimated from:

- **Previous studies**: Published research on similar populations often reports standard deviations that can be used for planning.
- **Pilot studies**: A small preliminary study can provide a rough estimate of σ.
- **Expert judgment**: Subject-matter experts may have reasonable estimates based on experience.
- **Conservative estimates**: When in doubt, use a larger estimate of σ. This produces a larger (more conservative) sample size, ensuring the margin of error target is met even if the true variability is somewhat less than assumed.

Using s from pilot data. If you estimate σ from a pilot study with a small sample, the estimate may be unreliable. When using pilot data, it is wise to inflate the estimate somewhat — for example, using the upper end of a confidence interval for σ — to provide a safety margin.

Referring to the student loan debt example, suppose the \$12,000 standard deviation came from a pilot survey of 30 graduates. The sample standard deviation was $s = 12{,}000$. Should the researcher use exactly $\sigma = 12{,}000$ in the sample size formula?

With only 30 observations, the estimate of σ is itself uncertain. A 95% confidence interval for σ would be roughly \$9,600 to \$16,000. To be safe, the researcher might use $\sigma = 14{,}000$ or even $\sigma = 16{,}000$ in the formula, which would give $n = (1.96 \times 14{,}000/1{,}500)^2 = 335$ or $n = (1.96 \times 16{,}000/1{,}500)^2 = 437$. Using the larger value ensures the margin of error target is likely met.

28.3 Sample size for estimating a proportion

When estimating a population proportion p, the confidence interval takes the form:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

The margin of error is:

$$ME = z^* \sqrt{\frac{p^*(1-p^*)}{n}}$$

where p^* is a planning value for p (since we don't know p yet — that's what we're trying to estimate). Solving for n:

$$n = \left(\frac{z^*}{ME}\right)^2 \cdot p^*(1-p^*)$$

The **sample size for estimating a proportion** with margin of error ME at confidence level C is:

$$n = \left(\frac{z^*}{ME}\right)^2 \cdot p^*(1 - p^*)$$

where z^* is the critical value for confidence level C and p^* is a planning estimate of the proportion.

28.3.1 Choosing p^*

The value $p^*(1 - p^*)$ is largest when $p^* = 0.5$ and smallest when p^* is near 0 or 1. This means:

- If you have a reasonable estimate of p from prior research, use it as p^*.
- If you have no idea what p might be, **use** $p^* = 0.5$. This gives the largest (most conservative) sample size, ensuring the margin of error target is met regardless of the true proportion.

A political polling firm wants to estimate the proportion of voters who support a ballot measure to within 3 percentage points with 95% confidence. They have no prior information about voter support. How many voters should they survey?

With no prior information, we use $p^* = 0.5$ for the most conservative estimate:

- $ME = 0.03$ (3 percentage points)
- $z^* = 1.96$
- $p^* = 0.5$

$$n = \left(\frac{1.96}{0.03}\right)^2 \times 0.5 \times 0.5 = (65.33)^2 \times 0.25 = 4{,}268.4 \times 0.25 = 1{,}067.1$$

Rounding up, they need $n = 1{,}068$ **voters**.

This is why political polls typically survey about 1,000 people — it gives roughly a $\pm 3\%$ margin of error at 95% confidence.

A quality control manager wants to estimate the proportion of defective items in a production run to within 1 percentage point with 95% confidence. Based on historical data, the defect rate is usually around 4%. How large a sample should she inspect?

Show answer

With $ME = 0.01$, $z^* = 1.96$, and $p^* = 0.04$: $n = (1.96/0.01)^2 \times 0.04 \times 0.96 = (196)^2 \times 0.0384 = 38{,}416 \times 0.0384 = 1{,}475.2$. Rounding up, she needs $n = 1{,}476$ items. Note: if she had used $p^* = 0.5$ instead, she would have gotten $n = 9{,}604$ — a much larger and unnecessarily conservative sample. Using prior information about p can save a lot of resources.

Compare the required sample sizes when $p^* = 0.5$ versus $p^* = 0.10$ for a 95% confidence interval with $ME = 0.04$.

With $p^* = 0.5$:

$$n = \left(\frac{1.96}{0.04}\right)^2 \times 0.5 \times 0.5 = (49)^2 \times 0.25 = 2{,}401 \times 0.25 = 600.25 \to n = 601$$

With $p^* = 0.10$:

$$n = \left(\frac{1.96}{0.04}\right)^2 \times 0.10 \times 0.90 = 2{,}401 \times 0.09 = 216.09 \to n = 217$$

Using prior knowledge that p is near 0.10 reduces the required sample from 601 to 217 — a 64% reduction. This illustrates why it is valuable to use a good planning estimate when one is available.

28.4 Sample size for comparing two groups

When comparing two group means, the sample size calculation involves the power framework from Chapter 27. The goal is to find n such that the test has at least 80% power (or another target) to detect an effect of a specified size.

28.4.1 Conceptual approach

The idea is the same as in the one-sample case, but now we need to account for two sources of variability (one from each group). For a two-sample t-test comparing means with equal group sizes (n per group), equal standard deviations (σ), and $\alpha = 0.05$ (two-sided):

$$SE = \sigma\sqrt{\frac{2}{n}}$$

To achieve 80% power, we need the true effect size to be about $2.8 \times SE$ away from the null value.[1] Setting the minimum detectable effect δ equal to $2.8 \times SE$ and solving for n:

$$\delta = 2.8 \times \sigma\sqrt{\frac{2}{n}} \quad \Longrightarrow \quad n = 2\left(\frac{2.8 \cdot \sigma}{\delta}\right)^2$$

Returning to the blood pressure example from Chapter 27: a new drug is expected to lower blood pressure by at least 3 mmHg compared to a standard medication. The standard deviation is $\sigma = 12$ mmHg. How many patients per group are needed for 80% power with $\alpha = 0.05$?

Using the formula:

$$n = 2\left(\frac{2.8 \times 12}{3}\right)^2 = 2\left(\frac{33.6}{3}\right)^2 = 2 \times (11.2)^2 = 2 \times 125.44 = 250.88$$

Rounding up, $n = 251$ **patients per group** — a total of 502 patients.

This matches the result we found in Chapter 27 when we computed the power for various sample sizes and found that $n \approx 250$ per group gave approximately 80% power.

A psychologist plans a study comparing reaction times under two conditions. She expects a difference of 25 milliseconds, with a standard deviation of 60 ms in each group. How many participants does she need per group for 80% power at $\alpha = 0.05$?

Show answer

Using the formula: $n = 2(2.8 \times 60/25)^2 = 2(6.72)^2 = 2 \times 45.16 = 90.32$. Rounding up, she needs $n = 91$ per group, or 182 total participants.

28.4.2 Different power targets

The constant 2.8 in the formula above is specific to 80% power and $\alpha = 0.05$ (two-sided). For other targets:

Table 28.1: Multiplier values ($z_{\alpha/2} + z_\beta$) for common power and significance level combinations.

Power	$\alpha = 0.05$ (two-sided)	$\alpha = 0.01$ (two-sided)
80%	2.80	3.36
90%	3.24	3.86

[1] The 2.8 comes from $z_{0.025} + z_{0.20} = 1.96 + 0.84 = 2.80$, where $z_{0.025}$ defines the rejection boundary and $z_{0.20}$ defines the 80% power point on the alternative distribution.

Power	$\alpha = 0.05$ (two-sided)	$\alpha = 0.01$ (two-sided)
95%	3.60	4.20

Using Table 28.1, how many patients per group would be needed in the blood pressure example ($\sigma = 12$, $\delta = 3$) for **90% power** at $\alpha = 0.01$?

Show answer

Using the multiplier 3.86: $n = 2(3.86 \times 12/3)^2 = 2(15.44)^2 = 2 \times 238.4 = 476.8$. Rounding up, $n = 477$ per group, or 954 total — nearly double the 502 needed for 80% power at $\alpha = 0.05$. Higher power and stricter α both demand more observations.

28.5 Practical considerations

Formulas give a starting point, but real-world sample size planning involves several additional considerations.

28.5.1 Budget and resources

The formula might say you need 500 participants, but your budget only supports 200. In this case, you have several options:

- **Accept lower power** and acknowledge the study may not detect small effects.
- **Narrow the research question** to focus on larger effects that require fewer observations to detect.
- **Reduce variability** through more precise measurements, a more homogeneous study population, or a paired design (if applicable).
- **Seek additional funding** if the research question is important enough.
- **Don't run the study** if the available sample size gives unacceptably low power. A study that is almost certain to produce null results wastes everyone's time and resources.

Sometimes the best decision is not to run the study. If a power analysis shows that the achievable sample size gives power below 50%, the study is more likely to miss a real effect than to detect one. Researchers should seriously consider whether conducting such a study is worthwhile.

28.5.2 Recruitment and attrition

Not everyone you recruit will complete the study. Participants drop out, miss appointments, provide unusable data, or are lost to follow-up. The sample size formula tells you how many *completed* observations you need. You should plan to recruit more participants to account for expected attrition.

A longitudinal health study requires $n = 300$ completed observations. Based on similar studies, the researchers expect 15% attrition. How many participants should they recruit initially?

If 15% will be lost, then 85% of the recruited participants will complete the study. Let n_{recruit} be the number to recruit:

$$0.85 \times n_{\text{recruit}} = 300 \quad \Longrightarrow \quad n_{\text{recruit}} = \frac{300}{0.85} = 352.9$$

They should recruit **353 participants** to end up with approximately 300 completed observations.

28.5.3 Pilot studies

A **pilot study** is a small preliminary study conducted before the main study. Pilot studies serve several purposes relevant to sample size planning:

- **Estimate** σ (or p) for use in the sample size formula.
- **Test procedures**: Identify problems with the survey instrument, experimental protocol, or data collection process.
- **Estimate attrition rates**: How many participants drop out or provide unusable data?
- **Check feasibility**: Is it realistic to recruit the number of participants the formula requires?

Pilot studies typically involve 10-30 participants — enough to get rough estimates but not so many that significant resources are consumed before the main study begins.

Pilot study data should not be combined with main study data (at least not without careful statistical adjustment). The pilot study is used to *design* the main study. Combining the two can bias the results because the main study design was chosen based on the pilot data.

28.5.4 When formulas don't apply: simulation

The sample size formulas in this chapter assume specific test procedures (z-tests and t-tests) and specific distributional assumptions (normality). When these conditions don't hold — for example, when using a bootstrap or permutation test, or when the data are highly skewed — simulation provides an alternative.

The simulation approach for sample size planning is an extension of the simulation-based power estimation from Chapter 27:

1. **Choose a candidate sample size** n.
2. **Simulate** many datasets of size n under the assumed alternative.
3. **Run the analysis** on each simulated dataset.
4. **Compute the proportion** that reject H_0 — this is the estimated power for that n.
5. **Repeat** for different values of n until you find the smallest n that achieves the desired power.

This approach is computationally intensive but extremely flexible. It can handle any test procedure, any data distribution, and any study design.

A researcher wants to use a permutation test (Chapter 7) to compare two groups, but the data are expected to be heavily right-skewed. Formulas for the t-test may not apply. How could she determine the sample size?

She would use simulation:

1. Specify the alternative: for example, Group A has median 50 with a right-skewed distribution, and Group B has median 60 with the same shape.
2. For a candidate n (say, $n = 40$ per group), simulate 10,000 experiments: draw 40 values from each group's assumed distribution, run a permutation test, and record whether H_0 is rejected.
3. Compute the proportion of rejections — this is the estimated power at $n = 40$.
4. Repeat for $n = 50, 60, 80, 100, \ldots$ and find the smallest n where power ≥ 0.80.

This avoids any assumption about normality and accounts for the specific properties of the permutation test.

StatLens: Power Simulation Tool (Sample Size Slider). Use the Power Simulation Tool's sample size slider to explore the relationship between sample size and power interactively. Set the true effect size and variability, then slide the sample size up and down to see how power changes. The tool will highlight when you cross the 80% power threshold.

28.6 Summary of sample size formulas

The table below collects the key formulas from this chapter for quick reference.

Table 28.2: Summary of sample size formulas. In all cases, round the result up to the next whole number.

Goal	Formula	Key inputs
Estimate a mean to within ME	$n = \left(\frac{z^* \cdot \sigma}{ME}\right)^2$	z^*, σ, ME
Estimate a proportion to within ME	$n = \left(\frac{z^*}{ME}\right)^2 \cdot p^*(1-p^*)$	z^*, p^*, ME
Compare two means (80% power, $\alpha = 0.05$)	$n = 2\left(\frac{2.8 \cdot \sigma}{\delta}\right)^2$ per group	σ, δ (min. detectable effect)

These formulas are starting points, not final answers. Always adjust for attrition, consider practical constraints, and verify assumptions. When in doubt, simulation provides a robust alternative.

28.7 Chapter review

28.7.1 Summary

Sample size planning is a critical step in study design. For estimating a population mean, the required sample size is $n = (z^* \cdot \sigma / ME)^2$, where ME is the desired margin of error. For estimating a proportion, $n = (z^*/ME)^2 \cdot p^*(1-p^*)$, where p^* is a planning value for the proportion (use $p^* = 0.5$ if unknown). For comparing two group means with 80% power, $n = 2(2.8\sigma/\delta)^2$ per group, where δ is the minimum detectable effect.

In all cases, the formula requires prior information about variability (σ or p), which may come from previous studies, pilot data, or conservative estimates. Results should always be rounded up and adjusted for expected attrition.

When standard formulas do not apply — for example, with non-normal data, complex study designs, or non-standard test procedures — simulation-based power analysis offers a flexible alternative. The researcher simulates many datasets under the assumed alternative, runs the planned analysis on each, and identifies the sample size that achieves the desired power.

Practical considerations — budget, recruitment feasibility, attrition, and ethical constraints — always play a role in the final sample size decision. Sometimes the most honest conclusion of a power analysis is that the study cannot be conducted at an adequate scale, and resources should be directed elsewhere.

28.8 Exercises

Exercises for this chapter are under development.

Chapter A

Datasets

This appendix will catalog all datasets used in the textbook with descriptions, sources, and download links.

Datasets used in this textbook are drawn primarily from:

- The openintro R package (400+ datasets)
- The IMS dataset collection
- Public sources (Data.gov, UCI Machine Learning Repository)
- Local and regional datasets from the University of Wisconsin-La Crosse

All datasets are available in CSV format for use with any statistical software.

Chapter B

StatLens Quick Reference

StatLens is a free, browser-based statistics toolkit used throughout this textbook. This appendix provides a quick reference for all available tools.

B.1 Distribution Calculators

Tool	URL	Use For
Normal	Normal Calculator	Probabilities and percentiles for normal distributions
t	t Calculator	t-distribution probabilities (Ch 11-12)
Chi-Square	Chi-Square Calculator	Chi-square probabilities (Ch 14–15)
F	F Calculator	F-distribution probabilities (Ch 16)
Binomial	Binomial Calculator	Binomial probabilities (Ch 24)

B.2 Simulation Tools

Tool	URL	Use For
Bootstrap CI: One Sample	Bootstrap Mean	Confidence intervals for a mean (Ch 8)
Bootstrap CI: One Proportion	Bootstrap Proportion	CIs for a proportion (Ch 8, 13)
Bootstrap CI: Paired	Bootstrap Paired	CIs for paired differences (Ch 8)
Bootstrap CI: Two Samples	Bootstrap Two Means	CIs for difference in means (Ch 8)
Randomization: Diff in Means	Randomization Means	Hypothesis test for two means (Ch 7)
Randomization: One Proportion	Randomization Prop	Hypothesis test for one proportion (Ch 7)
Randomization: Diff in Proportions	Randomization Props	Hypothesis test for two proportions (Ch 7)
Randomization: Chi-Square	Randomization Chi-Sq	Randomization test for independence (Ch 15)
Bootstrap CI: Slope	Bootstrap Slope	CI for regression slope (Ch 19)

B.3 Conceptual Demos

Tool	URL	Use For
Sampling Distributions	Sampling Dist	Visualizing sampling variability (Ch 6, 9)
CI Coverage	CI Coverage	Understanding confidence level (Ch 8)

B.4 Explore Data

Tool	URL	Use For
Descriptive Statistics	Descriptive	Summary stats, histograms, boxplots (Ch 3-4)
Regression	Regression	Scatterplots, regression lines (Ch 18–20)
Categorical Data	Categorical	Bar charts, contingency tables (Ch 3, 14–15)

B.5 Keyboard Shortcuts

All StatLens pages support keyboard navigation:

Key	Action
`1`	Generate +1 sample/resample
`2`	Generate +10
`3`	Generate +100
`4`	Generate +1000
`0`	Reset simulation
`Space`	Play/pause animation
`?`	Show keyboard help
`Esc`	Close dialog

B.6 URL Parameters

StatLens supports URL parameters for reproducible demonstrations:

```
?seed=42          Seeded random number generator (deterministic results)
?n=30             Sample size
?data=1,2,3,4,5   Inline data (comma-separated)
?ci=95            Confidence level (%)
```

Example: Bootstrap CI with seed

Chapter C

Statistical Tables

This appendix will include standard normal (z), t-distribution, and chi-square tables for reference. With StatLens's distribution calculators available, tables are provided as a convenience for exams and offline use.

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